User Manual

Reporting & Data Visualization

This manual provides an overview of reporting and visualization features, including Financial Reporting, IA ASK AI, SmartInsight, and Distribution Reports.

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The reporting system

Introduction

CCH Tagetik has a reporting system based on specifically designed Excel sheets called *forms*. Using these forms, users can interact with the application databases to which they have access to insert, extract, consult and compare data.

These forms can be used both from the CCH Tagetik web interface and in Excel directly. Some jobs, like planning and designing a form, are carried out exclusively in Excel, while others, like running a form to navigate data, can be run both from Excel and from the CCH Tagetik Web interface. For more details on web forms, see "Web forms" on page 12.

A dedicated add-in must be installed to use reporting functions in Excel (see "Excel add-in: technical features" on page 14).

Ways of working on forms

In CCH Tagetik, forms can be used:

- as reports (also called output forms), to consult data from the database
- as data entry forms, to enter data in the database

However, regardless of the purpose for which the form was designed, the run mode and actions that can be run in the form depend on the tool being used.

Element	Working mode	Actions on the form
Excel Add-in	Allows you to run: • all types of form in design mode • data entry forms in data entry mode • all types of form in navigation mode	• Creation and management of the structure of standard and transactional forms. • Data entry and form navigation.
Web interface	Allows you to run: • only data entry forms in data entry mode from the contributor cockpit • all types of form in navigation mode from the Forms tile.	• Creation and management of the structure of transactional forms. • Data entry and form navigation. • Inclusion of data entry forms as process tasks.

***Note**: however, the general settings on forms are run in the CCH Tagetik settings (see Settings window).*

Form types

Depending on how they are used, forms can ideally be divided into two fundamental types:

Type	Description
Report	These forms are used to extract and view data that is directly present in the database or calculated data. They are not editable and can be either shared or ad hoc (Form accessibility ("Form accessibility" below).
Data entry forms	These forms are used to enter data in application databases. The use of these forms depends on the user's role and restriction on data handled by the form. They are divided into two groups: • **Standard forms**: allow data to be entered or modified in the database related to the dimensions of and can also be used to extract and consult data, i.e. as reports.. • **Transactional form** :they allow new elements to be created and data to be inserted or modified in the database. They also make it possible to manage dimensions other than standard dimensions and custom dimensions (e.g. SQL formulas), in order to increase the level of detail of the data entered. (see "Elements of the transactional forms model" on page 283).

Form accessibility

Forms can be saved, and therefore made accessible, in several ways.

Accessibility	Description
Shared	Report or data entry form saved in the application database and therefore accessible to all users with the relative rights. They can be exported and saved locally, thus becoming ad hoc.
Ad hoc	Report forms saved on a local computer by a specific user and therefore accessible to that user only. They can be loaded onto the application database and thus become shared. They are created specifically for data consultation.

The function of forms

Depending on their settings, forms can be classified as follows:

- *system form*, i.e. forms predefined by the administrator which can be used as models for developing customised reports. They are not editable, but can be copied and edited to be adapted to various business needs.
- *user form*, or forms created and managed entirely by the user according to their needs.

Web forms

Web forms are forms which can both be opened in the Excel add-in and be run from the CCH Tagetik web interface. They allow the contributor user to access information contained in the database and to perform data entry jobs without having to use the Excel add-in.

All forms are created in Excel and potentially enabled so they are also available as web forms (see "Enable web forms by default" on page 19).

***IMPORTANT**: compared with forms accessible from Excel, web forms have certain limitations (see "Features supported by web forms" on page 19).*

Form advanced selection

Advanced selection rules can be set in the form.

Selection type	Description
Static forms	Manage a fixed structure, i.e. data from the database called up by the filters set in the matrix (see "Form matrices" on page 100).
Dynamic forms	Manage a structure that can change over time, i.e. elements calculated and called up when the form is run using advanced selection rules (see "Dynamic matrices" on page 132).

Origin of the data

Depending on the project's needs, Financial Workspace data, on CCH Tagetik dimensions, or Analytical Workspace data can be managed in forms. Analytical Workspaces are used when the project is useful for creating datasets containing customised dimensions that supplement the CCH Tagetik dimensions.

For more details on Financial Workspaces, see The dimensions of CCH Tagetik.

For more details on Analytical Workspaces, see "Forms on Analytical Workspace data" on page 324.

Reporting: user roles

User roles associated with reporting

A user's ability to design, modify and navigate forms of any type depends on the roles associated with the user in the specific application database. A list of the system roles, correlated with reporting jobs, is provided below. For more details, see Roles.

Role	Description	Licence requested
$ADMINISTRATOR	Has administrator privileges for an application database and can: • design data entry forms and reports • define financial statement templates	Administrator
$USER	Has contributor privileges and, based on the reading and editability rights for the entities for which they are responsible, can: • enter data from data entry forms • run reports and view the data they contain	Power User
$ALL	Can access all $ADMINISTRATOR and $USER functions.	Administrator
$DEREPORT	Can do the following: • run reports • enter data in data entry forms • run data processing included in data entry forms or configured in a process workflow.	Data Entry & Reporting or higher

***Note**: custom roles can be created for specific needs (e.g. create a contributor role with the possibility of creating ad hoc forms).*

Specific dimensions for reporting

What they are for

With the reporting module, data can be fine tuned using the module's specific Amount Type and Period Length dimensions.

Amount type

This dimension specifies the type of data displayed by the matrix and, for data entry forms, the type of data that the user can insert. This dimension is not mandatory.

The dimension's attributes are provided below.

Element	Description
Original in Entity currency	Indicates an amount expressed in the entity currency. This is the value indicated in the process for the entity and, if data are gathered outside of Tagetik, corresponds to the currency inserted by the user in their own system (ERP or Data Warehouse). Default value.
Original amount in transaction currency	Indicates an amount expressed in the currency with which the transaction is carried out (may be different from the entity's default currency).
Description	Allows you to insert notes or comments.

Element	Description
Proportional in Entity currency	Indicates amount types calculated by the system during the consolidation process and allows you to see the corresponding amount for each step. To generate the proportional data, the data processing applies the entity's proportionality percentage, as deployed in the consolidation area, to the original data (expressed in the entity currency).
Converted into Consolidation currency	The amount converted into consolidation currency is generated on the basis of the proportional amount type by multiplying the values by the consolidation currency FX rate.
Consolidable in Consolidation currency	The consolidable amount type is calculated by copying the converted amount, as is the case with consolidated entities using the integrated or proportional method, or by generating the equity evaluation journals for consolidated entities using the equity method.
Converted in Transaction currency	Indicates the amount types calculated by the system during the consolidation process and allows you to see the corresponding amount in the transaction currency for each step.
Proportional in Transaction currency	
Consolidable in Transaction currency	

The Period Length dimension

By default, CCH Tagetik saves data entered by the user on a cumulative basis, i.e. through the cumulative addition of amounts. However, data may be available to the user by period (monthly, bimonthly, etc.). In such cases, the user can calculate and then enter the cumulative data. However, this method is also liable to possible calculation or data entry errors. To prevent this situation, the specific Report dimension can be used, namely the **Period Length** dimension. The user inserts the period data in the data entry forms and CCH Tagetik calculates the corresponding cumulative values.

Example

This example shows the conversion of period data into cumulative data.

Period data	Cumulative data
• Q1 (January - March) = 100 • Q2 (April - June) = 150 • Q3 (July - September) = 250 • Q4 (October - December) = 300	• March = Q1 = 100 • June = Q1 + Q2 = 250 • September = Q1 + Q2 + Q3 = 500 • December = Q1 + Q2 + Q3 + Q4 = 800

To use the Period Length dimension, it is necessary to use length LUN_3 - Quarter.

Period Length values

The values for this dimension are as follows:

Element	Description
Cumulative (LUN_0)	Default value in the CPM (Corporate Performance Management). The system does not carry out any calculation and the amount shown coincides with the amount in the database. In both the CPM and Analytical Workspace multidimensional matrices, the cumulative data are calculated based on the figures from the end of the previous period to the current period.
Monthly (LUN_1)	The cumulative data are calculated taking into account the amount of period n (the period set as a filter in the template) and the amount of period n-1. Generally used for scenarios whose open periods are 01, 02, 03, ... 12.
Bimonthly (LUN_2)	The cumulative data are calculated taking into account the amount of period n and the amount of period n-2. Generally used for scenarios whose open periods are 02, 04, 06, 08, 10 and 12.
Quarterly (LUN_3)	The cumulative data are calculated taking into account the amount of period n and the amount of period n-3. Used for scenarios whose open periods are from 03, 06, 09 and 12.

Element	Description
Every four months (LUN_4)	The cumulative data are calculated taking into account the amount of period n and the amount of period n-4. Generally used when weeks are managed as part of the periods.
Every five months (LUN_5)	The cumulative data are calculated taking into account the amount of period n and the amount of period n-5. Generally used when weeks are managed as part of the periods.
Half Year (LUN_6)	The cumulative data are calculated taking into account the amount of period n and the amount of period n-6. Generally used for scenarios whose open periods are 06 and 12.
Annual (LUN_12)	The cumulative data are calculated taking into account the amount of period n and the amount of period n-12. Generally used for scenarios that only have period 12.
Quarter to date (LUN_QTD)	The cumulative data are calculated from the end of the previous quarter to the current period.
Semester to date (LUN_STD)	The cumulative data are calculated from the end of the previous half-year period to the current period.
Life to date (LUN_YTD)	The cumulative data are calculated from minus infinity to the current period. Useful for multi-year projects.

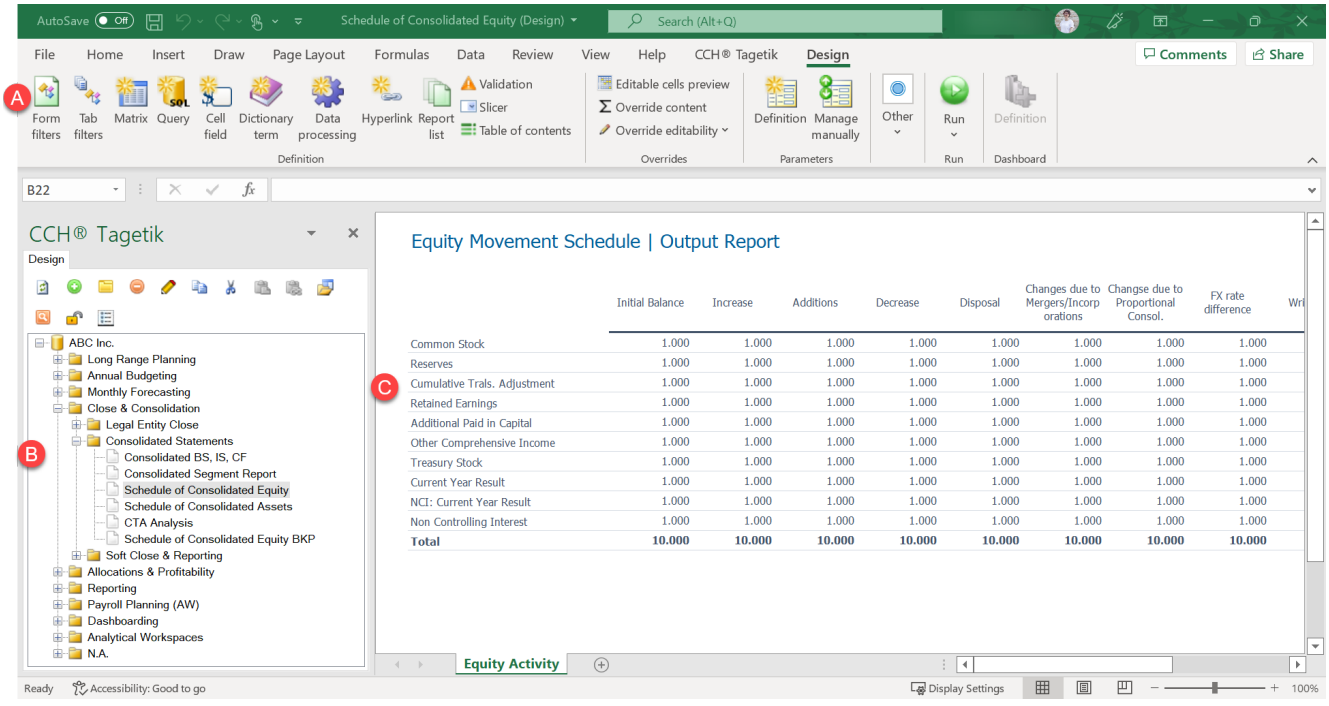
Period Length is not an obligatory dimension. If not specified, by default the system uses cumulative (LUN_0) in the CPM and monthly (LUN_1) in Analytical Workspaces.

Introduction to the Excel interface

Introduction

Once the dedicated add-in is installed, the **CCH ® Tagetik** tab appears among the standard Excel tabs. Other tabs and windows can be opened from the toolbar on this tab, to run all the reporting functions enabled for the connected user.

What it looks like



Element	Description
A	Tool bar
B	Left-hand panel
C	Work desktop

Tool bar

Allows you to access all of CCH Tagetik's reporting functions and view only the buttons related to the functions enabled for the connected user.

Left-hand panel

Shows the elements needed to work in the various reporting function macros contained in the **View** section of the toolbar. In particular, you can view the following tabs:

Element	Description
Design	Displays the folders of the application databases for which the user is responsible, where the administrator can create, edit and delete forms. If the user is connected to several databases, it also allows them to select the one they want to work on.
Navigation	Displays the list of available reports, allows you to run them and consult data extracted from the database. If the user is connected to several databases, it also allows them to select the one they want to work on.
Data Entry	Displays the list of data entry reports created previously, available for the selected process, scenario and period. Allows you to run them to enter data in the database. If the user is connected to several databases, it also allows them to select the one they want to work on.
Open items	Displays the documents currently opened on the work desktop, including those from different operational databases, organised hierarchically, and allows you to access them quickly. The currently selected item has a grey background. By right clicking an individual item, you can access any functions available for that item.
workflow	Only available if enabled, allows you to view and follow the workflow to enter data into the database.

Buttons on the left-hand panel

Depending on the selected tab, the left-hand panel shows some of the following buttons.

Element	Description
	Update the structure of the listed forms.
	Create a form.
	Run the form and open the windows to select the necessary parameters.
	Create a node within the selected node or at the same level as the selected form.
	Delete the selected form or node.
	Edit the name of the selected form or node.
	Copy the selected form or item.
	Cut the selected form or item so you can paste it elsewhere.
	Paste the previously cut or copied form or item into the selected node or item.
	Paste only the link to the previously copied form or item into the selected node or item.
	Open the selected form in design mode to be able to edit it.
	Only if there is a Report List, run data collection only for the reports selected in the list.

Element	Description
	Open a search window to find specific forms.
	Allows you to rename the selected item.
	Move the selected item up.
	Move the selected item down.
	Close the report that is selected and open on the work desktop.
	Run the selected task.
	Mark the selected task as completed.
	Mark the selected as not completed.
	Allows you to manually remove the temporary locks of a form generated by any connection problems that has not been automatically removed by the system. ***Note**: this function does not allow you to remove locks on forms in work off-line mode. To do this, you must use the **Locked data entry forms** function, which is only accessible to the administrator from the web interface in **Forms management**.*
	Allows the user to select the display mode for forms on the panel: with description only, with code only or with both description and code. ***Note**: the default view is defined in CCH Tagetik's general operational settings, see Finestra Impostazioni.*

Status of open forms

The mode in which forms are opened is shown in the **Open items** tab as follows.

Element	Description
	The form is open in design mode.
	The form is open in navigation mode.
	The form is open in data entry mode.

The constituent elements of every form open in design mode are listed below the form.

Work desktop

Area in which reporting operations are run and data extracted from the database is displayed.

Buttons on the Tagetik toolbar

This toolbar is always visible. Depending on the mode in which a form was opened, other toolbars may also be displayed. See Barra degli strumenti di data entry, "Design toolbar" on page 34 and "Navigation toolbar" on page 335.

The buttons are grouped into five different areas: **Login, Forms, Show, Tools and Info**.

Element	Description
Login	
	Allows you to login.
	Allows you to logout.

Element	Description
	Opens a window to exchange messages with other connected users.
	Opens a window to monitor the data processing in progress to check which processes are being run and avoid system overloading.
Forms	
	**Ad hoc form**: allows you to run the following functions: • create a new ad hoc form • convert a normal Excel spreadsheet into an ad hoc form. This allows you to treat it like a normal Tagetik report and insert a matrix, for example, or other elements.
Show	
	**Design**: opens the**Design** tab in the left-hand panel and therefore allows you to define the structure of the various forms.
	**Navigation**: opens the **Navigation** tab on the left-hand panel and therefore allows you to view the data present in the CCH Tagetik database.
	**Data Entry**: opens the **Data Entry** tab in the left-hand panel and therefore allows you to edit data present in the CCH Tagetik database or enter new data.
	**Open items**: shows the **Open items** tab in the left-hand panel and thus allows you to rapidly access the various items present in the open form.
	**Workflow - Table**: shows the **Workflow** tab in the left-hand panel.
Tools	
	**Report list**: opens the **Report list wizard** window to select and configure the publication of the set of reports called Report list. See "Report List Wizard window" on page 366.
	**Parameters Definition**: opens the **Parameters Definition** window to create, edit or delete shared parameters and, if the user is working in Design mode, also ad-hoc parameters.
	**Refresh custom functions**: allows you to run the following operations to refresh the custom formulas used to calculate the values of individual cells: • refresh all custom functions present in the current workbook (default option) • only refresh the custom functions present in the current sheet • only refresh custom functions present in the currently selected area.
	**Batch export**: locally exports a set of forms and transforms them into ad hoc forms. Also allows you to create a folder structure that mirrors the structure of the nodes in the group of forms.
	**Batch import**: imports a set of ad hoc forms into the application database. Also allows you to create a form group structure that mirrors the folder structure used locally by the user.
	**Forms - Import dictionary**: imports the dictionary items from one or several forms in the shared dictionary.
	**Tagetik style**: opens the **Tagetik style** window to edit style sheets.
	**Refresh Dimension tree**: refreshes the dimension tree.
	**Parameters selection mode**: allows you to define how to pre-populate the parameters used when the form is opened in **Navigation** and **Data entry**: • **Point of View**: If a value for every parameter has been specified in the POV (see Impostazioni utente), so the system can identify a specific form, then the form will be opened automatically. • **Last used parameters**: the parameters last set during the session will be applied. • **Defined by the system** : equivalent to **Point of View** or **Last used parameters** depending on the value assigned to the **Default Parameters selection mode** property in the advanced configuration (see Finestra Impostazioni).
	**Generate structure report**: generates a report on the structure of the selected form.
Info	

Element	Description
	**Options**: opens a window to configure connections to the server and the settings of some of the Reporting function's local parameters. See "Options window" below.
	**Information**: shows information on the Add-in and the client to which the user is connected.
	Shows the name of the application database on which the user is working.

Buttons on the work desktop

Element	Description
	Expends rows with subtotals.
	Collapse rows with subtotals

Common behaviours of the Excel interface

How to access the windows

In the Excel add-in, you can run the same action through different paths or with different modes. Some actions are listed below.

Action	Paths
Open a form in design mode	• (Left-hand panel, **Design** tab) Double click the desired form. • (Left-hand panel, **Design** tab) Select form >
Open a form in the navigation or data entry mode	• (Left-hand panel, **Navigation/Data Entry** tab) Double click the desired form. • (Left-hand panel, **Navigation/Data Entry** tab) Select form >
With a form already open in design mode (Design)	
Create a matrix	• **Matrix** • (Work desktop) Right click an empty cell > **Tagetik** > **New matrix** • **Open items** > (Left-hand panel, **Open items** tab) Right click a sheet > **Matrices** > **New matrix**
Open a matrix	• (Work desktop) Double click a cell of the matrix • (Work desktop) Right click the cell of the matrix > **Tagetik** > **Modify matrix** • **Open items** > (Left-hand panel, **Open items** tab) Double click the matrix • **Open items** > (Left-hand panel, **Open items** tab) Right click a matrix > **Matrices** > **Modify matrix**
In the Matrix Wizard	
Add row/column/matrix filters	• (Dimensions tree) Select elements > Click next to the local filters tree • (Dimensions tree) Select elements and drag them to the local filters tree
Add headers	• (List functions of predefined rows or columns) Select headers > Click • (List functions of predefined rows or columns) Select headers > drag chosen headers to the area of pre-defined headers

Options window

Excel > CCH® Tagetik > Options

Purpose of window

This window allows you to configure CCH Tagetik's reporting add-in in Excel. It is displayed automatically when the add-in is first used, but can be opened at any time.

Tabs in the window

Element	Description
Connection	Allows you to configure the connection to web services.
Local parameters	Allows you to set the following: • the specific local parameters of the Reporting module • the log level of the Reporting module

Buttons in the window

Element	Description
	Add a connection.
	Change the selected connection.
	Delete the selected connection.
	• Save the changes made to a pre-existing connection. • Save a new connection.
	Undo the last action.
	Run a connection test on the URL of the selected connection and view the result in a dedicated window.
	Set the selected connection as the working connection.

Fields in Local parameters

Element	Description
Enable automatic updating of Add-in	When the add-in is opened, the system shows any available updates and allows you to download them.
Temporary folder	Folder in which temporary files generated by the add-in are saved. This can be the Windows temporary files folder or a folder chosen by the user.
User settings folder	Folder containing the add-in configuration files. See "Configure local reporting parameters" on page 15.
Enable Delete temporary files	Enables the automatic deletion, from the folder specified as the **Temporary folder**, of temporary files that are older than the age specified in the **Number of days** field.
Use the new communication mechanism	Enables the use of the mechanism of communication with the Chromium-based server.
Cache	
Enable cache	Enables the use of the cache on add-in data.
Cache folder	Folder in which the cache is saved.

***IMPORTANT**: The settings in the Log section are reserved for specialist CCH Tagetik personnel. For more information, contact support.*

Configure the add-in for Excel

Install the add-in

This procedure allows you to download, install and connect the Excel add-in.

1. To start the download, on the CCH Tagetik Home page, click the user name, then select **Download**, then **Excel .NET client**.
2. To install the add-in, open the .ZIP file, extract the contents to a folder, double click the .EXE file and run the guided installation procedure.
3. Open Excel and click the **CCH Tagetik** tab: the **Options** window will open.
4. In the **Connection** tab, select a connection from the list on the left, or click to create a new one.
5. If a new one is created, type the address of the connection to the desired server in **Web Server**.
6. Click to set the selected connection as the current connection.
7. Click to save the connection.

Configure a new connection to the server

Excel > CCH® Tagetik > Options

1. In the **Connection** tab, click and assign a name to the connection.
2. In **Web Server**, type the address or the connection to the desired server.
3. (Optional) Click to set the selected connection as the current connection.
4. Click to save the connection.

Edit a connection to the server

Excel > CCH® Tagetik > Options

1. In the **Connection** tab, click the desired connection, then click .
2. Edit the URL as desired.
3. (Optional) Click to set the selected connection as the current connection.
4. Click to save the changes.

Configure local reporting parameters

Excel > CCH® Tagetik > Options

1. In the **Local settings** tab, click to activate the fields.
2. To view any available updates when authenticating the add-in, click **Enable Update Add-in automatically**.
3. (Optional) Enter the desired file paths in **Temporary folder** and **User settings folder** in which to save the add-in's temporary working files and configuration files.
4. To enable the automatic deletion of temporary files, click **Enable Delete temporary files** and specify a number of days.
5. If necessary, enable the new communication mechanism . ***Note: In the case of single sign-on to Tagetik, using the Chronium-based communication mechanism, access by entering a user name and password is supported.*** *Automatic access without providing credentials is not supported.*
6. If necessary, enable the cache and enter a file path to save the files.
7. Click to save the changes.

Access reporting functions in Excel

Excel > CCH® Tagetik > Login

Enter the user name and password used to access CCH Tagetik and click **Sign in**.

***Note**: if the system detects available updates for the add-in on authentication, the update will be requested. The user name and password must be entered again once the update is finished.*

Web forms

Definition

In CCH Tagetik, it is also possible to work on all types of form, except for ad hoc forms, from the web interface, on the condition that they are enabled as *web forms* (see "New form window" on page 38). There are two ways to open a web form:

- in navigation mode from the CCH Tagetik web interface
- in data entry mode, when related to processes, in the Contributor's processes cockpit (see Contributor Cockpit page).

***Note**: web forms also remain available for navigation and data entry from the Excel add-in.*

Forms can only be designed via the Excel interface.

If you try to open a form constructed on a Datasource that uses DTP parameters, an error message will be displayed and the form will be locked for opening.

Bookmarks in web forms

A bookmark allows you to save the running of a form in navigation or data entry mode. Once a bookmark has been created, the form can be run directly from the Home Page, maintaining the previously defined filters and settings.

***IMPORTANT**: if the form settings are edited, and the form can no longer be opened as a web form, any bookmarks pointing to the form remain active.*

Row drill down function

The *row drill down* functionality ( ) expands the starting row, generating as many rows as there are children of the starting hierarchy node (or, for "parent" items, as many rows as there are related accounts). The original row is transformed into the subtotal of the details below and formatted as required for subtotals.

***Note**: row drill down is not supported in the presence of OR filters and cannot be used on columns.*

If there are several nodes on the selected row, select the corresponding header cell to run the row drill down on an individual node. To run the drill down on all nodes, select the cell containing the value to be expanded.

Copy and quick insert function

This function is only available in Data Entry mode. In this mode, in web forms with transactional matrices, new rows or columns can be copied and pasted using the quick insert function. To do this, when adjacent cells are selected in the work desktop, the system shows action buttons for copying and pasting with quick insert.

Button	Description
	Copies the selected cells and changes to .
	Inserts new rows below the selected rows or new columns to the right of the selected columns, copying the content of the initial cells in the new ones. ***Note**: the created rows or columns are shown with a blue border until the form is saved.*

***Note**: the **Don't expand** option defined when the matrix is designed causes entire rows or columns to be inserted in web forms 2.0.*

When the quick-insert copy and paste function is used, the pasted rows or columns are editable, while those already present in the form cease to be editable.

Rows inserted in the matrix have the same functionalities as the rows already present and can be editable, protected or non-editable based on the matrix configuration.

Wrap Text option

This option is only available in Data Entry mode. The option can be enabled on the columns of Line by Line transactional matrices. When text is entered, or a form containing the matrix is opened in data entry or

navigation mode, the column cells expand to show the all the text. To use this option, a CCH Tagetik style must be applied to the columns.

Comments and attachments in web forms

Both comments and attachments can be associated with web form cells, to provide additional information on the data entered. This function is only available for Line by Line matrices and multidimensional matrices from the Analytical Workspace, and only for users with function 0002642 enabled.

For Line by Line matrices, several documents can be attached to a single comment. For multidimensional matrices, only one attachment per comment can be inserted from the Analytical Workspace.

Comments are visible in all forms for which the defined filters are the same as those of the cell to which the comment was linked.

Cells with which a comment or attachment is associated are identifiable by a red triangle in the top-right corner.

Excel add-in: technical features

Requirements

The Excel add-in can be installed on any version of Microsoft Office. To use the add-in, the device must only have one version of Microsoft Office (the same for every module of the suite) installed. To use the add-in in Office 365, the suite must be installed locally.

Configuration files

The add-in always creates two configuration files: Appsettings.xml e Usersettings.ini. These files are saved in the same folder in which the add-in is installed and contain the following information:

- details of the configuration of the add-in's connections to the web server
- summary of the add-in configuration preferences
- temporary files generated by the add-in

For more details on the contents of the configuration files, see "Options window" on page 9.

Display language

The language in which add-in interface elements (buttons, fields, etc.) are displayed is CCH Tagetik's display language.

Tracking user authentications

Authentication operations carried out by the .NET client are registered in the audit trail if the "Web Login/Logout" option is enabled in the audit configurations (see Basic configuration procedures). The following information is registered and allows you to perform statistical analyses on the versions of Excel and Windows installed on clients that connect to CCH Tagetik.

- Client.Office.Version, or the version of Excel with which the user logged in (12.x= Excel 2007, 14.x= Excel 2010, 15.x=Excel 2013, 16.x= Excel 2016).
- Client.Office.Build, or the Excel build.
- Client.OS.Version, or version of Windows with which the client is authenticated.
- Client.NetFramework.Version, or the .NET Framework version and build number.

Configure the add-in for Excel

Install the add-in

This procedure allows you to download, install and connect the Excel add-in.

1. To start the download, on the CCH Tagetik Home page, click the user name, then select **Download**, then **Excel .NET client**.
2. To install the add-in, open the .ZIP file, extract the contents to a folder, double click the .EXE file and run the guided installation procedure.
3. Open Excel and click the **CCH Tagetik** tab: the **Options** window will open.
4. In the **Connection** tab, select a connection from the list on the left, or click to create a new one.
5. If a new one is created, type the address of the connection to the desired server in **Web Server**.
6. Click to set the selected connection as the current connection.
7. Click to save the connection.

Configure a new connection to the server

Excel > CCH® Tagetik > Options

1. In the **Connection** tab, click and assign a name to the connection.
2. In **Web Server**, type the address or the connection to the desired server.
3. (Optional) Click to set the selected connection as the current connection.
4. Click to save the connection.

Edit a connection to the server

Excel > CCH® Tagetik > Options

1. In the **Connection** tab, click the desired connection, then click .
2. Edit the URL as desired.
3. (Optional) Click to set the selected connection as the current connection.
4. Click to save the changes.

Configure local reporting parameters

Excel > CCH® Tagetik > Options

1. In the **Local settings** tab, click to activate the fields.
2. To view any available updates when authenticating the add-in, click **Enable Update Add-in automatically**.
3. (Optional) Enter the desired file paths in **Temporary folder** and **User settings folder** in which to save the add-in's temporary working files and configuration files.
4. To enable the automatic deletion of temporary files, click **Enable Delete temporary files** and specify a number of days.
5. If necessary, enable the new communication mechanism . ***Note: In the case of single sign-on to Tagetik, using the Chronium-based communication mechanism, access by entering a user name and password is supported.*** *Automatic access without providing credentials is not supported.*
6. If necessary, enable the cache and enter a file path to save the files.
7. Click to save the changes.

Access reporting functions in Excel

Excel > CCH® Tagetik > Login

Enter the user name and password used to access CCH Tagetik and click **Sign in**.

***Note**: if the system detects available updates for the add-in on authentication, the update will be requested. The user name and password must be entered again once the update is finished.*

How to work with configuration windows

Define a filter

In the dimensions tree, click on the desired element or node and drag it to the definition area.

Change the attributes of a filter

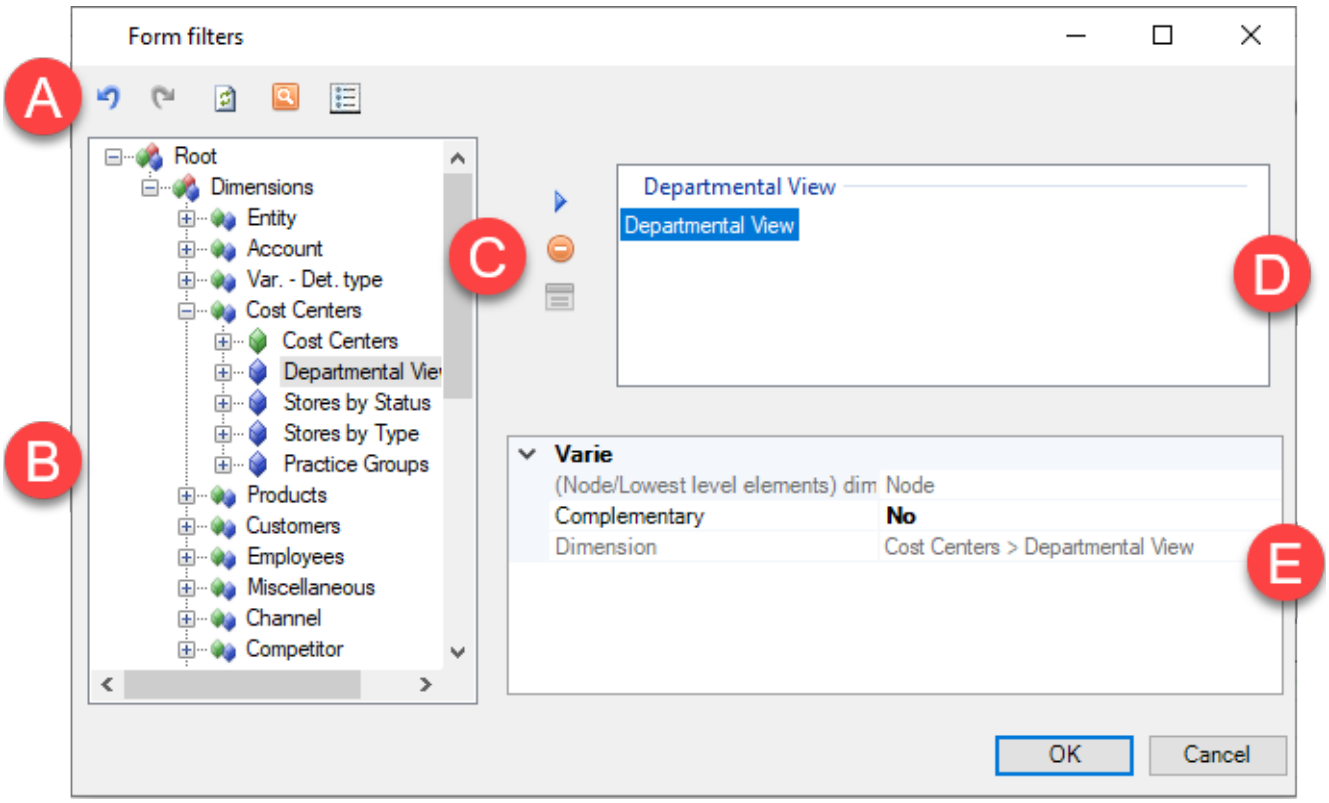
1. In the filters tree, click on the desired dimension.
2. In the filter properties area, click on the attribute and make the desired change.

Configuration window structure

Introduction

Several configuration windows are used in the creation of forms, and they have a structure similar to the one described below.

What it looks like



Element	Description
A	Action buttons
B	Dimensions tree
C	Buttons to manage dimensions in the definition area
D	Definition area
E	Table of properties of the selected element in the definition area

Action buttons

There are various buttons depending on the type of page and user. Below is a list of frequently-used buttons:

Element	Description
	Undo the last action.
	Repeat the last action.

Element	Description
	Refresh the dimensions tree.
	Allows you to search the dimensions tree.
	Allows the user to select the display mode for forms on the panel: with description only, with code only or with both description and code.

Dimensions tree

The dimensions tree shows the available dimensions organised into levels of nested nodes.

***Note**: elements identified as **N.A.** are specific elements not related to any node, but available for reporting.*

Level	Element	Description
Node type		Financial dimensions node
		Analytical Workspaces measures node
Node dimensions		Dimensions for which aggregation structures can be defined
		Dimensions for which aggregation structures cannot be defined
Dimensions content type		Aggregation structures
		Time-dependent hierarchies
		Elements of the dimension
Elements of the dimension		Element of the dimension

For time-dependent hierarchies, the dimensions tree does not show the lowest level elements of the nodes, namely the specific elements to which the hierarchy refers, because the relationships of the lowest level elements the first hierarchical level depends on the scenario/period. To make it easier to understand the elements related to nodes of time-dependent hierarchies, an *N.A.* node is displayed, under which all the elements of the dimension are listed and identified with the icon.

Properties of the selected element

Element	Description
Rows of properties	• Allows you to set the element's properties. • When several elements are selected, the user can make changes en masse to the properties that the selected elements have in common.

Properties table section

The content varies based on the selected dimension or filter. When multiple filters are selected, the table only displays the properties that the filters have in common, and allows you to manage these properties only.

Element	Description
	Group of properties related to the same aspect to be configured (e.g. the **Hyperlink** item enumerates all the properties related to hyperlinks). See "Row/column filters" on page 151.

Shared behaviours

Regardless of the purpose of the specific window, configuration is generally done by moving a set of elements or nodes from the dimensions tree to the definition area.

Filters can be added using the buttons, or by dragging and dropping. Filters can also be added individually or by multiple selection, by holding down CTRL or SHIFT.

In windows in which it is possible to define the order of the element (the **Tab Filters** window, for example), the filters are shown in the same order in which they were selected in the dimensions tree.

Features not supported

Features not supported in Excel

The following features are only available for web forms, but not in the Excel add-in.

- Hyperlink: the opening of the form specified in the next row/column is not supported (see "Hyperlinks" on page 220).
- On-demand editability (see "Enable Editability On Demand" on page 131).

Features not supported by web forms

Advanced selection and subtotals: the running of forms with intersection on rows/columns of dynamic nodes with subtotals enabled is not supported (see "Advanced selection" on page 114).

Azure Information Protection Label for Excel

Introduction

If the user belongs to a domain where the use of Sensitivity labels is mandatory, a label must be associated with the form template.

The label can either be explicitly selected when creating or editing a template, or the default label defined in the system configuration can be used (see chapter CCH Tagetik Overview > Settings Window)

How to apply a label

In order to apply a label to a form, it is necessary to open the template and select a label that does not require the file to be encrypted from among the labels made available to the user by the system. Encrypted labels are indicated by the image of a padlock in the label icon. The assignment procedure can be performed either when creating a new template or each time a template is opened in design mode. The selected label will be saved within the file only upon saving.

IMPORTANT: *Encrypted* and/or *non-existing* labels are NOT supported.

Default Label

The default label is meant to ensure correct operation when opening a form that does not have a label but is nevertheless launched in a domain where the label is mandatory.

Cases in which the default label is used in a way that is completely transparent to the end user are

- Template without label
- Templates with labels initially consistent and then no longer so, due to changes in the label definition
- Bach import of forms without label or with inconsistent label
- Import of ad hoc forms without label or with inconsistent label

***IMPORTANT**: If the default label is not consistent, the user is warned by an on-screen message when the default label should be applied. If the form is opened in Design mode, the opening process is not blocked, vice versa, if the form was opened in Navigation or Data Entry mode, it will not be opened.*

Enabling web forms

Admin | <application database>

Background

Forms can be made available as web forms individually or en masse. Both operations are reserved for the administrator user.

Enable web forms by default

This procedure allows you to enable all forms as web forms.

Settings > Form tab > Show

Select the **Enable all forms as web forms** option.

Enable individual web forms

Forms management tile > Elements > Attributes tab

Excel > CCH® Tagetik > Design > select form > (in create form phase) /(in edit form phase)

In **Enable as web form**, the following values are available:

- **Yes**: makes a form available as a web form
- **No**: to exclude a form from web forms
- **From configuration**: to apply the mode configured in the settings for all forms

Features supported by web forms

Introduction

Forms can be enabled for use as web forms in the form settings in the Excel interface (see "New form window" on page 38) or from the table of forms in the web interface (see Pagina Forms).

***IMPORTANT**: in web forms, the character "!" cannot be used in template names. If inserted, the system will remove it, preventing everything that points to these templates from functioning (e.g. hyperlinks).*

Supported formulas

Web forms support the following Excel formulas. For the detailed list, please refer to the official SpreadJS list. Anything that is not expressly indicated is not supported.

Cell Formats and Locale

Separators for thousands and decimals for numeric formats, as well as date/time formats, depend on the nationality set in user configuration in CCH Tagetik.

A non-exhaustive list of the differences detected in the test phase is provided below:

- For Japanese nationality, the "short date" format; in contrast to Excel, it always uses two characters for the month (e.g. Excel: "1976/8/30"; Web form: "1976/08/30").
- For the UK and Italian nationalities, the format "30/8/76 16:28"; in contrast to Excel, it uses two figures for the year (e.g. Excel: "30/8/1976 16:28"; Web form: "30/8/76 16:28").
- For Polish nationality, the "long date" format could use different variants of the names of months and days from those used by Excel.
- For Turkish nationality, the thousands separator is a space.

- Formats with fixed localisation set from Excel design (e.g. 14-Mar-18) are adapted to the user's locale, excluding cases in which the set fixed localisation is American English (EN_US), Chinese (ZH_CH) or Japanese (JA_JN).

Since release SP15, the "short date" and "long date" formats identified with "*" in the Excel add-in are overwritten by the user's local configurations.

Supported navigation features

A list of the supported navigation features is provided below:

Element	Description
Single and Multi-Template	Forms defined with several templates including tab filters are rendered. There are restrictions in data entry mode. Formulas and charts that refer to names (Excel named range) may not be defined correctly in sheets after the first of each template (e.g., formulas that refer to names generated by CCH Tagetik together with custom queries), so we advise against using them. Charts could lose some of their formatting and colour palette (where the standard palette has not been used) in sheets after the first.
Changing the value of parameters and the Refresh function	The value of parameters can be changed from the relevant bar after the form has been run and the form can be refreshed using the dedicated button on the toolbar.
Further information	• Comments are enabled to be displayed in navigation mode and attachments can be downloaded via the comment details window. • If the automatic loading of comments is disabled, they can be loaded using the dedicated button.
Edit IC	• Intercompany declarations can be inserted directly from an editable cell of a web form. • Pre-existing declarations can be edited and deleted.
Advanced selection	• Advanced selection on rows and columns is supported. For correct formatting, we recommend using Tagetik styles, without which the same formatting used by the Excel add-in cannot be guaranteed. • All matrix expansion options are supported, i.e. "Move cells down/right", "do not expand" and "entire row/column". ***Note**: for the double expansion of multidimensional matrices, with a double "Move cells" option (for both rows and columns), the cells involved in the expansion will be first be moved down, where possible, and then the remaining cells will be moved right. To the extent that it is possible to have advanced selection on both rows and columns simultaneously, some features are not yet fully supported with the double "Move cells" option. Some cells are not moved and, in particular, some Excel components, such as named ranges and notes, are not fully recalculated.*
Matrix layout options	• The automatic adjustment of columns "by value" is supported, but adjustment "from the template" is not supported. Matrices with automatic adjustment "from the template" are treated as if they had automatic adjustment "by value". • The "Hide row/column" matrix option is supported.
Tagetik styles	CCH Tagetik styles are supported, except for the numeric format.
User queries	User-defined queries are supported. These include the following options: • Show Query Name • Show headers • Create name for the result • Apply the format of the first row to the whole Output area • Automatic adjustment of columns Copy the formulas to the left/right of the query to the Output area. Queries that refer to Excel cells via A1 syntax are not supported, e.g. ${A1}. User-defined validation queries are supported, including the "Enable free text insertion" option and the ability to define dependent queries. Drill-through queries are supported.

Element	Description
CCH Tagetik hyperlinks	• CCH Tagetik hyperlinks are now supported to run data processing jobs and drill-through queries and to open modules. Hyperlinks can only be defined as "Open form specified in the next row/column" for web forms and for matrix hyperlinks. • CCH Tagetik hyperlinks defined in a numeric cell are not yet supported. Standard Excel hyperlinks are not supported. • CCH Tagetik hyperlinks support the running of processes with parameters defined on Excel cells. • The reference to the **Excel Cell** must only indicate cells of the current sheet in H30 format (for cell H30). The reference to the Excel cell is not supported for opening another form via hyperlink.
Bookmark	Bookmarks can be created for running forms in navigation or data entry modes. These bookmarks can be used to run forms directly from the Home Page with the current filters, without them being requested.
Tab filters	Filters that have an effect on all matrices defined in the template are supported.
Hide rows or columns	This functionality is supported, except where it depends on the result of Excel formulas.
Drill Down	Double clicking the headers of rows or columns not at the maximum level of detail activates the drill down, i.e. displays detailed rows or columns. The following features are not supported after a drill down: • it cannot be guaranteed that all the formulas that refer to the row subject to drill down will always be refreshed correctly • hide rows or columns • validation checks • comments Row and column drill down follows the expansion methods defined when the matrix was designed. In relation to "Move cells down/right", malfunctions have been detected in the lateral grouping and in the application of styles on column deployments.
Overridden content	The ability to override the content of static matrix cells is supported.
Linked templates and matrices	The ability to use links to matrices and templates of other forms is supported.
Multi-selection	For transactional matrices, applying multi-selection in the counter options allows you to obtain the column with cells for selecting check boxes for every record (row/column) where it will be possible to select several records (rows/columns) simultaneously. These boxes can be used in Excel formulas, taking on a positive value when selected, or a negative value otherwise.
Wrap Text	For transactional matrices, in the event that a column has the Wrap Text style option enabled, when a form containing that matrix is opened, the cells belonging to that column widen automatically based on the text that the cells contain. At the moment, this is only supported on Line by Line transactional matrices to which a CCH Tagetik style is applied.

Element	Description
Grouping	Grouping supports the following formulas: • SUBTOTAL • Algebraic sum of cells (does not support the native SUM function) The grouping's orientation is determined by the **Subtotal before the detail** flag, with the following behaviour: • If not selected, the grouping's orientation is standard and only formulas which have adjacent items (or items that end up adjacent after lower-level groupings have been completed) above the formula will be grouped. • If selected, the grouping's orientation is the opposite of standard and only formulas which have adjacent items (or items that end up adjacent after lower-level groupings have been completed) below the formula will be grouped. For Synthetic FSTs, however, the grouping's orientation is determined by the first formula to be used: • If the formula refers to items positioned above, the grouping's orientation is standard and only formulas that use adjacent items above the formula will be grouped. • If the formula refers to items positioned below, the grouping's orientation is opposite to the standard and only formulas that use adjacent items below the formula will be grouped.
Rich Text	Supported only on transactional Line-by-Line matrix using a formatted text type field (see Pannello proprietà campo and TESTO), it allows formatted text to be displayed on a specific cell. • An unformatted preview of the text entered in Data Entry can be displayed on the cells of the column related to the formatted text type field by means of an editor. • To view the entire formatted text, which is available in read-only mode, it is necessary to open the panel containing the editor, which can be accessed via a button in the ribbon bar (see "The web forms page (Navigation)" on page 350).

Supported data entry functionalities

Element	Description
Single and Multi-Template	Forms defined with several templates including tab filters are rendered.
Editability	• Lock editing for sheets in both data entry and navigation modes (when the "Lock output sheets in navigation" option is active) and unlock cell editability. • Conditional editability • Only unlock editable cells and areas explicitly defined as "Unlocked" by the user in design mode • Different colouring for editable or unlocked cells, protected cells, editable cells with IC, editable cells with edit transaction currency (IC editor and transaction currency not available at the moment) • Save edited values (without updating values) Validation checks on numeric values use a tolerance threshold defined when the form is defined, in the **Form options** window (see "Form options window" on page 55).

Element	Description
Open new rows/columns	New rows or columns can be inserted. • New rows are formatted as per the applied Tagetik style. • The formulas present on the relative columns are propagated on the new rows, with the exception of formulas with bidimensional references {row,column}, formulas with offsets {+x}, {-x} and Server-type formulas. Where such formulas are present, an error message will be shown. • Inserted records can be deleted using the delete button in the Status column when selecting the row. • For transactional matrices, the number of rows/columns to add is not required, as the system always adds them individually. • For multidimensional matrices, the refreshing of validation checks, validation queries and comments is supporting.
Multi-selection	For transactional matrices, with multi-selection active in the counter options, you can obtain the column with cells for selecting check boxes for every record (row/column) in which it will be possible to select several records (rows/columns) simultaneously. These boxes can be used in Excel formulas, taking on a positive value when selected, or a negative value otherwise. Multi-selection is not supported in Line By Line matrices of a validation query is used. This behaviour differs from the Excel client.
Spreading	The buttons for locking and unlocking cells can be accessed from the sheet's context menu: • the **Lock** button locks the selected cells • the **Unlock** button unlocks the selected cells • the **Unlock all** button unlocks all cells in all sheets Arithmetic operators to be used directly in the cell are not supported. Editing a cell containing a formula whose valuation is X to apply a spread, using X as the desired value, may make the form inconsistent (e.g. loss of the cell formula). The value of a cell containing a formula, even if locked, could be modified by a spread operation on another cell with a formula when the two formulas have at least one cell on which they both depend in common. When a cell with a formula is edited, only the values of the cells on which it depends are reconfigured, even when another locked formula depends on the cell. Even in this case, the value of the cell with a locked formula may change. When the two spread methods (formula, as it contains a formula, or inverse, as a locked cell with a formula depending on it) are applicable to a cell with a formula, both methods are applied in sequence. After the value in the cell with the formula has been changed, applying spreading activates the **Cancel spreading** option (). This allows you to restore values from prior to the activation of the Spreading method on all edited sheets. Where Spreading is applied several times to the same cell without the form ever having been saved, when the Spreading action is cancelled the restored values are the original values, shown when the form was opened. The following option is deactivated in the following situations: • If Spreading has not been applied at least once, or if the cell with the formula has not been edited and the user has pressed Enter or moved on to another cell. • If another item from the main menu (Home/Analysis/Utility) is selected during the Spreading step with the option active. • If, in Spreading mode with the option active, the user clicks in the form on a hyperlink that entails actions including "Save".

Element	Description
DatePicker	For editable date-type cells, double clicking will reveal a DatePicker to select the desired date. The appropriate Tagetik style for these cells must be set in Excel, otherwise cells in which a date is selected will be shown in numeric Excel format. When a date-type cell is copied, using the copy-paste function or by dragging the source cell, the DatePicker will be deactivated in the target cell. Therefore, copied dates will not be editable. ***Note**: the DatePicker is not currently supported on the Safari browser.*
Further information	• The display of comments in data entry mode and the download of attachments via the comment details window are enabled. • If the automatic loading of comments is disabled, they can be loaded using a specific button. • It is also possible to edit a pre-existing comment and add a new comment (which will also be visible in navigation mode if the relevant option is enabled in the **Comments** section of the Finestra Impostazioni).
Validation queries	Validation queries are supported, even with dependent rows/columns, including the **Enable free text insertion** option present in the Query Wizard (see "Query Wizard window" on page 255). ***Note:** in the presence of Dictionary and Validation Query insisting on the same textual measure, the dropdown shows all values in the validation query as suggestions, but if data validation is active, only those values belonging to the Dictionary are saved. An exception is made if the 'Out of dictionary values allowed' flag has been enabled on the measure.* When the elements that are returned by the query and can be selected in the cell are figures (e.g. 01, 02, etc.), the cell must have General or Text formatting; otherwise, the system could overlook the zero at the start and treat the selected value as a number. ***Note**: in contrast, the Excel client always treats the selected value as text.*
Dictionary	Dictionaries are supported • on the columns of transactional matrices related to Analytical Workspace datasets • on textual measures in multi-dimensional matrices related to Analytical Workspace datasets with non-incremental storage **Validation in the presence of a dictionary on Multi-Dimensional matrices** Automatically enabled once the dictionary is configured to the textual measure. Validation is based on the display mode combined with the set of values defined with the dictionary. • If the display mode of the dictionary is *Code* and the value you want to save belongs to the range of codes defined in the dictionary, then validation is successful and the data is saved. • If the display mode of the dictionary is *Description* and the value you want to save belongs to the range of descriptions defined in the dictionary, then validation is successful and the data is saved. • If the display mode of the dictionary is *Code and Description* and the value you want to save belongs to the *code* range or the range of *Codes and Description* defined in the dictionary, then validation is successful and the data is saved. • If the flag *Out of dictionary values allowed* is enabled, validation is always successful and all values are saved ***Note**: in the presence of Dictionary and Validation Query insisting on the same textual measure, the dropdown shows all values in the validation query as suggestions, but if data validation is active, only those values belonging to the Dictionary are saved. An exception is made if the 'Out of dictionary values allowed' flag has been enabled on the measure.* ***Note: the dictionaries described above are also supported on the Excel client.***

Element	Description
Validation checks	Validation checks are supported, both inside and outside the matrix. The **On save** and **On close** options have no effect in web forms because the checks are run continually and messages relating to failed checks are shown in a pane at the top-right of the interface. The occurrence of blocking expression breaches prevents data from being saved. The general functioning of validation checks positioned below or to the right of added rows or columns is guaranteed. In financial forms, added rows are not subject to evaluation. Validation checks on numeric values use a tolerance threshold defined in the form configurations, in the **Form options** window (see "Form options window" on page 55).
Wrap Text	Only supported on transactional Line by Line matrices to which a CCH Tagetik style is applied. When text is inserted on columns of a transactional Line by Line matrix on which the Wrap Text type option is enabled, or when a form containing the matrix is opened, the cells belonging to that column expand automatically based on the text that the cells contain.
Running ETLs	In data entry mode, ETLs can be run via the button present on the toolbar in the Home section. To be able to run an ETL, the user must select a previously configured job that has been associated with the process for which data entry is being performed. One or more parameters needed for running the ETL can be associated with the job and, by default, those parameters are defined with values inherited from the form execution context (e.g. Workflow). The value of those parameters can also be changed via the Filters panel and then the form can be updated before the ETL is run. In this case, the values of the parameters will be those of the form filters, and not those of the execution context. For an Entity-type parameter, it is possible to select hierarchy elements (and therefore all the entities belonging to the hierarchy). As it acts on the **Filters panel**, at the moment the ETL does not manage multiple scenarios, so only one scenario must be selected.
Rich Text	Supported only on transactional Line-by-Line matrices using a formatted text type field (see Pannello proprietà campo and TESTO), it allows formatted text to be inserted on a specific cell. • Formatted text can be entered using the relevant editor. Once the cell of interest has been selected, the editor is opened via a button in the ribbon bar (see Pulsanti di azione). • The text is only seen in formatted mode in the relevant panel, whereas an unformatted preview of the entered text is displayed on the matrix cells. • It is possible to insert text using cut and paste from other editors (e.g. Word), in these cases the formatting will be maintained within the limits of the editor's possibilities. • Only text input in the editor is permitted, images or other multimedia elements are not supported. • To close the editor, either use the 'x' button in the editor itself or select the button in the ribbon bar again. • It is recommended to insert a Wrap-Text style to allow better reading of the formatted text preview

Features not supported in web forms

Introduction

The features, or the options of some features, that are available in the Excel add-in but not supported – or supported with certain differences – in web forms are listed below.

Features supported with exceptions

Element	Description
Spreading	Does not support advanced syntax (e.g. "+10%").
Matrix Layout options	For the *autofit* function on columns, only the *by value* option is supported. This option is also used if *from the template* is selected in the autofit settings.
Drill Down	Only supports row/column drill down. ***Note**: Cell drill down is partly covered by ad hoc analysis.*
Slicer	Features not supported.  *Use of the web form filter panel to edit the values of parameters at runtime, and objects of the cell field to display the values of the parameter within the form.*
Autofit rows	Only supported for Line by Line transactional forms, not for multidimensional forms.
Autofit columns	Inherits the column autofit rules from the template. Supports *autofit by value*.
Validation checks	Supports verification checks, both inside and outside the matrix. Options *on save* and *on close* do not work in web forms because the verification checks are continuous and, in the presence of expressions that cannot be verified, the system prevents data from being saved. ***Note**: implementation differs when compared with the Excel add-in.*
CCH Tagetik hyperlinks	CCH Tagetik hyperlinks are supported, but references to an Excel cell can only lead to cells of the current sheet.
Validation queries	If elements generated by the query that can be selected in cells include numbers only (e.g. "01", "02", etc.), the cell formatting type must be "General" or "Text", otherwise, when the value is selected, the system could remove the zero and treat the selected value as a number. In contrast, the Excel client always treats the selected value as text. Multi-selection is not supported in Line by Line matrices. In contrast, multi-selection is possible on the Excel client.
Download forms	Only supported in XLSX format.
Subtotals	It does not support the intersection of row and column SUBTOTAL formulas in a dynamic matrix. As a result, the values in these cells are incorrect.

Features not applicable/not implemented

- VBA macros or objects
- Work offline
- Report list
- Cell-based reporting (TGKSend, TGKRet, etc.)
- Hidden comments
- Hidden sheets
- Save/load formulas
- Show formulas used
- Reference to Spreadsheet Cell in Query Form (e.g. using the syntax ${A1} to refer to the spreadsheet cell A1 in the selection).
- Excel table defined in the query result area
- Management of dictionary items outside the matrix
- Edit intercompany declarations for matrices with Analytical Workspace datasets
- Non-incremental saving of multidimensional matrix data
- Cell properties functionality in the presence of cells with overridden content
- Excel 'Data Validation' functionality (use Dictionaries instead)
- Validation queries are not supported in web forms

Unsupported Microsoft Excel features

The use of Excel's named ranges feature in formulas and charts is not recommended when applying tab filters to the form. When the form is opened as a web form, in all templates the named ranges may not be correctly defined in spreadsheets after the first (e.g. formulas with names generated by CCH Tagetik using

customised queries). In sheets after the first, charts could lose some of their formatting and, if used, any customised colour palette.

On web clients, in cells that are intersections of SUBTOTAL formulas in rows and columns, the formula is incorrect, and consequently, the cell values are also incorrect.

Types of reporting engine

Standard engine

CCH Tagetik Financial Reporting's standard reporting engine runs complex SQL queries based on the choice of filters. The data are transformed and refined directly as part of the SQL queries.

Every transformation (on-the-fly conversion, segment, etc.) is run directly in the database.

This engine is used under the following conditions:

• If minimum transformations must be applied to the block of data to be extracted.

• If a SAP HANA database is being used.

• If a large number of filters have been applied.

In-memory engine

The *in-memory engine* is a reporting engine that optimises the read/write process performed by standard engines. It has the following advantages:

- It ensures greater scalability: queries run by the in-memory engine are simpler and make it possible to shift the computational burden from the database manager to the application server.
- It reduces the runtime of forms. Especially for forms that use advanced reporting functionalities like segments, FX rates and time-dependence, the query generated by the standard engine takes a long time. The in-memory engine takes advantage of the benefits linked to a simpler query and server application concurrency.

By changing the *Reporting engine* option from **Standard** to **In memory**, or vice versa, in the form settings (see "Form options window" on page 55), the system warns the user that the concurrency settings will be reset. The concurrency settings are in fact linked to the reporting engine type. While for the standard engine the system tends to keep the settings defined in the configuration step, for the in-memory engine the system sets the degree of concurrency to one, because the in-memory engine has concurrency mechanisms that mean that a high degree of concurrency is superfluous.

When this engine is used, the following information must be specified: **Block size rows**, **Size of the data loading threadpool** and **Size of the data processing threadpool** (see "Run settings details" on page 56).

Currency conversion algorithm

On-the-fly currency conversion calculation

The reporting engine can convert values read from the database based on two conversion options specified in the form:

- The " FX rate" on page 153 option, specified on the *Currency* dimension (or via the " Reference FX rates" on page 153), which determines the FX rate to be applied.
- The " Periodic value calculation" on page 153 option, included among the *Period Length* dimension properties, which determines the periodic value calculation method. Valid for the Financial Workspace only

The conversion algorithm is defined by the combination of the *FX rate* and *Periodic value calculation* options.

IMPORTANT: On-the-fly conversion does **not** support the following FX rate types: **Previous Final FX** rate and **Historical FX** rate

Calculation of the equivalent value of a periodic amount

There are two possible algorithms for converting a **periodic amount**:

- Convert the monthly figure at the period FX rate: (YTD(n)-YTD(n-1)) * Fx(n). This algorithm is applied when the periodic figure is calculated as *Aggregation of periodic values*.
- Convert the YTD figure at the period FX rate and then calculate the periodic value: YTD(n) * Fx(n) - YTD (n-1) * Fx(n-1). This algorithm is applied when the periodic figure calculated as *Net Year-to-date values*.

Calculation of the equivalent value of a cumulated amount

When the equivalent value of a cumulated amount is calculated, in June, for example, there are two possible conversion algorithms:

- conversion of the cumulated value for the period (June)
- calculation of the converted amounts of the individual periods from the start of the year that are then added together (the monthly contributions from January to June).

In both cases (periodic figure and cumulated figure), the type of algorithm to be applied is determined by the *Periodic value calculation* option, present among the *Period Length* dimension properties.

Example

Details of the algorithms applied in the presence of the value in the example are provided below.

Type/Period	Jan	Feb	Mar	Apr	May	Jun
Cumulated amount	86	242	531	761	798	1166
Conversion rate	1.1853	1.1251	1.0014	1.0026	1.1776	1.1822

To calculate the *net year-to-date values* on the quarterly figure for June, the following algorithm is used:

1166 * 1.1822 - 531 * 1.0014 = 846,7018

To calculate *the aggregation of periodic values* on the quarterly figure for June, every monthly contribution is converted at the FX rate of the period to which it makes reference using the following algorithm:

(761 - 531) * 1.0026 + (798 - 761) * 1.1776 + (1166 - 798) * 1.1822 = 709,2188

Conversion algorithm for matrices with analytic datasets

On-the-fly conversion for matrices with analytic datasets

In multidimensional matrices with data from analytic datasets containing the currency as a dimension, conversion can be performed on the fly.

When a matrix is being designed, the following properties are available on row/column or matrix filters related to the dimension.

Property	Description
None	The figure is not converted.
Based on measure	• The figure is converted based on the conversion type defined for the measure in the dataset definition step.
Based on account	For datasets with the account dimension only. The data conversion method corresponds with the account conversion method. See "Details of conversion based on account" on the next page.

List of conversion algorithms

The following conversion algorithms are supported for matrices with analytic datasets:

- Average conversion type. Applies the average FX rate (year to date average FX rate: average FX rate from the start of the year to the period taken into consideration). In this case, the conversion algorithm subtracts from the cumulative value of the current period—converted at the average FX rate of the current period—the cumulative value of the previous period, converted at the average FX rate of the previous period.
- Period average conversion type. Applies the period average FX rate (Average FX rate of the month taken into consideration). In this case, the conversion algorithm multiplies the value of the current period by the period average FX rate (of the same period).
- Final conversion type. Applies the final FX rate (FX rate of the last day of the period taken into consideration). In this case, the conversion algorithm subtracts from the cumulative value of the current period—converted at the final FX rate of the current period—the cumulative value of the previous period, converted at the final FX rate of the previous period.

The type of monthly splitting applied to matrices with analytical datasets is *Aggregation of periodic values*, by virtue of the monthly saving of the analytical data. The period lengths supported for the conversion are all those that are active in the matrix design, with the exception of the*Life to date* period length.

Measure conversion algorithms

The conversion types that can be defined for a measure in matrices with analytical datasets are:

- average FX rate
- period average FX rate
- final FX rate
- none

With the option *do not convert*, the data are read and aggregated based on the measure policy without filtering or numeric conversion.

Details of conversion based on account

Conversion *based on account* is only applicable when the dataset contains the account dimension.

For conversion based on account, when the account's conversion type is not supported by analytical matrices, i.e. it is not *average FX rate*, *period average FX rate* or *final FX rate* (e.g. historic FX rate), the system applies the *based on measure* FX rate type.

The aggregation logic of the account, according to its nature, and of the measure, based on its configuration, are applied before conversion.

Conversion on datasets with no account

For a tab or form filter on the currency, the system behaves as follows:

- If the matrix has no overrides on the currency dimension, the system considers the currency of the tab or form filter: if this specifies a *final*, *average* or *period average* conversion type, the system applies the *based on measure* conversion type. If the conversion type is *do not convert*, then only data in the chosen currency are displayed.
- If the matrix has overrides on the currency dimension, global filters are not considered.

Conversion on datasets with Account

Where present in the tab/form filters, if the matrix has no overrides on the conversion the matrix settings are ignored and the *based on account* conversion type is applied.

Conversion details in cell properties

The cell properties report the details of the conversion type applied to each cell. For the *based on account* conversion type, if the account's conversion type is not supported for analytical matrices, the system applies the *based on measure* conversion type and cell properties report *based on account* as the applied conversion type.

Reporting functionalities and engines

Comparison of supported functionalities

A comparison between some of the functionalities supported by the two types of engine is provided below:

Functionality	In-memory engine	Standard engine	Note
Monthly splitting option on currency FX rate (see "Setup & Admin - Data Processing window" on page 246)	-	x	-
Support for the type of monthly splitting for positional reference FX rates	x	-	-

Functionality	In-memory engine	Standard engine	Note
Time-dependent entity hierarchies by consolidation method	-	x	Error message at runtime, if not supported
Transactional forms (see "Introduction to transactional forms" on page 281)	-	x	-
Maximum drill down into detail of rows and consolidable amounts (see "Run drill down" on page 342)	-	x	Disabled in case of in-memory engine with FX rate type Based on account or Period. However, the in-memory engine does support other FX rate types.
Web drill down	x	-	-
Comment query (see "Query" on page 251)	-	x	-
Formula query (see "Query" on page 251)	-	x	-
Management of approval levels	-	x	-
Intersegment in the presence of OR fields or in case of complementary specifics	-	x	-
Interactions with the CFP part	x	-	-
MD formulas in financial statement templates	x	-	Error message at runtime
MD formulas in financial statement templates: matching defined according to 'Control Type'.	-	x	
MD formulas in financial statement templates: matching defined according to 'Custom dimension' (without Account).	-	x	

Functionalities not implemented

A list of functionalities that are not implemented by either of the two reporting engines is provided below.

- Initial balance calculation for the Quarter To Date (QTD), Semester To Date (STD) or Life To Date (YTD) period length
- Filter (matrix) by ORIGIN, USER, JOURNAL NUMBER (NUM_RETTIFICA)
- FX rate on the fly in all types of dataset under the following conditions:
  - the Life to date (LUN_YTD) period length is applied
  - OR filters are used for any dimension, excluding the Account dimension.

    ***Note**: on-the-fly conversion with OR filters for the Account Dimension is supported for both OR leaf filters and OR node filters.*

  - several configured Currency filters exist on the same cell.

When the on-the-fly FX rate is used in a dataset by applying the *life to date* period length, the system behaves as follows both then the form is saved and when the form is opened in data entry mode:

- In design mode, when the Matrix Wizard is closed, a warning appears notifying the user of incorrect configuration.
- In data entry mode, an incorrect configuration warning only appears if the user clicks on the warning button on the Data Entry toolbar.

Design: basic concepts

Contents

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Structure of the data in the forms and filters

Introduction

In CCH Tagetik , the global definition of the data managed by the form depends on the setting of filters on specific data for different levels of the form's structure.

Level	Description
Form	Data handled at the global level in the form
Sheet	Data handled at the template level
Matrix	Data handled and viewed in a specific area of the sheet
Row/column	Data handled and viewed in the rows/columns of the matrix

Applying filters therefore means limiting the data handled in the form, the sheet and the matrix to certain dimensions and categories. When a filter is applied to one of the form's structure levels, it has a domino effect on the levels below.

It is in any case possible to override the exclusion of a filter from a level of the structure of the form that inherited it. For more details, see "Overrides" on page 107.

Saving the Point of View (POV)

Once the run filters have been set and a form has been run, it is possible to save the run settings used via the **Save Point of View** function ( ). In this way, the filters will be set automatically when the form is run. The filter settings window is no longer shown if the POV contains all the filters required by the form.

> ***Note**: in data entry mode, this function also allows you to save the form code so it can be reopened automatically, always in data entry mode, when next accessed.*

Filter settings can be changed and a different POV can be saved, or a POV can be deleted, using the **Delete Point of View** function (). See "Parameters window" on page 338.

Designing forms

Introduction

In CCH Tagetik, forms can be designed for reporting purposes, i.e. to consult and browse the data in the application database, and for data entry purposes, i.e. to edit or enter data in the application database. The design of the structure and the content is identical, but there are specific restrictions on the dimensions, as necessary for the data to be displayed and entered in forms correctly.

Mandatory dimensions

Depending on the selected form type and dataset, some dimensions are mandatory.

Form type	Mandatory dimensions
Report with Financial Workspace data	• scenario • period • amount type • accounts
Data entry with Financial Workspace data	• scenario • period • amount type • accounts • entity • category • all enabled custom dimensions If the details of a custom dimension are not available for certain data, the generic element can be used (see Impostazioni nuova dimensione). For example, if a certain cost is available by product but not by cost centre, when designing the data entry form, a filter can be set on the product dimension and, for the cost centre, the generic element.
Report with Analytical Workspace datasets	• scenario • period • a measure of the dataset • (only with datasets partitioned by business cycle and entity) entity • (only with datasets containing the account dimension) account ***Note**: the scenario and period filters must be specified and must be present in the dataset or inferable from the partition type.*
Data entry with Analytical Workspace datasets	The set of mandatory dimensions for a Data Entry form are the same as those required for an Output Form of the same type.

Specific behaviours per dimension for multidimensional Analytical Workspace matrices

A multidimensional Analytical Workspace matrix can only be run using the in Memory Reporting engine (see "Types of reporting engine" on page 28).

Only matrices with the following characteristics can be used:

- Only with numerical and/or text-based measures. This measure is only managed if the dataset has the data update mode set to "Update the existing record".
- Without Advanced selection (pruning at 0 + Lowest Level)

Also note the following behaviours:

- When the form is saved, if one or more mandatory dimensions are missing from a matrix, the system shows a message with the requirements that are not met.
- If used in at least one of the original datasets, the Account- and Reason-type dimensions, must be used explicitly in the matrix or as global filters on the form.
- When an optional dimension is not specified in the form filters, the system shows aggregate data for that dimension, i.e. it shows the sum of the value of all its elements. For example, if the Product dimension is present in the dataset but not set as one of the form filters, the system shows the sum

of all its products. If the form is a data entry form, it is not possible to write over these aggregate dimensions.

Data entry configuration

A data entry form must contain at least one of the following elements:

- cells editable by user (see "Cell editability" on page 78)
- other formulas (see "Formula filters" on page 159)
- query on cells (see "Query" on page 251)

For more details, consult the relevant pages.

Design toolbar

Excel > CCH Tagetik > Design > select existing form/create new form

Introduction

The design toolbar is shown when a form has been opened from the **Design** tab in the worksheet.

Buttons on the Design toolbar

Element	Description
Definition	
	**Form filters**: opens a window to set the filters to be applied to all templates included in the selected form. See "Form filters window" on page 139.
	**Tab filters**: opens a window to set the filters to be applied to the selected template. See "Tab filters window" on page 142.
	**Matrix**: runs the **Matrix Wizard** to create a matrix based on the cell selected in the work desktop. See "Matrix Wizard - Window structure" on page 109.
	**Query** opens the **Query Wizard** to define an SQL query in the selected cell in the work desktop. See "Query Wizard window" on page 255.
	**Cell field**: opens the **Cell fields wizard** to create a cell field in the selected cell in the work desktop. See "Cell Fields Wizard" on page 41.
	**Dictionary item**: opens the **Dictionary item** window to create a dictionary item in the selected cell in the work desktop. See "Dictionary Items window" on page 50.
	**Data processing**: allows you to link data processing to the form and set parameters for it. See "Setup & Admin - Data Processing window" on page 246.
	**Hyperlink**: opens the **Action List Wizard** to create a hyperlink in the selected cell in the work desktop. See "Action List Wizard window" on page 225.
	**Report list**: opens the **Report List Wizard** to manage Report lists. See "Report List Wizard window" on page 366.
	**Validation**: opens the **Validation** window to define a validation expression in the selected cell in the work desktop. See "Validations window" on page 44.
	**Slicer**: opens a window to define a slicer in the selected cell in the work desktop. See "New Slicer window" on page 98.
	**Index**: opens the **Report List Wizard** to insert an index of the sheets in the current form. Useful for Report lists which publish the result on a single file and for all forms containing more than one template. See "Report List Wizard window" on page 366.
Overrides	

Element	Description
	**Editable cells preview**: highlights cells on which editability rules have been set. The cell's colour depends on the editability rule set on it. For more details on colour allocation, see "Layout options details" on page 56.
	**Override content**: overrides the content of an element of the matrix with the value entered by the user in the selected cell. See "The Override content on cells function" on page 108.
	**Override editability**: changes the editability of the cell of the selected matrix in the work sheet. For more details on the available options, see "Editability" on page 116.
Parameters	
	**Definition**: opens the **Parameters Definition** window. See "Parameters Definition window" on page 68.
	**Management**: opens the **Parameters Configuration** window. See "Parameters Configuration window" on page 71.
Miscellaneous	
	**Cell properties**: in the left-hand panel, opens the **Cell properties** tab showing information on the selected cell and filters in place.
	**Form options**: opens the **Form options** window to define the form's graphical and functional properties. See "Form options window" on page 55.
	**Export as an ad hoc form**: exports the current form as an ad hoc form, in a folder chosen by the user.
	**Structure check**: checks that the form's structure is correct.
	**List of filters used**: opens the **List of filters used** window showing a list of the form's filters, templates and matrices.
	**Save a copy in the table**: only available in the design phase for ad hoc forms. Save a copy of the current form in the table.
Run	
	**Run**: generates a preview of the form in navigation mode.
	**Partial run**: only available for Report lists. Runs and updates the data for some of the Report list's form only.
	**Print Filters**: generates a preview of the form, highlighting cells containing specific filters requested by the user.
	**Run Web preview**: only for forms available as web forms. Generates a preview of the form on the web interface and activates navigation mode.
	**Run pivot preview**: generates a preview of the form, displaying the content in the form of a pivot table. See "Pivot tables" on page 349.
Dashboard	
	**Definition**: only available for forms with the **Enable dashboard** option active. Opens a window to define a dashboard based on the data present in the form template matrices. See "Define Dashboard window" on page 97.

Creating forms

Create a standard form

Excel > CCH® Tagetik > Design > select form grouping >

This procedure explains how to create a standard form from CCH Tagetik's Excel add-in.

1. In the **New form** window, enter a code and one or several descriptions for the form, based on the languages supported by the CCH Tagetik project.
2. Select the desired options.
3. To exclude the form from data entry and navigation, click **Do not display**.
4. To enable/disable the form as a web form, choose the desired option in **Enable as web form**.
5. To enable the form as a Dashboard report, select **Enable dashboard**.
6. To enable the form as a Pivot table report, select **Run as pivot table**.
7. To make the form non-editable, select **Lock Structure**.
8. To use an ad hoc form as a model for the new form, click next to **Ad hoc form** and select the desired file.
9. Click **OK**: the form will be created and opened in Design mode.

Create a multi-template form

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

It is possible to create multi-template forms both in the standard form creation step and from pre-existing forms, by adding one or more templates to the default template.

1. Run the desired form in design mode.
2. Click the button at the bottom-left next to the template name.
3. Enter the template name.
4. Alternatively, click on the toolbar, click in the column to the left of the opened document: the new template will appear below the existing template in the form.
5. Design the template with matrices, filters, etc., following the general procedures for reporting.

   ***Note**: we advise against adding spreadsheets that are not associated with a CCH Tagetik template, because they are not managed correctly when the form is run.*

Create an ad hoc form

Excel > CCH® Tagetik > Ad hoc form > New ad hoc form

Excel opens a new spreadsheet in design mode, the structure of which can be defined in accordance with the standard procedures for forms.

***IMPORTANT**: to save the ad hoc form, proceed as you would for a normal Excel document.*

IMPORTANT: In order to save the ad hoc form on OneDrive, at least one of the following variables must be configured with the path of the OneDrive folder:

- *One Drive folder* configured on the excel addin by following the path, *Options -> Local Settings*
- *OneDriveTagetik* configured as Custom environment variable
- *OneDrive* configured as an environment variable

The system follows the following logic:
If the variable One Drive Folder is populated, the relevant path will be evaluated. If the path is correct, the form can be saved correctly, but if the path is wrong, the user will be notified when opening the form that the wrong path has been used and the form will not be retrievable.
If the One Drive Folder variable is not populated, the system checks whether the OneDriveTagetik custom environment variable is present. The evaluation of the correctness of the path is identical to the previous variable.
Only if the One Drive Folder and OneDriveTagetik variables are not populated, the OneDrive environment variable is evaluated with the same logic as the previous ones.

The three variables must have the given names otherwise they are not considered by the system. In the absence of all three variables, the user is informed when saving that one of the variables needs to be set.

Create the structure of a transactional form

Admin | <application database>

Application database > Forms management tile > Elements > ⊕

This procedure explains how to create the structure of a transactional-type form from CCH Tagetik's web interface.

1. In the **Attributes** tab, insert a code and a description for the form.
2. In **Type**, choose the type of form to be created.
3. In **Input type**, select the type of information to be managed in the form. In the **Input properties** section, the system automatically selects input properties that are consistent with the selected input type.
4. (Optional) Change the selected input properties as desired.
5. In **ETL**, select the desired job and period length for running the ETL routine.
6. (Optional) In the **Relationships** tab, select a group of forms with which to associate the form.
7. Click **Save**.

Create a transactional form

1. Create a data entry form, see "Create a standard form" on page 35.
2. Insert the COUNTER element, contained in the **Counter** folder in the dimensions tree, as a row filter.
3. Insert the columns of one of the transactional forms present in the **Column form** folder in the dimensions tree as a column filter: the columns defined in the web interface for that form will be shown.

   ***Note**: the same (measure or dimension) column cannot be inserted on an editable basis more than once.*
4. To be able to subsequently delete data present in a row of the transactional form, the DELETE column must also be included.

   ***Note**: if the DELETE column is not inserted, it will only be possible to edit the data already in the form or insert new data.*

Create a form with a Line by Line matrix

After having created the form structure, it is possible to insert a transactional matrix into the form to create the finalised form.

Excel > CCH® Tagetik > Design > select form grouping > ⊕

1. In the **New form** window, enter a code and one or several descriptions for the form.
2. Select the desired options for the form.
3. Select the cell in which to insert the matrix, the click .
4. In the **Choose the dataset** window, click the desired dataset from the Analytical Workspace.
5. Select the **Transactional** option.
6. Click **OK**.

Once the matrix has been created, the system assigns a default name (e.g. Matrix00, Matrix01...), which can be edited in the matrix wizard by the **Matrix options** tab in the **Code** field.

Create and multidimensional form

Excel > CCH® Tagetik > Design > select form > Matrix

***Note**: a multidimensional form is a form containing multidimensional matrices, which manage data on both rows and columns. For more details, see "Multidimensional matrices" on page 101.*

1. In the **Choose the dataset** window, click the desired dataset.
2. Click **OK**.

Once the matrix has been created, the system assigns a default name (e.g. Matrix00, Matrix01...), which can be edited in the matrix wizard by the **Matrix options** tab in the **Code** field.

***IMPORTANT**: multidimensional matrices cannot be designed on datasets for which the "Show user data only" property is active.*

New form window

Excel > CCH® Tagetik > Design > select form grouping >

Purpose of window

This window allows you to create a new form.

Options in window

Element	Description
Code	Form code.
Description	Form description. In multilingual projects, there is a description field for each supported language.
Run as pivot table	In navigation mode, this allows the form to be opened in pivot mode. This allows you to optimise the matrix formatting and cell rendering processes in Excel. When it is active and several matrices are present, then the form is opened in Excel it becomes mandatory to choose a matrix to display. In the case of Web Forms, only forms containing a single matrix can be run.
System	When selected, this indicates that the form is a system form, i.e. a pre-defined form that can be used as a template to develop customised forms depending on the business at hand. ***IMPORTANT**: once created, these forms cannot be edited. To edit them, this property must be disabled.*
Use as dashboard	This allows you to enable the form to be used as a dashboard. See "Dashboard reports" on page 95.
Lock structure	Only the user who created the form can edit it. Forms with this option active are identified by the icon. It can only be enabled and disabled by users who have function 0000415 – Reporting Excel – Lock templates active in their role.
Do not display	Do not display the form in data entry or navigation modes. This option can be used to prevent forms from being seen by users in the configuration or editing phases.
Enable as web form	Defines whether the form is available as a web form. • **From configuration**: depends on how the *Enable all forms as web forms* option is set in CCH Tagetik's general settings. See Finestra Impostazioni. • **Yes** • **No**
Ad hoc form	Allows you to select an **Ad hoc form** to be used as the template for the new form. The new form will inherit the ad hoc form's structure and configuration.

Batch import window (Import Wizard)

Excel > CCH® Tagetik > Design > select form > Batch import

Purpose of window

This window allows you do load one or more ad hoc forms into the CCH Tagetik application database and save them as standard forms.

Options in window

Element	Description
Input folder	Allows you to select the path in the application database indicating the location from which to read the ad hoc forms.
Grouping	Allows you to select the rule according to which forms will be saved in the application database. • **Imports all forms in the selected node or the *N.A. - Unrelated* node of the grouping**: all ad hoc forms will be saved under the node selected in the left-hand panel, or the *N.A. - Unrelated* node, if no specific mode is selected. • **Recreates nodes using the folder structure based on the selected node or the *N.A. - Unrelated* node of the grouping**: based on the node selected in the left-hand panel, the system creates the nodes in which it saves the ad hoc forms by copying the folder structure in which the forms to be imported are organised in the file system's input folder. If no specific node is selected, the new nodes are saved in the *N.A. - Unrelated* node.

Batch export window (Export Wizard)

Excel > CCH® Tagetik > Design > select form > Batch export

Purpose of window

This window allows the user to download one or more forms from the CCH Tagetik application database onto their device and save them as ad hoc forms.

Options in window

Element	Description
Input folder	Allows you to select the path in the file system indicating the location in which to save the ad hoc forms.
Create grouping	This allows you to recreate, when the form is saved, a folder structure in which the forms will be saved, that is the same as that of the nodes of the grouping to which the forms to be exported are related.

Managing forms

Open a form in Design mode

Excel > CCH® Tagetik > Design

In the left-hand panel, double click on the desired form, or click .

Copy an existing form

Excel > CCH® Tagetik > Design

This procedure describes how to create a form with a similar structure to that of an existing form, by creating a copy of the existing form. After the form has been created, it is possible to make changes to it.

1. In the left-hand panel, select the form to be copied and click , or right click, then click **Copy**.
2. Select the desired folder, then click to paste the new form: the **New form** folder will open.
3. Set the basic configurations for the new form as desired (see "Create a standard form" on page 35).

Transform an ad hoc form into a shared form

Excel > CCH® Tagetik > Design

This procedure describes how to load an ad hoc form onto the application database, to make it accessible to other users.

1. In the left-hand panel, select the grouping node in which you intend to save the form.

   ***Note**: if the grouping node is not selected, the system saves the form in the "N.A. - Unrelated" node.*

2. On the **Design** toolbar, click : the **Import** window will open.
3. Configure the general form settings (see "New form window" on page 38).
4. Click **OK**: the shared form will open.

   ***Note**: if the form is not shown in the left-hand panel when opened, click .*

Transform several ad hoc forms into shared forms

This procedure describes how to load several ad hoc forms onto the application database, to make them accessible to other users.

1. On the **CCH Tagetik** toolbar, click : the **Import Wizard** window will open.
2. In **input folder**, select the path leading to the ad hoc forms to be saved onto the application database.
3. In **Grouping** select the rule according to which the forms must be saved, choosing whether or not to recreate the system of subfolders that may be present in the input folder.
4. (Optional) If you also want to copy the system of subfolders present in the input folder, select **Recreate nodes using the folder structure from the selected node or...**.
5. Click **Next**.
6. Insert the code and description of the form (these are shown in the table of CCH Tagetik forms).
7. Click **Next**: the **Export Wizard Logs** will appear showing details of the exported forms and/or groupings.

   *Click **Copy** to copy the log in text-based format.*

8. Click **End**.

Transform a shared form into an ad hoc form

Excel > CCH® Tagetik > Design

This procedure describes how to export a shared form to create a local copy of it, i.e. an ad hoc form.

1. Open the form.
2. On the **Design** toolbar, click : the **Save As** window will open.
3. Select the file system path to indicate where the form should be saved.
4. Click **Save**.

Transform several shared forms into ad hoc forms

This procedure describes how to export several shared forms in ad hoc format, to create a local copy of them that is only accessible to the user who exported them.

1. On the **CCH Tagetik** toolbar, click : the **Export Wizard** window will open.
2. Select the desired forms or groupings of forms.
3. Click next to **Output folder** and select the file path leading to the location in which the selected forms and/or groupings are to be saved.
4. (Optional) Select **Create groupings** to copy any hierarchies into which the selected forms and/or groupings are organised.

5. Click **OK**: the **Export Wizard Logs** will appear showing details of the exported forms and/or groupings.
6. Click **End**.

Cell fields

Function of cell fields

In the matrix configuration phase, when a dimension element is not defined in the form as a row or column filter, it is clearly not possible to retrieve its description from among the matrix headers. To do this, it is useful to insert *Cell fields* directly into the spreadsheet to allow you view the desired attribute, related to the dimensions used in the form or otherwise, in a header cell (e.g. details of elements defined as matrix filters, form filters or tab filters). Cell fields are also useful for audit purposes.

For example, when the user wants to view, in a cell, information on the last user to open the form and the date on which it was opened, it is possible to create a cell field with the following phrase, for example: User USER1 ran the form on 10/08/2020. This phrase contains two text-based parts ("User" and "ran the form on"), one piece of parametric information that refers to the user name (User function) and one that refers to the date on which the form was opened (runDate function).

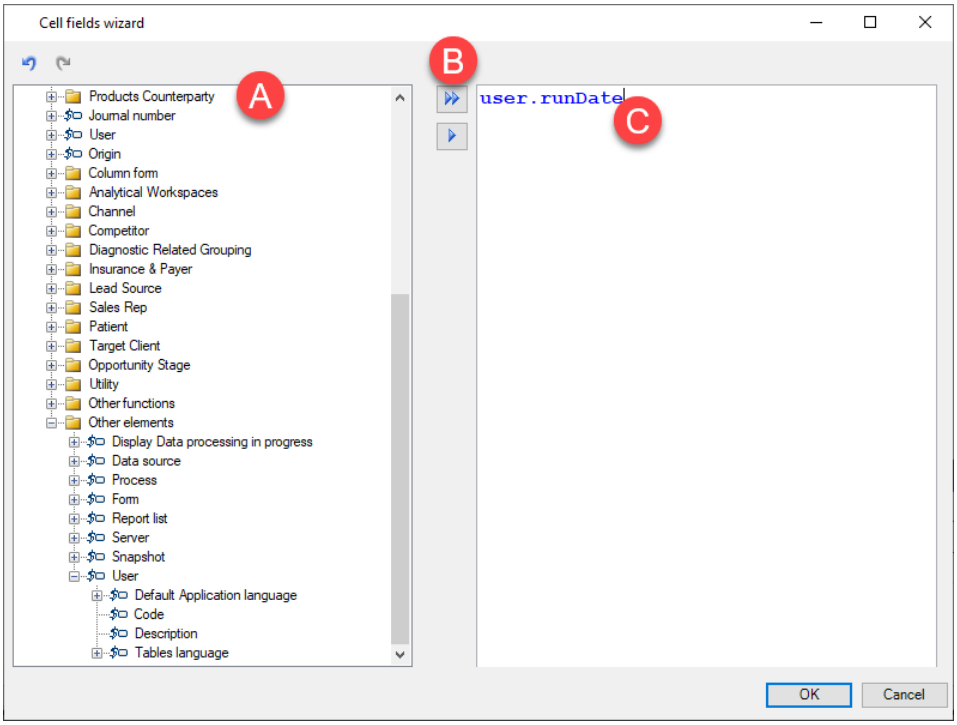
Cell Fields Wizard

Excel > CCH® Tagetik > Design > open form > select cell > Cell field

Purpose of window

The Cell Fields Wizard allows you to insert into the form cell fields in which to display specific descriptions.

What it looks like



Element	Description
A	Tree of functions with attributes that can be inserted as cell fields ( )
B	Action buttons
C	Syntax associated with the cell field

Buttons in the window

Element	Description
	Inserts the whole expression associated with the cell field.
	Inserts the current function only, not the entire expression. Useful for manually building cell fields not available in the predefined functions.

Configuring cell fields

Excel > CCH® Tagetik > Design > open form > select cell > Cell field

Inserting a dimensional cell field

1. Select the desired dimension element in **Predefined functions** and click . The syntax for the selected element will be shown in the cell definition area.
2. Repeat step 1 to insert other dimension elements. Elements are separated by the "." symbol.
3. Click **OK**: the cell will be inserted into the form.
4. If useful, edit the cell format.
5. If useful, add a brief description of the cell field's content in the cell next it.

When the form is run, the cell field is defined with the chosen element. By changing the entity or node when opening the form, the cell value changes depending on the selected element.

Inserting text with parametric elements

This procedure shows how to insert parametric text into a cell of the work desktop using parts of text and cell fields.

1. In the definition area of the cell fields wizard, insert the desired text between double quotation marks (e.g. "text"), and concatenate it to the text with operators such as + or -.

   ***Note**: if no operator is inserted by default, they are automatically concatenated using the operator + ", " +.*

2. To insert parametric information, select the desired element from the functions tree and click . The syntax for the selected element will be shown in the cell definition area, next to the inserted text.
3. (Optional) Edit the text-based content as desired and repeat step 2 for every desired dimension element.
4. Click **OK**.

When the form is run, the inserted text and the values of the parametric information to which the syntax refers are shown in the cell.

Delete a cell field

Excel > CCH® Tagetik > Open items

In the left-hand panel **Open items**, select the desired cell field and click .

Validations

Purpose of data validation

When using data entry forms, it is useful to perform *validation* to check that the entered data are correct, before saving them. To do this, CCH Tagetik uses *validation checks*, i.e. conditions defined by the user that the data must fulfil in order to be saved.

There are the following types of validation:

- **Free validations**, performed on the content of an individual Excel cell in a template
- **Matrix validations** performed on the content of each of the cells belonging to a row or column of the matrix

For example, in the presence of a matrix in which users enter the prices of software products sold, if all the products are priced at between EUR 200 and EUR 4000, validations can be used to prevent users from accidentally entering prices outside of the correct range.

The check can be set to run when a form is opened or saved. When the outcome of the check is negative, depending on the configuration, the data will be prevented from being saved, or the system will display non-blocking error messages (warnings) containing details of the cells containing errors.

Levels of matrix validation

The following validation levels can be set:

- On all columns/rows, i.e. the system only saves the data when all columns/rows are correct.
- On at least one column/row, i.e. it only saves the data when the check is successfully passed on at least one of the cells of the column/row.
- On a formula containing the sum of the values of all the columns/rows, i.e. the system only saves the data when the sum of the data remains within a specific value limit.

Validation tolerance threshold

For every form, it is possible to set a threshold below which the system does not report a validation as incorrect (see "Other options details" on page 57).

By default, the value of this threshold is 0.00000001.

Configuring validations

Define a free validation on a cell

Excel > CCH® Tagetik > Design > open form > select cell > Validation

1. Select the operator.
2. Select the data type.
3. Enter a reference value for the check.
4. (optional) Click to add a term to the validation expression.
5. (optional) Change the relationship between the terms of the validation expression.
6. Repeat steps 1 and 2 for the added term.
7. Insert the warning message to be shown when validation is impossible, for every system language.
8. Define the rules for running the validation check.
9. To save the validation, click **OK**.

The validation element will appear in the **Open items** panel, below the node of the current template.

Define matrix validation on a row/column

Excel > CCH® Tagetik > Design > select form > Matrix

1. Select the row/column on which to set the validation check.
2. In the properties table, click ..., next to **Validation**: the **Validation** window will appear.
3. Select the operator.
4. Select the data type.
5. Enter a reference value for the check.
6. (optional) Click to add a term to the validation expression.
7. (optional) Change the relationship between the terms of the validation expression.
8. Repeat steps 3 and 4 for the current term.
9. Insert the warning message to be shown when validation is impossible, for every system language. The message can be defined by applying shared dictionary items (see "Dictionaries" on page 48).

10. Define the rules for running the validation check.
11. To save the validation, click **OK**.

Validations window

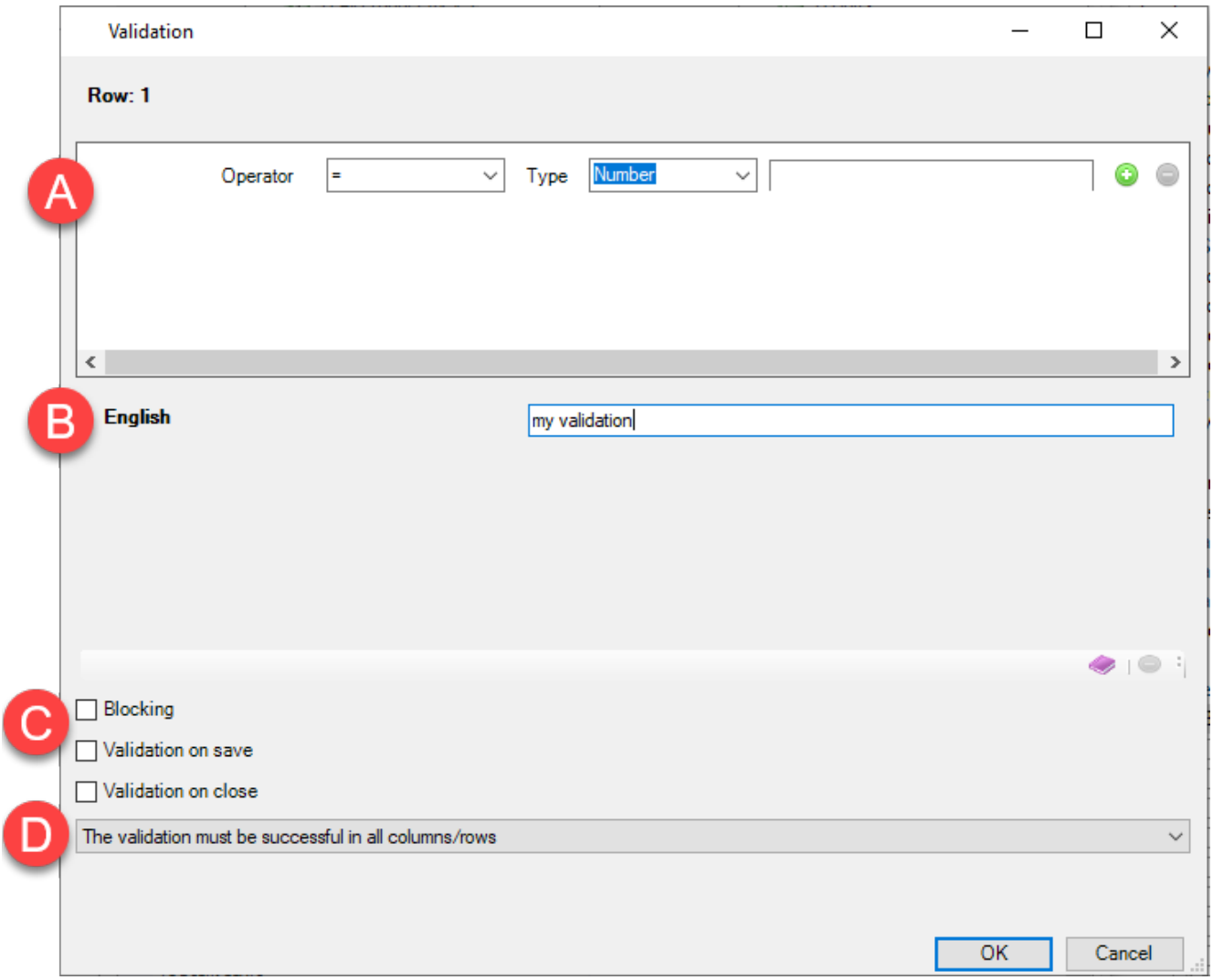
Excel > CCH® Tagetik > Design > open form > select cell > Validation

Excel > CCH® Tagetik > Design > open form > open matrix > select row/column filter > (Properties grid) Validation > ...

Purpose of window

This window allows you to define the validation expression based on a validation check.

What it looks like



Element	Description
A	Validation expression definition area
B	Error message definition area
C	Validation performance rules definition area
D	Selection of the validation run mode in matrix

Window sections

Element	Description
Validation check	Allows you to define the conditions of the validation expression.
Error message	Allows you to set, in all system languages, the error message shown in case validation is impossible.
Validation performance rules	The following options are available: • **Blocking**: if validation of the data is impossible, the system prevents it from being saved. If the option is not selected, the system shows a warning message and asks the user for confirmation before continuing. • **Validation on save**: in the data entry phase, validation is performed when the form is saved. • **Validation on close**: in the data entry phase, validation is performed when the form is closed. This option does not allow you to select the **Blocking** option.
Validation run mode in matrix	Only shown during the definition of a matrix validation. Allows you to define the rows/columns on which to run the validation check.

Validation check details

Element	Description
Operator	• **isTrue** and **isFalse**: operator referring to a cell containing a Boolian or numeric value (0=false, <>0= true). • **=, <>, >, >=, <, <=** • **Between** (included between): defines a range of numerical, string or data values within which the insertions must fall. • **IsEmpty**: the validation is passed if the cell is empty. • **IsNotEmpty**: the validation is passed if the cell is not empty. • **IsNumeric**: the validation is passed if the cell contains a numerical value. • **IsString**: the validation is passed if the cell contains a text-based value. • **IsInteger**: the validation is passed if the cell contains a whole number. Any decimal figures introduced by copying and pasting will not be saved.
Type	Defines the type of data on which to run the validation check. Is only required for some operators. The available options are: • **Number** • **Body** • **Date**
Reference value	Defines the reference value to run the validation check.
Concatenation of several conditions	Only with several conditions in the validation expression. Defines the relationship between the individual conditions. • **AND** • **OR**
	Adds a condition to the validation expression.
	Removes a condition from the validation expression.

Error message details

Element	Description
	s you to create a dictionary item for the error message.

Details of validation run mode in matrix

Only shown when the validation is defined on a property of a matrix row/column filter.

Element	Description
The validation must be successful in all columns/rows	Validation is run individually on every cell in the selected row/column and is only successful if the outcome is positive for all cells (there is an 'AND' relationship between the validations)
The validation must be successful in at least one column/row	Validation is run individually on every cell in the selected row/column and is only successful if the outcome is positive for at least one of the cells (there is an 'OR' relationship between the validations).
The validation is performed by summing up the values of the formula in all columns/rows	Validation is run on the sum of the values of all the cells in the selected row/column. This option is particularly useful when you want to define a condition on the total of a row/column without inserting a formula.

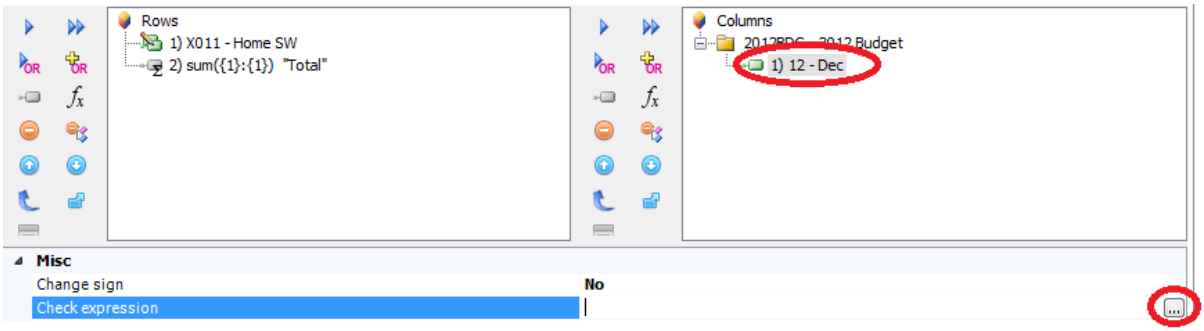
Buttons in the window

Element	Description
OK	Saves the validation.
Delete	Only for matrix validations, this is used when the validation definition window is opened and a previously defined validation is present, allowing the user to delete it.

Examples of validation

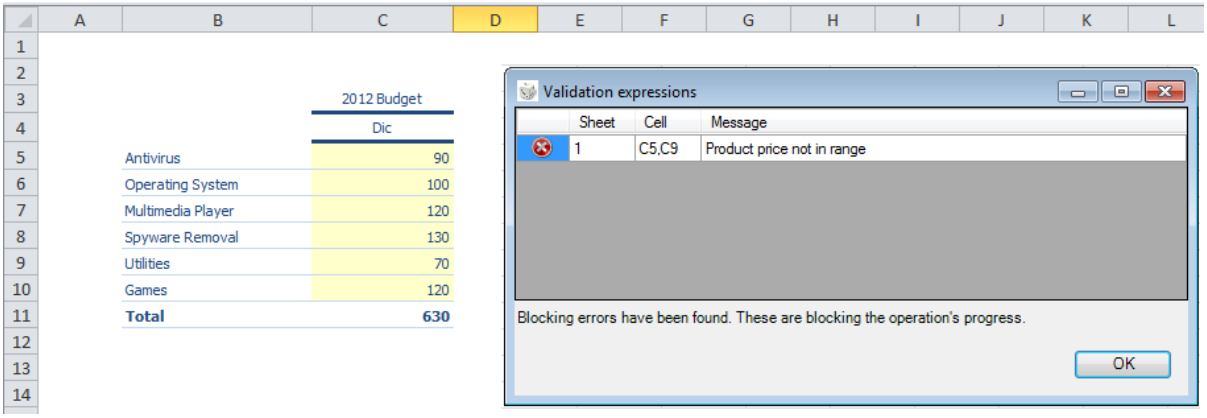
Introduction

Suppose you have a matrix in which the products sold by a certain company appear on the rows and the user must enter the prices for the scenario/period of budget 2012BDG/12 in the columns. All the products must have a price of between EUR 100 and 150. Given that the values must be entered in the matrix column, let's define a validation on the column.



Example 1: The validation must be successful in all columns/rows.

Let's suppose that incorrect prices of below EUR 100 have been entered. On saving, the system reports that there are errors in the saved data. In the window containing the error message, the system provides a description of the error and the cells in which incorrect values have been entered. In the below example, we have errors in cells C5 and C9.



To successfully save the entered prices, the data in those cells must be edited to ensure that the values fall within the range. Once the values have been changed, the system will correctly save the data.

	2012 Budget
	Dic
Antivirus	120
Operating System	100
Multimedia Player	120
Spyware Removal	130
Utilities	120
Games	120
Total	**710**

Example 2. The validation must be successful in at least one column/row.

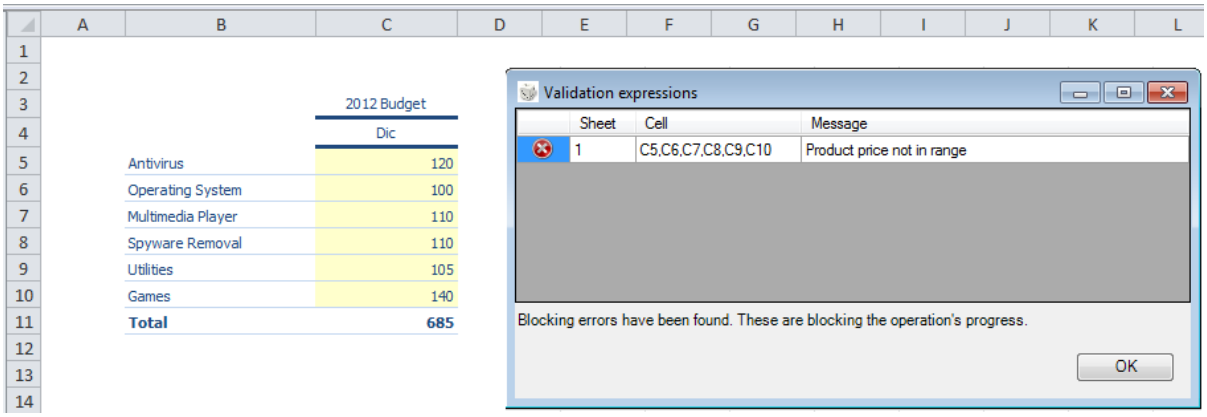
Suppose all the prices have been entered incorrectly. On saving, the system reports an error.

To successfully save the entered prices, at least one price must be edited. If one price falls within the range defined in the validation expression, the system saves the data.

	2012 Budget
	Dic
Antivirus	120
Operating System	70
Multimedia Player	155
Spyware Removal	95
Utilities	40
Games	160
Total	**640**

Example 3. The validation is performed by summing up the values of the formula in all columns/rows.

In this case, the data are validated not on the individual values within the column, but on the sum of the values present in it. In the below example, despite the fact that all the individual values are within the defined range, saving failed because the total of the value does not fall within the range of values specified in the validation.



To successfully save the entered prices, they must be edited so that the sum total falls within the specified range.

	2012 Budget
	Dic
Antivirus	10
Operating System	50
Multimedia Player	10
Spyware Removal	50
Utilities	10
Games	10
Total	**140**

Dictionaries

Purpose of dictionaries

To create a multilingual form, it is necessary to prepare the translation for all tables (accounts, reporting forms, FST items, custom dimension elements, form names), so that the descriptions can be shown in the language defined for the specific user, with which the form is accessed (see Impostazioni utente).

> ***Note**: in CCH Tagetik, it is possible to specifically set a reporting language that is different from the system language and the language set in the user profile. For more details, see Pagina impostazioni Localizzazione.*

However, a form may contain other useful fields that are not included among the tables. All these fields can be translated into several languages using a *dictionary item*.

In particular, a dictionary item is useful for obtaining the following outcomes:

- set a description for a placeholder or formula (e.g. "Total")
- override the description used as the existing header of an element within a matrix (e.g. when the filter on the column contains several 'OR' elements)
- insert multilingual text in the form

Dictionary items can be defined both within the matrix and outside of it. The same item can also be used in several forms by creating shared items. Each dictionary item is identified by two elements:

- a **code**
- a **description** in the configured languages.

Some dictionary items may be very frequently used (e.g. "Total"). To save the translations of these items once only and make them available for use in several forms, *shared dictionary items* are created. These items can then be retrieved when necessary.

The **Dict** cell field allows you to retrieve the description of the shared dictionary item in a certain language, based on the code. See "Non-dimensional functions" on page 195.

In particular, web forms support the translation of dictionary items for all Reporting languages, only for dictionaries set up on the web.

Whereas translation of dictionary items is not supported for local dictionaries entered directly into the Excel add-in.

Importing a dictionary

It is possible to create a table of shared items using the **Import dictionary** wizard, which allows you to generate items with descriptions corresponding to those used in the primary language forms, but with codes generated on the fly by the system. In particular, the new codes have the following characteristics:

- they only contain permitted characters (see Creare nuovi codici)
- they do not contain spaces at the start or end
- they concatenate words up to a maximum of 25 characters

Alternatively, it is possible to import items into the shared table by generating the relative codes based on progressive numbering.

Working on dictionary items

Create a dictionary item outside of a matrix

Excel > CCH® Tagetik > Design > open form > Dictionary item

For every managed language, enter the desired text then click **OK**.

***Note**: when the dictionary item is created outside of the matrix, to view the dictionary item in a cell of the form, select the cell before clicking .*

Create a dictionary item within a matrix

Excel > CCH® Tagetik > Design > open form > open or create matrix > Row/column filters

1. Select the element on which to set the description or override the header.
2. In the properties table, click next to **Dictionary**: the **Dictionary item** window will open.
3. For every managed language, enter the desired text then click **OK**: the language set is shown in bold.

Delete a dictionary item

Excel > CCH® Tagetik > Open items

In the left-hand panel, select the dictionary item to be deleted, then click or select the item to be deleted, right click and select **Dictionary items**> **Remove dictionary item**.

Alternatively, click the cell containing the item to be deleted, right click **Tagetik** > **Delete dictionary item**.

Edit a dictionary item

Excel > CCH® Tagetik > Open items

Select the item to be edited then right click and select **Dictionary items** > **Edit dictionary item**.

Alternatively, click the cell containing the desired item, right click **Tagetik** > **Insert/Edit dictionary item**.

Create shared dictionary items

Excel > CCH® Tagetik > Design > open form > Dictionary item

1. From the **Dictionary item** window, click : the **Shared dictionary** window will appear showing a list of the dictionary items that have already been shared.
2. To add a new item, click .
3. Enter a code and a description for the dictionary item.
4. Insert the translations in the system languages.
5. Click **OK**.

***Note**: to view the shared dictionary item in a cell of the form, select the cell before clicking .*

Create shared dictionary items within a matrix

Excel > CCH® Tagetik > Design > open form > open or create matrix > Row/column filters

1. Select the element on which to set the description or override the header.
2. In the properties table, click next to **Dictionary**: the **Dictionary item** window will open.
3. Click : the **Shared dictionary** window will appear showing a list of the dictionary items that have already been shared.
4. To add a new item, click .
5. Enter a code and a description for the dictionary item.

6. Insert the translations in the system languages.
7. Click **OK**.

Form - Import Dictionary window

Excel > CCH® Tagetik > Form - Import Dictionary

Purpose of window

This window allows you to import dictionary items to create a table of shared items.

Import options

Element	Description
Import table	Imports the dictionary items present in the selected forms into the table of shared items.
Replace the instances with a link to the dictionary items in the table	Replaces, in the selected forms, the dictionary items with links to the shared items.
Primary language	Select the language to be used to compare dictionary items.
Consider two items to be equal if	Select the rule according to which two items are to be considered equal so no duplicates are created.
Generate codes automatically	Select the rule according to which codes for the imported dictionary items will be generated.

Dictionary Items window

Admin | <application database>

Forms management tile > Elements > Form Dictionary items

Purpose of the page

This page allows you to create dictionary items shared among the forms. Changes made to dictionary items via this page are also visible in the Excel add-in.

Attributes tab

Element	Description
Code	Dictionary item code
Description <language>	Text of the dictionary item in the supported language.

Example: importing dictionaries

Import with code generation

Let's look at an example of how this wizard can be used. Suppose the user has the following form in which the displayed rows are almost all dictionary items,



report.desc

Entity

Investment Plan

				2013 Plan	
	Description	Fiscal Category	Type	Investment Cost	Need for L/T Financing
Line Item Nodes (for dynamici	1,000	1,000	1,000	1,000	100000%
Invested Total				**1,000**	

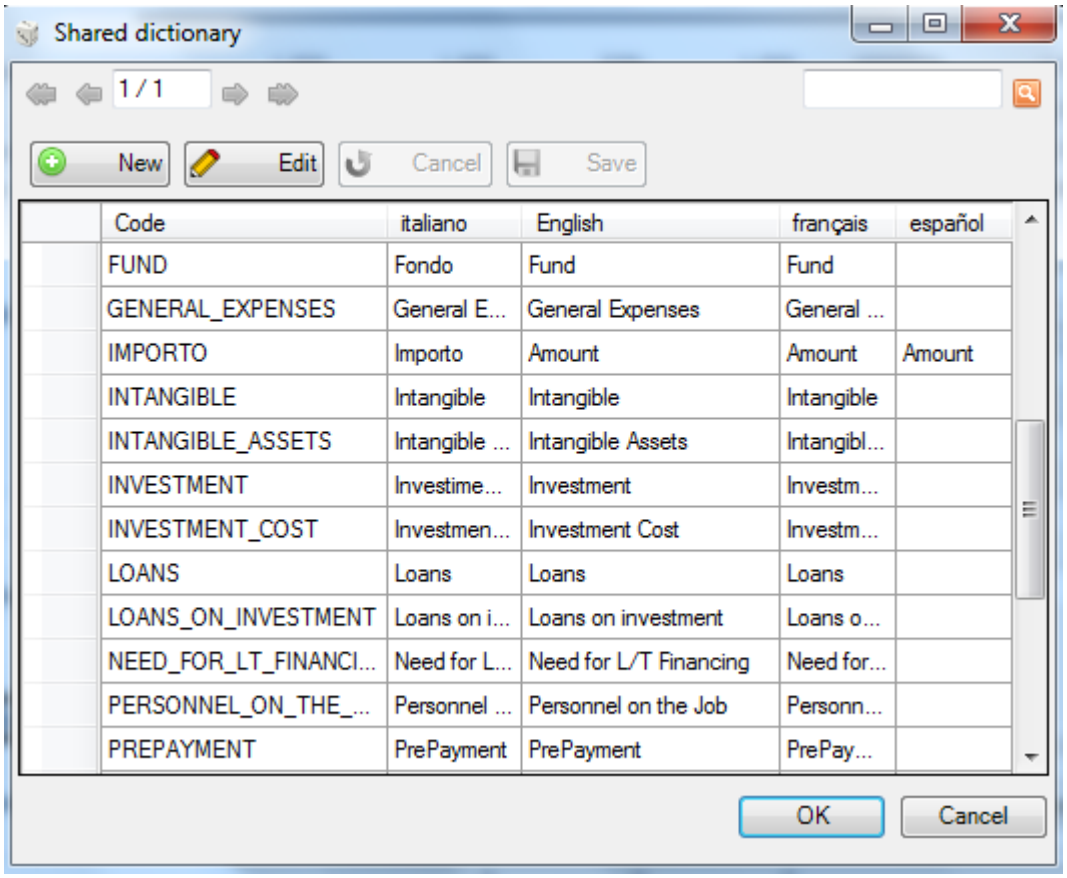
Long Term Financing Plan

	Existing Loans	Est. Prepayments on Ex. Loans		
	2012 Actual	2013 Plan	2014 Plan	2015 Plan
Loans	-1,000	999,000	-998,001,000	########
PrePayment	1,000	100000%	100000%	100000%
Debt Rate	-50.0%	-50.0%	-50.0%	-50.0%
Long Period Financial Exp	**1,000**	**499,000,500**	**#######**	**#######**

Depreciation Plan

	2013 Plan	2014 Plan	2015 Plan
Intangible Assets			
Rate	100000%	100000%	100000%
Investment	0	0	0
Amortization	0	0	0
Fund	0	0	0
Tangible Assets			
Rate	100000%	100000%	100000%
Investment	0	0	0
Amortization	0	0	0
Fund	0	0	0

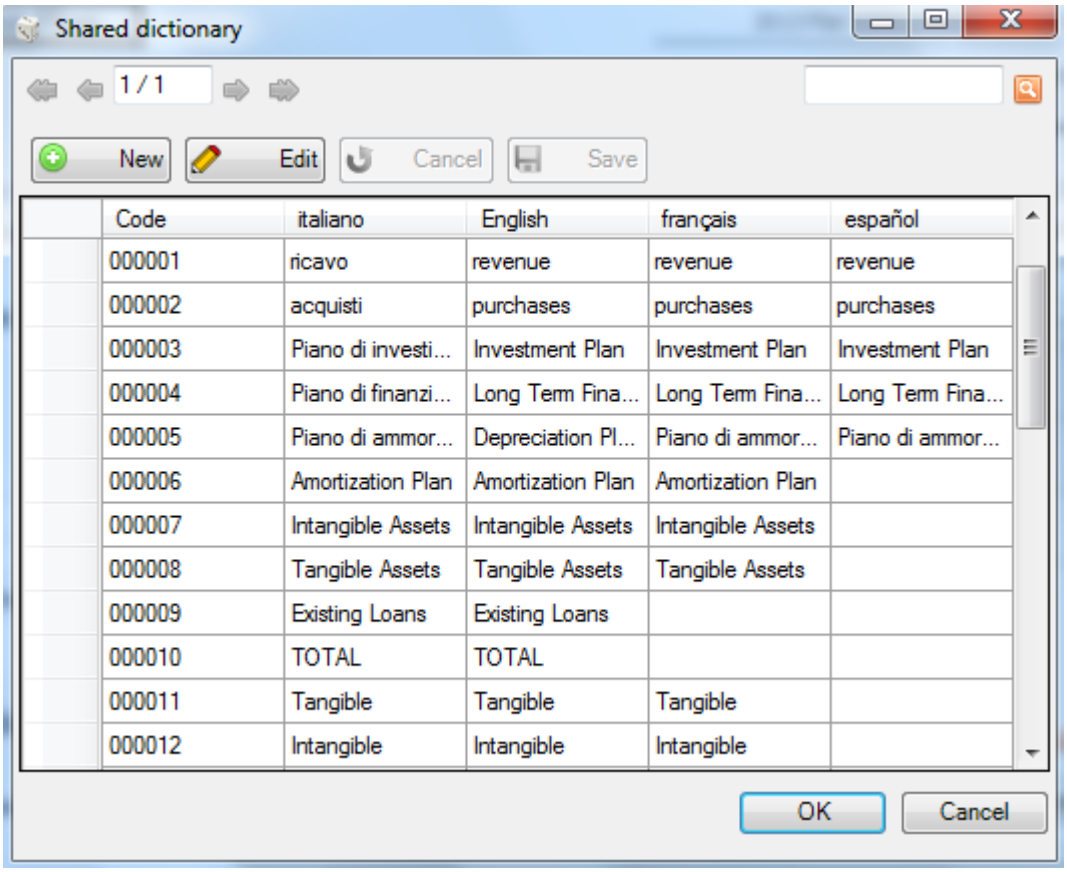
and that they want to import those items into the shared table, generating codes based on the primary language (English). The outcome is:



i.e. based on the description of the dictionary item in the primary language, the system tries to generate a mnemonic code by completing the following steps:

- deleting from the description any characters that are not permitted in codes, e.g. the "/" character from the "Need for L/T Financing" item
- deletes any spaces at the start or end
- concatenates the remaining words, removing any characters at the end if the concatenation exceeds 25 characters

If, however, the items were imported into the shared table with an indication that the codes of the new items must be determined on the basis of a progressive number, the outcome is:


Shared dictionary
1 / 1
New
Edit
Cancel
Save
Code
italiano
English
français
español
000001 ricavo revenue revenue revenue
000002 acquisti purchases purchases purchases
000003 Piano di investi... Investment Plan Investment Plan Investment Plan
000004 Piano di finanzi... Long Term Fina... Long Term Fina... Long Term Fina...
000005 Piano di ammor... Depreciation Pl... Piano di ammor... Piano di ammor...
000006 Amortization Plan Amortization Plan Amortization Plan
000007 Intangible Assets Intangible Assets Intangible Assets
000008 Tangible Assets Tangible Assets Tangible Assets
000009 Existing Loans Existing Loans
000010 TOTAL TOTAL
000011 Tangible Tangible Tangible
000012 Intangible Intangible Intangible
OK
Cancel

Design: form structure

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Structure of a form

Constituent elements

To design a form, the following elements must be defined in accordance with the purpose of the form.

Element	Description
Template	Spreadsheets, which represent the minimum structure of the form. A form may contain one or more templates (see "Multi-template forms" below).
Matrix	Tool for selecting data from the application database by setting filters on CCH Tagetik dimensions. A form may contain one or more matrices on one or more templates.
Filters	Selecting data from the application database to be managed in the form.

Multi-template forms

What they are

In CCH Tagetik, it is possible to create both reports and data entry forms that have several templates, in which every template contains specific matrices and filters.

In both data entry and navigation modes, when a multi-template form is opened, all the templates are run. Multi-template forms can be used for two purposes:

- to create reports (and therefore report books, at the highest level) containing several work folders that are useful in a specific operational context
- to create reports and data entry forms for complex data processing, using formulas that can target cells of other templates.

Example of a multi-template report

To create a report on data at the year-end, a form with three templates needs to be created:

- Profit and Loss
- Balance Sheet
- Indirect cash flow

Example of a multi-template form for data entry and reports

To create a data entry form in which to insert quantities sold by product (Q) and prices per product (P), the following templates must be included:

- one template for inserting the quantities sold
- one template for inserting prices
- one template the shows the result P*Q in order to calculate the total revenues for the products

Restrictions on the names of multi-template forms

The names of the sheets in the form cannot exceed 31 characters. When they are named automatically by the *Link the template name in the sheets* setting, the template name is concatenated to the sheet names, and the name generation algorithm removes excess characters (see "Form options window" below).

Form options window

Excel > CCH® Tagetik > Design > open form > Form options

Purpose of window

This window allows you to manage the general run settings and the layout and operational settings of the current form.

Window tabs

Element	Description
Run settings	Allows you to manage the form's run and save options, the data processing run mode and the query run mode.
Layout options	Allows you to manage the form's layout options and use Tagetik styles.
Other options	Allows you to manage specific operational functionalities of the form.

Run settings details

Element	Description
Data processing type	
Run **Saving** **Run data processing**	Defines how the form and the application server interact. • **Automatic** (default): the system determines how the form interacts with the application server. • **Synchronous**: form interacts with the application server in a synchronous manner. • **Asynchronous**: form interacts with the application server in an asynchronous manner. This option is advisable for slow forms that take more than one minute to run.
Reporting engine	Defines the reporting engine to be used to run the form. ◦ **From configuration**: the reporting engine is defined in advanced form settings (see Settings window). ◦ **Standard**: engine suitable for forms with a high number of filters and form runtimes in excess of one minute (see "Standard engine" on page 28). ◦ **In memory**: engine suitable for simpler forms that have advanced functionalities (see "In-memory engine" on page 28). ***Note**: when the In memory engine is selected, the system warns the user that the concurrency settings will be reset, because they are specific to every type of reporting engine (see "Types of reporting engine" on page 28).*
Concurrency	
Concurrency level	Defines the maximum number of queries that can be run concurrently. • **Default**: the number of queries run concurrently is determined by the relative settings at the data processing level (see Data Processing page). • **1 overridden**: manages queries as if they were concurrent, but runs them individually. • **1 (2, 4, 8, ...32)**: the system runs queries one at a time, two at a time, etc. The maximum number of queries cannot exceed the number set for the specific data process in the data processing table (see Data Processing page).
Elements to run concurrently	Defines what elements to run concurrently. • **Templates + Queries** • **Templates + Matrices** • **Templates + Matrices + Queries** • **Templates**
Advanced options: Run in memory	• **Block size rows**: maximum row size of the blocks of data retrieved from the database. • **Size of the data loading threadpool**: number of database connections established. • **Size of the data processing threadpool**: number of blocks processed concurrently.

Layout options details

Element	Description
Colours	
Editable	Defines the background colour of editable cells.
Protected	Defines the background colour of protected cells.

Element	Description
Editable with IC allowed	Defines the background colour of editable cells that allow intercompany data.
With edit Transaction currency	Defines the background colour of editable cells that allow data in the transaction currency.
Layout rules	
Lock output sheets in navigation	Allows you to make the form non-editable when opened in navigation mode.
Tagetik style	Allows you to select the Tagetik style to be used in the form (see "Tagetik styles " on page 230).
Fields separator	Allows you to select the separator to be used in row/column headers when they are made up of a concatenation of several fields.
Generate sheet names according to	Only for forms in which tab filters have been set up. Defines the rule according to which the names of the various sheets are generated. • **Code**: sheet names correspond to the code of the element used as the tab filter. • **Description**: sheet names correspond to the description of the element used as the tab filter.
Link the template name in the sheets	(Only for multi-template forms) Defines the conditions under which the template name must be concatenated to the sheet names. • **Always** • **Never**
Additional information on the headers	(Only for matrices with headers on dimensions for which additional Segment, FX rate, Reference FX rate or Weight type information has been defined) Allows you to configure the headers on which the additional information is displayed. • **On all headers** (default): the additional information is shown on all headers of the dimensions that support it, e.g. Code (SEG.) and Description (SEG.). • **Only on the first header**: the additional information is shown on the first header of every dimension that supports it, e.g. Code (SEG.) and Description. • **Only on the last header**: the additional information is shown on the last header of every dimension that supports it, e.g. Code and Description (SEG.). • **On no header**: additional information is not shown on any header. ***Note**: additional information is shown between parentheses after the relative header.* ***Note**: the additional information is not shown if a dictionary item has been set on the row/column.*

Other options details

Element	Description
Handle custom event callbacks with unprotected sheets	(Only with VBA macros that are associated with a custom event and modify protected sheets generated when the form is run) Allows you to temporarily unlock sheets and lock them again after the macro has been run. For more details on VBA macros, see Custom macros.
Refresh value after saving	Sets the form to automatically refresh when data entered in data entry mode are saved.
Automatically close the form after saving data	Sets the form to automatically close when data are saved.
Disable custom functions	Disables the use of custom functions when the form is run (see "Parameters" on the next page).

Element	Description
Enable Save zero values	Manage cell values of zero. • Active: when a 0 is entered in a form cell, the system creates the corresponding 0 value in the database. • Not active: the system ignores the 0 value and does not edit the database. ***IMPORTANT**: it is not possible to enable the Enable Save zero values option and show data on a non-cumulative basis in the same form (see "Periodic value calculation" on page 113).* ***IMPORTANT**: when both the Enable Save zero values option and the Show zero values if the cells aren't defined option are enabled (see "Matrix options tab" on page 128), the system will also enter and save a zero value in cells with no value, provided they have not been edited directly by the user.* ***IMPORTANT**: When both the Enable Save zero values option is enabled and, on the process for which the form is being run, the Save data in form incrementally option is active (see Pagina Regole Saldi e IC), values that have already been saved on cells cannot be deleted, even if the value is zero.*
Enableschanges made in data entry mode to be tracked.	Enables changes made in data entry mode to be logged. When the form is saved, information on the added/edited rows is saved on the LOG_PROSPETTO_DATA table. The system does not track changes related to transactional forms, intercompany declarations and gross amounts made via the interface, or data processing run when the form is saved. For more details on the table, see Log di tracciamento del data entry.
Drill down	Enables drill down functionality (see "The drill down function" on page 340).
Allow Drill down only on used dimensions	Enables drill down only on the dimensions used in the form (see "The drill down function" on page 340).
Tolerance threshold for the check operations	Allows you to set a threshold within which a validation check will not be reported as invalid (see "Validations" on page 42). The default value is 0.00000001.
Validation of Transactional forms when saving data	For transactional forms with which dictionaries are associated, this enables the validation of values assigned to columns when the form is saved. • **Configuration**: validation depends on the value assigned to the relative setting in form settings (see Form tab Details). • **Yes**: allows you, for the current form, to override the setting defined in advanced form configuration and enable validation on saving. • **No**: allows you, for the current form, to override the setting defined in advanced form configuration and disable validation on saving.
Disable lock when reading values	Manage the data processing lock in the database. • **Configuration**: the lock depends on the value assigned to the relative setting in form settings (see Settings window). • **Yes**: disables the database lock when reading data. • **No**: enables the database lock when reading data. ***IMPORTANT**: removing the data lock allows the form to be run faster, but the accuracy of the data displayed cannot be guaranteed, e.g. if several users make changes to the same data simultaneously.*

Parameters

Parameter functionality

In general, designing a form means defining filters on the data in the operational database, on several levels. If fixed filters are defined, to display different elements of a specific dimension it would be necessary to change the form structure, or design several copies of the same form, every time. To make forms easier to read and allow them to be reused, the user can set *parameters*, i.e. filters whose value is not defined when the form is designed, but when it is run. Thus, the form can be run to show different data without its structure being changed, because parameters allow you to select the desired dimensional elements and/or aggregation structures on a dynamic basis. When running, the form therefore only shows the data that the user has chosen via the parameter. For more details on configuring parameters when a form is opened, see "Parameters Configuration window" on page 71.

Parameters can also be used in forms as row/column filters, matrix filters or tab filters. Using parameters to design forms makes increases flexibility when updating the form structure as every change made to the set of elements included in the parameter is automatically taken into account by the same parameter.

It is possible to use *system parameters*, managed by CCH Tagetik, or *ad hoc* or *shared custom parameters*, managed by the user.

Details of the existing types of parameters are provided below.

Supported types of data

Parameters can be based both on dimensional data and on non-dimensional data.

Parameter	Description
Dimensional	The value obtained from a set of elements of a specific dimension: • *Dimension*: element or node of a CCH Tagetik dimension/hierarchy • *Advanced Financial Statement Template*: parameter for advanced FSTs structured by account/custom dimension and/or control type • *Standard Financial Statement Template*: Parameter for standard FSTs structured by account
Non-dimensional	Parameters used for specific configurations typical of advanced selection scripts. • *Flag* • *Date*: date parameter selected from a calendar • *Dictionary*: dictionary item parameter • *Numeric*: number parameter • *Text*: free text parameter • *Process*: process parameter (see)

System parameter details

System parameters are created automatically when forms are deployed (*Application database* > *General deployment tile* > select *Forms* > *Detailed selection* > select *Data model* > *Other tables* > *Forms* > *Update Reporting dimension parameters*). These parameters allow you to choose any element of the dimensions or any of the nodes of the aggregation structures.

System parameters are available in the dimensions tree (in blue) and have a code generated in accordance with the following rule:

- Parameters related to elements of a dimension: *$ + <name assigned to the dimension by Tagetik> + _ P*. For example, the code of the parameter related to the entity dimension is *$AZI_P*, while the code of the origin scenario is *$SCE_P*.
- Parameters related to nodes of an aggregation structure: *$ + <name assigned to the dimension by Tagetik> + _ + <aggregation structure code> + _P*. For example, the code of the parameter related to entity hierarchy G01 is *$AZI_G01_P*, while the code of the group of scenarios is *$SCE_$_P*.

The description of a parameter corresponds with the description of the dimension or aggregation structure from which it originated.

System parameters are identified with the prefix $ and cannot be created, modified or deleted manually.

When the data model is modified, e.g. a new hierarchy is created, the application needs to be deployed in order to use the new system parameter.

Specific features of custom parameters

Custom parameters are created by the user and can contain hierarchy nodes and/or selected individual elements of a dimension. The user can choose a subset of a dimension's elements or nodes, e.g. all of a hierarchy's nodes except for one, lowest level elements only, certain specific elements and nodes, and so on. Creating custom parameters therefore means that you do not have to create specific system hierarchies for individual forms.

Custom parameters are available for all users and can be:

- *shared*, can be used on all forms.
- *ad hoc*, can only be used in the forms in which they have been defined.

When the custom parameter is created or modified, the application needs to be deployed in order to use the new parameter or updated parameter.

Specific features of ad hoc parameters

Ad hoc parameters have the same characteristics as shared parameters but they can only be used in the forms in which they were defined.

The use of parameters is only advisable in the following contexts:

- The parameter is not needed in other forms, e.g. due to overly specific restrictions.
- The form contains a high number of shared parameters and the parameter in question is only necessary for the specific form.
- The parameter does not need to be shared with another form.

In ad hoc parameters, process parameters or dimensional parameters that allow OR clauses cannot be defined on the period, because they are not supported by forms.

Restrictions on parameters

Certain restrictions can be defined on parameters, allowing you to fine tune the behaviour of the specific parameter.

Element	Description
Standard restrictions	These allow you to restrict the elements that can be selected as the parameter value when the form is opened. In particular, standard limitations allow you to do the following: • defined whether it is possible to select lowest level elements, nodes or several OR elements • select values manually from a dimensions tree, specifying them as explicit restrictions • exclude any restrictions defined at the process level on the dimension to which the parameter is linked
Dynamic restrictions	These allow you do define the list of values that can be assigned to the parameter. In particular, dynamic limitations allow you to do the following: • Define the list of values using a dynamic rule (e.g. insert children, insert lowest level elements). • Set dependent parameters using advanced selection script (see "Details of advanced selection scripts for calculating the default value" on page 63).

The concept of dependency allows you to restrict the set of values for a given parameter depending on the value taken on by another parameter.

> ***Note**: the parameter selection window shown when the form is opened (see "Parameters window" on page 338) undoes the selection of a dependent parameter when the parameter on which it depends is changed. Dependent parameters are shown below the parameters on which they depend.*

Types of parameter

Multi-hierarchy parameter

When a parameter is defined on a hierarchy, the user must specify the hierarchy in question. When the form is opened, it will only be possible to choose values belonging to that hierarchy.
In certain types of configuration, it may be necessary to define multi-hierarchy parameters. These parameters are defined on a dimension which, when the form is opened, allows you to choose a hierarchy from among all those belonging to the dimension. Multi-hierarchy parameters can only be defined on the Entity and Target dimensions.

Multi-hierarchy parameters cannot be used within filters (form, tab, matrix, row, column) defined as OR clauses. When the multi-hierarchy parameter is used, it is not possible to select several elements belonging to different hierarchies as OR items.

If a multi-hierarchy parameter is created with a hierarchy that is already being used in the same position (rows, column, matrix filter), the system does not allow the form to be run.

When the form is run, to obtain the hierarchy to which the chosen dimension element belongs, the syntax *@MHP_PARAM.Dimension.HierarchyCode*, where *MHP_PARAM* is the code of the defined multi-hierarchy parameter, can be used in cell fields or in elements that allow advanced selection.

If you decide to use multi-hierarchy parameters on the rows/columns of a matrix, for the description and/or code to be displayed in the headers correctly, the parameter syntax must be added to the headers configured in the Matrix Wizard. For example, with a multi-hierarchy parameter on the Entity dimension, the syntax to use to obtain the description is *$Entity(false).desc*.

Process parameter

The process parameter allows you to view the list of processes included in the table.

It cannot be used in Reporting forms, but it can be used by DTP modules, QDL and Jobs. The following properties can be configured:

- code and description in the Properties tab
- the *Only show active processes* parameter in the Restrictions tab, which limits the display to the table's active processes available for data entry

***Note**: Process parameters are not shown in the dimensions tree for defining form, tab and matrix filters, because they are not supported.*

Dictionary parameter

The dictionary items available for the parameter depend on the dictionary associated with the form.

Dictionary parameters are associated with a transactional dictionary defined on the Financial layer, or an analytical dictionary belonging to an Analytical Information Hub workspace.

In the parameters definition window, after selecting the type "dictionary", you will be able to select its source by choosing whether to use a Financial Layer dictionary or a dictionary from a specific Analytical Workspace.

When Financial Workspace is selected, the system loads all the transactional dictionaries whereas when an Analytical Workspace is selected, the user can choose a dictionary from among all those belonging to the selected workspace.

Dictionary parameters cannot be defined in the following cases:

- The selected dictionary parameter uses an analytical dictionary belonging to a workspace with no dataset.
- The selected dictionary parameter uses an analytical dictionary defined using an SQL query containing dimensional attributes.

The running of the analytical dictionary query is not affected by user restrictions on dimensions.

Working with custom parameters

Create a shared custom parameter

Excel > CCH® Tagetik > Parameters Definition

1. Click : a row will appear in the **Shared parameters** list.
2. Enter a code and a description.
3. In the **Type** section, select the desired data type. If you choose a dimensional data type, select the desired node or dimension in the **Dimensions** pane.
4. In the **Attributes** section, define the properties and default input mode for the parameter when the form is run in the parameters selection window.
5. Click the **Restrictions** tab and, if desired, in the **Attributes** section, set the input options for parameters omitted and restrictions on selectable data in the parameters selection window.
6. In the **Advanced selection** section, define the rules for dynamic parameter input. To define an advanced selection script, select **Advanced selection** and insert the script in the dedicated box.
7. (Optional) In the **Long descriptions** section, set a detailed description of the parameter for the system languages.
8. Click .

Create an ad hoc custom parameter

Excel > CCH® Tagetik > Design > (Left-hand panel) open form > > Parameters Definition

1. Click the **Ad hoc parameters** tab.
2. Click : a row will appear in the list
3. Enter a code and a description.
4. In the **Type** section, select the desired data type. If you choose a dimensional data type, select the desired node or dimension in the **Dimensions** pane.
5. In the **Attributes** section, define the properties and default input mode for the parameter when the form is run in the parameters selection window.
6. Click the **Restrictions** tab and, in the **Attributes** section, set the input options for parameters omitted and restrictions on selectable data in the parameters selection window.
7. In the **Advanced selection** section, define the rules for dynamic parameter input. To define an advanced selection script, select **Advanced selection** and insert the script in the dedicated box.
8. (Optional) In the **Long descriptions** section, set a detailed description of the parameter for the system languages.
9. Click .

Enable automatic input of default values

Application database > (Settings) > Forms tab

When an optional run parameter is not defined by the user, it can be defined automatically by the system by enabling the automatic input of default values at the system settings level.

To do this, in the **Display** section, click **Automatically define parameters if omitted at the start** (see Finestra Impostazioni).

At runtime, based on the parameter settings, the system defines the parameter with a *default value*, or a value defined by a *script* associated with the parameter.

Set a default value for a parameter

Excel > CCH® Tagetik > Parameters Definition

1. Create a parameter (see "Create a shared custom parameter" on the previous page or " Create an ad hoc custom parameter" above).
2. In the **Properties** tab, next to **Input mode**, select **Set using a default value**. A drop-down menu will appear to select the default value.
3. Click and select the default value.
4. Click .

Set a default value using a script

Excel > CCH® Tagetik > Parameters Definition

1. Create a parameter (see "Create a shared custom parameter" on the previous page or " Create an ad hoc custom parameter" above).
2. In the **Properties** tab, next to **Input mode**, select **Script for default value**. A drop-down menu will appear.
3. Click . The **Cell Fields Wizard** will open.
4. Insert the script indicating the desired dimensional value (see "Inserting a dimensional cell field" on page 42).
5. Click .

Automatic parameter definition

Introduction

A form's run parameters can be defined *manually*, where the user chooses the desired value from a list of values, or *automatically*, where the system defines the parameter based on behaviours defined by the administrator user.

Automatic parameter definition methods

When the user does not define an optional parameter, it can be defined automatically according to two methods.

Method	Description
Input default value	The system automatically selects the value indicated as the *default value* in the parameter settings.
Define using dynamic script	The system reads a dynamic script defined by the user in the parameter settings to calculate the default value.

Details of dimensional parameters with default values

When a default value is has been set on a CCH Tagetik dimensional parameter in the definition step, the system will behave in one of the following ways:

- In the presence of parameters with a default value selected by the user or with a default value comprising the output of a TGKML script, when the *Do not display when the default value is specified* option is active, the parameter will not be shown in the parameters selection window (see "Parameters window" on page 338), but the value will be shown directly in the form. When this happens for all the parameters that can be selected in the parameters selection window, the window will not be show.
- For web forms launched from workflow, from which the scenario, period and entity can be unambiguously inferred, when the form is run the default value will be ignored and the parameter will be defined with the value used in the context.
- In the presence of parameters set with default values, when the *Advanced restriction type* option is active, when the form is run the default value will be ignored and the parameter will be defined with the value selected in the adjacent dimensions tree.

The described behaviours are also valid when parameters are selected on opening a DTP (see Parametric DTPs).

Details of advanced selection scripts for calculating the default value

The requirements for defining an advanced selection script to define run parameters are as follows:

- The script must return a single dimensional element (if it returns more than one, the default parameter value will be the OR of all elements).
- The script does not depend on the value of other parameters that have already been set or filters present in the form.

When the script returns an error, an element that is not visible, or an element that is not compatible with the user restrictions and/or characteristics of the parameter, the default value will not be set.

***IMPORTANT**: if the element code set as default is changed, the attribute will be invalidated.*

For more details on the advanced selection, see "Dynamic matrices" on page 132.

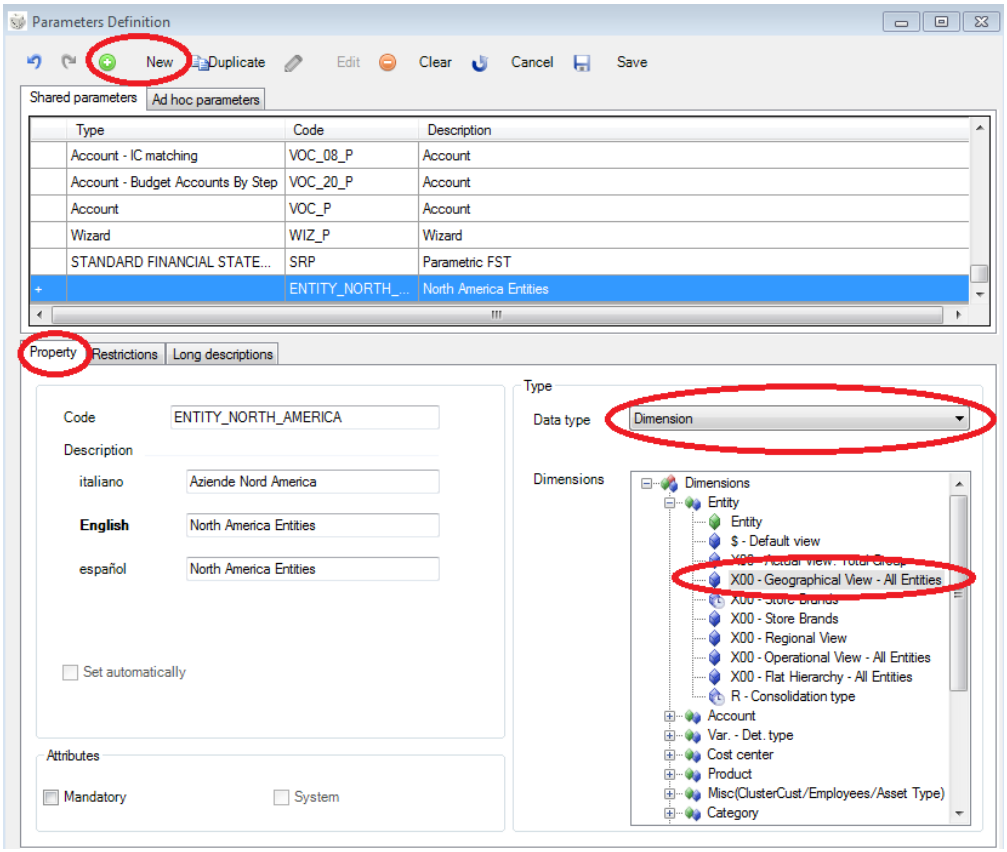
Examples of custom parameters

Creating a custom parameter

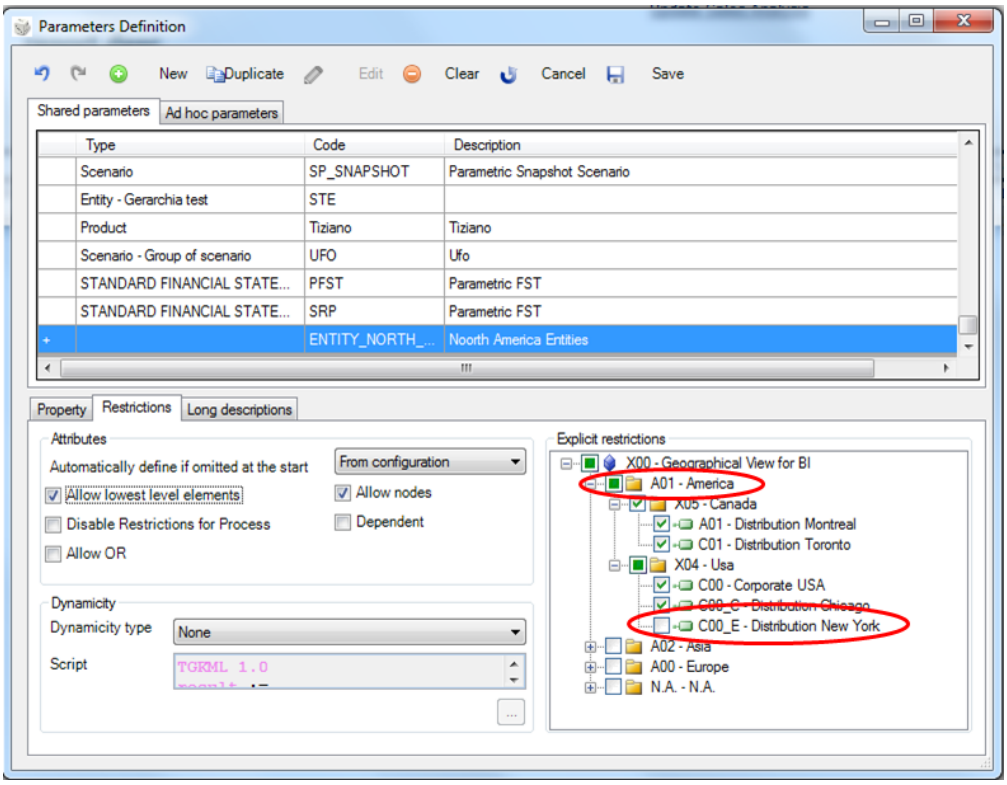
Suppose we need to configure a form that can only be selected by American entities and we have a hierarchy that groups entities based on a geographical rule, i.e. based on countries and regions. To configure a form based on those requirements, we can create a shared custom parameter to restrict the choice to the North America node and all of its child nodes.

To create the parameter after the form has been opened in Design mode:

- click the **Parameters Definition** button on the Design ribbon bar;
- click the **New** button;
- insert a code (ENTITY_NORTH_AMERICA), a description (North America Entities), assign Dimension as the parameter type and select the hierarchy to be associated with the parameter (X00 - Geographical View).



At this point, we can set restrictions on the selected hierarchy. We only want the child elements of the America node to be selectable. To do this, we need to select the **Restrictions** tab and then select the node A01 America in the **Explicit restrictions** section. If a default node is selected, all of that node's child elements will also be selected. An element can be excluded by deselecting it from the hierarchy tree. In the example below, we have selected the America node and all of its children, except for element C00_E. Thus, when the user runs the form they will not be able to assign C00_E as the parameter value.

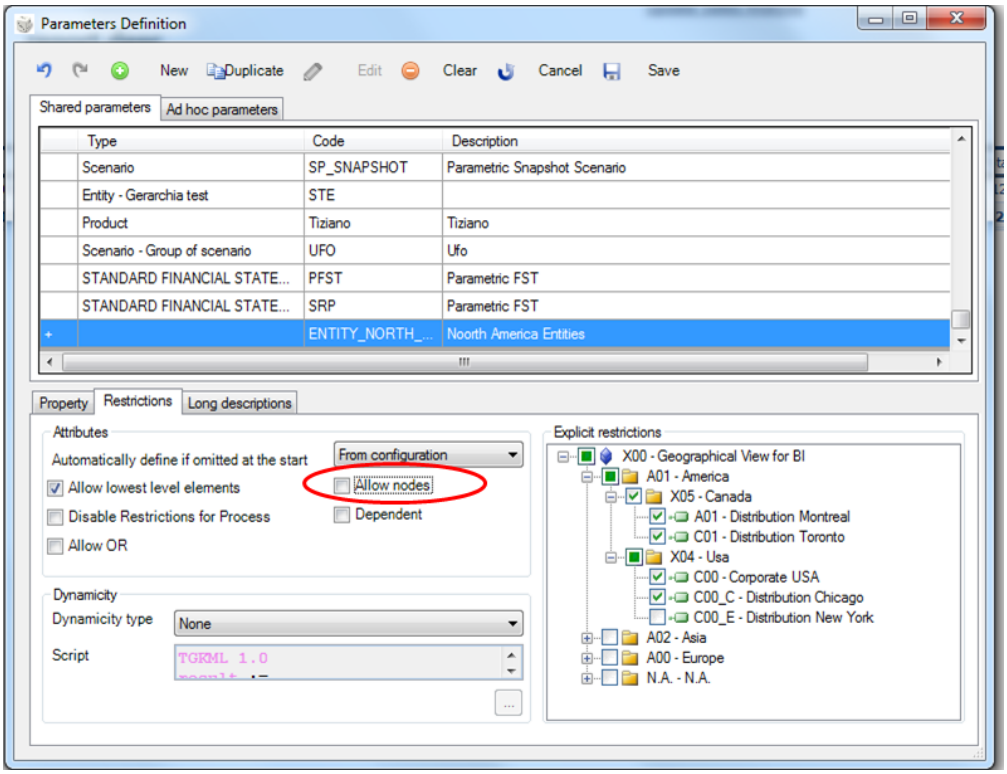


Since this parameter is not mandatory, if the user does not assign a value when running the form and automatic input is NOT active for that parameter, the system will not filter on the dimension associated with the parameter when reading the data to be shown.

However, if automatic input is active and the parameter is restricted to nodes X05 - Canada and X04 - USA with the exception of C00_E - Distribution New York then, when reading the data, the system will filter on the dimension associated with the parameter by establishing an OR relationship between the elements, in this case A01, C01, C00, C00_C.

Set restrictions on the selection of dimensional elements

Suppose we need to edit the form created previously for American entities and ensure that the results can only be seen at the individual entity level and not at the country node level. Or, for example, we want the individual entity managers to be able to see their entity's results but not the results for the whole country. We can restrict our previously created parameter by further limiting the set of data that the user can choose for that parameter. If the **Allow Nodes** option is deselected, it will not be possible to choose node elements. Therefore, the user will be forced to choose a single entity for this parameter at runtime.

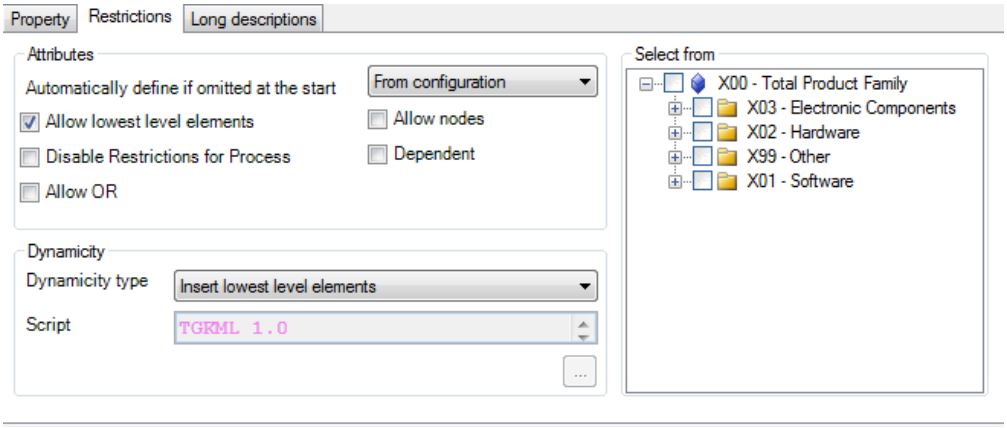


Set up a dynamic parameter

Suppose we want to create a form that has the products sold by an entity on the rows. Since the products sold by the entity change continually (they are created and withdrawn very frequently) the best thing to do is to create a dynamic parameter so that it doesn't need to be maintained too often.

The process of creating a parameter follows the same steps as in the previous example, with the only difference being that the hierarchy to be associated with the parameter is the product hierarchy.

To make the parameter dynamic, a **selection type** that is not "None" must be defined in the **Restrictions** tab; more specifically, because we want the user to be able to select all the elements related to the products hierarchy, it must be defined as **Insert lowest level elements**.



At this point, this parameter can be used within matrices without the need for updates. When the elements in the hierarchy are changed (new elements added, nodes added/changed), the parameter does have to be changed.

Create a parameter that is dependent on another parameter

For example, supposing that, in the presence of an entity restricted to certain cost centres, once the user has specified the entity on which they want to work, the system will only show the cost centres enabled for that entity.

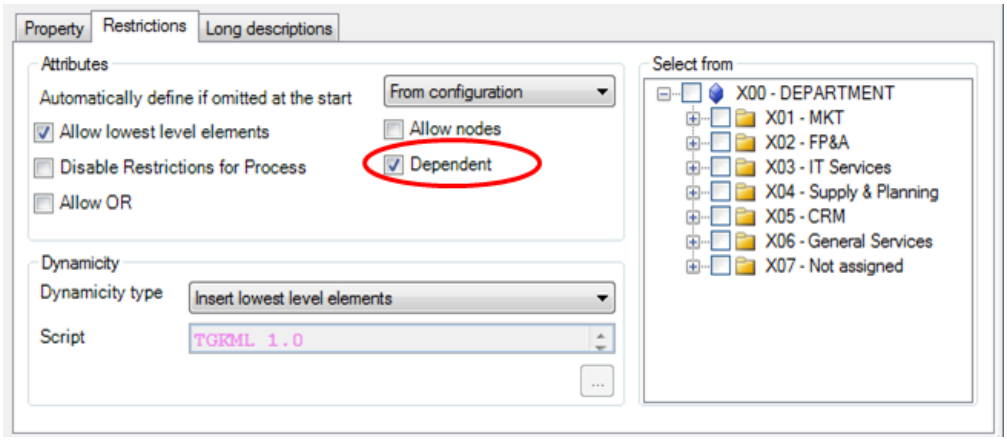
To obtain an outcome of this kind, the user must:

- create a shared Dimension-type custom parameter associated, for example, with hierarchy X00 - Department of the Cost centre dimension;

- tell the system that, as the parameter can have variable values, it is dynamic, i.e. define a **Selection type** that is not "None" in the **Restrictions** tab;
- activate the **Dependent** option in the **Restrictions** tab.

***Note**: the order in which the parameters are set within the form is important: a dependent parameter must always be placed after the parameters on which it depends.*

A dependent parameter must necessarily be inserted in the form using the Parameters management and positioned in the **Other parameters** group.



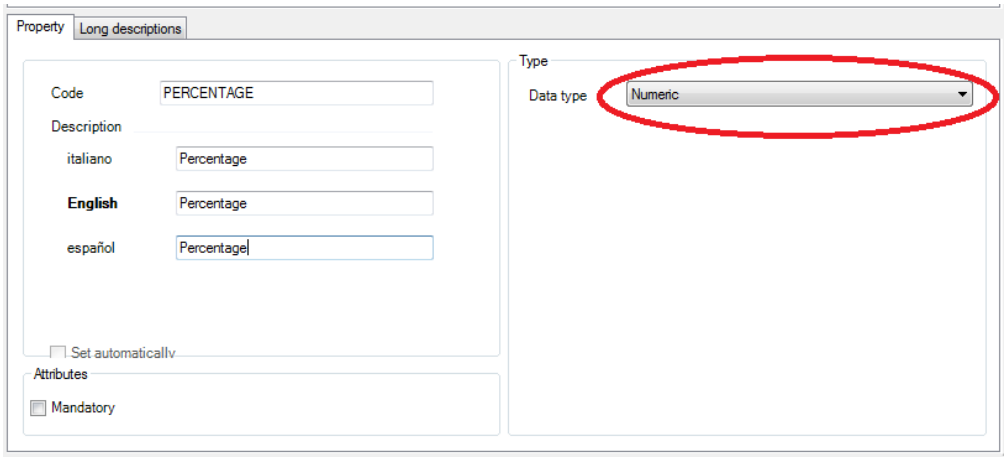
In the SP14 release, the concept of dependence was also introduced to the DTP run window, where there are parameters that depend on others. At the moment, only explicit dependency is supported in these situations (i.e. dependence that uses the TGKML 1.0 advanced selection scripts) whereas, if only the dependence check is being used (implicit dependence), then it has no effect on the parameters present in the run window and therefore the results are therefore not influenced.

***Note**: explicit dependence between Scenario-Period type parameters and between Process-Scenario type parameters has been added to both the Parametric Jobs run window and the DTP window.*

Create a parameter for a percentage value

Suppose we need to build a form in which users have to enter a percentage value to run what-if analyses. For example, you want to determine the revenues from a product based on a discount inserted using a parameter.

To obtain an outcome of this kind, a Numeric-type parameter can be used. As parameters of this type are not Dimension parameters, they are not visible within the dimensions tree and must therefore be inserted into the form using Parameters Configuration window.



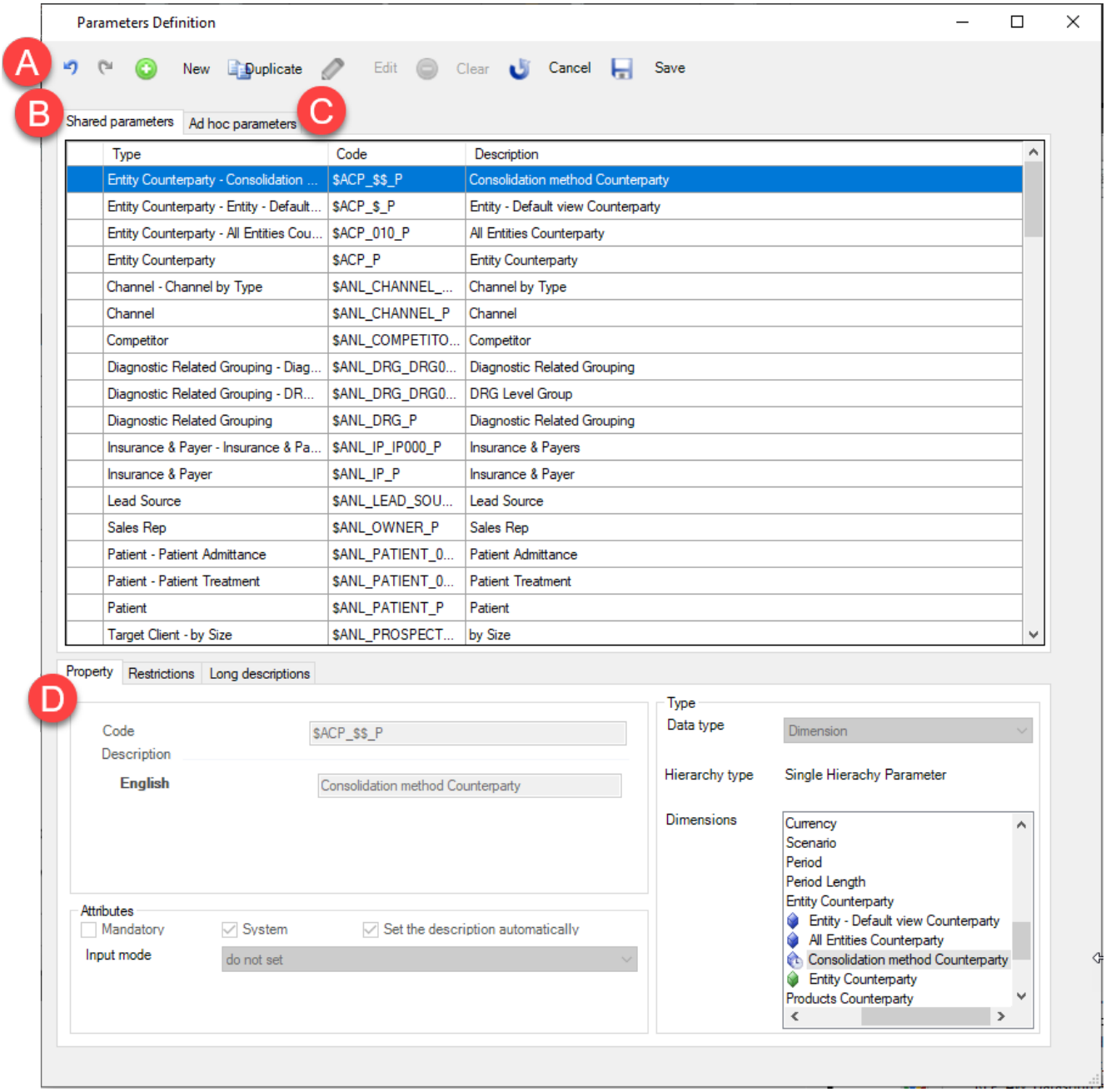
Parameters Definition window

Excel > CCH® Tagetik > Parameters Definition

Purpose of window

This window allows you to manage shared and ad hoc parameters.

What it looks like



Element	Description
A	Buttons on the page
B	Shared parameters tab
C	Ad hoc parameters tab
D	Configuration tabs

Buttons on the page

Element	Description
	Undoes/repeats the last action.
	Add a parameter.
	Edit the parameter.
	Delete the parameter.
	Duplicate the parameter.
	Undo all unsaved changes to the current parameter.
	Save the current parameter.

Page sections

Element	Description
Shared parameters tab	Shows a list of shared parameters.
Ad hoc parameters tab	Shows a list of ad hoc parameters.
Properties tab	Allows you to define the type of parameter, give it a name and certain attributes. If the type is dimensional, it also allows you to define the dimension and hierarchy.
Restrictions tab	Allows you to define specific restrictions for defining omitted parameters at runtime, advanced selection and explicit restrictions.
Long descriptions tab	Allows you to insert an in-depth description of the parameter.

Properties tab

Element	Description
Code	Parameter code.
Description	Parameter description.
Type section	
Data type	Allows you to define the data type of the parameter. The data can be dimensional or non-dimensional. It is possible to choose dimensional parameters the value of which is obtained from a set of elements of a specific dimension: • **Dimension**: allows you to select elements or nodes of a CCH Tagetik dimension/hierarchy. • **Advanced financial statement templates**: parameter for advanced FSTs structured by account/custom dimension and/or control type, in which a dimension has been included. • **Standard Financial Statement Template**: Parameter for standard FSTs structured by account, with no dimension included When a dimensional option is selected, the **Dimensions** pane opens. It is possible to choose non-dimensional parameters for specific advanced selection script configuration. • **Flag**: yes/no parameters; • **Date**: Date parameter selected from a calendar; • **Dictionary**: Parameter selected from a list of elements; • **Numeric**: Number parameter; • **Text**: Free text parameter. • **Process**: Process parameter.
Parameter origin	Only for **Dictionary**. Shows all existing workspaces.

Element	Description
Dictionary	Only for **Dictionary**. Shows a list of the dictionaries available in the system.
Hierarchy types	Only for **Dimension**. Specifies the type of parameter selected in the dimensions tree below. • **Multi-hierarchy parameter**: to define a multi-hierarchy parameter, the root of the tree dimension (Entity or Target) must be selected. • **Single Hierachy Parameter** • **Flat**: all elements of the parameter (without distinction between hierarchies or groupings)
Dimensions	Only displayed with **Dimension** or **Advanced/Standard Financial Statement Template** in **Data type**. Shows the dimensions tree and allows you to select the desired dimension and hierarchy. For Advanced/Standard financial statement templates, only the account dimension can be selected.
Attributes section	
Mandatory	Makes it mandatory to define the parameter when the form is run. For non-mandatory parameters that are not defined, the system displays aggregated data. ***IMPORTANT**: parameters on the Scenario, Period and Account dimensions are always mandatory.*
Set the description automatically	Only for dimensional parameters. Enables the automatic insertion of the description of the dimension or grouping chosen in the parameter description.
System	Indicates whether the parameter is a system parameter or custom parameter.
Input mode	Defines the default method of defining parameters in the parameter selection window when the form is opened. • **Do not set**: do not enter any value. • **Set with a default value**: (only for **Dimension**-type parameters) allows you to specify a value to be shown by default at runtime. The available values depend on the restrictions defined on the parameter. ***IMPORTANT**: if the element code set as default is renumbered, the attribute will be invalidated.* • **Script for default value**: (only for **Dimension**-type parameters) allows you to specify a value to be shown by default at runtime using an advanced selection script (see "Dynamic matrices" on page 132).
...	(With **Input mode** set to **Set with default value**) Allows you to set the default value for the parameter.
Hide when a default value is specified	(With **Input mode** set to **Set a default value**) If selected, the system hides the parameter from the form run window, and inputs the default value. When the user does not have access to the default value due to restrictions on their user profile, it is not used and the system asks the user to define the parameter. ***IMPORTANT**: if active, the system will use the configuration option:**Automatically define if omitted at the start**.*

Restrictions tab

Element	Description
Explicit restrictions	Allows you to manually define the set of values that the parameter can take on by selecting elements from the dimensions tree.
Attributes	

Element	Description
Automatically define if omitted at the start	Indicates the input mode for parameters not defined by the user: • **From configuration**: the system behaves as specified by the *Automatically define parameters if omitted at the start* parameter, in advanced form configuration (see Dettaglio tab Form). • **No**: the system does not define the parameter. • **Yes**: the system automatically defines the parameter based on the restrictions specified in the definition of that parameter (OR), or as the root of the dimension to which it belongs, if it has no restrictions.
Allow lowest level elements	When the form is run, this allows you to select the lowest level elements of the hierarchy as the parameter's value.
Allow nodes	When the form is run, this allows you to select the nodes of the hierarchy as the parameter's value.
Disable Restrictions for Process	Excludes any restrictions defined by default at the process level from the dimension related to the parameter.
Dependent	(**Selection type** not equal to None) Makes the parameter dependent on the value of another parameter.
Allow OR	When the form is run, this allows you to select several values for the parameter simultaneously.
Advanced selection	
Selection type	Defines the parameter's advanced selection. • **None**: the parameter is not dynamic. **Insert children**: with the parameter, all the parameter's nodes and children are also selected. • **Insert lowest level elements**: with the parameter, all lowest level elements (leaves) of the node whose advanced selection is defined are also selected. • **Advanced selection**: enables an advanced selection script to be written.
Script	Only with **Selection type** set to **Advanced selection**. Allows you to insert an advanced selection script for the parameter.

Long descriptions tab

Element	Description
<languages supported by the system>	List of table languages managed by the system.
Insert text field	Allows you to insert in-depth information on the parameter for the relative language. Descriptions that will be shown in a tooltip when the user positions the cursor over the parameter in the parameters selection window.

Parameters Configuration window

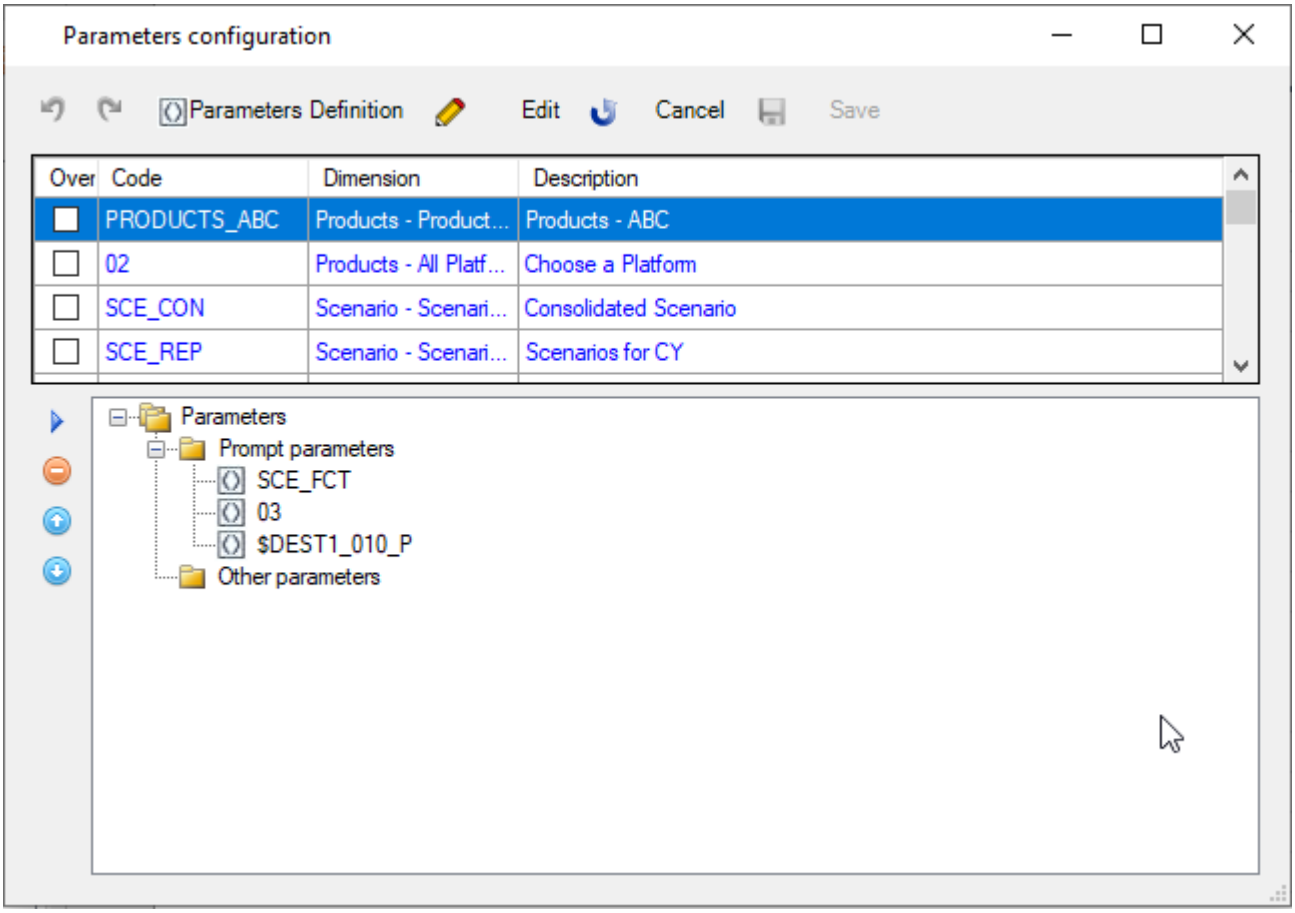
Excel > CCH® Tagetik > Design > open form > Manage Parameters

Purpose of window

This window allows you to manage the run parameters of the current form.

***Note**: this window does not show process-type parameters or dimensional parameters on the period that allow OR clauses, as they are functional to the configuration of the parameters but not directly supported by the forms.*

What it looks like



Window sections

Element	Description
List of parameters	Shows a list of the current form's system, shared and ad hoc parameters and allows you to insert any parameters that are not present into the form as filters. The system automatically labels a manually inserted element as *Overridden*.
Parameter management section	Allows you to manage the parameters associated with the current form. The default parameters are grouped as follows: • **Prompt parameters**: these affect the editability of a cell when a form is run in data entry mode. When the form is run, in the presence of an entity parameter, for example, all the cells that refer to that parameter are locked for the chosen entity and other users cannot edit the data for that same entity. • **Other Parameters**: parameters that do not affect cell editability. For example, if an entity parameter of this type is defined, the entity will not be locked for other users in data entry mode

Parameter management buttons in the form

***IMPORTANT**: the order in which the parameters are displayed has an effect on the parameters that have a relationship of dependency with other parameters. A dependent parameter must necessarily come after the parameter on which it depends.*

Element	Description
	Adds the selected parameter to the list of form parameters, at the bottom of the list.
	Removes the selected parameter from the section.
	Moves the selected parameter up/down.

Buttons in the window

Element	Description
	Undoes/repeats the last action.
	Opens the **Parameters definition** window.
	Allows you to add parameters not included among the form filters as overridden parameters.
	Cancels all unsaved changes and closes the window.
	Saves the list of form parameters.

Advanced selection: scenario and period parameters

Using parameters in the Matrix Wizard

In CCH Tagetik, parameters can be used as row/column filters and matrix filters to create dynamic matrices.

In the dimensions tree, parameters are shown in blue and, once inserted as filters, their properties can be configured (see "Table of row/column filter properties" on page 112 and "Matrix filter properties table" on page 119).

For scenario and period parameters, the properties table can be used to manage additional properties for navigating between the possible values of the parameter.

> ***Note**: Dimensional parameters on the period that allow OR clauses are not shown in the dimensions tree for defining form, tab and matrix filters, because they are not supported.*

Navigating scenario parameters

Parameters on the scenario dimension allow you to create dynamic forms for comparing data saved on different scenarios. In particular, the Parameter **Navigation** property (see "Navigation" on page 118) can be used to create rows/columns between the scenarios of which there exist temporal and comparison logics.

Details and potential uses of the logics according to which navigation can be designed are provided below.

Logic type	Example
Temporal	Uses the temporal relationships defined in the **References** section of the scenarios table (see Anagrafica scenari). For example, in the presence of the *2012ACTUAL* scenario, which collects actual data for 2012, it is possible to create a *Previous*-type relationship with scenario *2011ACTUAL*, and a *Following*-type relationship with scenario *2013ACTUAL*. Chains of scenarios can be crated in relation to the scenario before the previous scenario or even previous/following scenarios in relation to scenarios set as *References #1-5*. For more details, see "Table of row/column filter properties" on page 112 and "Matrix filter properties table" on page 119.
Comparison	Uses the relationships defined in the **References** section of the scenarios table (see Anagrafica scenari), to compare data in forms. The user can indicate up to five *Reference scenarios* numbered from 1 to 5.

The order in which the elements are added to the Navigation list is very important because the system navigates the scenarios in the order in which they are added.

Navigating period parameters

Parameters on the period dimension allow you to see, on the rows/columns, periods calculated on the basis of an offset from the run period. For example, it is possible to define columns that show data for 6 months prior to the reference period.

> **Example**
>
> Choose period 12 if you want to show data for period 6
>
> To do this, set:
>
> - a row/column for the reference period parameter
> - a row/column for the period parameter with the Period Difference property set with the desired offset (-6).
>
> If the user runs the same report for period 11, the system will show period 5 for comparison, and so on.

If the difference is greater than the number of periods in the run scenario, the system obtains the excess periods from the scenario set as the previous or following (depending on whether you are going forwards or backwards).

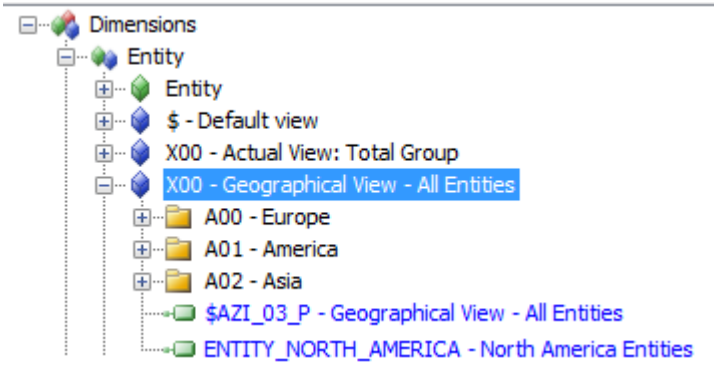
***Note**: in Data Entry mode, the period navigation feature is not fully supported for matrices with Analytical Workspace data. The system does not calculate the value of the scenario dimension correctly if, due to a difference greater than the number of periods present in the run scenario, it is necessary to use the scenario set as the previous or following. This is the case in relation to the editability of multidimensional matrices, transactional matrices and data extracted from transactional matrices.*

Examples of using dynamic parameters

Using parameters

All parameters present in Tagetik (both system parameters and custom parameters) can be used within the Matrix Wizard to make the matrices dynamic.

In Tagetik, there is a parameter for every dimension and hierarchy. Parameters can be selected from the dimensions tree and are distinguished from the other elements by the colour blue. Both shared parameters and ad-hoc parameters can be chosen in the dimensions tree.



Once a parameter from the dimensions tree has been chosen, it can be inserted into the matrix as a row filter, a column filter or a matrix filter. Once inserted into a matrix, a parameter behaves the same way as a fixed value. They can also be nested into other elements and the properties that can be modified in the properties table are the same as for a fixed value (non-parametric) element.

For dynamic scenario and period parameter elements, there are additional properties that are activated in the properties table when inserted in the matrix as row filters or column filters. We will look at these properties in the next sections.

The dimensions tree used to define the Form, Tab and Matrix filters does not show Process-type parameters or dimensional parameters on the period that allow 'OR' operations, as these are not supported by the forms.

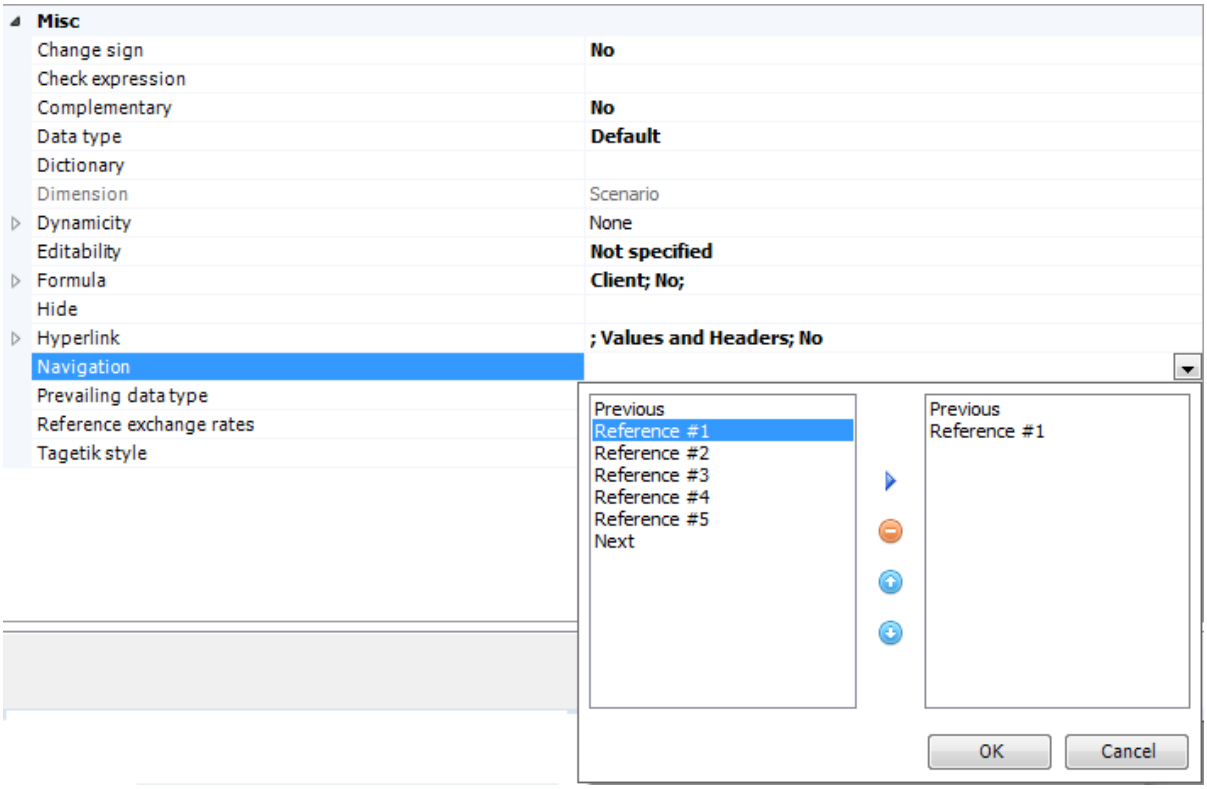
Navigating scenario parameters

Navigating scenarios allows the user to create dynamic forms for comparing data saved in the system in different scenarios. When scenarios are defined in Tagetik, temporal links and comparative links can be created between them.

For example, if I defined a scenario called 2012ACTUAL, which collects actual data for the year 2012, I could create a "Previous" type link to scenario 2011ACTUAL and in turn create a "Next" type link between scenario 2011ACT and 2012ACT.

In addition to temporal relationships, comparative relationships between different scenarios can also be defined. For example, a scenario called 2012BDG, which is a forecast for the year 2012, must be compared with the actual data for that year, e.g. 2012ACT. In Tagetik, up to 5 reference-type relationships between scenarios can be defined.

These chains of scenarios can be used within the Tagetik reporting client when scenario-type parameters are selected. When a scenario-type parameter is inserted in a row or column filter, the "Navigation" property is activated in the properties table.

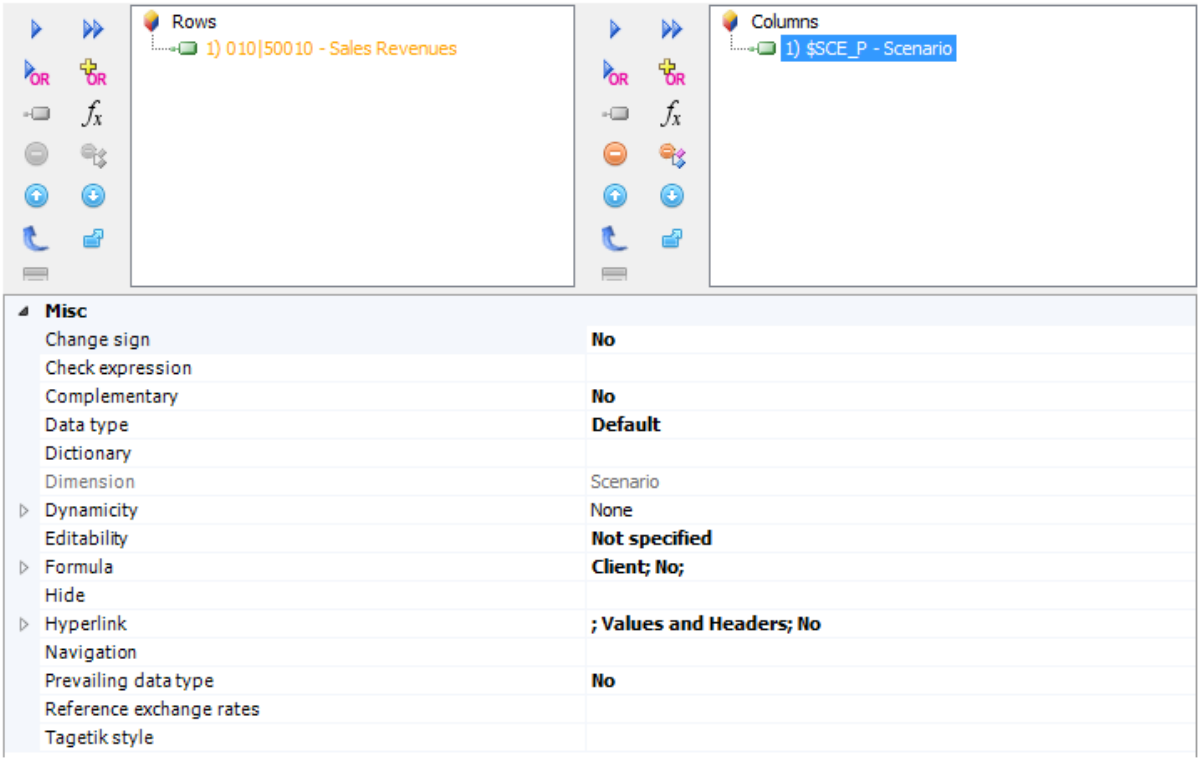


In this window, we need to define the navigation chain for our scenario. Navigation starts from the element chosen by the user when the form is run. For example, if a user chooses scenario 2012ACT and a "Previous" navigation has been defined, scenario 2011ACT will be used.

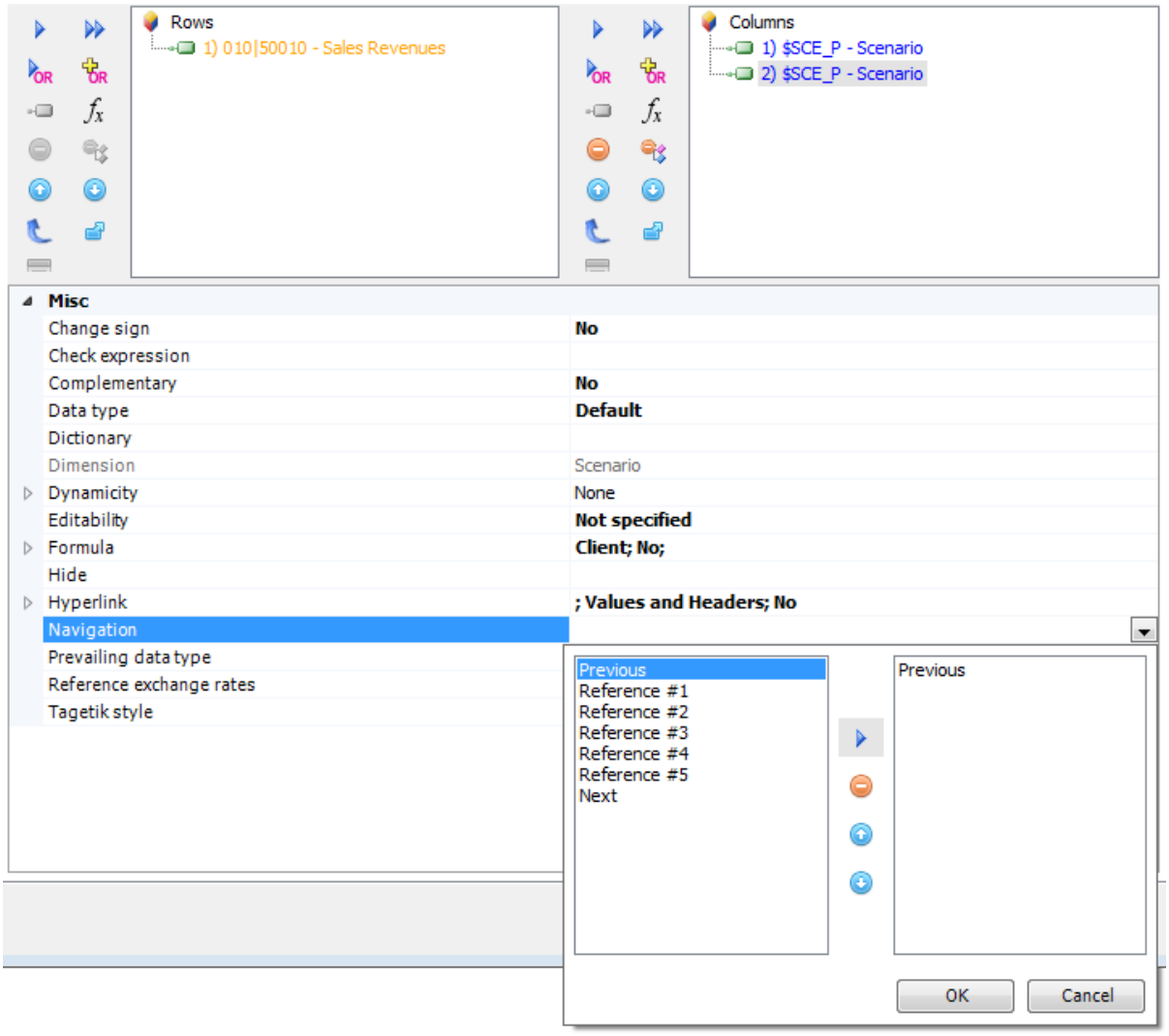
Scenarios are navigated by choosing an element from the list and clicking the button. The element can be removed using the button. The order of the various scenarios can be changed using the and buttons. Several elements can be inserted to create navigation chains, e.g. the Previous --> Previous scenario or the Previous --> Reference #1 scenario.

The order in which the elements are added to the list is very important as the system navigates the scenarios according to the order in which they are added.

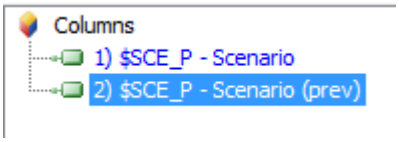
Let's look at an example in which a dynamic matrix is created to compare two actual data scenarios. First, we need to insert a parametric scenario within our matrix. This element will be the element chosen by the user, so we will not set any navigation for the element.



At this point, we must insert the second scenario, which will be the comparison scenario. In this column, we want to see the data for the scenario before the one selected by the user. So let's insert a parameter with the same code as the one inserted previously and set the navigation for the element.



Once navigation has been set for a parameter, the text in the filter window is changed to indicate which chain has been set.



At this point, the dynamic matrix is ready and can be used. Navigation chains between scenarios can be changed within the web interface without having to change the structure of the created form.

Navigating period parameters

The mechanism for navigating between period-type parameters is similar to navigating between scenarios. Offsets related to the parameter chosen as the filter can be defined between dynamic periods.

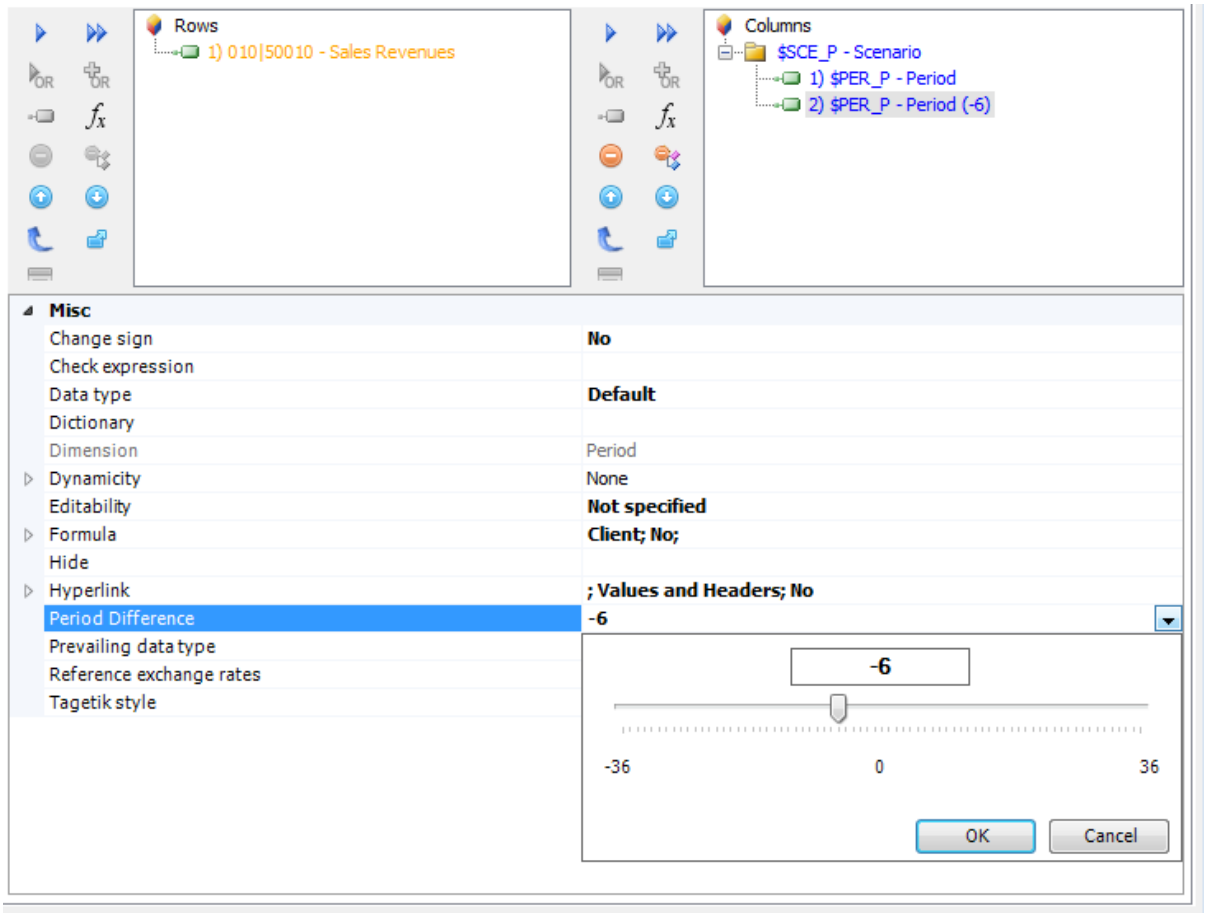
For example, I want to define a column which, depending on the chosen period, shows me data relative to 6 months earlier. If i choose period 12, the column will be defined with period 6, if I choose period 11, it will be defined with period 5, and so on.

A parameter of this type can be set using the "**Period Difference**" property present in the attributes table when we insert a period-type parametric element in a row or column filter. When we set a period difference, we must set the number of periods back or forward that we want to shift in relation to the chosen period. If the difference is greater than the number of periods in a scenario, the system obtains the periods from the scenario set as the previous or next scenario.

La navigazione del periodo non è totalmente supportata nell'AW, in particolare il sistema non ricava correttamente il valore della dimensione Scenario se, a causa dello scostamento maggiore del numero di periodi, è necessario usare lo scenario impostato come precedente o successivo. Questo si riscontra:

- nell'editabilità delle matrici multidimensionali
- nell'editabilità delle matrici LBL (Line By Line)
- nei dati estratti dalle matrici LBL (Line By Line)

Differences can be inserted directly within properties or using the scroll bar shown when properties are opened. The chosen difference is shown in the title of the edited parameter within the row or column filter.



Cell editability

Introduction

To configure data entry on the form, the user must define editable cells, i.e. cells in which the user can enter data that will be written in the database when saved. To do this, the *Editability* property on row/column filters can be configured in the Matrix Wizard. For details of the property values, see "Editability" on page 116.

The effective editability of a cell depends both on the filter properties and on the rights that the user has in their own role (see I ruoli) and the user's data restrictions. The editability rule on a cell can be overridden using the **Override editability** button on the design toolbar (see "Design toolbar" on page 34).

The cells of data entry forms are differentiated by colour:

Color	Cell status
White	Not editable
Yellow	Editable, but does not allow intercompany data
Green	Editable, and allows intercompany data
Blue	Protected

These colours can be changed for the individual form, via the **Form options** window (see "Layout options details" on page 56).

Editability in intersection cells

On the intersection cell between, for example, an editable row and a non-editable column, the system calculates the intersection between the row editability rules and the column editability rules and makes the cell editable or non-editable based on the rules shown in the table below.

		Column			
		Unspecified	Editable	Protected	Non-editable
Row	Unspecified	NON-ED.	EDITABLE	PROTECTED	NON-ED.
	Editable	EDITABLE	EDITABLE	PROTECTED	NON-ED.
	Protected	PROTECTED	PROTECTED	PROTECTED	NON-ED.
	Non-Editable	NON-ED.	NON-ED.	NON-ED.	NON-ED.

Placeholder element editability

For placeholder elements on the rows/columns, editable status is replaced with "unlocked". A placeholder for which editability has not been specify will be non-editable by default (see "Placeholder elements" on page 151).

Conditional editability

Conditional editability expressions allow you to subordinate the editability of a row/column filter to the occurrence of a condition on other rows/columns of the matrix. Thus, it is possible to unlock the editability of cells set as editable until the occurrence of a certain condition on another cell.

When the form is run in data entry mode, every time the content of the cells to which the expression refers is edited, the system runs a check on the condition.

When a row/column included in the conditional expression is the outcome of a formula, the system only considers the formulas contained in the same sheet that do not use the INDIRECT function.

In general, conditional editability allows you to edit the content of a cell, but not insert comments or attachments. Comments and attachments can only be inserted on multidimensional forms, see Commenti su matrici con editabilità condizionale.

Conditional editability and spreading in web forms

When a conditional editability expression contains one or several cell operands the value of which is changed by the running of spreading, a check is run on the expression's conditions after spreading or only

after any changes have been made to the spreading method.

The interaction between conditional editability and spreading is only managed in web forms (see Spreading).

Conditional editability expression window

Excel > CCH® Tagetik > Design > open Matrix Wizard > select a row/column filter > (Properties grid) Conditional editability expression > ...

Purpose of window

This window allows you to define the condition that needs to be met to make a matrix row/column filter editable.

Window sections

Element	Description
Minimum editability level	Defines the editability type to be assigned to the cell when the condition is not met: • **Protected**: the cell is not editable for direct entry, but only via formulas. • **Not editable**(default): the cell is not editable.
Row/column	Allows you to define the logical expression on rows/columns based on the condition to be met.

Details of the expression for the condition to be met

Element	Description
Row/column	Row/column on which to verify the condition. The row/column to which the check relates must be static.
Operator	• **isTrue** and **isFalse**: operator referring to a cell containing a Boolian or numeric value (0=false, <>0= true). • **=, <>, >, >=, <, <=** • **Between** (included between): defines a range of numerical values, a string or data within which the insertions must fall. The first and last values are included in the range. • **IsEmpty**: the condition is met if the cell is empty. • **IsNotEmpty**: the condition is met if the cell is not empty. • **IsNumeric**: the condition is met if the cell contains a numerical value. • **IsString**: the condition is met if the cell contains a text-based value. • **IsInteger**: the condition is met if the cell contains a whole number. Any decimal figures introduced by copying and pasting will not be saved.
Type	Defines the type of data on which to verify the condition. Is only required for some operators. The available options are: • **Number** • **Body** • **Date**
Reference value	Defines the reference value to run the check.
	Adds an expression to the condition.
	Removes an expression from the condition.
Concatenation of several conditions	Only with several conditions based on several expressions. Defines the relationship between the individual conditions. • **AND** • **OR**

Forms grouping

Definition and function

It is possible to create form aggregation structures, called *groupings*. These allow you to group forms according to specific business logics, depending on user restrictions on forms. Working with groupings in a node structure also allows you to optimise form consultation, both in the web interface and in the Excel interface, and to manage collected data more easily.

In CCH Tagetik, there is a predefined grouping called $ - *Forms*, which corresponds to the parent node, and to which all newly created forms are related by default. The grouping can be customised later.

It is also possible to create groupings that are useful for simplifying the management of restrictions on forms and delimiting the list of forms allocated to each individual process. Groupings can be hidden for both data entry and navigation.

Groupings can be defined both in the web interface, by the administrator user (see "Working with grouping nodes (web interface)" on the facing page), and in the Excel interface, by authorised users only (see "Working with grouping nodes (Excel add-in)" below).

Working with grouping nodes (Excel add-in)

Introduction

The following procedures describe how to work on grouping nodes via the Excel add-in.

Create a grouping node

Excel > CCH® Tagetik > Design

1. In the left-hand panel:

If...	Then...
you want to insert the grouping within a node	click on the desired node
you want to insert the grouping at the level of a specific form	click on the desired form

2. Click .
3. In the **New folder** window, insert the desired code and description for the folder.
4. (Optional) To hide the grouping node from the left-hand panel, select **Do not display**.
5. Click **OK**.

Move a form to another grouping node

Excel > CCH® Tagetik > Design

In the left-hand panel: select the form to be moved and drag it to the desired node, or select the form, click , then on the desired node and right click, then click **Paste**.

Add a form to a grouping node

Excel > CCH® Tagetik > Design

In the left-hand panel, proceed as follows:

1. Select the desired form and click , or right click, then click **Copy**.
2. Select the desired grouping node, then click to refresh the form.

***Note**: a form can be related to several grouping nodes.*

Working with grouping nodes (web interface)

Forms management tile > Groupings

Introduction

The following procedures describe how to work on grouping nodes via the web interface.

Create a grouping node

1. In **Forms grouping**

If...	Then...
you want to insert the grouping within a node	click on the desired node
you want to insert the grouping at the level of a specific form	click on the desired form

2. Click . a row will appear in the list of form groupings.
3. In **Attributes**, insert the desired code and description for the grouping.
4. (Optional) To exclude the grouping node from data entry, navigation and reporting jobs, select **Do not display in data entry and reporting modes**.
5. Click **Save**.

Move a form to another grouping node

1. In **Forms grouping**, select the form to be moved and drag it to the desired node, or select the form, click , then on the desired node and right click, then click .
2. Click **Save**.

Add a form to a grouping node

1. Select the desired form and click .
2. Select the desired grouping node, then click to refresh the form.
3. Click **Save**.

***Note**: a form can be related to several grouping nodes.*

Remove a form from a grouping node

1. In **Forms grouping**, select the desired form and drag it onto the **All elements** node.
2. Click **Save**.

Forms grouping page

Admin | <application database>

Forms management tile > Groupings

Purpose of the page

This page makes it possible to manage form groupings.

Page sections

Element	Description
Switch view	**Element/node relationships**: opens the "Element/node relationships page" on the facing page page. **Structure**: opens the "Form groupings page (Structure view)" below page.
Report Grouping	Shows the hierarchy of form groupings.
List of groupings and forms	Shows the form groupings and forms.
Attributes tab	Allows you to manage form groupings or individual forms.

Buttons on the page

Element	Description
	Adds a form.
	Adds a grouping.
	Duplicates a form/form grouping.
	Deletes a form/form grouping.

Attributes tab

Element	Description
Click 	
See Pagina Forms.	
Click 	
Code	Define a name for the grouping.
Description	Define a description for the grouping.
Form default relationship	When selected every new form is automatically related to the grouping.
Do not display in data entry and reporting	Allows you to exclude the selected grouping of forms from data entry and reporting jobs. Useful for excluding forms created for configuration purposes only, or obsolete forms.

Form groupings page (Structure view)

Admin | <application database>

Forms management tile > Groupings > Forms grouping > Switch view > Structure

Purpose of the page

This page allows you to create form nodes. To associate the form with the node, see "Element/node relationships page" on the facing page.

Attributes tab

Element	Description
Node	Defines the node code.
Description	Defines the node description.
Node type	Indicates the node type. • **Root**: the node is the grouping's parent node. • **Normal**: the node is one of the grouping's nodes.

Element	Description
Parent node	Only with **Node type** set to **Normal**. Allows you to select the parent node of the current node.
Form default relationship	When selected, the system automatically relates every new form to the node. Can be selected on several nodes of the grouping, in which case the form is related to all nodes.
Exclude from reporting	Allows you to exclude the node from reporting. Useful for excluding forms created for configuration purposes only, or obsolete forms.
Sorting	Allows you to configure the order with which the forms and nodes are displayed in the forms tree.

Element/node relationships page

Admin | <application database>

Forms management tile > Groupings > Forms grouping > Switch view > Element/node relationship

Purpose of the page

This page allows you to associated individual forms with nodes of the grouping of forms.

***Note**: once saved, the relationship can no longer be changed from this window.*

Attributes tab

Element	Description
Form	Selects a form.
Forms node	Selects the group with which to associate the form.

Financial statement templates

What is a financial statement template

A financial statement template (FST) is a structure used to aggregate accounts or custom accounts/dimensions that can represent financial statements (IAS profit and loss statements and balance sheets, management reports, direct/indirect financial reports), KPIs and other indices for reporting. A financial statement template can be added to a matrix in its entirety or in part as a filter to be able to design its structure more quickly and efficiently.

***Note**: financial statement templates without accounts or with control type or multi-dimensional formulas cannot be added to matrices drawn on the Analytical Workspace.*

For each financial statement template, relationships can be defined based on account dimension, custom dimension, account/custom dimension pair, account/analytical dimension pair or account/control type pair.

Relationships can be defined synthetically by specifying, in the definition, the individual accounts (or dimensional elements) and/or hierarchy nodes that must be added to or removed from the various FST items: the effective resulting relationships are identified by applying the specifications according to the defined order.

Structure of a financial statement template

A financial statement template is structured into two levels:

Element	Description
First level	May contain the following elements, which constitute the *FST items*: • *Parent element*, or a node to which one or more accounts can be related, showing the sum of the amounts of the relative accounts. • *Formula*, or a formula-type node defined on the basis of a relatable items and/or other formulas. • *Description*, or a description-type node which allows you to customise the template header.
Second level	Contains all the accounts associated with relatable elements.

Specific FST items can be recalled from other financial statement templates.

Financial statement templates in row/column filters

The FST can be added to other row/column filters of the matrix in the following ways.

Element	Description
Analytic	Adds the whole FST structure to the row/column/matrix filters tree, organised over two levels according to the levels of the original template.
Synthetic	Adds the FST structure to the row/column/matrix filters tree, organised over one level, i.e. the first.

Accounts in the financial statement template

For every financial statement template, relationships can be defined based on the following criteria:

- account dimension
- custom dimension
- analytical dimension
- control type

Relationships can be defined synthetically by specifying, in the definition, the individual hierarchy nodes and/or dimensional elements that need to be added to or removed from the FST items.

The order in which relationships are specified in FST items determines the final content of the FST item.



For forms that use FST items with relationships with Accounts or Account/Custom Dimension nodes or Custom Dimension nodes, especially in the presence of relationships with Initial balance accounts/Custom Dimensions, it is advisable to use the in memory reporting engine.

***Note**: financial statement templates containing multidimensional formulas or defined on analytical dimensions can only be used in forms based on the in-memory engine.*

Multiple account relationships

By default, the account or account/custom dimension pair can only be related to a single FST item. Where there are several relationships between one account or account/custom dimension pair and several FST items, the system uses the most recently defined relationship on which deployment has been run. If multiple relationships are activated, the system considers all the FST items related to the account or the account/custom dimension pair and adds the amounts of all related FST items together.

This option may be useful in the following cases:

- For indirect cash flow forecasts in which, based on the annual profit, the depreciation account must be reclassified as two report items in order to obtain the *flow of net working capital generated by operations* and *flow of net working capital generated by investments and disposals*.
- To define an FST item with several relationships and specific items with details of the individual components.

Several FST items related by OR clauses can be concatenated in a formula. In this case, the final result is not equivalent to the sum of the amounts of the accounts involved, but the sum of the amounts of the FST items used. Thus, one single account can contribute several times to the final value of the amount.

***Note**: the in memory engine and the standard engine manage OR clauses in the same way, with the exception of OR clauses between FST items with multiple relationships and FST items without multiple relationships: for forms containing such clauses, the standard reporting engine is not recommended, as it cannot correctly manage multiple contributions for a single account.*

Tagetik styles on FST items

To highlight FST item data (formulas, subtotals), a specific formatting style can be related to FST items. This relationship can be established at the Tagetik style level, and therefore applied to all forms to which it is associated, or at the individual FST item level, and therefore applied to the specific FST item. For more details on styles, see "Tagetik styles " on page 230.

Financial statement template on Analytical Workspaces

For matrices drawn on analytical workspaces, it is possible to use:

- Standard FST defined for the account dimension
- Advanced FST with custom dimension and account dimension

The following are not supported:

- FST without account dimension
- FST with control type
- FST with multi-dimensional formulas
- FST on which on-the-fly conversion is applied
- FST on which onScenarioPeriod is applied

FSTs & KPIs page

Admin | <application database>

Dimensions tile > Account tile > ⋮ > FST & KPI

Purpose of the page

This page allows you to create headers for financial statement templates.

Attributes tab sections

Element	Description
Code and description	Allows you to assign a code and a description to the financial statement template.
Dimensionality	Allows you to associate accounts, custom dimensions and control types to the financial statement template, and to manage their relationships with items.
Advanced	Allows you to define options that regulate the use of the financial statement template: • use of the financial statement template in reporting • they system's behaviour in the event of a sign change on an FST item • types of formula supported in the financial statement template
Valid accounts	Allows you to define the types of accounts that can be related to the financial statement template.

Dimensionality details

Element	Description
Account	Enables FST items to be calculated as the sum of the accounts related to them. Financial statement templates created with this option are saved in the **Standard financial statement templates** folder.
Custom Dimensions	Allows you to select a custom dimension in order to calculate FST items as the sum of the elements of the selected custom dimension. With **Account** selected: FST items are calculated as the sum of the account/custom dimension element combinations. With **Account** not selected: FST items are calculated as the sum of the accounts that are valid on the custom dimension. Financial statement templates created with this option are saved in the **Advanced financial statement templates** folder.
Control type	Enables FST item accounts to be related based on the control type existing between the reference entity (the parent company for consolidated data, the entity for statutory data) and the counterparty entity. For example, the FI account can be related to the FI FST items in *subsidiaries*, *associates* or *other* mode, based on the type of relationship existing between the entity pairs. The Control type differentiates the relationships between an account and the FST items. An account declared to an owned entity can thus be linked to different FST items depending on the defined control type, and on the scenario/period. Financial statement templates created with this option are saved in the **Advanced financial statement templates** folder.
Multiple relationships	By default, every account or account/custom dimension pair can be related to one FST item only. If enabled, it is possible to relate a single account or account/custom dimension pair to several FST items. ***Note**: for this option to be valid on the financial statement template, dependence on the account dimension must be activated in dimensionality. Alternatively, if dependence is only active for the custom dimension, an account must be included in the custom dimension definition.*

Advanced details

Element	Description
Exclude from reporting	Excludes the financial statement template from being used in reporting and data entry forms.
Override Sign for class on the Account by FST item	Enables, on all accounts, the use of the sign class set on the item itself regardless of the class defined on the account.
Formula type	Defines the formula type to be used in the financial statement template for formula-type FST items. • **Standard Formula**: allows you to define standard formulas only (default value). • **MD formula**: allows you to define multidimensional formulas only. Once set, this can only be changed if the financial statement template contains no formula-type items. For more details, see "Formulas in financial statement templates" on the facing page.

Valid accounts details

Element	Description
Select accounts	Defines the characteristics of the accounts that can be related to the financial statement template. • **Account Type**: **Normal**, **Detail**, **Variation** • **Account Nature**: **Profit and Loss**, **Balance Sheet**, **Memorandum**, **Other Stocks**, **Other Variations** By default, the system includes standard accounts, P&L accounts, BS accounts and memorandum accounts. ***Note**: for financial statement templates that depend on the account, at least one account type and nature must be selected.*
Direct cash flow/Indirect cash flow	Only if no options have been selected in **Account types** and **Account nature**. Allows you to define a direct/indirect cash flow financial statement template. The system automatically opens all the corresponding accounts.

Buttons on the page

Element	Description
	• **FST items**: this opens the FST items definition page. • **Copy from**: opens a window to copy FST items between financial statement templates. • **Diagnostics**: downloads a PDF containing a list of all Accounts/Custom dimension elements/Control types that have not been related to any FST item. • **Set formula for cube**: starts the data processing to calculate the formula for cube on FST items containing standard formulas.

Formulas in financial statement templates

Standard formulas

When defining financial statement templates, *standard formulas* can be set on FST items. A standard formula can have the following elements as arguments:

- other FST items of the current template
- FST items of other templates with the same dimensionality (see "Dimensionality details" on the previous page)
- accounts, only in financial statement templates that depend on the account dimension only

For more details, see "Standard formula syntax" on the next page.

IMPORTANT: It is not possible to refer to an FST with multidimensional formulas (FST MD) from a standard FST.

Multidimensional (MD) formulas

Multidimensional formulas allow you to run operations between accounts and/or include other dimensions in operations. To create multidimensional formulas, called *TGKML MD*, the multidimensional script syntax TGKML is used. This syntax derives from the MultiDimensional eXpressions (MDX) language dedicated to OLAP analysis on the SQL Server. For more details on MGKML MD syntax, see Calculation logics of "Multi-Dimensional Script" type.

Financial statement templates containing multidimensional formulas can be used in forms based on the in-memory engine. For more details on multidimensional formulas, see "Particular features of multidimensional formulas" on page 92.

Formula for cube

Standard formulas are suitable for use in relational databases. However, a formula for cube can be created for every standard formula, to be used by external sources such as Microsoft Service. To create formulas for cube, the multidimensional script syntax TGKML is used. This syntax derives from the MultiDimensional eXpressions (MDX) language dedicated to OLAP analysis on the SQL Server. For more details on MGKML MD syntax, see Calculation logics of "Multi-Dimensional Script" type.

Standard formula syntax

Introduction

When a formula is created in a financial statement template, its arguments, i.e. the FST items and/or accounts, are defined, together with the logical operators that establish relationships between the arguments. In the formula definition area, once the desired element type has been indicated, CCH Tagetik provides suggestions to help with its creation.

Syntax of arguments in formulas

The definition, in formulas, of elements based on FST items or accounts is guided by CCH Tagetik.

***Note**: when you type [in the formula definition area, depending on the origin the system will show the elements of the chosen dimension of this financial statement template, other financial statement templates, or accounts depending on what is selected in the "add element from" field.*

If you want to add...	Then...	Example
an FST item of the same template	typing *[* in the formula definition area will open a drop-down menu showing a list of FST items in the template. The syntax for the element is *[<current FST code>|<FST item code>]*	With current financial statement template *010 - Sales revenue*, selecting item *015 - Discount & Promotions* will create the formula element *[010|015]*.
an FST item from other templates	typing *[* in the formula definition area will open a drop-down menu showing a list of all the other existing financial statement templates. The syntax for those items is *[<FST code>|<FST item code>]*.	With current financial statement template *010 - Sales revenue*, selecting item *030 - Depreciation & Amortization* for template *020 - Balance sheet* will create the formula element *[020|030]*.
an account	typing *[* in the formula definition area will open a drop-down menu showing a list of existing accounts in the table. The syntax for the element is *[<account code>]*.	With current financial statement template *010 - Sales revenue*, selecting account *10010 - Cash in Bank Accounts* will create the formula element *[10010]*.

Operators and functions in formulas

A list of supported operators is provided below.

Element	Description
Algebraic operators	• + • - • * • /
Logical operators	• = • <> • > • < • >= • <= • && • ||
Functions	• IF (logical_test, value_if_true, value_if_false) • OR (logical1, logical2) • AND (logical1, logical2) • NOT (logical) • 1 (= TRUE) • 0 (= FALSE)

***Note**: when items are saved, the system highlights any syntax errors.*

Scenario and period elements

The string syntax *onScenarioPeriod()* can be used to calculate KPIs with reference to scenarios and periods other than those present in the form to which the financial statement template belongs. This can be applied to FST items and accounts, and accepts the following arguments:

- a positive or negative number indicating the time difference in terms of period (e.g. *-2* = two previous periods, *2* = two subsequent periods)
- two strings between double inverted commas (") for the scenario code and period code
- reference to a scenario/period using cell fields (see "Cell fields" on page 41).

To separate the account code or account/item and the *onScenarioPeriod* element, "." is used as a separator.

Reference scenarios and periods	Element description	Example of syntax
Previous period of the current scenario	• financial statement template *PL* • *item 0010* • *2 previous periods of the current scenario*	*[PL|0010].onScenarioPeriod(-2)*
current scenario/period	• financial statement template *PL* • item *0010* • current scenario/period	*[PL|0010].onScenarioPeriod($Scenario.code, $Period.code)*, or *[PL|0010]* because the default scenario and period are the current scenario and period
a specific scenario (e.g. 2014BDG), period (e.g. 12):	• financial statement template *PL* • item *0010* • scenario *2014BDG* • period *12*	*[PL|0010].onScenarioPeriod("2014BDG","12")*
last period of the current quarter	• account *0010* • last period of the current quarter	*[00010].onScenarioPeriod(quarter().lastSP())*
last period of the third to last quarter 3	• account *0010* • last period of the third to last quarter 3	[00010].*onScenarioPeriod(quarter().offsetSP(-3).lastSP())*

Example

In a formula with reference to account *00010* read from the current period for period *01*, and read from the previous period for all other periods, the syntax is as follows:

IF($Period.code == "01", [00010], [00010].onScenarioPeriod(-1))

How to create a financial statement template

Admin | <application database>

Dimensions tile > Account tile > ⋮ > FST & KPI

Sequence of procedures to create a financial statement template

1. "Create the header of a financial statement template" on the next page
2. "Create FST items" on the next page
3. "Relate accounts, custom dimensions and control types to FST items" on the next page

Create the header of a financial statement template

1. Click ⊕. A new header will appear.
2. In the **Attributes** tab, insert a code and a description for the header of the FST.
3. In the **Dimensionality** section, select the type of element to relate to the financial statement template (filtered by accounts and/or custom dimensions and/or control type) and, if desired, enable **Multiple relationships**.
4. In the **Advanced** section, select the desired options for the use of the FST, for sign change management on FST items and for the supported types of formula.
5. In the **Valid accounts** section, define the types of accounts that can be related to the financial statement template.
6. Click **Save**.

Create FST items

Dimensions tile > Account tile > ⋮ > FST & KPI > ⋮ > FST Items

1. Click ⊕. A new FST item will appear.
2. In the **Attributes** tab, insert a code and a description for FST item.
3. In **Item type**, select the desired type of FST item.
4. In **Class**, select the FST item class.
5. In **Formatting type**, select the level of formatting for the desired Tagetik style.
6. Click **Save**.

Relate accounts, custom dimensions and control types to FST items

Dimensions tile > Account tile > ⋮ > FST & KPI > ⋮ > FST Items > Relationships

This procedure allows you to create relationships, provided elements that can be related to the form exist.

***IMPORTANT**: the order in which relationships are specified in FST items determines the final content of the FST item.*

1. In the **Attributes** tab, select the desired action for the account.
2. In **Account**, select the desired account or node of accounts.
3. In **FST item**, select the desired FST item.
4. If present, select the desired custom dimension and/or control type.
5. Click **Save**.
6. Run the procedure for all the desired relationships.
7. Click **Deployment**.

Setting up a financial statement template

Admin | <application database>

Relate an account to a parent FST item

Dimensions tile > Account tile > ⋮ > FST & KPI > ⋮ > FST Items

1. Select the desired FST item.
2. In **Description**, insert the desired description.
3. In **Item Type**, select **Parent Item**, then click **Save**.
4. Click the **Relationships** link and add the desired accounts to the FST item.
5. Click **Save**.

Insert a description on a description-type FST item

Dimensions tile > Account tile > ⋮ > FST & KPI > ⋮ > FST Items

1. Select the desired FST item.
2. In **Description**, insert the desired description.
3. In **Item Type**, select **Description**, then click **Save**.

Define a standard formula on an FST item

Dimensions tile > Account tile > ⋮ > FST & KPI > ⋮ > FST Items

Only when **Item type** is set to **Standard formula**.

1. Click ✎: the formula management fields will appear.
2. In **Add item from**, select a financial statement template, an account or a node of accounts to be inserted into the formula.
3. In the formula definition area, type *[*: a list of FST items or accounts available to the relationship will appear.
4. Double click the desired FST item or account: the argument will appear in the definition area.
5. Repeat the previous steps for every desired argument.
6. Create the formula logic using the desired logical connectors and functions

   ***Note**: click **Guide** for more details on formula syntax, or consult the dedicated page in the manual (see "Standard formula syntax" on page 88).*

7. Click **Save**.

Override a formula for cube on an FST item

Dimensions tile > Account tile > ⋮ > FST & KPI

***IMPORTANT**: to define a formula for cube, an FST item containing a standard formula from which to override the formula for MSService cubes must have already been created. For more details, see "Formula for cube" on page 87.*

1. Click ⋮ > **Set formula for cube**: a popup progress bar will open showing the progress of the data processing launched on the entire financial statement template. Wait for the data processing complete message.
2. Click ⋮ > **FST items** and select the desired FST item.
3. In the formula for cube definition area, check the syntax of the formula calculated by the data processing.
4. To edit the formula, select **Override formula for cube** and make the desired changes.
5. Click **Save**.

   ***Note**: when the formula for cube is edited manually, the edited formula is not edited by the subsequent running of data processing to calculate formulas for cube.*

Define a multidimensional formula

Dimensions tile > Account tile > ⋮ > FST & KPI > select an FST that allows the MD formula > ⋮ > FST items

1. Select a row with **Item type**: MD Formula.
2. In the **Formula** definition area, insert the multidimensional formula.

How to exclude individual items from a financial statement template

Dimensions tile > Account tile > ⋮ > FST&KPI > ⋮ > FST Items > Relationships

Normally, all the accounts that belong to the account nodes inserted in an FST item will be included in the relative financial statement template. To exclude some specific items, proceed as follows:

1. In the table, select the account not to be included.
2. In the **Attributes** tab, select the **Exclude** action, then click **Save**: the account is no longer related to the FST item.

How to check the accounts present in the FST items

Dimensions tile > Account tile > ⋮ > FST&KPI > ⋮ > FST Items > Relationships

To check which accounts have been inserted in an FST item after a formula has been applied, it is useful to follow the procedure below.

Select the desired FST item, then click **Definition result**: the **Definition results** page will open, showing all the accounts that have actually been inserted in the FST item.

Particular features of multidimensional formulas

Syntax

For multidimensional formulas, the TGKML MD syntax is used (see Logiche calcolo conti di tipo "Script Multi-Dimensionale"). Some specific elements are described below.

Element	Description
TGKML 1.0 header	Syntax required at start of script. It has the following characteristics: • mandatory in an MD Calculation Logic. • to be omitted in MD-type FST formulas.
The reserved @output clause	This identifies, on the cell containing the MD formula, the result returned by the calculation on a temporary account. This clause is reserved and case-sensitive. For example, *@output :={010, 50010}+{010, 50020};*.
Parent FST items associated with relationships dependent on the account dimension only	It is the only type of FST item to which the MD formula can refer.

Constructing multidimensional formulas

When constructing multidimensional formulas, the following restrictions must be respected:

- It is not possible to use temporary accounts as entire parameters of the 'time skip' or 'number of months' functions, for example *Lag()*, *Span()*.
- As an MD financial statement template can also be used on reporting forms (which are therefore not linked to a specific process), when reference is made to a company node (*Entity.[|NN]* syntax), the user must specify the organisational hierarchy being referred to, i.e. the organisational hierarchy specified in the definition of the data collection process.
- The conversion functionality (*ConvertOriginal*) is not available.
- Assignments to physical accounts to write to the database cannot be defined within a multidimensional formula, but it is possible to use assignments to temporary accounts, which can be reused to calculate the final result (identified by the @output specification. If there are several records in the temporary *@output* account at the end of the script, the amounts will be added up.
- The final result is always a dimensionless number, and so the output Pro directive (conversion of periodic result into *Year-to-date* value) does not need to be specified.

Particular features of cells with multidimensional formulas

Cells containing multidimensional formulas have the following characteristics:

- The script for a multidimensional cell is run separately for every cell in which it appears, applying at runtime all the specific filters set for the cell.
- The currency is always the one indicated for the cell of the form to which the calculation is being applied.
- The amount in the company currency coincides with the amount in the original currency.
- Data are always read from the cube identified by the definition of the matrix in which the financial statement template is being used.
- The result of a multidimensional formula is always a number shown on the relative cell of the report, and not an n-tuple saved in the database.

FST items page

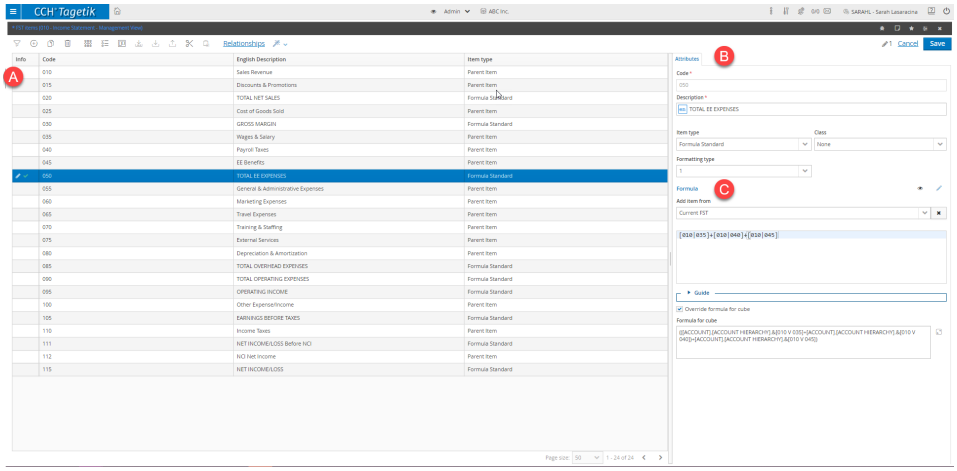
Admin | <application database>

Dimensions tile > Account tile > ⋮ > FST & KPI > ⋮ > FST Items

Purpose of the page

This page allows you to create FST items.

What it looks like



Element	Description
A	List of FST items
B	FST item definition tab
C	Formatting section of formula-type FST items

Attributes tab

Element	Description
Code	FST item code
Description	Description of the FST item
Item type	Define the item type: • **Parent Item**: the item can be related to one or more accounts • **Description**: header line, cannot be related to accounts • **Standard formula**: item determined by a formula
Class	Only with **Parent Item**. Define the FST item class: • **None** • **Cost**: changes the sign. • **Revenue**: does not change the sign (default) • Active • Passive For details, see "Sign class" on page 102.
Formatting type	Allows you to indicate the formatting type of the desired Tagetik style for FST items. See "Levels of style application in the matrix" on page 230.

Formula section

Only with **Item type** set to **Standard formula**.

Element	Description
	Shows the **Add item from** field.
	Hides/shows the **Add item from** field.
Add item from	Indicates the desired dimension on which you want to insert the formula, to view relevant syntax suggestions in the formula definition area. • **Current FST**: items of the current FST (default) • **Other Financial statement templates**: items of other FSTs • **Accounts**
Formula definition area	Allows you to insert the formula text.
Guide	Shows information on the syntax to be used in the **Formula definition area**.
Override formula for cube	Enables the formula definition area for overridden cube.
Formula for cube	(Only with **Override formula for cube** selected) MDX operation to be run on formula-type FST items.

Buttons and/or links on the page

Element	Description
	Copies the selected FST item.
	Pastes a copied FST item below the selected FST item.
Relationships	Opens a window to define the relationships between accounts and FST items.

Relationships page

Admin | <application database>

Dimensions tile > Account tile > ⋮ > FST & KPI > ⋮ > FST Items > Relationships

Purpose of the page

This page allows you to relate accounts, custom dimensions and/or control types to FST items. The fields shown on this page vary based on the settings configured in the **Dimensionality** section of the FST header definition page (see "FSTs & KPIs page" on page 85).

Relationship definition tab

Element	Description
Action	• **Include**: includes the account/custom dimension/control type. • **Exclude**: excludes the account/custom dimension/control type.
Account	Select an account or a node of accounts.
FST item	Select an FST item from the current FST.
Control type	Select a control type.
Custom dimension	Select a custom dimension.

Buttons and/or links on the page

Element	Description
Deployment	Runs the data processing to relate the accounts/custom dimensions/control types to the FST items.
Definition result	Opens a window showing details of the existing relationships.

Copy From window

Admin | <application database>

Dimensions tile > Account tile > ⋮ > FST & KPI > ⋮ > Copy From

Purpose of window

This window allows you to copy FST items from a source FST to the selected FST items.

Options in window

Element	Description
Target FST	Indicates the FST on which to copy the items.
Source FST	Indicates the FST from which to copy the items.
Copy accounts mapping also	Enables the copying of account/custom dimensions/control type relationships on the FST items of the source template.

Dashboard reports

Definition

Charts can be designed to visually represent the data included in a report. To do this, dashboard reports can be prepared.

In a dashboard report, each template can correspond to a dashboard and can, in turn, contain several matrices. The ability to generate charts can be enabled for all matrices.

Only users who have function 0003616 - Reporting .NET - Navigation - Edit dashboard" associated with their own role can configure Dashboard reports.

Filters on charts: slicers and shared slicers

The data shown in charts can be filtered both with filters applied to the form and with filters that can be modified by the user at runtime (see "Slicer" on page 98). However, while the filters applied to the module are applied to the whole report, slicer-type filters are applied to the individual chart.

Slicer filters are defined during the definition of the individual chart but the chosen values can be edited later to obtain a better result in the chart. However, editing a slicer does not run a new query on the database. Changes to the slicer therefore have no effect on the selection of data shown in the chart, but only affect the way in which they are displayed.

Slicers can be shared among matrices. If a slicer is applied to a specific dimension, all the matrices that share the slicer must contain the dimension in question.

Example

A dashboard report contains matrix1, matrix2 and matrix3. The Entity dimension is present on all the matrices. To define a shared slicer on the Entity dimension for every matrix, select **Link the dimensions** in the dashboard configuration window and enable **Share** on the Entity dimension. Then proceed as follows:

- For matrix1, select **matrix2**.
- For matrix2, select **matrix3**.
- For matrix3, select **matrix1**.

Available page layouts

Charts can distributed within pages according to the following predefined layouts:

- Full screen
- Two horizontal panes
- Two vertical panes

- Three horizontal panes (type 1 and type 2)
- Three vertical panes (type 1 and type 2)
- Four panes

***Note**: the system tries to optimise the position of charts. For example, if a template contains 5 matrices and a layout with 4 panes for every page has been chosen, the system will create two pages: the first will contain 4 charts and the second will contain the fifth in full-screen mode.*

Types of chart

A chart from among those contained in the following macro areas can be defined for every matrix in the report: column and bar, pie, line, combinations, maps.

Working with dashboard reports

Define the layout of a dashboard report

For more detailed instructions on how to create a form, see "Create a standard form" on page 35.

Excel > CCH® Tagetik > Design

1. When creating a form (or editing an existing form), activate the **enable dashboard** option.
2. Open the desired report, then click in the design toolbar: the **Define Dashboard** window will open.
3. In **Select a layout**, select the desired layout: a dedicated section will be shown for every template.

   ***Note**: if necessary, new pages will be created automatically to contain all the charts.*
4. In the specific section for each matrix, from **Select a chart**, choose the chart type and dimensions to be shown depending on the chosen type of chart.

   ***Note**: only dimensions defined on (matrix) row and column filters and tab filters can be selected.*

Change the order of charts in a dashboard report

Excel > CCH® Tagetik > Design

1. Open the desired dashboard report, then click in the design toolbar: the **Define Dashboard** window will open.
2. In **Select a layout**, select the desired layout: a dedicated section will be shown for every template.

   ***Note**: if necessary, new pages will be created automatically to contain all the charts.*
3. In **Position charts in the layout panes** it is possible to define which matrix will generate the chart for every pane of the layout.

Navigating dashboard reports

User | <application database>

Dashboard tile > click grouping > select the report

1. Set the required filters, then click **OK**: the charts prepared in the report will be filled in.
2. To view the charts of a specific sheet, select the corresponding tab.
3. To edit the value of the chart filters, click the button and edit the slicer values: only the filtered data will be shown in the chart.

4. To refresh the data in the charts, click .

   ***Note**: if the value of a non-shared slicer is edited, only the data for the current chart will be recalculated. If the value of a shared slicer is edited, the data will be recalculated and all the charts that share the slicer will be refreshed.*

5. To generate pivot tables based on the matrices used to generate charts, click to open a preview.

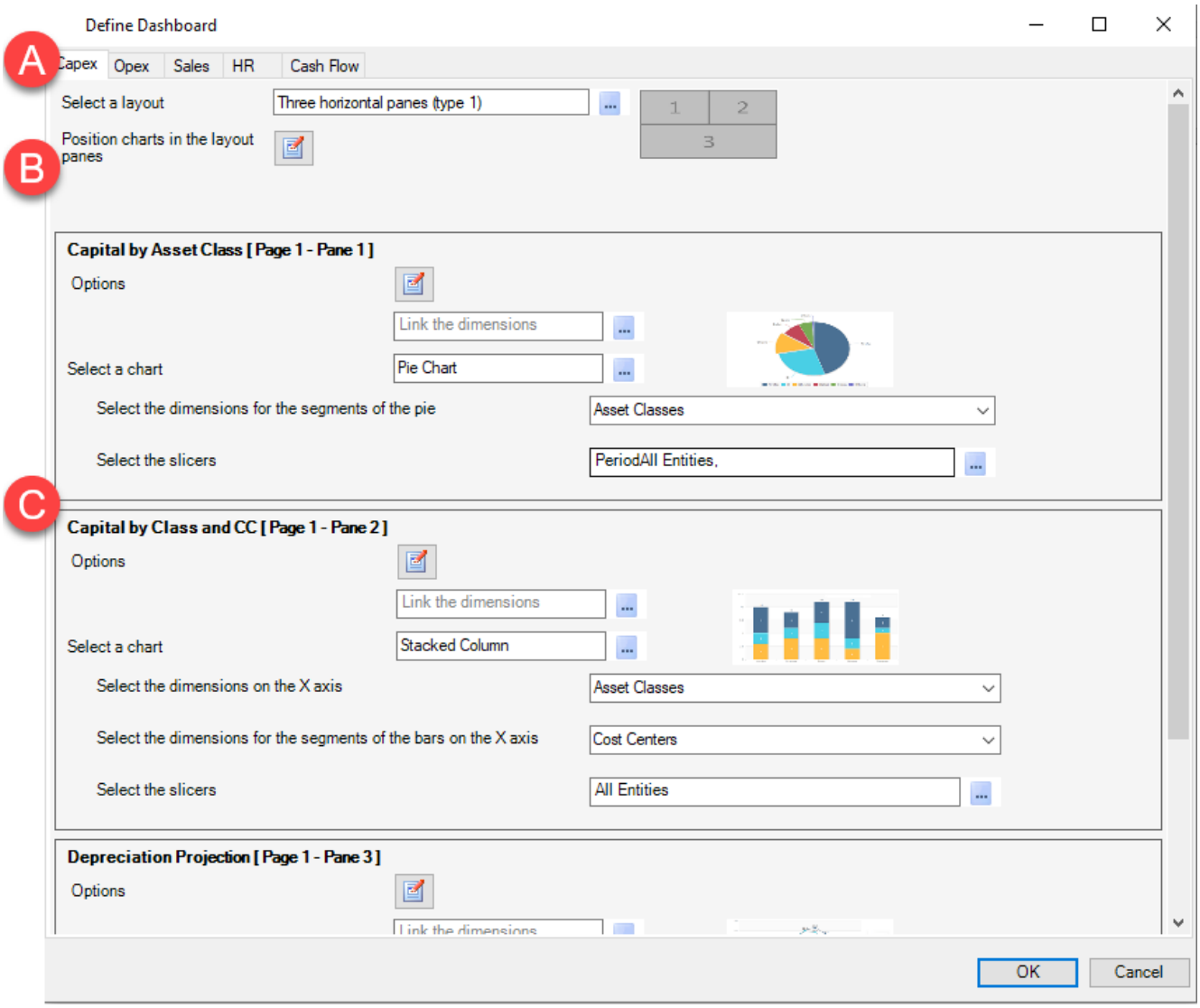
Define Dashboard window

Excel > CCH® Tagetik > Design > open dashboard form > Definition

Purpose of window

This window allows you to define the appearance of the dashboard report, or choose the types of chart to be displayed, their distribution within the template and the data shown. The default order of the charts within the pages mirrors the order of the matrices within the relative template.

What it looks like



Element	Description
A	List of sheets generated to contain all charts defined for the report
B	Layout section to set how charts are arranged on the page
C	Chart section to define a chart for every matrix

Slicer

Definition

In CCH Tagetik, the *Slicer* functionality allows you to change, directly in navigation mode or data entry mode, the parameters defined on the form without having to rerun the same form or open the form in design mode to edit it.

This functionality allows you to create, in a cell of the form, a slicer that displays a parametric element selection menu. It also allows you to change the slicer directly at runtime.

Working with slicers

Define a slicer

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

***IMPORTANT**: before inserting a slicer in the cell, to improve the visibility of the slicer when the form is run, it may be necessary to change the size of the cell in which it will be inserted, or merge it with other cells.*

1. Select the desired cell.
2. Click : the **New slicer** window will open.
3. Select the parameter on which to define the slicer.

   ***Note**: the user can select both parameters defined in the (standard or ad hoc) form, and parameters that are not used in the form but are defined in the system. Unused parameters can be shown by activating the **Show unused parameters** option.*

4. Click **OK**.

The inserted slicer is shown in the cell in the work desktop and in the **Open items** tree, with a description of the selected parameter.

Select a parameter in a slicer

Excel > CCH® Tagetik > Data Entry > open form

Excel > CCH® Tagetik > Navigation > select form >

1. Select the cell with the slicer and click .
2. Select the desired parameter, or click ... to open the selection window and choose the parameter. In this window, the user can define advanced options (e.g. reference FX rates for the selected values).
3. Click **OK**.
4. In the toolbar, click to refresh the form.

New Slicer window

Excel > CCH® Tagetik > Design > open form > select cell > Slicer

Purpose of window

This window allows you to select a parameter on which to create a slicer in the current form.

Columns in window

Element	Description
Used	Indicates whether the parameter is used in the form.
Ad Hoc	Indicates whether the parameter is an ad hoc parameter, i.e. only available in the current form.
Code	Parameter code
Description	Parameter description

Options in window

Element	Description
Show unused parameters	Shows the parameters that exist in the system but are not used in the form.

Design: matrices

Contents

This section covers the following topics:

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Form matrices

What is a matrix?

A form's matrix is the tool with which the user defines the data managed by that form, the ways in which the data are managed, and the ways in which the data are displayed when the form is run. Creating a matrix therefore means deciding how to filter the data of the CCH Tagetik dimensions on the basis of the form's objectives.

When the form is run, the matrix appears as a normal data table in Excel, the row/column structure of which depends on the data managed, but its appearance can be customised using Tagetik styles (see "Tagetik styles " on page 230).

Matrix elements

Matrix configuration involves configuring the following elements via the Matrix Wizard ("Matrix Wizard - Window structure" on page 109).

- "Row/column filters" on page 151 (necessary)
- "Matrix filters" on page 148
- "Headers" on page 103
- "Overrides" on page 107
- Additional management options (see "Matrix options tab" on page 128)

Example matrix

This example describes the creation of a matrix which shows a comparison, for entity 001, of the budget data and the actual data for 2010 relative to the month of December, including the following revenues and costs:

- sales revenue of cost centre "Software sales", product "Antivirus"
- sales revenue of cost centre "Hardware sales", product "Memory cards"
- costs of cost centre "IT", product "Databases"

To do this, the following filters must be set:

- the account, cost centre, product dimensions on rows
- the scenario/period dimension on columns
- entity A01 and the amount type (mandatory dimension specific to Reporting) on the matrix (i.e. on both rows and columns).

The created matrix appears as below.

Entity A01	2010BDG December	2010ACT December
Product Revenues – Sales SW - Antivirus	- 1000	- 980
Product Revenues – Sales HW – Memory cards	- 900	- 950
Product Costs – IT – Databases	2000	2500
TOTAL	**100**	**570**

Transactional matrices

Transactional matrices only manage data on columns. When a transactional matrix is set in the form, the form is called a transactional form. Transactional matrices can manage both Financial Workspace data and Analytical Workspace data.

For more details, see "Introduction to transactional forms" on page 281.

Multidimensional matrices

Multidimensional matrices manage data on both rows and columns.
In particular, multidimensional matrices can be used to manage data from Analytical Workspace datasets. When creating the matrix, the user can select a dataset, a datasource or a virtual dataset for every Analytical Workspace.

The creation of matrices with Analytical Workspace data is subject to the following limitations:

- Multidimensional matrices cannot be designed on datasets with the "Show user data only" property enabled.
- "Comments and Attachments" fields cannot be added to multidimensional matrices, even if the associated dataset contains an equivalent field.
- Matrices cannot be defined on datasources that retrieve data via a different connection from the one used by the user.

Sorting of data in matrix trees

The display mode and order of elements in the dimensions tree and in the local row/column filter and matrix filter trees is defined in CCH Tagetik's general settings (see Settings window).

Display rules can also be set directly from the Excel interface, via the  **Display mode** icon in the design toolbar (see "Tool bar" on page 6).

Sorting entity, category and custom dimensions data

The order of elements related to the entity, category and custom dimensions is defined in the **Sorting** tab in the general settings for each dimension. For more details, see How to work with dimensions.

***Note**: whatever the configuration, the order is always ascending.*

The Origin dimension

What is it for

The *Origin* dimension allows you to obtain information on the origin of data. It can be used as a row/column filter, a matrix filter, a tab filter or a form filter, statically by selecting a lowest level element, or dynamically by using a system parameter (see "System parameter details" on page 59).

Specific features for use in parametric forms

Both during the design and during the running of parametric forms, the origin dimension defines the comparison condition with which to filter the data based on their origin. The comparison is on textual data via comparison operators:

- the operator "=" specified a single origin to be displayed
- the operator "<>" excludes the specific origin from the comparison

For example, by specifying "=" as the comparison operator, and *INPUT_DEFORM* as the comparison value on an origin filter, you can obtain the sum of all the figures with origin *INPUT_DEFORM*.

For more details on the possible values for the origin dimension, see Provenienza dei dati.

Sign class

Introduction

The sign with which amounts are shown in forms depends on general settings made in the system and on any specific overrides made at the individual account or FST item level, therefore in the report and/or template definition.

Depending on the needs of the project, it is possible to show the amounts with their original sign inverted in relation to the sign they have in the database, or as an absolute value (all amounts shown as positive).

In particular, setting the *Debit* sign determines the relative sign for the *Credit* sign (which will have the opposite sign). The sign setting for *Asset/Liability* and *Asset/Liability Increase* and *Asset/Liability Decrease* balance sheet amounts is valid in reporting for all the related amounts.

Both in data entry and navigation, if a value with a different sign has been inserted on the individual account or FST item, the system inverts the sign based on the accounting classes associated with the account/FST item in question.

***Note**: the following accounting classes are available in CCH Tagetik: Revenue, Cost, Asset, Liability, Asset increase, Asset decrease, Liability increase, Liability decrease.*

Details of signs

The sign attributed to the various accounting classes determines the sign with which the amount, if inserted from a data entry form, is saved on the database. When, at the account or FST item level, the sign inserted is the opposite of the sign defined on the accounting class in general settings, to save on the database the system recalculates the amount to obtain the sign indicated in the class settings.

A positive or negative value can be attributed to the Debit sign. The behaviour of the accounting classes, depending on the sign, is described below.

Debit sign	Description
Positive	On the application database, revenue and liability values are saved with a negative sign, while cost and asset values are saved with a positive sign. Therefore, if the signs of the values entered on this type of account are consistent with the signs specified above, the system will not change the sign.
Negative	On the application database, revenue and liability values are saved with a positive sign, while cost and asset values are saved with a negative sign. Therefore, if the signs of the values entered on this type of account are consistent with the signs specified above, the system will not change the sign.

Example: to create a balance sheet FST in which every line needs to have a positive sign, the signs of the accounting classes Liability and Liability (Decrease) must be set to positive in general settings. At the specific FST item level, it is then necessary to set the liability sign class for all liability accounting items.

Override the sign class on the account from FST item

When it is not necessary to change the sign on the individual accounts, or when a single logic is used for the FST items and the accounts related to them, the *Override Sign for class on the Account by FST item* option in the FST settings allows the accounts related to an FST item to inherit the sign class set on the same item (see "Financial statement templates" on page 83).

***Note**: In the case of dynamic matrices, in the presence of an account with a sign class inherited from an FST item and a dimension with "Insert children" selection type, the system will also apply the sign change to the generated subtotals.*

***IMPORTANT**: If the FST item contains a Formula, it does not inherit the sign from a hierarchical parent; the sign is derived from:

- the **Class**, if set on the FST item containing the formula;
- the **calculation expression**, i.e. the way in which the operators (+, -, ×, ÷) combine members that already have their own sign, including any reverse sign (sign inversion by convention) and aggregation signs (the sign contribution in hierarchical totals), according to the **basic rules of arithmetic**.
- any **overridden rules at the report level**

Override the sign in the Matrix Wizard

In reporting, the class settings given on the tables are visible in the dimensions tree in the matrix creation step, both for FSTs and for accounts.

In the matrix definition step it is possible to manually insert change sign settings for the sign that the system proposes as the result of attributing a certain sign class.

Based on the sign changes, the system shows the filter with specific colours.

If...	Then...
the class-sign combination present in configuration on the *debit* account entails a sign change	the filter is shown in orange.
the sign changes due to a manual override	the filter is shown in red.
due to the overridden sign change, the filter returns back to the sign that it had before the class was applied	the filter is shown in black.

For more details, see "Sign change" on page 113.

Headers

What they are for

By default, when the form is run, in the matrix table the row/column filters are identified by the table description of the dimension to which the dimensional elements of the filters refer.

***Note**: if the additional information Segment, FX rate, Reference FX rate, Weight is defined for dimensions on which a header is defined, then the system shows those properties at the end of the header (see "Layout options details" on page 56).*

For example, for the scenario filter *2021 Actuals*, in the Matrix Wizard the predefined header is *Description ($Scenario.desc)*, which recovers the description in the scenario table.

To facilitate the reading of tables and/or supplement the row/column descriptions, customised headers can be set for every filter. In particular, the predefined description can be replaced or supplemented with one or more descriptions, linked in a chain, from those available for the dimension.
Alternatively, it is possible to replace the description with a dictionary item which, once created on the row or column, is available among the headers.

When multiple descriptions are set, it is possible to choose whether to display the information in a single cell, separated by the "|" sign, or on several different cells on the same row, see "Setting headers" below.

Advanced Headers

In some cases, it is necessary to compose more complex headers, the definition of which depends not only on the dimensions used as filters, but also on all other dimensions available in the system. In this case, it is possible to use an advanced editor, the Cell Fields Wizard, which allows you to set how the text of attributes not related to the dimension is displayed in the cells (see "Cell fields" on page 41).

Descriptive information regarding the entire matrix cannot be inserted as headers, but only using cell fields.

Header structure

Headers show information as follows.

Element	Description
Segment	• *(SEGM.)* for *Per Segment* • *(PARENT SEGM.)* for *Per Segment Parent* • *(INTERSEGM.)* for *Per Intersegment* • *(PARENT INTERSEGM.)* for *Per Intersegment Parent* For more details, see "Tab filter properties" on page 144.
FX rate type	• *(Average FX rate)* • *(Final FX rate)* • *(Based on account)*
Dividing factor	• *(Periods)* in case of period factor • *(Days)* in case of daily factor
Reference FX rate	The FX rate's time reference with the relative FX rate type. For example, for a reference FX rate on the May period with average FX rate, it shows (REF.CH. 05 - May(Average FX rate)).

No additional information is shown if a dictionary item is set on the row/column.

Setting headers

Excel > CCH® Tagetik > Design > select form > Matrix > Headers

Edit an individual header of a dimension

1. In the list of predefined headers, expand the **Predefined functions**, then expand the desired nodes of headers, then select the desired header.
2. Click .
3. (Optional) In the list of defined headers, remove the header initially set by the system (the description of the dimension).

Create multiple headers to be shown on several rows/columns.

1. In the list of predefined headers, click **Predefined functions**, then click the desired hierarchy nodes, then select one or more desired headers.
2. Click .

Create multiple headers to be shown in a single cell.

1. In the list of predefined headers, click **Predefined functions**, then click the desired hierarchy nodes, then select one or more desired headers.
2. Click .

Headers inserted in this way will appear in a single cell, separated by a comma.

Edit the order of multiple headers

1. In the list of defined headers, select the desired header then click . The multiple header editor will open.
2. Change the header order.
3. Click **OK**.

Create advanced headers

1. In the list of defined headers, select the desired header then click . The **Cell Fields Wizard** will open (see).
2. Set the desired header (see "Cell fields" on page 41).
3. Click **OK**.

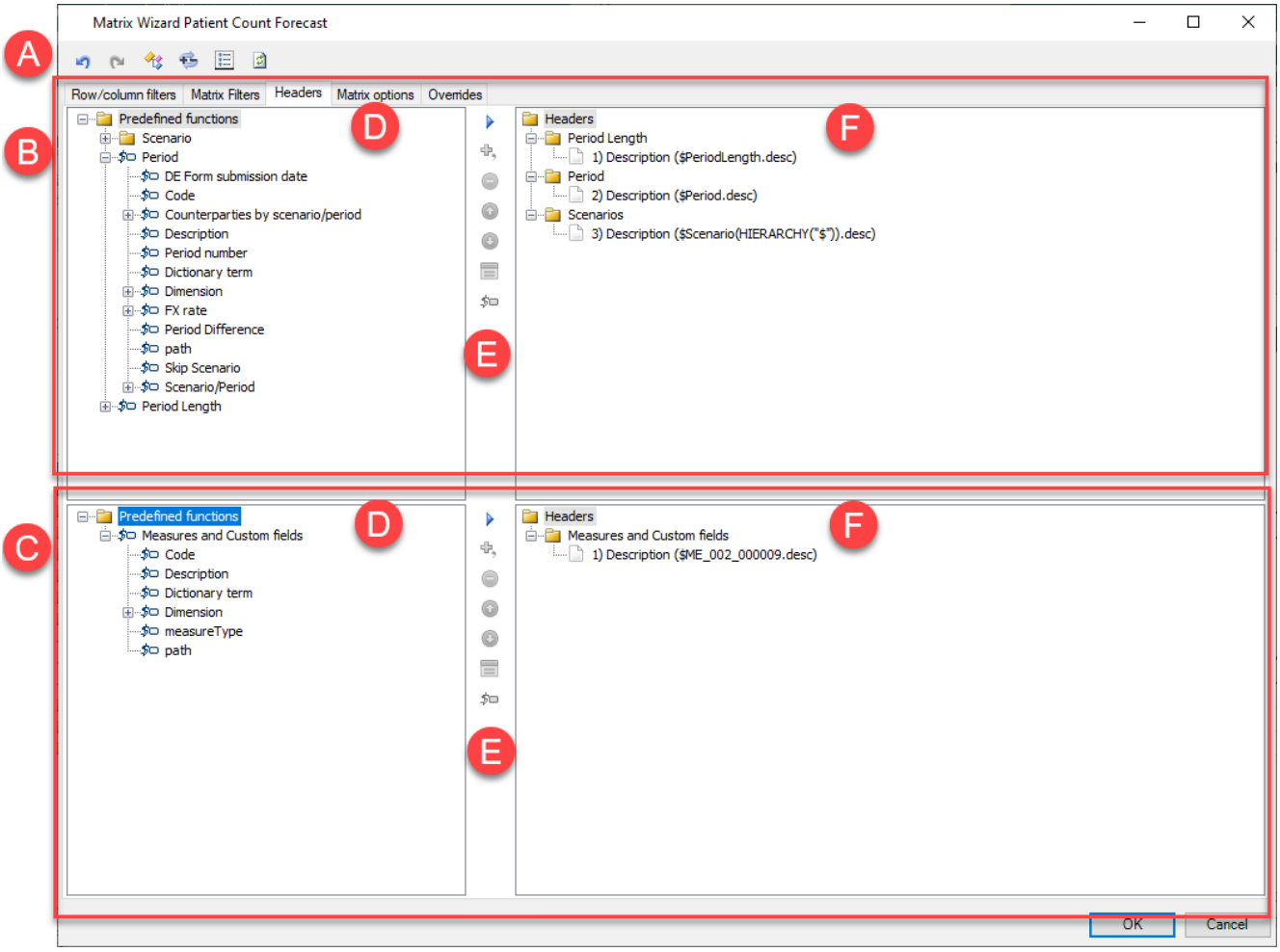
Headers tab

Excel > CCH® Tagetik > Design > select form > Matrix > Headers

Purpose of the tab

This tab allows you to customise the row and column headers of the matrix using predefined headers in order to make it easier to read the data in the table.

What it looks like



Element	Description
A	Action buttons in all tabs. See "Configuration window structure" on page 16.
B	Row headers section
C	Column headers section
D	List of available headers
E	Buttons specific to the tab
F	List of defined headers

Areas of row/column header sections

Element	Description
List of predefined headers	Displays the list of predefined headers, indicated by .
List of defined headers	Displays the list of defined headers. Every defined header presents the following information: • matrix row/column number on which the header is defined • text shown in the matrix headers

Buttons in tab

Element	Description
	Adds one or more selected predefined headers to the list of defined headers.
	Adds, through concatenation, one or more selected predefined headers to a header that has already been defined. In the matrix, such headers are shown as a concatenation.
	Only with headers already defined. Removes one or more selected headers from the list of defined headers.
	Only with headers already defined. Moves the selected header up one position.
	Only with headers already defined. Moves the selected header down one position.
	Only with headers already defined with a OR logical function. Allows you to remove or reorganise one or several headers of a group of concatenated headers with the OR logical function.
	Opens the Cell Fields Wizard. See "Cell Fields Wizard" on page 41.

Overrides

Introduction

The Matrix Wizard's *Override* tab allows you to create, at the matrix level, overrides on the dimensions included in the form by form filters and tab filters. In fact, restrictions defined by form filters and tab filters on certain dimensions can be *excluded* from the matrix, or elements of dimensions originally excluded by the form filters and tab filters can be *included* in the matrix (see "Structure of the data in the forms and filters" on page 32).

***Note**: for the override to include functions, the elements selected in the matrix's override tab must belong to the same hierarchy used to insert the elements overridden in the matrix.*

Example

This example describes a form containing a sheet with filters on entities *A00*, *B00* and *A01* and two matrices respectively showing:

- data for a given scenario selected by the user when the form was run
- data for the previous scenario

To compare the data for one scenario on entities A00, B00 and A01 with data for the previous scenario related to the entities node *EU-Europe* (parent node of entities A00, B00 and A01), overrides for the tab filters must be applied on the second matrix, so as to define the EU node as a matrix filter for the second matrix. To do this:

1. In the **Matrix Filters** tab, in hierarchy *X00 - Geographical area*, select the filter *EU-Europe*.
2. In the **Override** tab in the Matrix Wizard, select entity hierarchy *X00 - Geographical area* which contains all the entities grouped geographically.

   ***Note**: for the override to take effect, the hierarchy selected in the **Overrides** tab must be the same as the one in the initial matrix filter.*

Overrides on Analytical Information Hub matrices

As regards matrices based on both multidimensional and Line by Line transactional Analytical Information Hub datasets, the override system is simplified. It is not possible to define an override on individual hierarchies of a dimension, but only on the dimension itself.

The Override content on cells function

Overrides on formulas

In CCH Tagetik, it is possible to insert formulas into form cells, via the Matrix Wizard (see "Matrix Wizard - Window structure" on the facing page).

The *formula* filters are filters on matrix rows or columns which, while the form is being run, function like Excel formulas and make it possible to process data based on the data handled by the matrix itself.

For more details, see "Formula filters" on page 159.

In the Matrix Wizard, the formula is set on the entire row/column of the matrix. However, in some contexts, it is necessary to define specific formulas for specific cells. Some examples are provided below:

- It is necessary to refer to cells that belong to other matrices or other cells of the form.
- Formulas must have different behaviour for certain cells of the matrix (whereas standard formulas are defined on the matrix rows and/or columns).
- Different constant values must be assigned for every cell of the matrix.

To create formulas at the cell level, it is possible to use the *Override content* function, which allows you to insert an Excel formula directly into a cell in the area occupied by a matrix.

It is important to consider the following behaviours:

- Inserting ordinary Excel formulas means that, when the matrix is generated as the form is run, the formula will be overwritten with what is present in the matrix.
- The content of cells can only be overridden in static matrices.
- If the data resulting from the formulas needs to be saved in the database, the cells containing the formulas in the matrix must have editability set to **Editability protected**

To remove an override, right click on the cell to which the override is applied and select Tagetik > Remove "Override content"

Override Headers

The *Override content* functionality also allows you to override header-type content in a cell. This override can be run in the following ways:

- by using a dictionary item on the row/column of the Matrix Wizard
- by setting the override on the matrix headers

Setting up an override on matrix headers is useful, for example, for inserting content in the matrix's starting cell, or the first cell at the top-left, at the intersection between the first row and the first column of the matrix. In order for the text content inserted directly into this cell in Excel not to be removed as the matrix is regenerated when the form is run, it can be added via overrides.

Overriding cell content

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

Set a content override

1. In the work desktop, select the cell in the area of the matrix on which to override content
2. Insert the desired formula or value in the cell.
3. In the toolbar, click **Override content**.

The system will change the format of the cell and show its value in red. In the **Open items** panel, the presence in the form of cells with overridden content is indicated by the element **Cells with overridden content**.

Remove a content override

In the work desktop, right click on the cell and select *Tagetik > Remove "Override content"*.

Setting overrides

Excel > CCH® Tagetik > Design > select form > Matrix > Overrides

See "How to access the windows" on page 9 for alternative paths.

Exclude restrictions of form/tab filter on dimensions from the matrix

In the dimensions tree, select the filters on one or several dimensions known to be originally defined at the form/tab level. This allows you not to apply the restrictions set by the filters at the form/tab level to the matrix.

***Note**: to add elements of the dimension to the row/column filters and matrix filters, see "Set row/column filters" on page 152 and "Setting matrix filters" on page 148.*

Overrides tab

Excel > CCH® Tagetik > Design > select form > Matrix > Overrides

See "How to access the windows" on page 9 for alternative paths.

Purpose of the tab

This tab shows the tree of dimensions included in the form and allows you to override the exclusion of the matrix from the form filters and tab filters action on one or several dimensions.

***Note**: once the matrix has been excluded from the restrictions on the dimensions set by the form filters and tab filters, it is possible to include dimensional elements of those dimensions in the row/column filters and matrix filters. For this to work, the form filter and selected overrides must belong to the same dimensions hierarchy.*

Matrix Wizard - Window structure

Excel > CCH® Tagetik > Design > select form > Matrix

Excel > CCH® Tagetik > Design > select form > right click the cell on the work desktop > Tagetik > New Matrix or Modify matrix

Excel > CCH® Tagetik > Design > select form > double click the cell of the matrix in the work desktop

Excel > CCH® Tagetik > Open items > right click the left-hand panel > Matrices > New Matrix

Purpose of window

The Matrix Wizard allows you to create, edit and manage matrices in forms.

What it looks like

The structure of the Matrix Wizard tabs mirrors the general structure of the Excel add-in windows. For more information, see "Configuration window structure" on page 16.

Window tabs

Element	Description
Row/column filters	Allows you to define CCH Tagetik dimensions as filters in the matrix rows and columns. See "Row/column filters tab" on page 156.
Matrix Filters	Allows you to define filters on dimension data at the matrix level. See "Matrix Filters tab" on page 149.
Headers	Allows you to define a description for the matrix rows and columns to make them easier to read. See "Headers tab" on page 105.
Matrix options	Allows you to manage the following: • form formatting • how cells without values are saved in the database • how to dynamically open rows/columns • Break back options See "Matrix options tab" on page 128.
Overrides	Allows you to define overrides in the matrix. See "Overrides tab" on the previous page.
Transactional form editor	For transactional forms only. Allows you to define the layout of the form. See "Transactional form editor tab" on page 131.
Custom filters	Only for transactional forms based on Analytical Workspace datasets. Allows you to create custom filters on the measures of the dataset associated with the form.

Buttons in the window

Element	Description
	Undo the last action.
	Repeat the last action.
	Shows a list of the filters used.
	If the class attribute has been changed, it realigns the signs of the accounts on the class of which it has been applied.
	Allows the user to select the display mode for forms on the panel: with description only, with code only or with both description and code.
	Refresh the dimensions tree.

Display options

Element	Description
Double click the top border	Displays the window full screen/resizes the window if already being displayed full screen.

Dimensions tree in the Matrix Wizard

Introduction

In the Matrix Wizard, the **Row/column filters**, **Matrix filters**, **Overrides** and **Forms** tabs allow you to set filters on reporting data based on a dimensions tree.

Elements of the dimensions tree

Depending on the selected tab, the dimensions tree shows the following dimensions:

Tab	Description
Row/column filters	• system dimensions (scenario, period, account, control groups, entity, etc.) • custom dimensions (e.g. : products, cost centres, etc.) • specific reporting dimensions (e.g. period length), see "Specific dimensions for reporting" on page 3. • specific dimensions of some reporting functionalities (e.g. Column form and Counter, used in the management of transactional data entry forms) • predefined filters, see "Parameters" on page 58. • Wizard > Control groups, see I gruppi di controllo: concetti di base.
Matrix filters	• system dimensions (scenario, period, account, entity...) • custom dimensions (e.g. : products, cost centres, etc.) • specific reporting dimensions (e.g. period length), see "Parameters" on page 58.
Overrides	• system dimensions (scenario, period, account, entity...) • custom dimensions (e.g. : products, cost centres, etc.) • specific reporting dimensions (e.g. period length), see "Parameters" on page 58. • predefined filters, see "Parameters" on page 58. • Wizard > Control groups, see I gruppi di controllo: concetti di base.

Account dimension utility nodes

In the dimensions tree, the account hierarchy and following utility nodes, which allow you to quickly select several accounts at the same time, are shown by default for the account dimension.

Node	Description
Financial statement templates	Aggregation structure of the accounts or accounts/custom dimensions generally used to create financial statements (IAS profit and loss statements and balance sheets, management reports, direct/indirect financial reports), KPIs and other indices necessary for reporting. See "Financial statement templates" on page 83.
Control groups	Groups of accounts organised on the basis of balancing rules between accounts, variations that document how they change over time, or elements that allow the user to detail an account. See I gruppi di controllo: concetti di base.
$ - All Accounts	Default node that allows you to select one or several elements or nodes from the group of accounts
$-All Control Groups	Node that allows you to select one or several elements or nodes from the group of control groups.

Parameter-type elements

In the dimensions tree, for each dimension and hierarchy there is a *parameter* shown in blue in the dimensions tree, the name of which always starts with $ (e.g. $AZI_P for entity elements, $PER_P for period elements, etc.). This allows you to select the desired dimension by choosing from among the available elements deriving from the context and compatible with the relative user restrictions.

In addition to dimensional parameters, when defined, *system parameters* and *custom parameters* can be used as row/column filters or matrix filters.

For more details on parameters, see "Parameters" on page 58.

Display rules

The data displayed in the dimensions tree vary based on the following criteria:

- user rights on dimensions
- **Exclude from reporting** option active on some hierarchies or nodes of a group because they are only useful for configuration, otherwise they are obsolete

Click to refresh the dimensions tree immediately after the deployment of changes in the dimensions table.

Working in the properties table

Procedures for setting the values of properties

Depending on the property, it is possible to apply one of the following methods.

- Click and select the desired element from the drop-down menu (e.g. sign change definition).
- Click on the desired field and insert the value in the dedicated window (e.g. formula insertion).
- Click the row containing the property and then on to open the editor the guides the insertion of values (e.g. reference FX rates definition).

Table of row/column filter properties

Introduction

The properties table allows the user to manage the filter properties to customise the matrix. When several filters are selected, only the shared properties are shown and the values set on them will be applied to all the selected filters.

Reference FX rates

This property manages the FX rate when a different FX rate from the one specified in the table is needed, or if it must be read from a different scenario or period from those of the data to be converted.

The possible values are provided below:

Element	Description
average	Average FX rate of the period in which it is calculated. These are the FX rates generally used for P&L accounts.
period average	Applies a different FX rate for every period into which the year is divided. Allows the user to perform more accurate conversions than the average FX rate.
final	Period-end FX rates. These are used for the balance sheet accounts, with the exception of the equity accounts, which are typically converted at the historical FX rate.
based on account	FX rate used for conversions and defined in the accounts table

FX rate

Currency dimension properties.

Element	Description
For matrices on the financial layer	
From Configuration	Value, displayed in brackets next to the active property, previously configured in **Default FX rate** for the new filters on the currency in the **Forms** section of CCH Tagetik 's general settings. See Valuta..
None	The Currency dimension is used as a dimension filter: only records whose currency is the selected currency will be read by the database.
Final FX rate	The records read by the database are converted into the selected currency using the final FX rate, as defined in the FX rate table.
Average FX rate	The records read by the database are converted into the selected currency using the average FX rate, as defined in the FX rate table.
Period average FX rate	The records read by the database are converted into the selected currency using the period average FX rate, as defined in the FX rate table. The **FX rate** property of the Currency dimension, as well as the equivalent property in **Reference FX rates**, does not support the **Period average FX rate** option if the form uses the standard reporting engine. Where this configuration is used, at the run form step, this option is automatically set to **Based on account**. A warning is sent and the user must correct the form configuration.

Element	Description
Based on account	In the accounts table, it is possible to define the FX rate to be used in the various data processing to covert the relevant amounts (**Conversion** property, with the following possible values: **None**, **Final**, **Average** or **Period average**). With this setting, the form converts records read from the database into the selected currency, using the relative FX rate, as defined in the table, for each of them.
For matrices on the analytics layer	
None	The Currency dimension is used as a dimension filter: only records whose currency is the selected currency will be read by the database.
Based on the measure	The option to be applied is read from the numerical measure configuration in which it is possible to define the desired FX rate for that measure from the web interface. (See Dataset fields).
Based on account	With this setting, the form converts records read from the database into the selected currency, using the relative FX rate, as defined in the table, for each of them. If the account dimension is not present in the dataset, the **Based on the measure** FX rate is applied.

Particular features of conversion based on account

Conversion based on account is only applicable when the dataset has the account dimension. If the “based on account” conversion type is specified and the matrix does not have the account dimension, the form is not run. If the account has a conversion type that is not supported by the analytical matrices, i.e. it is not the average FX rate, period average FX rate or final FX rate, the **Based on the measure** FX rate type is applied.

Forms in which global filters (form filters or sheet filters) on the currency are used will have the following behaviours:

When...	Then...
the dataset does not contain the account dimension	if the matrix has no overrides on the currency dimension, the system considers the currency indicated in the global filter: • if the conversion type is **final**, **average**, or **period average**, the system applies the **Based on the measure** conversion type. • if the conversion type is **None**, then no filter is applied and only the values in the selected currency are read.
	of the matrix specifies an override on the Currency dimension, then the global filters are not applied.
the dataset contains the Account dimension	the **Based on Account** conversion type is applied, regardless of what is specified at the matrix level, save for matrix overrides.

Periodic value calculation

Period length dimension properties.

Element	Description
Automatic	The properties’ behaviour varies based on the engine used for reporting. Default value. See "Types of reporting engine" on page 28. ***Note**: option used when the Period length is not specified in any of the form’s filters.*
Aggregation of periodic values	Calculates the cumulated data as the sum of periodic figures. For example, the quarterly figure for the second quarter is calculated as the sum of the monthly figures read from periods 4, 5 and 6.
Net Year-to-date values	Calculates the cumulated data as the net year-to-date values. For example, the quarterly figure for the second quarter is calculated as the difference between the *year to date* figure for period 6 and the *year to date* figure for period 3.

Sign change

This account dimension property allows you to change the sign with which the account is displayed. When active, this inverts the sign of the data.

***Note**: the elements for which the sign change is active are shown in red in the Matrix Wizard.*

Line by Line matrices do not support the native sign change relative to the account dimension unless additional filters on the account dimension are present (Form/Sheet/Matrix filters).

For multidimensional AIH matrices related to an AW dataset configured with data refresh mode set to **Refresh existing record**, the sign change is not present for text-based measures.

Complementary

This property allows you to exclude a specific filter or node from a node of filters in a hierarchy. Thus, the system displays the entire hierarchy of the filter's parent node or excluded node, except for the specific filter or node. This type of filter is recognisable from the **<NOT>** prefix.

***Note**: the additional filter works on the hierarchies but not groupings.*

Data Validation

This property allows the user to associated a validation query with the selected filter to limit their choice to a restricted set of elements.

Size

This property is not editable and indicates the dimension to which the selected element belongs.

Advanced selection

Element	Description
Allow new rows/columns wizard in data entry	Only for transactional forms with **Pruning** set to **To zero**. When active, in data entry it allows you to view and edit rows/columns that are open but not shown because they are empty. ***Note**: the effects of the changes are shown in the form automatically (e.g. changes on amounts).*
Additional rows to open	With the Counter dimension, it is possible to set the number of additional rows to be opened in transactional forms. Set this value in the **Additional rows to open** field in the **Advanced selection** attribute included in the counter properties in the matrix. (See "Details of the Counter element" on page 328). Can only be set if **Allow new rows/columns wizard in data entry** is set to "false".
Detail subtotal	Displays the subtotal for every dimension nested in the filter node. *In the presence of three or more nested dimensions, activate the option to prevent potential style application errors.*
Generate subtotals up to	Allows the user to define the starting node for calculating subtotals. • **Root**: generates subtotals based on the hierarchy root (default value). • **Starting node**: generates subtotals based on the selected intermediate node. ***Note**: this has no effect if the elements returned as the result of the Advanced Selection script do not belong to the starting node's sub-tree. See "Details of advanced selection scripts for calculating the default value" on page 63.*
Restrictions	The system allows you to insert in the table restrictions on dimensions of the row/column, primarily between entity/custom dimensions, in order to allow, in data entry, data to be read/written only on the relative custom dimensions specified in the table, to improve performance. This option allows you to override these restrictions on dynamic elements, in order to see the data without the restrictions set in the table. **Hierarchy Node Intersection**: by default, if a single dimension of the matrix has several hierarchies, the system presents the intersection of the two hierarchies, i.e. the data related to elements belonging to both hierarchies. By activating this option, the system shows all combinations of elements of both hierarchies. See Entity elements.

Element	Description
Pruning	Hides empty rows/columns. • **Not valued**: hides from the table rows/columns which are found to have no data in the database during the run form phase. The value zero is considered as data. • **To zero**: hides from the table rows/columns which are found to have no data in the database during the run form phase or have the value 0 entered. • **None**: shows all rows/columns.
Ranking	Defines the criteria for sorting data within the table. • **No rule**: sorts the row data based on the values of a reference column. • **Top elements**: brings the elements with the highest values to the top. • **Bottom elements** brings the elements with the lowest values to the top. • **Pareto analysis 80/20**: selects the minimum set of rows/columns with the highest value in the reference column and which make up a defined % of the column/rows total. By default, the **threshold** is 80%, but it can be changed. • **Only elements that are not zero**: removes from the display rows/columns with no value entered or with the value 0 in the reference column/row. • **Reference column/row**: selects the matrix column/row (depending on where the ranking property is being set) on which basis the data is to be sorted. • **Number of elements to display**: sets the maximum number of rows to display. • **Accept more elements than requested**: if there are several rows with the same value, all rows will be displayed. If disabled, the system chooses the rows at random, so that it displays the best rows while respecting the limit indicated in **Number of elements to display**. • **Sorting rule**: defines the row sorting criteria (**increasing**, **decreasing**, **none**).
Script	Allows the user to set a query on CCH Tagetik 's application database in the cell using a TGKML syntax, see Advanced selection script.
Subtotals	Only with dynamic filters. When the form is run, at every hierarchy level interruption, it displays the subtotal calculated on the lowest level elements. The subtotal therefore takes the name of the parent node of the lowest level elements on which the subtotal is calculated (for example: Europe is the name, and therefore the parent node, of the subtotal calculated on the lowest level elements Italy, France, Spain etc.). For more details on the general settings for subtotals, see "Matrix options tab" on page 128.
Selection type	Define which hierarchy levels to display. • **None** • **Insert all children**: only displays the child nodes of the selected filter. • **Insert lowest level elements**: displays all the lowest level elements present below the selected node, to the last level. • **Insert Parent**: displays the parent node of the selected filter. • **Advanced selection**: activates the "Script " above property. Allows the user to insert dynamic elements that cannot be inserted via standard selection. • **Insert analytical FST** : (with Account type elements inserted via items of an FST) allows the user to define an analytical FST in which both the FST items and the related accounts are displayed. For more details, see "Financial statement templates" on page 83. • **Insert synthetic FST**: (with Account type elements inserted via items of an FST) allows the user to define a synthetic FST in which only the FST items are displayed. For more details, see "Financial statement templates" on page 83. • **Control group**: (with Account type elements entered via control groups) allows the user to define the accounts with reference to a group.

Dictionary

Element	Description
	Allows the user to select an existing dictionary item. See "Dictionary Items window" on page 50.

Editability

Element	Description
Editable	Makes the filter editable.
Not editable	Makes the filter non-editable.
Protected	Makes the filter editable using the formula, the result of which is not directly editable by is saved in the database when the form is saved.
Unspecified	Default value.

Transactional form editor

Only for transactional forms with "**Use Transactional form editor** active.

Element	Description
Tab	Defines the editor tab in which to view the selected column filter. • **Excluded from the editor**: does not show the filter in any tab • **<tab>**: shows the filter in the selected tab
Column	Defines the column of the tab on which the column filter is shown (2 = second column of the tab).
Row	Defines the row of the tab on which the column filter is shown (2 = second row of the tab).
Filled columns	Defines the number of columns occupied by the filter.
Label space	Defines the dimension in percentage terms with which to show the column label in the editor.

Validation

Only for data entry forms. Allows you to set checks on the correctness of the data entered. See "Validations" on page 42.

Conditional editability expression

Only on editable parameters or parameters with protected editability. Allows you to subject the editability of the current filter to conditions and logical expressions in relation to other filters. See "Validations" on page 42.

Dividing factor

Account dimension properties. Not supported for matrices designed on the Analytical Workspace.

Element	Description
Day	Calculates the average of the amounts based on the number of days that have passed from the start of the year to the specific month, e.g. 180 days from January to June.
Period	Calculates the average of the amounts based on the number of days in the specific month, e.g. 30 days for the month of June.

Tagetik style

Only available if a Tagetik style is defined for the matrix.

Element	Description
	Opens the style selection window for the matrix elements. See "Tagetik styles " on page 230.

Formula

Element	Description
	Opens a window to write the calculation formula.
Prevailing	Makes the formula the prevailing formula in case of intersection with others.

Element	Description
Calculation logic type	• **Client**: the formula refers to dimensional elements contained in the matrix and is solved in the local client. • **Server**: the formula refers to elements of the dimensions present in other matrices or other elements of the form, the formula is solved on the server and the client receives the resulting value.

Hyperlink

Element	Description
Action list	Opens the **Action List Wizard** window to define the actions to be run when a hyperlink is selected. See "Action List Wizard window" on page 225.
Prevailing	Makes the link defined for the row/column prevail over any other hyperlinks on intersection rows/columns on the same matrix which, at runtime, could create more than one value on the same cell.
Values or Headers	Allows you to specify the part of the matrix on which the hyperlink can be inserted. • **Values and headers**: the link is available both on headers and on row/column values. • **Values**: the link is only available on row/column values. • **Headers**: the link is only available on row/column headers.

Multi-selection

Property only available from the web interface on row/column filters with counter dimension in transactional matrices of the Analytical Workspaces.

When active, this shows tick boxes for selecting multiple rows/columns when the form is run.

Hide

Element	Description
Do not hide in the presence of editable cells	Shows the hidden row/column in the presence of editable cells.
Hide row or column	
Always	Synthetically hides the row/column. In contrast to Excel's native functionality, this option also remains anchored to the element when the form is dynamic and the position can change.
Not defined	Hides the row/column without data.
To zero	Hides the row/column the amounts of which are zero.
chained to	Hides the row/column when this is subject to the display of another row/column that can in turn be hidden.
Ignore the cells that contain formulas	Only with **Undefined**/**To zero**. In the presence of cells containing formulas, the value of the cells resulting from the formula is not considered when the system assesses of whether or not to hide the row/column.
Ignore the matrix headers	Only with **Undefined**/**To zero**. Does not consider the content of the values in the matrix header areas in the decision of whether or not to hide the row/column. Enabled by default.
Do not consider cells that contain text	Only with **Undefined**/**To zero**. In the presence of cells containing non-numeric values (amount type, description, etc.), these are not considered when the system assesses whether or not to hide the row/column. ***Note**: when this option is disabled, the cells containing non-numeric values are considered non-empty cells with values other than 0.*
Also evaluate other matrices on the same row/column	Only with **Undefined**/**To zero**. In the presence of several matrices containing cells on the same row/column, this only hides the row/column if all the matrices allow it to be hidden. The outcome of this check is therefore applied to all matrices. Cells outside of the matrix are ignored.
Reference row	Only with **Chained to**. Allows you to select the row/column to which to subordinate the current row/column.

Segment

Properties of the entity and accounts dimension

Allows you to view intercompany data related to the entities of an entity node called a *segment*. This property's default value is defined in advanced form settings and used on all the tab filters set on entity dimensions, custom dimension 2 or the counterparty entity. This value can be overridden when designing the matrix.

Element	Description
from configuration	Reads and applies the values indicated in the application's general settings to the Default section of the Form tab. SeeSettings window.
none	Does not apply any filters to journals
Parent segment	The value shown includes all consolidation journals (IC eliminations, financial investment eliminations, other corrections) between legal entities belonging to the same "parent" node.
Segment	The value shown includes all consolidation journals (IC eliminations, financial investment eliminations, other corrections) between legal entities belonging to the same node.
Intersegment	Identifies the values relative to all journals referring to legal entities belonging to different segments. This function is activated ONLY if the selected node has at least TWO levels below it. Data can be read according to this method both when creating "Tab filters" and when selecting "Prompt Filter", respecting all the restrictions on the depth of the levels for the choice of "Intersegment".
Parent intersegment	Identifies the values relative to all journals referring to legal entities belonging to different segments and, starting from a level below, takes all the journals that start from the node on which the function is activated.

Data type

Element	Description
Date	Applies the date format to the row/column.
Number	Applies the number format to the row/column.
Body	Applies the text format to the row/column.
Default	Applies the data type defined in the cell properties.

Prevailing data type

When active, this makes the data type set on the cell predominant over the data type defined for the entire row/column.

Navigation

Only for parametric filters on the scenario dimension. See "Navigating scenario parameters" on page 73.

Element	Description
Previous	Selects the scenario indicated in the scenarios list as the *previous scenario* to the selected scenario. See "Navigating scenario parameters" on page 73.
Next	Selects the scenario indicated in the scenarios list as the *next scenario* following the selected scenario. See "Navigating scenario parameters" on page 73.
Reference	Selects the scenario indicated in the scenarios list as the selected scenario's *reference scenario*. See "Navigating scenario parameters" on page 73.

Period Difference

Only for parametric filters on the period dimension.

Allows the user to navigate periods, shifting forwards or backwards from the period selected when the form is run.

See "Navigating period parameters" on page 74.

Matrix filter properties table

Introduction

The properties table allows the user to manage the filter properties to customise the matrix. When several filters are selected, only the shared properties are shown and the values set on them will be applied to all the selected filters.

Reference FX rates

Element	Description
average	Average FX rate of the period in which it is calculated. These are the FX rates generally used for P&L accounts.
period average	Applies a different FX rate for every period into which the year is divided. Allows the user to perform more accurate conversions than the average FX rate.
final	Period-end FX rates. These are used for the balance sheet accounts, with the exception of the equity accounts, which are typically converted at the historical FX rate.
based on account	FX rate used for conversions and defined in the accounts table

FX rate

Currency dimension properties.

Element	Description
For matrices on the financial layer	
From Configuration	Value, displayed in brackets next to the active property, previously configured in **Default FX rate** for the new filters on the currency in the **Forms** section of CCH Tagetik 's general settings. See Valuta..
None	The Currency dimension is used as a dimension filter: only records whose currency is the selected currency will be read by the database.
Final FX rate	The records read by the database are converted into the selected currency using the final FX rate, as defined in the FX rate table.
Average FX rate	The records read by the database are converted into the selected currency using the average FX rate, as defined in the FX rate table.
Period average FX rate	The records read by the database are converted into the selected currency using the period average FX rate, as defined in the FX rate table. The **FX rate** property of the Currency dimension, as well as the equivalent property in **Reference FX rates**, does not support the **Period average FX rate** option if the form uses the standard reporting engine. Where this configuration is used, at the run form step, this option is automatically set to **Based on account**. A warning is sent and the user must correct the form configuration.
Based on account	In the accounts table, it is possible to define the FX rate to be used in the various data processing to covert the relevant amounts (**Conversion** property, with the following possible values: **None**, **Final**, **Average** or **Period average**). With this setting, the form converts records read from the database into the selected currency, using the relative FX rate, as defined in the table, for each of them.
For matrices on the analytics layer	
None	The Currency dimension is used as a dimension filter: only records whose currency is the selected currency will be read by the database.
Based on the measure	The option to be applied is read from the numerical measure configuration in which it is possible to define the desired FX rate for that measure from the web interface. (See Dataset fields).
Based on account	With this setting, the form converts records read from the database into the selected currency, using the relative FX rate, as defined in the table, for each of them. If the account dimension is not present in the dataset, the **Based on the measure** FX rate is applied.

Particular features of conversion based on account

Conversion based on account is only applicable when the dataset has the account dimension. If the "based on account" conversion type is specified and the matrix does not have the account dimension, the form is

not run. If the account has a conversion type that is not supported by the analytical matrices, i.e. it is not the average FX rate, period average FX rate or final FX rate, the **Based on the measure** FX rate type is applied.

Forms in which global filters (form filters or sheet filters) on the currency are used will have the following behaviours:

When...	Then...
the dataset does not contain the account dimension	if the matrix has no overrides on the currency dimension, the system considers the currency indicated in the global filter: • if the conversion type is **final**, **average**, or **period average**, the system applies the **Based on the measure** conversion type. • if the conversion type is **None**, then no filter is applied and only the values in the selected currency are read.
	of the matrix specifies an override on the Currency dimension, then the global filters are not applied.
the dataset contains the Account dimension	the **Based on Account** conversion type is applied, regardless of what is specified at the matrix level, save for matrix overrides.

Periodic value calculation

Period length dimension properties.

Element	Description
Automatic	The properties' behaviour varies based on the engine used for reporting. Default value. See "Types of reporting engine" on page 28. ***Note**: option used when the Period length is not specified in any of the form's filters.*
Aggregation of periodic values	Calculates the cumulated data as the sum of periodic figures. For example, the quarterly figure for the second quarter is calculated as the sum of the monthly figures read from periods 4, 5 and 6.
Net Year-to-date values	Calculates the cumulated data as the net year-to-date values. For example, the quarterly figure for the second quarter is calculated as the difference between the *year to date* figure for period 6 and the *year to date* figure for period 3.

Size

Specifies the dimension and group/hierarchy from which the filter has been taken. Not editable.

Navigation

Only for parametric filters on the scenario dimension. See "Navigating scenario parameters" on page 73.

Element	Description
Previous	Selects the scenario indicated in the scenarios list as the *previous scenario* to the selected scenario. See "Navigating scenario parameters" on page 73.
Next	Selects the scenario indicated in the scenarios list as the *next scenario* following the selected scenario. See "Navigating scenario parameters" on page 73.
Reference	Selects the scenario indicated in the scenarios list as the selected scenario's *reference scenario*. See "Navigating scenario parameters" on page 73.

Period Difference

Only for parametric filters on the period dimension.

Allows the user to navigate periods, shifting forwards or backwards from the period selected when the form is run.

See "Navigating period parameters" on page 74.

Details of FX rate properties

Description

This property allows you to convert the currency (of the entity or transaction) of every figure in the matrix into the selected currency, using the FX rate defined for the scenario/period in use. To do this, it is necessary to define an element of the Currency dimension (row/column filter, matrix filter, tab filter or form filter) and set its FX rate property with the FX rate to be used. This operation may be useful for comparing the turnover of entities with different currencies, for example.

Property details

Depending on the type of matrix, the FX rate property has several available options:

Matrix type	Options
Financial layer	• **From configuration**: default value, which appears between brackets. The FX rate is determined by the **Default FX rate for the new filters on the currency** parameter in the general form settings in CCH Tagetik*Setup & Admin > Settings > Forms tab > Default section). Navbar > Settings > Forms tab > Default section)* See Settings window. • **None**: only reads records with the selected currency from the database. • **Final FX rate**: the records read by the database are converted into the selected currency using the final FX rate, as defined in the FX rate table. • **Average FX rate**: the records read by the database are converted into the selected currency using the average FX rate, as defined in the FX rate table. • **Period average FX rate**: the records read by the database are converted into the selected currency using the period average FX rate, as defined in the FX rate table. **Based on account**: in the accounts table, it is possible to define the FX rate to be used in the various types of data processing to covert the amounts (see Anagrafica conto). With this setting, the form converts records read from the database into the selected currency, using the relative FX rate, as defined in the table, for each of them. ***Note**: The FX rate property of the Currency dimension (as well as the equivalent property in Reference FX rates) does not support the **Period average FX rate** option if the form uses the standard reporting engine. In this case, when the form is run, the system changes this option, setting it to **Based on account**. The user receives a message inviting them to correct the form's configuration.*
Analytical layer	**None**: the Currency dimension is used as a dimension filter. Only records whose currency is the selected currency will be read by the database.
	Based on the measure: the option to be applied is read from the numeric measure. The desired FX rate for measures can be defined from the web interface. See Dataset fields
	Based on account: with this setting, the form converts records read from the database into the selected currency, using the relative FX rate, as defined in the table, for each of them. If the account dimension is not present on the dataset, the **Based on the measure** behaviour will be applied.

Matrices on multiple workspaces

If a sheet contains both a matrix on a financial workspace and a matrix on an analytical workspace, when a currency filter is applied to several matrices, such as a tab filter or form filter, the available options are the same as for the financial workspace. In order for the filter options to be also applied to the multidimensional matrix of the analytical workspace, the global filter options are converted as follows:

- none -> none
- final FX rate, average FX rate, period average FX rate, based on account -> based on measure

For more details on the logics that regulate these conversions, see "Currency conversion algorithm" on page 28.

Details of Reference FX rate properties

Description

By default, the system converts the amounts in the form using the FX rate defined in the table for the scenario/period applied to the cell containing the data (the FX rate type is defined on the row/column filter properties, or on the matrix filter, tab filter or form filter).

To define a different FX rate from the one in the table, it is possible to edit the reference FX rate property and set FX rate type of the scenario/period in use or another scenario/period. Data for a scenario/period can be used after being converted with different FX rates, for a "what if" type analysis.

> ### Example
>
> To carry out a comparison of the data of an account on the scenarios *2011 Actual* and *2012 Actual*, it is possible to calculate the value of the *2012 Actual* data "as if" it were subject to the *2011 Actual* FX rate. To do this, when managing the *2012 Actual* properties, *2011 Actual* and the FX rate type to be applied are selected as the reference FX rate.

For details on FX rate types, see "Reference FX rates" on page 112.

Details of Periodic value calculation properties

Description

This property allows you to specify the calculation method according to which CCH Tagetik transforms periodic figures into cumulated progressive figures.

Calculation options

The available calculation options are described in detail in "Periodic value calculation" on page 113.

Calculation options syntax: standard or in-memory reporting engine

Details of the calculation syntax of the periodic value in relation to the various reporting engines:

<table>
<tr><th>Element</th><th>Description</th></tr>
<tr><td>Standard reporting engine</td><td>The system identifies the process from the scenario/period pair of the figure to be converted and, based on the FX rate specified on the process, defines the calculation method for the periodic value and the relative algorithm.
<table>
<tr><th>If...</th><th>Then the periodic value is calculated as...</th></tr>
<tr><td>the process is set to Average FX rate</td><td rowspan="2">Net Year-to-date values</td></tr>
<tr><td>the scenario/period is not related to any process</td></tr>
<tr><td>the process is set to Period average FX rate</td><td>Aggregation of periodic values</td></tr>
</table>
IMPORTANT: if a single scenario includes periods belonging to processes with different FX rate configurations, in the presence of cumulated period lengths, the standard reporting engine may return incorrect results.
For more details on the FX rate calculation algorithm, see "Currency conversion algorithm" on page 28.</td></tr>
<tr><td>In-memory engine</td><td>The system identifies the FX rate to be applied to the individual cell and defines the periodic figure calculation method, regardless of where this has been derived overridden in the form, from the account or the process.
<table>
<tr><th>If...</th><th>Then the periodic value is calculated using the method:</th></tr>
<tr><td>the process is set to Final FX rate or Average FX rate</td><td>Net Year-to-date values</td></tr>
<tr><td>the process is set to Period average FX rate</td><td rowspan="4">Aggregation of periodic values</td></tr>
<tr><td>the process is set to Based on account</td></tr>
<tr><td>non conversion is applied and Periodic value calculation is set to Automatic</td></tr>
<tr><td>the period length is indicated as a form filter or tab filter, and the multidimensional matrix on the Analytical Workspace is influenced by this filter</td></tr>
</table></td></tr>
</table>

***Note**: the **Automatic** option is applied when the period length is not specified in the form.*

Tagetik style property details

Description

This property allows you to define a style on the selected row/column filter. When a style is set, the way in which it is applied to the filters will depend on the level of the tree of the sections of the matrix in which it is set. In fact, the style will be applied starting with the highest level of the tree. In other words, the sections belonging to a given level of the tree inherit those of the higher levels, unless another style has explicitly been defined for them.

This property is only visible in the presence of a Tagetik style defined for the matrix.

By default, when the form is run, the style set on the last level of filters, or the style set on an intermediate filter that is then propagated on the last level, is applied.

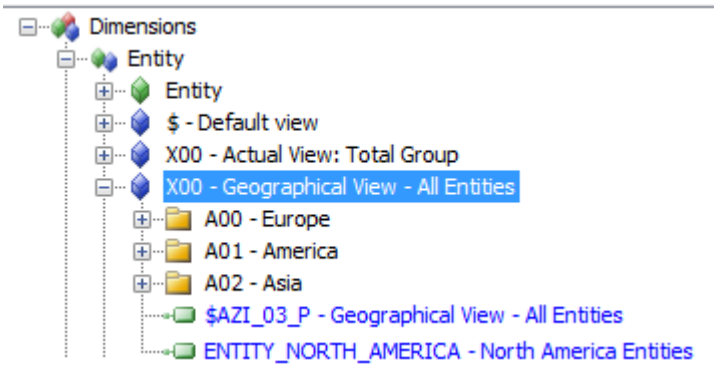
In the presence of subtotals generated at runtime, these only inherit the style if they are subtotals of the last level and no style has been set on that level.

Specific features of scenario and period parameter properties

Using parameters

All parameters present in Tagetik (both system parameters and custom parameters) can be used within the Matrix Wizard to make the matrices dynamic.

In Tagetik, there is a parameter for every dimension and hierarchy. Parameters can be selected from the dimensions tree and are distinguished from the other elements by the colour blue. Both shared parameters and ad-hoc parameters can be chosen in the dimensions tree.



Once a parameter from the dimensions tree has been chosen, it can be inserted into the matrix as a row filter, a column filter or a matrix filter. Once inserted into a matrix, a parameter behaves the same way as a fixed value. They can also be nested into other elements and the properties that can be modified in the properties table are the same as for a fixed value (non-parametric) element.

For dynamic scenario and period parameter elements, there are additional properties that are activated in the properties table when inserted in the matrix as row filters or column filters. We will look at these properties in the next sections.

The dimensions tree used to define the Form, Tab and Matrix filters does not show Process-type parameters or dimensional parameters on the period that allow 'OR' operations, as these are not supported by the forms.

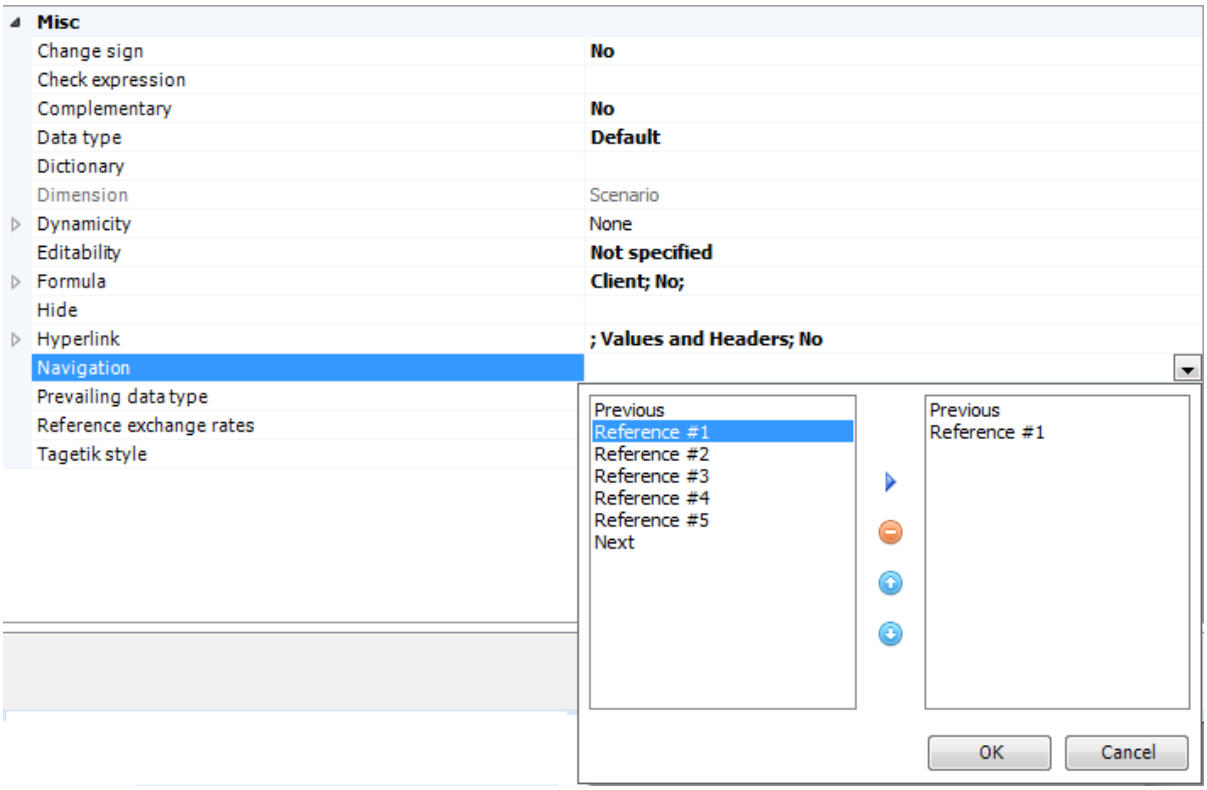
Navigating scenario parameters

Navigating scenarios allows the user to create dynamic forms for comparing data saved in the system in different scenarios. When scenarios are defined in Tagetik, temporal links and comparative links can be created between them.

For example, if I defined a scenario called 2012ACTUAL, which collects actual data for the year 2012, I could create a "Previous" type link to scenario 2011ACTUAL and in turn create a "Next" type link between scenario 2011ACT and 2012ACT.

In addition to temporal relationships, comparative relationships between different scenarios can also be defined. For example, a scenario called 2012BDG, which is a forecast for the year 2012, must be compared with the actual data for that year, e.g. 2012ACT. In Tagetik, up to 5 reference-type relationships between scenarios can be defined.

These chains of scenarios can be used within the Tagetik reporting client when scenario-type parameters are selected. When a scenario-type parameter is inserted in a row or column filter, the "Navigation" property is activated in the properties table.

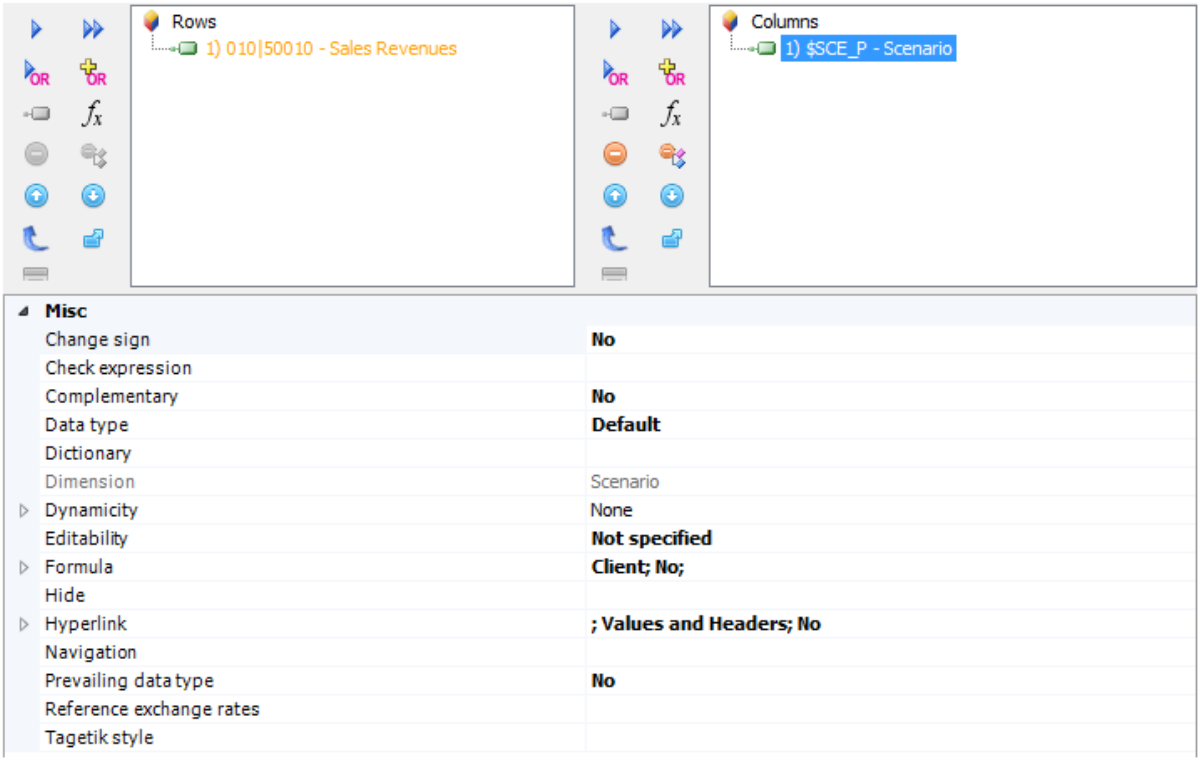


In this window, we need to define the navigation chain for our scenario. Navigation starts from the element chosen by the user when the form is run. For example, if a user chooses scenario 2012ACT and a "Previous" navigation has been defined, scenario 2011ACT will be used.

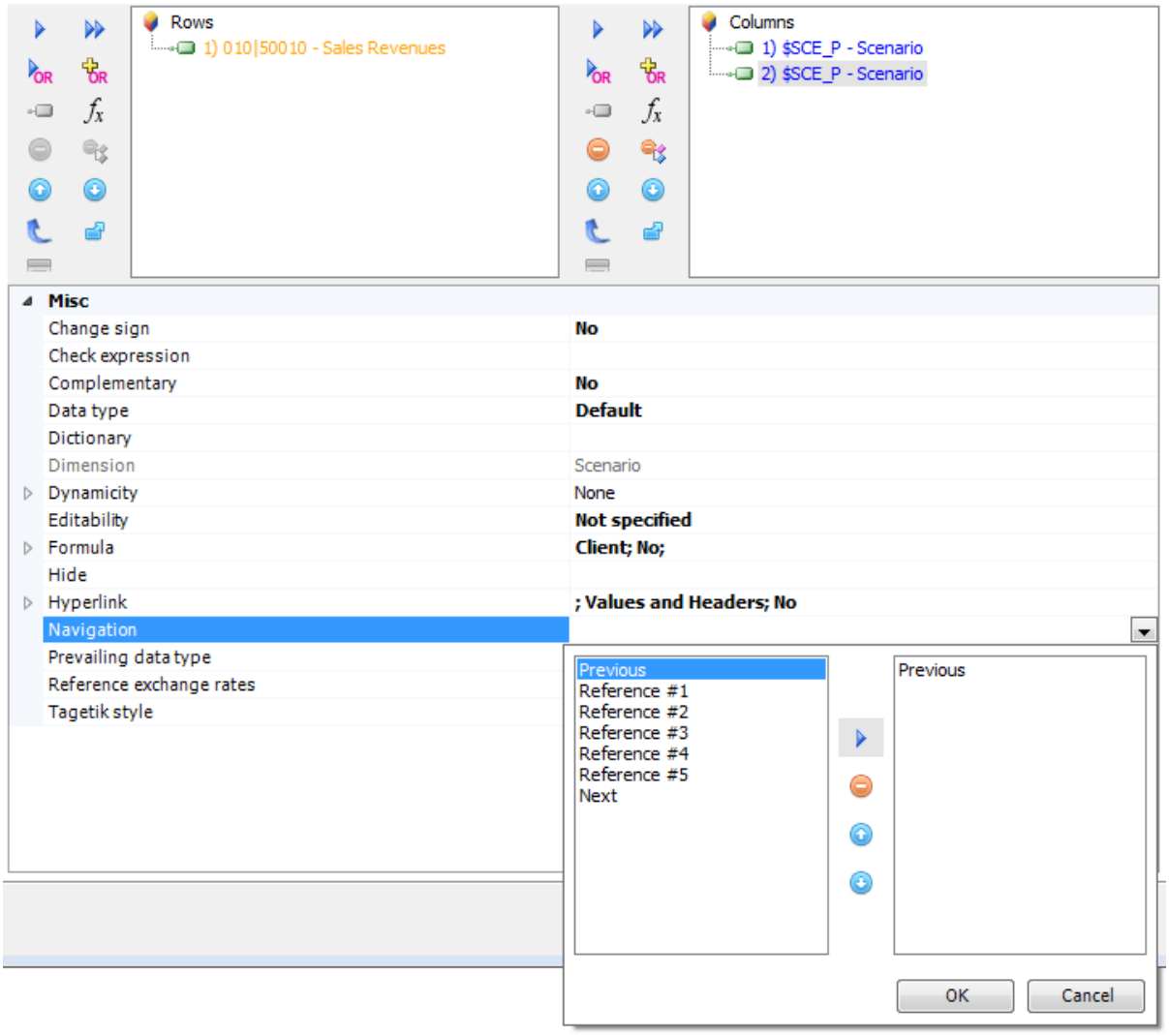
Scenarios are navigated by choosing an element from the list and clicking the button. The element can be removed using the button. The order of the various scenarios can be changed using the and buttons. Several elements can be inserted to create navigation chains, e.g. the Previous --> Previous scenario or the Previous --> Reference #1 scenario.

The order in which the elements are added to the list is very important as the system navigates the scenarios according to the order in which they are added.

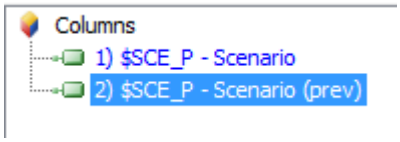
Let's look at an example in which a dynamic matrix is created to compare two actual data scenarios. First, we need to insert a parametric scenario within our matrix. This element will be the element chosen by the user, so we will not set any navigation for the element.



At this point, we must insert the second scenario, which will be the comparison scenario. In this column, we want to see the data for the scenario before the one selected by the user. So let's insert a parameter with the same code as the one inserted previously and set the navigation for the element.



Once navigation has been set for a parameter, the text in the filter window is changed to indicate which chain has been set.



At this point, the dynamic matrix is ready and can be used. Navigation chains between scenarios can be changed within the web interface without having to change the structure of the created form.

Navigating period parameters

The mechanism for navigating between period-type parameters is similar to navigating between scenarios. Offsets related to the parameter chosen as the filter can be defined between dynamic periods.

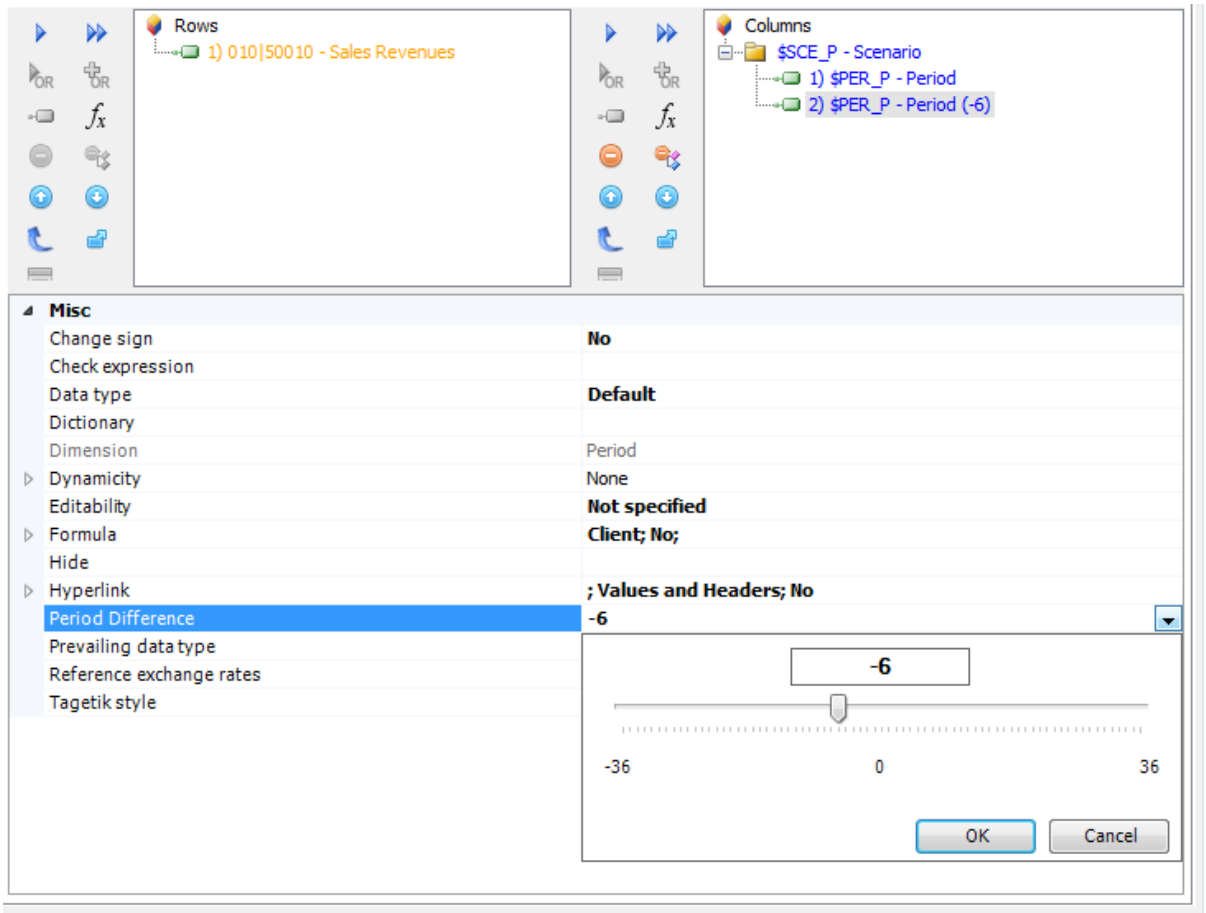
For example, I want to define a column which, depending on the chosen period, shows me data relative to 6 months earlier. If i choose period 12, the column will be defined with period 6, if I choose period 11, it will be defined with period 5, and so on.

A parameter of this type can be set using the "**Period Difference**" property present in the attributes table when we insert a period-type parametric element in a row or column filter. When we set a period difference, we must set the number of periods back or forward that we want to shift in relation to the chosen period. If the difference is greater than the number of periods in a scenario, the system obtains the periods from the scenario set as the previous or next scenario.

La navigazione del periodo non è totalmente supportata nell'AW, in particolare il sistema non ricava correttamente il valore della dimensione Scenario se, a causa dello scostamento maggiore del numero di periodi, è necessario usare lo scenario impostato come precedente o successivo. Questo si riscontra:

- nell'editabilità delle matrici multidimensionali
- nell'editabilità delle matrici LBL (Line By Line)
- nei dati estratti dalle matrici LBL (Line By Line)

Differences can be inserted directly within properties or using the scroll bar shown when properties are opened. The chosen difference is shown in the title of the edited parameter within the row or column filter.



Search window

Excel > CCH® Tagetik > Open items > double click matrix > Row/column filters or Matrix filters >

Purpose of window

This window allows you to search for elements using textual criteria, in the table of the dimensions of the environment that can be used in forms.

Options in window

Element	Description
Search bar	Allows you to insert the text to search for among the filter codes and/or descriptions. ***Note**: the text can contain the special characters "?" and "*".*
Type	Limits the search • **Code** • **Description** • **Code and description**
Regular Expression	Enables the search for regular expressions.
Case sensitive	Enables distinction between uppercase/lowercase.
Results display area	Displays the search results and allows you to select them to identify them in the dimensions tree.

Choose the dataset window

***IMPORTANT**: this window is only shown if the system used Analytical Workspaces and if at least one dataset has been set on them.*

Purpose of window

This window allows you to choose the dataset on which to define the matrix. The system shows the default Financial Workspace dataset and the list of all the available datasets for Analytical Workspaces. Once the dataset has been selected, the Matrix Wizard will open, showing the data of the dataset in the filter setting tabs (see "Matrix Wizard - Window structure" on page 109).

Options in window

Element	Description
Line item management	This makes the matrix transactional and allows you to create a Line by Line transactional form. See "Form matrices" on page 100.

Matrix options tab

Excel > CCH® Tagetik > Design > select form > Matrix > Matrix options

See "How to access the windows" on page 9 for alternative paths.

Purpose of the tab

This tab allows you to manage the following:

- form formatting in areas of the matrix
- how empty matrix cells are saved in the database
- how to dynamically open rows/columns
- Break back options

Sub-tabs

Element	Description
Layout options	Allows you to do the following: • matrix formatting • matrix subtotal display mode • dynamic matrix display mode • how matrix row and column data are saved in the database
Calculation options	Allows you to manage elements related to optimisations and how matrix data are saved in the database.
Edit amounts	Allows you to manage the opening of new rows in data entry mode, for transactional forms.
Form - Break Back	Allows you to manage break back mode at the matrix level.

Note: *Calculation options, Edit Amounts, Form - Break Back are not support on transactional forms or multidimensional Analytical Workspaces.*

Options in the Layout options tab

Element	Description
Code	Allows you to edit the matrix code.
Formatting type	Allows you to enable/disable the use of Tagetik styles. See "Tagetik styles " on page 230. • **None**: the matrix is formatted using Excel's formatting functionalities. • **Apply Tagetik style**: the matrix is formatted using the Tagetik style associated with the form.
Subtotals	Allows you to manage the subtotal display mode in dynamic matrices. • **Subtotal before the detail**: shows the subtotal value after the total and before the lowest level elements. • **Deploy the lowest elements then the nodes of the same level**: in the presence of nodes to which both child nodes and elements of a dimension are related, this allows you to manage the order in which the form rows are generated. • **Group subtotals**: groups subtotals using Excel's group functionality. When enabled, every node and their children are grouped into an Excel level. ***Note****: transactional forms do not support the management of subtotals.*
Matrix	Allows you to change the matrix display mode in navigation and data entry. • **Do not repeat duplicate information on every row/column**: in the presence of dimensions nested on rows/columns, this allows you to prevent the description of those dimensions from being repeated in every row/column header, so it is only shown on rows where one of the nested dimensions changes. • **Show zero values if the cells aren't defined**: manages how empty cells are displayed. ◦ **Not defined**: items for which no value exists in the database are shown as empty cells in the form, ◦ **To zero**: items not defined are saved in the database and shown in the form with value 0. ***Note****: When this second option is used in combination with the form option "**Enable Save zero values**", the system saves a 0 in the database for every cell left empty when the form is submitted.* • **Allow matrix splitting**: no longer in use but present for compatibility with the client's previous versions. • **Show Documentation link**: allows you to create hyperlinks to web pages, to insert additional information on elements of dimensions. The links can only be configured in the web interface. The inserted link is shown in navigation and data entry modes inside the row/column header. ***Note****: transactional forms do not support the management of links.*

Element	Description
Rows and columns	This allows you to configure the way in which the dynamic matrix expands when the form is opened in navigation or data entry mode. • (Not enabled for transactional forms and Multidimensional Analytical Workspaces)**Maximum Number of Rows/Columns**: defines the maximum number of rows and columns that the system creates when it generates a dynamic matrix (**0**: unlimited). • **Row/column expansion mode**: allows you to configure way in which the rows or columns of a dynamic matrix can be generated ◦ **Shift cells down/right**: creates cells by moving existing cells for rows down and moving existing cells for columns to the right. ◦ **Entire row/column**: adds a row/column to the sheet. ◦ **Don't expand**: disables row/column expansion. • **Autofit columns**: allows you to set the column width. ◦ **None**: the column width is not adapted to the content. ◦ **From template**: the column width corresponds to the width set in the form design. ◦ **By value**: the column width adjusts to its content.
Row/column behaviour preview area	Shows the row/column expansion mode.

Options in the Calculation options tab

Element	Description
Enable overriding	• No longer in use but present for compatibility with the client's previous versions.
Save the original data and currency data separately	Allows you to save the original data and the currency data separately. By default, the system converts original currency data according to the scenario/period FX rate, then saves the result over the original data in the entity currency.
Exclude from evaluation of zero value sheets	Allows you to exclude a matrix from pruning (see "Pruning" on page 115). By default, with form filters with *pruning* set to *not defined/to zero* and matrices in the template with values of the same filters to zero/not defined, the system hides the sheet when the form is run.
Keep the IC amount in synch with the Gross amount	
Active	Automatically updates the gross figure every time the intercompany data are edited (e.g., if the gross figure is 100 and the intercompany figure is changed from 70 to 80, increasing it by 10, the system saves the gross figure as 110).
Not active	Does not update the gross figure every time the intercompany data are edited (e.g., if the gross figure is 100 and the intercompany figure is changed from 70 to 80, the system leaves the gross figure as 100).
Unspecified	The save mode is determined by the value of **Copy intercompany to gross amounts** in the advanced settings of the data entry rules configuration (*Setup & Administration > Data Entry & Submission Rules* > Amount & Intercompany rules) (*Navigation Panel > Processes & Workflows* > Amount & Intercompany rules). See Le regole di data entry.
Hints database	
Enable	Enables the use of database hints in queries run when the form is run.
Enable calculation on data	Enables the use of database hints in queries run when the form is run based on the content of the data tables.
Disable	Disables the use of database hints in queries run when the form is run.
Automatically	The system automatically activates/deactivates the use of database hints in queries run when the form is run.
Use optimization in report data entry	
Yes	When set to Yes, the matrix loading is optimized for calculating editability in the financial workspace, allowing for more efficient handling of high cardinality of accounts (or account/category combinations), and improving performance when dealing with thousands of accounts.
No	When set to No, the matrix loading is not optimized for calculating editability.
From configuration	When set to 'From configuration', the optimisation options are determined by the value "Use optimization in report data entry" in the System Settings window, under the Form -> Data Entry tab.

Options in the Edit amounts tab

Element	Description
"Instead of asking the user for a filter, automatically open the first one available"	(Only valid in forms that contain dynamic nodes with the advanced selection option **Allow new rows/columns wizard in data entry** enabled) When the user clicks on +, a new row/column opens on the first available element of the filter.
In the case of transactional forms, ask how many rows or columns to open	In a transactional form in which the **Allow new rows/columns wizard in data entry** option is enabled on the advanced selection options of the Counter dimension, it allows the following behaviour: • If active, and the form is run in data entry mode, when the user clicks +, it allows new rows to be opened manually. • If not active, when the form is run in data entry mode, the system opens a fixed number of rows specified on the attribute of the **Additional rows to open** matrix counter.
Enable Editability On Demand	(Only visible with transactional matrices) When the form is run in data entry mode, the cell editability calculation is not performed at the start, but afterwards, when the cell is clicked. See "Editability on Demand" on page 330.

Options in the Form - Break Back tab

Allows you to override break back mode on the matrix.

Element	Description
Break back dimensions	• **Default**: applies the break back mode specified in the data entry rules related to the process. • **break back deactivated**: deactivates break back, so the form's cells which correspond to a node are not editable. • **targets**: enables break back on custom dimensions only.
Save on the reference element	With the break back functionality active, indicates whether, in the data entry step from a form, the values inserted on the nodes must always be attributed to the reference lowest element. • **Default**: applies the break back mode specified in the data entry rules related to the process. • **Yes**: the values inserted on the nodes are attributed to the reference lowest element. • **No**: when the values that are inserted on the node can be distributed over lowest level elements related to the same node in proportion to the old value, they are automatically attributed to the nodes' reference lowest element.

Transactional form editor tab

Excel > CCH® Tagetik > Design > open form > Matrix > Transactional form editor

See "How to access the windows" on page 9 for alternative paths.

***IMPORTANT**: this tab is only shown for transactional forms, when at least one element has been added to the local column filters tree in the Row/column filters tab.*

Purpose of the tab

This window allows you to define the characteristics of the detailed data entry window of transactional forms. In particular, it allows you to manage the following:

- Define the window's tabs, give them labels and set the number of rows and columns.
- Enable buttons to manage the form's rows in the data entry phase.
- Define the dimensions in terms of details window pixels.

The data entry window thus defined allows you to run data entry much more flexibly.

Once the characteristics of the details window have been defined from this tab, the user must define the tab of the window in which every column filter should be shown, and the column and row by setting the **Transactional form editor** column property (see "Table of row/column filter properties" on page 112).

Layout management options

Element	Description
Use Transactional form editor	Enable data entry management via the detail window.
Show buttons	
Add	In the data entry step, this shows the button on the row, allowing you to add a row below the selected row.
Edit	In the data entry step, this shows the button on the row, which opens a detail window to edit the data on the row. See "Transactional data entry details window" on page 289.
Delete	In the data entry step, this shows the button on the row, allowing you to delete the selected row.
Editor dimensions	
Width	Allows you to define the width in pixels of the detailed data entry window.
Height	Allows you to define the height in pixels of the detailed data entry window.

Layout management buttons

Element	Description
Add tab	Allows you to create a new tab in the window, define its label and number of rows/columns.
Remove tab	Removes the selected tab.
Rename tab	Renames the selected tab.
Preview	Shows a preview of the detail window. See "Preview window" below.

Tab management tabs

Element	Description
Rows	Defines the number of rows of the detail window.
Columns	Defines the number of columns of the detail window.

Preview window

Excel > CCH® Tagetik > Design > open form > Matrix > Transactional form editor > Preview

See "How to access the windows" on page 9 for alternative paths.

Purpose of window

This window allows you to view a preview of the final layout of the detailed data entry window of transactional forms (see "Transactional form editor tab" on the previous page).

Dynamic matrices

What they are

In matrices, it is always possible to change certain properties of filter nodes to make them dynamic. Thus, it is possible to take account of how the composition of the nodes changes over time without having to edit them in the matrix. These matrices have a variable row and column structure, which is calculated at runtime when the form is run. The system recovers the indicated node, considering all elements contained within it when the form is run.
For example, in a matrix with an account of the sales of products over the month, creating a dynamic matrix allows you to retrieve, when the form is run, dimensional data on the desired products and period without having to define them manually as filters in the matrix.

It is also possible to set only some of a matrix's elements as dynamic, even if they belong to the same dimension.

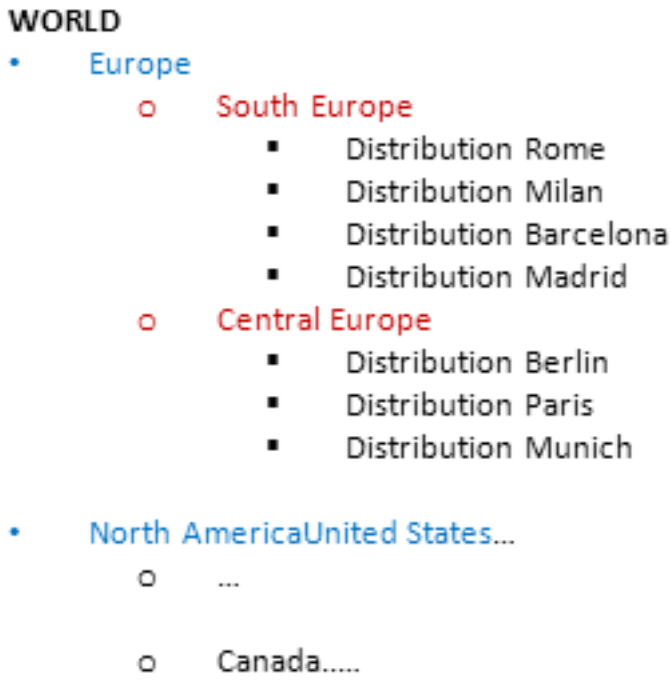
***IMPORTANT**: in dynamic matrices, elements inserted as dynamic row/column filters have Insert lowest level elements as their default selection type on creation. This setting can be changed.*

Selection type

This property allows you to define the dynamic element that must be shown depending on the various levels of the selected hierarchy.

Example

This example shows a hierarchy on the entity dimension that groups entities by geographical area, where the subsidiaries are lowest level elements (leaf elements) while regions are aggregations of elements (node elements).



Dynamic forms can be constructed to see several entities based on the selected node. If the node inserted as a row filter is *Europe*, the following results are obtained based on the selected selection type:

Advanced selection	Outcome
Insert Parent	At runtime, the parent node of the element indicated as the filter is inserted, so the matrix only shows the row corresponding to the node of the *WORLD* hierarchy containing the total of all nodes below it. 2012 Budget Dec Budget View - All Entities 3.306.274.160
Insert children	At runtime, all the rows corresponding to the child nodes of the node indicated as the filter are inserted, so the matrix only shows the rows corresponding to the *South Europe and North Europe* nodes, containing the relative subtotals. 2012 Budget Dec South Europe 75.976.479 Central Europe 59.611.203
Insert lowest level elements	At runtime, all the lowest level elements of the node indicated as the filter are inserted, so the matrix shows the rows corresponding to the lowest level elements, and therefore *Distribution Rome, Distribution Milan, Distribution Barcelona, Distribution Madrid, Distribution Berlin, Distribution Paris and Distribution Munich.* 2012 Budget Dec Distribution Barcelona * 57.522 Distribution Rome 83.492.435 Distribution Milan * Distribution Madrid 26.379.558 Distribution Berlin 38.192.537 Distribution Paris 21.418.666 Distribution Munich *

***IMPORTANT**: If advanced selection is applied to a hierarchy of time-dependent elements, the scenario and period must be determined unambiguously. The scenario and period elements must therefore be inserted either in the matrix, in the filters themselves (row or column) but above the dynamic elements, or in the form filters.*

Creating matrices

Introduction

Setting up matrices means creating a structure in which to define the data managed by the form. Once that structure has been created, it is necessary to define the database data that must be managed by the matrix. To do this, it is therefore necessary to set *filters* on the database data at several configuration levels. Matrix filters define which data are shown in a specific matrix. Form filters and tab filters are applied to all matrices present in the individual form or sheet, respectively. This module shows how to create matrix elements. For more details, see "Matrix filters" on page 148.

Define the matrix starting cell

Before creating a matrix, it is necessary to select a cell in the work desktop of the spreadsheet that will become the first cell at the top-left of the matrix table.

***IMPORTANT**: Do not select cell A1 of the work desktop to allow any information on the form to be displayed at runtime. Choose the starting cell in such a way that the created matrix never overlaps with other matrices, especially in the presence of dynamic matrices: overlapping could cause errors and lead to data being incorrectly displayed.*

The matrix can always be moved inside the template.

Create/modify a matrix with Financial Workspace data

Excel > CCH® Tagetik > Design > select form > Matrix

See "How to access the windows" on page 9 for alternative paths.

1. In the **Choose the dataset** window, click **$CPM-Financial Workspace**.
2. Click **OK**.

Once the matrix has been created, the system assigns a default name (e.g. Matrix00, Matrix01...), which can be edited in the matrix wizard by the **Matrix options** tab in the **Code** field.

Create/modify a matrix with Analytical Workspace data

***IMPORTANT**: only with the Analytical workspaces module active and at least one workspace configured and online. Offline linked Analytical Workspace forms will not be accessible in either data entry or navigation mode.*

Excel > CCH® Tagetik > Design > select form > Matrix

See "How to access the windows" on page 9 for alternative paths.

1. In the **Choose the dataset** window, click the desired dataset.
2. (Only if an Analytical Workspaces dataset is selected) To create a transactional matrix, click **Transactional**.
3. Click **OK**.

Once the matrix has been created, the system assigns a default name (e.g. Matrix00, Matrix01...), which can be edited in the matrix wizard by the **Matrix options** tab in the **Code** field.

Create a data entry matrix for the user

Excel > CCH® Tagetik > Design > open form > open or create matrix > Row/column filters

For a matrix to be a data entry matrix, some of its cells must be made available for data entry for the user, i.e. they must be made editable. To do this, once the desired matrix has been created, do the following:

1. In the **Row/column filters** tab, set the desired row/column filters (see "Set a filter on rows/columns" on page 152).
2. Select the row/column filter to be made editable and, in the filter properties table, set the filter's editability property to **Editability**.

***IMPORTANT**: matrices whose fields, with **Protected** editability, are fed by formulas or queries that calculate an overall figure based on the reading of data existing in the database, are not considered data entry matrices. For more details, see "Formula filters" on page 159 and "Query" on page 251.*

Managing matrices

Open a matrix to edit it

Excel > CCH® Tagetik > Design > select form > double click the cell of the matrix in the work desktop

See "How to access the windows" on page 9 for alternative paths.

To edit a matrix recalled via a link, edit the source matrix directly. See "Create a link to a matrix" below.

Create a copy of a matrix

Excel > CCH® Tagetik > Open items

See "How to access the windows" on page 9 for alternative paths.

This procedure describes how to create a matrix with attributes similar to those of an existing matrix, by creating a copy of the existing matrix. A matrix can also be copied to several templates or forms. After a matrix has been created, it can be edited.

1. In the left-hand panel, select the matrix to be copied, then click .
2. In the work desktop, select the development cell of the new matrix and click .
3. Enter the name of the new matrix.
4. Click **OK**.

Create a link to a matrix

Excel > CCH® Tagetik > Open items > select matrix

See "How to access the windows" on page 9 for alternative paths.

In Excel, it is possible to create links to existing matrices in other forms or another template of the same form.

1. Click .
2. In the work desktop, select the development cell of the new matrix.
3. Click .
4. Enter the name of the matrix called up.

   ***Note**: the matrix recalled via the link cannot be opened in design mode or edited. To edit that matrix, the source matrix must be edited.*

Change the position of a matrix

Excel > CCH® Tagetik > Open items

See "How to access the windows" on page 9 for alternative paths.

1. Select the desired matrix. In the work desktop, the system shows the matrix in a box.
2. Position the cursor on one of the box borders and, when it becomes , drag the matrix to the desired position.

Delete a matrix

There are several ways to delete matrices.

- *Excel > CCH® Tagetik > Design > select form > right click a cell of a matrix on the work desktop > Tagetik > Delete matrix*
- *Excel > CCH® Tagetik > Open items > select matrix > right click > Matrices > Delete matrix*
- *Excel > CCH® Tagetik > Open items > select matrix >*

  ***Note**: deleting a matrix recalled via a link will not delete the source matrix. However, when the source matrix is deleted, the recalled matrix will also be deleted and marked as . The form containing it becomes invalid.*

Design: filters

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Form filters

What they are for

Form filters define the data displayed for all form template matrices. It is therefore useful to define a form filter when there are several matrices and the data to be shown in the form refer to a specific element of a dimension. For example, if the form only displays data related to the *Original* amount type, it is useful to define the *Original* amount type as a form filter.

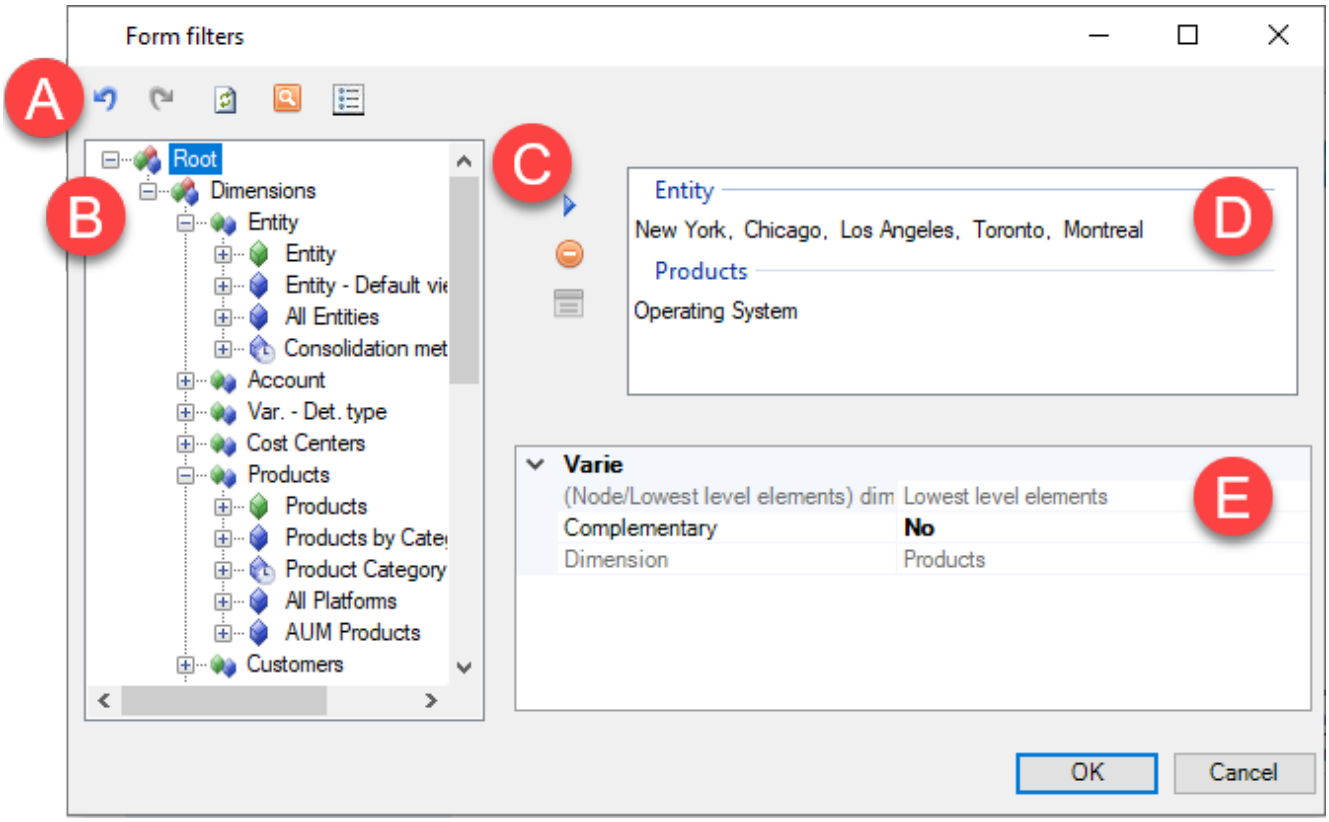
Form filters window

Excel > CCH® Tagetik > Design > select form/create new form > Form filters

Purpose of window

This window allows you to set filters on forms. These filters are elements and/or nodes of the dimensions tree that are added and modified in the form filters definition area and determine the data managed in the form.

What it looks like



Element	Description
A	Action buttons in all windows. See "Configuration window structure" on page 16.
B	Dimensions tree
C	Buttons specific to the window
D	Form filters definition area
E	Table of properties of the selected element in the definition area

Window sections

Element	Description
Dimensions tree	Show the list of available dimensions. • system dimensions (scenario, period, account, entity...) • custom dimensions (customisable, up to a maximum of 5. E.g. products, cost centres...) • specific reporting dimensions (e.g. period length)
Form filters definition area	Show the list of filters currently set on the form.
Properties table	Allows you to manage the editable attributes of the selected dimension or filter.

Buttons in window

Element	Description
	Adds one or several elements or nodes selected in the dimensions tree to the form filters definition area.
	Only with filters with the OR function already defined. Allows you to remove one or more elements from filters defined as OR.
	Removes the selected filter.

Properties table

Element	Description
	Expands a set of details which change a filter property.
	Expands the list of possible values for the property.
	Opens the property configuration window.

For details of the possible options in relation to properties, see "Form filter properties" below.

Form filter properties

List of form filter properties

***Note**: the list of properties displayed varies depending on the type of dimension used as the filter.*

Element	Description
FX rate	For details on this property, see "Details of FX rate properties" on page 121.
Reference FX rates	Allows you to indicate a different FX rate from the one defined in the table, for the entity for the scenario/period in use, to be used in the form navigation step.
Complementary	This allows the user to exclude a specific filter or node from a node of filters in a hierarchy. Thus, the system displays the entire hierarchy of the filter's parent node or excluded node, except for the specific filter or node. ***Note**: the additional filter is identified by the prefix <NOT> and works on hierarchies but not groupings.*
Size	Indicates the dimension to which the selected element belongs. Non-editable property.
Dimension (node/lowest level element)	Indicates whether the filter is a node or a lowest level element.
Dividing factor	Allows you to set the factor for the calculation of the average amount for a defined number of days or by period.
Periodic value calculation	Allows you to define the calculation of cumulated data as the sum of period figures (e.g. the quarterly figure for the second quarter is calculated as the sum of the monthly figures read from periods 4, 5 and 6), or as the net year-to-date values (e.g. if the *period length* is *monthly*, the quarterly figure for the second quarter is calculated as the difference between the values for period 6 and the values for period 3).
Comparison condition	It allows the user to define a condition for comparison, or an operator and a value with which to compare the filter value.

Setting form filters

Excel > CCH® Tagetik > Design > select form/create new form > Form filters

Set a form filter

1. In the dimensions tree, select one or more elements of the desired dimension to be used as filters, then click next to the definition area, or drag them into the area: details of the reference dimension and the dimension elements will appear in the definition area.
2. Set the filter properties as desired.
3. Repeat the previous step to add other elements to the dimension or create filters related to other dimensions.

Tab filters

What they are for

Tab filters have an effect on all matrices included in the template. It is useful to define a tab filter when you want to manage several dimension elements of the same type and view one sheet per element in the form. For example, to view data for entities *A00*, *A01*, *A02*, if these entities are set as tab filters, the form will display one sheet for each entity.

The generated sheets are named as defined in the *Generate sheet names according to* setting in the general form options (see "Generate sheet names according to" on page 57).

Tab filters on forms with several templates

In forms with several templates, several tab filters can be set for each template. When the form is run, the system creates a copy of the template for every tab filter set. For example, in a form with the templates *PL - Profit & Loss* and *BS - Balance Sheet*, applying the tab filters on entities *A00*, *A01*, *B00* for *PL - Profit & Loss* and applying the tab filters on nodes *X04*, *X05* and *X06* (of hierarchy *X00-Departments* defined on the cost centre custom dimension) to *BS - Balance Sheet*, will mean that the system generates a form with 6 sheets.

Tab filters with formulas

It is possible to create tab filters containing formulas. The system creates a sheet containing the result of the formula.
As is the case for any other formula, it is possible to define a dictionary item to show a description for the sheet containing the values. When several sheets are shown in one template, it may be useful to create a sheet containing totals or statistics on data managed in the other sheets. To do this, a formula can be set as a tab filter argument.

> **Example**
>
> In a form with tab filters on entities A00, A01 and B00, by creating a fourth filter with the formula {1} + {2} + {3}, when the form is run the system will display the three sheets with entity filters and a fourth sheet containing the total of the amounts of the three entity sheets.

Setting tab filters

Excel > CCH® Tagetik > Design > select form/create new form > Tab filters

Set a tab filter

1. In the dimensions tree, select one or more of the dimension's elements to be used as tab filters, then click next to the definition area, or drag them into the area: details of the reference dimension and the dimension elements will appear below the node.
2. Set the filter properties as desired.

3. Repeat the previous step to add dimension elements or nodes from other dimensions.

   ***Note**: when the form is run, the system assigns a name to each sheet, following the rules established in the form options; see "Form options window" on page 55.*

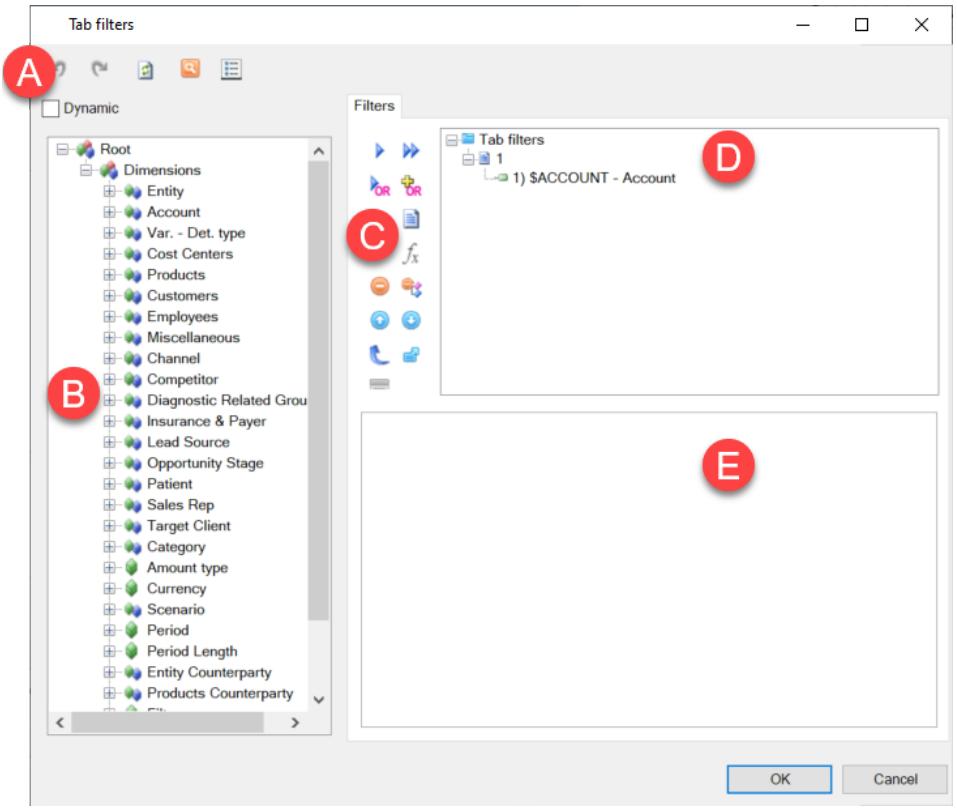
Tab filters window

Excel > CCH® Tagetik > Design > select form/create new form > Tab filters

Purpose of window

This window allows you to set filters applied to the form to subdivide data into separate sheets. These filters are elements and/or nodes of the dimensions tree that are added and modified in the local filters tree and determine the number of sheets in the report and the data shown in each of them.

What it looks like



Element	Description
A	Action buttons in all windows. See "Configuration window structure" on page 16.
B	Dimensions tree
C	Buttons specific to the window
D	Tab filters definition area
E	Table of properties of the selected element in the definition area

Window sections

Element	Description
Dimensions tree	Show the list of available dimensions. • system dimensions (scenario, period, account, entity...) • custom dimensions (products, cost centres...) • specific reporting dimensions (e.g. period length)
Tab filters definition area	Show the list of filters currently set.
Properties table	Allows you to manage the editable attributes of the selected dimension or filter.

Buttons in the window

Element	Description
	Adds one or several elements or nodes selected in the dimensions tree under the selected node in the filters tree.
	Only with filters already defined, adds one or several elements or nodes selected in the dimensions tree to all filters already defined. ***Note**: if there are no pre-defined filters in the column, it functions like the*  *button.*
	Adds one or several filters selected in the dimensions tree to a selected filter, creating a OR type logical relationship. OR type logical relationships can only be created with elements or nodes belonging to the same dimension as the filters already defined. The added filter is labelled with the concatenation of the individual element codes separated by ",". The value displayed is equal to the sum of the elements entered.
	Adds one or several elements selected in the dimensions tree belonging to the same filter dimension to a node of filters linked by an OR logical relationship. The added filter is labelled with the concatenation of the individual element codes separated by ",". The value displayed is equal to the sum of the elements entered.
	(Only for control groups) Adds the control group selected in the dimensions tree to the filter. See I gruppi di controllo: concetti di base.
	(Only for control groups) Adds the control group selected in the dimensions tree to all filters already defined. See I gruppi di controllo: concetti di base.
	(Only for FST items) Analytically adds, under the selected filter, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	(Only for FST items) Analytically adds, under all filters, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	(Only for FST items) Synthetically enters the FST under the currently selected filter. See "Financial statement templates" on page 83.
	(Only for FST items) Synthetically adds, under all filters, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	Adds a sheet on which a logical formula can be inserted. See "Formula filters" on page 159.
	Allows you to create a new template.
	Pastes the copied filters under the selected filter.
	Pastes the copied filters under every filter.
	Allows you to copy the structure of a filter.
	Only with filters already defined. Moves the selected element up one position.
	Only with filters already defined. Moves the selected element down one position.
	Only with filters already defined on several levels. Moves the selected level up one level.

Element	Description
	Only with filters already defined. Replaces en bloc the selected elements of a filter with elements selected in the dimensions tree. Useful for reusing a filter by replacing its elements, but not its properties.
	Only with filters already defined in the rows/columns with an OR logical relationship. Deletes one or several elements from the filter containing the OR logical relationship.
	Only with filters already defined. Deletes the selected element or node.
	Only with filters already defined. Deletes the entire level of the selected filter.

Properties table

Element	Description
	Expands a set of details which change a filter property.
	Expands the list of possible values for the property.
	Opens the property configuration window.

For details of the possible options in relation to properties, see "Row/column filter properties" on page 153.

Filter colour legend

Only for account dimensions.

Element	Description
	• Green: the sign is the same as the dimension class sign. • Orange: the sign has been changed automatically based on its class. See "Sign class" on page 102. • Red: the sign has been changed in the filter properties. See "Sign change" on page 113.

Tab filter properties

List of tab filter properties

***Note**: the list of properties displayed varies depending on the type of dimension used as the filter.*

Element	Description
Reference FX rates	Allows you to indicate a different FX rate from the one defined in the table, for the entity for the scenario/period in use, to be used in the form navigation step.
Sign change	This allows the user to change the sign in which the account is displayed. If the sign of a filter included in a formula changes, the formula will automatically account for the sign change.
Complementary	This allows the user to exclude a specific filter or node from a node of filters in a hierarchy. Thus, the system displays the entire hierarchy of the filter's parent node or excluded node, except for the specific filter or node. ***Note**: the additional filter is identified by the prefix <NOT> and works on hierarchies but not groupings.*
Size	Indicates the dimension to which the selected element belongs. Non-editable property.
Dividing factor	Allows you to set the factor for the calculation of the average amount for a defined number of days or by period.
Dictionary	Allows the user to associate dictionary items in the languages enabled in the environment with rows. When the form is run, instead of the description, the sheet name shows the dictionary item translated into the user's language. For more details, see "Dictionaries" on page 48.

Element	Description
Advanced selection	This allows the user to define a set of elements that the system must dynamically recover when running the form. For more details, see "Dynamic matrices" on page 132.
Navigation	Only visible if a parametric filter on the Scenario dimension has been selected. Allows the user to set criteria for navigating between scenarios (previous, next and reference). For more details, see "Navigating scenario parameters" on page 73.
Period Difference	Only visible if a parametric filter on the Period dimension has been selected. Allows the user to set criteria for navigating between periods. For more details, see "Navigating period parameters" on page 74.
Formula	Allows you to define elements calculated as formulas within the sheet. For more details, see "Formula filters" on page 159.
Hide	Allows you to hide sheets when the form is run.
Segment	Allows you to view intercompany data related to the entities of an entity node called a *segment*. This property's default value is defined in advanced form settings (see Settings window) and used on all the tab filters set on entity dimensions, custom dimension 2 or the counterparty entity. This value can be overridden when designing the form.

Tab filter properties table

Description

This table shows certain properties, the number and type of which depend on the selected filter or filters. The name and value are shown for every property. The properties needed for standard tab filter functionality are defined with a default value. Properties which refer to the same aspect of the setup and represent the details thereof are grouped below a main item identified with the icon.

If several filters are selected at the same time, the table will only show the shared properties and it will be possible to define the same property for all the selected filters with a single action.

Reference FX rates

Element	Description
average	Average FX rate of the period in which it is calculated. These are the FX rates generally used for P&L accounts.
period average	Applies a different FX rate for every period into which the year is divided. Allows the user to perform more accurate conversions than the average FX rate.
final	Period-end FX rates. These are used for the balance sheet accounts, with the exception of the equity accounts, which are typically converted at the historical FX rate.
based on account	FX rate used for conversions and defined in the accounts table

Sign change

Account dimension properties. When active, this inverts the sign of the data.

Complementary

When active, this excludes a filter or node from the node of filters containing it. The value shown is the total value of the parent node excluding the filter or node on which the property is active

Size

Not editable. Indicates the dimension to which the selected element belongs.

Advanced selection

Element	Description
Detail subtotal	Displays the subtotal for every dimension nested in the filter node.  *In the presence of three or more nested dimensions, activate the option to prevent potential style application errors.*
Generate subtotals up to	Allows the user to define the starting node for calculating subtotals. • **Root**: generates subtotals based on the hierarchy root (default value). • **Starting node**: generates subtotals based on the selected intermediate node. ***Note**: this has no effect if the elements returned as the result of the Advanced Selection script do not belong to the starting node's sub-tree. See "Details of advanced selection scripts for calculating the default value" on page 63.*
Restrictions	The system allows you to insert in the table restrictions on dimensions of the sheet, primarily between entity/custom dimensions, in order to allow, in data entry, data to be read/written only on the relative custom dimensions specified in the table, to improve performance. This option allows you to override these restrictions on dynamic elements, in order to see the data without the restrictions set in the table. **Hierarchy Node Intersection**: by default, if a single dimension of the matrix has several hierarchies, the system presents the intersection of the two hierarchies, i.e. the data related to elements belonging to both hierarchies. By activating this option, the system shows all combinations of elements of both hierarchies. See Entity elements.
Pruning	Hides empty rows/columns. • **Not valued**: hides from the table rows/columns which are found to have no data in the database during the run form phase. The value zero is considered as data. • **To zero**: hides from the table rows/columns which are found to have no data in the database during the run form phase or have the value 0 entered. • **None**: shows all rows/columns.
Script	Allows the user to set a query on CCH Tagetik 's application database in the cell using a TGKML syntax, see Advanced selection script.
Subtotals	Only with dynamic filters. When the form is run, at every hierarchy level interruption, it displays the subtotal calculated on the lowest level elements. The subtotal therefore takes the name of the parent node of the lowest level elements on which the subtotal is calculated (for example: Europe is the name, and therefore the parent node, of the subtotal calculated on the lowest level elements Italy, France, Spain etc.). For more details on the general settings for subtotals, see "Matrix options tab" on page 128.
Selection type	Define which hierarchy levels to display. • **None** • **Insert all children**: only displays the child nodes of the selected filter. • **Insert lowest level elements**: displays all the lowest level elements present below the selected node, to the last level. • **Insert Parent**: displays the parent node of the selected filter. • **Advanced selection**: activates the "Script " above property. Allows the user to insert dynamic elements that cannot be inserted via standard selection. • **Insert analytical FST** : (with Account type elements inserted via items of an FST) allows the user to define an analytical FST in which both the FST items and the related accounts are displayed. For more details, see "Financial statement templates" on page 83. • **Insert synthetic FST**: (with Account type elements inserted via items of an FST) allows the user to define a synthetic FST in which only the FST items are displayed. For more details, see "Financial statement templates" on page 83. • **Control group**: (with Account type elements entered via control groups) allows the user to define the accounts with reference to a group.

Dictionary

Element	Description
	Allows the user to select an existing dictionary item. See "Dictionary Items window" on page 50.

Formula

Element	Description
	Opens a window to write the calculation formula.
Prevailing	Makes the formula the prevailing formula in case of intersection with others.
Calculation logic type	• **Client**: the formula refers to dimension elements contained in the tab filter and is solved in the local client. • **Server**: the formula refers to elements of the dimensions present in other sheets or other elements of the form, the formula is solved on the server and the client receives the resulting value.

Hide

Element	Description
Never	In this case, the sheet is displayed when the form is run.
Always	In this case, the sheet is hidden when the form is run.

Segment

Properties of the entity and accounts dimension

Element	Description
(from configuration)	Reads and applies the values indicated in the application's general settings to the Default section of the Form tab. See Settings window.
None	Does not apply any filters to journals
Per Segment parent	The value shown includes all consolidation journals (IC eliminations, financial investment eliminations, other corrections) for all nodes or legal entities belonging to the same "parent" node.
By segment	The value shown includes all consolidation journals (IC eliminations, financial investment eliminations, other corrections) between legal entities belonging to the same node.
Intersegment	Identifies the values relative to all journals referring to legal entities belonging to different segments. This function is activated ONLY if the selected node has at least TWO levels below it. Data can be read according to this method both when creating "Tab filters" and when selecting "Prompt Filter", respecting all the restrictions on the depth of the levels for the choice of "Intersegment".
Per Intersegment Parent	Identifies the values relative to all journals referring to legal entities belonging to different segments and, starting from a level below, takes all the journals that start from the node on which the function is activated.

Navigation

Only for parametric filters on the scenario dimension. See "Navigating scenario parameters" on page 73.

Element	Description
Previous	Selects the scenario indicated in the scenarios list as the *previous scenario* to the selected scenario. See "Navigating scenario parameters" on page 73.
Next	Selects the scenario indicated in the scenarios list as the *next scenario* following the selected scenario. See "Navigating scenario parameters" on page 73.
Reference	Selects the scenario indicated in the scenarios list as the selected scenario's *reference scenario*. See "Navigating scenario parameters" on page 73.

Period Difference

Only for parametric filters on the period dimension.

Allows the user to navigate periods, shifting forwards or backwards from the period selected when the form is run.

See "Navigating period parameters" on page 74.

Matrix filters

What they are for

The data that the matrix manages are defined using row/column filters. When a logic that groups the data to be set on row/column filters can be identified, then the data can be filtered at the global level on the matrix, by defining *matrix filters*. For example, if the matrix needs to manage data of a single entity only, it is useful to set the entity filter on the entire matrix. Thus, without the need to apply the filter to every row/column in the matrix, when the form is run it shows, for the entire matrix, only data related to the entity indicated in the matrix filter.

Setting matrix filters

Excel > CCH® Tagetik > Design > select form/create new form > Matrix > Matrix Filters

Introduction

Setting matrix filters means managing filters that have an effect on the whole matrix, not just on certain rows/columns. It is possible to define filters made up of individual dimensions and filters with more complex structures.

Set a filter

1. In the dimensions tree, select one or more elements to be used as a filter, then click next to the matrix filters definition area, or drag them into the area. Details of the reference dimension and the selected dimension elements will appear in the definition area.
2. Set the filter properties as desired (see "Matrix filter properties" on the facing page).
3. Repeat the previous step to add elements of the same dimension or create filters related to other dimensions.

Add a dimension element to a filter

1. In the dimensions tree, select the desired dimension element and click : the element will be added to the elements node relative to the dimension to which it belongs.
2. Set the filter properties as desired.

Customise a filter

In the filters definition area, click on the desired filter and change its properties in the properties table (see "Matrix filter properties table" on page 119).

Matrix filter properties

List of matrix filters

Element	Description
Complementary	This allows the user to exclude a specific filter or node from a node of filters in a hierarchy. Thus, the system displays the entire hierarchy of the filter's parent node or excluded node, except for the specific filter or node. ***Note**: the additional filter is identified by the prefix <NOT> and works on hierarchies but not groupings.*
Comparison condition	It allows the user to define a condition for comparison, or an operator and a value with which to compare the filter value. *To specify an individual element of the filter to be compared, use the operator "=". To exclude one of them, use the operator "<>".* *For example: with operator "=" and value "INPUT_DEFORM" on an Origin type filter, the user can obtain the sum of all the figures with origin "INPUT_DEFORM".*
Reference FX rates	This allows the user to define a different FX rate from that which is specified in the table, i.e. whether it needs to be read from a scenario or period other than those of the data to be converted. See "Details of Reference FX rate properties" on page 122.
FX rate	Allows the user to view the entity or transaction amounts in a different currency. See "Details of FX rate properties" on page 121.
Periodic value calculation	Allows the user to specify the calculation method for the periodic value. The calculation methods vary depending on the reporting engine used (see "Form options window" on page 55).
Size	Indicates the dimension to which the selected element belongs. Non-editable property.
Navigation	Only visible if a parametric filter on the Scenario dimension has been selected. Allows the user to set criteria for navigating between scenarios (previous, next and reference). For more details, see "Navigating scenario parameters" on page 73.
Period Difference	Only visible if a parametric filter on the Period dimension has been selected. Allows the user to set criteria for navigating between periods. For more details, see "Navigating period parameters" on page 74.

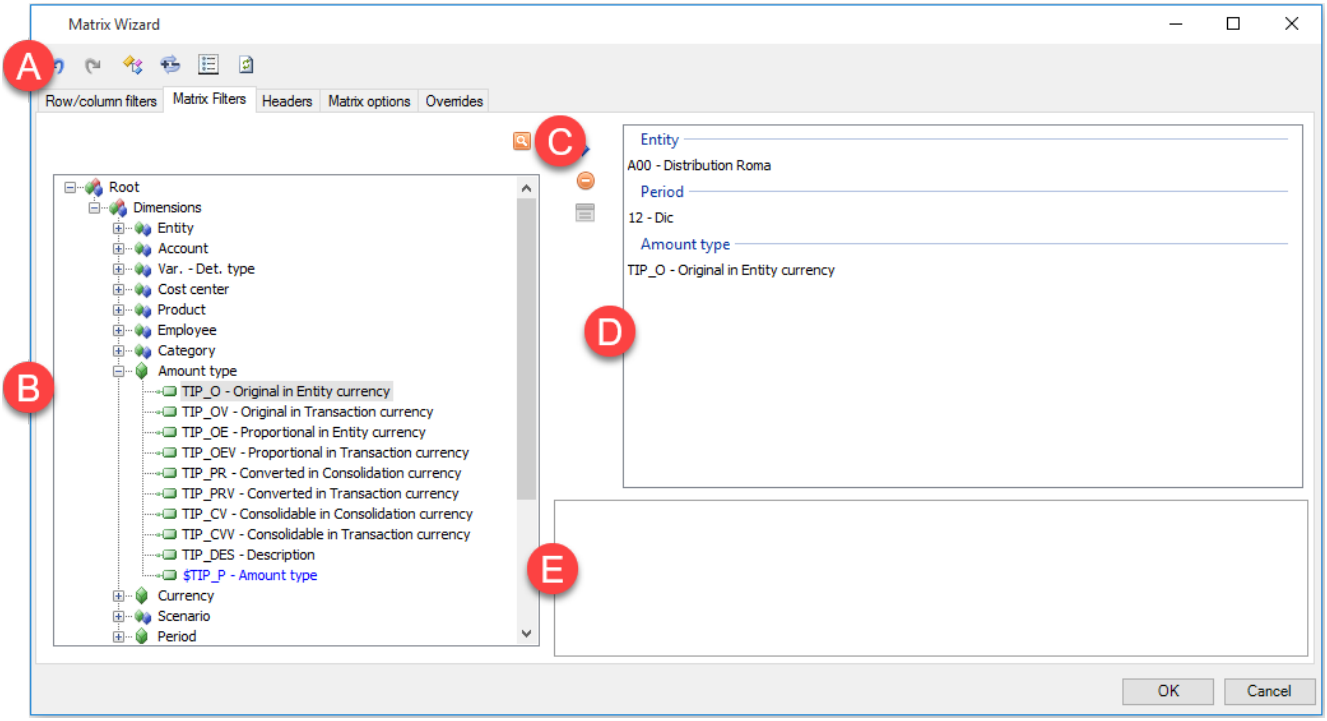
Matrix Filters tab

Excel > CCH® Tagetik > Design > select form/create new form > Matrix > Matrix Filters

Purpose of the tab

This tab allows you to define filters on dimensions that are valid over the entire matrix.

What it looks like



Element	Description
A	Action buttons in all tabs. See "Configuration window structure" on page 16.
B	Dimensions tree
C	Buttons specific to the tab
D	Matrix filters definition area
E	Table of properties of the selected element in the definition area

Tab sections

Element	Description
Dimensions tree	Show the list of available dimensions. • system dimensions (scenario, period, account, entity...) • custom dimensions (custom dimensions: e.g. products, cost centres...) • specific reporting dimensions (e.g. period length)
Matrix filters definition area	Show the list of filters set on the matrix.
Properties table	Allows you to manage the editable attributes of the selected dimension or filter.

Buttons in tab

Element	Description
	Adds one or several elements or nodes selected in the dimensions tree to the matrix filters definition area.
	Only with filters with the OR function already defined. Removes one or more elements from filters defined as OR clauses. This button is only activated when matrix filters with the OR logical relationship have already been defined.
	Removes the selected filter.

Properties table

Element	Description
	Expands a set of details which change a filter property.
	Expands the list of possible values for the property.
	Opens the property configuration window.

For details of the possible options in relation to properties, see "Matrix filter properties" on page 149.

Row/column filters

Introduction

To define the content of the matrix, filters must be set on the rows, columns and potentially the entire matrix. In particular, in order to define a filter, select elements of the dimensions, said dimension elements and any aggregation structures the user wants to view or process the data of.

Specific features of filters on account dimensions

For the account dimension, the definition of filters can be simplified using *financial statement templates* or *control groups* (see "Financial statement templates" on page 83 and I gruppi di controllo: concetti di base).

A financial statement template (FST) is a structure used to aggregate accounts or custom accounts/dimensions that can represent financial statements (IAS profit and loss statements and balance sheets, management reports, direct/indirect financial reports), KPIs and other indices for reporting. A financial statement template can be added to a matrix in its entirety or in part as a filter to be able to design its structure more quickly and efficiently.

Similarly, when the structure of a control group conforms to the specifics of the form, it can be added as a filter so all the accounts contained in it are added to the form.

For more information, see "Financial statement templates" on page 83 and I gruppi di controllo: concetti di base.

Row/column filter properties

When a filter is selected, the properties table displays all the properties of the filter with the relative values. Mandatory properties are set automatically to the default value. The user can edit filter properties to adapt the matrix configuration to their own business needs.

Depending on the type of property, the user can choose an option from among pre-set values, or enter formulas, dictionary items and so on.

When a row/column filter is added to an account dimension, the system changes the sign depending on the class assigned to the account (see "Sign class" on page 102).

For details of filter/column properties, see "Table of row/column filter properties" on page 112.

Placeholder elements

To add empty rows/columns to the matrix, elements called *placeholders* can be added between row/column filters. The purpose of these elements is to add empty rows/columns directly into the matrix structure and regulate the layout of the form. In fact, any empty rows/columns added manually in Excel are overwritten when the form structure is changed.

Formulas

To add formulas to cells of the matrix, *formula*-type elements can be added between row/column filters. The purpose of these elements is to add formulas directly into the matrix structure. Any formulas added manually in Excel are overwritten when the form structure is changed.

Formulas can be used to insert a row total or a variance between two columns. For example, in a matrix with costs on the rows and two scenarios on the columns, it is possible to view in runtime an additional row containing the total costs for the two scenarios and the annual variance, calculated as the difference

between the totals for the two scenarios. To obtain this result, a dedicated formula between the column filters must be prepared.

Set row/column filters

Excel > CCH® Tagetik > Design > open form > open or create matrix > Row/column filters

Introduction

Setting row/column filters means defining what data will be shown in the matrix's rows/columns. It is possible to define filters made up of individual dimensions and filters with more complex structures.

Set a filter on rows/columns

This procedure explains how to define filters on rows. Proceed in the same way to define column filters.

1. In the dimensions tree, select an element or node of the desired dimension to be used as a filter for the rows, then click next to the local filters on the rows, or drag it directly into the rows section. The filter will appear below the **Rows** node.
2. Set the filter properties as desired (see "Row/column filter properties" on the facing page).
3. Repeat the previous steps to add dimension elements or nodes from other dimensions.

Customise a filter

In the local row/column filters tree, click on the desired filter and edit its properties in the properties table (see "Matrix filter properties table" on page 119).

Create OR filters in the local row/column filters tree

This procedure explains how to create a filter containing several elements of a hierarchy of dimension elements directly in the row/column filters tree, by linking them to a cumulative relationship. To create such a filter, the OR function must be used ( ). At runtime, that hierarchy is run in the matrix just like any other node. The procedure to be followed is:

1. In the desired local filters tree, select the Rows (or columns) parent node.
2. In the dimensions tree, select the desired dimension elements and click . The elements are added on the same level.

Add OR values to an existing filter

1. In the filters tree, select the desired filter.
2. In the dimensions tree, select the desired dimension elements and click . The elements are added on the same level.

Add a placeholder

Click the row/column filter that will precede the placeholder, then click . The placeholder will appear below the selected filter.

Add a formula

1. Click on the row/column filter or on the placeholder that will precede the formula, then click . The formula placeholder will appear below the selected filter and the formula editor will open.
2. Select the desired options for the formula and enter the text of the formula. For instructions on how to write formulas, see "Formula filters" on page 159.

To view a list of the pre-existing filters on rows/columns and select the desired ones, position the cursor in the formula writing area and right click.

3. Click **OK**.

Row/column filter properties

List of row/column filters

The row/column filters are listed below.

Element	Description
Reference FX rates	This allows the user to define a different FX rate from that which is specified in the table, i.e. whether it needs to be read from a scenario or period other than those of the data to be converted. See "Details of Reference FX rate properties" on page 122.
FX rate	Allows the user to view the entity or transaction amounts in a different currency. See "Details of FX rate properties" on page 121.
Sign change	This allows the user to change the sign in which the account is displayed. If the sign of a filter included in a formula changes, the formula will automatically account for the sign change.
Periodic value calculation	Allows the user to specify the calculation method for the periodic value. The calculation method varies depending on the reporting engine used (see "Form options window" on page 55).
Complementary	This allows the user to exclude a specific filter or node from a node of filters in a hierarchy. Thus, the system displays the entire hierarchy of the filter's parent node or excluded node, except for the specific filter or node. ***Note**: the additional filter is identified by the prefix <NOT> and works on hierarchies but not groupings.*
Comparison condition	It allows the user to define a condition for comparison, or an operator and a value with which to compare the filter value. *To specify an individual element of the filter to be compared, use the operator "=". To exclude one of them, use the operator "<>".* *For example: with operator "=" and value "INPUT_DEFORM" on an Origin type filter, the user can obtain the sum of all the figures with origin "INPUT_DEFORM".*

Element	Description
Data Validation	Only with validation queries set in the form. Associates a validation query with the selected element to limit the user's choice to a restricted set of elements. ***Note:*** in the presence of Dictionary and Validation Query insisting on the same textual measure, the drop-down shows all values in the validation query as a suggestion; if data validation is active, only those values belonging to the Dictionary are saved. An exception is made if the 'Out of dictionary values allowed' flag has been enabled on the measure.
Size	Indicates the dimension to which the selected element belongs. Non-editable property.
Advanced selection	This allows the user to define a set of elements that the system must dynamically recover when running the form. For more details, see "Dynamic matrices" on page 132.
Dictionary	Allows the user to associate dictionary items in the languages enabled in the environment with rows. When the form is run, instead of the description, the sheet name shows the dictionary item translated into the user's language. For more details, see "Dictionaries" on page 48.
Editability	Allows you to define row/column editability. For more details, see "Cell editability" on page 78.
Validation check	Allows you to define accuracy checks on data entered via the Validation Checks Wizard. For more details, see "Validations" on page 42.
Conditional editability expression	Allows you to subject the editability set on the row/column to one or several conditions that depend on the content of other rows/columns of the current matrix. For more details, see "Validations" on page 42.
Division factor	Allows you to calculate the average of a value at a certain date, according to the following calculation methods. • *day*: the average is calculated on the basis of the number of days from the start of the year to the specific month, e.g. 180 days from January to June. • *period*: the average is calculated on the basis of the number of days in the specific month, e.g. 30 days for June. ***Note****: this functionality is not supported for matrices designed on Analytical Workspace.*
Tagetik style	Allows you to define a style for the selected row/column filter. See "Tagetik style property details" on page 123.

Element	Description
Formula	Allows you to define elements calculated as formulas on the row/column. For more details, see "Formula filters" on page 159.
Hyperlink	Allows you to define links between forms. For more details, see "Hyperlinks" on page 220.
Multi-selection	Property only available from the web interface on row/column filters with counter dimension in transactional matrices of the Analytical Workspaces. Allows you to see check boxes for multiple selection, in Navigation and Data Entry and for every row/column.
Less	Allows you to hide rows/columns when the form is run. In particular, it allows you to: • create specific layouts for each output sheet • create useful rows/columns linked to matrix formulas • hide superfluous rows/columns in specific contexts in which the form is used ***Note**: in data entry, rows/columns cannot be hidden if they contain at least one editable cell.*
Segment	Allows you to view intercompany data of the entities of a chosen node of entities, called a *segment*. This property's default value is defined in advanced form settings (see Finestra Impostazioni) and is used on all the row/column filters on which the segment is set on entity dimensions, custom dimension 2 or the counterparty entity. This value can be overridden during design or when editing the matrix from the properties table. ***Note**: matrices designed on Analytical Workspace do not support this property.*
Data type	Allows the user to define the row/column data type (string, date, numerical, etc.) to optimise how it is displayed in the run form phase.
Prevailing data type	Allows the user to choose, for an individual cell, a different data type from the one defined for the row/column.
Navigation	Only visible if a parametric filter on the Scenario dimension has been selected. Allows the user to set criteria for navigating between scenarios (previous, next and reference). For more details, see "Navigating scenario parameters" on page 73.
Period Difference	Only visible if a parametric filter on the Period dimension has been selected. Allows the user to set criteria for navigating between periods. For more details, see "Navigating period parameters" on page 74.

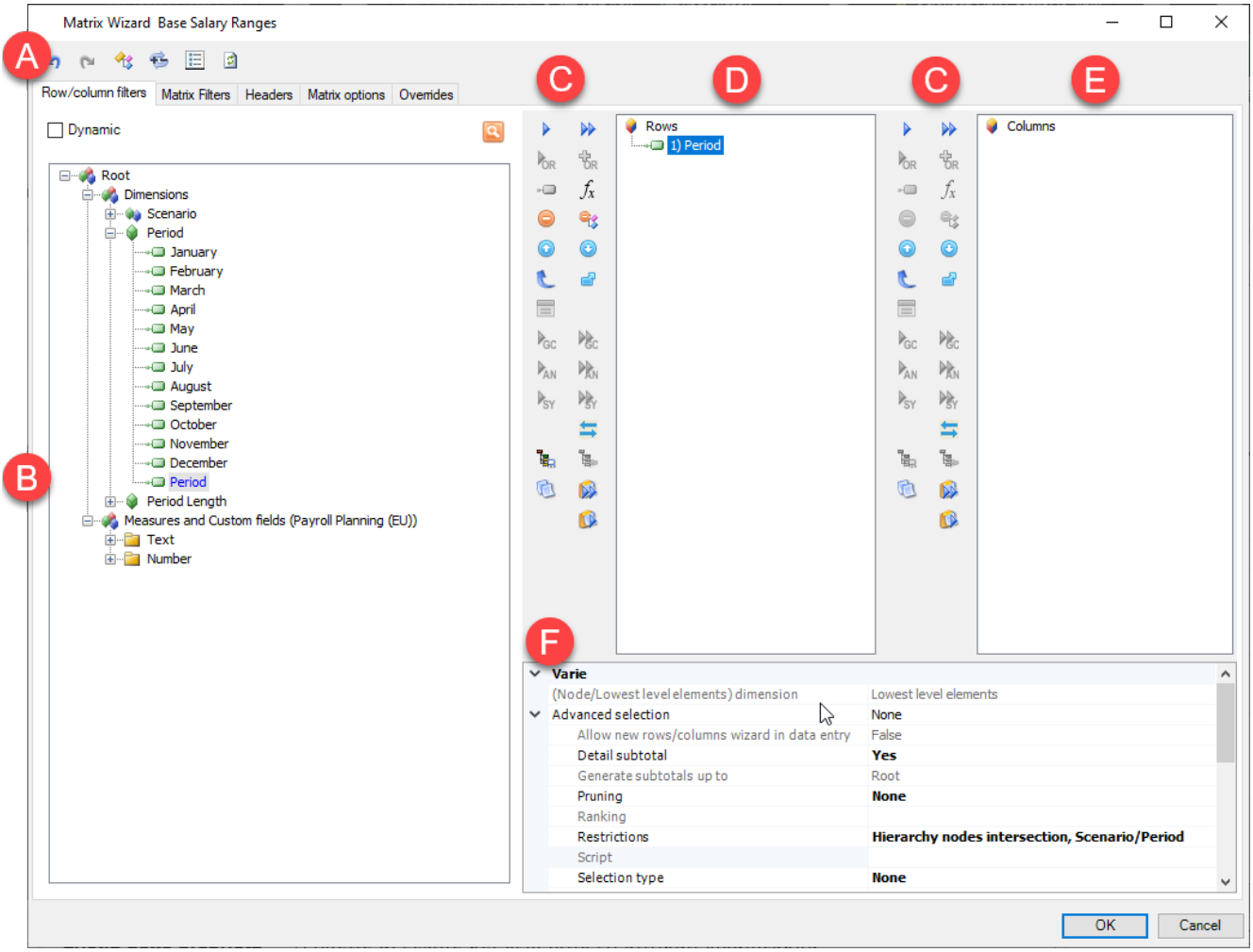
Row/column filters tab

Excel > CCH® Tagetik > Design > open form > open or create matrix > Row/column filters

Purpose of the tab

This tab allows you to set filters on the rows and columns of the matrix. These filters are elements and/or nodes of the dimensions tree that are added and modified in the local row/column filters tree and determine the selection of data managed in the rows and columns.

What it looks like



Element	Description
A	Action buttons in all tabs. See "Configuration window structure" on page 16.
B	Dimensions tree
C	Buttons specific to the tab
D	Local row filters tree
E	Local column filters tree
F	Table of properties of the selected element in the definition area

Tab sections

Element	Description
Dimensions tree	Show the list of available dimensions. • system dimensions (scenario, period, account, entity...) • custom dimensions (e.g. "Products", "Cost centres" ...) • specific reporting dimensions (e.g. period length) • specific dimensions of some reporting functionalities (e.g. Column form, Counter and Status, used in the management of transactional data entry forms)
Local row filters tree	Shows the list of filters set on rows.
Local column filters tree	Shows the list of filters set on columns.
Properties table	Allows you to manage the editable attributes of the selected dimension or filter.

Buttons in the tab

Element	Description
	Adds one or several elements or nodes selected in the dimensions tree under the selected node in the filters tree.
	Only with filters already defined, adds one or several elements or nodes selected in the dimensions tree to all filters already defined. ***Note**: if there are no pre-defined filters in the column, it functions like the*  *button.*
	Adds one or several filters selected in the dimensions tree, creating a OR type logical relationship. OR logical relationships can only be created with elements or nodes belonging to the same dimension. The added filter is labelled with the concatenation of the individual element codes separated by ",". The value displayed is equal to the sum of the elements entered.
	Adds one or several filters selected in the dimensions tree to a selected existing filter, creating a OR type logical relationship. The added filter is labelled with the concatenation of the codes of the individual existing and added elements, separated by ",". The value displayed is equal to the sum of the elements entered.
	Adds the control group selected in the dimensions tree to a filter in the rows/columns.
	Adds the control group selected in the dimensions tree to all filters already defined.
	Analytically adds, under the selected filter, the FST selected in the dimensions tree. See .
	Analytically adds, under all filters, the FST selected in the dimensions tree.
	Synthetically enters the FST under the currently selected filter.
	Synthetically adds, under all filters, the FST selected in the dimensions tree.
	Save a filter as predefined. The new filter is then added to the dimensions tree. See "Predefined filters" on page 162.
	Update a predefined filter. The new filter is then added to the dimensions tree. See "Predefined filters" on page 162.
	Add an empty placeholder element.
	Adds a placeholder on which a logical formula can be inserted. See "Formula filters" on page 159.
	Pastes the copied filters under the selected filter.
	Pastes the copied filters under every filter.
	Allows you to copy the structure of a filter.
	Only with filters already defined. Moves the selected element up one position.

Element	Description
	Only with filters already defined. Moves the selected element down one position.
	Only with filters at several levels already defined in the row/column columns. Moves the selected level up one position.
	Only with filters already defined. Inverts the row filters and column filters.
	Only with filters already defined. Replaces en bloc the selected elements of a filter with elements selected in the dimensions tree. Useful for reusing a filter by replacing its elements, but not its properties.
	Only with filters already defined in the rows/columns with an OR logical relationship. Deletes one or several elements from the filter containing the OR logical relationship.
	Only with filters already defined. Deletes the selected element or node.
	Only with filters already defined. Deletes the entire level of the selected filter.

Properties table

Element	Description
	Expands a set of details which change a filter property.
	Expands the list of possible values for the property.
	Opens the property configuration window.

For details of the possible options in relation to properties, see "Row/column filter properties" on page 153.

Filter colour legend

Only for account dimensions.

Element	Description
	• Green: the sign is the same as the dimension class sign. • Orange: the sign has been changed automatically based on its class. See "Sign class" on page 102. • Red: the sign has been changed in the filter properties. See " Sign change" on page 153.

Examples of row/column filter settings

Example of simple filters

To create a data entry form to insert quantities of products managed, the following must be set:

- the product custom dimension on the rows
- the quantity account dimensions on the columns

Example of complex filters

To create a form for comparing data between the budget P&L and the actual P&L on periods June and December, the following must be defined:

- the P&L financial statement template for the row filters
- the scenario and the period for the column filters.

In the run step, the following elements will therefore be generated:

- one row for every type of data in the P&L financial statement template
- a total of four columns for the column filters, corresponding to the combinations of the periods with scenarios: 2019BDG/December, 2010ACT/December, 2019BDG/June, 2010ACT/June.

Entity A01	2010BDG December	2010ACT December
Product Revenues – Sales SW - Antivirus	- 1000	- 980
Product Revenues – Sales HW – Memory cards	- 900	- 950
Product Costs – IT – Databases	2000	2500
TOTAL	**100**	**570**

Formula filters

What they are

The *formula* filters are filters on matrix rows or columns which, while the form is being run, function like Excel formulas and make it possible to process data based on the data handled by the matrix itself.

There are advantages to inserting a formula during the design of a matrix, rather than into the matrix area directly. Formulas inserted in this way are also stored when the matrix is regenerated, or whenever the form's structure is changed. On the contrary, formulas inserted directly into the matrix area in Excel are overwritten every time the matrix is regenerated.

Formula format

The supported syntax for formulas is English with separator ",", regardless of the display language used for CCH Tagetik, the add-in or the Excel installation. In Matrix Wizard formulas, references are made to filters in rows/columns rather than cell coordinates like in Excel. The syntax to be followed is detailed below.

- {n}: Row formulas: nth row, current column
- {m}: Column formulas: mth column. current row
- {n,m}: Row/column formulas: nth row, mth column.

E.g. Excel function "=SUM(A1: A2)" is "SUM({1}:{2})" in the Matrix Wizard.

In the form run step, the system transforms the Matrix Wizard formula into an Excel formula for all the row/column filters to which the formula refers. It then automatically adds the "=" symbol in Excel when it generates the matrix.

***Note**: in the local filters tree, formulas do not have descriptions except for formula-type FST items. In any case, descriptions can be allocated to formulas using dictionary items. See "Dictionaries" on page 48.*

Server formulas

To add information on a filter within the matrix, *server formulas* can be inserted. These formulas allow you to add cell fields that refer to other elements of the matrix and form (relative to a dimension present on the rows if working on columns and vice versa, or other matrices or tab filters, form filters) to show the dimension properties (e.g. description, company name, etc.)

Prevailing formulas

When a row and a column containing formulas intersect, the system considers the row formula by default.

To tell the system your preference for a specific row or column in case of intersections, the **Prevailing formulas** option can be enabled in the desired row or column.

***Note**: if both intersecting formulas are set as prevailing formulas, the default behaviour will be applied. The default behaviour and the prevailing formula option do not apply to cells with subtotals.*

Example of a subtotal on a row/column

In the presence of a subtotal calculation formula on both rows and columns, if the prevailing formula option is set as active in both formulas, the subtotal will be calculated on all rows and columns.

	January	February	March	April	Σ
A00	1000	1000	1000	1000	4000
A01	1000	1000	1000	1000	4000
A02	1000	1000	1000	1000	4000
A03	1000	1000	1000	1000	4000
A04	1000	1000	1000	1000	4000
Σ	5000	5000	5000	5000	20000

Example of different subtotals on rows/columns

By activating the prevailing formula option, several subtotal calculation logics can be defined in the same matrix and a selective display of the desired subtotals can be set in the form run step.

For example, in the following matrix, you want to see:

- the total of the intersecting column data in the cells of the last row
- a ratio to the data of the last column in the last cell of the same column

If the sum formula is set on the row, the matrix reports an unsatisfactory-type datum for the last cell of the last column.

	2030 Actual	2031 Actual	Variation
	FY	FY	%
Antivirus	35,700	128,403	259.7%
SW Utilities	5,511,664	3,327,259	-39.6%
Office Automation	8,825,100	12,119,264	37.3%
Monitor	1,176,689	3,482,844	196.0%
Video Projectors	392,230	600,255	53.0%
Total	**15,941,383**	**19,658,024**	**506.4%**

If the sum formula is set on the row and a prevailing ratio formula on the last column, when the form is run that formula will prevail over the sum formula on the row and the cell in question will report the desired value.

	2030 Actual	2031 Actual	Variation
	FY	FY	%
Antivirus	35,700	128,403	259.7%
SW Utilities	5,511,664	3,327,259	-39.6%
Office Automation	8,825,100	12,119,264	37.3%
Monitor	1,176,689	3,482,844	196.0%
Video Projectors	392,230	600,255	53.0%
Total	**15,941,383**	**19,658,024**	**23.3%**

Quick actions for inserting formulas

Quick insert actions can be used to optimise the insertion of formulas in forms, e.g. to insert formulas on the entire row or matrix. For more details, see "Managing formulas" on the next page.

Create reference to specific cells

On standard non-dynamic forms, it is possible to create formulas with references to specific cells in the form.

Element	Description
Bidimensional reference	Absolute reference to a specific cell. Created by indicating the cell coordinates, constructed as follows: • first term = row number • second term = column number For example, *{2,5}* corresponds to an absolute reference to the intersection cell between the second row and the fifth column.
Offset	Relative reference to a cell. Created by indicating the cell's position relative to the row/column on which the reference is being defined. • first term = number of row positions above or below the formula row/column (e.g. *-3* = 3 rows *above*, *+3* = 3 rows *below*) • second term = number of column positions to the left or right of the formula column (e.g. *-3* = 3 columns *to the left*, +3 = 3 columns *to the right*) For example, *{+2, -3}* corresponds to a relative reference to the intersection cell between the second row below and the third column to the left of the row/column on which the formula is set.
Hybrid	Reference created by combining bidimensional and offset reference rules. For example, *{1,-3}* corresponds to a reference to the intersection cell between row *1* and the third column to the left of the row/column on which the formula is set.

Interrupting automatic ranges

When two or more of a formula's arguments include references to adjacent cells, on running the form the system transforms those references into a range. Where the syntax of specific functions does not accept ranges, the system generates error messages. In such cases, the adjacent cells need to be separated with a placeholder row or column, which can be hidden. An example is provided below:

Syntax with range	Syntax on running the form
NETWORKDAYS({1},{2}, Holidays_Dates)	NETWORKDAYS(A1:A2, Holidays_Dates) The syntax includes a range that is not permitted.

Syntax with placeholder between rows	Syntax on running the form
NETWORKDAYS({1},{3}, Holidays_Dates)	NETWORKDAYS(A1, A3, Holidays_Dates) The syntax does not include ranges.

Supported Excel formulas

The add-in supports the following Excel formulas:

Formula	Formula	Formula
Address	Hyperlink	Rank
Bin2Dec	If	Rate
Choose	Iferror	Replace
Code	ImReal	Rept
Column	Index	Roman
Columns	Indirect	Row
Complex	Intercept	Rows
Count	IPMT	Slope
Coounta	Irr	Substitute

Formula	Formula	Formula
Countbank	Lookup	Subtotal
Countif	Match	Sumif
Countifs	Mina	Sumifs
Days360	Maxa	Sumproduct
Dec2Bin	Mirr	Sumx2my2
Dec2Hex	Mode	Sumx2py2
Variance	Na	Sumxmy2
EDate	Now	T
Error.type	Npv	Text
Even	Oct2Dec	Time
Fact	Odd	Today
FactDouble	Offset	Value
Hex2Dec	PPMT	Vlookup
Hlookup	Quotient	Weekday

***Note**: to calculate the value of formulas before viewing them, e.g. when sorting the matrix elements based on the value of a formula, it is necessary to use Excel functions related to advanced selection scripts (see "Excel functions" on page 216).*

Managing formulas

Insert a formula on all cells in a row/column

This procedure shows you how to insert a formula on a whole row/column from a matrix cell.

1. To select the cell:

If you want to copy the formula...	Then...
on the row	select the cell of the first column relative to the row
on the column	select the cell of the first row relative to the column

2. Insert the desired formula in the cell in accordance with Excel syntax.
3. Press ENTER: the formula will be propagated over the whole row/column.

In the matrix, the system automatically converts Excel syntax into CCH Tagetik syntax.

When inserting elements in braces, make sure you edit them in or click on the {} button at the top-right, otherwise the number will be considered a pure number, not an element of the matrix.

To insert the elements involved in the formula, position the cursor in the Reporting Client Formula window and right click; a list will appear showing all the rows/columns of the matrix in which to select the desired element. This prevents errors and allows you to write the formula quicker.

Predefined filters

Description

To optimise the design of the form structure, the definition of sets of row/column filters and the property settings of the filters that they contain can be saved in a single *predefined filter*. The predefined filter is then shown in the matrix dimensions tree together with the dimensions (in the **Filters** folder) and can be used like a normal filter. Predefined filters can be modified or removed at any time. For example, this functionality is useful in the case of several forms with the same or similar structures; saving a set of filters as a single filter allows you to quickly replicate the structure for all forms, avoiding the need to design a new one for each of them.

The predefined filters functionality is only available to users for whom function "0002914 - Reporting .Net - Param - Predefined filters" is enabled in their role.

Predefined filters can be used as a parameters (see "Parameters" on page 58).

Static or dynamic use of predefined filters

Predefined filters can be used statically or dynamically.

Use	Description
Static	The filter is defined on the matrix in the row/column filters definition phase.
Dynamic	The filter is used as a form run parameter, so it is defined dynamically when the form is opened (see "Parameters" on page 58). All changes made to dynamic filters are automatically propagated to all forms in which the filter is used.

Particular cases in the use of predefined filters

The following particular cases may be encountered when using predefined filters:

- Several predefined filters can be inserted dynamically (when the form is run, the system checks that the various predefined filters on which the deployment is performed contain the same dimensions in the same order).
- Placeholders can be inserted between predefined filters.
- It is possible to add elements of other dimensions below predefined filters inserted dynamically (when the form is run, the system checks that the added dimensions do not include any of the dimensions already present in the predefined filter).
- Predefined filters can be inserted statically or dynamically below pre-existing elements of other dimensions (when the form is run, the system checks that the filter dimensions do not include any of the dimensions already present in the filters tree).

Working with predefined filters

Excel > CCH® Tagetik > Design > open form > open or create matrix > Row/column filters

Create a predefined filter in a matrix

This procedure shows you how to create a predefined filter based on a combination of filters set on rows of the matrix, but it is also valid if done on the basis of filters set on columns.

1. Click next to the local row filters tree: the **Predefined filters** window will open.
2. Insert a code for the new predefined filter and a description for every language supported by the system.
3. Click **OK**.

Saved predefined filters are visible in the **Filters** node of the Matrix Wizard dimensions tree.

***Note**: in local filter trees, the predefined filter is identified with the icon if it is dynamic or the icon if it is static.*

Update a predefined filter

Modify the structure of the filter as desired and click .

Edit the description of the predefined filter

1. In the dimensions tree, select **Filters**, right click the desired predefined filter and select **Edit header**.
2. Edit one or more descriptions as desired.
3. Click **OK**.

Delete a predefined filter

In the dimensions tree, select **Filters**, right click the desired predefined filter and select **Delete**.

***IMPORTANT**: when the predefined filter is deleted, the system does not check whether it is used in one or several forms.*

Insert a predefined filter dynamically

1. Open the matrix (see "How to access the windows" on page 9).
2. Select **Dynamic** at the top-left.
3. Insert the predefined filter as a row/column filter or a matrix filter.

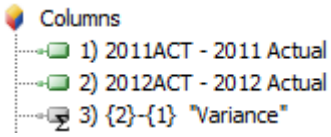
***Note**: in filter trees, the predefined filter is identified with the*  *icon.*

Examples of predefined filter settings

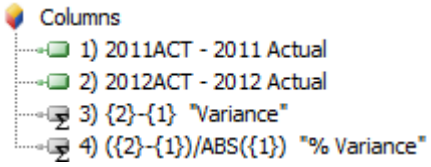
Using predefined filters

Suppose you have the following predefined filters available:

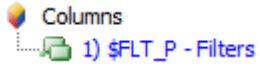
- Fixed_Var: allows you to compare scenarios 2012ACT and 2011ACT and to calculate the relative variance



- Fixed_Var_%: allows you to compare scenarios 2012ACT and 2011ACT, and calculate the relative variance and percentage variance

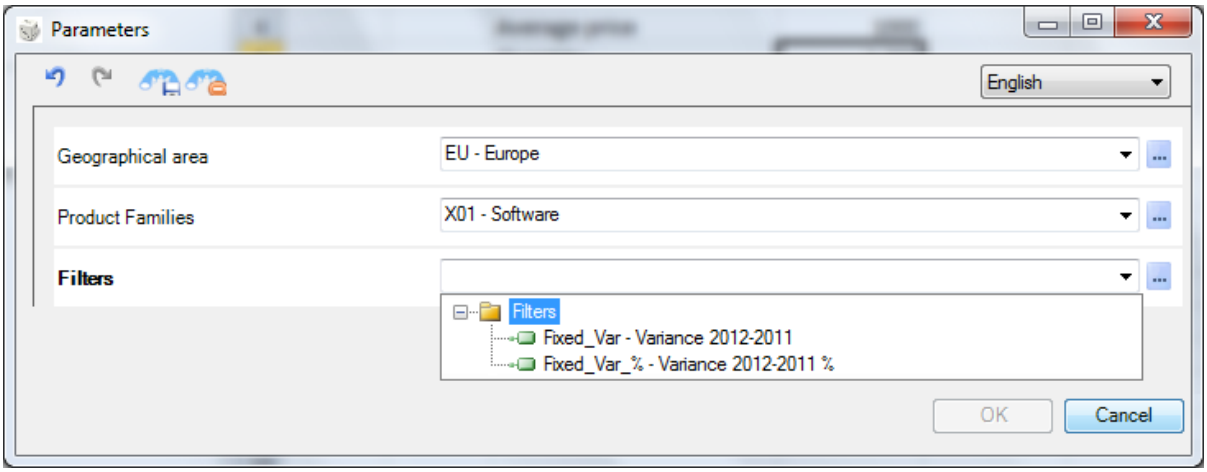


Tagetik allows the user to decide which predefined filter to use when the form is run, by inserting (in this case in the columns) the Predefined filter parameter "$FLT_P - Filters".



The Predefined filter parameter must be inserted as Dynamic, by activating the "Dynamic" option at the top-left of the Matrix Wizard.

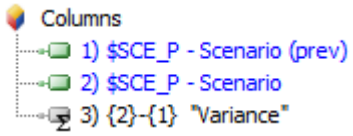
When the form is run, the user selects the predefined filter to be used (in addition to the other available parameters).



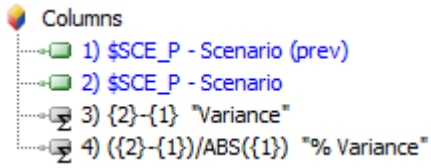
Using predefined filters with parametric elements

In the previous example, the predefined filter did not contain any parametric elements. Now let's look at an example in which the predefined filter contains a parametric scenario "$SCE_P". Suppose you have the following predefined filters:

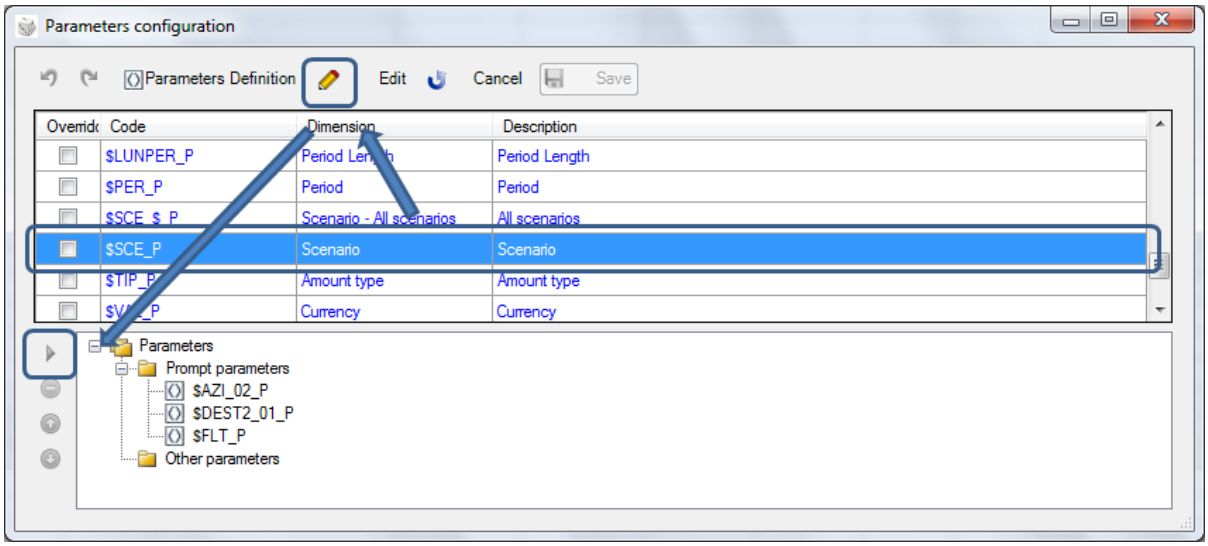
- Param_Var: allows you to compare the current and previous scenarios and calculate the relative variance



- Param_Var_%: allows you to compare the current and previous scenarios and calculate the relative variance and percentage variance



If the Predefined filter parameter "$FLT_P - Filters" is inserted in the columns, in the run step the system will ask the user to indicate which predefined filter to use, but will not require the scenario. This is because the form structure is only known after it has been run. It is therefore necessary to override the scenario parameter in the form. To do this, the user must click the **Manage parameters** button in the Design bar ribbon.



In the parameters configuration window, the user must

- click the **Edit** button;
- select parameter $SCE_P from the "Other parameters" folder and move it to the definition area using the button.

This means that the user will be asked for the scenario when the form is run.

Custom filters

What they are

In transactional Analytical Workspace matrices, custom filters can be defined on dataset measures.

When the form is run, custom filters are applied by running a query on the database.

***Note**: the filter semantics can be made conditional on the implementation of the specific vendor.*

***Note**: when an Analytical Workspace is duplicated, and reports are included in the duplication, any custom filters on measures will be cloned, but not filters on measures with which a Analytical Workspace dictionary type is associated.*

Custom filters are defined using the Matrix Wizard (see "Custom filters tab" below).

Custom filter restrictions

Custom filters are subject to the following restrictions:

- they only use the equals operator (=)
- they use AND as the logic operator to join filters. This allows the user to obtain the intersection of the results of filters applied in this way.

Application of filters when cloning workspaces

If the function is used to duplicate a workspace (see Clone window) that includes forms containing filters on measures, the cloned forms will have the same filters. However, filters on measures with which an AW dictionary-type parameter is associated will not be cloned.

Locking mechanism

The report locking mechanism also considers parametric custom filters. In fact, it blocks access in data entry mode if the set of data retrieved with the application of custom filters is the same.

Custom filters tab

Excel > CCH® Tagetik > Design > open form > Matrix > Custom Filters

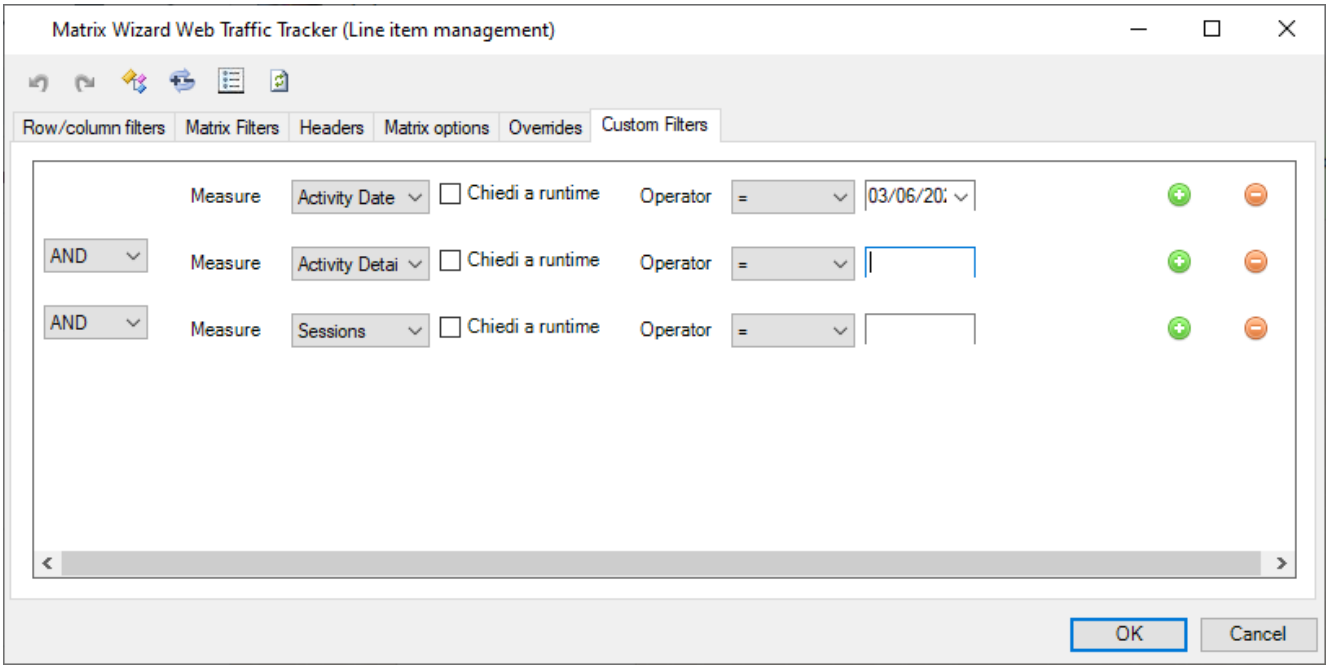
See "How to access the windows" on page 9 for alternative paths.

***IMPORTANT**: this tab is only shown when a transactional matrix refers to an Analytical Information Hub dataset.*

Purpose of the tab

This tab allows you to create a custom filter based on the measures data of the dataset associated with the form.

What it looks like



Page sections

Element	Description
Custom filter conditions definition area	Shows rows for defining the conditions of the filter on the dataset's measures.
Filter management buttons	Allows you to manage filters on measures.

Fields for defining custom filter conditions

Element	Description
Measure	Allows you to select a dataset measure.
Ask at runtime	Allows you to set, on the measure data, a filter on a fixed value or a value obtained from the definition of a custom parameter when the form is run (see "Parameters" on page 58).
Parameter	Only visible with **Ask at runtime** selected. Shows the list of non-dimensional parameters that are compatible with the selected measure and allows you to select one of them to filter the measure data. ***Note**: parameters are compatible when the data type is the same as the measure type (text, number, date), or, for dictionary-type parameters, when the parameter refers to a dictionary belonging to the same Analytical Workspace as the measure and the measure type is text.*
Operator	Allows you to select the operator to be applied to the filter.
Text box	Only visible with **Ask at runtime** not selected and when the measure type is text or numeric. Allows you to enter the value of the filter.
Date selection box	Only visible with **Ask at runtime** not selected and when the measure type is date. Allows you to select a date for the filter.

Filter management buttons

Element	Description
	Adds a filter.
	Removes a filter

The Print Filters function

What is it for

The *Print Filters* function allows you to see a version of the form in which the matrix cells show details of the filters to which the amount is subject, instead of the amount itself. This functionality is useful for the configurator user, because it allows you to consult, in the context of the work desktop, information on the origins of what is set in the form, or in the matrix, form filters or tab filters.

In particular, it makes it easier to consult the dimensions used in the matrix when, in the presence of complex configurations, they cannot be immediately inferred from the rows/columns of the form. For example, the dimensions present on the form run cannot be immediately inferred in a form that displays variations with control groups inserted on the columns and variation accounts inserted on the rows.

Print Filters Wizard

Excel > CCH® Tagetik > Design > open form > Run > Print Filters

Purpose of window

This window allows you to select the attributes of dimensions relative to the form filters to start printing the filters. Once the desired dimensions have been selected, the system generates the filters printout by overlaying on the matrix cell fields showing the requested attributes of the dimensions involved. For more details on cell fields, see "Cell fields" on page 41.

Page sections

Element	Description
Predefined functions tree	Shows the existing dimensions and the relative attributes.
Dimensions area to be printed	Shows a list of the chosen dimensions, which will populate the cell fields on the structure of the matrix in the print filters step.

Buttons on the page

Element	Description
	Adds the selected dimension to the dimensions area.
	Deletes the selected dimension from the dimensions area.
	Allows you to access an advanced editor for the cell field relative to the dimension selected in the dimensions area.
	Moves the selected dimension up/down in the dimensions area.

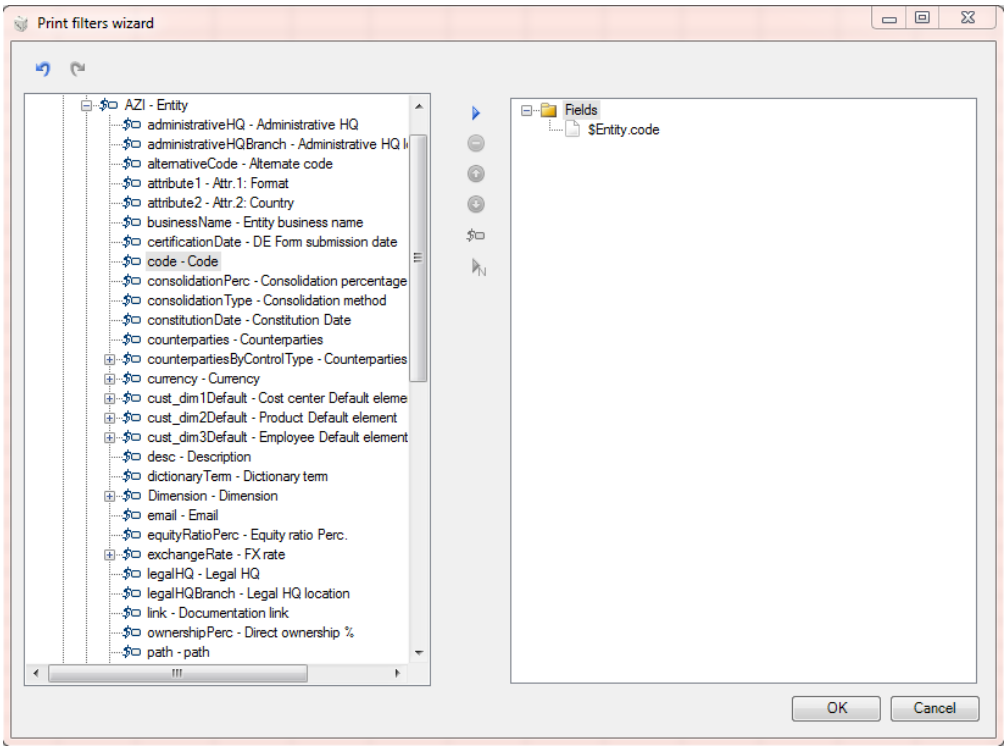
Print Filters examples

Example 1

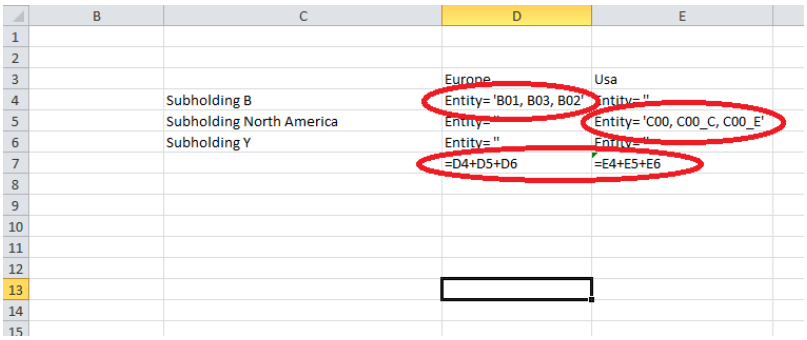
Suppose you have a form in which a filter on a hierarchy of the entity dimension is used as a filter both on the rows and on the columns. The matrix present in the form is therefore of the type shown below

	A	B	C	D	E	F
1						
2						
3				Europe	Usa	
4			Subholding B	1,000	1,000	
5			Subholding North America	1,000	1,000	
6			Subholding Y	1,000	1,000	
7				3,000	3,000	
8						
9						
10						
11						

We insert the $Entity.code cell field in the print filters wizard window.

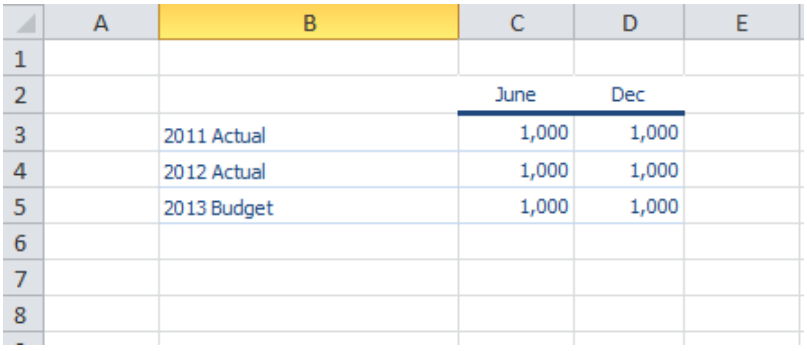


When the OK button is clicked, the system displays the following

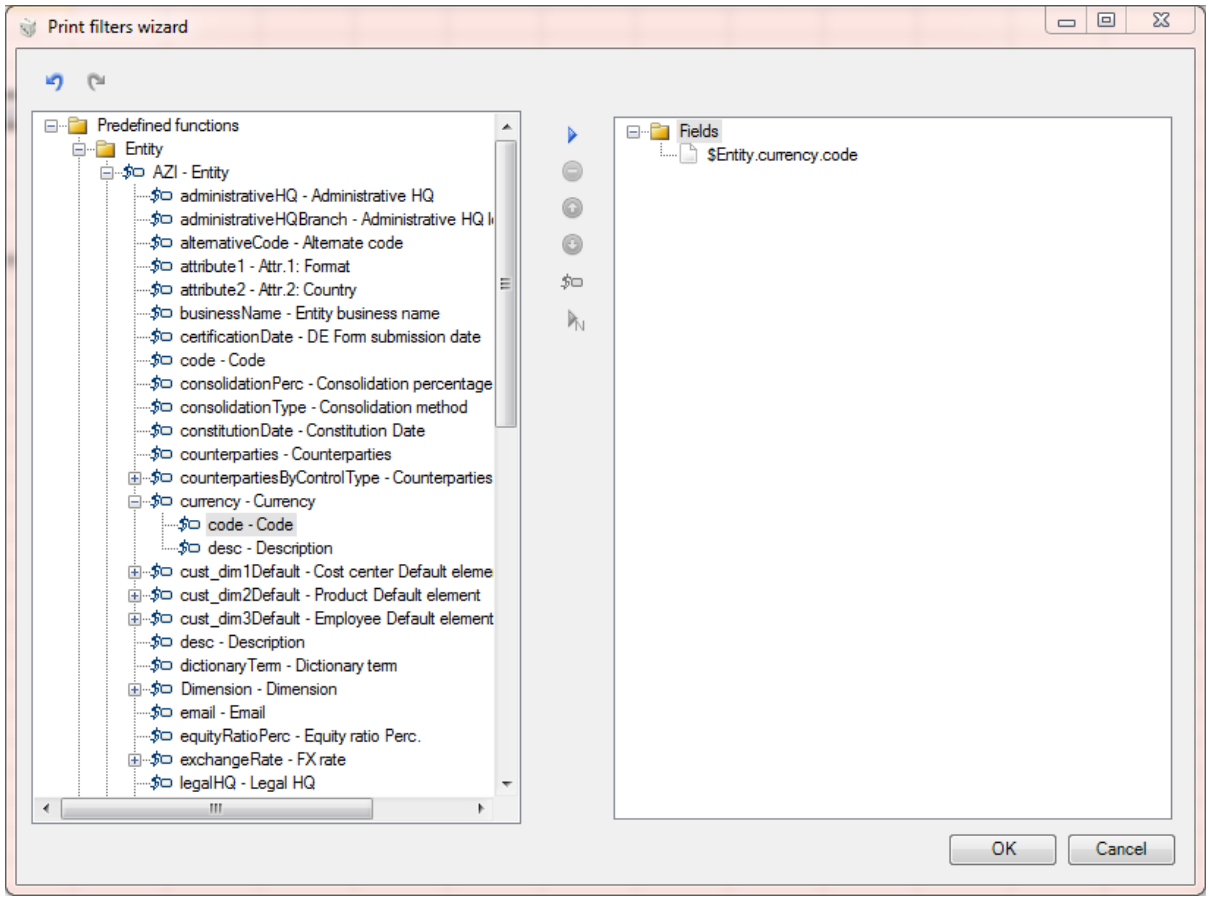


Example 2

Suppose you have a form in which scenarios are on the rows and periods are on the columns, and a matrix filter has also been defined on the entity dimension. The matrix present in the form is therefore of the type shown below



We insert the relevant cell field in the print filters window as necessary to show the entity currency



When the OK button is clicked, the system displays the following

	A	B	C	D
1				
2			June	Dec
3		2011 Actual	Entity= 'EUR'	Entity= 'EUR'
4		2012 Actual	Entity= 'EUR'	Entity= 'EUR'
5		2013 Budget	Entity= 'EUR'	Entity= 'EUR'
6				
7				
8				

Generate a filters printout

Excel > CCH® Tagetik > Design > open form > Run > Print Filters

Select the dimensions for the cell fields of the filters printout

To find out the dimensions of the matrix's row/column filters, click : the List of filters used window will appear (see "List of filters used window" on the facing page).

1. Select the desired dimension or the attribute of the dimension and click to add it to the dimensions area.
2. Repeat step 1 for all desired dimensions.
3. (Optional) Click to run advanced configurations for the cell field relative to the dimension selected in the dimensions area to be printed (for more details on advanced cell field configuration, see "Cell fields" on page 41. Run the advanced configuration and click **OK**.
4. Click **OK**: in the work desktop, the matrix will show the cell fields of the filters printout.

List of filters used window

Excel > CCH® Tagetik > Design > open form > List of filters used

Excel > CCH® Tagetik > Design > open form > open the matrix wizard of one of the matrices > List of filters used

Purpose of window

This window shows a list of all the filters set on the current form.

Data in window

Element	Description
Code	Form code.
Description	Form description.
Form filters	List of filters applied to the form.
<template name> Template	List of filters contained in the template, grouped by filter type. • **Tab filters** • **Matrix Filters** • **Row filters** • **Column filters**
Used parameters	List of opening parameters set on the form.

Buttons on the page

Element	Description
	Copy of the information shown in the window.

Design: functions

Contents

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Dimensional functions

Description

These functions return dimensional objects like nodes of aggregation structures or lowest level elements (leaves) if these form part of a filter (row/column, matrix, sheet, form) to which advanced selection is applied. They are grouped into nodes, one for every CCH Tagetik dimension.

The "$" symbol followed by the dimension name represents the function that returns the dimension object used.

Example

var := $Account; saves the used accounts in the "var" variable.

var := $Scenario; saves the used scenarios in the "var" variable.

Description of dimensional functions

The $ Function

Element	Description
Availability	From CCH Tagetik 3.0912
Applicability	To all dimensions
Syntax	*$<DimensionName>[(boolean)]*
Parameters	(optional) Boolean: • TRUE: allows you to obtain the lowest level elements of the indicated dimensional object. • FALSE: allows you to obtain the indicated dimensional object.
Value returned	Filter defined on the following dimensions: • **$Entity.**: entity dimension • **$Account.**: account dimension • **$Causal.**: causal dimension • **$Cust_Dim1...5.**: custom dimensions • **$Category.**: category dimension • **$AmountType.**: amount type dimension • **$Currency.**: currency dimension • **$Scenario.**: scenario dimension • **$Period.**: period dimension • **$PeriodLength.**: period length dimension • **$ICEntity.**: counterparty entity dimension • **$ICCust_Dim.**: counterparty custom dimension

Example

`$Entity`

Returns the entity on which it is possible to operate with attributes.

Example

`$Entity.code` or `$Entity(TRUE).code`

With a parameter on a hierarchy of the entity dimension.

If the following is selected when running the form...	The function will return...
a lowest level element of the hierarchy	the code of the selected lowest level element
a node of the hierarchy	a string resulting from the concatenation of the codes of all the lowest level elements related to the selected node.

Example

`$Entity(FALSE).code`

With a parameter on a hierarchy of the entity dimension, if the user selects a node of the hierarchy, the code of the selected node will be obtained when the form is run.

Particular features of dimensional functions

For every dimensional function, there is a corresponding function without the $ symbol. With the $, the function returns the current value associated with a dimension; without the $, it returns a description of the dimension.

Examples

`$Entity.desc`: returns the string "Tagetik srl".

`Entity.desc`: returns the string "Entity".

These functions are also useful for recalling elements in forms where the involved dimension is not used. Descriptions are always available and return information on all the system's dimensions, including those not linked to the current filters.

Example

The **$Entity.getLeaf("A1").desc** function can be used to print a cell field with the description of entity "A1". However, this only works if a starting filter is present on the entity dimension and it is therefore possible to identify the lowest level element with code "A1". If there are no filters on the entity dimension, or if the entity dimension is not included in the form, the $Entity expression will not return anything. In this case, the **Entity.getLeaf("A1").desc** function will return the desired information.

Functions without the $ are also applicable to hierarchies: **hierarchy("G1").Entity.getNode("N1").desc** or **Entity("G1").getNode("N1").desc**

Hierarchy

Element	Description
Availability	From CCH Tagetik 3.0912
Applicability	To all dimensions in which hierarchies can be defined
Syntax	*Hierarchy("heirarchy_code")*
Parameters	*"heirarchy_code."*
Value returned	Filter on the dimension hierarchy indicated in "heirarchy_code"

Example

`$Entity(Hierarchy("G1"))`: returns the filter on hierarchy G1 defined on the entity dimension.

Dimensional attributes

Description

The attributes of dimensional functions provide information on the dimensional object called up by the function. The most common attribute is **code**, available for almost all functions. This attribute returns a string of text containing the code of the dimensional object.

Dimensional functions and attributes are typically used to define matrix headers and cell fields. Attributes can be valid for all dimensions or specific for one dimension.

Example

var := $Account.code; saves the code of the used account in the "var" variable.

var := $Account.desc; saves the description of the used account in the "var" variable.

Description of the attributes shared by all dimensions

Attribute 1, ..., Attribute 5

Element	Description
Availability	From 3.0912
Applicability	All dimensions for which custom attributes can be managed (Entity, Account, Custom Dimensions)
Syntax	*Attribute1*
Value returned	String with the value of the custom attribute defined in the elements table of the dimension to which it is applied.

Example

`$Entity.Attribute1`

Returns the value of the entity's attribute 1.

Code

Element	Description
Availability	From 3.0912
Applicability	All objects, dimensional and otherwise
Syntax	*code*
Value returned	String with the dimensional object element code

Example

`$Entity.code`

Returns the entity code specified in the filter (row/column, matrix, sheet, form).

Desc

Element	Description
Availability	From 3.0912
Applicability	All objects, dimensional and otherwise
Syntax	*Desc([language])*
Parameters	• *language*: (Optional) Code of the table language indicated as a whole number.
Value returned	String with the description of the dimensional (or non-dimensional) object element. The optional parameter is only supported if there are multiple descriptions. If omitted, the user's table language is used.

Example

`$Entity.desc(2)`

Returns the description of the entity, specified in the filter (row/column, matrix, sheet, form), in table language 2.

DictionaryTerm

Element	Description
Availability	From 3.1012
Applicability	All dimensional objects (lowest level elements and nodes)
Syntax	*DictionaryTerm([language])*
Parameters	• *language*: (Optional) Code of the table language indicated as a whole number.
Value returned	Any dictionary items associated with the filter (row/column, matrix, sheet, form) or an empty string. The optional parameter is only supported if there are multiple descriptions. If omitted, the user's table language is used.

Examples

- `$Entity.dictionaryTerm`

Returns the description of the dictionary item in the user's table language related to the filter (row/column, matrix, sheet, form) on the entity.

- `$Entity.dictionaryTerm(2)`

Returns the description, in table language 2, of the dictionary item related to the filter (row/column, matrix, sheet, form) on the entity.

Dimension

Element	Description
Availability	From 3.0912
Applicability	All dimensional objects (lowest level elements and nodes)
Syntax	*Dimension*
Attributes	• *code* • *desc* • *hierarchyCode*
Value returned	Dimension or hierarchy to which the filter (row/column, matrix, sheet, form) belongs

Examples

- `$Entity.dimension.code`

Returns the string "AZI", which identifies the entity dimension.

- `$Entity.dimension.desc`

Returns the name assigned to the entity dimension in configuration, e.g. "Legal Entity".

- `$Entity.dimension.hierarchyCode`

Returns the code of the hierarchy to which the entity specified in the filter (row/column, matrix, sheet, form) belongs.

Link

Element	Description
Availability	From 3.1012
Applicability	All dimensional objects (lowest level elements and nodes)
Syntax	*link*
Value returned	Any link to the documentation related to the element of the dimensional object

Example

`$Entity.link`

Returns any link to the documentation related in the table to the entity specified in the filter (row/column, matrix, sheet, form).

LowestReference

Element	Description
Availability	From 3.0912
Applicability	All nodes of a hierarchy
Syntax	*lowestReference*
Value returned	Reference lowest element relative to the node

Example

`$Entity(Hierarchy("01")).lowestReference.desc`

Returns the description of the reference entity relative to the node used as a filter (row/column, matrix, sheet, form) on the entity's hierarchy 01.

Path

Element	Description
Availability	From 4.2.0
Applicability	All dimensions and forms
Syntax	*Path([level], [separator], [cellField])*

Element	Description
Parameters	• *level*. (Optional) The number of levels to go up in the hierarchy (1 = parent, 2 = grandparent, etc.). If omitted or zero, it will go down to the root of the hierarchy. • *separator*. (Optional) Separator to be used between each element in the returned string. If omitted, the backslash "\" will be used. • *cellField*. (Optional) Cell field-type expression to be defined on each node of the hierarchy belonging to the *path*. If omitted, the code will be used. If the *path* function is applied to the form object, this field is not supported and only the code of the group's nodes belonging to the *path* will be used.
Value returned	String obtained from the concatenation of expressions defined on the nodes of the current element's path.

Examples

The following examples look at a filter on the "AG - Geographical area" hierarchy of the entity dimension and presume that the city of Lucca (LU) is related to the Tuscany (TOS) node followed by the Italy (ITA) / Europe (EU) nodes:

- `$Entity.path`

Returns the string "AG\EUR\ITA\TOS\LU".

- `$Entity.path(0)`

Returns the string "AG\EUR\ITA\TOS\LU".

- `$Entity.path(2)`

Returns the string "TOS\LU".

- `$Entity.path(".")`

Returns the string "AG.EUR.ITA.TOS.LU".

- `$Entity.path(3,".")`

Returns the string "ITA.TOS.LU".

- `$Entity.path(3,".","$Entity.desc")`

Returns the string "Italy.Tuscany.Lucca".

With a form related to the "DE - Data Entry" \ "P&L - Profit And Loss" nodes of form group "$", the following behaviour is obtained:

- `Report.path`

Returns the string "$\DE\CE".

- `Report.path(2)`

Returns the string "DE\CE".

- `Report.path(".")`

Returns the string "$.DE.CE".

Segment Type

Element	Description
Availability	From 3.1012
Applicability	All hierarchy nodes
Syntax	*segmentType*
Value returned	Segment type applied to the filter (row/column, matrix, sheet, form)

Example

```
$Entity(Hierarchy("01")).segmentType.desc
```

Returns the description of the segment type relative to the node used as a filter (row/column, matrix, sheet, form) on the entity's hierarchy 01.

Description of attributes for the Entity dimension

The AdministrativeHQ attribute

Element	Description
Availability	From 3.0912
Syntax	*AdministrativeHQ*
Value returned	Name of the entity's administrative headquarters

Example

```
$Entity.administrativeHQ
```

Returns the entity's administrative headquarters.

The AdministrativeHQBranch attribute

Element	Description
Availability	From 3.0912
Syntax	*AdministrativeHQBranch*
Value returned	City of the entity's administrative HQ

Example

```
$Entity.administrativeHQBranch
```

Returns the city of the entity's administrative HQ.

The AlternativeCode attribute

Element	Description
Availability	From 4.2.1
Syntax	*AlternativeCode*
Value returned	Alternative entity code

Example

Entity with code A00 and alternative code A00_ITA.

```
$Entity.alternativeCode
```

Returns the string "A00_ITA".

The BusinessName attribute

Element	Description
Availability	From 3.0912
Syntax	*businessName*
Value returned	The entity's business name

Example

```
$Entity.businessName
```

Returns the entity's business name.

The CertificationDate attribute

Element	Description
Availability	From 3.0912
Syntax	*certificationDate*
Value returned	The entity's submission date

Example

```
$Entity.certificationDate
```

Returns the entity's submission date/phase containing the form for the process to which scenarios/periods present in the filter (row/column, matrix, sheet, form) are related.

The ConsolidationPerc attribute

Element	Description
Availability	From 3.0912
Syntax	*consolidationPerc*
Value returned	The entity's consolidation percentage

Example

```
$Entity.consolidationPerc
```

Returns the entity's consolidation percentage.

The ConsolidationType attribute

Element	Description
Availability	From 3.0912
Syntax	*consolidationType*
Value returned	The entity's consolidation type

Example

```
$Entity.consolidationType
```

Returns the entity's consolidation type.

The ConstitutionDate attribute

Element	Description
Availability	From 5.0
Syntax	*constitutionDate*
Value returned	The date on which the entity was established

Example

```
$Entity.constitutionDate
```

Returns the date on which the entity was established.

The Counterparties attribute

Element	Description
Availability	From 3.0912
Syntax	*counterparties*
Value returned	Counterparty entities related to the entity

Example

`$Entity.counterparties.code`

Returns the codes of the counterparty entities related to the entity.

The CounterpartiesByControlType attribute

Element	Description
Availability	From 3.0912
Syntax	*counterpartiesByControlType*
Attributes	• *code* • *desc*
Value returned	Counterparty entities related to the current entity by scenario/period (control chart). Requires a filter (row/column, matrix, sheet, form) on the scenario and period dimensions.

***IMPORTANT**: this attribute always returns elements belonging to the "Counterparty Entity" dimension. The elements obtained can then only be used directly within advanced selection scripts related to the "Counterparty Entity" dimension (see "Details of advanced selection scripts for calculating the default value" on page 63).*

Example

`$Entity.counterpartiesControlType.code`

Returns the codes of the counterparty entities related to the entity within the control chart.

The Currency attribute

Element	Description
Availability	From 3.0912
Syntax	*currency([scenario],[period])*
Attributes	• *code* • *desc*
Parameters	• *scenario* (optional) Scenario code • *period* (optional) Period code
Value returned	Entity currency specified in the filter (row/column, matrix, sheet, form) for the process to which scenarios/periods present in the filter are related.

Examples

- `$Entity.currency.code`

Returns the currency code associated with the entity specified in the filter for the process to which scenarios/periods present in the filter are related.

- `$Entity.currency.code.getLeaf()`

Returns the filter on the currency associated with the entity

- `$Entity.currency(column("MATRIX1",1).$Scenario,column ("MATRIX1",1).$Period.code`

Returns the currency code associated with the entity specified in the filter for the process to which scenarios/periods present on the first column of "MATRIX1" are related.

The Cust_dim1Default, Cust_dim2Default, ... Cust_dim5Default attribute

Element	Description
Availability	From 4.2.1
Syntax	*cust_dim1Default*
Attributes	• *code* • *desc*
Value returned	Default element of the custom dimension associated with the entity specified in the filter (row/column, matrix, sheet, form) extracted from the elements table

Example

```
$Entity.cust_dim1Default.code
```

Returns the code of the default element of custom dimension 1 associated with the entity specified in the filter.

The Email attribute

Element	Description
Availability	From 5.0
Syntax	*email*
Value returned	Entity email

Example

```
$Entity.email
```

Returns the entity's email.

The EquityRatioPerc attribute

Element	Description
Availability	From 3.0912
Syntax	*equityRatioPerc([scenario],[period])*
Parameters	• *scenario* (optional) Scenario code • *period* (optional) Period code
Value returned	The entity's equity ratio percentage

Examples

- `$Entity.equityRatioPerc`

Returns the equity ratio percentage.

- `$Entity.equityRatioPerc(column("MATRIX 1",1).$Scenario,column("MATRIX 1",1).$Period)`

Returns the equity ratio percentage for scenarios/periods indicated as parameters, not those specified in the filter (row/column, matrix, sheet, form).

The ExchangeRate attribute

Element	Description
Availability	From 3.0912
Syntax	*exchangeRate([scenario],[period],[entity])*
Applicability	Entity or currency
Parameters	• *scenario* (optional) Scenario code • *period* (optional) Period code • *entity* (optional) Entity code
Attributes	• *average* Average FX rate • *final* Final FX rate • *period*: Period average FX rate
Value returned	FX rate of the currency used as the filter (row/column, matrix, sheet, form) or of the entity indicated as the parameter

Examples

- `$Currency.exchangeRate.average`

Returns the average FX rate of the entity currency for the scenario/period/entity defined in the filter.

- `$Currency.exchangeRate.final`

Returns the final FX rate of the entity currency for the scenario/period/entity defined in the filter.

- `$Currency.exchangeRate.periodic`

Returns the period average FX rate of the entity currency for the scenario/period/entity defined in the filter.

- `$Currency.exchangeRate(matrix("MATRIX 1").$Scenario, matrix("MATRIX 1").$Period, matrix("MATRIX 1").$Entity).average`

Returns the average FX rate of the entity currency for the scenario/period/entity defined in "Matrix 1".

The LegalHQ attribute

Element	Description
Availability	From 3.0912
Syntax	*legalHQ*
Value returned	The entity's legal headquarters

Example

`$Entity.legalHQ`

Returns the entity's legal headquarters.

The LegalHQBranch attribute

Element	Description
Availability	From 3.0912
Syntax	*legalHQBranch*
Value returned	City of the entity's legal HQ

Example

`$Entity.legalHQBranch`

Returns the city of the entity's legal HQ.

The OwnershipPerc attribute

Element	Description
Availability	From 3.0912
Syntax	*ownershipPerc([scenario],[period])*
Parameters	• *scenario* (optional) Scenario code • *period* (optional) Period code
Value returned	The entity's ownership percentage

Examples

- `$Entity.ownershipPerc`

Returns the ownership percentage.

- `$Entity.ownershipPerc(column("MATRIX 1",1).$Scenario,column("MATRIX 1",1).$Period)`

Returns the ownership percentage relative to the scenario/period defined as a parameter of the function and not for the scenario/period specified in the filter.

The Phone attribute

Element	Description
Availability	From 5.0
Syntax	*phone*
Value returned	Entity telephone number

Example

```
$Entity.phone
```

Returns the entity's telephone number.

The ProcessStep attribute

Element	Description
Availability	From 3.0912
Syntax	*processStep*
Attributes	• *code* • *desc*
Value returned	Step related to the entity for the process to which the scenario/process present in the filter (row/column, matrix, sheet, form) is related

Example

```
$Entity.processStep.code
```

Returns the code of the step associated with the entity specified in the filter.

The ProportionalityPerc attribute

Element	Description
Availability	From 3.0912
Syntax	*proportionalityPerc*
Parameters	• *scenario* (optional) Scenario code • *period* (optional) Period code
Value returned	The entity's proportionality percentage

Examples

- `$Entity.proportionalityPerc`

Returns the entity's proportionality percentage.

- `$Entity.proportionalityPerc(column("MATRIX 1",1).$Scenario,column("MATRIX 1",1).$Period)`

Returns the entity's proportionality percentage for scenarios/periods processed as parameters of the function, not for those specified in the filter.

The Snapshot attribute

Element	Description
Availability	From 3.0912
Syntax	*snapshot*
Value returned	Information related to the entity/scenario/period triplet when a snapshot scenario is created

Examples

- `$Entity.snapshot.comment`

Returns the comment inserted when the snapshot scenario for entity/scenario/periods present in the filter was created.

- `$Entity.snapshot.desc`

Returns the description inserted when the snapshot scenario for the entity/scenario/period present in the filter was created.

The Variable attribute

Element	Description
Availability	From 3.0912
Syntax	*variable(code)*
Parameters	• *code* Classifier code
Attributes	• *code* • *desc*
Value returned	The Variable element associated with the entity specified in the filter

Example

`$Entity.variable("V01").desc`

Returns the description of variable "V01" associated with the entity specified in the filter.

Description of attributes for the Account dimension

The AccountsToDetail attribute

Element	Description
Availability	From 3.0912
Syntax	*accountsToDetail*
Value returned	Set of parent account of the control group to be detailed. In case of: • **filter on a control group**, it returns the group's parent account • **filter on the account** it returns the parent accounts of the control group to which the account belongs

Example

`$Account.accountsToDetail`

Returns all accounts to be detailed of the control group specified in the filter (row/column, matrix, sheet, form) or of the control group to which the account specified in the filter belongs.

The CausalsGroup attribute

Element	Description
Availability	From 3.0912
Syntax	*causalsGroup*
Attributes	• *code* • *desc*
Value returned	Any node of causals used to create the control group specified in the filter

Examples

- `$Account.causalsGroup.code`

Returns the node of causals used to create the control group specified in the filter.

- `$Account.causalsGroup.lowest`

Returns all individual causals of the node of causals used to create the control group specified in the filter.

The GroupType attribute

Element	Description
Availability	From 3.0912
Syntax	*groupType*
Attributes	• *code* • *desc*
Value returned	Type (code and description) of the control group specified in the filter. The following values are returned: • D - Details • F - Variation • Q - Balancing • C - Coherence In case of: • **filter on control group** returns the control group type • **filter on account** returns the type of the control group to which the account belongs

Example

With filters on a Variation-type control group:

`$Account.groupType.code`: returns string "F"

`$Account.groupType.desc`: returns the “Variation" string or equivalent string, depending on the application language set by the user

The OriginalAccounts attribute

Element	Description
Availability	From 3.0912
Syntax	*originalAccounts*
Value returned	Set of accounts related to a FST item. In case of: • **filter on account**, it returns the same account • **FST items** and **nodes of a hierarchy defined on the account dimension**, it is equivalent to the children() function and returns the related accounts • **formula-type FST items**, it progressively returns the accounts related to all FST items involved in the formula, including those involved indirectly, depending on the items to which other filters are related in a chain, up to the one to which the accounts are related. • **control group/causal intersection**, it returns the account present in the intersection

The OriginalControlGroup attribute

Element	Description
Availability	From 3.0912
Syntax	*originalControlGroup*
Value returned	Control group associated with a filter on an account. In case of: • **filter on a control group**, it returns the control group • **filter on account** taken from **within a control group**, it returns the control group to which the account belongs • **filter on account taken from elsewhere** (hierarchies, FSTs), it returns all control groups that detail the account

Examples

- `$Account.originalControlGroup.code`

Returns the control group to which the account specified in the filter belongs

- `$Account.originalControlGroup.controlGroup`

Deploys the control group to which the account specified in the filter belongs, like in the Matrix Wizard.

- `$Account.originalControlgroup.causalsGroup.code`

Returns, given a filter on an account, the code of the detail-type node related to the control group to which the account belongs.

The Schema attribute

Element	Description
Availability	From 3.0912
Syntax	*template*
Attributes	• code • desc • taxonomy
Value returned	Financial Statement Template to which the account specified in the filter belongs. If the account was chosen from a hierarchy defined on the account dimension, it returns an empty string

Examples

`$Account.schema.code`

Returns the code of the FST to which the account specified in the filter belongs.

`$Account.schema.desc`

Returns the description of the FST to which the account specified in the filter belongs.

The ShortDesc attribute

Element	Description
Availability	From 3.0912
Syntax	*shortDesc*
Value returned	Short description in the language of the user of the specified account

Example

`$Account.shortDesc`

Returns the description of the account specified in the filter.

The XBRLName attribute

Element	Description
Availability	From 3.1012
Syntax	*xbrlName*
Value returned	Value of the XBRL name field associated with a filter on the account. In particular, in case of: • **a filter on an FST item**, it returns the value of the XBRL name field associated with the item • **a filter on account** taken from **within an FST**, it returns the value of the XBRL name field associated with the FST item to which the account is related • in all other cases, it returns an error

Example

`$Account.xbrlName`

Returns the value of the XBRL name field associated with the account.

Description of attributes for the Scenario dimension

The CertificationDate attribute

Element	Description
Availability	From 3.0912
Syntax	*certificationDate*
Value returned	The entity's submission date

Example

```
$Entity.certificationDate
```

Returns the entity's submission date for the process to which scenario and period present in the filter are related.

The ConsolidationScenario attribute

Element	Description
Availability	From 3.0912
Syntax	*consolidationScenario*
Attributes	• *code* • *desc*
Value returned	If applied to an original scenario, it returns the associated consolidation scenario or set of consolidation scenarios. If applied to a consolidation scenario, it returns the same scenario.

Examples

- `$Scenario.consolidationScenario.code`

Returns the code of the consolidation scenario associated with the original scenario specified in the filter.

- `Scenario.consolidationScenario("true").code`

Returns the code of the consolidation scenario associated with the original scenario specified in the filter. In the event that the scenario is included in more than one group, it is shown below every group in which it has been inserted.

- `$Scenario.consolidationScenario("false").code`

Returns the code of the consolidation scenario associated with the original scenario specified in the filter

The CurrencyScenario attribute

Element	Description
Availability	From 3.1012
Syntax	*currencyScenario*
Attributes	• *code* • *desc*
Value returned	Currency of the consolidation scenario specified in the filter. If applied to an original scenario, it returns an empty string.

Example

```
$Scenario.currencyScenario.code
```

Returns the currency code of the consolidation scenario specified in the filter.

The ExchangeRate attribute

Element	Description
Availability	From 3.0912
Syntax	*exchangeRate([scenario],[period],[entity])*
Applicability	Entity or currency
Parameters	• *scenario* (Optional) Scenario code • *period* (Optional) Period code • *entity* (Optional) Entity code
Attributes	• *average* Average FX rate • *final* Final FX rate • *period*: Period average FX rate
Value returned	FX rate of the currency used as the filter or of the entity indicated as the parameter

Examples

- `$Currency.exchangeRate.average`

Returns the average FX rate of the entity currency for the scenario/period/entity defined in the filter.

- `$Currency.exchangeRate.final`

Returns the final FX rate of the entity currency for the scenario/period/entity defined in the filter.

- `$Currency.exchangeRate.periodic`

Returns the period average FX rate of the entity currency for the scenario/period/entity defined in the filter.

- `$Currency.exchangeRate(matrix("MATRIX 1").$Scenario, matrix("MATRIX 1").$Period, matrix("MATRIX 1").$Entity).average`

Returns the average FX rate of the entity currency for the scenario/period/entity defined in "Matrix 1".

The Holding attribute

Element	Description
Availability	From 3.0912
Syntax	*holding*
Attributes	• *code* • *desc*
Value returned	Holding of the consolidation scenario specified in the filter

Example

`$Scenario.holding.code`

Returns the code of the holding.

The Next/Prev/Ref1/Ref2/.../Ref5 attribute

Element	Description
Availability	From 3.0912
Syntax	*next, prev, ref1, ref2, ... ref5*
Attributes	• code • desc
Value returned	Value of the relative field for the scenario specified in the filter. In the scenarios table, the attributes and fields correspond as follows: • **next** - next scenario • **prev** - previous scenario • **ref1** - reference scenario 1 • **...** • **ref5** - reference scenario 5

Examples

- `$Scenario.next.code`

Returns the code of the scenario following the one specified in the filter.

- `$Scenario.prev.code`

Returns the code of the scenario prior to the one specified in the filter.

- `$Scenarioref1.code`

Returns the code of reference scenario 1 specified in the filter.

- `$Scenario.next.next.desc`

Returns the description of the scenario following the scenario following the scenario specified in the filter.

- `$Scenario.prev.ref1.desc`

Returns the description of reference scenario 1 of the scenario prior to the one specified in the filter.

The NumCopia attribute

Element	Description
Availability	From 3.1109
Syntax	*numCopia*
Attributes	• code • desc
Value returned	Version of the snapshot scenario

Example

With an original 2012ACT scenario for which three snapshot scenarios are defined:

- Version 01: 2012ACT_V1
- Version 02: 2012ACT_V2
- Version 03: 2012ACT_V3

If scenario 2012ACT_V2 is used in the filters, then the `$Scenario.numCopia` syntax returns string "2".

The Opening attribute

Element	Description
Availability	From 4.2.0
Syntax	*opening*
Attributes	• *CumulatedDays* • *endDate* • *MonthDays* • *startDate*
Value returned	Given a scenario, it returns the opening scenario/period.

Example

`$Scenario.opening.startdate`

Returns the start date of the opening period.

The OriginalScenario attribute

Element	Description
Availability	From 3.0912
Syntax	*originalScenario*
Parameters	("true/false")

Element	Description
Attributes	• *code* • *desc*
Value returned	If applied to a consolidation scenario, it returns the associated original scenario. If applied to an original scenario, it returns the same scenario.

Example

- `$Scenario.originalScenario.code`
  - Returns the code of the original scenario associated with the consolidation scenario specified in the filter.

    `Scenario.originalScenario("true").code`

Returns the code of the original scenario associated with the consolidation scenario specified in the filter. In the event that the scenario is included in more than one group, it is shown below every group in which it has been inserted.

- `$Scenario.originalScenario("false").code`

Returns the code the original scenario associated with the consolidation scenario specified in the filter.

The OriginalSnapshotScenario attribute

Element	Description
Availability	From 3.0903
Syntax	*originalSnapshotScenario*
Attributes	• *code* • *desc*
Value returned	Given a snapshot scenario, it returns the linked original scenario.

Example

`$Scenario.originalSnapshotScenario.code`

Returns the code of the original scenario linked to the snapshot scenario.

The ScenarioPeriod attribute

Element	Description
Availability	From 3.0912
Syntax	*scenarioPeriod*
Attributes	• *CumulatedDays* • *endDate* • *MonthDays* • *Quarter* • *startDate* ***Note**: the "quarter" attribute is used to refer to the current quarter. In the editor window, it is also possible to navigate between quarters using the offsetSP, firstSP and lastSP functions.*
Value returned	Scenario/period pair associated with the filters on the scenario and period dimensions. Can be applied to both the scenario and the period dimension.

Examples

- `$Scenario.ScenarioPeriod.endDate`

Returns the scenario/period end date associated with the scenario specified in the filter.

- `$Period.ScenarioPeriod.endDate`

Returns the scenario/period end date associated with the period specified in the filter.

- `$Period.ScenarioPeriod.(Period.getLeaf("04")).endDate`

Returns the period end date with code 04.

"Quarter" attribute examples

With scenario "2014" and period "02" as input filters, the following behaviours may take place:

- *quarter.firstSP.endDate*

Returns the date of the last day in the first period of the quarter containing the current period ("31/01/2014").

- *quarter.lastSP.endDate*

Returns the date of the last day in the last period of the quarter containing the current period ("31/03/2014").

- *quarter.offsetSP(1).firstSP.EndDate*

Returns the date of the last day in the first period of the quarter following the current quarter ("30/04/2014").

- *quarter.offsetSP(-1).firstSP.EndDate*

Returns the date of the last day in the first period of the quarter prior to the current quarter ("31/10/2013").

The ScenarioPeriod attribute

Element	Description
Availability	From 5.1.0
Syntax	*scenarioPeriod*
Attributes	• *desc*
Value returned	Description of the period of the scenario specified in the filter

Example

```
$Scenario.scenarioPeriod.Description
```

Returns the description of the period of the specified scenario.

The Periods attribute

Element	Description
Availability	From 5.3.SP14
Syntax	*Periods*
Attributes	• *code* • *desc*
Value returned	List of periods of the scenario specified in the filter

Example

```
$Scenario(@NOME_PARAMETRO).Periods
```

Returns the list of periods of the scenario specified in the filter.

The SnapshotScenarios attribute

Element	Description
Availability	From 3.0903
Syntax	*snapshotScenarios*
Attributes	• *code* • *desc*
Value returned	Set of all snapshot scenarios related to the scenario specified in the filter

Example

```
$Scenario.snapshotScenario.item(1).code
```

Returns the code of the first snapshot scenario related to the scenario specified in the filter.

The Version attribute

Element	Description
Availability	From 3.0912
Syntax	*version*
Value returned	Value of the "Version" field in the scenarios table for the scenario specified in the filter

Example

```
$Scenario.version
```

Returns the version of the scenario specified in the filter.

The Year attribute

Element	Description
Availability	From 3.0912
Syntax	*year*
Value returned	Value of the "Year" field in the scenarios table for the scenario specified in the filter

Example

```
$Scenario.year
```

Returns the year of the scenario specified in the filter.

Description of attributes for the Period dimension

The CertificationDate attribute

Element	Description
Availability	From 3.0912
Syntax	*certificationDate*
Value returned	The entity's submission date

Example

```
$Entity.certificationDate
```

Returns the entity's submission date for the process to which scenarios/periods present in the filter are related.

The DescNumPeriod attribute

Element	Description
Availability	From 3.1109
Syntax	*descNumPeriod*
Value returned	Value of the "Period number" field in the periods table for the specified period

Example

```
$Period.descNumPeriod
```

Returns the number of the period specified in the filter.

The ExchangeRate attribute

Element	Description
Availability	From 3.0912
Syntax	*exchangeRate([scenario],[period],[entity])*
Applicability	Entity or currency
Parameters	• *scenario* (Optional) Scenario code • *period* (Optional) Period code • *entity* (Optional) Entity code
Attributes	• *average* Average FX rate • *final* Final FX rate • *period*: Period average FX rate
Value returned	FX rate of the currency used as the filter or of the entity indicated as the parameter of the function

Examples

- `$Currency.exchangeRate.average`

Returns the average FX rate of the entity currency for the defined scenario/period/entity.

- `$Currency.exchangeRate.final`

Returns the final FX rate of the entity currency for the defined scenario/period/entity.

- `$Currency.exchangeRate.periodic`

Returns the period average FX rate of the entity currency for the defined scenario/period/entity.

- `$Currency.exchangeRate(matrix("MATRIX 1").$Scenario, matrix("MATRIX 1").$Period, matrix("MATRIX 1").$Entity).average`

Returns the average FX rate of the entity currency for the scenario/period/entity defined in "Matrix1".

Offset attribute

Element	Description
Availability	From 3.1109
Syntax	*offset*
Parameters	• *offset* Deviation from the starting period
Value returned	Period obtained based on the period indicated when applying the offset. If the shift is greater than the number of periods present in the scenario, prior or subsequent to the indicated period, the system creates a chain of periods, taking account of the previous and subsequent relationships of the scenarios.

Example

With filter on Period = April and Scenario = 2013ACT and 2012ACT = previous scenario. Scenarios are made up of 12 periods.

- `$Period.offset(-3).code`

Returns the code of the "January" period

- `$Period.offset(-5).code`

Returns the code of the "November" period

The ScenarioOffset attribute

Element	Description
Availability	From 3.1109
Syntax	*scenarioOffset*
Parameters	• *offset* Deviation from the starting period
Value returned	Scenario obtained from the "offset" of the periods indicated in the parameter. Returns the scenario indicated in the filter if the shift does not give rise to scenario offsets.

Example

With filter on Period = April and Scenario = 2013ACT and 2012ACT = previous scenario. Scenarios are made up of 12 periods.

- `$Period.scenarioOffset(-6).code`

Returns the code of the previous scenario (2012ACT)

- `$Period.scenarioOffset (-3).code`

Returns the code of the filter scenario (2013ACT)

The ScenarioPeriod attribute

Element	Description
Availability	From 3.0912
Syntax	*scenarioPeriod*
Attributes	• *CumulatedDays* • *endDate* • *MonthDays* • *startDate*
Value returned	Scenario/period pair associated with the filters on scenario and period dimensions. Can be applied to both the scenario and the period dimension.

Examples

- `$Scenario.ScenarioPeriod.endDate`

Returns the scenario/period end date associated with the scenario.

- `$Period.ScenarioPeriod.endDate`

Returns the scenario/period end date associated with the period.

- `$Period.ScenarioPeriod.(Period.getLeaf("04")).endDate`

Returns the period end date (overridden) with code 04.

Description of attributes for other dimensions

Attributes for the Column form dimension (of a transactional form)

The ColAttribute attribute

Element	Description
Availability	From 3.1109
Syntax	*colAttribute(codAttr)*
Parameters	*codAttr* Attribute code
Value returned	Value of the codAttr attribute associated with the column of a transactional form

Example

`$Formcolonna.colAttribute(“XYZ”)`

Returns the value of the attribute that has code "XYZ" associated with the current column.

Form attribute

Element	Description
Availability	From 3.1109
Syntax	*form*

Element	Description
Attributes	• *code* • *desc*
Value returned	Form to which the current column belongs

> ### Example
> `$Formcolumn.form.code`
> Returns the code of the form to which the current column belongs.

Attributes for Measure dimensions

The MeasureType attribute

Element	Description
Availability	From 5.3.4
Syntax	*measureType*
Value returned	Measure type (numeric, date, text)

> ### Example
> If the form uses a measure with dimension code "ME_001_000001":
> • `$ME_001_000001.measureType`
> It returns the measure type indicated in the filter.

Non-dimensional functions

Description

Non-dimensional functions return non-dimensional objects like processes, users or forms.
They present their attributes.

> ### Examples
> • `var := User.code;`
> Saves, in the "var" variable, the code of the user who ran the form.
> • `var := Form.desc;`
> Saves, in the "var" variable, the description of the form in use in the language set in the user table.

Description of non-dimensional functions

The Dataprocessing function

Element	Description
Availability	From 3.1106
Syntax	*dataprocessing*
Attributes	• *desc* • *dataprocessingcount*

> ### Examples
> • `dataprocessing.desc`

Returns the string "Data Processing in progress", in the user's application language, in the presence of data processing operating on the set of dimensions defined by the form filters, otherwise it returns an empty string.

- `dataprocessing.dataprocessingcount`

Returns the number of data processes in progress on the set of dimensions defined by the form filters.

The Datasource function

Element	Description
Availability	From 3.1103
Syntax	*datasource*
Attributes	• *code* • *desc*

Examples

- `Datasource.code`

Returns the code of the datasource in use.

- `Datasource.desc`

Returns the description of the datasource in use.

The Process function

Element	Description
Availability	From 3.0903
Syntax	*process*([scenario],[period])
Attributes	• *code* • *desc* • *BusinessCycle* • *Scenarios* • *Periods*
Parameters	• *(Optional) scenario: scenario code* • *(Optional) period: period code*

Examples

- `Process.code`

Returns the code of the process on which data entry is carried out. If the form is not run in data entry mode, an error is generated.

- `Process.desc`

Returns the description of the process on which data entry is carried out. If the form is not run in data entry mode, an error is generated.

- `Process($Scenario.code,$Period.code).code`

Returns the code of the process to which scenarios/periods present in the form/tab filter are related. If the scenario-period combination has no related process, this function will generated an error.

- `Process($Scenario.code,$Period.code).desc`

Returns the description of the process to which scenarios/periods present in the form/tab filter are related. If the scenario-period combination has no related process, this function will generated an error.

The BusinessCycle function

Only used as a *Process* attribute. Returns the process submission type.

Element	Description
Availability	From 5.3.11
Syntax	*process.BusinessCycle*
Attributes	• *code* • *desc*
Parameters	No parameter.

Examples

- `Process.BusinessCycle.code`

Returns the process submission type code. For single submission, it returns "U", for scenarios/periods, it returns "S".

- `Process.BusinessCycle.desc`

Returns the description of the process submission type. For single submission, it returns "Single", for scenarios/periods, it returns "Scenario/Period".

The Scenarios function

Only used as a *Process* attribute. Returns a list of the scenarios associated with the process.

Element	Description
Availability	From 5.3.11
Syntax	*process.Scenarios*
Attributes	• *code*
Parameters	No parameter.

Example

`Process.Scenarios.code`

Returns the codes of the scenarios associated with the current process.

Element	Description
Availability	From 5.3.14
Syntax	*process([processo]).Scenarios*
Attributes	• *code*
Parameters	(Optional) process: process code

Example

`Process(@PARAMETRO_PROCESSO).Scenarios`

Returns the codes of the original and consolidation scenarios, associated with the process specified via a dynamic parameter. This function can be inserted in a scenario-type parameter based either on the flat dimension or on a hierarchical dimension.

The Periods function

Only used as a *Process* attribute. Returns a list of the periods associated with the process regardless of the scenario to which they belong (can only be used within DTPs and Parametric Jobs).

Element	Description
Availability	From 5.3.SP14
Syntax	*process.Periods*
Attributes	*code*
Parameters	(Optional) process: process code

Examples

- `Process.Periods`

Returns the codes of the periods associated with the current process.

- `Process(@COD_PROCESSO).Periods`

Returns the codes of the periods associated with the process specified as the parameter.

The Item(n) function

Function only used as a *Scenarios* attribute.

Element	Description
Availability	From 5.3.11
Syntax	*process.Scenarios.Item(n)*
Parameters	*n*: scenario to be targeted, ordered by code from 0 in ascending alphabetical order.
Value returned	Scenario n from among those associated with the process

The ProcessScenarioPeriods function

Function that can be used in the DTP run or Parametric Jobs window.

Element	Description
Availability	From 5.3.SP14
Syntax	*ProcessScenarioPeriods(codProcesso,codScenario)*
Parameters	• *codProcesso*: process code • *codScenario*: scenario code
Value returned	Available (not locked) periods of a scenario belonging to the selected process

Examples

- `ProcessScenarioPeriods("ACT","2019BDG")`

If there is a process with code "ATC" containing a scenario with code "2019BDG", it returns the non-locked periods of the scenario associated with the process.

- `ProcessScenarioPeriods(@COD_PROCESSO,@COD_SCENARIO)`

If a parameter with code @COD_PROCESSO and a scenario-type parameter with code @COD_SCENARIO are selected, and the scenario belongs to the process, it returns the non-locked periods of that specific scenario associated with the process.

The Server function

Element	Description
Availability	From 4.2.1
Syntax	*server*
Attributes	*url*

Example

`Server.url`

Returns the string containing the url of the server to which the client is connected.

The User function

Element	Description
Availability	From 3.0903
Syntax	*user*

Element	Description
Value returned	User who performed the login (current user)
Attributes	• *code* • *desc* • *appLang* ◦ *code* ◦ *desc* ◦ *langIndex* • *tablesLang* ◦ *code* ◦ *desc* ◦ *langIndex*

Examples

- `User.code`

Returns the code of the current user.

- `User.desc`

Returns the description of user.

- `User.appLang.code`

Returns the code of the user's application language.

- `User.appLang.desc`

Returns the description of the user's application language.

- `User.appLang.langIndex`

Returns the whole number by which the user's application language is identified.

- `User.tablesLang.code`

Returns the code of the user's table language.

- `User.tablesLang.desc`

Returns the description of the user's table language.

- `User.tablesLang.langIndex`

Returns the whole number identifying the user's table language.

The Dict function

Element	Description
Availability	From 4.3.2
Syntax	*dict(codVoceDizionario, [language])*
Value returned	String containing the description of the dictionary item identified by *codVoceDizionario*.
Parameters	• *codVoceDizionario*: shared dictionary item code • [*language*] (optional): language in which the dictionary item description is returned (numbers from 0 to 4). If omitted, it returns the description in the language in which the form was run.

The Datasetpartition function

Element	Description
Availability	From 5.3.SP7
Syntax	*DATASETPARTITION(codWorkspace, codDataset, [codProcesso], [codScenario], [codPeriodo], [codAzienda], [codTipologia], [valoreSeMancante])*

Element	Description
Value returned	String containing the Dataset physical partition name identified by the codDataset code and belonging to the workspace with the codWorkspace code. Based on the partitioning type, the other additional parameters need to be specified.
Parameters	• *codWorkspace*. Workspace code • *codDataset*. Dataset code • *codProcesso*. Process code • *codScenario*. Scenario code • *codPeriodo*. Period code • *codAzienda*. Entity code • *codTipologia*. Type code • *valoreSeMancante*. (optional) String returned if the partition does not exist

Particular features for parameters

Parameters can be specified using fixed values (between double inverted commas) or using the dedicated syntax. See "Details of advanced selection scripts for calculating the default value" on page 63.

The values of the parameters, separated by commas, are interpreted based on the position in which they are specified (the first string is interpreted as the Workspace code, the second as the Dataset code, etc.). To omit an optional parameter, specify an empty string in the relative position, between double inverted commas or otherwise.

- With **Dataset not partitioned**: set the first two parameters only. If specified, the other parameters will be ignored by the system.
- With **Dataset partitioned by Business Cycle**: also specify the Process. If the Process is submission by Scenario/Period, the Scenario and Period will also need to be specified, otherwise the *codScenario* and *codPeriodo* parameters are optional and will be ignored by the system if specified. If the process is single submission, only the Process needs to be specified, in addition to the reference dataset and workspace.
- With **Dataset partitioned by Business Cycle and Entity**: specify the Process and Entity. If the chosen Process is submission by Scenario/Period, the Scenario and Period will also need to be specified, otherwise the *codScenario* and *codPeriodo* parameters will be ignored by the system. These parameters must in any case be set, but this can be done using an empty value.
  If the **dataset is partitioned by entity**, then the entity must be specified.
  If some parameters are not specified because they are not required by the partitioning type, then they will be ignored. When such parameters are not necessary, the user can avoid specifying them, using several consecutive commas (as many as there are parameters to ignore) or empty strings.

To access data of a different type from the Reporting Layer: also specify the type parameter, in addition to the various parameters for the partitioning type used.

In general, access to a calculation type requires the user to specify the Entity (or the node of Entities - hierarchy code *pipe*node code) that they want to access: to dynamically resolve the lowest level element or the Entity node of the context, the entity parameter can be specified with the following syntax:
`if($Entity.isleaf,$Entity.code,$Entity.Dimension.hierarchyCode +"|" + $Entity(False).code)`

For example, it is possible to obtain the partition with DatasetPARTITION("001", "000007", "ACT", "2019ACT", "05", , "TYPOL") of the dataset partitioned by business cycle with code "000007" in the workspace with code "001", given the process is submission by scenario/process, relative to the "ACT" process, period "05" of scenario "2019ACT", with type "TYPOL" but not partitioned by entity. The same behaviour is obtained using the following wording, with the inverted commas: DatasetPARTITION("001", "000007", "ACT", "2019ACT", "05", "", "TYPOL").

If the specified parameters do not identify any partition existing on the system, the cell field will not return an error but an empty string and if this is used in a query as an extraction table then a syntax error message will be returned on reading. If the partition is not present but a value to return in case the partition is missing (*valoreSeMancante* parameter) is specified as the last parameter, then the function will return the value inserted as the parameter.

Examples

- `DatasetPARTITION("001", "000006")`

With a non-partitioned dataset with code "000006" in the workspace with code "001", it returns the name of the single physical partition, if present.

- `DatasetPARTITION("001", "000007", "BUD")`

With a dataset partitioned by business cycle with code "000007" in the workspace with code "001", and given the "BUD" single submission process, it returns the name of the single physical partition related to the "BUD" process, if present.

- `DatasetPARTITION("001", "000007", "ACT", "2019ACT", "01")`

With a dataset partitioned by business cycle with code "000007" in the workspace with code "001", and given the "ACT" scenario/period submission process, it returns the name of the single physical partition, if present, related to the "ACT" process, period "01" of scenario "2019ACT".

- `DatasetPARTITION("001", "000007", process("2019ACT","01"), "2019ACT", "01")`

With a dataset partitioned by business cycle with code "000007" in the workspace with code "001", it returns the name of the single physical partition, if present, related to period "01" of scenario "2019ACT" and to the process to which the scenario-period combination is related. That process is returned by the "process("2019ACT","01")" function.

- `DatasetPARTITION("001", "000008", "BUD", "", "", "CORP")`

With a dataset partitioned by business cycle - entity with code "000008" in the workspace with code "001", and given the "BUD" single submission process, it returns the name of the single physical partition, if present, related to the "BUD" process, entity "CORP".

- `DatasetPARTITION("001", "000008", "ACT", "2019ACT", "01", "CORP")`

With a dataset partitioned by business cycle - entity with code "000008" in the workspace with code "001", and given the "ACT" scenario/period submission process, it returns the name of the single physical partition, if present, related to the "ACT" process, period "01" of scenario "2019ACT", entity "CORP".

- `DatasetPARTITION("001", "000008", process("2019ACT","01") , "2019ACT", "01", "CORP")`

With a dataset partitioned by business cycle with code "000007" in the workspace with code "001", it returns the name of the single physical partition, if present, related to entity "CORP", period "01" of scenario "2019ACT" and to the process to which the scenario-period combination is related. That process is returned by the "process("2019ACT","01")" function.

```
DatasetPARTITION("001", "000007", "ACT", "2019ACT", "05", , , "TGK_DUAL")
```

With a dataset partitioned by business cycle with code "000007" in the workspace with code "001", and given the "ACT" scenario/period submission process, related to the "ACT" process, period "05" of scenario "2019ACT", and no partition has been generated for that scenario/period pair, then it returns the text reported as the last "TGK_DUAL" parameter.

Non-dimensional attributes

Description

Non-dimensional functions related to forms and reports have attributes that make it possible to obtain information on those objects.

The non-dimensional attributes of forms and report lists are described below.

The Code attribute

Element	Description
Availability	From 3.0912
Applicability	Non-dimensional "form" object
Syntax	*code*
Value returned	String with the code of the current form

Example

```
report.code
```

Returns the current form code.

The Desc attribute

Element	Description
Availability	From 3.0912
Applicability	Non-dimensional "form" object
Syntax	*desc([language])*
Parameters	*language* (optional) Code of the table language indicated as a whole number
Value returned	String with the description of the current form. The optional parameter is only supported with multiple descriptions. If omitted, the user language is used.

Example

```
report.desc(2)
```

Returns the description of the current form in table language 2.

The Path attribute

Element	Description
Availability	From 4.2.0
Applicability	Non-dimensional "form" object
Syntax	*Path([level], [separator])*
Parameters	• *level* (optional) The number of levels to go up in the form group. If omitted or zero, it goes to the root of the group • *separator* (optional) Separator to be used between each element in the string. Of omitted, a backslash "\" is used
Value returned	String obtained from the concatenation of expressions defined on the nodes of the current element's path.

Example

With a form related to the "DE - Data Entry" \ "CE - Profit And Loss" nodes of form group "$":

- `Report.path`

Returns the string "$\DE\CE".

- `Report.path(2)`

Returns the string "DE\CE".

- `Report.path(".")`

Returns the string "$.DE.CE".

The ReportFile attribute

Element	Description
Availability	From 4.2.1 (SP20)
Syntax	*reportFile*
Attributes	• *filename* • *filepath* • *fullName*
Value returned	If the current form is an ad hoc form, it returns some information related to the file on disk containing the form structure. If the form is shared, it returns an empty string.

Example

To find out whether a form is ad hoc:

```
if(report.reportFile.filename="", "Form shared", "Form Ad hoc")
Report.reportfile.filename
```

Returns:

- the string "myReport.xlsx" for "myReport.xlsx" ad hoc forms,
- an empty string for shared forms,
- "book1" for ad hoc forms that have not yet been saved.

`Report.reportfile.filepath`

Returns:

- the string "C:\Users\Username\Documents" string for "myReport.xlsx" for "myReport.xlsx" ad hoc forms saved in the user's documents folder,
- empty string for shared forms,
- an empty string for ad hoc forms that have not yet been saved.

`Report.reportfile.fullFileName`

Returns:

- the string "C:\Users\Username\Documents\myReport.xlsx" for "myReport.xlsx" ad hoc forms saved in the user's documents folder,
- empty string for shared forms,
- "book1" for ad hoc forms that have not yet been saved.

The RunDate attribute

Element	Description
Availability	From 3.0903
Syntax	*runDate([dataFormat],[timeFormat])*
Parameters	• *dataFormat*: can take one of the following values: ◦ -1: no format. The date is not printed ◦ 0: the Full format is used ◦ 1: the Long format is used ◦ 2: the dd-mmm-yyyy format is used ◦ 3 (or omitted): the Short format is used • *timeFormat*: can take one of the following values: ◦ -1 (or omitted): no format. The date is not printed ◦ 1: the Full format is used ◦ 2: the Long format is used ◦ 3: the hh.mm format is used
Value returned	Date and time at which the form was run in the time zone of the application server. Poiché l'attributo RunDate è valorizzato in base al contesto di esecuzione del form, è possibile che data e ora risultino differenti quando il form viene eseguito dal Web rispetto all'esecuzione tramite Excel Add-in. Optionally, it is possible to specify one format type for the date and one for the time. Effective formatting also will depend on the user's local formatting.

Example

A form run on 15 March 2013 at 15:27 with format setting = "Italian".

- `report.runDate`

Returns 15/03/13.

- `report.runDate(0)`

Returns Friday 15 March 2013.

- `report.runDate(1)`

Returns 15 March 2013.

- `report.runDate(2)`

Returns 15-Mar-2013.

- `report.runDate(1,1)`

Returns 15 March 2013 15.27.59 GMT+01:00.

- `report.runDate(1,2)`

Returns 15 March 2013 15.27.59.

- `report.runDate(1,3)`

Returns 15 March 2013 15.27.

- `report.runDate(-1,2)`

Returns 15.27.59.

The Code attribute

Element	Description
Availability	From 3.0912
Applicability	Non-dimensional "ReportList" object
Syntax	*code*
Value returned	String containing the code of the form containing the Report list

Example

`ReportList.code`

Returns the form code of the current Report list.

The Desc attribute

Element	Description
Availability	From 3.0912
Applicability	Non-dimensional "ReportList" object
Syntax	*desc([language])*
Parameters	• *language* (optional) Code of the table language indicated as a whole number
Value returned	String containing the description of the form containing the Report list. The parameter is only supported with multiple descriptions. If omitted, the user language is used

Example

`ReportList.desc`

Returns the description of the report containing the current report list.

The ReportListFile attribute

Element	Description
Availability	From 4.2.1 (SP20)
Syntax	*ReportListFile*
Attributes	• *filename* • *filepath* • *fullName*
Value returned	Form containing the current Report list. In the absence of a containing form, with the report list therefore created "on the fly", an error will be generated. Can only be used with parametric expressions inserted in the Report list creation wizard.

Example

`${reportList.reportListFile.filePath}\myOutput.xlsx`

Defines a "myOutput.xlsx" file in the same folder as the Report list structure as the output file of a report list.

`ReportList.reportListfile.filename`

Returns:

- the string "myReportList.xlsx" for "myReportList.xlsx" report lists,
- an empty string for report lists saved in the table,
- the string "book 1" for ad hoc report lists that have not yet been saved.

```
ReportList.reportListFile.filePath
```

Returns:

- "C:\Users\Nomeutente\Documents" for "myReportList.xlsx" report lists saved in the user's documents folder,
- an empty string for reports run in the table,
- an empty string for ad hoc reports that have not yet been saved.

```
Report.reportfile.fullFileName
```

Returns:

- "C:\Users\Username\Documents\myReportList.xlsx" for "myReportList.xlsx" report lists saved in the documents folder,
- an empty string for reports lists saved in the table,
- "book 1" for ad hoc report lists that have not yet been saved

Location functions

Description

Location functions allow you to access specific positions in a form (e.g. a row, a column, a matrix). They are used in combination with dimensional functions.

Example

```
var := row("MATRIX 1", 10). $Entity.code
```

Saves, in the "var" variable, the entity code defined in row 10 of "MATRIX1".

Description of location functions

The Column function

Element	Description
Availability	From 3.0912
Syntax	*Column(MatrixCode, ColumnNumber)*
Parameters	• *MatrixCode* Matrix code written between inverted commas • *ColumnNumber* Column number of the matrix
Value returned	Column "ColumnNumber" of the matrix with code "MatrixCode"

Example

```
column("MATRIX 1", 10).$Entity.code
```

Returns the code of the entity present in column 10 of "MATRIX1"

The Matrix function

Element	Description
Availability	From 3.0912
Syntax	*Matrix(MatrixCode)*
Parameters	• *MatrixCode* Matrix code written between inverted commas
Value returned	Reference to the matrix filters of the matrix with code "MatrixCode"

Example

```
matrix("MATRIX 2").$Entity.code
```

Returns the code of the entity present in the matrix filters of the matrix with code "MATRIX 2".

The Parameter function

Element	Description
Availability	From 3.0912
Syntax	*Parameter(ParameterCode)*
Parameters	• *ParameterCode* Parameter code written between inverted commas
Value returned	Reference to the parametric filter with code "ParameterCode"

Example

```
parameter("S01").$Scenario.code
```

Returns the code of the scenario associated with the parametric filter with code "S01".

The ParameterValue function

Element	Description
Availability	From 3.1012
Syntax	*ParameterValue*
Value returned	The value of a parameter

Example

```
parameter("PP").parameterValue
```

Returns the value of parameter "PP".

The PositionIndex function

Element	Description
Availability	From 3.0912
Syntax	*PositionIndex*
Value returned	Current sheet for cell fields, queries, data processing or deployment. Current row or column for conditional formatting expressions

Example

```
positionIndex.$Scenario.code
```

Returns the code of the scenario associated with the sheet, e.g. when the scenario is inserted as a tab filter.

The PrintFilters function

Element	Description
Availability	From 4.2.1
Syntax	*printFilters([Position],[cellField 1],[cellField 2],...,[cellField N])*
Parameters	• *Position* (optional) Reference to a position via location function • *cellField1,...,cellField N* (optional) List of field attributes for every filter, must be written between inverted commas
Value returned	String obtained by concatenating the field attributes. The separator between two fields of a filter is "-" (dash between two spaces) and the separator between two filters is "," (comma followed by space)

Examples

- `Sheet(1).Printfilters`

Returns the concatenation between code and description of the filters of the first sheet.

- `PrintFilters(row("MATRIX1", "5"),"code", "attribute1")`

Returns the concatenation between code and attribute 1 of the filters on row 5 of MATRIX 1.

The Prompt function

Element	Description
Availability	From 3.0912
Syntax	*Prompt*
Value returned	Reference to dynamic global filters for the VBA client (also left compatible for the .NET client)

Example

```
prompt.$Scenario.code
```

Returns the code of the scenario associated with the dynamic global filters.

The ReportFilters function

Element	Description
Availability	From 3.1006
Syntax	*reportFilters*
Value returned	Reference to the form filters for .NET Client Reporting.

Example

```
reportFilters.$Scenario.code
```

Returns the code of the scenario associated with the form filters.

The Row function

Element	Description
Availability	From 3.0912
Syntax	*Row(MatrixCode, RowNumber)*
Parameters	• *MatrixCode* Matrix code between inverted commas • *RowNumber* Matrix row number
Value returned	Row "RowNumber" of the matrix with code "MatrixCode"

Example

```
row("MATRIX 1", 10).$Entity.code
```

Returns the entity code of row 10 of the matrix with code "MATRIX1".

The Sheet function

Element	Description
Availability	From 3.0912
Syntax	*Sheet*
Parameters	*sheet* Sheet number between inverted commas
Value returned	Reference to the "sheet" sheet

Example

```
Sheet("13").$Scenario.code
```

Returns the scenario code read from sheet 13.

Aggregation structure navigation functions

Description

Navigation functions allow you to move within an aggregation structure to obtain an object or set of objects.

Examples

- `var := this.lowest;`

Saves, in the "var" variable, the lowest level elements (leaves) of the node on which advanced selection has been defined. Has the same outcome as "Insert lowest level elements".

- `var := this.lowest.parent;`

Saves, in the "var" variable, the parent nodes of the lowest level elements (leaves) of the node on which advanced selection has been defined.

- `var := this.children.onlyLeafs;`

Saves, in the "var" variable, the elements related to the node on which advanced selection has been defined.

Description of aggregation structure navigation functions

The Parent function

Element	Description
Availability	From 3.0903
Syntax	*parent*
Value returned	Parent node of the element or node on which advanced selection has been defined

Example

`$Entity.parent.code`

Returns the code of the parent node of the entity specified in the filter.

The Children function

Element	Description
Availability	From 3.0903
Syntax	*children([scenario],[period])*
Scenario	• *scenario* (optional) Scenario code • *period* (optional) Period code
Value returned	Elements and/or child nodes of the node to which the function is applied. If the function is applied to an element and not a node, it returns the element.

Examples

- `$Entity.children`

Returns the child elements of the node of the specified entity hierarchy.

- `$Entity.children(2013ACT,03)`

Returns the child elements of the node of the time-dependent entity hierarchy specified for period 03 of scenario 2013ACT.

The Lowest function

Element	Description
Availability	From 3.0903
Syntax	lowest*[scenario],[period],[restrictions]*
Parameters	• *scenario* (optional) Scenario code • *period* (optional) Period code • *restrictions* (optional) These "partialise" the result of the deployment based on the filters in place. They can be activated or deactivated. The syntax for activating or deactivating them is the following: *"nomeLimitazione1:=TRUE|FALSE;nomeLimitazione2:=TRUE|FALSE;..."* Restrictions that are not specified are deactivated by default. The following restrictions are available: • *EntityAccountCustomNLim* (e.g. EntityAccountCustom1Lim, EntityAccountCustom2Lim, ...). Restrictions between Entity/Custom dimension N and between Account/Custom dimension N • *CounterpartyAccountCustom2Lim*. Restrictions between Counterparty Entity/Custom dimension 2 and between Account/Custom dimension 2 • *IntersectionLim*. Intersection of filters in place
Value returned	Lowest level elements of the node to which the function is applied. If it is applied to an element instead, it returns the element.

Examples

- `$Entity.lowest`

Returns the lowest level elements of the node of the entity hierarchy specified in the filter.

- `$Entity.lowest(2013ACT,03)`

Returns the lowest level elements of the node of the time-dependent entity hierarchy specified in the filter for period 03 of scenario 2013ACT.

- `$Entity.lowest(EntityAccountCustom1Lim := FALSE;IntersectionLim := FALSE)`

Returns the lowest level elements of the node of the entity hierarchy specified in the filter not considering the restrictions between entity and custom dimension 1. Ignores possible filters on the entity dimension originating from other hierarchies.

The Root function

Element	Description
Availability	From 3.0903
Syntax	*root*
Value returned	Root of the hierarchy to which the node or element belongs

Example

```
$Entity.root.code
```

Returns the code of the root node of the entity hierarchy specified in the filter.

The All function

Element	Description
Availability	From 3.0903
Syntax	*all*
Value returned	Nodes and lowest level elements of the dimension specified in the filter

Example

```
$Entity.all.code
```

Returns the codes of all lowest level elements and nodes of the entity hierarchy specified in the filter.

The Ancestor function

Element	Description
Availability	From 3.0903
Syntax	*Ancestor(ancestorLevel)*
Parameters	*ancestorLevel* Level of hierarchy
Value returned	Ancestor level *ancestorLevel*. If the parameter exceeds the hierarchy root, it returns an empty set.

Examples

- `$Entity.ancestor(1).code`

Returns the code of the parent node of the entity. Ancestor(1) is equivalent to the parent function.

- `$Entity.ancestor(2).code`

Returns the code of the grandparent node of the entity on which it is applied.

The AllAncestors function

Element	Description
Availability	From 3.0903
Syntax	*AllAncestors(ancestorLevel)*
Parameters	*ancestorLevel* Level of hierarchy
Value returned	All ancestors up to the *ancestorLevel*. If the parameter exceeds the hierarchy root, the system returns all ancestor nodes up to the root, inclusive

Example

`$Entity.allancestors(2).code`

Returns the code of the parent node and grandparent node of the entity on which it is applied.

The Descendants function

Element	Description
Availability	From 3.0903
Syntax	*Descendants(descLevel)*
Parameters	*descLevel* Level of hierarchy
Value returned	Descendent levels *descLevel*. If the parameter exceeds the level of the hierarchy's lowest level elements, the system returns an empty set

Examples

- `$Entity.descendants(1).code`

Returns the codes of the elements and/or child nodes of the node of entities on which it is applied. Descendants(1) is equivalent to the children function.

- `$Entity.descendants(2).code`

Returns the codes of the elements and/or grandchild nodes of the node of entities on which it is applied.

The AllDescendants function

Element	Description
Availability	From 3.0903
Syntax	*AllDescendants(descLevel)*

Element	Description
Parameters	*descLevel* Level of hierarchy
Value returned	All descendants up to the *descLevel* level. If the parameter exceeds the level of the hierarchy's lowest level elements, the system returns all the descendant nodes and leaves up to the lowest level elements, inclusive.

Example

```
$Entity.alldescendants(2).code
```

Returns the codes of the elements and/or child and grandchild nodes of the node of entities on which it is applied.

The Complementary function

Element	Description
Availability	From 3.0903
Syntax	*Complementary(ancestorLevel)*
Parameters	*ancestorLevel* Level of hierarchy
Value returned	Complementary elements of the element present in the filter, calculated in relation to the ancestor node specified by the *ancestorLevel* parameter.

Example

```
$Entity.complementary(1).code
```

Returns the codes of the entities belonging to the sub-tree of the parent node of the entity specified in the filter.

The Siblings function

Element	Description
Availability	From 3.0903
Syntax	*siblings*
Parameters	*ancestorLevel* Level of hierarchy
Value returned	The siblings of the element to which it is applied, including the element itself. If you want to exclude the starting element, you can remove it from the returned set using the difference operator between sets.

Examples

- `$Entity.siblings.code`

Returns the code of the siblings of the element to which it is applied, including the element itself.

- `$Entity.siblings.code - $Entity.code`

Returns the code of the siblings of the element to which it is applied, excluding the element itself.

The OnlyLeafs/OnlyNodes function

Element	Description
Availability	From 3.0903
Syntax	*onlyLeafs, onlyNodes*
Value returned	Only the leaves (lowest level elements) and nodes of the set

Examples

- `$Entity.children.onlyLeafs`

Returns only the leaves from among the children of the current entity node, excluding the nodes.

- `$Entity.children.onlyNodes`

Returns only the nodes from among the children of the current entity node, excluding the leaves.

The GetLeafs/GetNode function

Element	Description
Availability	From 3.0903
Syntax	*getLeaf(code), getNode(code)*
Parameters	• *code.* Code of the leaf and node to be retrieved
Value returned	Leaf or node, with the code indicated as parameter, in the same hierarchy as the node to which the functions are applied

Examples

- `$Entity.getLeaf("10")`

Returns only the leaf with code "10" from among the leaves of the hierarchy to which the current entity belongs.

- `$Entity.getNode("01").lowest`

Returns all the leaves of the node with code "01" of the hierarchy to which the current entity belongs.

- `$Category.getLeaf("C1")`

Returns category C1.

***IMPORTANT**: several nodes could be related in the groupings of a leaf. In such cases, the expression returns leaf duplicates for every node to which it is related. To take one of them, use the trimSet function.*

- `$Category.getLeaf("C1").trimSet(1).code`

Returns only the first occurrence of the leaf with code C1

The IsLeaf function

Element	Description
Availability	From 3.0903
Syntax	*isLeaf*
Value returned	1 if it is a leaf, otherwise 0. If the function is applied to a set with more than one element, the system returns an error

Example

`$Entity.isLeaf`

Returns 1 if the entity dimension is a leaf.

The Level function

Element	Description
Availability	From 3.0903
Syntax	*Level*
Value returned	Whole number indicating the level of nesting within the hierarchy. The root of a hierarchy is level 1, its children are level 2, etc. This function is useful for applying different formatting to the levels of a hierarchy with conditional expressions via a style sheet. From version 1109P, the function returns the value 0 if applied to: • row or column placeholders • "+" rows/columns created automatically when the possibility of inserting new elements in data entry is enabled

Example

```
$Entity.level
```

Returns 3 if the entity dimension is a leaf of a child node of the hierarchy root.

The This function

Element	Description
Availability	From 3.0903
Syntax	*this*
Value returned	Node specified in the row/column filters tree to which advanced selection has been applied

Example

```
this.lowest
```

Returns the leaves of the node used as the row/column filter on which advanced selection has been defined.

The Select function

Element	Description
Availability	From 3.0903
Syntax	• *select* (field, operator, value) • *select* (condition)
Parameters	• *field* Dimensional attribute (can be managed using string manipulation functions) to be compared with *value* • *operator* Comparison logic operator • *value* String or variable to be compared with *field* • *condition* Logical expression that returns a true/false result for every element of the set to which the formula is applied. If it is necessary to use inverted commas within the string expression, use the constant QT.
Value returned	Elements of the set that satisfy the condition defined via the three parameters (field, operator, value) or a single parameter (condition)

Examples

- `$Entity.lowest.select("$Entity.attribute1", "=", "A")`

Returns the leaves of the starting node that have "A" as the value of attribute 1.

- `$Entity.children.select("$Entity.lowest.size > 3")`

Returns the children of the starting node that have more than 3 leaves related.

- `$Entity.children.select("LEFT($Entity.code,3)=" + QT + "000" +QT)`

Returns the children of the starting node with a code starting with "000".

The Sort function

Element	Description
Availability	From 3.0903
Syntax	*sort(expression, ["DESC"])*
Parameters	• *expression* Expression on the basis of which to run sorting • *"DESC"* (optional) The predefined sorting is ascending; it becomes descending if "DESC" is specified
Value returned	List of the elements of the set sorted according to the *expression* parameter

Examples

- `$Entity.children.sort("$Entity.attribute1"`

Sorts the children of the entity filter in ascending order based on the custom 1 attribute.

- `$Entity.children.sort("$Entity.attribute1", "DESC")`

Sorts the children of the entity filter in descending order based on the custom 1 attribute.

- `$Entity.children.sort("$Entity.children.size" , "DESC").sort( "$Entity.attribute1")`

Sorts the children of the entity filter in ascending order based on the custom 1 attribute. If the value of attribute 1 remains the same, it sorts the number of children of the involved elements in descending order.

The SetEditability function

Element	Description
Availability	From 3.0903
Syntax	• *setEditability(editMode, field, operator, value)* • *setEditability(editMode, condition)*
Parameters	• *editMode* Editability type to be assigned to elements. It can be set to "editable" or "e", "protected" or "p" and "readonly" or "ro" • *field* Dimensional attribute (can be managed using string manipulation functions) to be compared with *value* • *operator* Comparison logic operator • *value* String or variable to be compared with *field* • *condition* Logical expression that returns a true/false result for every element of the set to which the formula is applied. If inverted commas need to be used in the string, use the constant QT
Value returned	Assigns the specified editability type to the elements of the set that satisfy the condition defined via the three parameters (field, operator, value) or a single parameter (condition). Editability can only be overridden in accordance with the following rules: • If the starting node has been indicated as "not specified" or "editable" in the Matrix Wizard, it can be overridden with any editability type • If the starting node is indicated as "protected", it can only be made non-editable • If the starting node has been indicated as "not editable", its editability cannot be changed using this function.

Examples

- `myAccountsSet.setEditability("readonly", "LEFT($Account.code,3)", "<=", "000")`

Makes the accounts of the set whose codes starts with "000" non-editable.

- `myAccountsSet.setEditability("readonly", "$Account.code.length <= 5")`

Makes the accounts of the set whose code has less than 5 characters non-editable.

The OrFilter function

Element	Description
Availability	From 3.0903
Syntax	*orFilter(element1, element2, ... elementN)*
Parameters	• *element1* Will form part of the final filter. Can be defined as a string between inverted commas or dimensional function • *element2* As above
Value returned	A single filter obtained by combining, with OR relationships, elements processed as parameters. If one of the parameters is a set, OR relationships are established among all elements of the set.

Examples

- `OrFilter($Entity.getLeaf("01"), $Entity.getLeaf("02"))`

Establishes an OR relationship between entities 01 and 02.

- `OrFilter($Entity.children)`

Establishes an OR relationship among the children of the node of the entity hierarchy to which it is applied.

Can also be written as: $Entity.children.orFilter().

The ContainedFilter function

Element	Description
Availability	From 3.0903
Syntax	*containedFilters(orFilter)*
Parameters	*orFilter* Filter made up of a set of filters combined using OR relationships
Value returned	The inverse orFilter function. Returns the set of OR filters in the filter processed as a parameter

Example

```
$Dest1.containedFilters.lowest.code
```

Takes the OR clauses and obtains the originally involved lowest level elements from them.

The SyntheticSchema function

Element	Description
Availability	From 3.0903
Syntax	*syntheticSchema(FSTCode)*
Parameters	*FSTCode* Schema or FST item code
Value returned	Schema or FST item shown in synthetic mode

***IMPORTANT**: prevent 'partialising' the result of this script with the select functions and similar functions. If the script only returns some of the elements, some formulas present in certain FST items may be targeting elements that are not returned and this could be generating an error in the run step.*

Example

```
SyntheticSchema(@SP)
```

Generates the synthetic schema specified as the value of parameter SP.

The AnalyticalSchema function

Element	Description
Availability	From 3.0903
Syntax	*analyticalschema(FSTCode)*
Parameters	*FSTCode* Schema or FST item code
Value returned	Schema or FST item shown in synthetic mode

***IMPORTANT**: prevent 'partialising' the result of this script with the select functions and similar functions. If the script only returns some of the elements, some formulas present in certain FST items may be targeting elements that are not returned and this could be generating an error in the run step.*

Example

```
AnalyticalSchema(@SP)
```

Generates the analytical schema specified as the value of parameter SP.

The ControlGroup function

Element	Description
Availability	From 3.0903
Syntax	*controlGroupcontrolgroupCode*
Parameters	*controlgroupCode* Control group code
Value returned	Control group detailed in the accounts

Example

```
ControlGroup(@CG)
```

Generates details of the control group specified as the value of parameter CG.

FromHierarchy

Element	Description
Availability	From 3.0903
Syntax	*fromHierarchy(hierarchy_code)*
Parameters	• *hierarchy_code* Hierarchy code
Value returned	The same element to which the function is applied but taking it from the hierarchy specified as the parameter

Example

With a "Geographical area" hierarchy with code "AG" on the entity dimension and with a form filtered on another hierarchy, the geographical area to which the entity belongs can be obtained even if the "Geographical hierarchy" has not been used in the form.

- `$Entity.fromHierarchy("AG")`

Returns the same element used a s a filter but read by the "AG" hierarchy.

***IMPORTANT**: if it is a node and there is no node with the same code in the "AG" hierarchy, the system will return a warning and the cell will not be calculated.*

- `$Entity.fromHierarchy("AG").parent.code`

Returns the code of the node of the "AG" hierarchy to which the entity specified in the filter is related.

Excel functions

Description

These are Excel functions that can be used in the script's TGKML syntax. They are usually used to manipulate the content of a variable.

Example

```
var := LEFT($Entity.code, 1);
```

Saves, in the "var" variable, the first character of the entity code specified in the filter

Logical functions

Element	Description
AND(condition1, condition2...)	AND logic
OR(condition1, condition2...)	OR logic
NOT(condition)	Negation
IF(condition, Truevalue, Falsevalue)	Conditionlogic

Aggregate functions

Element	Description
SUM(argument1, argument2, ...)	Sum
AVERAGE(argument1, argument2, ...)	Average value
MAX(argument1, argument2, ...)	Maximum value
MIN(argument1, argument2, ...)	Minimum value
COUNT(argument1, argument2, ...)	Number of elements contained
PRODUCT(argument1, argument2, ...)	Value produced by multiplying the arguments
SUBTOTAL (function, argument1, argument2, ...)	Only supported in row/column/sheet formulas. Supports the following functions: • 1 - arithmetic mean • 2 - numbers count • 4 - max • 5 - min • 6 - product • 9 - sum Sum of specified arguments

Other arithmetic functions

Element	Description
SIGN(value)	Returns -1 for positive values, 1 for negative values
ABS(value)	Absolute value
MOD(dividend, divisor)	Rest of the division
INT(value)	Rounding down
ROUND(value, figure)	Rounding

String manipulation functions

Element	Description
CONCATENATE (argument1, argument2, ...)	Concatenation of strings. Equivalent to "String 1" + "String 2" + "String 3" ...
FIND(text, string, [start])	Excel's "Find" function
LEFT(text, numberCharacters)	Returns the number of characters indicated starting from the start of the string.
LEN(text)	Returns the length of the string.
LOWER(text)	Changes the text to lowercase.
UPPER(text)	Changes the text to uppercase
MID(text, start, numberCharacters)	Excel's "Extract String" function
SUBSTITUTE(text, occurrence, substitution)	Excel's "Substitute" function
REPLACE(text, start, numberCharacters, substitution)	Excel's "Replace" function
RIGHT(text, numberCharacters)	Returns the number of characters indicated starting from the end of the string.
REPT(text, times)	Excel's Repeat function
TRIM(text)	Excel's "Delete spaces" function
SEARCH(text, string, [start])	Excel's "Find" function (without distinction between uppercase and lowercase)
VALUE(text)	Changes the specified text into numeric text. Excel's "Value" function

Error management functions

Element	Description
ERR()	Error expression
IFERROR(expression, value)	Returns the processed value if the expression is an error expression, otherwise it returns the expression itself.
ISERROR(expression)	Returns 1 if the expression is an error expression, otherwise it returns 0.

Functions on sets

Description

Some functions, usually those used to navigate trees of aggregation structures, return sets of elements on which the user can operate with specific functions and operators.

The And function

Element	Description
Availability	From 3.0903
Syntax	* or &
Value returned	Intersections between sets

> ### Example
>
> `$Category.getNode("01").lowest * $Category.getNode("02").lowest`
>
> Returns the set made up of the lowest level elements simultaneously related to nodes "01" and "02" of the categories group.

The Or function

Element	Description
Availability	From 3.0903
Syntax	+ or /
Value returned	Union between sets. Duplicates are deleted.

> ### Example
>
> `$Entity.allAncestors(2) + $Entity.children`
>
> Returns the set made up of the parent, grandparent and children of the current node.

The Complement function

Element	Description
Availability	From 3.0903
Syntax	-
Value returned	Difference between sets (complementary set)

> ### Example
>
> `$Entity.allDescendants(5) - $Entity.children`
>
> Returns all descendants up to the fifth level, excluding the children of the entity node to which it is applied.

The Contains function

Element	Description
Availability	From 3.0903
Syntax	*contains(element)*
Parameters	*element*: element to be searched for in the set, defined as a string between inverted commas or dimensional function
Value returned	1 if the *element* parameter belongs to the set, otherwise 0.

Example

`Entity.lowest.contains($Entity.getLeaf("A01"))`

Returns 1 if the descendants of the entity node include entity "A01", otherwise it returns 0.

The Item function

Element	Description
Availability	From 3.0903
Syntax	*item(index)*
Parameters	*index*: index identifying the element belonging to the set
Value returned	Element of the set with index specified as a parameter (0-based). Returns an error if the set has an index lower than the indicated index+1 children.

Example

`$Entity.children.item(0)`: returns the first child of the node of entities.

`$Entity.children.item(1)`: returns the second child of the node of entities.

The Err() function

Element	Description
Availability	From 4.3.1
Syntax	*err()*
Value returned	Always returns an error.

The Iserror function

Element	Description
Availability	From 4.3.1
Syntax	*iserror(argument)*
Parameters	*argument*: generic function
Value returned	TRUE if the processed argument is an error, otherwise FALSE.

The Set Function

Element	Description
Availability	From 3.0903
Syntax	*set(element1, element2, ...elementN)*
Parameters	*elemento1* *elemento2* *element<...>*: elements that will form part of the set, defined as a string between inverted commas or dimensional function
Value returned	Creates a set with the parameters processed as elements. ***Note**: if one of the arguments is an error, the system will return a set containing only the elements that do not return errors.*

Example

```
Set($Entity.getNode("C1"), $Entity.getNode("C2"))
```

Set containing nodes C1 and C2 of the entity hierarchy specified in the filter.

The Size function

Element	Description
Availability	From 3.0903
Syntax	*size*
Value returned	Number of elements in the set

Example

```
$Entity.children.size
```

Returns the number of sheets in the node of the entity hierarchy specified in the filter.

The Sum/Intersect function

Element	Description
Availability	From 3.0903
Syntax	*sum(set1, set2, ... set<...>), intersect(set1, set2, ... setN)*
Parameters	*set <...>*: set of objects to be united or intersected with the sets specified in the subsequent parameters.
Value returned	Union and intersection between the sets indicated as parameters

Example

Let us consider two variables, a and b, defined as follows:

```
a := Set($Entity.getNode("C1"), $Entity.getNode("C2"));
b := Set($Entity.getNode("C3"), $Entity.getNode("C4"));
sum(a, b)
```

Returns a set containing nodes C1, C2, C3 and C4 of the entity hierarchy specified in the filter.

The TrimSet function

Element	Description
Availability	From 3.0903
Syntax	*trimSet(numElements)*
Parameters	*numElements*: number of elements, starting with the first, that you want to consider.
Value returned	The first "numElements" of the set. If the set contains fewer elements, they are all returned.

Example

```
$Entity.lowest.trimSet(3)
```

Returns the first three entities belonging to the sub-tree of the entity hierarchy node to which the specified entity belongs.

Hyperlinks

Purpose of the function

Hyperlinks allow you to link several forms belonging to the working application database. They are used for navigation and data entry purposes. For example, they can be inserted in a cockpit to guide access to forms or in an item of a form to allow the user to automatically see the source data of that item. A link can also activate a series of advanced actions, such as saving and printing files, and running macros or data processing.

Characteristics

Hyperlinks appear as hypertextual Excel links and can be positioned on matrix rows/columns or in any of the form's cells which does not contain a data structure (i.e. matrices, queries and cell fields).

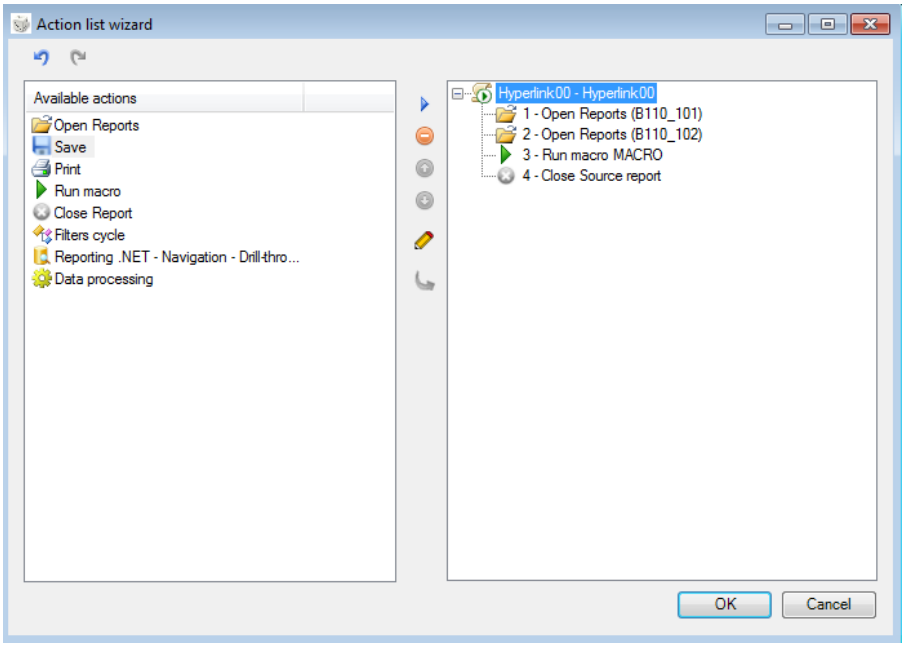
They are created from the **Action List Wizard** window by defining the hyperlink and the actions that must be run when the hyperlink is opened, together with the run order. It is possible to define sequential actions or nest one action within another. Nested actions are indicated with the identification number of the action in which they are nested and a decimal number corresponding to the run order.

If a node is selected rather than a single form, all the forms in that node will open when the hyperlink is selected.

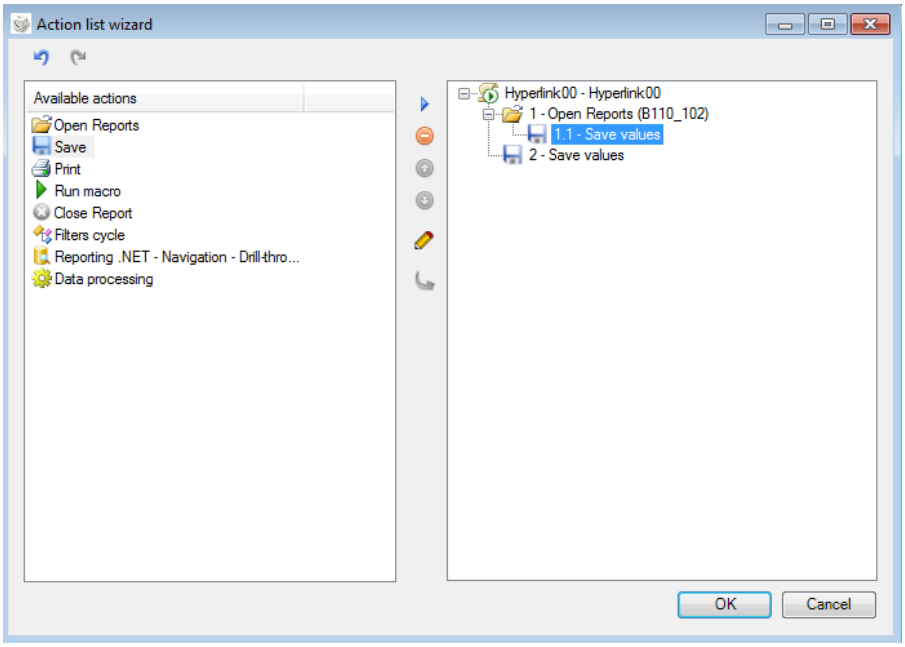
***Note**: several actions cannot be run sequentially via hyperlinks from web forms.*

Example

This example shows the **Action List Wizard** window in which the following sequence of actions, to be run once the hyperlink has been selected, was defined: open two reports, run a macro and close the source report.

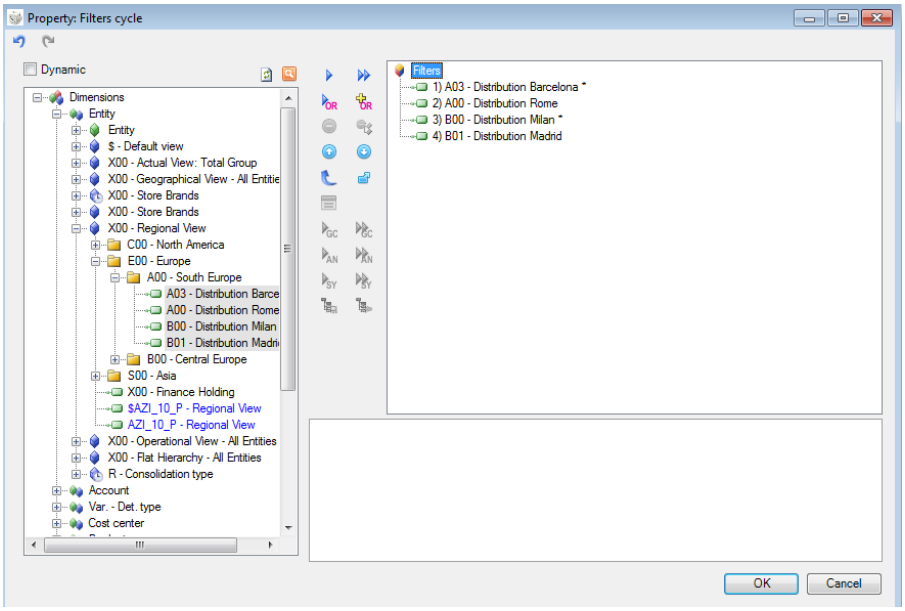


In this example, in contrast, two **Save** actions have been inserted. Action **1.1 - Save values** is nested in **Open report**. The values contained in form B110_102 are automatically saved when the report is opened. Action **2 - Save values** is not nested. This action saves the values of the source report after actions 1 and 1.1 have been run.



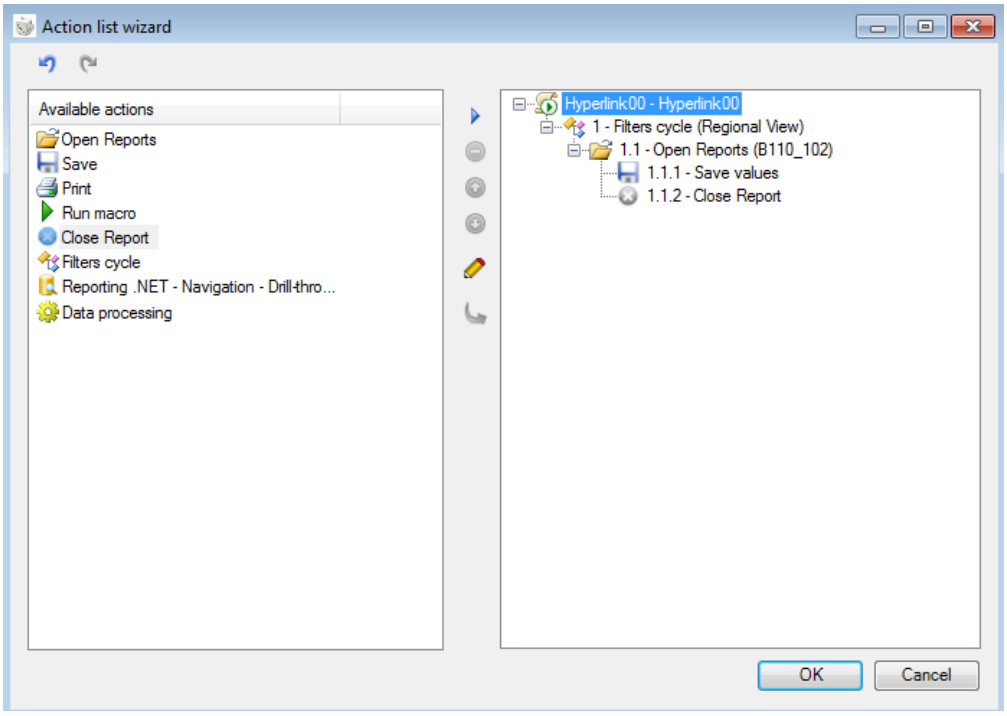
Filters cycle example

This example illustrates how to open the same report for all entities. To do this, first of all a **Filters cycle** action that has those entities as filters must be defined.



Next, the **Open report** action to be nested in **Filters cycle**, the action with which to save the values and the action for closing the report are defined. Thus, each of these actions is repeated four times, once for every entity of the filters set in the **Filters cycle** window.

This cycle works by running the report in data entry mode, as it involves the saving of values.



Hyperlinks in matrices

In matrices, hyperlinks can be positioned in header cells and cells containing values. The definition of the actions that must be run when that hyperlink is opened is the same as the definition described for the links inserted directly in the report. The only differences are as follows:

- If the **Inherit from the source report** option is activated for the **Open report** action, the parameters are inherited from the filters present on the cell in the report in use, from which the hyperlink is opened.

  For example, for a matrix that has several entities on its rows, the report opened from the hyperlink is filtered on the entity present in the row from which the hyperlink was opened. Specifically, if the entity of the first row were A00, the corresponding report would be opened with the filter on entity A00.

- The query of a **Reporting.NET Navigation - Drill-through** action is run according to the filters present in the cell in use, from which the hyperlink is opened.

For more details on the hyperlink settings, see Table of row/column filter properties.

Managing hyperlinks

Open the Action List Wizard window

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

There are several ways in which the Action List Wizard window can be opened to create hyperlinks:

- Select a cell and click .
- Right click the desired cell and select **Tagetik** > **New Hyperlink**.
- In the **Open items** tab, right click on the template in which to insert the hyperlink and select **Hyperlink** > **New Hyperlink**.

Create a hyperlink in a cell of the form using Open form

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

1. Click the empty cell in which to insert the hyperlink.
2. Open the **Action List Wizard** window. See "Open the Action List Wizard window" on the previous page.
3. In the list of runnable actions, select **Open form** and click : the **Properties: Open form** window will appear showing the forms present in the working application database.
4. Select the desired forms and click **Next**.
5. Set the way in which to value the parameters defined for all the forms selected on opening.
6. If several forms are opened to ad system or ad hoc parameters, select **+ System parameter** or **+ adHoc parameter** as required and repeat step 5.
7. Click **Next**.
8. Indicate the way in which to open the target form and click **OK**: the **Open form** action will be added to the list of actions to be run.

Create a hyperlink to open a form defined in the next row/column

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

***IMPORTANT**: in the Excel add-in, the opening of a form in a subsequent row/column can only be configured via hyperlinks. They can only be used in web forms.*

1. Click on an empty matrix cell in which to insert the hyperlink.
2. Open the **Action List Wizard** window. See "Open the Action List Wizard window" on the previous page.
3. In the list of runnable actions, select **Open form specified in the next row/column** and click .
4. Indicate the way in which to identify the target form and click **OK**: the **Open form specified in the next row/column** action will be added to the list of actions to be run.

At runtime, when the hyperlink is clicked, the system will open the form indicated in the cell of the next row/column.

***Note**: the parameters of the target form can only be inherited from the context.*

Using the Filters cycle action

Excel > CCH® Tagetik > Design > (left-hand panel) click form > > Hyperlink (in Design bar)

See "How to access the windows" on page 9 for alternative paths.

1. In the list of runnable actions, select **Filters cycle** and click : the **Properties: Filters cycle** window will appear.
2. Select the desired dimensions on which to filter and click .
3. Click **OK**.
4. Select the actions to be repeated and click .
5. Click **OK**.

Create a hyperlink in a matrix

Excel > CCH® Tagetik > Design > select form > Matrix

1. In the **Choose the dataset** window, click on the desired dataset and on **OK**.
2. Open an existing matrix, or create the matrix structure by inserting the desired filters on the rows and columns.
3. Select the (static or dynamic) row/column on which to insert the hyperlink.
4. In the properties table, click **Hyperlink** and define its properties.

   ***Note**: all the actions listed above can be inserted from the Action List Wizard.*

5. Click **OK**.

Action List Wizard window

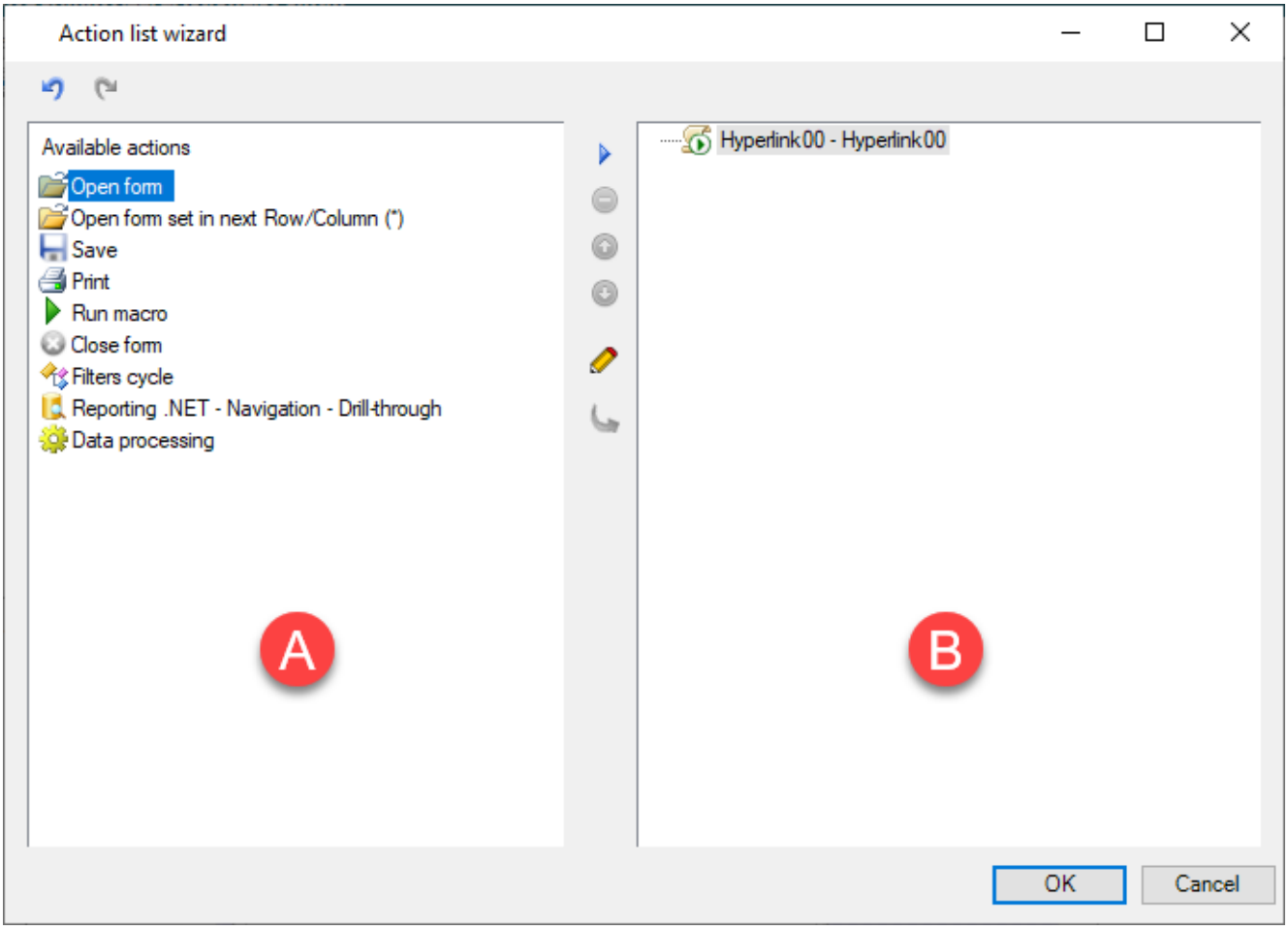
Excel > CCH® Tagetik > Design > (left-hand panel) click form > > Hyperlink (in Design bar)

See "How to access the windows" on page 9 for alternative paths.

Purpose of window

This window allows you to create the hyperlink and actions to be run once it has been opened.

What it looks like



Element	Description
A	List of actions that can be run when the hyperlink is clicked
B	List of sorted actions to be run when the hyperlink is clicked

Runnable actions

Element	Description
Open form	Allows you to set the opening rules for one or more target forms via a hyperlink on the source form. See "The Open form window (Action List Wizard)" on the next page.
Open form specified in the next row/column	(Only for web forms 2.0). Allows you to select one or more forms to be run, both in data entry and navigation modes, starting with the next row/column. Thus, a formula that returns the code of the form to be run can be inserted in the next row/column. In this mode, the parameters of the target form can only be inherited from the context.

Element	Description
Save	Automatically saves the values of the target or source form when the user clicks the hyperlink. The form that is saved will depend on the position of the **Save** action in the list of actions to be run. Any data processing to be run on saving, as defined in the form, are run automatically immediately afterwards.
Print	Not supported in web forms. Prints the target or source form on the predefined printer. The position of the **Print** action in the list of actions to be run determines the form to be printed Possible values for setting the print area to be printed by selecting the hyperlink: • **All**: all form content • **Areas**: only the content of the print areas, as defined in the form and indicated by the user • **Print area**: current print area defined for the form
Run macro	Running Visual Basic macros defined in the form. Macros can be run in both the target form and the original form only if the **Run macro** action is nested in a form opening action. Macros are called up in the form by calling up their names. If the **Run macro** action is followed by other actions in the list of actions to be run, then **Run macro** will need to return **True** for those subsequent actions to be run.
Close form	Closes a form when the hyperlink is selected. Depending on its position in the list of actions to be run, it closes the source or target form. It also locks the refresh form request set for other data processing inserted in the same hyperlink. If the form has been edited before the hyperlink is clicked, the form can be saved before closing.
Filters cycle	Not supported in web forms. Allows you to repeat actions for every element of a certain filter. The filter must belong to the same dimension/hierarchy as the parameters defined in the forms recalled by the hyperlink.
Reporting .NET - Navigation - Drill-through	Allows you to runs a drill-through query when the hyperlink is selected. See "Drill-through queries" on page 348.
Data processing	Allow you to run data processing defined in the form. The data processing is added to the actions to be run when it is selected with the name that was attributed to it when created. See "Data processing run from reporting" on page 244.

Buttons

Element	Description
	Adds the selected action to the list of actions to be run.
	Removes the selected action from the list of actions to be run.
	Moves the selected action up or down in the run order.
	Allows you to rename the link or change its settings for the selected action
	Nests the action selected from the list of actions to be run within the previous action.

The Open form window (Action List Wizard)

Excel > CCH® Tagetik > Design > (left-hand panel) click form > > Hyperlink > select Open form >

Purpose of window

This window allows you to select one or more forms to be opened via the hyperlink and to set their parameters and the modes in which they are run via the hyperlink.

Window structure

Based on the characteristics of the target form, the window can may show the following:

- Definition of run parameters (only if the selected forms contain run parameters)
- Definition of the mode in which the target form is opened

Options in the definition of run parameters

Option	Description
Inherit from context	The system retrieves the value of the parameter from the filters of the cell from which the hyperlink is opened or from the form/tab filters. *For example, if there are several sheets in a form and a different entity has been defined for each sheet, to open the target form the system uses the entity defined in the sheet from which the hyperlink is opened.* If the hyperlink is contained in a transactional form matrix, the dimensional parameters of the form being run can inherit their value from the row containing the hyperlink and the parameter dimension from the column containing the hyperlink. Values on dimensions with display mode set to **description only** are not inherited.
Inherit parameters	The system retrieves the value of the parameter from the value of the corresponding parameter in the source form. This is only possible if the source form parameters belong to the same dimension and hierarchy as the target form parameters.
Excel Cell	The system retrieves the value of the parameter from an Excel cell. This cell must be specified using the syntax **${*cell coordinates*}**. For example, **${A5}** corresponds to cell A5.
Select	Allows the user to select the value of the parameter from a list of possible values.
Cell field	The system retrieves the value of the parameter from a cell field. For a transactional Analytical Workspace form, the cell field is also calculated considering the dimensional field of the row that has the same dimension as the parameter to be defined. In this case, the dimensional field has priority over form or matrix filters. The only dimension fields considered are the ones that have **code** or **code and description** display mode.
+System parameter	(Only if the hyperlink opens several forms) Allows you to set system parameters that are valid for running all forms.
+AdHoc parameter	(Only if the hyperlink opens several forms) Allows you to set ad hoc parameters that are valid for running all forms.

Options when defining the mode in which the form opens

Option	Description
Data entry	The target form opens in Data Entry mode.
Inherit from the source form	The target form will open in the same mode in which the source form was opened. *For example, if the source form was run in Data Entry mode, the target form will also open in Data Entry mode.*
Navigation	The target form opens in Navigation mode.

Buttons in the window

Element	Description
Next	(Only with forms containing run parameters) Saves the defined run parameters and shows the options for defining the mode in which the target form is run.
Previous	(Only with forms containing run parameters) Saves the settings on the mode in which the target form is opened and shows the options for defining the run parameters.
OK	Saves the settings and closes the window.
Cancel	Closes the window without saving.

Properties Window: Filters cycle

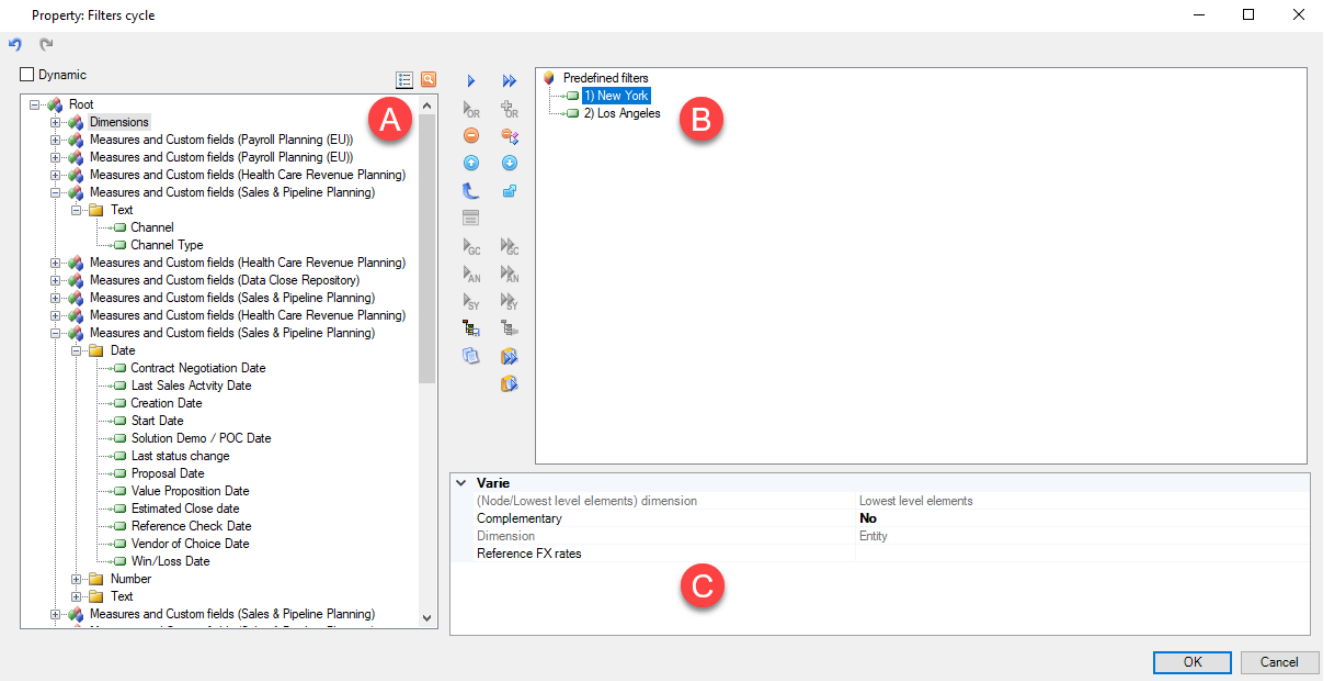
Excel > CCH® Tagetik > Design > (Left-hand panel) click form > > Hyperlink > Filters cycle

Purpose of window

This window allows you to set the actions to be repeated as many times as there are elements of a certain filter.

Options that can be inserted on tab filters can be inserted on option filters. See "Tab filter properties" on page 144 for a list of options.

What it looks like



Element	Description
A	List of available dimensions
B	List of sorted filters to be run
C	Details of selected filter

Buttons

Element	Description
	Changes the display mode.
	Allows you to search for the desired dimension.
	Adds one or several elements or nodes selected in the dimensions tree under the selected node in the filters tree.
	Only with filters already defined, adds one or several elements or nodes selected in the dimensions tree to all filters already defined. ***Note**: if there are no pre-defined filters in the column, it functions like the button.*
	Adds one or several filters selected in the dimensions tree to a selected filter, creating a OR type logical relationship. OR type logical relationships can only be created with elements or nodes belonging to the same dimension as the filters already defined. The added filter is labelled with the concatenation of the individual element codes separated by ",". The value displayed is equal to the sum of the elements entered.

Element	Description
	Adds one or several elements selected in the dimensions tree belonging to the same filter dimension to a node of filters linked by an OR logical relationship. The added filter is labelled with the concatenation of the individual element codes separated by ",". The value displayed is equal to the sum of the elements entered.
	Only with filters already defined. Deletes the selected element or node.
	Only with filters already defined. Deletes the entire level of the selected filter.
	Only with filters already defined. Moves the selected element up one position.
	Only with filters already defined. Moves the selected element down one position.
	Only with filters already defined on several levels. Moves the selected level up one level.
	Only with filters already defined. Replaces en bloc the selected elements of a filter with elements selected in the dimensions tree. Useful for reusing a filter by replacing its elements, but not its properties.
	Shows one or more elements of the filter containing the OR logical relationship, allowing you to see them and move or delete one or more of them.
	(Only for control groups) Adds the control group selected in the dimensions tree to the filter. See I gruppi di controllo: concetti di base.
	(Only for control groups) Adds the control group selected in the dimensions tree to all filters already defined. See I gruppi di controllo: concetti di base.
	(Only for FST items) Analytically adds, under the selected filter, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	(Only for FST items) Analytically adds, under all filters, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	(Only for FST items) Synthetically enters the FST under the currently selected filter. See "Financial statement templates" on page 83.
	(Only for FST items) Synthetically adds, under all filters, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	Allows you to use predefined filters.
	Updates the predefined filters.
	Copies the selected filter or dimension.
	Pastes the copied filter or dimension in all filters.
	Pastes the copied filter or dimension in the selected filter.

Design: style sheets

Contents

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Tagetik styles

Tagetik styles function

Tagetik styles are a reporting formatting tool that allows you to define and customise the appearance of a form: for example, it is possible to highlight certain data and improve the legibility of a table, by applying a specific font or changing the colour of the borders and cells.

To use Tagetik styles, style sheets must be created in the Excel add-in, their attributes must be defined and the style sheet must be applied to the desired forms or matrices.

Applying styles

Once a Tagetik style has been defined, it can be applied at two different levels:

- The form level. In this case, the user can choose whether to use the style defined as the default in the table or a customised style.
- At the matrix level. In this case, the user can choose to apply no style or to apply the style chosen for the form to the matrix's rows only or to its columns only

The style sheet is defined in Excel and therefore becomes available for all forms. Any changes to the style sheet affect all the forms with which it is associated.

Elements of the form to which to apply the styles

Styles can be set on the following elements:

- numbers
- alignment
- fonts
- borders
- fill
- protection

It is useful to create a different style for every attribute, e.g. one for the fonts, one for the numbers, one for alignment and one for the borders, rather than one style containing several definitions. Thus, it is possible to design a flexible layout and create several combinations of styles.

Levels of style application in the matrix

A customised style is applied by associating the styles on the matrix with the elements that make up its structure, shown as a hierarchical tree. The style is then applied downstream from the highest level; the

lower sections inherit the settings of the higher levels, unless a different style is specified.

***Note**: styles can be associated with all sections of the matrix.*

The details of the tree hierarchy and descriptions of its elements are provided below:

Element	Description
(Level 1) Matrix	Highest level of the tree: styles are applied to all elements within the matrix.
(Level 2) Rows and columns	Styles on this level are applied to all the matrix rows/columns.
(Level 3) Row/column structure	Styles on this level are applied on the areas of the row/column structure. The following details are available: • **Only values**: the style is only applied to the row/column values area (numbers). • **Only headers**: the style is applied to the row/column headers. • **Whole area**: the style is applied both to the values area and to the row/column headers.
(Level 4) Row/column content	This is a further detail of the level 3 elements. Here, styles are applied only on the basis of specific content in the row/column values. • **Amounts**: the style is applied to the values read from the database. • **Placeholders**: the style is applied to placeholders defined in the matrix. • **Formulas**: the style is applied to formulas defined in the matrix. • **Subtotals**: the style is applied to subtotal-type formulas defined in the matrix. • **Hyperlink for new elements wizard**: the style is applied to hyperlinks for opening new rows in dynamic matrices. • **FST items** the style is applied to items of the financial statement template used in the matrix. ***Note**: at this level, several styles can be associated with every element.* ***Note**: If there are several elements of this level (e.g. amounts and FST items) in the matrix's row/column filter tree, then the system will apply the style set on the lowest element in the row/column filter tree.*
(Level 5) Formatting type	At this level, styles manage the formatting of FST items, formulas and subtotals. ***Note**: when FST items are defined (see "FST items page" on page 93), it is possible to define the **Formatting type** attribute, which specifies the format that will be applied only to lines with the same formatting.* If there are several levels of formulas and subtotals, this level represents the level at which they are nested within the matrix. *For example:* *- Level 0 subtotals/formulas only depend on non-formula elements (amounts),* *- Level 1 subtotals/formulas depend at the most on level 0 subtotals/formulas.*

When a style is applied to a given section, that section is shown in bold in the tree.

Conditional formatting

The conditional formatting function allows you to specify a condition subject to which the style will be applied when the form is run. Conditions are defined by inserting expressions in cell field syntax (see "Cell fields" on page 41). Some examples are provided below:

Condition	Description
$Period.code = '06'	Apply the formatting for period 06 only
CONTAINS($Scenario.version, "ACT")	Apply the formatting for the current scenarios only
OR($Period.code = "06", $Period.code = "12")	Apply the formatting for periods 06 or 12 only
$Cust_Dim2.attribute5 = "Yes"	Apply the formatting only for elements that have Custom dimension 2 with attribute 5 set as "Yes".

*To insert a condition that needs to be met for the style to be applied, click ... next to **Condition**, or use the predefined functions present in the left-hand column.*

Leading and trailing styles

The **leading** and **trailing** options allow you to apply the style only to the first or last row/column of a matrix. They are useful when you only want to apply an external border to a matrix, for example. In this case, the style must only be applied to the outermost elements of the matrix, leading and trailing.

Styles for the index

The defined styles include specific default styles to format index sheets. These styles can be modified.

Style	Description
TGK_TOC_PAGE_COLUMN	Style applied to the column containing page numbers
TGK_TOC_REPORT_ROW	Style applied to index rows representing a form
TGK_TOC_TEMPLATE_ROW	Style applied to index rows representing a template
TGK_TOC_SHEET_ROW	Style applied to index rows representing a sheet

Tagetik styles for web forms

The style settings configured in the Excel add-in are also valid for web forms. However, when styles are set in Excel, it is possible to determine the elements to be managed in the style, e.g. only fonts, only borders, etc. On the contrary, for web forms the system reads all the existing settings. Consequently, if previously formatted elements have been deselected in the style within Excel, the web form will in any case read all the elements that make up the style.

Applying style sheets

Associate a style sheet with a form

Excel > CCH® Tagetik > Design > open form > Form options

Select the **Layout options** tab and select the desired style sheet in **Tagetik Style**.

***Note**: the appearance of the form elements displayed in any working mode is determined when a style sheet is associated with the form. The **Layout options** tab includes parameters that manage the appearance and certain cell formatting characteristics, specifically for using the form in data entry mode.*

Associate a style sheet with a matrix

When a style sheet is associated with the form, its use at the matrix level needs to be activated in any case.

1. Open the matrix (see "How to access the windows" on page 9).
2. In the **Matrix options** tab, select the **Layout options** tab, then in **Formatting type**, select **Apply Tagetik style**.

The matrix will then be formatted via the style sheet associated with the form in the **Form options** window (see "Associate a style sheet with a form" above).

Associate a style sheet with matrix rows/columns

Even if a style sheet is associated with the form and in use in the matrix, it is possible to define a specific style for one or several rows/columns.

1. Open the matrix (see "How to access the windows" on page 9).
2. In the **Row/column filters** tab, select the **Layout options** tab, then in **Formatting type**, select **Apply Tagetik style**.
3. In the row/column filters tree, select a row/column filter, then click ... next to **Tagetik style** in the properties table. The system will open a window of styles to be applied. The elements hierarchy only shows elements contained in the row/column.

4. In the elements hierarchy, select the desired element.
5. In the list of styles, select the desired style.
6. Click to create the relationship.
7. Click **OK**.

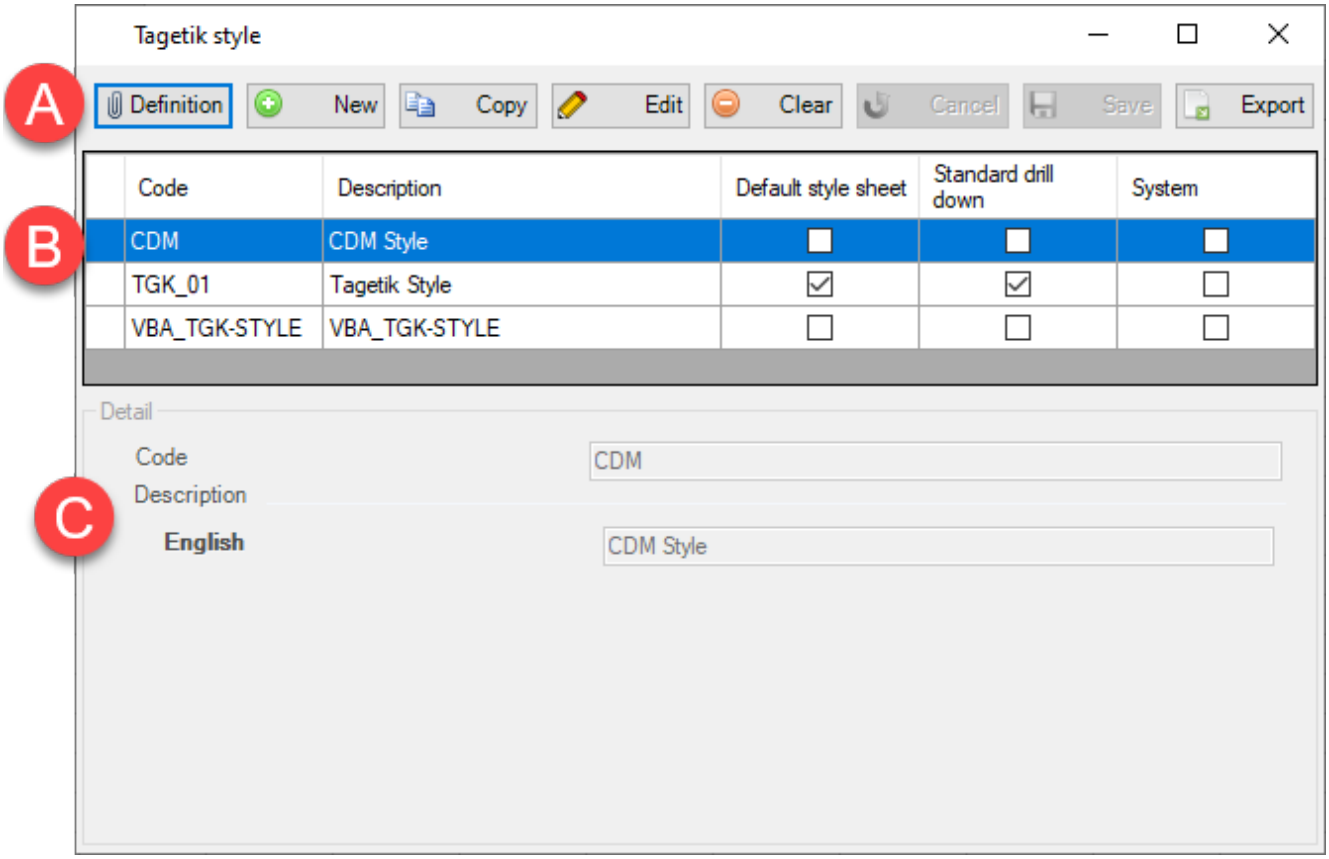
Tagetik style window

Excel > CCH® Tagetik > Tagetik style

Purpose of window

This window shows a list of the existing Tagetik styles, allows you to create and delete them and change some of their management settings.

What it looks like



Element	Description
A	Action buttons
B	List of Tagetik styles
C	Manage Tagetik styles section

Window sections

Element	Description
List of existing Tagetik styles	Allows you to view a list of the existing Tagetik styles and manage some of their operational properties
Manage Tagetik styles	This allows you to manage the information (code and description) on the selected style sheet, and enter a specific description for every language supported by the system.

Buttons in the window

Element	Description
	Opens a window to define details and formatting in the style sheet.
	Creates a new style sheet.
	Creates a style sheet as a copy of the selected one.
	Edits the description and some characteristics of the selected style sheet.
	Undoes edits to the style sheet that have not yet been saved.
	Deletes the selected style sheet.
	Saves changes made to the style sheet.
	Exports the styles defined in the selected style sheet and makes them available in the currently open Excel file.  *Take advantage of this function in the Collaborative Office environment to produce external reports in the same formats as CCH Tagetik reports. Exported styles can overwrite pre-existing styles or be added to them.*

List of existing Tagetik styles

The following information is shown for each style sheet:

Element	Description
Code	Style sheet code.
Description	Description of the style sheet.
Default style sheet	Sets the selected style sheet as the default style sheet for all new forms created after the current window is closed.
System	Indicates a system style sheet: not editable.
Standard drill down	Indicates the style sheet which, in the absence of default style sheets on the reference form, the system applies to the Excel sheet containing the drill down result (see "The drill down function" on page 340).

Styles window

Excel > CCH® Tagetik > Tagetik style > Definition

Purpose of window

This window allows you to define the characteristics of the selected style sheet.

Tabs in window

Element	Description
Defined styles	Shows a list of the defined styles for the style sheet. ***Note**: the list will appear empty when defining a new style sheet. See "Styles window - Defined Styles tab" on the facing page.*
Style to use	Shows the relationships between the matrix sections and the style to be used and allows you to create styles that are applicable to the matrix sections. See "Styles window - Styles to use tab" on the facing page.

Styles window - Defined Styles tab

Excel > CCH® Tagetik > Tagetik style > Definition > Defined Styles tab:

Purpose of the tab

This tab allows you to see and manage the styles that have been defined on the style sheet from which they are being accessed.

Action buttons in the Defined Styles tab

Element	Description
	Opens a window to select the managed elements to create a new style.
	Opens a window to select the elements managed in the selected existing style.
	Deletes the selected style.
Open index styles	Inserts system styles for index sheets.

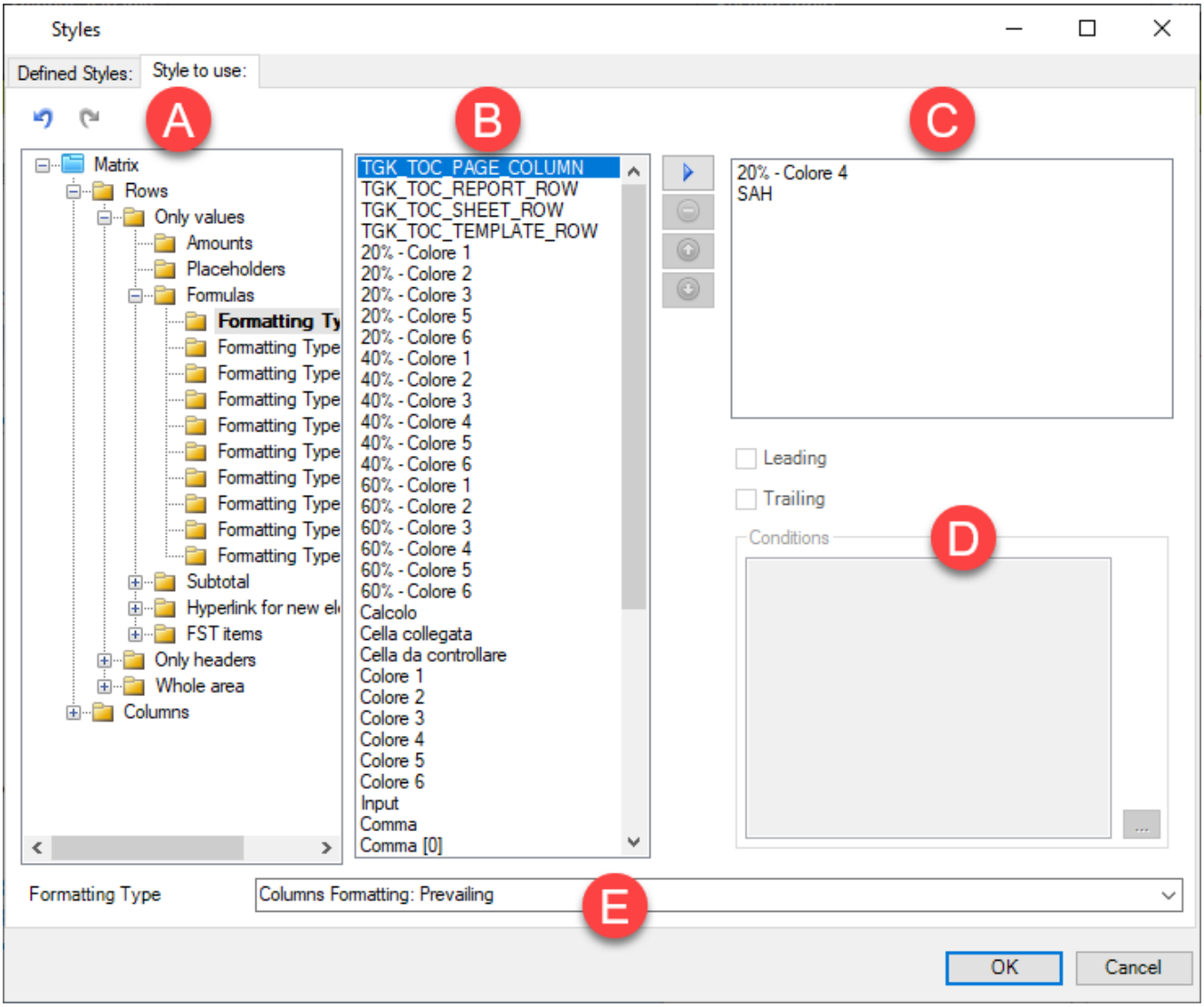
Styles window - Styles to use tab

Excel > CCH® Tagetik > Tagetik style > Definition > Style to use tab

Purpose of window

This window allows you to define the style to be used for every section of the matrix of the open form.

What it looks like



Element	Description
A	Hierarchy of matrix elements on which to define styles ***Note**: the elements with which a style is already associated are shown in bold.*
B	List of available default styles
C	List of styles associated with a section of the matrix. ***Note**: the order in which the styles are displayed corresponds to the order in which they are applied at runtime.*
D	Definition area for any conditional formatting
E	Prevailing formatting type selection

Buttons and fields in the window

Element	Description
	Relates the selected style to the selected matrix element.
	Removes the style/matrix element relationship.
	Moves the selected style up.
	Moves the selected style down.

Element	Description
Leading	Allows you to apply the style only to the first row/column of a matrix.
Trailing	Allows you to apply the style only to the last row/column of a matrix.

Prevailing formatting type

Element	Description
Rows Formatting: Prevailing	If several styles are defined on rows and columns for the same sections, the style set on the row will prevail.
Columns Formatting: Prevailing	If several styles are defined on rows and columns for the same sections, the style set on the column will prevail.

Style settings window (Styles)

Excel > CCH® Tagetik > Tagetik style > Definition > New

Purpose of window

This window allows you to create a style and define its characteristics.

Options on page

Element	Description
Style name	Allows you to set the style's name.
Format	Opens Excel's **Format Cells** window.
The style includes (according to selection)	Select one or more elements managed by the style. • **Number**: (default value: Standard) • **Alignment**: (default value: General, Bottom alignment) • **Font**: (default value: Calibri (Body) 11; text 1) • **Border**: (default: No borders) • **Fill**: (default: No colour) • **Protection**: (default value: Locked) When an element is selected, its default settings are shown, and can be customised (see "Styles window" on page 234).

Defining Tagetik styles

Introduction

Defining a Tagetik style involves several levels of configuration, to be carried out in the order indicated below

Create a style sheet

Excel > CCH® Tagetik > Tagetik style

1. Click . A row will appear in the list of Tagetik styles.
2. Insert a code and a description for the Tagetik style.
3. In the list, select the desired operational properties.
4. Click .

Create a style

Excel > CCH® Tagetik > Tagetik style > Definition

1. Click . The **Styles** window will open.
2. Insert a name for the style.
3. Select one or several elements to be defined in the style being created.

   ***Note**: to optimally manage Tagetik styles and correctly apply them in web forms, it is advisable to create one style for every element to be formatted.*

4. Click **Format...**: Excel's **Format Cells** window will open.
5. Set the formatting of the elements selected in step 3 in the relative tab in the window and click **OK**.
6. In the **Styles** window, click **OK**.

Create styles for index sheets

Excel > CCH® Tagetik > Tagetik style > Definition

1. Click **Open index styles**: the styles for index sheets will appear in the list of Defined styles.
2. Select the desired style
3. To change it, click : the **Styles** window will open.

   ***Note**: styles can be renamed*

4. Click **Format...**: Excel's **Format Cells** window will open.
5. Set the formatting of the elements selected in step 3 in the relative tab in the window and click **OK**.
6. In the **Styles** window, click **OK**.

Relate styles to matrix elements

Excel > CCH® Tagetik > Tagetik style > Definition > Style to use tab

1. In the hierarchy of matrix elements, select the desired element.
2. In the list of styles, select the desired style.

   ***Note**: the style related to the element is also applied to any nested elements, if possible and if no style has been associated with them. The sections to which the style is related are shown in bold.*

3. Click to create the relationship.
4. (Optional, if a style has been related to a row/column element) Select the **leading** option and/or the **trailing** option to apply the style to the first or last row/column of the matrix, respectively.
5. (Optional) To make the application of the style conditional, click ... and insert the desired parametric text in the **Cell Fields Wizard**. Alternatively, use the predefined functions present in the left-hand column (see "Inserting text with parametric elements" on page 42).
6. Click **OK** and close the **Styles** window.

Set the prevailing formatting type

Excel > CCH® Tagetik > Tagetik style > Definition > Style to use tab

In **Prevailing formatting type**, select the option to be applied if several styles are defined on rows and columns.

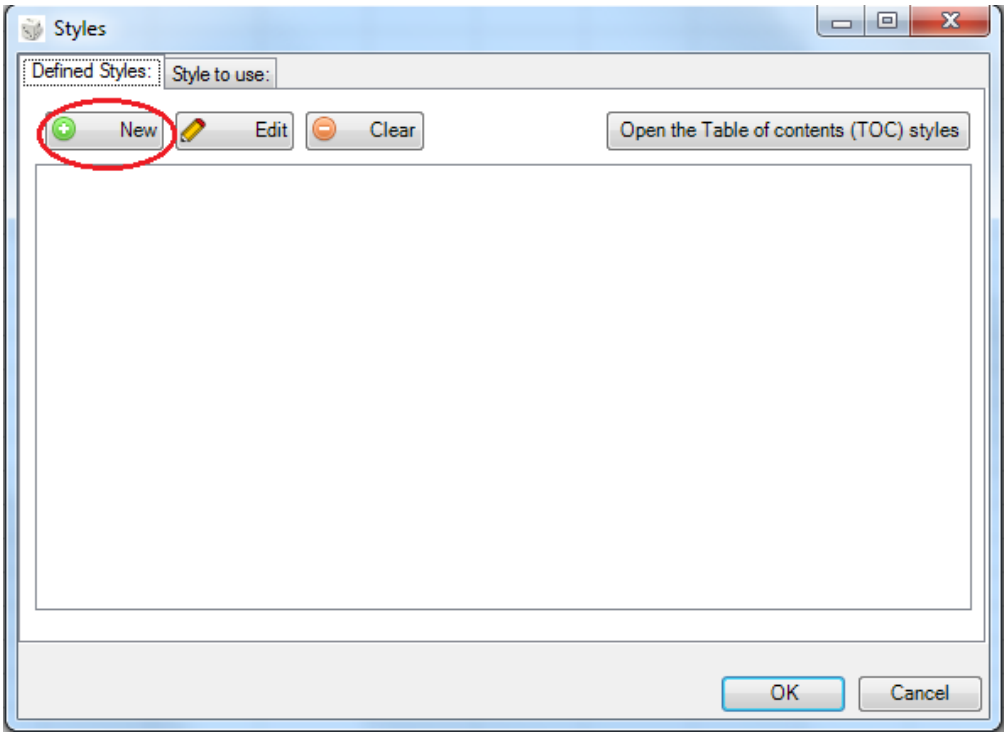
Example of the definition of style sheets

Defined styles

Suppose you want to define three different styles: one for the font to be used within the matrix, one for the numbers and one for the column headers.

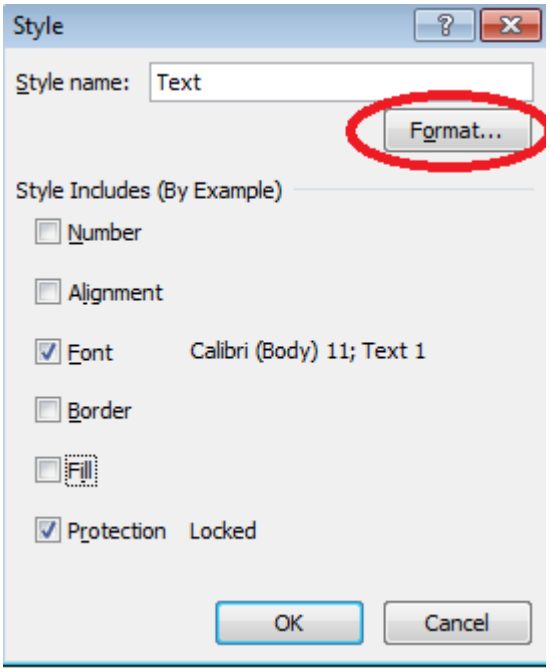
It is advisable to create one style for every attribute, i.e. one for alignment and one for borders, instead of creating one style containing both definitions. This allows you to design the layout with greater flexibility, applying various combinations of styles.

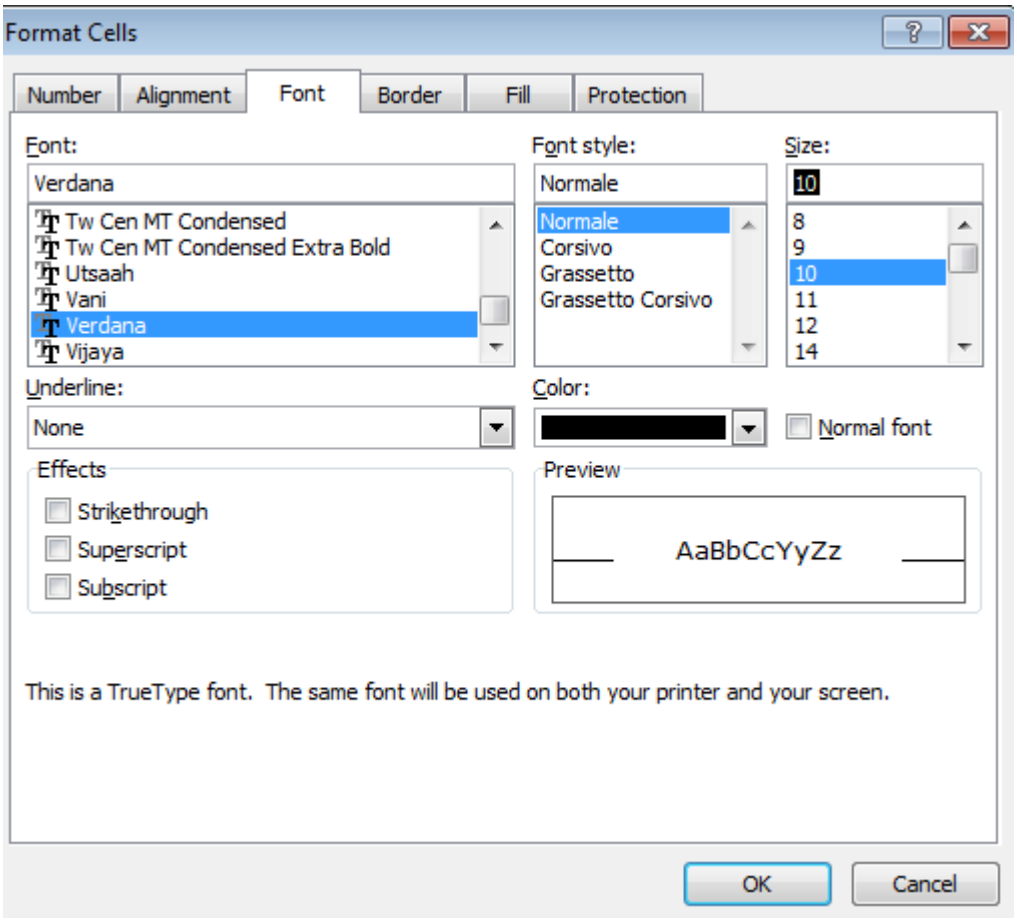
To create a new style, click the **New** button in the definition window.



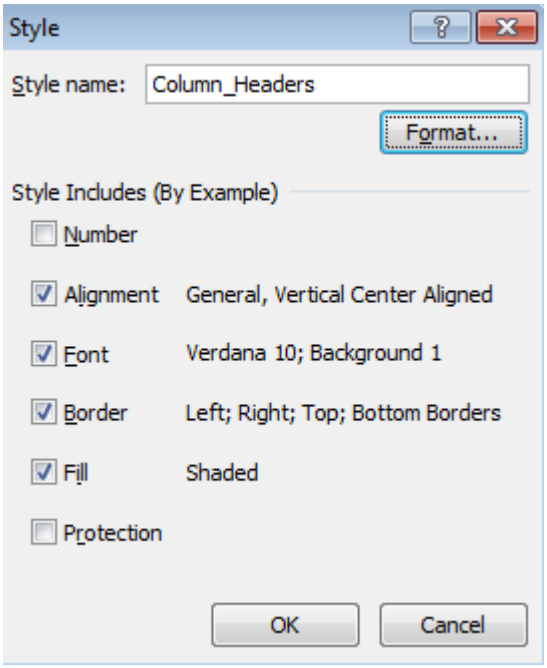
When the "New" button is clicked, the system will open a window in which a name must be assigned to the style being defined, e.g. Text, and the type of style to be applied must be selected. In this case, we have selected "Font" because we want to define the type of font to be used for the data displayed in the matrix. The format of the font can be changed by clicking on the **Format** button.

Web form 2.0 treats the options in the form below as if they were always selected. Consequently, if one of these options is not selected in Excel (e.g. the Fill flag), but one of the related settings has been previously customised (such as background colour or font), that setting will still be applied.

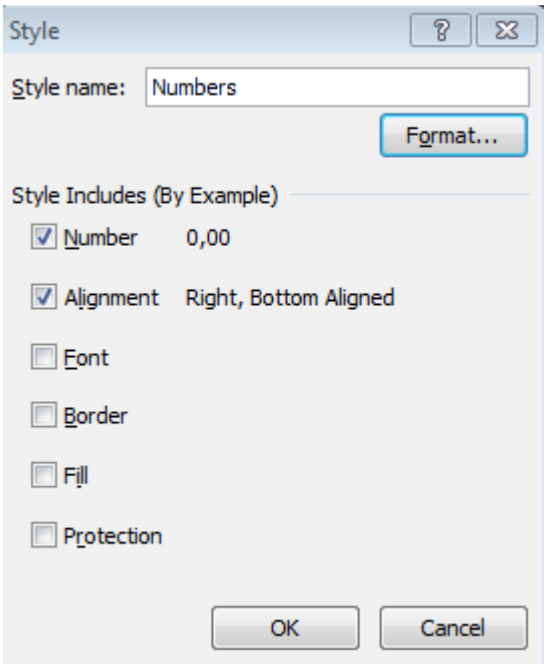




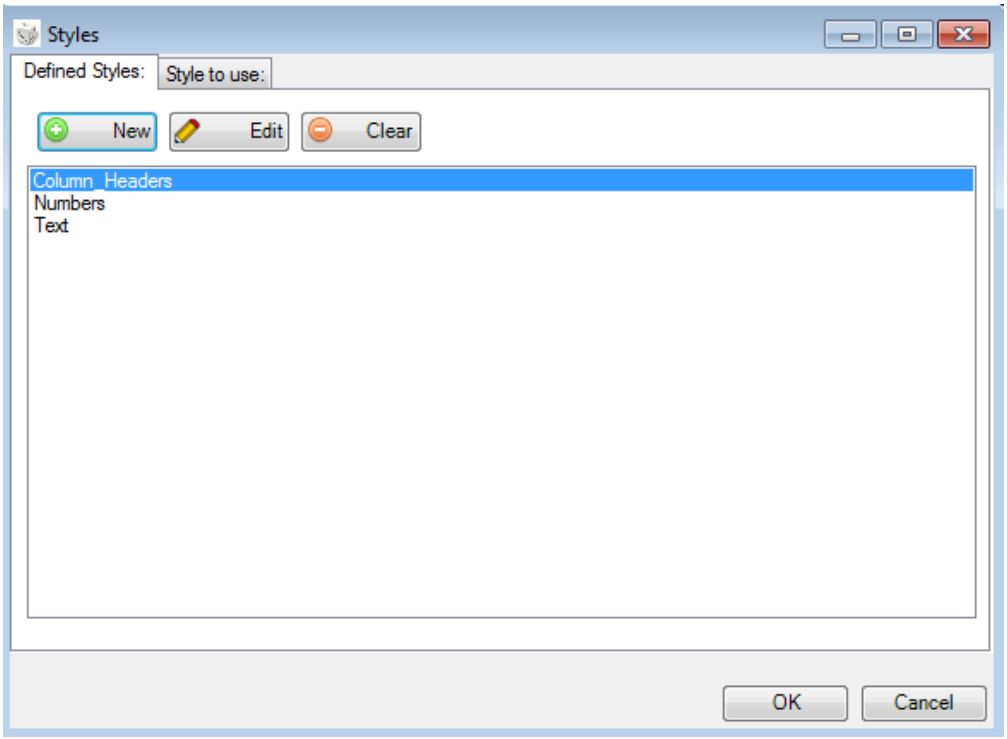
Now let's see how to create the second style to be applied to header columns in our matrix. Suppose you want the header columns to have a blue background, with white text aligned to the centre, and a black border. In this case, we will have to activate the Alignment, Border, Fill options and set the desired format for each of them



The last style to be defined is the one to apply to numbers. Suppose you want the numbers aligned to the right, with two decimal figures. In this case, we will need to activate the Number and Alignment options and set the desired format for each of them.



At this point, our style sheet contains three different styles. Any changes can be made using the **Edit** button.

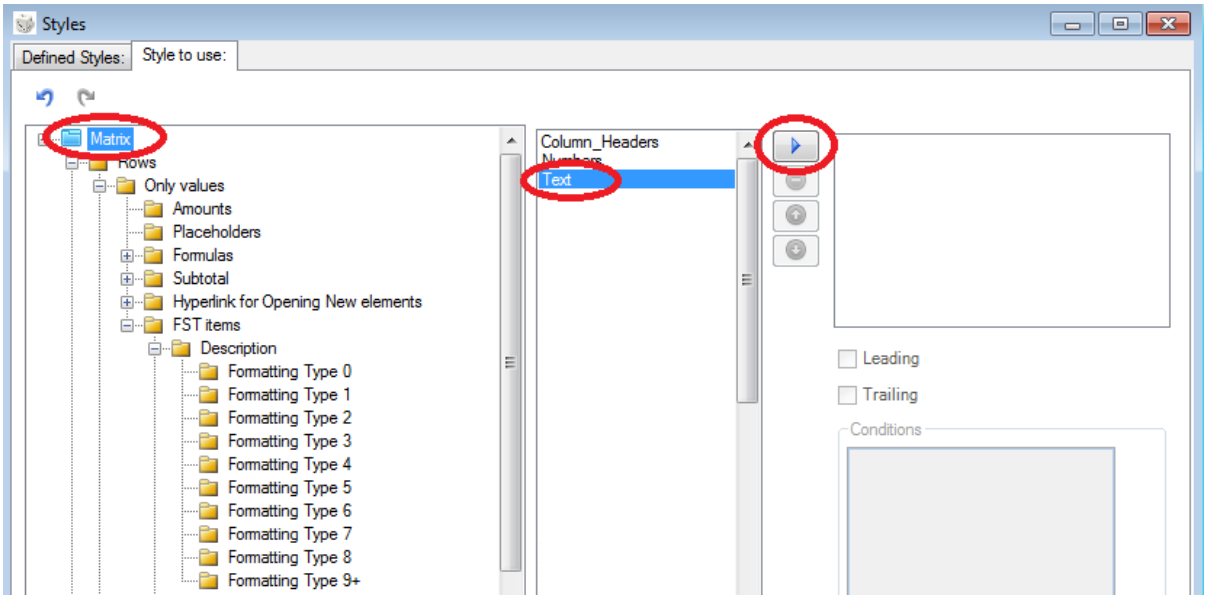


Style to use

Now let's see how to associated the three styles defined in the previous example – *Column_Headers*, *Numbers andText* – to the various sections of the matrix.

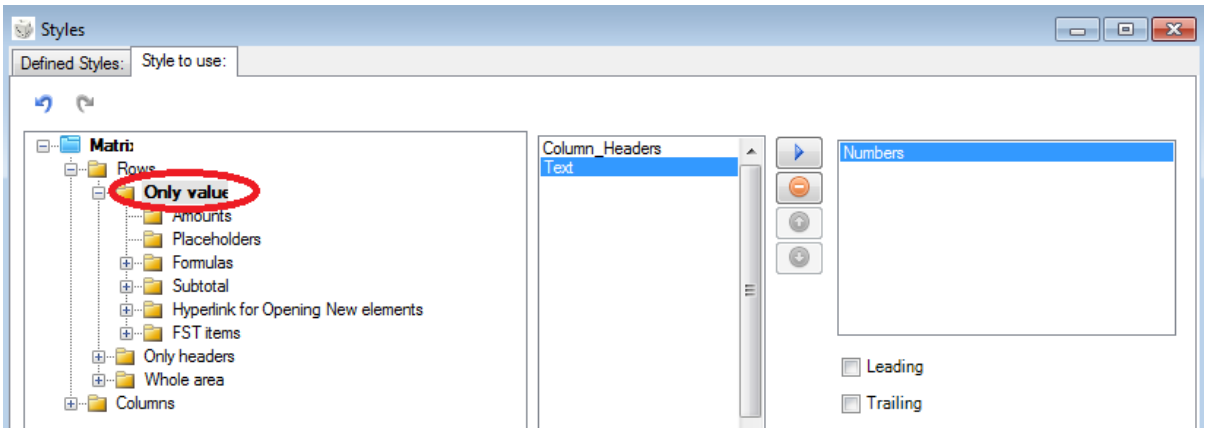
The *Text* style, which sets the font to be used for the entire matrix, must be associated with level 1 of the tree. Then the steps to take are:

- in section (1) select level 1 "Matrix" of the tree;
- in section (2) select the Text style
- click the button 



The *Numbers* style must be assigned to all areas of the matrix in which numbers are shown or to the Rows / Only values section. Then the steps to take are:

- in section (1) select level 3 "Matrix/Rows/Only values" of the tree;
- in section (2) select the Numbers style
- click the button 

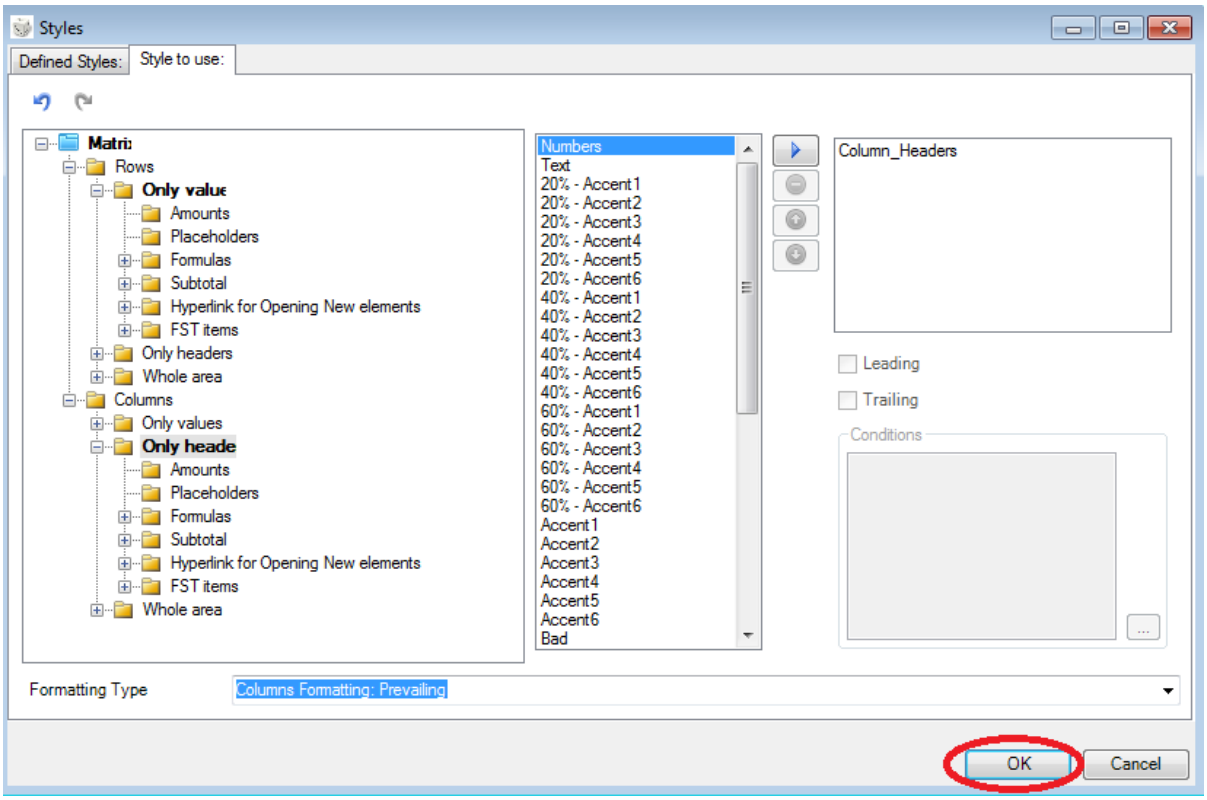


The *Column_Headers* style must be assigned to all column headers or to the Columns / Only headers section. Then the steps to take are:

- in section (1) select level 3 "Matrix/Columns/Only headers" of the tree;
- in section (2) select the Column_Headers style
- click the button 


The system shows the matrix sections for which a style has been defined are shown in bold

At this point, when the OK button is clicked, our new style sheet will be ready to be used



The results of applying the style sheet defined in the previous example are shown below. Matrix formatting can be further modified by fine-tuning our style sheet; for example, by adding formatting for row headers, borders and a different font/colour for subtotals and formulas.

	2012 Budget	2012 Actual
	Dec	Dec
Product Revenues	-1000,00	-1000,00
Sales Revenues	-1000,00	-1000,00
Other Revenues	-1000,00	-1000,00
Other Revenues	-1000,00	-1000,00
Total Gross Revenues	-2000,00	-2000,00
Cost of Good Sold	1000,00	1000,00
Variance to STD	1000,00	1000,00
Cost of Purchase	2000,00	2000,00
Direct Margin	-4000,00	-4000,00
Sales Deduction	1000,00	1000,00
Year-End Bonus	1000,00	1000,00
Sales Discounts	2000,00	2000,00
Royalties	1000,00	1000,00
Returned Goods	1000,00	1000,00
Advertising & Promotions	1000,00	1000,00
Salespeople Costs	1000,00	1000,00
Advertising and Promotions	4000,00	4000,00
Total direct costs	6000,00	6000,00
Operational Gross Margin	-10000,00	-10000,00

Reporting Data Processing

Contents

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Data processing run from reporting

Data processing from forms

The reporting system allows you to run data processing directly from forms, using the filters set in the same forms as run parameters. This data processing allows you to load data onto the system, edit them and run complex calculation algorithms (e.g. calculations related to the consolidation area or financial planning area).

This data processing must be inserted in the forms in the form design step and can only be run from data entry forms.

Data processing configuration

The configuration of data processing is guided and requires the following steps to be completed:

1. Select the data processing to configure.
2. Define the method for retrieving the value of the parameters needed to run the data processing.

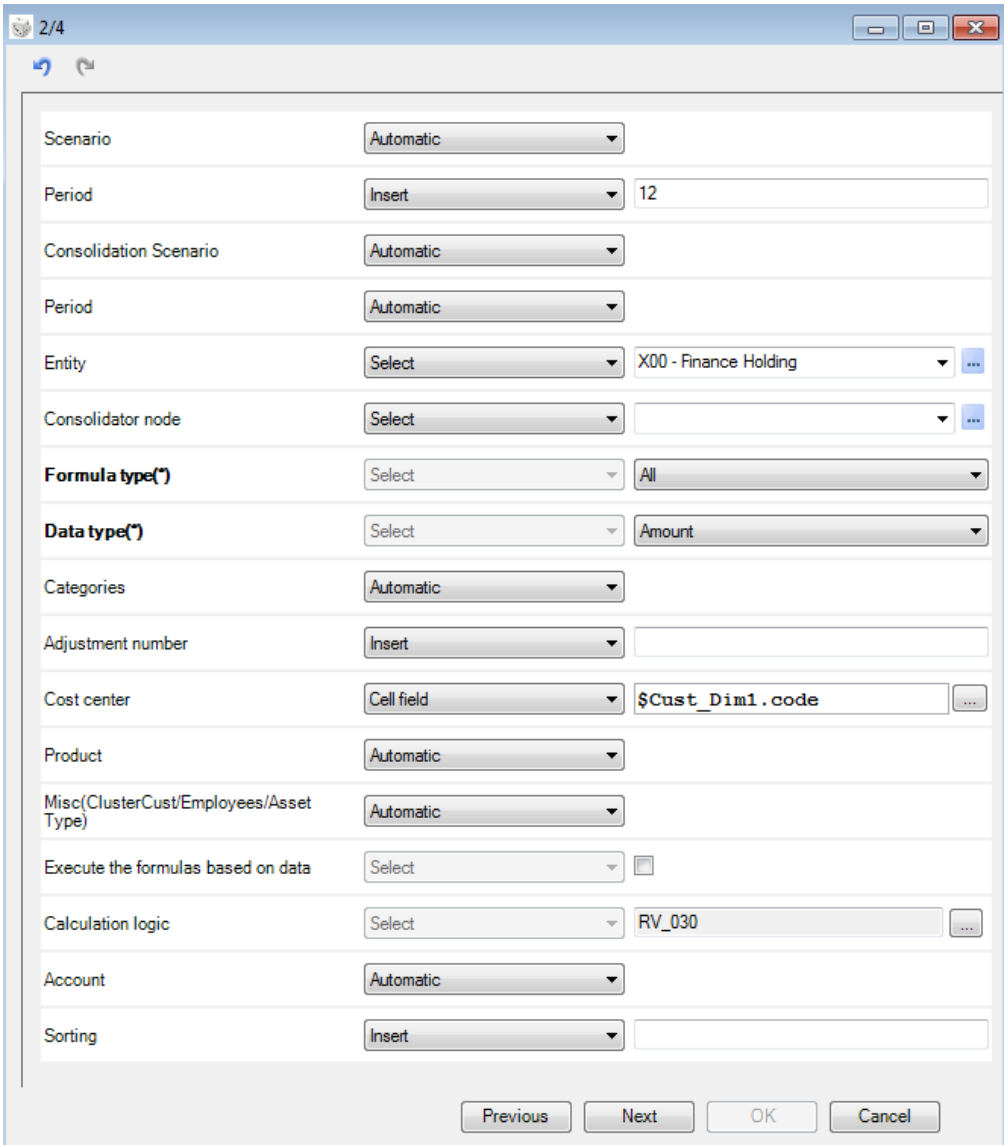
   ***Note**: the presence of errors in this step prevents the data processing from being run (e.g. if a cell field with which a parameter must be defined returns an error).*

3. Define the run settings.
4. Assign a code and description to the created run.

Example of options for retrieving the value of the parameters

To insert **Calculation logics** data processing in a form, it is necessary to define the method for retrieving the value of the parameters shown below. Some parameters refer to CCH Tagetik dimensions, such as **Scenario** and **Entity**, others refer to the type of calculation logic to be processed and the type of data to be processed.

The values of the **Scenario**, **Category**, and **Custom dimensions** parameters are retrieved automatically from the form's run parameters. The **Period** parameter was typed, the **Entity** parameter was selected from the list of tables and the **Cost centre** parameter was recovered from a cell field. **Calculation logic** requires the user to indicate the calculation logic to be run.



Manage data processing in reporting

Insert data processing in a form

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

Click : the **Setup & Admin - Data Processing** window appears. Otherwise, in the **Open items** tab, right click the form template in which to insert the data processing and select **Run Data processing** > **New Data processing**.

Configure data processing

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

1. Click : the **Setup & Admin - Data Processing** window appears. Alternatively, in the **Open items** tab, right click the form template in which to insert the data processing and select **Run Data processing** > **New Data processing**.
2. Select the desired data processing and click **Next**.
3. Define how the value of the parameters needed for the data processing should be retrieved and click **Next**. If a DTP referring to reporting parameters is chosen, repeat this step for those parameters.
4. Select the run settings and click **Next**.
5. Assign a name and description to the data processing and click **OK** to save.

Setup & Admin - Data Processing window

Excel > CCH® Tagetik > Design > (Left-hand panel) click form > > Data Processing

Purpose of window

This window allows you to configure the data processing to be run by the form.

Elements of the data processing selection step (1/5)

Element	Description
ETL job for tables	Data processing that loads metadata from an external source to the CCH Tagetik database via an ETL Job
ETL job for values	Data processing that loads data from an external source to the CCH Tagetik tables of values via an ETL Job.
Preliminary data processing on Original scenarios	Data processing to be run on original scenarios before collecting the data
Preliminary data processing on Consolidation scenarios	Data processing to be run on consolidation scenarios before starting data collection and consolidation operations
Closing	Closing data processing (allocations)
Entity data processing	Data processing to be run on original scenarios during data collection
Consolidated	Data processing to be run on consolidation scenarios during the consolidation process
Cash Flow Planning	Specific data processing for the Cash Flow Planning module
Calculation logics	Data processing that allows you to run classic and advanced calculation logics (e.g. MD scripts)
Monthly splitting	Data processing that runs the new monthly splitting procedure. This procedure allocates an amount of a scenario/period on a range of scenarios/periods according to the set rules.
Form - Break Back	Break-back data processing that distributes the values present at the node level on the lower levels
Process cube	Data processing that updates CCH Tagetik cubes
Algebraic operations	Data processing that allows you to copy, organise progressively, delete and compress the data of a scenario
Monthly splitting	Data processing that runs the old monthly splitting procedure. This procedure allocates an amount of a scenario/period on a range of scenarios/periods based on a driver.
Forecast	Data processing that generates a forecast scenario based on an actual scenario and a budget scenario

Element	Description
LOANS_PLANNING - Bank loans depreciation calculation plan	Data processing that creates the entity's bank loans depreciation plan
Data Transformation	Data processing that runs the Data Transformation Package of one or several Analytical Workspaces. Applicable on all Analytical workspaces in the environment which have at least one associated DTP.
Tagetik model process	Data Discovery: Qlik View Model process
Send generic email	Data processing that allows you to send an email on saving or via a link
Update Financial Workspace from Analytical Workspace	Data processing that allows you to automatically synchronise the data of an Analytical Workspace dataset and the Financial Workspace data structures. To do this, Financial Workspace mapping, which defines the loading method and the relationship between the dimensions and the CCH Tagetik logics must have been configured (see Quick Data Loader - Pagina Datasource mapping).
Loading to Analytical Workspace	Data processing that allows you to load data from external datasources onto physical partitions of Analytical Workspace Datasets (e.g. the form content).
SUBMISSION_MD - Multi-organisation submission	Multi-organisation submission data processing on nodes for the analytical dimension by process / scenario / period / workflow / step
APPROVAL_MD - Multi-organisation approval	Multi-organisation approval data processing on nodes for the analytical dimension by process / scenario / period / workflow / step
REJECTION_MD - Multi-organisation rejection	Multi-organisation rejection data processing on nodes for the analytical dimension by process / scenario / period / workflow / step
Stored Procedures	Data processing that runs a stored procedure. See Stored procedure.
ML Training	Data processing that trains a Machine Learning Model
ML Execute	Data processing that runs the prediction of a previously trained Machine Learning Model
Deploy	Data processing that runs a deployment for all users, scenarios and periods as if it were run from a workflow task. To run the deployment for all processes, **Enable all processes** must be selected. Otherwise, the deployment will only concern the process used to run the report. The visible deployments correspond to those reported in the general deploy item in the Admin menu.

Options in the parameters configuration step (2/5)

This step shows the available parameters for the data processing selected in the first step. Those shown in bold are mandatory. If errors are returned during this step, the data processing will be interrupted.

> ***Note**: if the data processing selected during the first step is **Data Transformation**, the mandatory **DTP** field will appear with the DTPs of the Analytical Workspaces chosen in that step.*

The configuration options for each parameter are detailed below.

Option	Description
Automatically	Only for dimensional parameters The system retrieves the value from any input parameters in the form, the global filter, the tab filter (tab active at launch) and the matrix filter (the latter only if there is only one matrix in the sheet). The value returned will be the same as that returned by the cell field on the flat dimension of the parameter. For example, if the parameter is of Entity type, then the value will be equal to that taken through the $Entity cell field. See the cell field definition mode below for further understanding
Excel Cell	The system retrieves the value of the parameter from the desired Excel cell of the current sheet, specified with the syntax **cell H30**.
Insert	Allows the user to enter the parameter value.

Option	Description
Select	Allows the user to select the value of the parameter from a list of possible values.
Cell field	The system retrieves the value of the parameter from a cell field. If this option is used for a dimensional parameter, it is preferable to type in an expression that, when defined, returns a node or leaf, not its code or description. *For example, for an entity-type parameter, the expression $Entity must be used, not $Entity.code.* The cell field is resolved by taking into account the form input parameters, the global filter, the tab filter (tab active at launch) and the matrix filter (the latter only if there is only one matrix in the sheet)

If the chosen data processing is a **DTP**, and if this DTP refers to reporting parameters , then the system displays a further screen with the following options

Option	Description
Automatically	Only for dimension parameters. The system retrieves the value from any input parameters in the form, the global filter, the tab filter (tab active at launch) and the matrix filter (the latter only if there is only one matrix in the sheet). The value returned will be the same as that returned by the cell field on the flat dimension of the parameter. For example, if the parameter is of Entity type, then the value will be equal to that taken through the $Entity cell field. See the cell field definition mode below for further understanding
Insert	Only for text-type parameters. Allows the user to enter the parameter value.
Select	Only for dimension parameters. Allows the user to select the value of the parameter from a list of possible values.
Cell field	The system retrieves the value of the parameter from a cell field. If this option is used for a dimensional parameter, it is preferable to type in an expression that, when defined, returns a node or leaf, not its code or description. *For example, for an entity-type parameter, the expression $Entity must be used, not $Entity.code.* The cell field is resolved by taking into account the form input parameters, the global filter, the tab filter (tab active at launch) and the matrix filter (the latter only if there is only one matrix in the sheet)

Note: **Boolean-** and **dictionary**-type parameters are not supported.

Options in the execution configuration step (4/5)

Option	Description
Force refresh	Reloads the form with data updated after running the data processing.
Override refresh dimensions tree	Reloads the dimensions tree with metadata updated after running the data processing.
Run when saving data	The system automatically runs the data processing on saving the form opened in data entry to ensure that the data processing is run on updated data.
skip tab filters	Excludes the values of the tab filters present in the form from the **Automatic** definition of the parameters.
"When saving data, please use modified cells for setting automatic parameters"	The system only considers the filters of the edited cells for the purpose of**Automatic** definition of the parameters. *For example, if two periods of a matrix containing 12 periods have been edited, the system only considers the filters of those cells to define the automatic parameters.*
"At Runtime, show a link for the run"	On opening a form in data entry mode, the system creates a link to the data processing from which it can be run.

Buttons in the save step (5/5)

Element	Description
	Copies the string in the Visual Basic macro to run the data processing. See Macro custom.
	Uses a shared dictionary item in the description.
	Removes the shared dictionary item in use.

Setup and Administration Window - Data processing on custom/analytic organisations

Excel > CCH® Tagetik > Design > (Left-hand panel) click form > > Data Processing

Purpose of window

This window allows you to configure the data processing to be run from form when launching it from a process with dimension by step, in the case where the step is on a custom/analytic dimension.

Elements of the data processing selection step (1/5)

Element	Description
Data Transformation	Data processes running the Data Transformation Package of a single Analytical workspace. Applicable on all Analytical workspaces in the environment which have at least one associated DTP.

Options in the parameters configuration step (2/5)

This step displays the parameters available for the data processing selected in the first step. Those shown in bold are mandatory.

***Note**: if the data processing selected during the first step is **Data Transformation**, the mandatory **DTP** field will appear with the DTPs of the Analytical Workspaces chosen in that step.*

***Note** data processing performed on processes for analytical/custom dimensions requires enabling the 'context-independent run' parameter*

The configuration options for each parameter are detailed below.

Option	Description
Automatically	Only for dimensional parameters The system retrieves the value from any input parameters in the form, the global filter, the tab filter (tab active at launch) and the matrix filter (the latter only if there is only one matrix in the sheet). The value returned will be the same as that returned by the cell field on the flat dimension of the parameter. For example, if the parameter is of Entity type, then the value will be equal to that taken through the $Entity cell field. See the cell field definition mode below for further understanding
Excel Cell	The system retrieves the value of the parameter from the desired Excel cell of the current sheet, specified with the syntax **cell H30**.
Insert	Allows the user to enter the parameter value.

Option	Description
Select	Allows the user to select the value of the parameter from a list of possible values.
Cell field	The system retrieves the value of the parameter from a cell field. If this option is used for a dimensional parameter, it is preferable to type in an expression that, when defined, returns a node or leaf, not its code or description. *For example, for an entity-type parameter, the expression $Entity must be used, not $Entity.code.* The cell field is resolved by taking into account the form input parameters, the global filter, the tab filter (tab active at launch) and the matrix filter (the latter only if there is only one matrix in the sheet)

The parameter ORGANISATIONAL DIMENSION shows all custom and analytical dimensions. The selected dimension will be the dimension of the step on which the data processing will be performed.

Options when setting up the parameters (3/5)

This step displays additional data processing parameters. They are:

- DIMENSION ELEMENT CODE: represents the element belonging to the dimension chosen in 2/5 as the value of the DIMENSIONE_ORGANIZZATIVA parameter
- Any reporting parameters defined within the **DTP**

The configuration options for each parameter are detailed below.

Option	Description
Automatically	Only for dimensional parameters The system retrieves the value from any input parameters in the form, the global filter, the tab filter (tab active at launch) and the matrix filter (the latter only if there is only one matrix in the sheet). The value returned will be the same as that returned by the cell field on the flat dimension of the parameter. For example, if the parameter is of Entity type, then the value will be equal to that taken through the $Entity cell field. See the cell field definition mode below for further understanding For the DIMENSION ELEMENT CODE parameter the system retrieves the value of the parameter from a form, tab or matrix filter if there is only one matrix in the sheet. In the case of two or more filters on the same dimension, the parameter will not be defined.
Excel Cell	The system retrieves the value of the parameter from the desired Excel cell of the current sheet, specified with the syntax **cell H30**. Option not available for the DIMENSION ELEMENT CODE parameter
Insert	Allows the user to enter the parameter value.
Select	Allows the user to select the value of the parameter from a list of possible values.
Cell field	The system retrieves the value of the parameter from a cell field. If this option is used for a dimensional parameter, it is preferable to type in an expression that, when defined, returns a node or leaf, not its code or description. *For example, for an entity-type parameter, the expression $Entity must be used, not $Entity.code.* The cell field is resolved by taking into account the form input parameters, the global filter, the tab filter (tab active at launch) and the matrix filter (the latter only if there is only one matrix in the sheet)

Note: **Boolean-** and **dictionary**-type parameters are not supported.

Options in the execution configuration step (4/5)

Option	Description
Force refresh	Reloads the form with data updated after running the data processing.
Override refresh dimensions tree	Reloads the dimensions tree with metadata updated after running the data processing.
Run when saving data	The system automatically runs the data processing on saving the form opened in data entry to ensure that the data processing is run on updated data.
skip tab filters	Excludes the values of the tab filters present in the form from the **Automatic** definition of the parameters.
"When saving data, please use modified cells for setting automatic parameters"	The system only considers the filters of the edited cells for the purpose of**Automatic** definition of the parameters. *For example, if two periods of a matrix containing 12 periods have been edited, the system only considers the filters of those cells to define the automatic parameters.*
"At Runtime, show a link for the run"	On opening a form in data entry mode, the system creates a link to the data processing from which it can be run.

Buttons in the save step (5/5)

Element	Description
	Copies the string in the Visual Basic macro to run the data processing. See Macro custom.
	Uses a shared dictionary item in the description.
	Removes the shared dictionary item in use.

Query

Utility

Queries allow you to obtain information from internal or external databases, in addition to that extracted from matrices, such as FX rates, tax rates and other rates saved in customised tables. To obtain this information, *SQL queries* can be created in the form.

Queries can be of the following types: **Normal**, **Data Validation** and **Drill-through**.

***Note**: details of the CCH Tagetik Data Model can be consulted in the Tagetik Data Model Repository document, available to download from CCH Tagetik's main toolbar > > **Download** (see Barra degli strumenti principale).*

Normal query

Query defined using standard SQL language. This is used to extract data from the database and view it in the form in table format. The table generated by the query can be edited using standard Excel functionalities.

Cell fields can be inserted into the query to make it parametric. This allows you to filter the data to be extracted and write queries that can be used in several contexts.

Example

Query defined to extract data from the table of FX rates: *SELECT cod_valuta, cambio_medio, cambio_finale FROM dati_cambio*. Since the data are not filtered by scenario/period, the query returns all the FX rates defined for all scenarios/periods.

If, for a form, you want to only extract the FX rates of a specific scenario/period, the query must be made parametric using cell fields: *SELECT cod_valuta, cambio_medio, cambio_finale FROM dati_cambio where cod_scenario = ${$Scenario.code} and cod_periodo = ${$Period.code}*.

If the query is run in a form that has the scenario and period as parameters, the expressions *${$Scenario.code}* and *${$Period.code}* will be replaced with the selected values.

> To make the scenario and/or period value fixed, the expressions *${$Scenario.code}* and *${$Period.code}* must be replaced with the desired value written between double quotation marks (" ").

Data Validation query

Query which allows you to define the value of a cell by selecting it from a list of predefined values. This list is the result of a *Data Validation* query associated with an element of the matrix. *Data Validation* queries do not appear in the Excel sheet.

> ### Example
>
> Given a matrix to enter information related to products managed on custom dimension 1. Every row must contain the product code and the respective price. To ensure that the inserted products are those actually being produced, the user's choice can be limited and a query that reads data from the custom dimension 1 table can be defined.
>
> If **Enable free text insertion** is selected, the system disables the validation notice and the user can also insert values not included in the results of the *Data Validation* query.

It is possible to make the results of a data validation query dependent on various types of filter defined on the form. The greater the number of filters on which the query results depend, the longer it will take to run the form. See "Types of filter for Data Validation queries" below.

Drill-through

Query that allows you to analyse the data that make up a value that has been imported into CCH Tagetik via ETLs. For more details on how to use the query in Navigation mode, see "Drill-through queries" on page 348.

This type of query is filtered on the basis of what is defined in the cell in which it is positioned. It can be used to show details of several values in the matrix, according to the selected cell. Drill-through queries are not run when the form is run, but are run explicitly by the user. They are run based on a cell of the matrix. This allows you to take advantage of all the matrix filters, not just the tab filters and form filters.

> ### Example
>
> Suppose the generic Insurance account has been fed with values imported from accounting records. Only the total figure is present in CCH Tagetik, not the details of the various types of insurance (e.g. auto, fire, legal). Creating a drill-through query allows you to query the staging table used by the ETL (as an intermediate step for loading) to obtain those details.
>
> The query must be run on the MAP_DATI_TRASFORMATI table, which contains the details of every type of insurance and every invoice. To do this, the system must know which MAP_DATI_TRASFORMATI fields correspond to the fields for which we want to show the values. The result is as follows:
>
> *SELECT campo1 as "GL ACCOUNT", campo2 as "ACCOUNT CODE", campo3 as "INVOICE No', campo4 as "AMOUNT" FROM MAP_DATI_TRASFORMATI where cod_mappatura='ETL_IMPORT' and cod_import='ETL_ROUTINE and campo5=${$Account.code} and campo6=${$Scenario.code}'*
>
> In the example, the values are filtered based on the accounting account, which corresponds to field 5, and on the scenario, which corresponds to field 6.

Of a form contains a drill-through query and the **0002868 - Reporting .NET - Navigation - Drill-through** function is enabled for the user, then the  button will appear when the form is opened in Navigation or Data entry mode. This button allows you to open the left-hand **Drill-through** panel, in which the system shows the drill-through queries present in the form, grouped by template.

Types of filter for Data Validation queries

The possible types of filter for *Data Validation* queries can be selected in **Affected by** and are as follows:

- Cell filters
- Row/column filters
- Parameters/Tab filters/Matrix filter
- Parameters/Tab filters

Cell filters

The query recalls the cell fields related to dimensions present in the row and column filters. The system runs the query once for every editable cell with which it is associated.

Example

Suppose we have defined one filter on the scenario dimension in the query and one filter on the period dimension in a matrix with the following row and column filters:

Row Filter	Column Filter
Scenario 1	Period 1
Scenario 2	Period 2
Scenario 3	Period 3
	Period 4

In this case, to generated the list of selectable values, the system runs 12 queries, i.e. one for every scenario/period intersection.

Row/column filters

The query recalls the cell fields related to dimensions present in the row filters or column filters. The system runs the query for every row or column with which it has been associated.

Example

Suppose a *Data Validation* query has been defined in the matrix from the previous example and that matrix has been filtered on the period dimension only. In this case, the system runs four queries, i.e. one for every element present in the filter (row or column).

Parameters/Tab filters/Matrix filter

The query recalls the cell fields related to dimensions present in the parameters and tab filters or matrix filters. The system runs the query for every sheet/matrix combination.

Parameters/Tab filters

The query calls up the cell fields related to dimensions present in the parameters and tab filters. The system runs the query for every sheet.

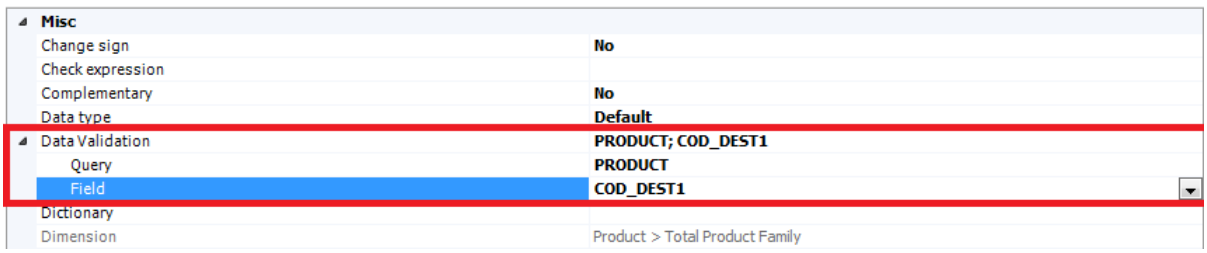
Data Validation properties

After a Data Validation query has been created, the *Data Validation* property is enabled in the Matrix Wizard properties table. This property allows you to link a validation query to a row or column element defined in the matrix and shows the following options:

- **Query**, to specify the identifier of the query to be associated with the selected row or column element
- **Field**, to specify the column of the table generated by the query to be used to populate the list of selectable elements

Example

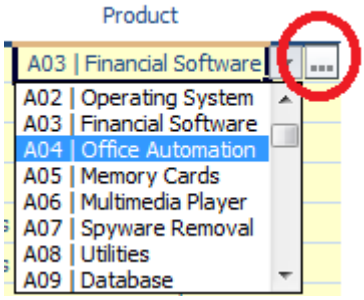
We have created the PRODUCT query, which returns all products actually in production. To associate that query and see the product code in the list, the **Query** option must be defined as **PRODUCT** and the **Field** option must be defined as **COD_DEST1.**



In the data entry step, the following appears in every cell with which a *Data Validation* query is associated:

- a drop-down list, to select the value to be assigned to the cell
- the ... button, to search for the value to be defined on the cell from among several values in the list,

by typing * and ?



Dependent cells can be obtained by selecting the same validation query, but different fields, on two or more of the matrix rows or columns. In data entry, selecting a value in the first field on a cell of the first row allows you to select, on the corresponding cell in the second row, only the values of the second field which generate combinations that exist in the query results.

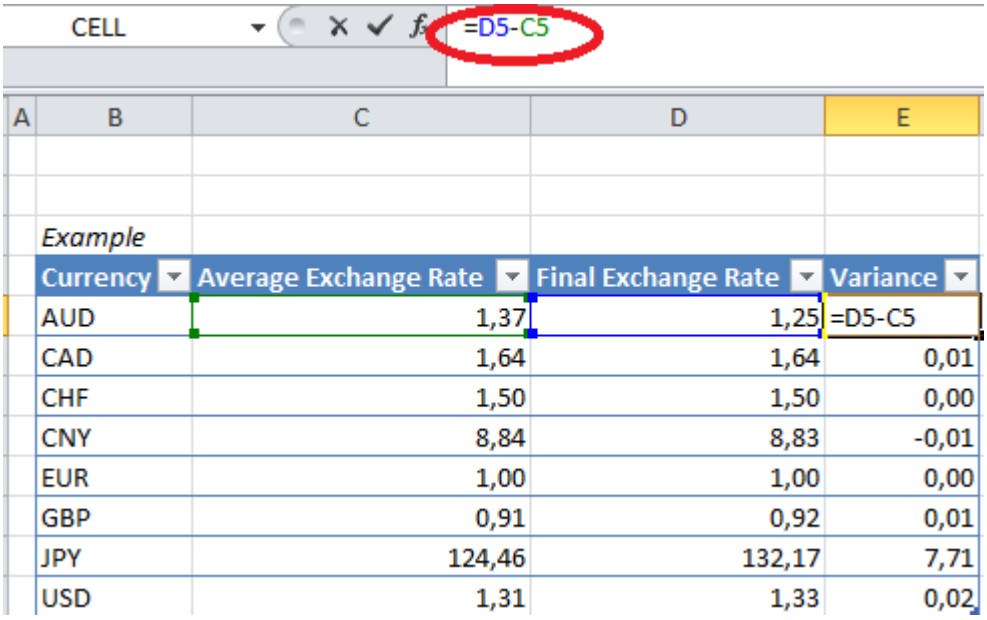
If you want to enable multiselection on a cell that has both a Data Validation query and an associated dictionary, multiselection must be configured on the dictionary.

Use of Layout options for formulas

It is possible to define Excel formulas that have the elements of a query as arguments. For example, this is useful for calculating the difference between two columns of the query. Moreover, with specific layout options, CCH Tagetik allows you do automatically copy formulas inserted to the right or left of the query for all rows generated by the same query.

Example

Suppose a query that extracts the average and final FX rates and an Excel formula for a subtraction between the two fields have been defined. When the query is run, the formula is replicated for all rows for all pairs of FX rates extracted by the query.



Data visibility settings for users

To prevent unauthorised access to data, data visibility must be defined for each user in the role setup phase. This allows the system to know whether to apply user rights when an SQL query is run in a form and whether to allow access to the tables of the ETL module. The following functions are available:

- **0003100**: allows unrestricted access to the data tables via SQL queries contained in shared forms.
- **0003101**: allows unrestricted access to the data tables via SQL queries contained in ad hoc forms.
- **0003102**: allows access to the tables of the ETL module via SQL queries contained in shared forms.
- **0003103**: allows access to the tables related to the ETL via SQL queries contained in ad hoc forms.

***Note**: if functions **0003102** and **0003103** are disabled, an error message indicated that access is not permitted will appear when a query referring to the table of the ETL mode is run.*

Query Wizard window

Excel > CCH® Tagetik > Design > (Left-hand panel) click form > > Query

See "How to access the windows" on page 9 for alternative paths.

Purpose of window

This window allows you to create and defined new queries.

Tabs in the window

Field	Description
Edit	Allows you to define the type of query and the text of the query in SQL language.
Columns	Allows you to set the attributes of the columns returned by the query.
Layout options	Allows you to customise the layout of the table with the query results.

Managing queries in forms

Creating a query in a form

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

1. Click on the empty cell in which to insert the query.
2. Open the **Query Wizard** window. See "Open the Query Wizard window" below.
3. Write the query in the text box as desired.

To obtain suggestions for the names of the tables and the relative fields, type the start of the name and press CTRL+SPACE BAR: a list of suggestions will appear.

Open the Query Wizard window

Excel > CCH® Tagetik > Design > select form >

See "How to access the windows" on page 9 for alternative paths.

Choose one of the following methods:

- Click .
- Right click the desired cell and select **Tagetik** > **New Query**.
- In the **Open items** tab, right click on the template and select **Queries** > **New Query**.
- In a cell of the Excel sheet that does not belong to any matrix, type **SELECT FROM** and press ENTER.

Edit tab

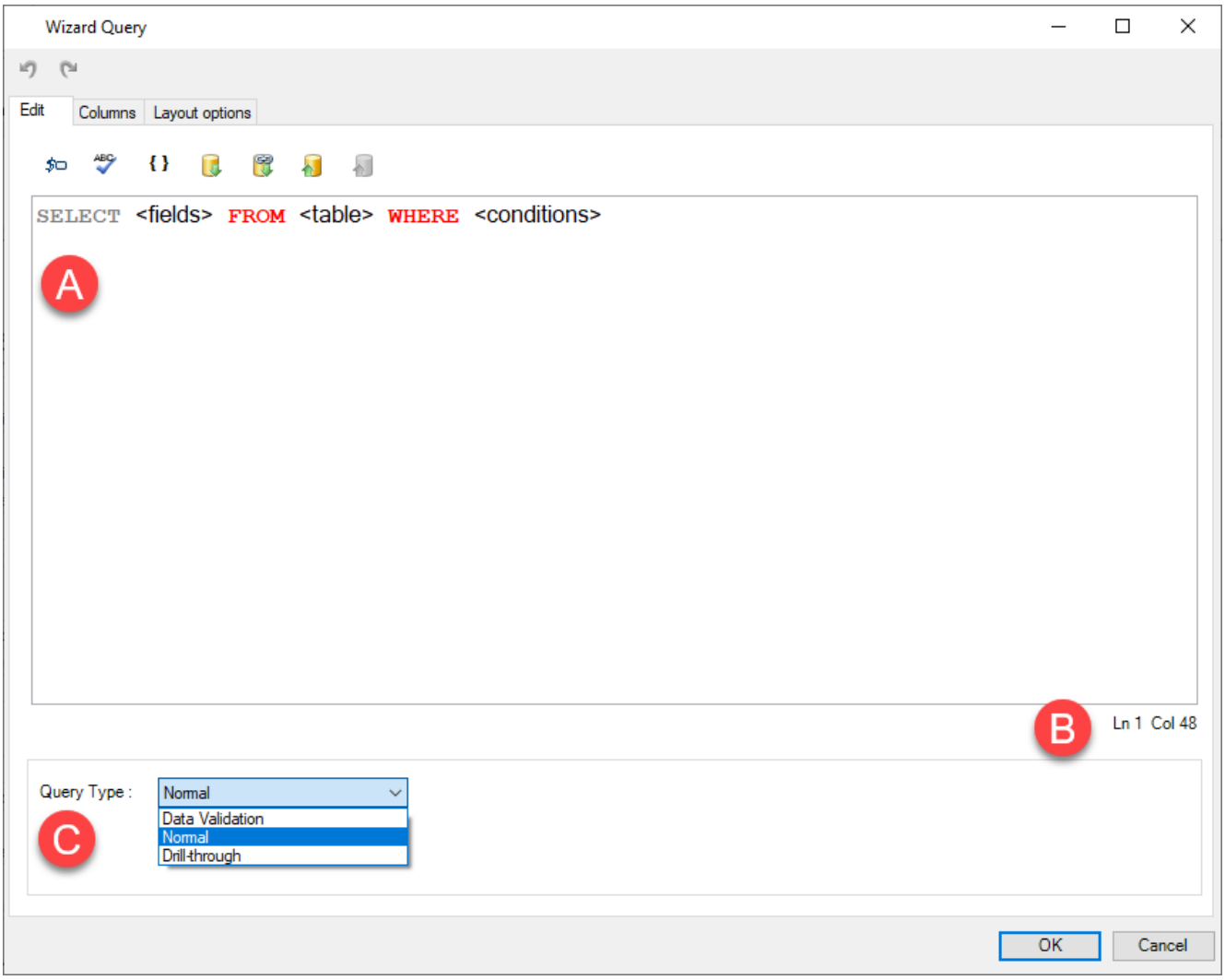
Excel > CCH® Tagetik > Design > (Left-hand panel) click form > > Query

See "How to access the windows" on page 9 for alternative paths.

Purpose of the tab

This tab allows you to define the type of query and the text of the query in SQL language.

What it looks like



Element	Description
A	Text box with default query, in which to type the query.
B	Position of the selected cell in the form.
C	Definition of the query type

Buttons for managing the query

Element	Description
	Advanced fields. Insert a cell field into the query (see "Cell fields" on page 41). This allows you to define parametric queries to be filtered according to the content of the form and the parameters chosen when it is run.
	**Syntax check**. Checks that the query syntax is correct.
{ }	**Insert cell fields container**. Adds the { and} symbols to the text box, as required to contain a cell field.
	**Import query from a shared table**. Imports a previously defined query into the database as a shared query. The imported query will appear in the window and can be edited. When the imported query is changed from the query in the table, its structure will be saved in the form.

Element	Description
	**Import query as link from a shared table**. Imports a link to a query previously defined in the database as a shared query. The query imported as a link will appear in the window as ${query:QUERY_NAME}. Therefore, the body of the query cannot be edited and will not be saved in the form.
	**Export query to a shared table**. Exports the defined query to share it with other users. To export it, a code and a description must be specified. Once shared, the query can be used in drill down.
	**Save changes to the {QUERY_NAME} query**. Saves the changes defined on the selected query in the shared table.

Query structure conventions

The following conventions are applied within the query text:

- In blue: words related to the SQL syntax
- In red: table column names

Columns tab

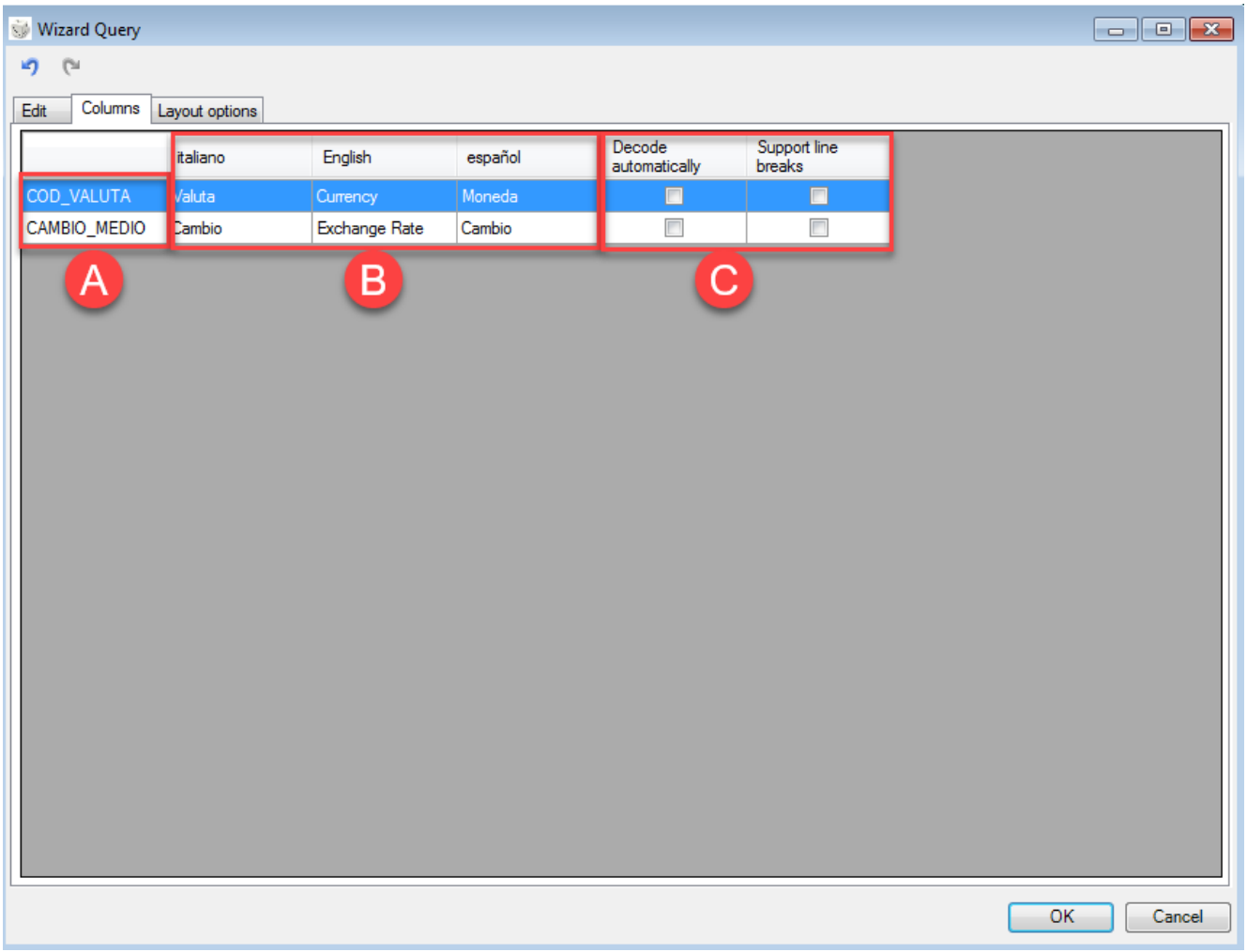
Excel > CCH® Tagetik > Design > (Left-hand panel) click form > > Query

See "How to access the windows" on page 9 for alternative paths.

Purpose of the tab

This tab shows the columns of the database table and allows you to set, after defining the **Edit** tab, certain attributes of the columns returned by the query.

What it looks like



Element	Description
A	Headers of the columns of the database table called up or the columns resulting from the query. They are non-editable and can help to refine the query itself, by calling them up to define a value
B	Row with the row and column headers shown in the query results, editable. The available languages are those that have been configured in the system.
C	Configurable attributes of the database column.

Fields in the tab

These fields correspond to the attributes that can be set.

Field	Description
Decode automatically	The system automatically decodes any special characters (e.g. accent marks).
Support line breaks	If selected, the system keeps the row breaks in the form cell, as defined in the text extracted from the query. Otherwise, the text remains on the same row.

Layout options tab

Excel > CCH® Tagetik > Design > (Left-hand panel) click form > > Query

See "How to access the windows" on page 9 for alternative paths.

Purpose of the tab

This tab allows you to customise the layout of the table with the results of the **Normal** and **Drill-through**-type query.

Options in tab

Option	Description
Code and description	Code to assign to the query and description in every language managed in the table. The description in the table language associated with the current user is shown in bold.
	**Use shared dictionary item**: use a shared dictionary item for the query description.
	**Remove shared dictionary item**: removes the shared dictionary item used to define the query description.
Show Query Name	Shows the query name above the table. The query name corresponds to the description assigned to the query in **Code and description** in the language in which the form is run.
Show Header	Shows the headers of every column in the first row of the table. The description of the columns is as defined in the **Columns** tab.
Fit column width	Adapts the column width to the length of the data extracted by the query. If the query contains many rows and columns and Unicode characters, activating this option may cause the system to slow down.
Apply the format of the first row to the whole Output area	Applies the format defined for the first row of data in the generated table (not the row containing the column headers) to all rows generated by the query. *For example, if the first row has been given a blue background, that background colour will also be applied to the remaining rows of the table.*
Remove the formatting from rows that are no longer displayed after refresh	Removes the formatting of rows that do no longer belong to the query after it has been refreshed.
Copy the formulas to the right of the query to the Output area	Copies an Excel formula found to the right of the query for all rows generated by the same query.
Copy the formulas to the left of the query to the Output area	Copies an Excel formula found to the left of the query for all rows generated by the same query.
Create name for the result	Creates an Excel print area associated with the rows generated by the query. This allows you to call it up in pivot tables. The name of the Excel area coincides with the code assigned to the query.

Report list

Utility of the function

A Report list allows you to create a set of forms in Excel format that can be exported, printed and sent by email. These forms contain relevant financial data extracted from the CCH Tagetik application database and can be shared within the company with users who do not have the Excel add-in installed. A Report list can also be inserted into ad hoc forms, empty sheets and Excel sheets not generated by CCH Tagetik.

Types of report list

There are two types of Report list:

- Ad hoc: allows you to run a set of forms quickly. It is created on-the-fly and cannot be saved for future use.
- Shared: once set, the settings are saved in a special form that allows you and other users to call up and rerun the same Report list subsequently. The Report list is run simultaneous to the form containing it. Some of Report list run parameters can be set to run when the form containing the Report list is run. This makes the Report list configurable. Forms containing shared Report lists may have a single template and do not support tab filters.

Creating a Report list

Report lists are created via the **Action List Wizard** window. This window can be opened from the **CCH® Tagetik** toolbar or the **Design** toolbar. It has the following distinctive features:

- Ad hoc and shared Report lists can be created from the **CCH® Tagetik** toolbar.
- From the **Design** toolbar, a shared report list can be created and saved on the open form or, alternatively, the settings of a Report list that is already present can be changed. In fact, there can be just one Report list in a form.

***Note**: a Report list's Office colour palate must be consistent with the one used in the forms.*

When the Report list is being created, it is also possible to customise the initially assigned default settings for each individual form.

Publication and saving

The system asks you to set the publication method and, if the **Disk or portal publication** method is chosen, to specify the type of file to be generated (Excel or PDF). If the user decides to generate separate files, by default the system will name the file with the code of the form being exported, i.e. **${Report.code}**. If there is just one file, by default the system will name the file with the Report list code, i.e. **${ReportList.code}**.

By default, the system saves the various files in the Tagetik subfolder in the user's Documents folder, with the syntax **${getEnv(<USERPROFILE>)}\Documents\Tagetik\My Report List\${report.path}**. The Tagetik folder is organised into subfolders corresponding to the form grouping structure.

The following functions are available to allow the publication path to be set:

- **getEnv(<NOME_VARIABILE_AMBIENTE>)** which returns the value of a client environment variable.

  *For example, **${getEnv("USERPROFILE")}\Documents** returns the current user's Documents folder.*

- **report.path** which returns the path of the report within the hierarchy. The nodes are indicated by the code and separated by '\'.

The system creates a single file or one file for each of the Report list's forms, depending on the settings in use, and saves it in the specified folder. Those files are visible to everyone.

If the user has the rights to load documents to SharePoint, the generated files can be published in shared folders in a SharePoint portal using the syntax **\\server\share**.

Role settings

Functionalities related to Report lists are only available to users who, in the role with which they are accessing the application database, have the following functions enabled:

- **0002761 - Reporting .NET - Report List - Report List**. If enabled in read/write mode, this allows the user to insert/edit/delete Report lists. If enabled in read only mode, this allows the user to access the wizard in view only mode on existing Report lists.
- **0002762 - Reporting .NET - Report List - Output to Printer**. This function allows you to print the output of a Report list.
- **0002763 - Reporting .NET - Report List - Output to mail**. This function allows you to send the output of a Report list via email.
- **0002764 - Reporting .NET - Report List - Output to system file**. This function allows you to save the output of a Report list on the disk.
- **0002765 - Reporting .NET - Report List - Store Report list in Reports**. This function allows you to save a Report list on a report to be used later.
- **0003692 - Client. NET - Design - Report list - Password protection**. This function allows you to set a security password.
- **0004158 - Reporting - Generate a Form from the Form Navigation tree and save on a default End Point** This function allows you to add the Report List tile to the home page.

Report Lists from the web

A dedicated tile for managing report lists can be added to the CCH Tagetik web interface (User Home page). Excel and PDF files required by report lists created by the Excel add-in can also be generated from this page. Depending on how the Report List has been configured from the Excel add-in, the system will generate a single Excel file or a single PDF with as many sheets as there are forms included in the Report List, or the same number of individual Excel or PDF files.

Managing Report lists from Excel

Create a Report list

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

1. Select the desired options.

   ***Note**: the Print and Disk or portal publication options are alternatives.*

2. If **Print** has been selected, set the printer to be used. If **Disk or portal publication** has been selected, set the publication to be generated, and the file name and path.
3. If useful, set the recipients of the emails (see "Indicate the email recipients." below).
4. Click **Next**.
5. Select the desired form or node of forms and click .

   ***Note**: each form can be inserted into the Report list several times.*

6. Click **Next**.
7. Defining and managing parameters for selected forms.
8. To override tab filters, select **Override tab filters**, click , override the filters as desired and click **OK**: **Ignore** will appear next to the overridden parameters.
9. Click **Next**.
10. To override the individual form's default settings, select the desired form and the desired settings, and customise them.
11. Click **Next**.
12. If **Allow to specify output sheets sorting** was selected in the previous window, a window will open to specify how the output sheets will be sorted. Select the desired sheet and drag it to the desired position, or use the buttons.
13. Click **Next**.
14. Set the Report list's run and save options and click **End**.
15. If **Schedule the run** was selected in the previous window, the **Schedule Wizard** window will open. Set the parameters as desired and click **Next**.
16. If the window for setting the parameters defined as **Run**, set them and click **Next**.

If the Report list scheduling is not successful, to define an Excel-type TGKService, see Pagina TGK Service.

Indicate the email recipients.

In **Send mail** > **Recipients**, choose one of the following options:

- Type the desired email addresses, separated by commas.
- Type the desired user's username from the relevant CCH Tagetik list.
- By selecting and the desired user, and clicking . If the user is not on the list, click and type the desired email address.
- To indicate a reference user, type **<name of reference user>**. The email will be sent to all users linked to the reference user.

  ***IMPORTANT**: the characters < and > are part of the syntax for inserting the name of the reference user.*

Changing the Report list's security password

In Navigation mode, click **Change Report List Password** on the multifunction bar.

Run a Report list

1. In Navigation mode, open the desired form containing the Report list.
2. If requested, enter the value of the necessary parameters: the system will run the Report list.

3. To update the entire Report list, click . To update specific parts of the Report list, click and select them.

Once the Report list has been run, the system prints the forms according to the print areas defined within the various forms.

Managing Report Lists from the Web

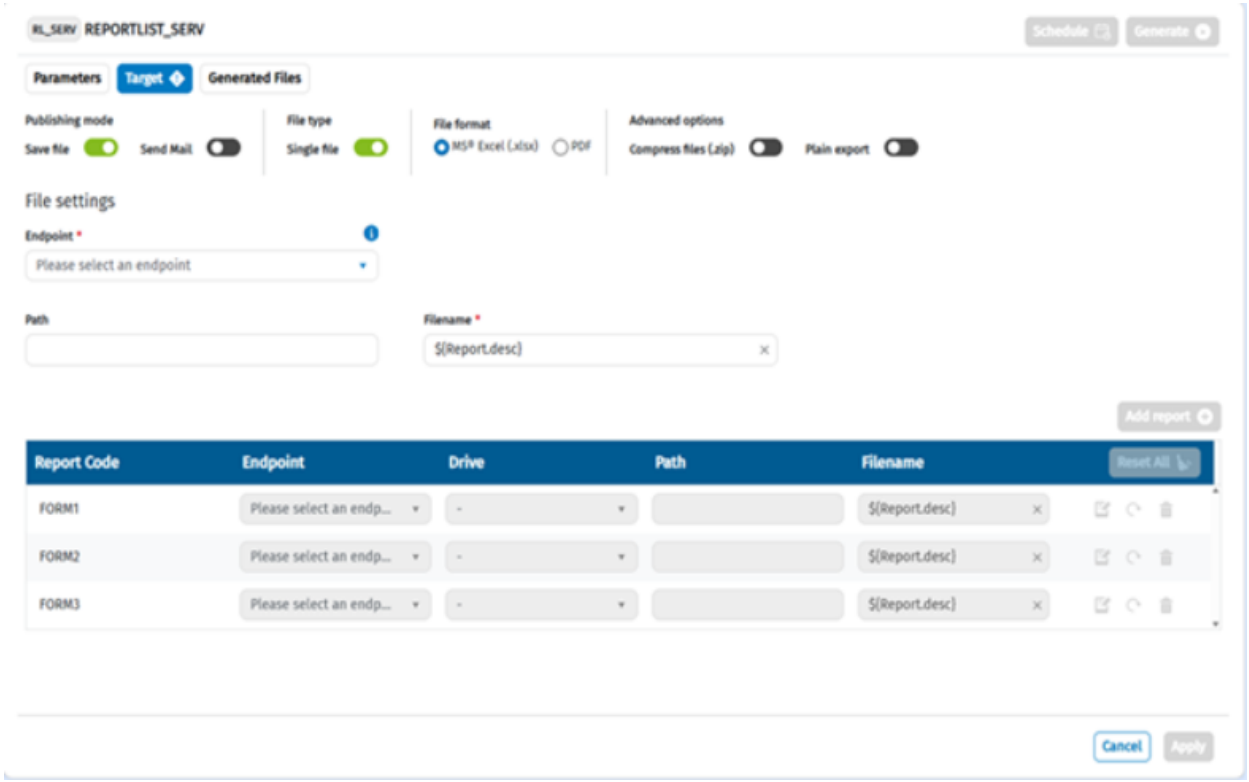
Configuration sets

Run Report List from the Web

Tile Report List

Navigation > Reports & Analysis > Report List

1. Select the desired Report List.
2. A new window is displayed with 3 tabs: 'Parameters', 'Target', 'Generated files':



3.

If...	Then...
The Report List requires parameters	the 'Parameters' section will show the selectable ones, then click on ... next to each dimension to select the desired items or nodes.
the Report List does not require parameters	the 'Parameters' section will remain empty.

4. In the ‘Target’ section, the user must enter all the necessary information to generate the output file, such as:

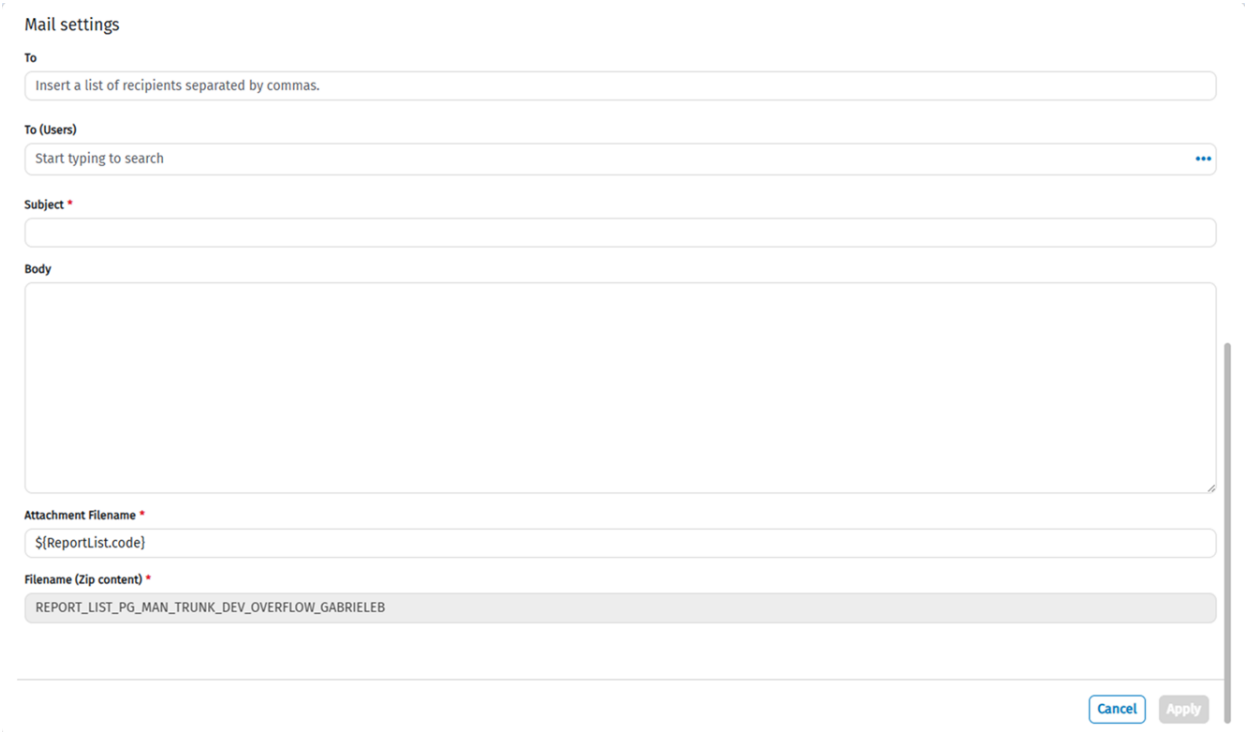
- Publication mode (**Save file** - option selected by default - or Send mail)
- File type (**Single file** - selected by default)
- File format (**MS Excel .xlsx** - selected by default - or **PDF**)
- Advanced options (**Compress file** - zip - and **Plain export**)
- Endpoint
- Path
- File name

5. In the 'File Settings' section, the user must enter the desired Endpoint, Path and File Name (set by default to ${Report.desc}).

   Admin will thus create Endpoints in the Repository on which the report will be saved:

   - Automatic creation of sub-folders in the endpoint when saving the RL
   - Handling of the automatic creation of a subdirectory in the endpoint when saving the RL via a dynamic script
   - Server-side support for generating a PDF or Excel
   - Generating files without .ZIP
   - Files generated at the end of LAUNCH_REPORT_LIST data processing

   When the user activates the 'Send mail' option, the e-mail settings are displayed below the report list.

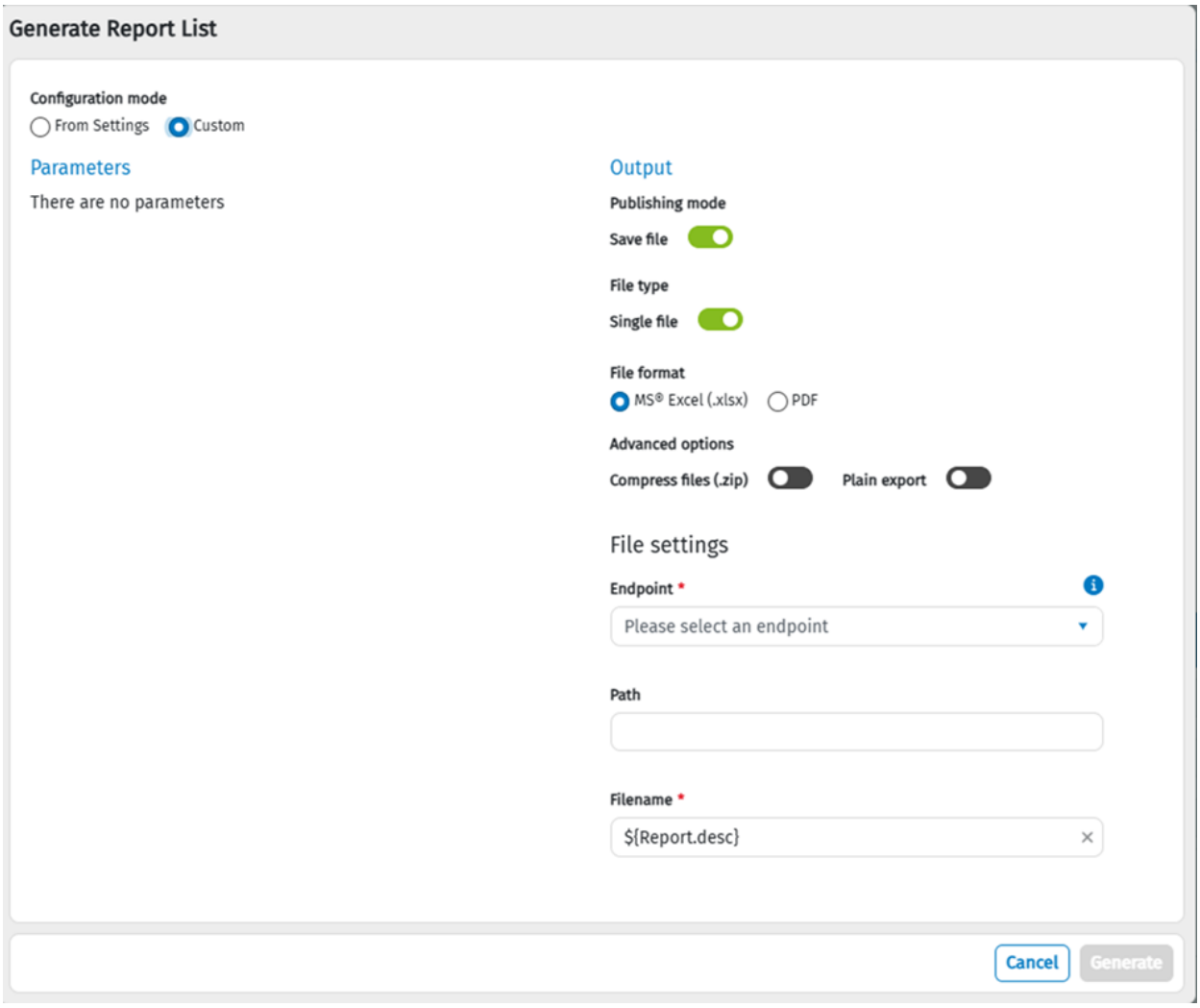


6. Clicking **Generate** will start the data processing that will produce the file for the selected Report List and the user will see a notification of data processing in progress.

   The files can be generated using the saved settings:



Alternatively, ad-hoc settings, valid for the specific execution only, may be used:



7. data processing:

If...	Then...
the user is still on the Report List page	A 'Results' pop-up window will appear containing two tabs: 'Exported files management' and 'Audit'. The 'Audit' tab will show the audit details of the data processing just performed.
the user is no longer on the Report List page	a notification of completed data processing will be displayed in the bottom right-hand corner, and via the **Show Result** button, the 'Results' window can be displayed.

***IMPORTANT**: If a Report List is generated several times with the same code and by the same user, it will be overwritten by the most recent file.*

Server-side support for generating a PDF or Excel

The configuration to generate only PDF files or only Excel files is also supported for the new Report List based on the server engine, instead of always generating both file types.

Types of parameters and functionalities/services supported on the Web

The types of parameters supported when launching a Report List from the Web are: standard dimensional parameters, AIH parameters, NON-dimensional parameters (Text, Flag, Date, Dictionary) and Multi-hierarchy parameters.

The features/services supported on the Web instead are:

- The mandatory flag
- Restrictions related to user rights
- Multi-selection within the parameter tree
- Quick search, on the dimension tree of the selected parameter, only active in the window containing the parameters in use of the Report List
- Lowest level elements enabling flag
- Nodes enabling flag
- Default Value, explicitly selected from the parameter definition, the default value defined via advanced dynamic script is not supported.
- Default Value, explicitly selected from the parameter definition, the default value defined via advanced dynamic script is not supported.
- Advanced selection (insert lowest elements, insert children elements)
- Advanced selection script
- Dependent parameters, where the dependency may be implicit (between two parameters belonging to the same dimension) or explicit (where the dependency is defined within the script)

Automatic creation of sub-folders in the endpoint when saving RL

The user is able to automatically generate sub-folders from a selected Endpoint, based on the definition of the Report List file name. Each section of the filename delimited by '\' will be a subfolder of the selected endpoint, the last section being the file name.

Managing the automatic creation of a subdirectory in the endpoint when saving the RL using a dynamic script

When launching a Report List from the web, the user can define a dynamic file name and path using the 'Cell field' language. In this way, the system generates a folder for each element divided by '\' recognising the script in the file name and translating it according to the Report configuration, within the Report List.

Generate files without .ZIP

When the user generates or schedules, the server side of the Report List will have the option of compressing the generated files. By default generated files are not compressed. This option can be accessed by a flag in the generation and scheduling window.

Tab for Managing Files Generated on Personal Report Type Endpoints

The Generated Files section, which has been renamed Personal Reports, is supported. This section, accessible from the Report List tile on the User Home Page, allows the user—after selecting a Report List—to view, download, or delete the latest file(s) generated for that Report List through the endpoint marked as Personal Report, by the currently logged-in user.

To activate this section, it is first necessary to create an Endpoint, and save it in the Settings of the application database, in the section *Form> Report List > Endpoint for Personal Reports*, and then generate the selected Report List using this Endpoint, which will be the first in the list.

The file name will follow a standard non-modifiable logic: ReportList_database ID_user code_ ReportListcode if it is a single file or ReportList_database ID_user code_ReportListcode_Form_code in the case of separate files.

Files generated at the end of LAUNCH_REPORT_LIST data processing

If you choose to generate Excel type files then the files produced by the data processing will be browsable, unless you have chosen the option "Generate non-refreshable Excel files disconnected from Tagetik" in report list design mode, in which case the files will be simple excel files.

Opening a file generated by the Report List data processing that does not have the option "Generate non-refreshable Excel files disconnected from Tagetik" will prompt the user to log in to the Excel add-in CCH Tagetik, if the user is not logged in. If the report list has the 'Single File' option, then the tools enabled in navigation mode will be 'Cell Properties' and 'Drill Down'.

***IMPORTANT**: For the navigation tools to work properly, the use of the prefix 'TMP_' in 'Name Management' in reports contained in a Report List is not recommended.*

Override options and override run Parameters

We introduce advanced options, applicable from the Report List settings in the Excel add-in, which allow you to override the run parameters for each of the reports used in the Report List. This functionality allows the user to change one or more of the run parameters of a report compared to the parameter settings of the Report List. It also allows you to run, server-side, a Report List that includes the same report several times, but with different run parameters set.

Filename and Endpoint Override

This function makes it possible to manage the replacement of one or more file and/or path names in the new Report List, even within several Reports contained in the Report List:

- The replacement options are already available in **Excel** and only apply to Report Lists generated with the Excel engine.
- The replacement options have been extended to the Tagetik **web** application, in this case the Report Lists are generated with the Tagetik server engine, which unlike Excel involves the use of Endpoints and does not support GetEnv() functions and saving files to local folders.

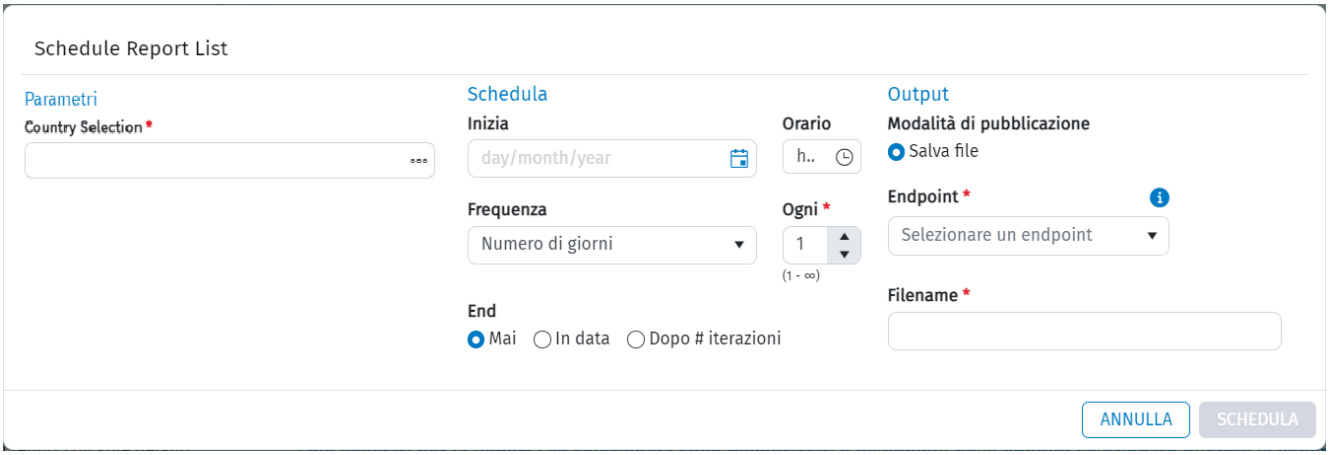
When the user selects a specific report included in the Report List and enables file name and/or target replacement:

If...	...then
A Report List generates a **single file**	reports that have replaced one or more save settings (Endpoint, File Name and Path) will be separated as single files and generated separately.
A Report List generates several **separate files**	the replacement affects only the target or the name of the overwritten files.

Scheduling a Report List from the Web

Tile Report List

Navigation > Reports & Analysis > Report List



1. Select the desired Report List, then click on **Schedule**.
2. A new window consisting of three sections will pop up: in addition to 'Parameters' and 'Output', similar to those in the generation window, there will also be a 'Schedule' section where the user can set the settings required for scheduling. The 'Start' field indicates the date on which the scheduling is to start. The 'Time' field indicates at what time each report list is to be launched. The 'Frequency' field indicates how often the scheduling is to be repeated:
   - Number of days -> The field 'Every' appears, in which the user indicates how often the scheduling is to be run
   - Days of the week -> A list of days of the week appears below where the user can select on which days the scheduling is to be run
   - Day of the month -> An 'On day' field will appear in which the user must indicate the day of the month on which to run the scheduling
   - Day of the year -> Two fields will appear in which the user must enter the month and day respectively on which to run the scheduling

   The 'End' field allows the user to indicate when the scheduling should end:
   - Never -> Scheduling will be repeated without an expiry date
   - On Date -> Scheduling will end on the day indicated below
   - After # iterations -> Scheduling will end after # times
3. When clicking on **Schedule** (only active if all the previously described sections have been filled in correctly) the generation of the selected Report List will be scheduled, based on the parameters entered during the scheduling phase.

   ***Note**: Mandatory fields will be marked with a red asterisk and the 'Schedule' button will remain disabled until the user has filled them all in.*
4. It is possible to **set up** a notification e-mail for each possible execution outcome directly from the scheduler.

   ***Note**: Currently, in the event of total or partial data processing success, the content and recipients of the e-mail are the same, but the subject line differs. It will be 'Data processing completed successfully' if the operation was successful, or 'Data processing completed with warnings' if there were warnings.*

Index explanation

***Note**: The following applies to both scheduling and generation.*

If the **Generate Index** option was selected in Excel during the creation of a single-file report list, the ZIP created following generation will contain an Excel file incorporating an Index as the first sheet marked with a table to navigate to all the forms in the report list.
Each table element on the **Sheet Description** section is in fact a hyperlink which, once clicked, navigates to the selected form.
Similarly, the ZIP file will also contain a Pdf file that also contains a table of contents on the first page with hyperlinks on the **Sheet Description** section, but in addition will contain the new item **Page Number** indicating the page on which the form in question is hyperlinked, respectively.

Report List Index on web

In the new Report List, the **Table of** Contents, which was automatically generated by adding the index to the list, is not supported. The index is still available, but in a different format and without **table of contents**. The format is tabular and includes a first column for the **Form Code**, a second for the **Form Description**, a third for the **Sheet Description**, and a fourth column (for PDF files only) for the **Page Number.**

Note: *The* ***Sheet Description*** *and* ***Page Number*** *contain a hyperlink to the corresponding page.*

Features not supported

When adding a form node to the Report List and not individual forms one by one, it will not be possible to override the filename and file path on this same node

Manage notifications when one or more reports fail in the Report List

When one or more reports, contained in a Report List, fail during data processing, a red light will light up in the end-of-processing notification, highlighting the fact that data processing has ended with errors. Further details on the error are then provided by the Audit.

Support of Explicit Restrictions on the New Report List

Explicit restrictions on the web, defined manually in the parameter definition performed on the Excel add-in, are supported.

Manage the index

Create an index

Excel > CCH® Tagetik > Design

1. Open the desired form.
2. In the multifunction bar, click . Otherwise, in the **Open items** tab, click the desired form > : the system will create an index of the existing sheets.

Edit an index

1. To open the **Index Wizard**, choose one of the following methods:
   - With the index open, double click one of its cells.
   - In Design mode with the index open, click in the multifunction bar.
   - In the **Open items** tab, double click below the item containing the index.
2. To enter cell fields, click **Report** or **Sheets** as desired, select the desired predefined functions from the list and click .
3. Click on the desired button.

Report List Wizard window

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Purpose of the Wizard

This Wizard takes the user through a guided procedure to create and set up a Report list. It is made up of six steps:

Phase	Description
1	Allows you to set the publication method.
2	Allows you to define the forms/Excel sheets to be included in the Report list.
3	Allows you to define and manage the run parameters for the selected forms. Parameters shared by several forms are only required once. System parameters or ad hoc parameters can be added.
4	Allows you to override, for individual forms, certain settings assigned as defaults in the previous phases. Specifically, this allows you to override the print or save settings, parameters, templates and output sheets if you do not want to publish an entire form. Allow to specify output sheets sorting.
5	Only present if **Allow to specify output sheets sorting** was selected in the previous phase. Allows you to define the order of the sheets generated by the Report list.
6	Allows you to set the Report list run and save options.

Report List Wizard window: first tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Fields

Element	Description
Print	Allows you to select the printer on which to print the forms. • **<Default Printer>**: the device's default printer • **<Current PDF Printer>**: any PDF printer in use Once the Report list has been run, the system prints the forms according to the print areas defined within the various forms. *Configure the print areas of each form before creating the Report list.*
Disk or portal publication	Allows you to specify the type, format and name of the file to be generated and where it will be saved.
Send mail	Allows you to send the reportlist.zip file containing a file for every form in the Report list to a list of recipients. This is only possible after the fields related to the email have been set in the Repository configuration (see Pagina impostazioni Email). A reference user from among the Tagetik users of the environment in which you are working can also be inserted as the recipient. This is identified by the user name between angle brackets, e.g. <ferrari>. In this case, the email is automatically sent to all users linked to it (see Users) The user must specify the name of the files to be attached to the email. To send several files, the default option **${Report.code}.xlsx** must not be changed. In this way, the system gives each file a name equal to the form code to be sent

Report List Wizard window: second tab

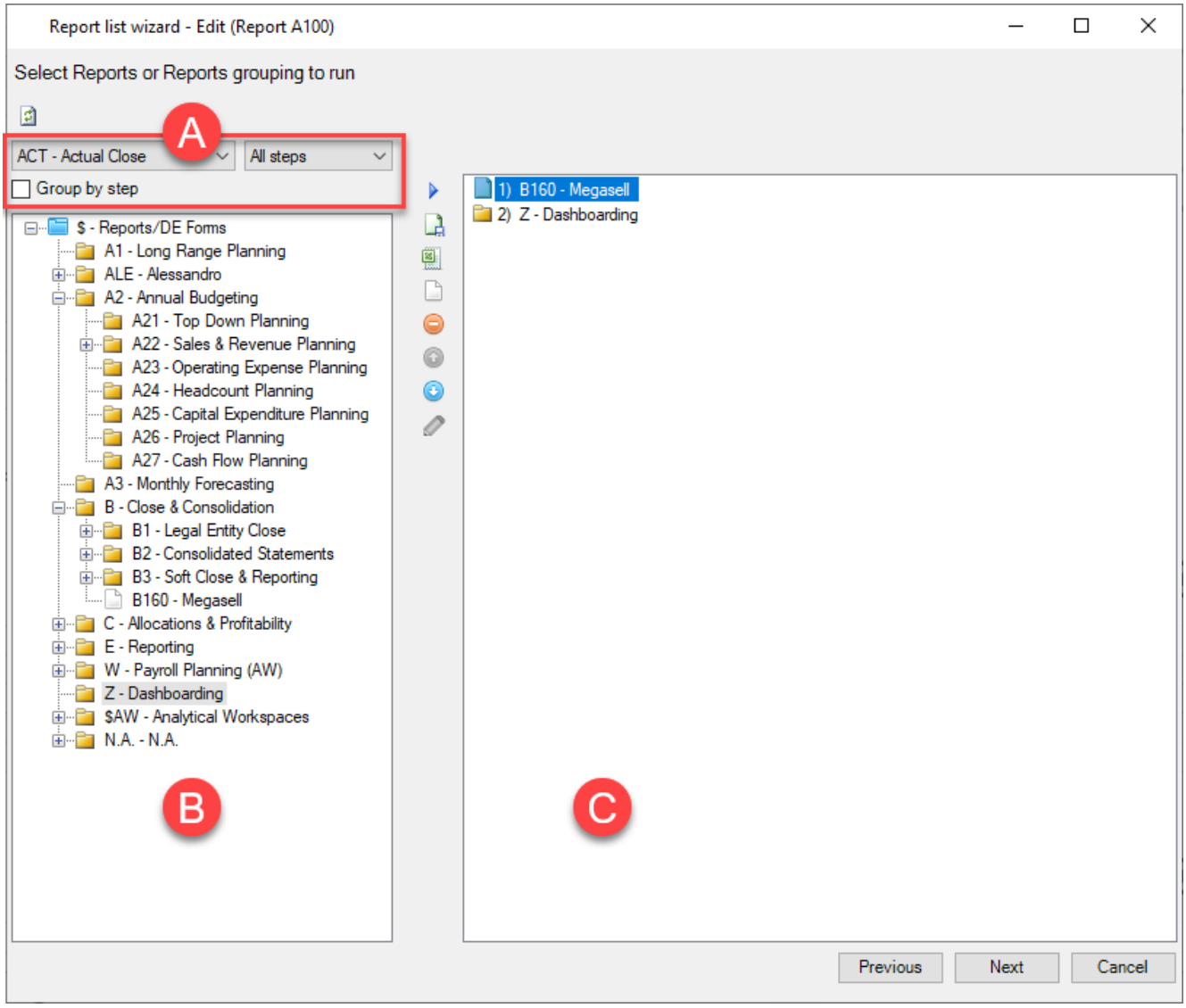
Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Purpose of the second step

This step allows you to define the forms/ad hoc forms/Excel sheets to be excluded from the Report list.

What it looks like



Element	Description
A	Form search filters by process and step to which they belong. **Group by step**: groups forms by step if the process includes several steps.
B	List of grouped forms based on the form grouping nodes
C	Forms in the Report list

Buttons

Element	Description
	Refreshes the list of forms.
	Adds the form selected in the list to the Report list.
	Adds an ad hoc form to the Report list.
	Adds an Excel file to the Report list.

Element	Description
	Adds an empty sheet to the Report list.
	Removes the selected element from the Report list.
	Moves the element selected in the Report list up or down the list.
	Renames the element selected in the Report list.

Report List Wizard window: third tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Purpose of the tab

This tab is dedicated to choosing parameters. If the Report list contains a single form, then the required parameters are those defined in the form. If the Report list includes several forms, then the parameters that are shared by several forms are only required once.

Parameter options

One of the following input modes must be set for every parameter.

Option	Description
Run	Allows you to insert the parameter value directly on running the Report list. If at least one parameter is set to "Run", it is not possible to treat the Report list as "ad hoc", but it must be saved within a form.
Select	Allows you to select the parameter value from the list immediately in the definition step. Therefore, this value will not be required when run.

Buttons

Element	Description
+ System parameter	Allows you to add system parameters.
+ adHoc parameter	Allows you to add ad hoc parameters.
	Only with **Override tab filters** selected. Opens a window to override tab filters. See "Override tab filters window" on page 370.

Fields

Element	Description
Override tab filters	Allows you to override tab filters present in the forms in the Report list.

Override tab filters window

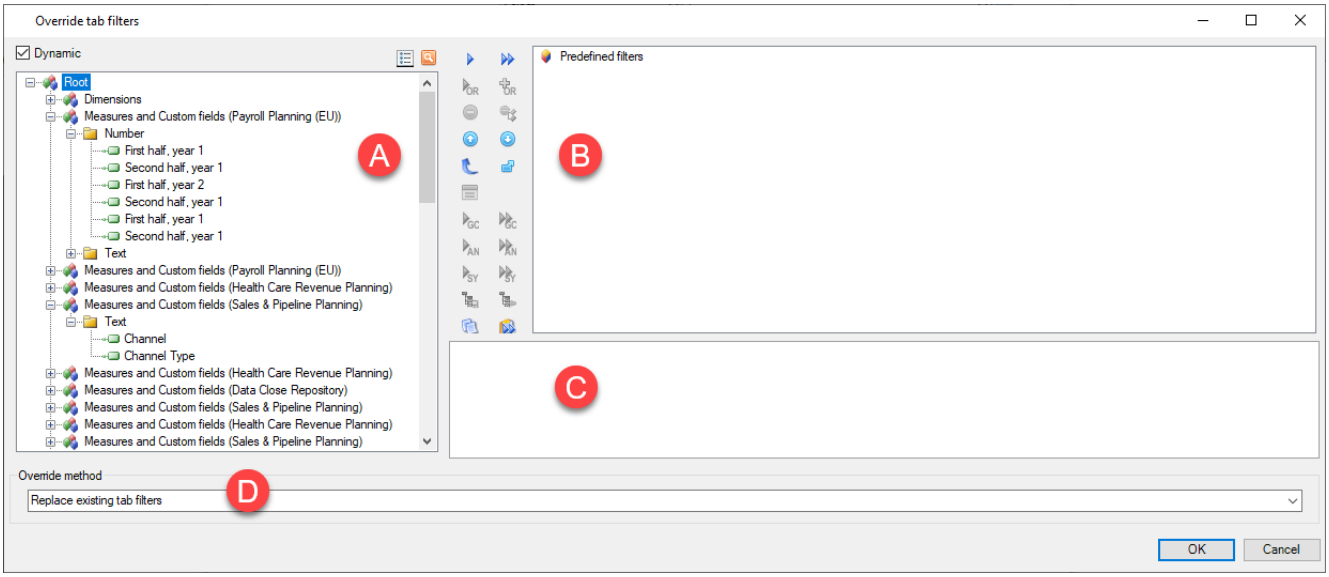
Excel > CCH® Tagetik > Report list > (third tab of the Wizard) select Override tab filters > ...

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list > (third tab of the Wizard) select Override tab filters > ...

Purpose of window

This window allows you to override tab filters. On closing the window, the system assigns **Ignore** as the definition method for the parameters shown in the run parameters configuration window. In this way, the user does not need to specify the value of the parameter, as it is overridden.

What it looks like



Element	Description
A	List of overridable tab filters (static, dynamic, parametric lists)
B	List of chosen tab filters to override
C	Details of the selected tab filter
D	Selected override method

Override method

Option	Description
Applies override only if there are no pre-existing tab filters.	The system adds the overridden tab filters only if it does not identify pre-existing tab filters.
Nest under existing tab filters	The system adds the overridden tab filters to any existing filters. If the overridden filter concerns a dimension already present in the existing filters, the system does not consider the overridden filter and returns a non-blocking warning message when running/refreshing the Report List.
Replace existing tab filters	The system inserts the tab filters, replacing the pre-existing ones.

Buttons in window

Element	Description
	Changes the display mode.
	Allows you to search for the desired tab filter.
	Adds one or several elements or nodes selected in the dimensions tree under the selected node in the filters tree.
	Only with filters already defined, adds one or several elements or nodes selected in the dimensions tree to all filters already defined. ***Note**: if there are no pre-defined filters in the column, it functions like the  button.*
	Adds one or several filters selected in the dimensions tree to a selected filter, creating a OR type logical relationship. OR type logical relationships can only be created with elements or nodes belonging to the same dimension as the filters already defined. The added filter is labelled with the concatenation of the individual element codes separated by ",". The value displayed is equal to the sum of the elements entered.
	Adds one or several elements selected in the dimensions tree belonging to the same filter dimension to a node of filters linked by an OR logical relationship. The added filter is labelled with the concatenation of the individual element codes separated by ",". The value displayed is equal to the sum of the elements entered.
	Only with filters already defined. Deletes the selected element or node.
	Only with filters already defined. Deletes the entire level of the selected filter.
	Only with filters already defined. Moves the selected element up one position.
	Only with filters already defined. Moves the selected element down one position.
	Only with filters at several levels already defined in the row/column columns. Moves the selected level up one level.
	Only with filters already defined. Replaces en bloc the selected elements of a filter with elements selected in the dimensions tree. Useful for reusing a filter by replacing its elements, but not its properties.
	Only with filters already defined in the rows/columns with an OR logical relationship. Deletes one or several elements from the filter containing the OR logical relationship.
	(Only for control groups) Adds the control group selected in the dimensions tree to the filter. See I gruppi di controllo: concetti di base.
	(Only for control groups) Adds the control group selected in the dimensions tree to all filters already defined. See I gruppi di controllo: concetti di base.
	(Only for FST items) Analytically adds, under the selected filter, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	(Only for FST items) Analytically adds, under all filters, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	(Only for FST items) Synthetically enters the FST under the currently selected filter. See "Financial statement templates" on page 83.
	(Only for FST items) Synthetically adds, under all filters, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.

Report List Wizard window: fourth tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Fields

Element	Description
Customize print settings	Allows you to define a different printer for the individual form.
Customize parameters	Allows you to customise the parameters defined in the third step for the individual form.
Customize templates and output sheets	Allows you to indicate templates and output sheets to be exported for the individual form. If not specified, the system exports all templates and output sheets in the Report list by default. **Override sheet name**: allows you to rename the selected template/output sheet, to which a progressive number is assigned by default. This may be inserted in any of the managed table languages. In the Report list run step, the system uses the name of the table language of the user that performed the run.
Allow to specify output sheets sorting	Allows you to define an arbitrary order for all sheets generated by the Report list, regardless of their order in the source forms or Report list. This is not possible in the following cases: • if the selected publication method is **Print** • if publication/send email takes place on different files • there is an override on the file name set on the element, template or tab filters. • if there is an override on the email recipients set on element, template or tab filters.

Report List Wizard window: fifth tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Purpose of the tab

This tab shows a list of the sheets of all the forms present in the Report list. It is only visible if the **Allow to specify output sheets sorting** option has been selected on the fourth tab.

Buttons

Element	Description
	Only with filters already defined. Moves the selected element up one position.
	Only with filters already defined. Moves the selected element down one position.
	Opens a window to change the order of the selected sheets. *Select the sheets of which you want to change the order and then click* .

Report List Wizard window: sixth tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Fields

The available fields vary depending on whether the Report list configuration window has been opened from **CCH® Tagetik**'s multifunction bar or the left-hand **Design** panel.

Element	Description
Run	Allows you to set the Report list run settings. See "Run" below.
Saving	Allows you to set the Report list saving methods. See "Saving" on the next page.
Scheduling	Allows you to schedule the running of saved Report lists or Report lists that need to be saved on a shared form. See "Scheduling" on the next page.
Protect generated files	Only if the file format is not .PDF. Allows you to set a security password for Report lists generated in **Disk or portal publication** or **Send mail** modes. See "Protect generated files" on the next page.

Run

Element	Description
Include in the Report list	Allows you to indicate what to include in the Report List: • **All** • **Only Templates that are defined**: the template of a form if at least one cell of a matrix is defined. • **Only Templates with not null matrix cells**: the template of a form if at least one cell of a matrix is defined with a value other than 0. • **Only Sheets that are defined**: the sheet of a form if at least one cell of a matrix is defined. • **Only Sheets with not null matrix cells**: the sheet of a form if at least one cell of a matrix is defined with a value other than 0.
Always consider templates without any matrix as valued	Disabled with **All** selected. Also include templates without matrices in the Report list. Such templates could contain headers, queries or other information that would otherwise be discarded.
Open the generated files at the end	Disabled if .PDF format has been chosen. If the Report list has **Single file** as its publication file type, the system will open all the generated files when the run finishes.
Always show the data processing logs when finished	After the Report list has been run, the log window will appear even if no errors have been detected.
Create the sheets index	If the Report list has **Single file** as its publication file type and is saved in a container form that does not have an index, the system will automatically create an index of the forms included in the Report list. It is created after the Report list has been defined.
Run now	The Report list is run once its definition is complete. If at least one parameter is set to "Run", it is not possible to treat the Report list as "ad hoc", but it must be saved in a form.
Generate non-refreshable Excel files disconnected from Tagetik	The Report list is saved as a standard Excel document. Consequently, it is not linked to the data in the Tagetik database and cannot be updated. This option is selected and cannot be changed if .PDF file format has been chosen.
Use form code for sheet names	The Report list uses a code generated on the basis of the code of the form for the name of the sheets that it contains.

Saving

Element	Description
Do not save	This prevents the Report list from being saved and called up in the future.
Save as a new shared form:	Saves the Report list in a form of the application database and allows other users to call it up. The code and description to be assigned to the new form must be specified. **Save and close the new form**: when the Report list has been defined, the system does not run the new form and, by default, saves the Report list under the node of the hierarchy of N.A. reports.
Save as a new ad hoc form	Saves the Report list locally in a form on the computer, preventing it from being called up by other users. The form appears after the Report list has been defined.
Save to the current form	Saves the Report list in the current form.

Scheduling

Element	Description
Schedule the run	Enables scheduling for the Report list and, after **End** has been selected, opens a window to set the relative parameters. See "Schedule Wizard window" on page 374.

Protect generated files

Element	Description
Password	Allows you to enter the password to protect the generated files. The syntax of the cell fields can be used, such as the scenario code chosen in the Report list parameters, for example.
Confirm password	Allows you to enter the **Password** to confirm it.

Schedule Wizard window

Excel > CCH® Tagetik > Report list > define Report list via Wizard

Excel > CCH® Tagetik > Design > (left-hand panel) select form > > Report list > define Report list via Wizard

Purpose of window

This window allows you to set the Report list scheduling parameters. Scheduling is only possible for Report lists that have already been saved or those which need to be saved on a shared form.

Report list scheduling can only be enabled if the **Schedule the run** option has been selected in the sixth tab. The window opens when the Report list definition sequence has ended.

Fields

Element	Description
Description	Description associated with the data processing to be scheduled. Makes it easier to identify data processing from the list of Scheduled items.
Recipient at start	List of users to be notified of the start of data processing. A user can be specified using their email address or user code.
Email note at start	Note to be added to the standard wording of the notification email
Recipient on success	List of users to be notified of the outcome of data processing completed successfully. A user can be specified using their email address or user code.
Email note in case of success	Note to be added to the standard wording of the notification email when the data processing is successfully completed

Element	Description
Recipients on failure	List of users to be notified of the outcome of data processing not completed successfully and not to be rerun. A user can be specified using their email address or user code.
Email note in case of failure	Note to be added to the standard wording of the notification email when the data processing fails and is not to be rerun.
Validity start date	Date from which the scheduled data processing is run. **Define Start date validity**: allows you to set the start date.
Minutes	Minute(s) in which to run the data processing. • **Every minute**: runs the data processing every minute. Equivalent to using ***** in the Crontab syntax. • **[*minute*]**: from 00 to 59 minutes
Hours	Hour(s) in which to run the data processing. • **Every hour**: run the data processing every hour. Equivalent to using ***** in the Crontab syntax. • **[*hour*]**: from 0 to 23 hours.
Days of the month	Day(s) of the month in which to run the data processing. • **Every day of the month**: runs the data processing every day. Equivalent to using ***** in the Crontab syntax. • **[*day of the month*]**: from 1 to 31 days.
Month	Month(s) in which to run the data processing. • **Every month**: runs the data processing every month. Equivalent to using ***** in the Crontab syntax. • **[*month*]**: from 1 to 12 months
Days of the week	Day(s) of the week in which to run the data processing. • **Every day of the week**: runs the data processing every day. Equivalent to using ***** in the Crontab syntax. • **[*day of the week*]**: days from Monday to Sunday
Limit number of repetitions	Limits the number of times the system must run the data processing. If the limit is not specified, the data processing is repeated ad infinitum according to what is specified in the Crontab syntax. **Number of repetitions**: number of time the system must run the data processing starting from the specified date/time
Maximum wait for booking	Allows you to set the wait time for booked data processing that cannot be run immediately due to a system overload or other data processing being run. The system assigns status **B** (Booked) to the booked data processing and automatically reattempts to run the data process every minute until the system accepts it or the maximum wait time set by the user has lapsed. **Maximum wait for the run (min)**: maximum wait time for running the data processing of the prebooked Report list.
Rerun of the data processing in the case of an exception	Allows you to configure a new run of the scheduled data processing that was not completed successfully. This is particularly useful when the failure is due to a temporary problem. • **Maximum number of iterations**: the maximum number of times that the data processing must be rerun in case of failure. • **Minutes to wait before the next attempt**: the number of minutes of wait time before the scheduled date processing is rerun.
Allowed servers	List of the servers on which the scheduling can be run (**All** or specific servers).

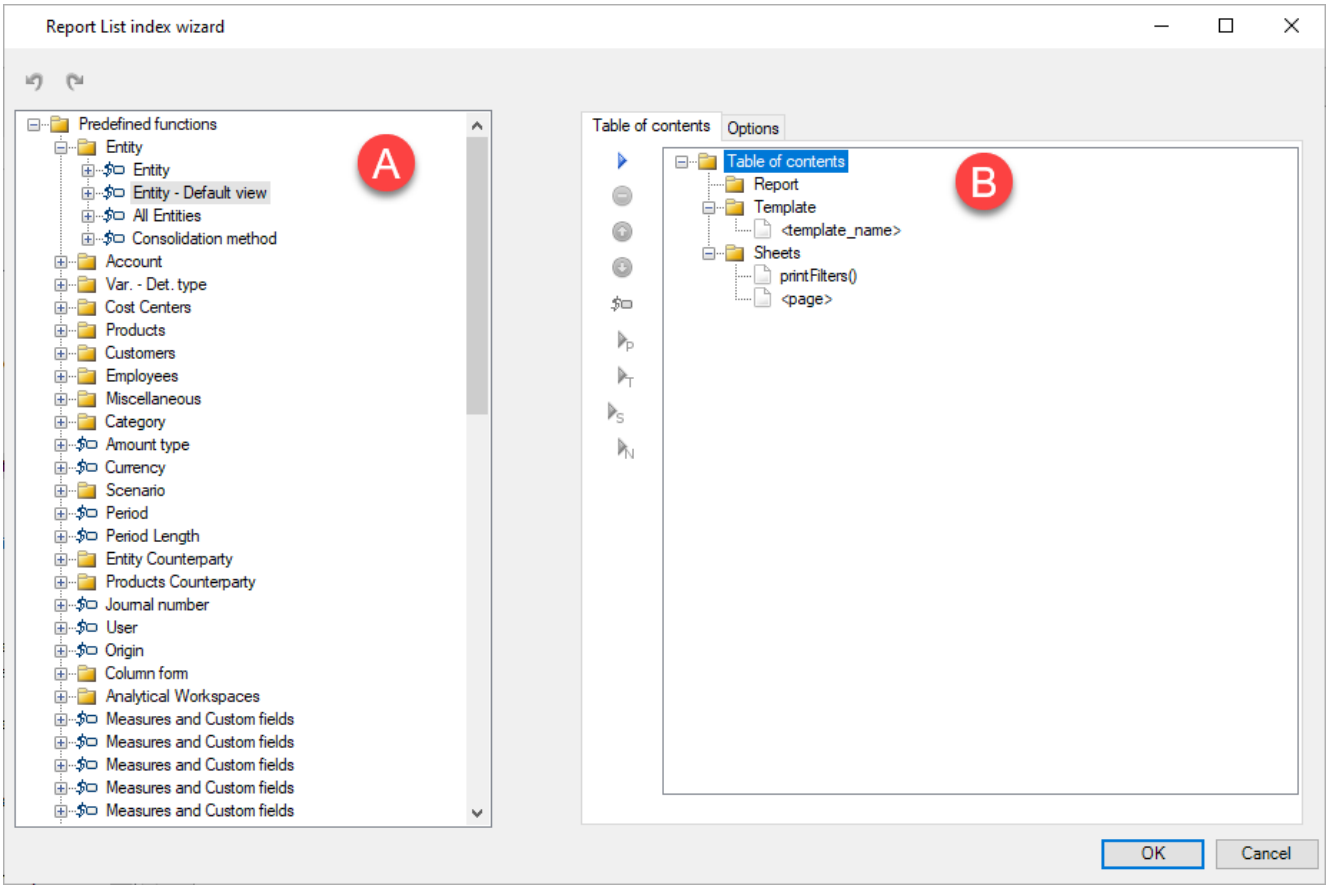
Report List Index Wizard window

Excel > CCH® Tagetik > Design > (Left-hand panel) click form > > Index

Purpose of window

This window allows you to create and then customise the structure and layout of the index of sheets contained in a form with several templates or a Report list that publishes the results in a single file.

What it looks like



Element	Description
A	List of predefined functions
B	Configuration tab

Table of Contents tab

In this tab, you can customise the structure of the table of contents, which always comprises three levels.

Element	Description
Report	• If the table of contents is a Report list, it shows the code and the description of the forms that comprise the Report list. • If the table of contents is a form comprising several templates, it does not show anything.
Template	Shows the name of the template.
Sheets	Shows the following: • the content of the **print Filters** cell fields • the report's tab filters • **<page>** the page number Hyperlinks to access the report's various sheets are created at this level.

Table of Contents buttons

Element	Description
	Adds the report selected in the list.
	Removes the selected element.

Element	Description
	Moves the selected element up or down.
	Adds an advanced cell field. See "Configuring cell fields" on page 42.
	Adds the specific cell field for the **Page number** tables of contents.
	Adds the specific cell field for the **Template name** indices.
	Adds the specific cell field for the **Sheet name** tables of contents.
	Adds the specific cell field for the **Page name** indices. Shows the actual name and any overridden name or the number of the sheet in the Report list.

Options tab

This tab allows you to customise the layout of the table of contents.

Element	Description
Show hyperlink in sheets	When the cursor is positioned over the table of contents page number, it shows a tooltip with the name of the sheet to which the hyperlink is linked.
Insert separator dots before the last column	Inserts dots to separate the content of the last column of the table of contents from that of previous columns.
Autofit columns	Automatically adapt the width of columns.
Format automatically	Applies the appropriate style sheet to the table of contents. See "Tagetik styles " on page 230. If a style sheet with the **TGK_TOC_PAGE_COLUMN, TGK_TOC_REPORT_ROW** or **TGK_TOC_TEMPLATE_ROW, TGK_TOC_SHEET_ROW** styles is associated with the report on which the table of contents is defined, the system will apply the defined style to the table of contents.
Also show the template name	Keeps the **Template** level in the table of contents. If deselected, this level will be removed.
In the case of the report list with sheets that are sorted manually, in the index:	Only with **Allow to specify output sheets sorting** selected in the Report list definition step. Allows you to specify the impacts of the sheet sorting override on the table of contents. • **Keep the sheets sorting and duplicate the rows related to reports/templates**: keeps the progressive order of the table of contents' page numbers and rows. The order corresponds to that which the user gives each sheet. If the table of contents allows you to move from one report/template to another, any report/template headers will be replicated. • **Keep the order of reports/templates and show unordered sheets**: the table of contents will show the **Report**, **Template** and **Sheets** levels, if active. The order of the reports is subject to the override set during the definition of the Report list. The order of templates and sheets is that of the respective reports to which they belong and could result in unsorted page numbers in the table of contents.
Rows	Allows you to configure the method by which the rows below the table of contents must be generated. • **Shift cells down**: insert the new cells, moving the existing cells down. • **Entire row**: inserts a new entire row into the Excel sheet. • **Don't expand**: prevents new rows from being inserted.

Report List Page (web)

User | <application database>

Tile Report List

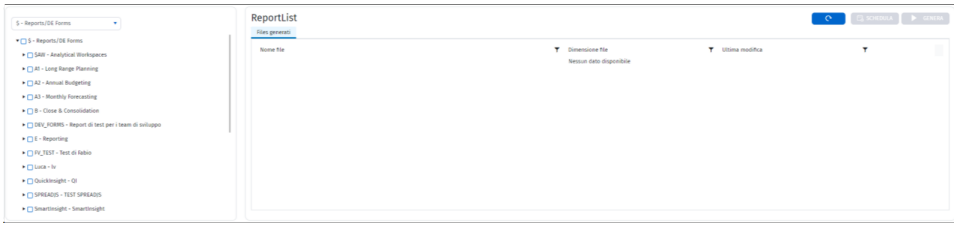
Navigation > Reports & Analysis > Report List

Purpose of the page

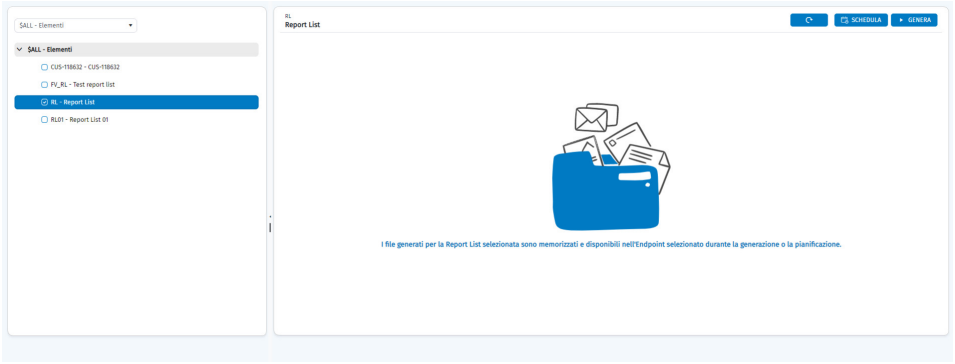
This page displays a list of Report Lists previously prepared in Excel and allows you to start the processes that generate the required files.

What it looks like

Before selecting a Report List from the list on the left, the page looks like this:



After selecting a Report List, the page looks like this:



Buttons on the page

Element	Description
	Updates the list of Report Lists shown on the page.
Schedule	Allows you to schedule the generation of the selected Report List
Generate	Starts the data processing that generates the required files from the selected Report List.

Design: transactional forms

Contents

This section covers the following topics:

Introduction to transactional forms 281
Transactional matrices 282
Elements of the transactional forms model 283
Setting the form structure from the web interface 284
Attributes of transactional form structure columns 286
Setting up forms with Line by Line matrices from the Excel add-in 288
Transactional data entry details window 289
Transactional form columns page 289
Transactional form dictionaries page 292
Transactional form elements page 292
Transactional form details page 293

Introduction to transactional forms

What they are

Transactional forms allow you to perform data entry both on standard CCH Tagetik dimensions and on "new dimensions" that are not included as standard and not defined as custom dimensions. This allows you to increase the level of detail of the data entered. In particular, transactional forms allow you both to enter data in the elements tables of dimensions and to enter values of dimension elements.

Since data entry in transactional forms reproduces the data tables created on the database, these forms are organised into columns. The user can insert columns containing CCH Tagetik dimension, text-type columns and amount-type columns.

Text-type columns can contain numeric-type default values, or default values with SQL queries (see "Transactional form columns page" on page 289), or dictionary-type filters.

Transactional forms can be used as web forms when the relative option is active on the form or in the application's general settings.

Transactional form creation process

To create a transactional data entry form, follow the procedure detailed below:

Phase	Description
Create an information model of the form in web interface	This phase requires the following actions: 1. "Create a transactional form" on page 285. ***Note**: the form created in this phase is a mere configuration element, and does not correspond to the data entry form to be used by the user.* 2. "Create the form columns" on page 285. 3. "Configuring column attributes" on page 285. ***IMPORTANT**: creating the information model in web interface also defines the structural elements with which working transactional data entry forms can be designed in the Excel add-in.*
Create a data entry form in the Excel add-in	This phase requires the following actions: 1. "Introduction to transactional forms" on the previous page. 2. "Introduction to transactional forms" on the previous page. ***Note**: the columns available for matrix configuration are those of the information model selected in the creation phase.* 3. (Optional) "Introduction to transactional forms" on the previous page. ***IMPORTANT**: Columns inserted in a form created in Excel can only be removed from the structure from the web interface if they no longer contain any values.*

Saving data in the database

When data entry is run on a transactional form, the data modified in the form are saved without distinction in a specific table in the database (FORM_DATI).
From this table, it is possible to transfer all data, based on type, to the specific table of values (e.g. the DATI_SALDI_LORDI, DATI_SALDI_IC tables) and to the specific dimensions table (e.g. DEST1, DEST2, etc.). To do this, an ETL routine must be run.

For more details on ETL routines, see Extraction, Transformation & Loading.

***Note**: it is still possible to relate the form to an ETL job provided the form was created with the utility in the Flex version of the application. On new forms the manual relating of an ETL job is now deactivated.*

On saving, the system systematically refreshes the form in the following cases:

- when elements table codes are generated automatically
- in the presence of columns with default values, with values calculated when the form is opened in Excel
- in the presence of columns with SQL formulas
- in the presence of read-only columns
- when at least one row is deleted

Transactional matrices

Transactional form matrices and Line by Line transactional matrices

Based on the origin of the data managed in the transactional form, the transactional matrix will have the following structures.

Origin of the data	Matrix structure
Financial workspace: **Matrix of a transactional form**	• Counter element on row filters (see "Counter element" on the facing page) • CCH Tagetik dimensions on column filters • *Delete* element on column filters (optional) (see "Delete element" on the facing page) • Dictionary filter on column filters (optional)
Analytical workspace: **Line by line transactional matrix**	• Counter element on row or column filters • Dataset fields on row or column filters • Dictionary filter on row or column filters (optional) • *Status* element on row or column filters

Counter element

The counter element possesses dynamic attributes that regulate the insertion of new rows when entering data in transactional forms. In particular, if the **Allow new rows/columns wizard in data entry** attribute is not active, the system will open a fixed number of rows specified on the **Additional rows to open when opening the form** attribute. However, if the **Allow new rows/columns wizard in data entry** attribute is enabled, the + symbol will appear on the left side of the work desktop, allowing you to open new lines manually. For more details, see "Advanced selection" on page 114.

The number of rows to open can be managed, up to a maximum of 5000. The limit can be set using the **In the case of transactional forms, ask how many rows or columns to open** option in matrix options.

For more details, see "Options in the Edit amounts tab" on page 131.

In Line by Line transactional matrices of the Analytical layer, when the counter element is positioned on rows, the system will not display the matrix correctly if it is positioned on the first column of the spreadsheet. Similarly, when the counter element is positioned on columns, the system will not display the matrix correctly if it is positioned on the first row of the sheet.

Delete element

The *delete* element allows you to delete the data present on a row of a transactional form based on Financial Workspace data. When present, when the form is run the system will show the **Delete** column as the first column, where the following actions can be performed:

Button	Function
	Deletes the data.
	Restores the most recently deleted data.

Status element

The *status* element allows you to delete the data present on a row or column of a transactional form based on analytical Workspace data. To ensure consistency among submitted data, in the presence of rows that are partially or entirely non-editable for submission reasons, the cell related to the status column is non-editable to prevent the user from deleting the row itself.

Transactional data entry details window management

In transactional forms, data are distributed over the table columns. When the form contains many columns, it may be difficult to navigate the sheet on which you are working. Data entry can be simplified by creating a customised details window. For more details, see "Transactional form editor tab" on page 131.

Elements of the transactional forms model

Characteristics of transactional form columns

The following elements can be specified for transactional form columns:

- *Attributes*, which supplement the column's definition with information on the format, the type of values that it can contain and potentially a table of those values. They facilitate data entry because they can generate drop-down menus in column cells to choose the value for the cell on which you are working.
- *Exceptions* to the column's mandatory nature and editability for specific accounts
- *Dictionary* on any column of the transactional form, once the dictionary type is defined, the same dictionary is selected in the form column relationship. Thus, it is possible to create columns that are not linked to CCH Tagetik dimensions only.
- *Filters*: indicate if a filter on the data related to the column must be set in the form's data entry step and what type the filter must be. The filter can be obligatory or optional, as a single choice (where only one value can be entered) or multiple choice (where one or several values can be entered).

Specific information on the default values of columns

A text-type default value or formula-type default value can be assigned to the column. Text-type values must be inserted manually. Formula-type values can be numeric or the results of SQL queries.

Value type	Description
Numeric	The formula determines the value shown in the form column. The formula can contain numeric constants, the four operations (+ - * /), round brackets and reference to a Field name (see "Transactional form columns page" on page 289).
SQL queries	The formula that determines the value shown in the column is an SQL-type query and allows you to indicate a value depending on the value of one or more fields of the form or a table in the CCH Tagetik database. Formulas with SQL queries have the following particular features: • Reference to a column of the form must be specified in braces. *{COD_DEST1} indicates the value of the "Custom dimension 1" column of the row being inserted.* • References to several numeric fields must indicate the relationship between the fields. *IMPORTO_1*IMPORTO_2/IMPORTO_3.*

When the form is run, the cells show the results of the formula.

Exceptions for columns with accounts

When columns are set up in the transactional forms information model, data entry rule (optional or mandatory) is defined, or the column can be made available in read only mode. A formula written in SQL language can be inserted on the column. The formula should be defined on the **Default value/Formula** field. When the form is run, the formula is not shown as an Excel formula but is calculated directly on the server.

If the column does not contain data related to CCH Tagetik dimensions, it is possible to insert **Exceptions in insertion for account**. This field indicates whether mandatoriness or editability exceptions are in place for the column, based on the account (see "Attributes tab" on page 290).

Dictionaries

A dictionary can be associated with any text-type or amount-type column; this allows the user, in the data entry step, to search for and select a value from a drop-down menu. This menu lists the dimension elements or dataset fields present on the column.

> #### Example
>
> In the presence of a column for control groups and another column for accounts, it is possible to create a single query-type dictionary in which the query returns the following results:
>
> - the code and description of the control group
> - the code and description of the accounts belonging to the control group
>
> These results must then be associated with the two columns.
>
> In data entry, the system proposes the list of possible control groups on the first column and the list of accounts belonging to that control group on the second column. For this to take place, once the query-type dictionary has been defined, it is necessary to specify which fields of the query correspond to the code and which correspond to the description using two free attributes: **Query code field** for the code and **Query description field** for the description.

Setting the form structure from the web interface

Admin | <application database>

Create a transactional form

Forms management tile > Elements

1. Click ⊕, a new row will appear in the table of forms.
2. In the **Attributes** tab, insert a code and a description for the form.
3. In **Input type**, define the types of data managed by the form.
4. In the **Input properties** section, select the desired data entry options for the form.
5. (Optional) in the **ETL** section, to activate the (manual or automatic) loading of data onto the data tables or onto specific dimensions tables, select the following:
   - In **Job**, select the desired ETL routine.
   - In **Period length**, select the mode in which the data should be saved on the target tables.
6. Click **Enable loading**

   ***Note**: this field is only valid for old forms created from the old flex interface, which has been discontinued. If enabled on a new form, it deactivates the field that allows you to relate an ETL job to the form.*

7. In the **Relationships** tab, click  in an empty field and select a group.
8. Click **Save**.

Create the form columns

Forms management tile > Elements > Form > select transactional from > (Attributes tab) Definition

1. Click ⊕: a new row will appear.
2. In **column #**, enter the column's unique identification number.
3. In **Description**, insert the text you want as the column header.
4. In **Field**, select the data to be associated with the column. If the data type is not shown, type it manually to search for it (e.g. amount, text).
5. In **Input type**, select the desired input rule.
6. (Optional) In **Dictionary**, select the desired dictionary filter.
7. (Optional) In **Default value/Formula**, insert a default value which, in data entry, the system associates with the column data, or insert the formula to be used to determine the default value to be inserted in the column.
8. (Optional) In **Sorting**, insert the sorting method for the form columns.
9. (Optional) In **Initial filter**, select the type of filter to be applied to the column when the form is opened.
10. (Optional) In **Attributes**, select the desired attributes for the chosen data.
11. Click **Save**.

    ***Note**: in transactional forms with **Input type** set to **Data only**, the following columns are mandatory: Scenario, Period, Entity, Account, Category, Custom dimension 1... 5 (if managed), Currency, Original currency, Amount, Original currency amount. The absence of these columns is only highlighted in the data entry step.*

Configuring column attributes

Forms management tile > Elements > Form > select transactional form > (Attributes tab) Definition > select column > ⋮ > Details

1. In **Properties**, select the desired type of attribute.
2. In **Value**, defined whether the desired value for the attribute must be selected from a list of values proposed based on the "Property", if the attributes are predefined, or whether it must be inserted by the manually by the user (YES).

Configuring a dictionary filter

Forms management tile > Elements > Form > Transactional Form Dictionaries

Forms management tile > Elements > select form > > Transactional Form Dictionaries

1. In **Code**, insert a code for the dictionary filter.
2. In **Description**, insert a description for the dictionary filter.
3. In **Dictionary type**, select the desired type of dictionary filter.
4.

If...	Then...
Dictionary type is set to **Dictionary by Entity**	in **Elements**, select the entities that you want to relate to that dictionary filter.
Dictionary type is set to **Dictionary independent of Entity**	in **Elements**, select the desired elements as values of the dictionary filter.
Dictionary type is set to **Predefined dimension**	in **Dimension**, select the desired entity, account, custom dimension or category.
Dictionary type is set to **SQL Query**	in **Elements selection query**, insert the SQL selection query for the values of the dictionary filter.

5. (Optional) In **Attributes**, insert up to 5 custom attributes to be shown in data entry next to the column with which the dictionary filter is associated, to supplement the relative information.
6. Click **Save**.

Define dictionary elements manually

Forms management tile > Elements > select form > > Transactional Form Dictionaries

1. Select the desired dictionary, then click **Elements**.
2. Insert a code and description for the element.
3. (Optional) Repeat steps 1 to 3 for every desired element of the dictionary filter.
4. Click **Save**.

Attributes of transactional form structure columns

What they are for

The content of the columns can be supplemented by defining attributes for the following aspects:

- the column data format
- the type of data contained in the column
- a table of element values

Defining attributes is particularly useful for text- and amount-type columns for which the only information available when the form is created is the format and data type (*field type*). The system uses those values to guide the data entry when the form is run.

For example, if the column is meant to contain the value of an element of a dimension, the system will show a window in the data entry form with a list of the available elements for that dimensions.

The use of attributes is also useful for managing dictionary-type filters recalled by SQL queries in text- or amount-type columns. In this case, the elements of a column's dictionary filter are filtered on the basis of the query syntax and may depend on the values of the elements of other columns (see "Dictionaries" on page 284).

Attribute details

A column's attributes can be accessed from the administrator user's web interface, following the path: *Setup & Admin > Forms management tile > Elements > link Definition > Actions* ( ) *> Details* and positioning the cursor over the **Attributes** section.

Every attribute is identified by the following elements:

- a property, which indicates the name of the attribute,
- a value, which can be selected from a list of values proposed by the system or inserted manually by the user depending on the chosen key.

The following attributes are available for transactional form columns.

Attribute	Description
New table elements	Only for text-type columns with Field *type* set to *DIMENSION* (dimension). Adds the elements inserted in the column in the data entry step as new items in the relevant elements table.
Generate code automatically	Only for text-type columns with Field *type* set to *DIMENSION* (dimension) and with *New table elements* selected. Allows you to specify whether the code of the new table element is generated automatically by the system based on the *FORM_<10 random uppercase letters>* convention or inserted manually by the user.
Size	Only for text-type columns with Field *type* set to *DIMENSION* (dimension). Allows you to select the available values for the column from a drop-down menu containing the CCH Tagetik dimensions (scenario, period, entity, etc.)
Field to which it refers	Only for text-type columns with Field *type* set to *DIM_DES* (Dimension description). Allows you to select, as values, the text-type field names to which the description refers.
Language	Only for text-type columns with Field *type* set to *DIM_DES* (Dimension description). This defines the table language on which the descriptions of the entered elements must be saved. Possible values: • **Write on all languages** (default value): saves the entered descriptions on all the managed table languages. • **Write on the user's language**: saves the entered descriptions only on the table language associated with the user. • **Specify**: saves the descriptions on the managed table languages specified by the user.
Field to which it refers	Only for text-type columns with Field *type* set to *HIERARCHY* (Hierarchy). Allows you to select, as values, the text-type field names to which the description refers.
Hierarchy code	Only for text-type columns with Field *type* set to *HIERARCHY* (Hierarchy). This defines the table language on which the descriptions of the entered elements must be saved. Possible values: • **Write on all languages** (default value): saves the entered descriptions on all the managed table languages. • **Write on the user's language**: saves the entered descriptions only on the table language associated with the user. • **Specify**: saves the descriptions on the managed table languages specified by the user.
Excel combo	Only for text-type columns with Field *type* set to *DIMENSION* (Dimension), that are associated with a dictionary and refer to a CCH Tagetik dimension. This allows you to associate a search window with the column to facilitate the selection of elements. ***Note**: a window for searching by code and description is present for dimension fields by default. To change this behaviour, manually insert the Excel "combo" attribute.*
Only description	Only for text-type columns with Field *type* set to *DIMENSION* (Dimension), that are associated with a dictionary and associated with a CCH Tagetik dimension. Allows you to see the description only, instead of the code/description pair, when reading the column.
Read only on already existing rows	Makes the column non-editable on the existing rows of the form.
Query code field	Only for columns with which a SQL Query-type dictionary is associated. Name of the query field used to define the dictionary to show on the code column.
Query description field	Only for columns with which a SQL Query-type dictionary is associated. Name of the query field used to define the dictionary to show on the description column.
Decode the Unicode text automatically	Only for text-type columns. Activates support for Unicode characters on the column.
Separator for multiple selection	Defines the separator value.
Monthly splitting	Only for amount-type columns

Setting up forms with Line by Line matrices from the Excel add-in

Introduction

The steps to be taken to set up forms containing Line by Line transactional matrices entirely in the Excel add-in are described below. Run the procedures in indicated order.

1. "Create a standard form" below.
2. "Insert a transactional matrix into the form" below.
3. (Optional) "Customise the transactional data entry details window" on the facing page.

1. Create a standard form

Excel > CCH® Tagetik > Design > select form grouping >

This procedure explains how to create a standard form from CCH Tagetik's Excel add-in.

1. In the **New form** window, enter a code and one or several descriptions for the form, based on the languages supported by the CCH Tagetik project.
2. Select the desired options.
3. To exclude the form from data entry and navigation, click **Do not display**.
4. To enable/disable the form as a web form, choose the desired option in **Enable as web form**.
5. To enable the form as a Dashboard report, select **Enable dashboard**.
6. To enable the form as a Pivot table report, select **Run as pivot table**.
7. To make the form non-editable, select **Lock Structure**.
8. To use an ad hoc form as a model for the new form, click next to **Ad hoc form** and select the desired file.
9. Click **OK**: the form will be created and opened in Design mode.

2. Insert a transactional matrix into the form

Excel > CCH® Tagetik > Design > select form > Matrix

1. In the **Choose the dataset** window, click the desired dataset.
2. (Only if an Analytical Workspaces dataset is selected) To create a transactional matrix, click **Transactional**.
3. Click **OK**.

Once the matrix has been created, the system assigns a default name (e.g. Matrix00, Matrix01...), which can be edited in the matrix wizard by the **Matrix options** tab in the **Code** field.

3. Design the transactional matrix

Excel > CCH® Tagetik > Design > select form > Matrix

This procedure is valid both for Line by Line transactional matrices based on an analytical workspace dataset and for transactional matrices based on a model designed with columns created by the application.

1. In the dimensions tree, select the **Counter** element then click next to the tree of local row filters, or drag it into the tree. The element will appear below the **Rows** node. In Line by Line transactional matrices, the **Counter** element can also be positioned on local filters on the **Columns**.
2. In the dimensions tree, select the desired dataset field or dimension element, then click next to the tree of local row filters, or drag it into the tree. The element will appear below the **Columns** node. In Line by Line transactional matrices, the field or dimension element can also be positioned on local filters on the **Rows**.
3. Set the filter properties as desired (see "Row/column filter properties" on page 153).

4. Repeat the previous steps 2 and 3 to add more dataset fields or dimension elements.
5. (Optional) To add the row content management column:

If the matrix's dataset is based...	Then, in the dimensions tree...
on the Financial Workspace	select the **Delete** element, then add it to the column filters.
on Analytical Workspaces (Line by Line transactional matrices)	select the **Status** element, then add it to the column filters.

4. Customise the transactional data entry details window

Excel > CCH® Tagetik > Design > open form > Matrix > Transactional form editor

1. Select **Use Transactional form editor** to enable the use of a data entry window for transactional form row details.
2. Select the desired row management buttons.
3. Select the desired height and width of the window.
4. Click **Add tab** to add a tab to the window and enter a name.
5. Set the desired number of rows and columns.
6. To associate a column with the template and define its position, click the **Row/column filters** tab and select the desired column filter.
7. In the properties table, click next to the **Transactional form editor** and select the tab with which to associated the column data.
8. In **Column** and **Row**, set the coordinates at which to position the data.
9. (Optional) In **Filled columns**, set the number of columns after the defined column in which to show the values of the selected column.
10. (Optional) In **Label space**, define the desired percentage dimension with which to view the column in the data entry window.
11. Repeat steps 8 to 11 for every desired column filter.
12. To see a preview of the details window, click **Preview**.
13. Click **OK**.

Transactional data entry details window

Excel > CCH® Tagetik > Data Entry > open transactional form > (click a cell of the transactional matrix)

Purpose of window

This window allows for the guided entry of data in a transactional form. The window's appearance is defined in the dedicated tab (see "Transactional form editor tab" on page 131).

Transactional form columns page

Admin | <application database>

Forms management tile > Elements > Form > select transactional from > (Attributes tab) Definition

Purpose of the page

This page allows you to manage the columns of the reference transactional form

***IMPORTANT:** a column can only be removed from the structure if it no longer contains any values.*

Attributes tab

Element	Description
# column	Defines the column's unique identification number. By default, this is the column number that exists in the elements table increased by 10.
Description	Defines the description shown in the form as the column header.
Field	Allows you to select or manually insert the field type to associate with the column. The following fields are supported: • dimension-type field. • text-type field (max. 50): columns with textual data that also allow the user to type amounts on any non-managed dimensions valid only in the FORM_DATI table. • amount-type field (max. 99): column with editable fields that may also contain formulas. With **Type** set to **Data only**, it is necessary to set columns with all the following fields: • Scenario • Period • Entity • Account • Category • Custom dimension 1... 5 (if managed) • Currency • Original currency • Amount • Original currency amount. ***IMPORTANT**: In the absence of one or more of these columns, the system will return an error in the data entry step.*
Type	Indicates the field type to calculate its attributes. • *DIMENSION*: (only for text-type fields) the field contains an element of a dimension (Account, Entity, etc.). • *DIM_DES*: (only for text-type fields) the field contains the description of an element of a dimension. • *HIERARCHY*: (only for text-type fields) the field contains a hierarchy. Only for free text-type fields. • *AMOUNT*: (only for amount-type fields) the field contains an amount. In this case, the column has the Monthly Splitting attribute by default.
Input type	Defines behaviour in the columns for data entry. • **Optional**: the entry of data associated with the column is optional. • **Mandatory**: the entry of data associated with the column is mandatory. • **Read only**: the column is visible but not editable. • **Formula**: the column contains a formula written in SQL language. When the form is run, the formula is not shown, but calculated directly on the server, and its result is shown.
Exceptions in insertion for account	With **Field** set to **Text** (from Text01 to Text50), **Amount** (from Amount01 to Amount99) or **Date** (from Date1 to Date 5). Allows you to define account-based exceptions on the input type defined on the column. If set to **Yes**, the **Definition** link is displayed to open the **Input exceptions** page to define the desired exceptions. ***Note**: it is not possible to define exceptions for columns associated with CCH Tagetik dimensions.*
Dictionary	(Only for columns with text- and amount-type fields) Allows you to associate a dictionary with the column (see "Transactional form dictionaries page" on page 292).

<table>
<tr><th>Element</th><th>Description</th></tr>
<tr><td>Default value/Formula</td><td>For all field values except for Date, Currency, Form, User, Entity, Origin. Allows you to define a default value for column fields.
If the Input type is Formula, the following behaviours may take place:
<table>
<tr><th>If...</th><th>Then...</th></tr>
<tr><td>the data associated with the column is numeric</td><td>this field contains the formula with which the value to be inserted in the form column must be determined.
Note: the formula can contain numeric constants, the four operations (+ - * /), round brackets and reference to a "Field name".</td></tr>
<tr><td>the data associated with the column is not numeric</td><td>this field contains a string in SQL language with which it is possible to indicate the value to be inserted in the column depending on the value of one or several fields in the form or in any other CCH Tagetik table.</td></tr>
<tr><td>you want the formula to refer to a column in the form</td><td>it needs to be written in braces {}.
{COD_DEST1} indicates that you want to consider the value of the "Custom dimension 1" column of the row being inserted.</td></tr>
<tr><td>you want the formula to refer to numeric fields</td><td>the relationship linking the various fields must be indicated.
IMPORTO_1*IMPORTO_2/IMPORTO_3</td></tr>
</table></td></tr>
<tr><td>Sorting</td><td>Defines the order in which the columns are displayed in the form run step.
• 1 Ascending
• 1 Descending
• 2 Ascending
• 2 Descending
• ...
• 9 Ascending
• 9 Descending
Note: the number (1, 2, 3, etc.) defines priority in the order: the system sorts column "1" (ascending or descending) first, then column "2" (ascending or descending), and so on.</td></tr>
<tr><td>Initial filter</td><td>Defines the rule for creating an initial filter on the column.
• No filter: does not create any initial filter.
• Optional filter: single choice: the filter is optional and requires only one value.
• Optional filter: multiple choice: the filter is optional and the user can set one or several values.
• Mandatory filter: single choice: the filter is mandatory and requires only one value.
• Mandatory filter: multiple choice: the filter is mandatory and it is possible to set one or several values.</td></tr>
<tr><td>Attributes to display</td><td>Defines the column attributes to be shown in the header.
• (Only for text- and amount-type columns associated with a dictionary, or columns with entity, account, custom dimensions) Displays the description of the element.
• (Only for text- and amount-type columns associated with a dictionary, or columns with entity, account, custom dimensions) Also displays the value of Attribute 1-5 associated with the element inserted in the column.</td></tr>
</table>

Transactional form dictionaries page

Admin | <application database>

Forms management tile > Elements > Form > Transactional Form Dictionaries

Purpose of the page

This page allows you to create a dictionary for transactional forms. Once a dictionary has been defined, it can be associated with a column of a transactional form (see "Transactional form columns page" on page 289).

Attributes tab

Element	Description
Code	Dictionary filter code.
Description	Description of the dictionary filter.
Dictionary type	Dictionary filter type selection: • **Dictionary by Entity**: allows you to define the dictionary filter element based on the entity. • **Dictionary independent of Entity**: allows you to define the same dictionary filter elements as valid for all entities. • **Predefined dimension**: the dictionary filter elements are those present in the elements table of an entity or account dimension, custom dimension or category specified in the **Dimension** field. • **SQL Query**: allows you to define the dictionary filter elements via an SQL query.
Elements	Only with **Dictionary by Entity** and **Dictionary independent of Entity**. Allows you to define the elements of the dictionary filter.
Size	Only with **Predefined dimension**. Allows you to select an entity, account, custom dimensions or category.
Elements selection query	Only with **SQL Query**. Allows you to insert an SQL query for the definition of the dictionary filter values. ***IMPORTANT**: When creating the dictionary, the descriptions of some data retrieved by the SQL query may be declared as significant for GDPR purposes. Such descriptions are then encrypted and stored in the application database and must be decrypted when the form is opened in data entry and navigation, regardless of whether the dictionary filter is associated with a matrix field or whether it is a parameter of the form.*
Attributes	
Attribute 1 ... 5	(Optional) Allows you to insert up to five custom attributes which, in the data entry step, are shown next to the column with which the dictionary filter is associated.

Transactional form elements page

Admin | <application database>

Forms management tile > Transactional form dictionaries > select dictionary > (Attributes tab) Elements

Forms management tile > Elements > select form > > Transactional Form Dictionaries > select dictionary > (Attributes tab) Elements

Purpose of the page

This page allows you to manually define the elements of a dictionary for transactional forms.

Attributes tab

Element	Description
Code	Dictionary element code.
Description	Description of the dictionary element.

Transactional form details page

Admin | <application database>

Forms management tile > Elements > Form > select transactional form > (Attributes tab) Definition > select column > ⋮ > Details

Purpose of the page

This page allows you to:

- define column attributes
- define attributes for query-type dictionary filters

Attributes tab

Element	Description
Property	Defines the attribute type. The values change depending on the type of starting column For a description of these attributes, see "Attributes of transactional form structure columns" on page 286.
Value	Defines the attribute value. Establishes whether it can be selected from a list of proposed values or inserted manually by the user (depending on the chosen key). • **Yes** • **No**

Advanced functions

Contents

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Custom functions

Introduction

In CCH Tagetik, data can be read/entered data in the database without defining matrices, by inserting *Custom functions* directly in the cells of forms.

Function	Description
TGKCURRDB	Retrieves the identification code of the database to which the user is connected during the current work session.
TGKRET/TGKRETF	Retrieves data from a certain database according to the arguments specified in the function invocation step.
TGKSEND	Retrieves a value from a certain database and allows the user to edit it and save it in the database again.
TGKFIELD	Retrieves the value of a cell field.
TGKAMOUNT	Retrieves the value present in the database according to the filters present on a cell.
TGKCALCSET	Allows you to optimise performance in the presence of forms containing many custom functions.
TGKOPTION	Allows you to specify the reporting engine with which to calculate TGKRET functions.

IMPORTANT: *custom functions only support uppercase characters.*

*If the forms do not contain custom functions, you can tell the system not to check for their existence in the form run step and then prevent them from being run via the "**Disable custom functions**" option (see "Form options window" on page 55). This reduces the time needed to run the form, particularly when it contains many cells.*

***Note**: When a CCH Tagetik form is exported to Excel, the resulting file is converted into a standard (raw) Excel file that is no longer linked to the CCH Tagetik Excel Add-in.*
As a consequence, the exported Excel file does not contain the information required to connect to the CCH Tagetik database, such as application context, dimensional metadata, or user authentication details.
For this reason, CCH Tagetik custom functions (for example TGKAMOUNT, TGKRET, TGKSEND, and related functions) are not available and cannot retrieve data when the form is used offline or when it is opened as an exported Excel file.

Working with custom functions

Excel > CCH® Tagetik > Data Entry > open form

Insert a custom function

1. Select the desired cell and click f_x to the left of the Excel formula bar or on the toolbar.
2. In the **Insert function** window, select **Tagetik function** as the category. All of CCH Tagetik's custom functions will appear in the selection area.
3. Select the desired function and click **OK**.
4. In the **Function arguments** window, insert the values of the desired dimensions for each of the function's arguments.
5. Click **OK**.
6. To recalculate the cell values according to the defined custom function, click f_x in the toolbar in the CCH Tagetik tab.

Save a value using the TGKSEND function

To save a value that has been inserted using the TGKSEND function, save the form opened in data entry mode, or click f_x.

Define the calculation set for TGKCALCSET

This procedure shows you how to define a calculation set.

1. In the work desktop, select the desired cells.

   ***Note**: it is advisable to select adjacent cells on which the same dimensions and/or hierarchies are defined, so that only one filter changes when they are moved horizontally or vertically.*

2. Add the TGKCALCSET function (see "Insert a custom function" above).

The TGKCURRDB function

Description

In CCH Tagetik, it is possible to access various databases in the same session. For example, you can have one environment for the budget and one for consolidation. This function retrieves the identification code of the database to which the user is linked at the time, or on which the current form was run.

Syntax

The function uses the following syntax: *=TGKCURRDB()*. The database code is used as the end in other custom formulas.

The TGKRET function

Description

From *TaGetik RETrieve*, the TGKRET function allows you to retrieve, according to filters on the dimensions, a value from a certain database, but not necessarily the one on which the form is run.

Syntax

The function uses the following syntax: *TGKRET(dataSource,"DimensionName1:[HierarchyCode:*]value1", "DimensionName2:value2", ...,"Convert")*

The value retrieved by the function is then shown in the cell in which the function is inserted.

Arguments

The items that can be added to the function are detailed below.

Datasource

Code of the database from which the value must be recovered. This code can be indicated in the following ways:

- with the database code, e.g. *=TGKRET("TCPM_DEMODEV_001", ...)*
- with the *TGKCURRDB* function, e.g. *=TGKRET(TGKCURRDB(), ...)*
- with reference to the Excel cell containing the database code, e.g. *= TGKRET(A25, ...)*

Selecting dimensions

To retrieve the value of the desired dimension, you must apply filters. Some examples of syntax are provided below:

- To retrieve a value of a dimension: *<name of dimension>:<value to filter>*, e.g. for entity A00, specify *Entity:A00*. To retrieve a value of a dimension on which aggregation structures exist, specify a node: *<name of dimension>:<aggregation structure code>:*<node code>*, e.g. for the entity *Entity:H_CODE:*H_NODE*.
- For the account dimension, it is also possible to refer to an FST: *<name of account dimension>:*<FST code|FST item code>*, e.g. *Account:*FST_CODE|FST_NODE*

The supported dimensions are listed below.

Dimension	Description
Account:	Account code. Mandatory dimension. Individual accounts, nodes of an account hierarchy or an FST item can be used. In order for the formula to be correct, the account must be present in the list and must meet one of the following conditions: • be related to an FST. • be present in a control group. • be related to a node of an accounts hierarchy.
AmountType	Amount type. Mandatory dimension. The possible values are: • *TIP_O*: (original) amount type of data entered in the system by the user. • *TIP_OV*: (in transaction currency) amount type that only represents the amounts expressed in a currency other than the default entity currency (for example, an amount in euros to be attributed to an entity whose currency is the dollar) • *TIP_OE*: proportional amount type • *TIP_OEV*: proportional amount type in transaction currency • *TIP_PR*: converted amount type • *TIP_PRV*: amount type converted into transaction currency • *TIP_CV*: consolidable amount type • *TIP_CVV*: consolidable amount type in transaction currency • *TIP_DES*: (description) amount type that allows the user to enter and save text-based information, such as notes or comments.
Category	Category code
Currency:	Currency code
Cust_Dim1..Cust_Dim5:	Element code of one of the custom dimensions. For example, if the products to recover the value of a certain product are handled on custom dimension 1, the code is Cust_Dim1.
Entity	Entity code
ICCust_Dim:	Element code of the intercompany custom dimension
ICEntity	Intercompany entity code
Period	Period code. Mandatory dimension.

Dimension	Description
PeriodLength:	Period length. The possible values are: • *LUN_0*: progressive • *LUN_1*: monthly • *LUN_2*: bimonthly • *LUN_3*: quarterly • *LUN_4*: every four months • *LUN_5*: every five months • *LUN_6*: half-yearly • *LUN_12*: annual • *LUN_QTD*: Quarter to date • *LUN_STD*: Semester to date • *LUN_YTD*: Year to date
Scenario	Scenario code. Mandatory dimension.

Data conversion

The *Convert* element allows the user to convert dimension data while running a form. The *Convert* values are:

- *Final*: final FX rate
- *Average*: average FX rate
- *Period*: period average FX rate
- *Account*: Conversion type associated with the account

The syntax is the following: *<("dimensionNamex:valuex")>* i.e. dimension name: Convert value.

Calculation by segment

In the TGKRET formula, it is possible to request the amount calculation in the following ways: by *segment*, *intersegment*, *parent segment* and *parent intersegment*.

To do this, the desired suffix (suffixes are case insensitive) must be concatenated to the desired dimension code (node).

Calculation method	Suffix
segment	• \s • \segment Example: <desired dimension code>\s
intersegment	• \i • \is • \intersegment Example: <desired dimension code> \i
parent segment	• \p • \ps • \parentSegment Example: <desired dimension code>\p
parent intersegment	• *pis* • *pi* • *parentIntersegment* Example <desired dimension code>\pis

Syntax conventions and specifics

- For dimension elements, codes can be entered in lowercase even if the application database on which the form was run does not support them (because the **Accept lower case codes** option is not active).
- To recover several elements or nodes of aggregation structures for the same dimension, values can be separated by ",". The system reads the data related to all specified elements. For example, to recover accounts 50010 and 40010, the syntax in the position relative to the account is *50010,40010*.
- To recover several nodes belonging to different aggregation structures for the same dimension, the system reads the intersection data related to the elements related to the nodes. For example, to recover the entities related to node N1 of hierarchy H1 and node N1 of hierarchy H2, the syntax is *Entity:H1:*N1,H2:*N1*.

Example

To retrieve the value of the revenue for the Antivirus product for the geographical area Italy and for scenario 2012BDG, apply the following filters:

- Scenario = 2012BDG
- Period = 12
- Account = all revenue accounts, i.e. FST item 500300 - Total Revenues in FST 010.
- Product (managed on custom dimension 2) = Antivirus, whose code is A01
- Entity = all entities related to the Italy node (X01) of hierarchy 03 which groups entities by geographical area
- Cost Center (managed on custom dimension 1) = generic element G01
- Employee (managed on custom dimension 2) = generic element G01

The function's syntax is *=TGKRET(TGKCURRDB(), "Account:*010|50030", "Scenario:2012BDG","Entity:03:*X01", "Cust_Dim1:G01", "Cust_Dim2:A01","Cust_Dim3:G01","Period:12","Category:$AMOUNT","AmountType:TIP_O","PeriodLength:LUN_0")*

To read the original data for account 50010 on December 2011 converted into EUR at the average FX rate, the function's syntax is *=TGKRET(TGKCURRDB(), "Scenario:2011ACT", "Period:12", "Account:50010", ... , "Currency:EUR", ... , "Convert:Average")*

The TGKRETF function

Description

From *TaGetik RETrieve Fixed*, the TGKRETF function allows you to retrieve, according to arguments set in a fixed order, a value from a database other than the one on which the form is run. In relation to the TGKRET function, the position of the filter parameters is fixed.

Syntax

The function uses the following syntax: *=TGKRETF("Scenario", "Period", "Account", ["AmountType"], ["Entity"], ["Category"], ["Cust_dim1"], ["Cust_dim2"], ["Cust_dim3"], ["Cust_dim4"], ["Cust_dim5"], ["ICEntity"], ["ICCustDim"], ["Currency"], ["PeriodLength"], ["Causal"], ["Convert"], ["DBCode"])*

The value retrieved by the function is then shown in the cell with which the function is associated.

Arguments

The items that can be added to the function are detailed below.

Supported dimensions

Dimension	Description
Account:	Account code. Mandatory dimension. Individual accounts, nodes of an account hierarchy or an FST item can be used. In order for the formula to be correct, the account must be present in the list and must meet one of the following conditions: • be related to an FST. • be present in a control group. • be related to a node of an accounts hierarchy.
AmountType	Amount type. The possible values are: • *TIP_O*: (original) amount type of data entered in the system by the user. Default value when the dimension is not defined. • *TIP_OV*: (in transaction currency) amount type that only represents the amounts expressed in a currency other than the default entity currency (for example, an amount in euros to be attributed to an entity whose currency is the dollar) • *TIP_OE*: proportional amount type • *TIP_OEV*: proportional amount type in transaction currency • *TIP_PR*: converted amount type • *TIP_PRV*: amount type converted into transaction currency • *TIP_CV*: consolidable amount type • *TIP_CVV*: consolidable amount type in transaction currency • *TIP_DES*: (description) amount type that allows the user to enter and save text-based information, such as notes or comments.
Category	Category code
Currency:	Currency code . If the dimension is not defined, it uses the entity currency.
Cust_Dim1..Cust_Dim5:	Element code of one of the custom dimensions. For example, if the products to recover the value of a certain product are handled on custom dimension 1, the code is Cust_Dim1.
Entity	Entity code
ICCust_Dim:	Element code of the intercompany custom dimension
ICEntity	Intercompany entity code
Period	Period code. Mandatory dimension.
PeriodLength:	Period length. The possible values are: • *LUN_0*: progressive • *LUN_1*: monthly • *LUN_2*: bimonthly • *LUN_3*: quarterly • *LUN_4*: every four months • *LUN_5*: every five months • *LUN_6*: half-yearly • *LUN_12*: annual • *LUN_QTD*: Quarter to date • *LUN_STD*: Semester to date • *LUN_YTD*: Year to date
Scenario	Scenario code. Mandatory dimension.
DB code	If the formula used in the current application database is not defined (similar to using the TGKCURRDB function)

***Note**: the system does not filter any data on undefined dimensions.*

Data conversion

The *Convert* element allows the user to convert dimension data while running a form. The *Convert* values are:

- *Final*: final FX rate
- *Average*: average FX rate
- *Period*: period average FX rate
- *Account*: Conversion type associated with the account

The syntax is the following: *<("dimensionNamex:valuex")>* i.e. dimension name: Convert value.

This is directly set without specifying its name (like the other parameters).

Calculation by segment

In the TGKRET formula, it is possible to request the amount calculation in the following ways: by *segment*, *intersegment*, *parent segment* and *parent intersegment*.

To do this, the desired suffix (suffixes are case insensitive) must be concatenated to the desired dimension code (node).

Calculation method	Suffix
segment	• \s • \segment Example: <desired dimension code>\s
intersegment	• \i • \is • \intersegment Example: <desired dimension code> \i
parent segment	• \p • \ps • \parentSegment Example: <desired dimension code>\p
parent intersegment	• *pis* • *pi* • *parentIntersegment* Example <desired dimension code>\pis

Syntax conventions and specific features

- The position of the parameters within the TGKRETF formula is fixed. If you want to indicate an optional parameter and that parameter comes after other optional parameters that you do not want to specify, they must each be identified with an empty space separated by ",".
- To retrieve several elements or nodes of aggregation structures for the same dimension, separate the values with ",". The system reads the data related to all specified elements. For example, to retrieve accounts 50010 and 40010, the syntax in the position relative to the account is *50010,40010*.
- To retrieve several nodes belonging to different aggregation structures for the same dimension, the system reads the intersection data related to the elements related to the nodes. For example, to retrieve the entities related to node N1 of hierarchy H1 and node N1 of hierarchy H2, the syntax is *H1:*N1,H2:*N1*.

Example

To retrieve the value of the revenue for the Antivirus product for the geographical area Italy and for scenario 2012BDG, apply the following filters:

- Scenario = 2012BDG
- Period = 12
- Account = all revenue accounts, i.e. FST item 500300 - Total Revenues in FST 010.
- Product (managed on custom dimension 2) = Antivirus, with code A01
- Entity = all entities related to the Italy node (X01) of hierarchy 03 which groups entities by geographical area
- Cost Center (managed on custom dimension 1) = generic element G01
- Employee (managed on custom dimension 2) = generic element G01

The function's syntax is *=TGKRETF("2012BDG",12,"*010|50030","TIP_O","03:*X01","$AMOUNT","G01","A01","G01", , , , , ,"LUN_0", , ,TGKCURRDB())*

To read the original data for account 50010 on December 2011, the function is *=TGKRETF ("2011ACT","12","50010")*

To read the original data for account 50010 on December 2011 on the $AMOUNT category, the function is *=TGKRETF("2011ACT","12","50010", , ,"$AMOUNT")*

To read the original data for account 50010 on December 2011 converted into EUR at the average FX rate, the function is *=TGKRETF("2011ACT", "12", "50010", , , , , , , , , , ,"EUR", , ,"Average",TGKCURRDB())*

The TGKSEND function

Description

The TGKSEND function allows you to retrieve, edit and save a value on a different database from the one on which the form is run. It can only be used if a form is opened in data entry mode and if the user has editability rights over the function's dimensions.

Syntax

The function uses the following syntax:*=TGKSEND(value, datasource, "DimensionName1:value1", "DimensionName2:value2....).* "Value" indicates the code of the Excel cell.

With this function, you must indicate the cell to be made editable as the first argument.

Arguments

The items that can be added to the function are detailed below.

Of the function's arguments have been defined correctly and the user has the necessary rights over the dimensions involved, the cell specified as the function's first parameter will be editable.

> ***Note**: for editability, the same arguments as the TGKRET function are permitted, but with individual elements (and not nodes); see "The TGKRET function" on page 295.*

Syntax conventions and specifics

- To recover several elements or nodes of aggregation structures for the same dimension, values can be separated by ",". The system reads the data related to all specified elements. For example, to recover accounts 50010 and 40010, the syntax in the position relative to the account is *Account:50010,40010*.
- To recover several nodes belonging to different aggregation structures for the same dimension, the system reads the intersection data related to the elements related to the nodes. For example, to recover the entities related to node N1 of hierarchy H1 and node N1 of hierarchy H2, the syntax is *Entity:H1:*N1,H2:*N1*.

Example

To define a cell as editable, only individual elements, not nodes, must be used as parameters. To retrieve the value of the revenue for the Antivirus product for the geographical area Italy and for scenario 2012BDG, apply the following filters:

- Scenario = 2012BDG
- Period = 12
- Account = 50010 - Product Revenues
- Product (managed on custom dimension 2) = A01 - Antivirus
- Entity = A00 - Distribution Rome
- Cost Center (managed on custom dimension 1) = G01 - Generic
- Employee (managed on custom dimension 3) = G01 - Generic

To make cell A5 editable, the function's syntax is *=TGKSEND(A5, TGKCURRDB(), "Account:50010", "Scenario:2012BDG","Entity:A00", "Cust_Dim1:G01", "Cust_ Dim2:A01","Cust_ Dim3:G01","Period:12","Category:$AMOUNT","AmountType:TIP_O","PeriodLength:LUN_0")*

The TGKFIELD function

Description

The TGKFIELD function allows you to obtain the value of a field cell.

Syntax

The function uses the following syntax: *=TGKFIELD(dataSource, "CellFieldExpression", "DimensionName1: value1", "DimensionName2: value2", ...)*

Arguments

The items that can be added to the function are detailed below.

Datasource

Code of the database from which the value must be recovered. This code can be indicated in the following ways:

- with the database code, e.g. *=TGKRET("TCPM_DEMODEV_001", ...)*
- with the *TGKCURRDB* function, e.g. *=TGKRET(TGKCURRDB(), ...)*
- with reference to the Excel cell containing the database code, e.g. *= TGKRET(A25, ...)*

Cell field

The cell field to be defined, e.g. *$Entity.code*.

For more details on cell fields, see "Cell fields" on page 41.

Selecting dimensions

To retrieve the value of the desired dimension, you can do as follows.

- To retrieve a value of a dimension: *<name of dimension>:<value to filter>*, e.g. for entity A00, specify *Entity:A00*.
- To retrieve a value of a dimension on which aggregation structures exist, refer to a node: *<name of dimension>:<aggregation structure code>:*<node code>*, e.g. *Entity:H_CODE:*H_NODE*.
- To retrieve a value of an account dimension: *<name of account dimension>:*<FST code|FST item code>*, e.g. *Account:*FST_CODE|FST_NODE*

The supported dimensions are listed below.

Dimension	Description
Account:	Account code. Mandatory dimension. Individual accounts, nodes of an account hierarchy or an FST item can be used. In order for the formula to be correct, the account must be present in the list and must meet one of the following conditions: • be related to an FST. • be present in a control group. • be related to a node of an accounts hierarchy.
AmountType	Amount type. Mandatory dimension. The possible values are: • *TIP_O*: (original) amount type of data entered in the system by the user. Default value when the dimension is not defined. • *TIP_OV*: (in transaction currency) amount type that only represents the amounts expressed in a currency other than the default entity currency (for example, an amount in euros to be attributed to an entity whose currency is the dollar) • *TIP_OE*: proportional amount type • *TIP_OEV*: proportional amount type in transaction currency • *TIP_PR*: converted amount type • *TIP_PRV*: amount type converted into transaction currency • *TIP_CV*: consolidable amount type • *TIP_CVV*: consolidable amount type in transaction currency • *TIP_DES*: (description) amount type that allows the user to enter and save text-based information, such as notes or comments.
Category	Category code
Currency:	Currency code . If the dimension is not defined, it uses the entity currency.
Cust_Dim1..Cust_Dim5:	Element code of one of the custom dimensions. For example, if the products to recover the value of a certain product are handled on custom dimension 1, the code is Cust_Dim1.
Entity	Entity code
ICCust_Dim:	Element code of the intercompany custom dimension
ICEntity	Intercompany entity code
Period	Period code. Mandatory dimension.

Dimension	Description
PeriodLength:	Period length. The possible values are: • *LUN_0*: progressive • *LUN_1*: monthly • *LUN_2*: bimonthly • *LUN_3*: quarterly • *LUN_4*: every four months • *LUN_5*: every five months • *LUN_6*: half-yearly • *LUN_12*: annual • *LUN_QTD*: Quarter to date • *LUN_STD*: Semester to date • *LUN_YTD*: Year to date
Scenario	Scenario code. Mandatory dimension.
DB code	If the formula used in the current application database is not defined (similar to using the TGKCURRDB function)

***Note**: the system does not filter any data on undefined dimensions.*

The TGKAMOUNT function

Description

This TGKAMOUNT conditional function allows you to read and edit a value in the database, identifying it on the basis of the filters present on the cell.

Syntax

The function uses the following syntax: *=TGKAMOUNT()*.

Example

=TGKAMOUNT({{if(condition1, function1, if(condition2, function2, <value calculated by DB>)) }})

The TGKCALCSET function

Description

The TGKCALCSET function allows you to improve performance in the presence of forms containing several custom functions. In particular, it allows you to group functions into *calculation sets* and set specific attributes for those sets.

Syntax

The function uses the following syntax: *=TGKCALCSET(range, [hints])*

Arguments

The items that can be added to the function are detailed below.

Calculation sets

Calculation sets are made up of the following arguments:

- *range*, which indicates the range of cells belonging to the set.
- *hints* (optional), which specifies whether to apply a hint to the calculation set query, and if so, what type of hint to apply

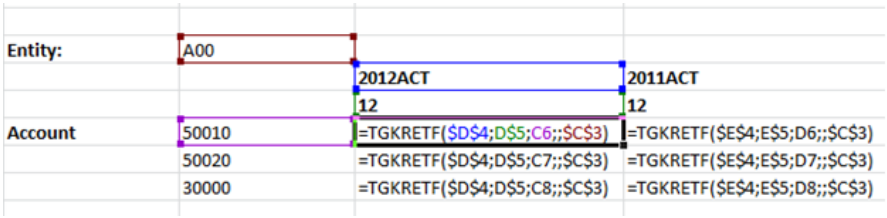
Hints

The possible values are:

Element	Description
• **E** • **ENABLED** • **<empty> (omit)**	Enables the use of hints.
• **D** • **DATA**	Indicates that the use of hints is based on the content of the data tables, optimising the running of the form.
• **NO** • **DISABLE**	Disables the use of hints.
• **A** • **AUTO**	Automatically activates/deactivates the use of hints.

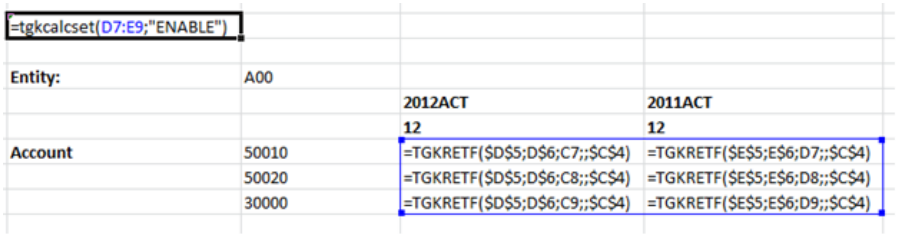
Example

In this example, several TGKRETF functions are present, including the entity, scenario and period dimensions.



Entity:	A00		
		2012ACT	2011ACT
		12	12
Account	50010	=TGKRETF(D4;D$5;C6;;$C$3)	=TGKRETF(E4;E$5;D6;;$C$3)
	50020	=TGKRETF(D4;D$5;C7;;$C$3)	=TGKRETF(E4;E$5;D7;;$C$3)
	30000	=TGKRETF(D4;D$5;C8;;$C$3)	=TGKRETF(E4;E$5;D8;;$C$3)

The formulas contain several shared dimensions, such as entity, scenario and period. Separate queries must be run for each cell, and this can have an impact on performance. To optimise the running of forms, therefore, it is possible to define a calculation set that groups linked elements.



=tgkcalcset(D7:E9;"ENABLE")			
Entity:	A00		
		2012ACT	2011ACT
		12	12
Account	50010	=TGKRETF(D5;D$6;C7;;$C$4)	=TGKRETF(E5;E$6;D7;;$C$4)
	50020	=TGKRETF(D5;D$6;C8;;$C$4)	=TGKRETF(E5;E$6;D8;;$C$4)
	30000	=TGKRETF(D5;D$6;C9;;$C$4)	=TGKRETF(E5;E$6;D9;;$C$4)

The result of the functions is as follows.

Entity:	A00		
		2012ACT	**2011ACT**
		12	**12**
Account	50010	1000	900
	50020	200	300
	30000	800	600

In general, it is advisable to define a TGKCALSET on cells defined via the same dimensions and/or hierarchies, trying – if possible – to identify "rectangular areas" in which to position the formula so that only one filter changes when they are shifted horizontally or vertically. In general, the more a set of cells to be grouped "is like a matrix", the more likely the user is to benefit from grouping them in a single "calculation set".

The TGKOPTION function

Description

The TGKOPTION function allows you to specify the reporting engine to be used to calculate the TGKRET functions defined on the form.

Syntax

The function uses the following syntax: *=TGKOPTION(engine)*.

Arguments

The items that can be added to the function are detailed below.

Engine

This argument supports the following values:

- *INM*, which indicates the reporting engine in memory
- *SQL*, which indicates the standard reporting engine

***Note**: if the TGKOPTION function is not specified, the system uses the reporting engine specified in the general form configurations; see Finestra Impostazioni.*

Editable categories pool

Function description

In CCH Tagetik, all data can be classified by assigning a category to it. This dimension allows you to segment and classify data, particularly journals, and thus makes it possible to keep track of the way in which data items are generated, i.e. whether it was entered manually, loaded from accounting records or generated automatically through processing).

It may be useful for some users to work on the same data, but on different categories without overlapping of the data entered on the node. For example, a contributor user may need to enter data, which will then be supervised by a second user, before finally being submitted by a consolidator user.

To do this, CCH Tagetik allows you to group several categories into *category pools*.By A category pool is essentially a node of the categories group that can be used within a form to show a value as the sum of several categories.

By using a filter on a node of categories labelled as a *pool* within a form, it will be editable for a given user provided the user has editability rights over just one of the categories related to the pool and visibility rights over all other categories related to the pool. If the user does not have these rights, the category pool is not editable within the form.

> #### Example
>
> To work on 3 levels, the following configuration is necessary to work with editable pools.
>
> 1. Include the AMOUNT, ADJUST_1, ADJUST_2 categories in a single node grouping categories.
> 2. Associate the AMOUNT category with the contributor user, as editable.
> 3. Associate the ADJUST_1 category with the responsible contributor user.
> 4. Associate the third ADJUST_2 category with the higher-level manager, i.e. the consolidator user.
> 5. Set visibility over the three categories for all users.
> 6. The data entry form must be structured in such a way that filters, at the category level, on the node.
>
> Data edited by the user is recorded in a unique category that only they can edit, so if the level one manager changes the aggregated value from 100 to 120, the difference of 20 is saved on the ADJUST_1 category. The contributor user and the second-level manager area able to see the total value of 120.

For more details on categories, see La Categoria: concetti di base.

Working with category pools

Define a node as a category pool

Operational database > Set up and Administration > Dimensions > Categories > Grouping > select node of categories > (Attributes tab) Editable categories pool

Operational database > Navbar > Dimensions > Categories > Grouping > select node of categories > (Attributes tab) Editable categories pool

Using a category pool in a form

1. Define the node grouping categories as an *editable category pool*.
2. Associate a category with every contributor user. If working on several levels, associate them with the category corresponding to their level in the hierarchy.
3. Associate the first-level category with the consolidator user.

The Break Back function

Function description

By default in CCH Tagetik, data entry takes place on the individual elements of dimensions, such as element *A00-Company* of the entity dimension, and not on nodes of aggregation structures, such as all entities belonging to the *Europe* node of the *Geographical Area* hierarchy defined on the entity dimension, for example.

However, in certain cases it may be useful to enter data on nodes. For example, when drafting a budget, it may be necessary to increase the total revenue of all entities belonging to the Europe node of the Geographical Area hierarchy. The *break back* functionality allows you to perform data entry via forms at the level of nodes of aggregation structures defined on the entity, account and custom dimensions, or on custom dimensions only.

With the **Save on the reference element** option set in the data entry step from form, the value entered on the node must always be attributed to the lowest reference element or distributed over the lowest level elements related to the same node in proportion to the old value

In particular, this makes the node cell editable and allows you to enter an amount manually. If the **Save on the reference element** option is not active, the value entered on the node is automatically distributed over all the elements of the node in proportion to the pre-existing values on the individual elements.
If one of the elements related to the node on which data entry is being performed does not have a value, the system excludes it from the distribution. If none of the elements related to the mode have a value, the system will show an error message and ask you to define a reference element for the node within the hierarchy defined on the dimension. That reference element will be used by the system as the target of the entered data on which to apply the break back. Any element related to the node can be used as a reference element.

Break back can be activated and configured for the entire application at the data entry rules level; however, it can be overridden for individual processes (see Campi nella sezione Break back). Overrides set on process can in turn be overridden for individual matrices (see "Matrix options tab" on page 128).

Using the Break Back function

Overriding Break Back in processes

Application database > Processes tile > select process > ⋮ > Process setup > Data Entry & Submission Rules section) > Gross & IC amounts > ✎

Application database > Processes & Workflows > Processes > select process > ⋮ > Process definition > (Data Entry & Submission Rules section) > Gross amounts & IC > ✎

Overriding Break Back in matrices

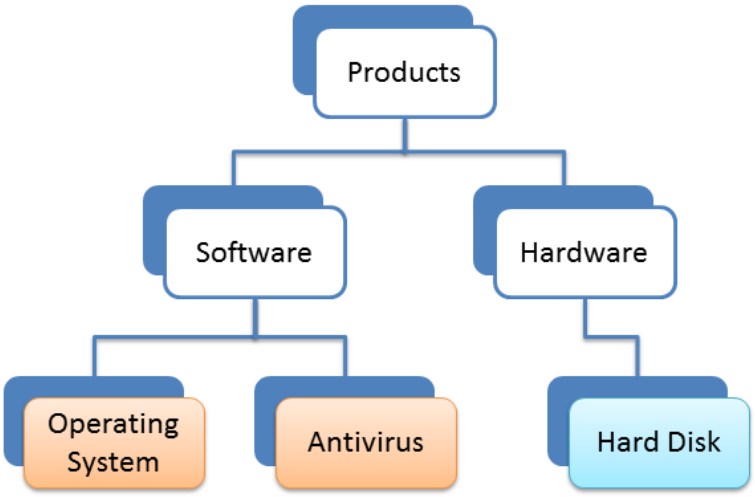
Excel > CCH® Tagetik > Design > select form > Matrix > Matrix options > Form - Break back

Use cases for the Break Back function

Break back can be useful in contexts such as the following:

The user wants to enter data for the 2014 forecast revenue on a form with the following characteristics:

- the form manages a Products custom dimension.
- the hierarchy of the Products custom dimension contains the Software node, which groups Operating Systems, Antivirus products and Hard Disk products
- in the form, revenue is detailed by product, e.g. Operating Systems, Antivirus products and Hard Disks.



By default, in a data entry form, the amount of the Software node corresponds to the sum of the Operating Systems and Antivirus amounts.

Product	Amount
Operating System	30
Antivirus	70
Software (node)	100

Enabling break back on custom dimensions allows you to enter data on the Software node instead of calculating it as the sum of the child elements. In this way, the Software cell in the form becomes editable and the entered amount is distributed on a proportional basis among the products included in Software.

Product	Amount before data entry on node	After entering 150 on Software node
Operating System	30	45
Antivirus	70	105
Software (node)	100	150

Example: the Always write on reference element option

By activating the *Always write on reference element* break back option, the system systematically saves the value entered at the node level on the reference element.

Product	Amount before data entry on node	After entering 150 on Software node (Write on ref. Element to FALSE)	After entering 150 on Software node (Write on ref. Element to TRUE)
Operating System	30	45	30
Antivirus	70	105	70
Software Reference	0	0	50
Software (node)	100	150	150

Case 1:

Process settings

Target dimensions = None

Write on reference lowest element = Irrelevant

Fields in Matrix Options		Result
Break back dimensions	**Save on the reference element**	
Default	Default	No Break back
Default	TRUE or FALSE	Error
Disabled Break back	Forced to Default	No Break back
Custom Dimensions	Any	Error

Case 2

Process settings

Target dimensions = Custom Dimensions

Write on reference lowest element = False

Fields in Matrix Options		Result
Break back dimensions	**Save on the reference element**	
Default or Custom Dimensions	FALSE or Default	Break back on elements of node that have some value
Default or Custom Dimensions	TRUE	Break back only on reference elements of node
Disabled Break back	Forced to Default	No Break back

Case 3

Process settings

Target dimensions = Custom Dimensions

Write on reference lowest element = True

Fields in Matrix Options		Result
Break back dimensions	**Save on the reference element**	
Default or Custom Dimensions	FALSE	Break back on elements of node that have some value
Default or Custom Dimensions	TRUE or Default	Break back only on reference elements of node
Disabled Break back	Forced to Default	No Break back

Case 4

Process settings

Target dimensions = Custom Dimensions - Entity - Account

Write on reference lowest element = False

Fields in Matrix Options		Result
Break back dimensions	**Save on the reference element**	
Default	FALSE or Default	Break back on elements of node that have some value
Default	TRUE	Break back only on reference elements of node
Custom Dimensions	FALSE or Default	Break back on elements of only Custom dimension nodes that have some value
Custom Dimensions	TRUE	Break back only on reference elements of only Custom dimension nodes
Disable break back	Forced to Default	No Break back

Reporting statistics

What they show

When it becomes necessary, in navigation or data entry modes, to assess the run times of a form, a summary report of the add-in's activities can be generated in Excel format.

In particular, the report is subdivided into sheets containing the details of the run times for the various operations completed when the form is run.

Element	Description
Tracer report	Contains details of the times taken to generate the various components of the form (e.g. sheets, matrices, cell fields), on both the server side and the client side.
Info	Contains information for debugging MDX scripts.
Data processing results	Contains details of the data processing run in the form (e.g. data processing name, parameters, extraction times for data to be shows, matrix generation times, layout preparation times).
Adv.dyn.scripts	Contains details of the advanced selection scripts present in the form (e.g. matrix in which they are present, value of classifiers).

Custom macros

Contents

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Custom macros

Introduction

In CCH Tagetik, custom macros are an integrated function which, through a series of scripts, allows you to customise or automate certain data entry operations. In particular, Custom Macros allow you to:

- write a VBA macro in an Excel form
- implement custom functions or routines to be run on the occurrence of events defined by the user

This module describes the details of the events that support custom macros and provides a description of the relative syntax.

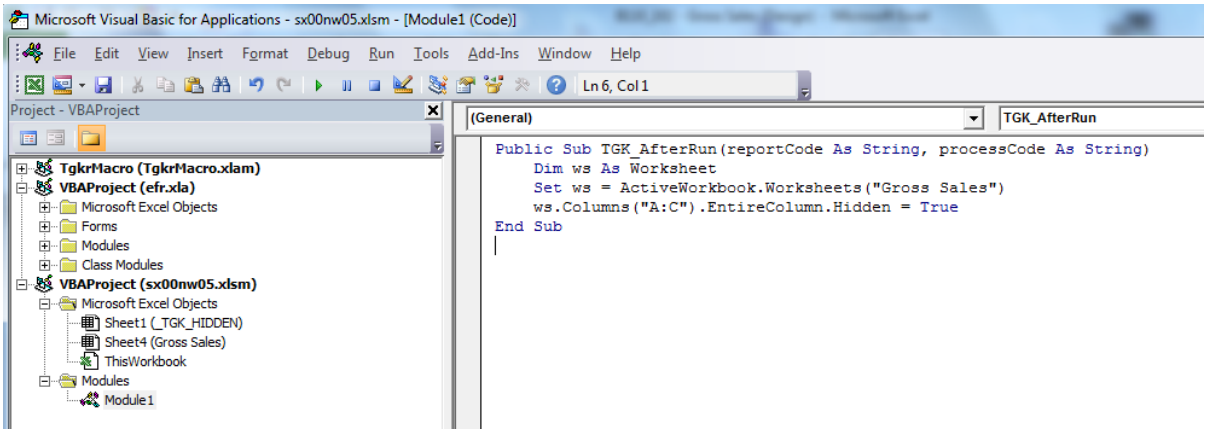
Supported events

The system supports the following events:

- running a form("TGK_BeforeRun" on the facing page / "TGK_AfterRun" on page 312)
- saving a form ("TGK_BeforeSave" on page 312 / "TGK_AfterSave" on page 312)
- updating a form ("TGK_BeforeRefresh" on page 313 / "TGK_AfterRefresh" on page 313)
- generating a deployed matrix ("TGK_MatrixCreated" on page 313)
- closing the details window to edit a transactional form ("TGK_BeforeFormEditorCommit" on page 314)

Example of execution after a form has been run

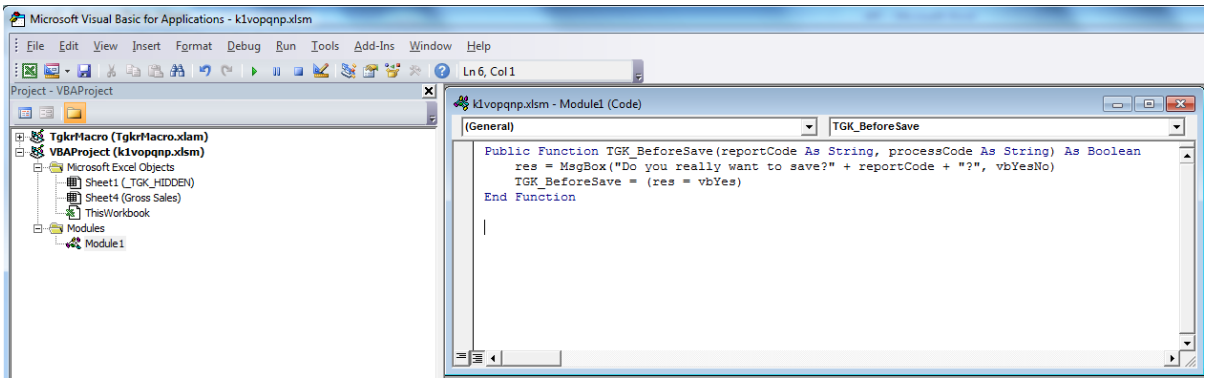
In this example, a subroutine recalled at the end of the form run step is defined ("TGK_AfterRun" on page 312).



After the form is run, columns, A, B and C are hidden.

Example of execution before a form is saved

In this example, a function recalled before the form is saved ("TGK_BeforeRun" below).



The function returns the following values:

If...	Then...	AND...
the user selects **Yes**	it returns the value TRUE	the system will allow you to save the data.
the user selects **No**	it returns the value FALSE	the system will not allow you to save the data.

TGK_BeforeRun

Element	Description
Syntax	`public function TGK_BeforeRun(reportCode as String, processCode as String, isRefill as Boolean) as Boolean`
Event description	This event is recalled before a form is run, immediately after the system asks if there are any filters. ***Note**: it is not possible to recall the running of data processing in this event, since data processing from forms can only be run if the form is already open in Data Entry mode.*
Necessary parameters	• *reportCode*. Code of the form run (not present in ad hoc forms) • *processCode*. Code of the current process (empty string if the form is run in Navigation mode) • *isRefill*.TRUE = next refresh, FALSE = initial run
Value returned	• TRUE if the run must proceed. • FALSE if the run must be cancelled.

TGK_AfterRun

Element	Description
Syntax	`public sub TGK_AfterRun(reportCode as String, processCode as String)`
Event description	This event is recalled at the end of the form run step. With the **Handle custom event callbacks with unprotected sheets** option active, this routine's code will be run with Excel spreadsheets temporarily unlocked, but they are locked again once the macro is completed.

Necessary parameters	• *reportCode*. Code of the form run (not present in ad hoc forms)
	• *processCode*. Code of the current process (empty string if the form is run in Navigation mode)
	When the user runs a cycle on all the workbook sheets, the system also considers hidden sheets containing information reserved for the application which may generate data processing errors. The following subroutine can be run to exclude hidden sheets:
	`Sub TGK_AfterRun (reportCode as String, processCode as String)`
	`numIndex = 1`
	`columnA = 22`
	`Do While numIndex < WorkSheets.Count`
	`if WorkSheets(numIndex).visible = xlSheetVisible`
	`WorkSheets(numIndex).Columns(columnA).entireColumn.Hidden = True`
	`end if`
	`numIndex = numIndex + 1`
	`Loop`
	`End Sub`

TGK_BeforeSave

Element	Description
Syntax	`public function TGK_BeforeSave(reportCode as String, processCode as String) as boolean`
Event description	This event is recalled before edited values are saved. With the **Handle custom event callbacks with unprotected sheets** option active, this routine's code will be run with Excel spreadsheets temporarily unlocked, but they are locked again once the macro is completed.
Necessary parameters	• *reportCode*. Code of the form the values of which you want to save • *processCode*. Code of the current process
Value returned	• TRUE if the save must proceed. • FALSE if the save must be cancelled.

TGK_AfterSave

Element	Description
Syntax	`public sub TGK_AfterSave(reportCode as String, processCode as String)`
Event description	This event is recalled after an edited form has been saved.
Necessary parameters	• *reportCode*. Code of the current form • *processCode*. Code of the current process

TGK_BeforeRefresh

Element	Description
Syntax	`public function TGK_BeforeRefresh(closeWorkbook as Boolean, action as integer) as Boolean`
Event description	This event is recalled before a form is refreshed. ***Note**: This is not used when a Report List is refreshed.* The function is run from the workbook on which the refresh was run. At the end of the update, the workbook can be closed by the macro. With the **Handle custom event callbacks with unprotected sheets** option active, this routine's code will be run with Excel spreadsheets temporarily unlocked, but they are locked again once the macro is completed.
Necessary parameters	• *closeWorkbook*. If TRUE at the end of the refresh, the current workbook is closed. • *action*. Indicates the event after which the refresh is invoked: ◦ 1 = Update requested from Excel using the button. ◦ 2 = Refresh requested using a VBA macro. ◦ 3 = Refresh requested after the form has been saved. ◦ 4 = Refresh requested at the end of data processing.
Value returned	• TRUE if the refresh must proceed. • FALSE if the refresh must be cancelled.

TGK_AfterRefresh

Element	Description
Syntax	`public function TGK_AfterRefresh(formCode as String, processCode as String) as Boolean`
Description	This event is recalled after a form is refreshed. ***Note**: This is not used when a Report List is refreshed.* The function is run from the workbook on which the refresh was run. At the end of the refresh, the workbook can be closed from the refresh action. With the **Handle custom event callbacks with unprotected sheets** option active, this routine's code will be run with Excel spreadsheets temporarily unlocked, but they are locked again once the macro is completed.
Necessary parameters	• *formCode*. This identifies the form on which the refresh was run. • *processCode*. This identifies the process, or an empty string if the form was run in Navigation mode.

TGK_MatrixCreated

Element	Description
Syntax	`public sub TGK_MatrixCreated(reportCode as String, matrixCode as string, matrixAddress as string)`
Description	This event is called up at the end of the deployment of a dynamic matrix. In this event, it is not possible to refer to output sheets or matrices that have not yet been generated by the system.
Necessary parameters	• *reportCode*. Code of the form run (not present in ad hoc forms) • *matrixCode*. Code of the matrix just created • *matrixAddress*. Address of the range occupied by the matrix just created

TGK_BeforeFormEditorCommit

Element	Description
Syntax	`public function TGK_BeforeFormEditorCommit(values() As Variant, rowIndex As Integer) as Boolean`
Description	This event is recalled by clicking on the **OK** button shown in the record details window of a transactional form editor ("Transactional data entry details window" on page 289).
Necessary parameters	• *values*. Set of values present in the current row of the transactional form when the **OK** button is clicked. • *rowIndex*. (Logical) index of the current row
Value returned	• TRUE: the details window closes and the new values inserted remain in the matrix. • FALSE: the original values present when the details window was opened are restored.

Managing macros

Link the VBA code to an event

Excel > CCH® Tagetik > Design

To link a VBA code to an event, it is necessary to create a public module in which to insert the event declaration and the VBA code to be run when the event occurs.

1. Open the form in which to insert the VBA code.
2. Click Excel's **Developer** tab, then click **Visual Basic**.
3. In the VBA project tree, right click the workbook of the desired form, then click **Insert** > **Module**.
4. In the module, create a *public function* if the result of the function must be used in other operations, or a *subroutine* if no result needs to be shown.

Call up an API within a macro.

There are two methods that can be used to recall an API within a macro, as described below.

- Using the following syntax:

```
dim valoreDiRitorno as <tipo>
```

```
valoreDiRitorno = Application.run("Tgkrmacro.xlam!nomeFunzione", "parametro1", parametro2", ...)
```

- Define a button in the form (e.g. CommandButton1) and insert the following example code in the event that manages button pressure:

```
Private Sub CommandButton1_Click()
```

```
Dim wb As Workbook
```

```
Set wb = Application.Run ("Tgkrmacro.xlam!TGK_Navigate", "B110_100")
```

```
End Sub
```

In the example, the instruction calls up the opening of the form with code "B110_100" in navigation mode.

Custom VBA macros via APIs

Introduction

CCH Tagetik provides public VBA APIs that can be used within the macros inserted in a form to automate operations such as opening a form, showing the mode in which a form was opened, and returning metadata on the form.

Parameter syntax

Run parameters can be specified in human-readable syntax. To save these elements, the system uses *variable string* and *constant string* VBA script. In the form, these variables can be used as arguments of run functions, or can be read automatically based on one or more cells of the current form.

An example of syntax specifying dynamic global filters is provided below:

```
Entity=A01;<CustDim1:G01>=*N1;CustDim2={DA,DB};
```

Global filters have the following particular features:

- the individual filter on a lowest level element is assigned using the syntax *DimensionName=LeafCode* (without double inverted commas). The dimension name is equivalent to the name used in the corresponding cell fields (without the initial "$")
- a hierarchy dimension is indicated between square brackets as *<DimensionName:hierarchy code>*
- a filter on a node is indicated by adding an asterisk as a prefix
- multiple filters are separated with a semicolon ";"
- an OR clause of multiple filters is indicated between braces, with values separated by a comma ",".
- to represent parameters (e.g. SP=2004C;PP=12)
  - the individual filter on an lowest level element is assigned using the syntax ParameterCode=LeafCode (without double inverted commas)
  - OR filter and filters on nodes are not permitted

VBA APIs to open/close/update forms

TGK_OpenTemplate

Element	Description
Syntax	`dim wb as workbook` `set wb = Application.run("Tgkrmacro.xlam!TGK_OpenTemplate", "REPORT_CODE")`
Description	Opens the form in design mode.
Mandatory parameters	REPORT_CODE: Code identifying the form to be opened
Value returned	Workbook associated with the open form. If the form has not been opened correctly, it returns *nothing* (e.g. if the code is non-existent or the user does not have permission to open the form in Design mode).

TGK_Navigate

Element	Description
Syntax	`dim wb as workbook` `set wb = Application.run("Tgkrmacro.xlam!TGK_Navigate", "REPORT_CODE", ["PROMPT_FILTERS"], ["PARAMETERS"])`
Description	Opens the form in navigation mode.
Mandatory parameters	• REPORT_CODE: Code identifying the form to be opened
Optional parameters	• PROMPT_FILTERS: Only present for compatibility with forms created using the VBA client • PARAMETERS: Parameters to be used when the form is run. Must be defined according to human-readable syntax. If omitted, the parameters are requested when the form is run.
Value returned	Workbook associated with the open form. If the form is not opened correctly, it returns *nothing* (e.g. if the code is non-existent or the user does not have permission to open the form in navigation mode).

Example

Example of syntax for running the form with code "PROSP01" in navigation mode, based on the following parameters:

- entity A01
- node N1 of hierarchy 01 of custom dimension 1
- parametric scenario (SP) 2004C
- parametric period (PP) 12.

```
dim wb as workbook
```

```
set wb = Application.run("Tgkrmacro.xlam!TGK_Navigate", "PROSP01",
"Entity=A01;<CustDim1:G01>=*N;SP=2004C;PP=12")
```

TGK_DataEntry

Element	Description
Syntax	`dim wb as workbook` `set wb = Application.run("Tgkrmacro.xlam!TGK_DataEntry", "REPORT_CODE", "PROCESS_CODE", ["PROMPT_FILTERS"], ["PARAMETERS"])`
Description	Opens the form in data entry mode.
Mandatory parameters	• REPORT_CODE: Code identifying the form to be opened • PROCESS_CODE: Code identifying the process on which data collection is performed
Optional parameters	• PROMPT_FILTERS: Only present for compatibility for forms created using the VBA client • PARAMETERS: Parameters to be used when the form is run. Must be defined according to human-readable syntax. If omitted, the parameters are requested when the form is run.
Value returned	Workbook associated with the open form. If the form is not opened correctly, for example the code is non-existent or the user does not have permission to open the form in Data Entry mode, it returns **nothing**.

TGK_OpenReportFile

Element	Description
Syntax	`dim wb as workbook` `set wb = Application.run("Tgkrmacro.xlam!TGK_OpenReportFile", "FILENAME.xls")`
Description	Opens an ad hoc form or a form opened in navigation mode and then saved as an Excel sheet.
Parameters	FILENAME: full XLS Path of the file to be used *e.g. "C:\Documents and Settings\....\myReport.xls"*
Value returned	Workbook associated with the open form. In the following cases, it returns *nothing*: • the file does not exist or is locked, or does not have read permission • user does not have permission to open ad hoc forms • the specified Excel file is not a valid form

TGK_CloseMySelf

Element	Description
Syntax	`Application.run("Tgkrmacro.xlam!TGK_CloseMySelf", WORKBOOK, [ASK_SAVE], [JUST_CLOSE])`
Description	Closes the current form (or the form containing the macro that calls up this function).
Mandatory parameters	• WORKBOOK: Workbook object corresponding to the form to be closed.
Optional parameters	• ASK_SAVE: the system only considers this parameter if the JUST_CLOSE parameter is FALSE. If it is TRUE, the system checks whether any changes have been made and asks the user if they want to save. If it is FALSE or omitted, the system saves and closes the form without displaying any messages. • JUST_CLOSE: if TRUE, it closes the form without saving or asking for confirmation. If FALSE or omitted, the parameter ASK_SAVE is taken into consideration.
Value returned	None

Example

Example of syntax for closing the form corresponding to the Excel document containing the macro, which asks for confirmation before saving data.

```
Application.run("Tgkrmacro.xlam!TGK_CloseMySelf", ThisWorkbook, true)
```

TGK_CloseOtherDocuments

Element	Description
Syntax	`Application.run("Tgkrmacro.xlam!TGK_CloseOtherDocuments", WORKBOOK, [ASK_SAVE])`
Description	Closes a different document from the one in which the macro is written.
Mandatory parameters	WORKBOOK: Workbook object corresponding to the form to be closed.
Optional parameters	ASK_SAVE: If TRUE, the system checks whether any changes have been made and asks the user if they want to save. If it is FALSE or omitted, the system saves and closes the form without displaying any messages.
Value returned	TRUE if the form was closed correctly; FALSE otherwise.

TGK_RunUnprotected

Element	Description

Syntax	`dim bResult as <tipo>`
	`bResult = Application.run("Tgkrmacro.xlam!TGK_RunUnprotected", WORKBOOK, "MACRO_NAME", _`
	`["macroParam_1"], ["macroParam_2"], ...,["macroParam_10"])`

Description	In all cases in which the form was opened in data entry mode, and is therefore protected, this allows you to run the specified macro by removing the protection from the sheet before running it and potentially protecting again at the end.
Mandatory parameters	• WORKBOOK: Workbook object corresponding to the form containing the macro to be run • MACRO_NAME: Name of the macro to be run. Must be a function present on a public module and belonging to the specified workbook.
Optional parameters	MACRO_PARAM_1,...,MACRO_PARAM_10: Any input parameters of the macro to be run.
Value returned	The macro name, as indicated in "MACRO_NAME"

Example

This example shows how to hid a specific row of a form, for example row 25, in a form containing a macro that hides the past row processed as an argument.

The details of the macro are provided below:

```
Public Function nascondiRiga(numRiga As Integer)
```

```
ActiveSheet.Rows(numRiga).Hidden = True
```

```
nascondiRiga = True
```

```
End Function
```

To hide row 25, the following syntax must be used:

```
dim bResult
```

```
bResult = nascondiRiga(25)
```

When the system runs this macro on a protected sheet, it returns the following error: "Impossible to set the 'Hidden' property for the 'Range' class". It must therefore be run with the sheets not protected, using the following syntax.

```
dim bResult
```

```
bResult = Application.run("Tgkrmacro.xlam!TGK_RunUnprotected", "nascondiRiga", 25)
```

The system closes the form corresponding to the Excel document containing the macro and asks for confirmation before saving data.

TGK_RunUnprotectedObj

Element	Description

Syntax	`dim objResult as <tipo oggetto>`
	`set obkResult = Application.run("Tgkrmacro.xlam!TGK_RunUnprotected", WORKBOOK, "MACRO_NAME", _`
	`["macroParam_1"], ["macroParam_2"], ...,["macroParam_10"])`

Description	The same as TGK_RunUnprotected, but to be used when the called macro returns an object rather than a primitive type.
Mandatory parameters	• WORKBOOK: Workbook object corresponding to the form containing the macro to be run • MACRO_NAME: Name of the macro to be run. Must be a function present on a public module and belonging to the specified workbook.
Optional parameters	MACRO_PARAM_1,...,MACRO_PARAM_10: Any input parameters of the macro to be run.
Value returned	The macro name, as indicated in "MACRO_NAME"

TGK_Save

Element	Description
Syntax	`dim bEsito as boolean` `bEsito = Application.run("Tgkrmacro.xlam!TGK_Save", WORKBOOK, [FILE_NAME])`
Description	Saves the file related to the specified form. Equivalent to **Save** in the button bar.
Mandatory parameters	WORKBOOK: Workbook object corresponding to the form to be saved.
Optional parameters	FILE_NAME: If specified, the system saves the form with the name indicated without asking for further confirmation. Otherwise, it follows the normal save Excel file request, i.e. it saves the changes to the current file or, if the document is a new document, asks the user for the name to be assigned to it.
Value returned	TRUE if the form was saved correctly; FALSE otherwise. Do not use the ThisWorkbook.save command within a macro. Replace with TGK_Save. TGK_Save with automatic refresh linked to saving may not function correctly. Disable this option and check that there is no linked data processing that requires it.

TGK_Refresh

Description	
Syntax	`dim wb as workbook` `set wb = Application.run("Tgkrmacro.xlam!TGK_Refresh")`
Description	Refresh the current form, already opened and run.
Value returned	Workbook updated. ***Note**: Dynamic forms will be closed and reopened in the refresh phase. Therefore, it is likely that any rows of codes subsequent to the TGK_Refresh function call will never be run, as the file is closed during the refresh. The report is closed and reopened in the refresh phase. It is therefore necessary to check that the workbook returned by TGK_Refresh is not Nothing even if any rows subsequent to the call will not be run.*

TGK_CountAttachments

Element	Description
Syntax	`Application.run("Tgkrmacro.xlam!TGK_CountAttachments", cell)` `=TGK_CountAttachments({cella})`
Description	Returns the number of attachments present in a cell or range of cells. Function supported in both Excel and Web. This function can also be used as a spreadsheet formula.
Mandatory parameters	CELL. • Single cell or range whose number of attachments is to be retrieved (if used within the form) • Column/row whose number of attachments is to be retrieved for each of its cells (if inserted in a formula within the Matrix Wizard)
Value returned	Number of attachments inserted in a cell or range of cells. ***Note**: To display the number of comments entered on a cell using this function, the option **Automatically display comments in reports data entry** must be enabled (Admin Home Page > Settings > form tab > Comments section). See Finestra Impostazioni.*

VBA APIs to extract data on forms

TGK_isAdHocTemplate

Element	Description
Syntax	`dim bRes as boolean` `bRes = Application.run("Tgkrmacro.xlam!TGK_isAdHocTemplate", WORKBOOK)`
Description	Determines whether the specified form is an ad hoc form template.
Parameters	• WORKBOOK. Workbook object corresponding to the form to be checked.
Value returned	TRUE if the current form is the structure of an ad hoc form; FALSE otherwise.

TGK_isNavigated

Element	Description
Syntax	`dim bRes as boolean` `bRes = Application.run("Tgkrmacro.xlam!TGK_isNavigated", WORKBOOK)`
Description	Determines whether the specified form is a form open in navigation mode.
Parameters	• WORKBOOK. Workbook object corresponding to the form to be checked.
Value returned	TRUE if the current form is a form open in navigation mode; FALSE otherwise.

TGK_isAdHocNavigated

Element	Description
Syntax	`dim bRes as boolean` `bRes = Application.run("Tgkrmacro.xlam!TGK_isAdHocNavigated", WORKBOOK)`
Description	Determines whether the specified form is an ad hoc form template open in navigation mode.
Parameters	• WORKBOOK. Workbook object corresponding to the form to be checked.
Value returned	TRUE if the current form is an ad hoc form open in navigation mode; FALSE otherwise.

TGK_isDataEntry

Element	Description
Syntax	`dim bRes as boolean` `bRes = Application.run("Tgkrmacro.xlam!TGK_isDataEntry", WORKBOOK)`
Description	Determines whether the specified form is a form open in Data Entry mode.
Parameters	• WORKBOOK. Workbook object corresponding to the form to be checked.
Value returned	TRUE if the current form is a form open in Data Entry mode; FALSE otherwise.

TGK_isReport

Element	Description
Syntax	`dim bRes as boolean` `bRes = Application.run("Tgkrmacro.xlam!TGK_isReport", WORKBOOK)`
Description	Determines whether the specified workbook is a form.
Parameters	• WORKBOOK. Workbook object corresponding to the form to be checked.
Value returned	Determines if the specified workbook is a form; FALSE otherwise.

TGK_getReportCode

Element	Description
Syntax	`dim sCodice as string` `sCodice = Application.run("Tgkrmacro.xlam!getReportCode", WORKBOOK)`
Description	Determines the code of the form associated with the workbook processed as an argument.
Parameters	• WORKBOOK. Workbook object corresponding to the form whose code is to be obtained.
Value returned	Form code or an empty string if the form is ad hoc or the specified workbook is not a form.

TGK_getReportProcess

Element	Description
Syntax	`dim sCodice as string` `sCodice = Application.run("Tgkrmacro.xlam!getReportProcess", WORKBOOK)`
Description	Determines the code of the process on which the user is performing data entry via the form specified as an argument.
Parameters	WORKBOOK: Workbook object corresponding to the form whose code is to be obtained.
Value returned	Process code or an empty string if the form is not open in data entry mode.

TGK_getUserName

Element	Description
Syntax	`dim sCodice as string` `sCodice = Application.run("Tgkrmacro.xlam!getUserName")`
Description	Determines the name of the current user.
Parameters	None
Value returned	Current user name or an empty string if the user has not logged in.

TGK_getReportCodes

Element	Description
Syntax	`dim arrCodici` `arrCodici = Application.run("Tgkrmacro.xlam!getReportCodes", [COD_NODO], [COD_PROCESSO], [COD_FASE])`
Description	Determines the list of codes identifying the forms related to a given node of the group of forms in the dimensions tree.
Optional parameters	• COD_NODO: If this parameter is specified and not an empty string, the function returns a list of codes identifying the forms related to the node, otherwise it returns a list of codes identifying all forms related to the nodes of the group of forms in the dimensions tree. • COD_PROCESSO: If this parameter is specified and not an empty string, the search is limited to the forms visible in the dimensions tree for the specified process. • COD_FASE: If this parameter is specified and not an empty string, the search is limited to the forms visible in the dimensions tree for the specified COD_PROCESSO process and for the specified step.
Values returned	List of codes of the related forms.

Example

This example shows the syntax for obtaining a list of codes of the forms related to node "001" for process "ACT2007":

```
dim arrCodici

arrCodici = Application.run("Tgkrmacro.xlam!getReportCodes", "001", "ACT2007")

for i = lbound(arrCodici) to ubound(arrCodici)

MsgBox "Form: " + cstr(arrCodici(i))

next i
```

TGK_getFilters

Element	Description

Syntax	`dim filters as String`
	`dim cella as range`
	`set cella = ....`
	`filters = Application.run("Tgkrmacro.xlam!TGK_getFilters", cella)`

Description	Obtains the set of filters that affect a certain cell or a certain range of cells and returns their expression in accordance with the corresponding human-readable syntax.

If the processed cell forms part...	Then the following are considered as filters:
of the values area of a matrix,	the parameters, tab filters, matrix filters, and row and column filters
of the row headers area,	the parameters, tab filters, matrix filters and row filters.
• of the column header area,	• the parameters, tab filters, matrix filters and column filters.

If the processed cell is not part of any matrix, CCH Tagetik will return an exception when the form is launched.

If a range of cells is processed, the set of filters will contain the union of all the filters that affect the cells involved. If there are different filters on various cells corresponding to the same dimension, OR filters will be returned with all the original values.

***Note**: also functions on cells of matrices containing field expressions.*

Parameters	• CELL. Individual cell or range from which to read filters.

TGK_getParameters

Element	Description
Syntax	`dim parameters as String` `parameters = Application.run("Tgkrmacro.xlam!TGK_getParameters")`
Description	Set of run parameters of the current form
Value returned	Returns the expression in accordance with the corresponding human-readable syntax. ***Note**: also functions on cells of matrices containing field expressions.*

VBA APIs for running actions on forms

Details of the API services for running actions on forms

TGK_refreshTree

Element	Description
Syntax	`Application.run("Tgkrmacro.xlam!TGK_refreshTree")`
Description	Overrides the dimensions tree refresh. • In the processing wizard, the refresh at the end of data processing can be overridden without using custom macros. • When edited values in transactional forms are saved, the dimensions tree is refreshed automatically, if necessary.

TGK_ManipulateFormRow

Element	Description

Syntax	`Dim bResult as Boolean`
	
	`bResult = Application.run("Tgkrmacro.xlam!TGK_ManipulateFormRow", range, "<CODICE_OPERAZIONE>")`

Description	Allows you to manipulate a row of a transactional form.
Necessary parameters	• RANGE. Range relative to the cell in the row to be manipulated and in the column relative to the deletion filter. • CODICE_OPERAZIONE. String that identifies the type of operation to be performed on the row identified by the previous parameter. The parameter can have the following values: ◦ ADD. Adds a row at the bottom of the matrix and opens the details form for the newly created row. ◦ EDIT. Opens the details for to edit the transactional form matrix row. ◦ DELETE. Deletes the row. ◦ UNDELETE. Undoes the deletion of the row.
Value returned	TRUE if the operation was completed successfully; FALSE otherwise. An operation may fail for the following reasons: • The active report is not open in data entry mode. • The specified range does not correspond to a deletion cell of a transactional matrix. • An invalid operation has been specified. • An ADD or EDIT operation has been specified and the editor has not been enabled for transactional forms. • A DELETE action has been specified on a row that has already been deleted. • An UNDELETE action has been specified on a row that has not yet been deleted.

VBA APIs to invoke CCH Tagetik email services

Details of API services to invoke CCH Tagetik email services

TGK_sendEMail

Element	Description

Syntax	`dim bResult as boolean`
	
	`dim cella as range`
	
	`set cella = ....`
	
	`bResult = Application.run("Tgkrmacro.xlam!TGK_sendReportingEMail", cella , rcpt,cc,bcc,subject, body)`

Description	Sends an email to a list of recipients.
Necessary parameters	• RCPT: List of Recipients (array of strings). • SUBJECT: Mail subject. Cannot be empty. • BODY: Email body. Cannot be empty. ***Note**: the recipient can be a CCH Tagetik username or an external email address in the form recipient@domain.*
Optional parameters	• CC: List of recipients CC'd. • BCC: List of recipients BCC'd. • CELL: Not used. Only present for the purposes of compatibility with previous versions. • BODY: Email body. Cannot be empty.
Value returned	TRUE if the email was sent successfully, FALSE otherwise.

TGK_sendEMailWithFilters

Element	Description

Syntax	`dim bResult as boolean`
	`dim filters`
	`bResult = Application.run("Tgkrmacro.xlam!TGK_sendReportingEMail", filters , arrDest,cc,bcc)`

Description	Sends an email to a list of recipients.
Parameters	• FILTERS: Description of the filters to be used to generate the email according to human-readable syntax. • ARRDEST: List of recipients • CC: List of recipients CC'd. • BCC: List of recipients BCC'd. ***Note**: The recipient can be a CCH Tagetik username or an external email address in the form recipient@domain.*
Value returned	TRUE if the email was sent successfully, FALSE otherwise.

Forms on Analytical Workspace data

Introduction

In the Analytical Layer of CCH® Tagetik it is possible to define customised data models called Analytical Workspaces. Every workspace allows the user to add multidimensional data tables, called Datasets, based on an unlimited number of dimensions, measures and custom fields. This allows for a greater calculation capacity and flexibility than the Financial Workspace. For more details on Analytical Workspaces, see Analytical Workspaces.

By selecting a specific dataset, it is possible to define matrices, and therefore forms, related to Analytical Workspaces.

When creating a form, the user can choose whether to manage CCH Tagetik dimensions data from the Financial Workspace, including custom dimensions, or analytical dimensions data from the chosen Analytical Workspace dataset.

The reports and data entry forms containing Analytical Workspace data are created, configured and managed in the same way as those containing Financial Workspace data. Some particular features are described below.

Form logics

Forms can be defined by accessing datasets, in both navigation and data entry modes, with a *transactional* logic, also called *Line by Line* logic, or with a *multi-dimensional* logic.

Element	Description
Transactional (line by line) logic	The dataset data are in rows only or in columns only. The counter element is placed on the other element, columns or rows, respectively. In the cells, the data shown or entered are filtered according to the filters set at the row or column level.
Multidimensional logic	The dataset data are organised into rows and columns. In the cells, the value of the data shown or entered by the user is filtered by the values set in the existing filters on the cell. Therefore, in this case, the matrix filters, tab filters and form filters also have an influence.

Virtual datasets

Line-by-line or multidimensional forms can be defined on the basis of virtual datasets.

A virtual dataset associates one or more physical datasets. In particular, it allows you to select only the desired fields from the original datasets and add formula fields whose operands are the fields of the original datasets.

On the dimensions tree, the formulas are identified with f_x.

In reporting, virtual datasets are used in read only mode, and it is not possible to change cell editability nor define conditional editability expressions in matrices based on virtual datasets.

Datasource

It is also possible to define Line by Line forms via a datasource. Datasources are used in read only mode for the sole purpose of reporting. It is not possible to change cell editability nor define conditional editability expressions in matrices based on datasources.

Data Transformation Package Data Processing

For every Analytical Workspace, a **Data Transformation** node is shown within the data processing tree. When a particular DTP is associated with a form, all the parameters defined for the Analytical Workspace in question are also shown in the data processing configuration and the user can set them (with all the available input modes). The value set in this way will be the value associated with that parameter when the DTP is run. If the Analytical Workspace is deleted, the data processing associated with the form will no longer function.

***Note**: a node not associated with any DTP is also present in the data processing tree. This node does not show parameters.*

For more details on DTP, see Data Transformation Package (DTP).

Particular features of multidimensional forms

How multidimensional matrices work

To create an analytical multidimensional matrix, the form must be opened in design mode (see "Open a form in Design mode" on page 39) and a multidimensional matrix that refers to the chosen Dataset must be added (see "Multidimensional matrices" on page 101). When a form with a multidimensional matrix related to a Analytical Workspace dataset is accessed in data entry mode, the matrix functionality is standard: the matrix retrieves from the dataset only data corresponding to the specified filters.

***Note**: datasets containing the "Show user data only" property cannot be associated with multidimensional matrices.*

Multidimensional matrices are only run by the in-memory engine and subject to the following conditions:

- They must contain numerical and/or text-based measures only.

***Note**: text-based only if the dataset has refresh data mode set to "Update the existing record".*

- They must not contain advanced selections (pruning at 0 + Lowest Level)

Matrices on datasets partitioned by entity

For matrices that refer to datasets partitioned by entity, it is possible to run the form, in the navigation or data entry phases, only filtering its data on a single element of the Entity dimension.

This requirement is not applicable for datasets that are not partitioned by entity.

If the entity is only identified by the partition and not an entity field, the table will not contain an Entity column. However, the information can in any case be retrieved from the details of the OWS_TABELLA_ PARTIZIONE partition.

Data in cells

In multidimensional matrices, the data of each cell corresponds to the value entered in the cell by the user, modified by the row and column filter values. In these matrices, the user can only explicitly edit data from custom dataset fields. The dimension field data are calculated on the basis of the filters in place on the cell.

Multidimensional matrices only manage numerical-type measures (custom fields) and text-based measures if the data has the update data mode set to "Update existing record". Text-based measures are not GDPR Relevant).

Default element

In CCH Tagetik, a *default element* is associated with all the analytical dimensions and custom dimensions of the Financial layer that are being used. In the data entry phase, in the presence of a multidimensional matrix, the system enters values in the dataset dimension fields not defined with the default element (if

one exists), on the condition that the relative dimension allows default values and the value is visible to the user.

The following dimensions do not allow default values:

- CTP entity
- Account
- Category
- Variation / Detail type
- CTP Custom dimension 2

Dimensions tree of the multidimensional matrix

The dimensions tree in the Matrix Wizard (see "Matrix Wizard - Window structure" on page 109) shows all the measures and dimensions used by the dataset, with their hierarchies, together with the following dimensions:

- Scenario and Period and Entity if a dataset that has been partitioned by entity has been used,
- the Period Length reporting dimension

The presence of the Scenario, Period and Entity dimensions is due to the partition logics, because they guide the determination of the partition from which the data are read and written in the navigation or data entry phase, respectively.

The Period Length dimension determines the time window from which to read the data to be aggregated. The following period lengths are supported for analytical multidimensional matrices:

- LUN_0: cumulated
- LUN_1 monthly
- LUN_2: bimonthly
- LUN_3: quarterly
- LUN_4: every four months
- LUN_5: every five months
- LUN_6: half-yearly
- LUN_12: annual
- LUN_QTD: quarter to date
- LUN_STD: semester_to_date
- LUN_YTD: Life_to_date

The default Period Length for analytical multidimensional matrices is LUN_1 monthly, because the data are saved on a monthly basis by default.

The **Calculate period values** property of the period length dimension is set to **Aggregate period values** by default and cannot be changed.

Particular features of dimension filters

If a dimension is specified using a hierarchical node rather than a lowest level element, the system defines the dimension using the reference lowest element. If the reference lowest element is not defined, or if the user does not have the rights to view that lowest level element, then the data cannot be saved.

Data entry on the reference lowest element is supported for the following dimensions:

- Account
- Entity
- Custom dimensions
- Analytical dimensions
- Category

When the node is associated with a lowest level element, the data entry uses that lowest level element to define the dataset field.
In the presence of two or more nodes referring to the same dimension, data cannot be saved because the system cannot determine which reference lowest element to use.

The only Period length supported in data entry mode is the monthly period length. Cells referring to different period lengths are not editable.

Cell editability

In the multidimensional matrix design phase, the user can configure the cells and define cell editability conditions on rows/columns, and the system calculates the cell value based on the intersection of these (see "Editability" on page 116). Every cell is subject to the same editability checks as the process.

The system will also make a cell non-editable under the following conditions:

- If one of the filters that the user has specified is a node (despite having an associated lowest reference element). If, in addition to the node, a lowest level element belonging to the node is also used, then the lowest level element will be editable, while for the node, the message "Node editability not supported" will appear in editability information.
- If any of the dimensions present in the dataset and in editability information are not specified in the matrix, then the message "Dimension missing: COD_DIMENSIONE" will appear.
- if the intersection to which the cell relates is composed of several records then the message "Cell has multiple values" will appear in editability information.

However, the editability set on a cell may not be effective when the matrix dataset contains one or more dimensions without a default element and there is no filter on the cell with reference to the specific dimension.

*In the cell properties panel, click **Editability info** to check the reason why a cell's editability is locked.*

For more information on editability checks, see "Checks on multidimensional matrices" on page 331.

Particular features of Line by Line forms with transactional matrices

Background

Line by Line forms allow you to perform data entry in a more flexible and customisable way. A form containing transactional matrices related to Analytical Workspace datasets has the particular feature of comprehensively showing all the records during the access phase in data entry mode: the retrieved records are filtered on the basis of the form filters (Form filters, Tab or matrix filters) and user rights. fields and editability settings are shown on the basis of their structure management.

How transactional matrices work

When a Line by Line form with a multidimensional matrix related to a Analytical Workspace dataset is accessed, the matrix functionality is standard for the user: the matrix retrieves from the dataset the fields corresponding to the filters on the fields present in the same matrix and, depending on the form type and user rights, writes, via the matrix, on the same fields.

The matrix structure shows the following particular features:

- The dataset fields can be set either on rows or on columns of the matrix; however, filters cannot be set on both rows and columns.
- The matrix contains the *Counter* element which, when the form is run, corresponds to a row or column that enumerates the records read by from the dataset (see "Details of the Counter element" on the next page).
- The matrix may contain the *Status* element which, when the form is run, makes it possible to delete the record on a row or column (see "Details of the Status element" on the next page).

To correctly manage the partitioning of a dataset's data, filters or parameters related to the Scenario, Period and Entity dimensions must be added to the reports and data entry forms, as per the following guidelines:

If...	Then...
the user wants to open, in the data entry phase from a process that can be submitted by Scenario/Period, a form containing a matrix associated with a partitioned dataset	the specific Scenario and Period elements, valid for the entire matrix, must be specified in the filters applied on opening. ***Note**: this requirement is not applicable for non-partitioned datasets.*
the user wants to open, in data entry or navigation mode, a form containing a matrix associated with a dataset partitioned by Entity	the Entity element must be specified in the filters applied on opening. ***Note**: this requirement is not applicable for datasets not partitioned by Entity.*

These conditions are also valid for datasets that do not contain the Scenario, Period or Entity dimension fields.

***Note**: the transactional matrix's dataset may include both analytical dimension fields and Financial Workspace fields.*

Details of the Counter element

The *Counter* element, which can be selected in the dimensions tree of the **Row/column filters** tab, is useful for data entry forms in which the number of records shown in the table can vary. When the form is run, it adds a column containing a numbered list of the rows displayed, to make it easier to read the form.

How the new rows are managed will depend on the settings of the selection attributes on the counter element, which can be accessed in the Matrix Wizard's properties table (see "Matrix filter properties table" on page 119).

Details of the Status element

The *Status* element, which can be selected in the dimensions tree of the **Row/column filters** tab, allows contributors to delete records from the dataset. When the form is run, for every dataset field it shows the Status column, containing a button to delete the record.

***Note**: if the status column is not present, then data can only be entered or edited.*

Creating Line by Line forms using transactional matrices

Transactional forms based on a Line by Line matrix can be created in the following ways:

- Automatically, with the **Data entry** command from the actions menu in the dataset definition. In this case, the form will contain a column for every field of the selected dataset.
- Manually, directly within the Excel add-in. In this case, it is possible to define, in the matrix, the dataset fields to be shown in the form.

Fields that are not managed will be ignored when the dataset is read. On saving, if any (form, tab or matrix) filter refers to fields that are not managed in the matrices, then they are set and filled out, if they can be taken from a form filter (form, tab, or matrix filter), or simply left empty.

For further details on the Analytical Workspace settings, see Finestra delle impostazioni del dataset.

Further information

In transactional forms that refer to an Analytical Workspace dataset, one or more comments can be added in the data entry phase, under the following conditions:

- This cell must be editable.
- The cell refers to a measure field of comments and attachments type.
- If the cell is part of a newly added row, at least one other cell (dimension or measure) of the new row must be populated.

For more details on inserting comments, see Finestra Commenti.

To view comments inserted on a cell during the navigation and data entry phase, the **Display automatically comments in reports data entry** and **Display automatically comments in navigation** options (*Home page - Administrator > Settings > forms tab > Comments section*) must be enabled. See Finestra Impostazioni.

***Note**: With the new function 0004073 it is possible to insert comments on both LBL and MultiDimensional matrices. As a result, edit/delete operations are only possible on comments inserted after submission and only if the function is enabled. However, these operations cannot be performed on comments inserted pre-submission. Moreover, editing a comment inserted during the submission phase means that the comment becomes pre-submitted when the data item is rejected, and it will no longer be editable at the time in which the user decides to resubmit the data.*

In transactional forms, it is possible to configure, via the validation window, a check for the presence of attachments on a comment cell (HasAttachment). The operator to be selected within the validation window is 'Contains Attachments' see Validation Window.

Note: If this type of validation is configured, the option Automatically display comments in reports data entry *(Administrator Home Page > Settings > form tab > Comments section) is enabled by default, regardless of whether it is enabled in the* Finestra Impostazioni.

Particular cases regarding editability

In general, the editability of form cells is determined by the settings on the matrix row/column editability attributes (see "Editability" on page 116). However, cell editability in Line by Line transactional forms depends on the following combination of conditions:

- The cell must be defined as editable in the matrix.
- The editability property of the dataset fields must be enabled (see Editabilità).
- An Account-type dimension field subject to the submission process must be present.
- One or more measure-type fields subject to the submission process must be present.

For more information on editability checks, see "Checks on multidimensional matrices" on page 331.

If a non-partitioned dataset has a field defined as key in the **Field Properties** panel, on saving the form on which cells have been edited, the system checks that the newly added keys are unique and will potentially display an error message. The message indicates the field and the keys that infringe the non-duplication requirement.

Particular cases regarding editability for the processes with dimensions by step

Depending on the dimensions used in the dataset or partition logics, a matrix may or may not be subject to editability checks that verify coherence between accounts and measures and the process definition and their potential submission.

The system therefore performs checks on accounts when the dataset contains, in the fields themselves or due to partition logic, the *Scenario, Period, Organization dimension and Account* or *Scenario, Period, Organization dimension, Account and Category*, *Scenario, Period, Organization dimension and Account* or *Scenario, Period, Organization dimension, Account and Category* dimensions.

***Note**: for running checks, the Category dimension is only needed if the process submission is by Account/Category.*

The system performs the checks related to a measure when, together with the Scenario, Period and Entity dimensions, the dataset contains the Measure dimension in question in the fields themselves, or due to partition logic. If the dataset contains data which are subject to submission in the process, the dimension fields of these data will be non-editable on the rows already defined.

Example

In the presence of a dataset in which the Scenario, Period and Entity dimension fields and an analytical dimension are editable, since the Scenario/Period/Entity combination identifies the dimensions subject to submission in the process, those fields are not editable on rows already defined. However, it is possible to edit other dimension elements on rows that have already be saved, the analytical dimension in this example, since the value of those dimensions does not impact the data's submission status.

The system also made all the cells of a row that has already been saved non-editable if at least one of the following conditions is met:

- The original scenario/period doesn't belong to the process;
- The scenario/period belongs to the process but is locked;
- The scenario/period belongs to the process but is not active.
- The period does not belong to the scenario.
- The entity does not belong the process.
- The matrix shown on the form refers to a dataset containing a key measure field (PK).

- When building the matrix, two different filters, two time-dependent nodes or one time-dependent node and another dimension belonging to the same category as the time-dependent node are used.
- When the datasets contain the Account and are subject to submission, and the Entity-Account pair is not valid for the process.
- When datasets contain the Account and the Category, are subject to submission, and are opened on a process with Account-type submission, the Category does not fall within the scope of the Categories filter specified in the process.
- When the datasets contain the Account are subject to submission and are opened on a process with Account/Category-type submission, and the Entity/Account/Category combination is not valid for the process.
- When the datasets contain the Account and are subject to submission, and the Scenario Period Entity Account combination refers to a submitted step.
- When the datasets contain the Account, are subject to submission, are opened on a process with Account/Category type submission, and the Scenario/Period/Entity/Account/Category combination refers to a submitted step.
- When, due to limitations on the dataset, the user views "Database not visible to the user" in Editability Info.
- When the user is limited to read only mode on one or more dimensions, "Row not editable due to limited visibility rights (read only)" will be shown in Editability Info. When the restriction concerns entities, Analytical dimensions and custom dimensions, the user cannot enter data for certain entities of specific organisation dimension. The message then shows the entity dimension and the codes on which edit/save operations cannot be performed. When the user is limited on several dimensions then all the dimensions and codes that violate the read-only restriction are specified in Editability Info. (See Users rights: Restrictions on dimensions and User Rights: Other Restrictions).
- When the validity range specified on a lowest level element of a custom dimension or an analytical dimension is not consistent with the scenario/period pair present on the row related to the same lowest level element in the matrix.

The system makes the cells containing measures non-editable if at least one of the following conditions is met:

- The measure does not belong to the process.
- The measure refers to a step submitted for the current entity.

***Note**: if at least one of the two conditions is met on a row that has already been saved, all of the row's dimension fields become non-editable.*

To ensure that submitted data are coherent, when the row is either fully or partially non-editable for submission reasons, the cell related to the Status column is non-editable to prevent the user from deleting the specific row.

To find out why a cell is not editable, in the cell properties panel > editability section > Editability Info button.

For more information on editability checks, see "Checks on multidimensional matrices" on the facing page.

If coherence checks are disabled, it is not possible to verify submission status. When coherence checks are disabled, submission checks are not carried out either.

Editability on Demand

This option is present in Line by Line matrices when the **Editability on Demand** option is enabled. With the On Demand editability mode, cell values are not editable and are not calculated when the form is launched but later, at the user's request. With this function enabled (see "Options in the Edit amounts tab" on page 131), the user can request to edit a cell that is at that time not editable, by clicking the cell in the work desktop. The button appears at the top for by-column matrices and to the left of the row for by-row matrices.

The button calculates the editability of all the cells in the selected row or column. It is therefore possible to request the editability of adjacent or non-adjacent blocks of cells at the same time. In by-row matrices, the button appears in the last row of the selected cells but, when an adjacent block is selected, the last selected cell is the same as the first selected cell. Similarly, in by-column matrices, the button appears in the last column of the selected cells.

Updating transactional forms

In some cases, the form updates automatically after being saved. In particular, it is updated in the following cases:

- When a predefined number of rows is set to be opened in data entry and the update is necessary to add the additional rows (see Row/Column Filter Properties Grid).
- When the **Editable only on new rows** option is active on some dataset fields and the update is necessary in order for the saved records to be defined as non-editable.

Checks on multidimensional matrices

Checks on opening

The attempt to open a Data Entry form (whether Line by Line or Multidimensional) on a Dataset partitioned by Scenario/Period/Entity is blocked and the user receives a warning message.
At the moment, on the Multidimensional Process, the partition policy is understood to be by Scenario/Period only.

Editability checks

In a dataset design step, the user specifies an editability value for each field: *Editable*, *Editable only on new rows* or *Read only*.

Checks on saving

There are 4 types of editability checks on opening:

- consistency checks
- submission checks
- checks for locked scenario/periods
- checks for limitations in read only mode for user and dataset

Therefore, on saving data, when the form refers to a dataset or partition logics containing all the dimensions needed to identify the process submission status (Scenario, Period and Entity or Scenario, Period or Organization Dimension for multidimensional processes), the system performs consistency checks first of all. In other words, it verifies that the data entered are coherent with the definition of the process.

In particular, the type of control varies based on the following conditions:

Condition	Check
The dataset fields allow for the identification of the submission status of the entities for the various steps of the process.	The system performs a check on the data entered for the various entities in the step for which the form was run.
The dataset or partition logic contains the account dimension in the dataset fields.	The system verifies that, by Scenario, Period and Entity (or organization dimension) the Account (or Account/Category if the process is submitted by Account/Category) belongs to the process, the data have not been submitted and there are no limitations for the user.
The dataset or partition logic contains one or several Measures in the dataset fields.	The system verifies that, for Scenario, Period, Organization Dimension and Measures, the data are editable, belong to the process and have not been submitted and that there are no limitations for the user.

It also verifies that the following conditions are met:

- In the event that the Dataset contains the Category dimension, and it is accessed via a process with submission based on the Account, the Category, if specified, must belong to the categories filter configured in the process.
- An Account/Organization Dimension pair (or Account/Category/Organization Dimension triplet) belongs to a process if, in the definition for the entity's Process Data Model, the account has been associated with a step of the process.

When the outcome of a check is negative, the form will not be saved and the user will receive an error message.

In case of Data Entry on category nodes containing several sub-nodes, the system fails to determine the appropriate lowest element node for data entry when the underlying sub-nodes refer to different lowest elements structures.
As a result, it returns the error message 'Category - Conflicting Value' and prevents the data entry operation. Data entry on a single node works correctly.

It is not possible to check submission status if consistency checks have been disabled. If this happens, submission checks are not performed either.

For datasets that do not contain the dimensions needed to identify the process submission status, the system does not perform any check and saves the form, regardless of the form structure and the process submission status.

For further details on check configuration, see the Control section in Operazioni.

Checks on datasets with the Account dimension

Introduction

On opening in data entry mode and saving form data, the system verifies that the data saved in dataset fields via the form are coherent with the dimension elements of the process. Details of the checks available in the presence of the Account dimension are provided below.

Dimension coherence checks

In the presence of Scenario, Period, Entity and Account (or Account and Category) dimensions or non-mandatory dimensions in the dataset or in the partition logics, when the form is accessed via a process with submission based on the Account (or Account/Category), these checks verify that all the dataset's dimensions are present in the form.

> ***Note**: a data entry form can be subject to submission checks for a certain process and not for others, depending on whether or not the Category dimension or the process submission type are present, based on Account or Account/Category.*

Process coherence checks

In the presence of Scenario, Period, Entity and Account (or Account/Category) dimensions in the dataset or partition logics, these checks verify that the saved data refer to Scenarios/Periods, Entities and Accounts (or Accounts/Categories) that belong to the process.

> ***Note**: an Account/Category pair (or Account/Category/Entity triplet) belongs to a process if, in the definition for the entity's process data model, the account has been associated with a step.*

When the dataset contains the Category dimension, and the form is accessed via a process with submission based on the Account, any specified Category must belong to the categories filter specified in the process.

Coherence checks on process submission

In the presence of Scenario, Period, Entity and Account (or Account/Category) dimensions in the dataset or partition logics, these checks verify that the saved data refer to Scenarios/Periods, Entities and Accounts (or Accounts/Categories) that have not yet been submitted.

Checks on datasets with the Measure dimension

Introduction

On opening in data entry mode and saving form data, the system verifies that the data saved on dataset fields via the form are coherent with the dimension elements of the process. Details of the checks available in the presence of the Measure dimension are provided below.

Dimension coherence checks

Where Scenario, Period, Entity and Measure dimensions are present in the form, the system checks that those dimensions correspond to those of the dataset fields.

For datasets that do not contain, as dataset fields or in partition logics, all the dimensions needed to identify the process submission status, no check is carried out and data can still be saved, regardless of the form structure and the process submission status.

Process coherence checks

For datasets containing at least one Measure and Scenario, Period or Entity, the saving of data must refer to scenarios, periods, entities and measures belonging to the process.

A Measure/Entity pair belongs to a process if, in the definition of the entity's process data model, the measure has been associated with a step.

Coherence checks on process submission

In the presence of Scenario, Period, Entity and Measure dimensions in the dataset or partition logics, these checks verify that the saved data refer to specific scenarios, periods, entities and measures that have not yet been submitted.

Queries on datasets

Introduction

Data related to a physical dataset can only be accessed in read only mode by entering SQL queries in the form's cells.

For more details on queries, see "Query" on page 251.

Dataset access syntax

The syntax for accessing a dataset partition is as follows: *SELECT <fields> FROM <table> WHERE <conditions>*.

The function to retrieves the <table> value is: *DATASETPARTITION(codWorkspace, codDataset, [codRaccolta], [codScenario], [codPeriodo], [codAzienda])*.

For example, with the syntax *SELECT * FROM ${DATASETPARTITION("001", "000015")}*, elements *001* and*000015* are a workspace code and a dataset code, respectively.

***IMPORTANT**: data read via queries are visible to all users. The user defining the query must therefore ensure that it does not retrieve data that violates the user rights set in the application.*

Execution of queries

Queries set on the form's cells are run when the form is opened. If the *DATASETPARTITION* function does not find any partition corresponding to the query parameters, the query cannot be run and the system displays an error message.

To prevent errors of this type, it is possible to specify a table or partition name to be used for the query as the last parameter of the *DATASETPARTITION* syntax.

For more details on the function, see "The Datasetpartition function " on page 199.

On-the-fly conversion

On-the-fly conversion is a functionality that can also be applied to multidimensional matrices, and specifically to matrices that refer to analytical datasets that have currency as a dimension.

To be able to use this function, the **Make the Forms editable for currency conversion on the fly** option must be selected (Home page - Administrator > Setup & Admin > Settings > Forms tab > Data Entry Section).Navigation panel > *Settings > Form tab > Data Entry section*) > View Settings window.

If, in the data entry phase, a change is applied to the matrix on the fly via the combination of the currency and measure filters, all of the matrix cells on which a single currency type is defined will be editable. In the presence of several filters with different currencies, the cells will not be editable.

For more details on changes in matrices with Analytical Workspace data, see "Matrix filter properties table" on page 119.

For more details on setting up on-the-fly conversion, see "Currency conversion algorithm" on page 28.

IMPORTANT: On-the-fly conversion is not supported on a transactional matrix referencing datasets.

IMPORTANT: On-the-fly conversion is not supported on a multidimensional matrix when an analytical dataset with non-incremental saving is used along with the Currency dimension defined as a form parameter.

IMPORTANT: On-the-fly conversion does **not** support the following FX rate types: **Previous Final FX** rate and **Historical FX** rate

Navigation from Excel

Contents

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Navigation toolbar

Excel > CCH Tagetik > Navigation > select form > set run parameters

Introduction

The navigation toolbar is shown when a report is run in navigation mode.

Buttons on the Navigation toolbar

Element	Description
Tools	
>	**Refresh**: runs the report again, potentially updating the data extracted from the database. ***Note**: this refresh operation could delete any edits to the run report made locally, especially if the report contains a dynamic matrix. To create areas that will not be updated, assign a name starting with "TGKNOTREFRESH" with the Excel AREAS function. The edits remain in place locally and are not visible to other users who are not in possession of that particular file.*
>	Only available for Report lists. Runs and updates the data for some of the Report list's form only.
>	**Refresh comments**: updates the comments linked to the report cells. See Finestra Commenti.
	Exports the report in a non-navigable Excel file, i.e. typical navigation functions (drill down, view cell properties, refresh, etc.) are not available. *To create a navigable Excel file, use the standard **Save as** function.*
>	**Show**: shows cells containing hidden comments, or comments associated with the lowest level elements if the selected filter is the node of an aggregation structure. In this case, only the row related to the node is displayed in the report.
>	**Details**: displays the comment in a dedicated window.
>	**Hide**: hides the displayed comments.

Element	Description
	**Warning**: Shows any non-blocking warnings found at runtime. Only visible if there are warnings to be shown.
	**Hide rows/columns if 0**: shows/hides any rows or columns given the property “Hide” in the design phase.
	Shows/hides any formula defined in the selected cell.
	Only available for Report lists. Sets, or allows you to change, the security password for the current Report list.
	Runs an Excel file containing data on the form’s performance. See "Reporting statistics" on page 309.
Show	
	**Cell properties**: in the left-hand panel, opens the **Cell properties** tab, showing information on the cell and filters.
	**Predefined filters**: in the left-hand panel, opens the **Filters** tab which shows the filters applied to the form parameters and allows the user to edit them.
	**Cell**: runs the drill-down on the selected cell. See "The drill down function" on page 340.
	**Column**: runs the drill-down on the selected column. See "The drill down function" on page 340.
	**Row: runs the drill-down on the selected row. See "The drill down function" on page 340.**
	Shows the value of the cell with the level of detail with which it had been inserted in CCH Tagetik from the original external data source. See "Drill-through queries" on page 348.
	**Data Processing**: runs the data processing associated with the form. See "Data processing run from reporting" on page 244.
	**Pivot table**: shows the contents of the selected matrix as a pivot table. See "Pivot tables" on page 349.
Edit	
	Generates a report in Excel format listing all the intercompany entries declared on the accounts present in the report. See Finestra Edit intercompany.

Navigation mode: filters

Data displayed in navigation mode

Navigation mode allows you to read data from the CCH Tagetik database. These data are shown on the basis of filters set on the form during the design step and the user's visibility rights over the CCH Tagetik dimensions.

Defining parameters

If parameters are inserted when a form is designed, when that form is run in navigation mode the system shows a window in which those parameters can be defined. This allows you to filter the subset of data to be shown in the form.

Changing the dimensions on which to filter

While the form is open, the selected dimensions can be changed, without having to close and rerun the same form, using the **Predefined filters** function (see "Predefined filters" on page 162).

Working on forms for navigation

Open a form for navigation

Excel > CCH® Tagetik > Navigation

1. In the left-hand panel, select the desired form.
2. Double click: the form will open, or if parameters were set when the form was designed, the **Parameters** window will open.

For details on how to work in the **Parameters** window, see "Define form parameters" below.

***Note**: if the user tries to reopen a form, the system will repropose the parameters inserted when it was last accessed.*

Define form parameters

This procedure allows you to specify the subset of data to be shown in the form.

1. Open the form ("Open a form for navigation" above).
2. For each desired parameter, so as follows;

If you want...	Then...
select a single element	type the code of the desired dimension or click and select it from the list that appears.
select several items of a dimension	1. Click ... 2. Hold **CTRL** and select the desired items from the tree. 3. Click : the selected elements will be added to the list of elements to be used. 4. Click **OK** to confirm.

3. Click **OK** to confirm: the form will open with the desired data.

Edit the value of the parameters of an open form

1. Open the form ("Open a form for navigation" above).
2. Set the desired run parameters.
3. While still in **Navigation**, click : the **Predefined filters** tab will open in the left-hand panel with the parameters currently set.
4. Edit the value of the desired parameters.
5. Click in the left-hand panel or in **Navigation** to see the form with the updated data.

Edit the value of parameters using a slicer

Excel > CCH® Tagetik > Navigation

1. Open the form, setting the desired run parameters ("Open a form for navigation" above).
2. Click in the cell with the slicer.
3. Select the desired dimension.
4. Click in the left-hand panel or in **Navigation** to see the form with the updated data.

Enable drill down

1. Open the form in **Design** mode (see "Open a form in Design mode" on page 39).
2. Click : the **Form options** window will open.
3. Select the **Other options** tab and do as follows:

If you want to enable drill down...	Then...
on the entire form,	select **Enable Drill down**.
on the dimensions used,	select **Allow drill down only on used dimensions**.

4. Click **OK**.

Disable drill down

1. Open the form in **Design** mode (see "Open a form in Design mode" on page 39).
2. Click : the **Form options** window will open.
3. Select the **Other options** tab and do as follows:

If you want to disable drill down...	Then...
on the entire form,	deselect **Enable Drill down**.
on the unused dimensions,	select **Allow drill down only on used dimensions**.

4. Click **OK**.

Run the drill down

1. Open the form in navigation mode ("Open a form for navigation" on the previous page).
2. Do as follows:

If you want to run the drill down for...	Then...
a cell	double click the desired cell. Alternatively, select the cell, click and .
a row	double click the header of the desired row. Alternatively, select the cell, click and .
a column	double click the header of the desired column. Alternatively, select the column and click and .

3. Set the desired dimensions.
4. Click .

Create a pivot table

1. Open the form ("Open a form for navigation" on the previous page).
2. Position the cursor over the matrix whose data you want to see.
3. Click : the pivot table will open in a new spreadsheet.
4. If useful, reorder the data using various filters.

Parameters window

Excel > CCH® Tagetik > Navigation > select form >

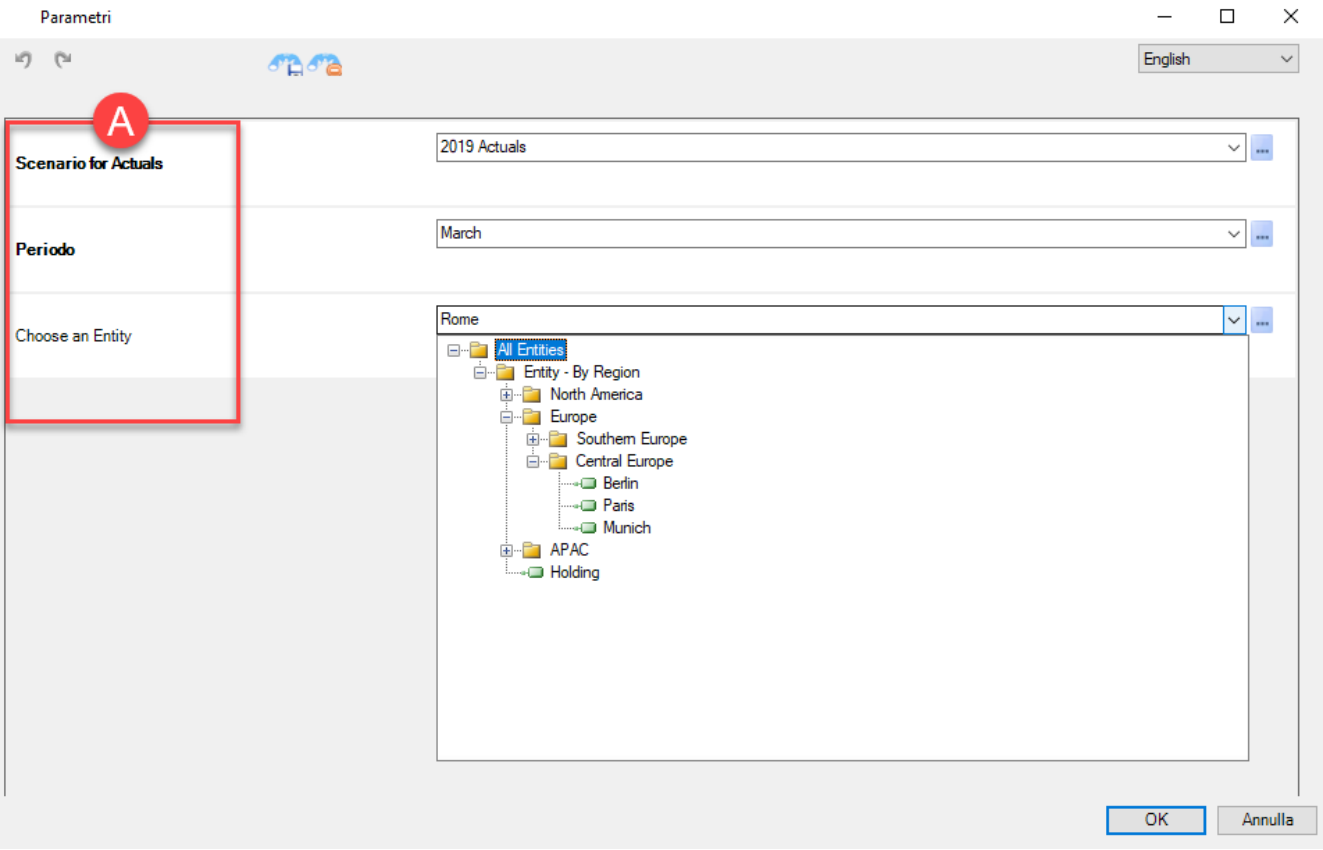
***IMPORTANT**: this window is only displayed if the form asks the user to define parameters when it is opened.*

Purpose of window

This window shows and allows you to define the parameters set when the form was designed.

***Note**: parameters shown in bold are mandatory, those shown in red indicate that the selected dimension is not valid.*

What it looks like



Element	Description
A	Parameters inserted in the form structure selected when it was designed.

Buttons in the window

Element	Description
	Opens a menu to select the elements of the dimension for which to filter. : node : node element
	Opens a window to select the desired elements from the dimensions tree of the parameter on which to filter. Useful for multi-selection. See "Window for selecting the values of a parameter" below.
	Undoes the last action performed.
	Redoes the last undone action.
	Saves the Point of View (see "Saving the Point of View (POV)" on page 32).
	Deletes the Point of View (see "Saving the Point of View (POV)" on page 32).

Window for selecting the values of a parameter

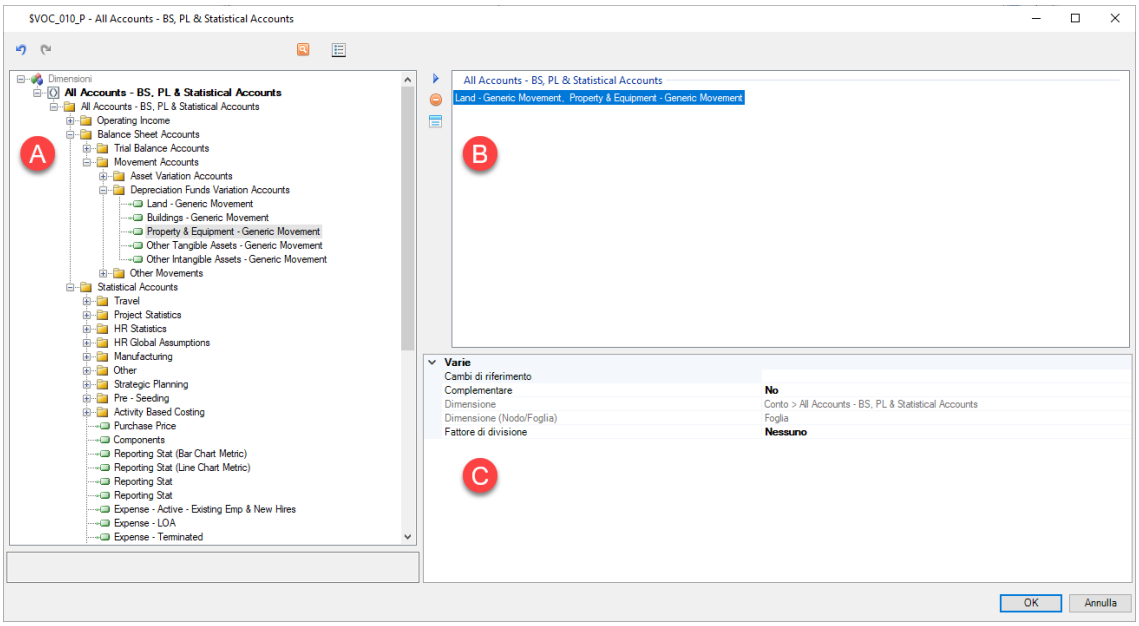
Excel > CCH® Tagetik > Data Entry > select form > (desired parameter) ...

Purpose of window

This window allows you to select the dimensions on which to filter.

***Note**: several elements of the same dimension can be selected in the tree; several can be selected at once by holding the CTRL button when selecting them.*

What it looks like



Element	Description
A	Dimensions tree
B	List of elements of the selected dimensions
C	Details of the elements of the selected dimensions

Buttons in the window

Element	Description
	Undoes the last action performed.
	Redoes the last undone action.
	Allows you to search for a specific dimension.
	Change the display mode (**Code and description**, **Code**, **Description**).
	Adds the selected dimensions to the list of those to be used.
	Removes all dimensions from the list of dimensions on which to filter.
	(Only with several OR dimensions selected) Opens a window to choose which dimension to remove from the OR row.

The drill down function

Description

The drill-down function allows the user to view the components that make up an aggregated figure.

Cell, **Row** or **Column** drill downs can be run. The functionality can be deactivated on individual forms.

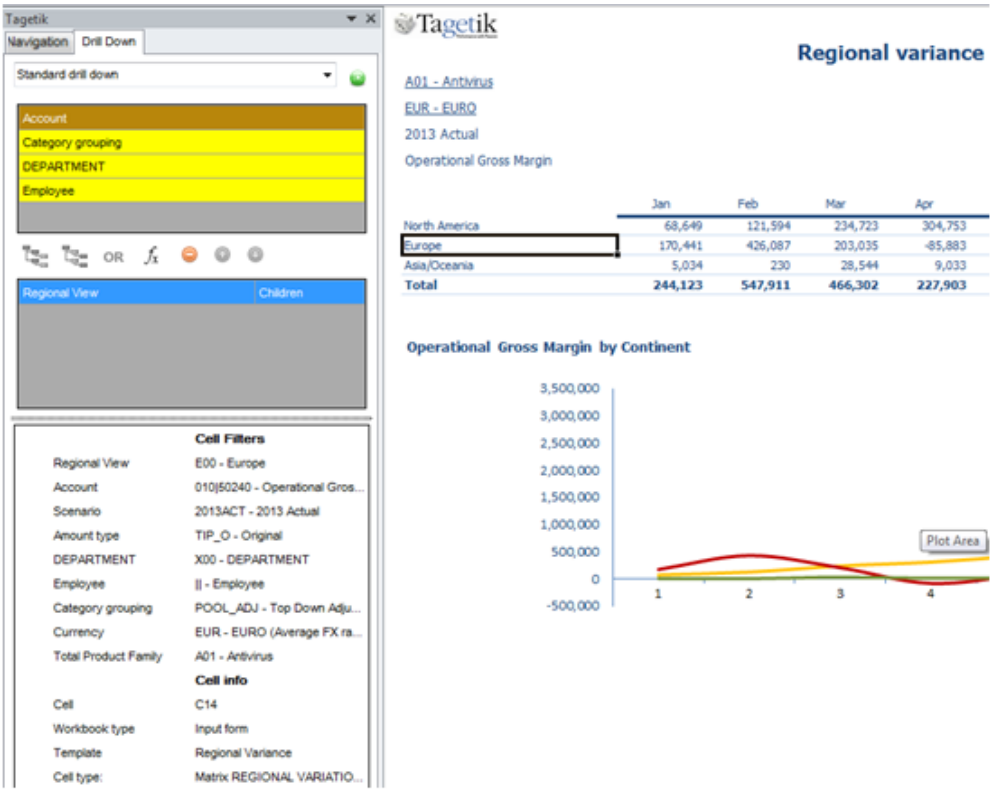
Example

The value 1000 is reported in a cell of European revenue. The cell drill down shows that the value is made up of:

Account	Amount
Revenues - Italy	80
Revenues - France	150
Revenues - Spain	70
Revenues - UK	330
Revenues - Germany	370
Revenues - Europe	**1000**

Example of row/column drill down

In this example, the Excel file shows a row of sales revenues for Europe and a **Regional View** hierarchy that groups entities by geographical area. With drill down on the row, the user can see how the individual European areas (**South Europe** and **Central Europe**) contribute to the generation of the European monthly revenues.



Those figures are shown in a new Excel sheet, which can be saved. In this example, the source cell properties are reported in the **Drill Down** column at the bottom left within **Cell info**. If the starting form had a header area, then this would also be reported in the results sheet.

		Cell Filters				
	Regional View	E00 - Europe				
	Account	010|50240 - Operational Gross Margin				
	Scenario	2013ACT - 2013 Actual				
	Amount type	TIP_O - Original				
	DEPARTMENT	X00 - DEPARTMENT				
	Employee	|| - Employee				
	Category grouping	POOL_ADJ - Top Down Adjustments				
	Currency	EUR - EURO (Average FX rate)				
	Total Product Family	A01 - Antivirus				
		Cell info				
	Cell	C14				
	Workbook type	Output form				
	Template	Regional Variance				
	Cell type:	Matrix REGIONAL VARIATION - Row Headers ($Entity(HIERA				
	Styles	R_formula, Amounts_Board				
		Jan	Feb	Mar	Apr	May
	South Europe	-14,584	53,708	97,926	127,258	178,217
	Central Europe	185,024	372,379	105,109	-213,140	-95,740
	Europe	**170,441**	**426,087**	**203,035**	**-85,883**	**82,477**

Run drill down

Introduction

The various drill down modes and their specific features are described below.

Standard drill down

This mode allows you to select the aggregation structures and dimensions to be used to view the composition of data. It generally makes sense to run drill down on the aggregation structures on which basis the data was aggregated (the dimensions currently used in the selected cell are shown in yellow in the list).

Once the user has selected the aggregation structure on the basis of which the drill down is to be run, they can specify whether it should be run on all child elements or on the lowest level elements only.

Example 1

The following hierarchy groups entities by geographic area and the **Europe** node is selected as a filter for the entity dimension. The hierarchy appears as below:

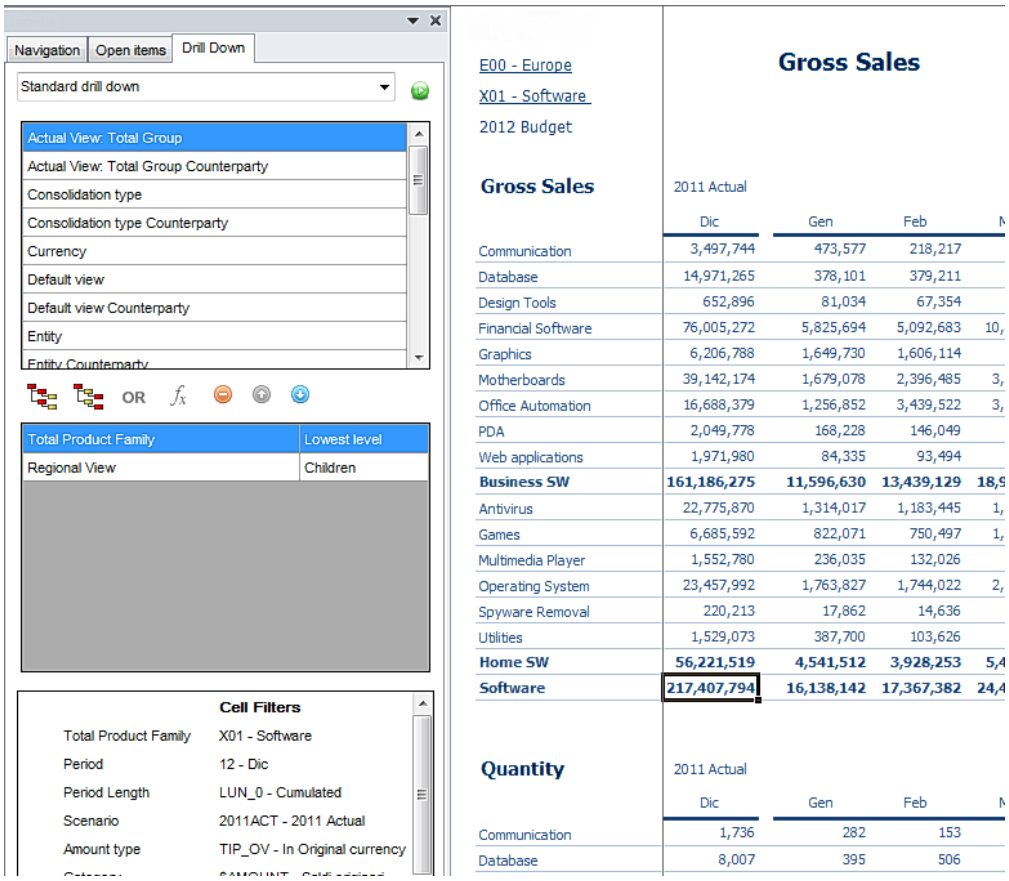
- Europe
  - South Europe
    - Italy
    - Spain
    - Greece
  - North Europe
    - UK
    - Germany
    - France
- North America
  - United States
    - ...
    - ...

- Canada
  - ...
  - ...

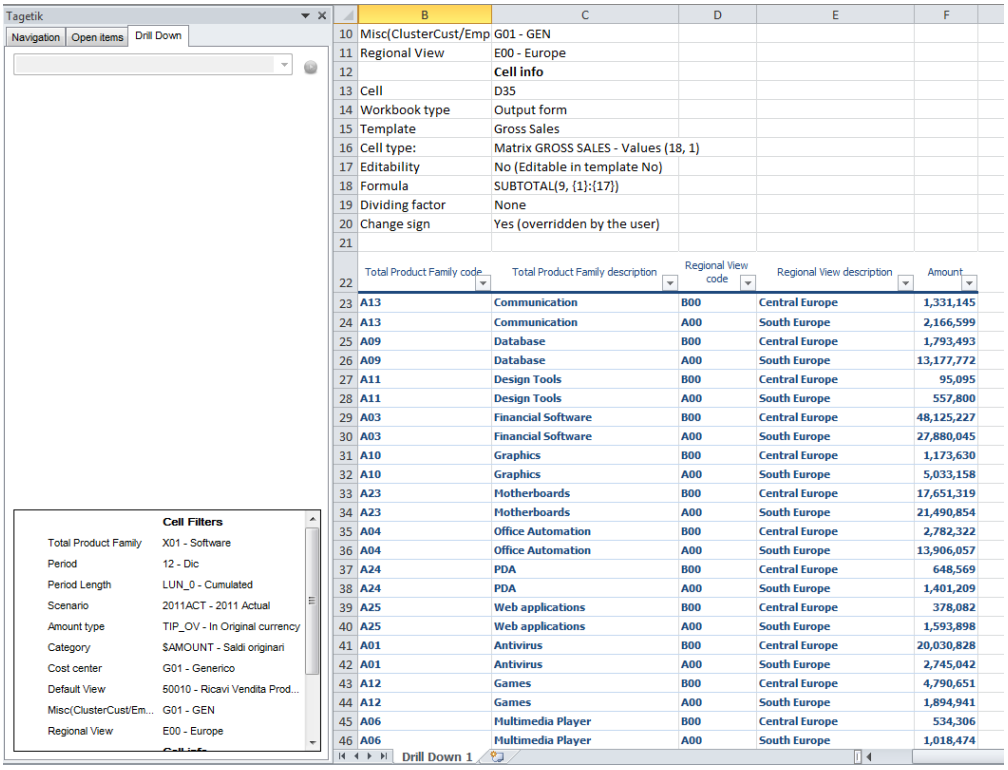
If the user runs the drill down on the child elements, the system shows the overall data for the **South Europe** and **North Europe** nodes. If the user runs the drill down on the lowest level elements, the system shows the data for **Italy**, **Spain**, **Greece**, **UK**, **France** and **Germany**.

Example 2

In this example, the Excel file shows the account for sales revenue produced by Europe, a **Regional View** hierarchy that groups entities by geographic area and a **Product types** hierarchy that groups products by type. If the user runs the drill down on the child elements of the **Regional View** hierarchy, the system displays the data distributed among the countries of the Europe area. If the user also runs the drill down on the lowest level elements of the **Product Types** hierarchy, the system displays the data for the individual products of a given type.

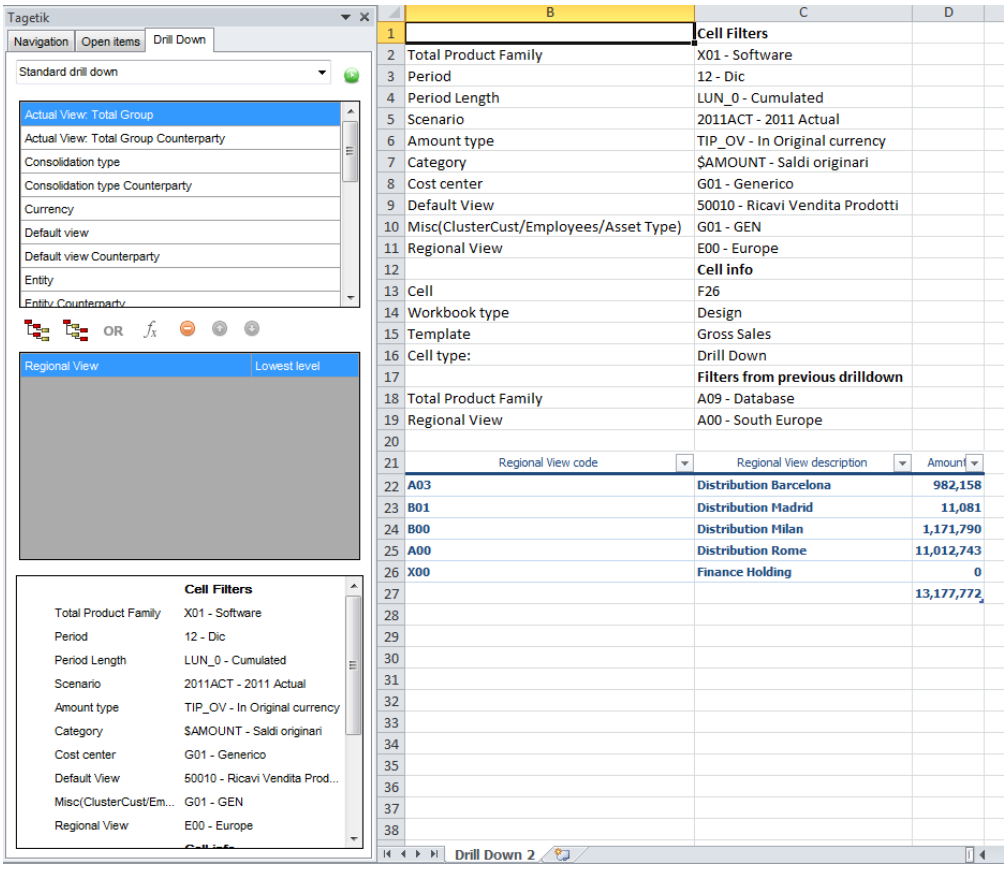


Those figures are shown in this second Excel sheet, which can be saved.



The system exports, in two different columns, the code and description of the elements of the tables on which it ran the drill down. Thus, it also allows you to reorder the rows based on the description. In this specific example, it is possible to continue the drill down to see the distributed data for the countries of south Europe and central Europe, for example.

***Note**: if several drill downs have been performed, it is possible to navigate between "Drill down" sheets.*



"OR" drill down

This mode allows you to see details of the contributions to the value corresponding to OR row/column filters, if present.

Formula drill down

This mode is activated using the button f_x in the drill down column and is only available on cells related to FST items that can be related or formula-type FST items. Shows the individual accounts that make up the FST item.

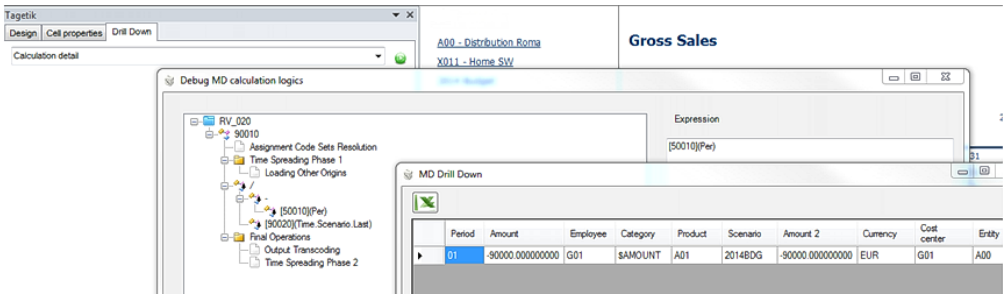
Calculation details

This mode allows you to obtain any basic accounts calculation logic that has fed the account defined in the selected cell and shows the following:

- if the adopted calculation logic type is **Sum**, the individual addenda already multiplied by the appropriate coefficient
- if the adopted calculation logic type is **Multi Dim. Script**, the syntax tree of the calculated account

The following conditions are necessary to see such information:

- The accounts calculation logic type must be **Sum**. This allows you to see the values of the individual addenda multiplied by the appropriate coefficient. The system only supports the simple case, in which only the account, and no other dimensions, is filtered on input. Moreover, only one calculation logic must be defined for the account on which drill down is being run. The output of the drill down always applies the filters of the starting form, except for the filter on the account, and ignores any filters originating from overrides.
- The accounts calculation logic type is **Multi Dim. Script**. In this case, the system shows the syntax tree of the calculated account in a new window.



- The cell only contains one account/category. That account must not have been generated by several accounts with the **Sum** or **Multi Dim. Script** calculation logic types with the same account as the outcome. In fact, in such cases the system would not know which account to show during the drill down.

To see all the available details, including the amounts of the various components, double click on the individual addenda and lowest level elements of the syntax tree shown. To see the calculation logic that generated the account present in the cell in question, simply click 🔍 ***Cell properties****, see "Design toolbar" on page 34.*

Lowest level elements: Dimensions

In the presence of aggregated data, this mode shows the figures for the individual elements of all dimensions for which the cell's current value has been aggregated.

Example

Applying the **Lowest level elements: Dimensions** mode to the previous example allows us to see the figure for each individual European country and for each individual software product. Standard drill down should therefore be run on the lowest level elements of the **Budget View** hierarchy and on the lowest level elements of the **Product Types** hierarchy.

System query *[query name]*

This mode allows you to run an SQL query defined by an administrator user and then run the drill down on information that has been stored in other tables of the CCH Tagetik Data Model. See "Query" on page 251.

This mode is not supported in AIH.

Headers

This mode allows you to run the drill down on the origin of changes made to a value. This allows you to revise the final value of a number stored in CCH Tagetik. This mode is not supported in AIH

Adjustment detail:

This mode provides more details than are shown with **Headers** adjustments, such as the user that made the adjustment and the date and origin of the adjustment. It therefore makes the final value fully traceable.

This mode is not supported in AIH.

Drill-down on Consolidation Scenarios

When a user performs a drill-down on a consolidation scenario, the displayed data may sometimes not match the original value of the selected cell.

In such cases, it is advisable to check the category filter applied to the scenario in the data model, as it may be the source of the discrepancy.

To perform this check, go to the **Exception > Category Filter** section on the consolidation scenarios page.

From this page, you can define which categories are allowed as input for consolidated data.

If some of the categories expected in the drill-down are missing from the filter, it is recommended to add them manually.

Once the filter has been updated, simply rerun the report and the drill-down to verify whether the issue has been resolved.

Unsupported features

Cell drill-down on placeholders containing formulas inserted through the matrix wizard is enabled both in AIH and in the Financial Workspace. However, there are some unsupported scenarios—for example, cases where the formula refers to numerical values and/or elements outside the matrix.

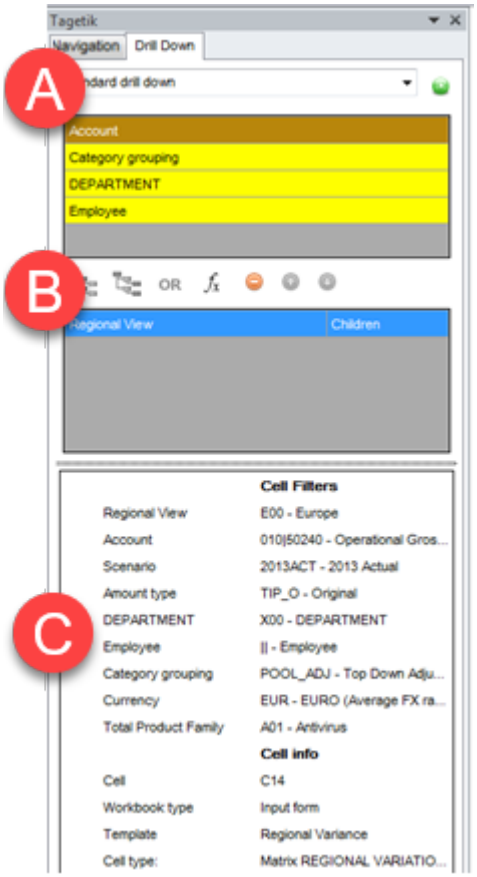
Drill Down tab

Excel > CCH® Tagetik > Navigation > open form > Drill Down > Cell/Column/Row

Purpose of the tab

This window allows the user to view the components that make up an aggregated figure.

What it looks like



Element	Description
A	Drill down selection menu
B	Only for standard drill down. Dimensions management pane.
C	List of filters applied to the selected cell.

Tab elements

Element	Description
Drill down selection menu	Allows you to select the way in the which drill down is run on the data. • **Details**: only on cell. Shows details of the adjustments. • **Lowest level elements: Dimensions**: only on cell. Shows the highest level of detail on all dimensions involved. • **Standard drill down**: allows you to select the aggregation structures used to aggregate the data and dimensions to be used to view the composition of that data. • **Lowest level: rows**: only on cell. Shows the highest level of detail on all dimensions involved, distributing the values by user/insertion date.(*not supported in AIH*) • **Headers**: only on cell. Shows the journal headers.(*not supported in AIH*)
Dimensions management pane	Allows you to manage the selection of dimensions on which to run the drill down. The aggregation structures and dimensions currently used in the selected cell are shown in yellow.
Filters applied to the cell	Shows the properties of the selected starting cell. See "Cell properties panel" on page 352.

Actions on dimensions

The actions that can be run from the dimensions management pane are described below:

Element	Description
	Runs the drill down on child elements.
	Runs the drill down on lowest level elements.
OR	In the presence of OR row/column filters, shows details of the contributions of the value corresponding to that filter. For example, for a value deriving from the **Sales revenues** and **Other revenues** accounts defined as a OR row filter, the values of each of the two individual accounts can be seen.
	Only available on cells related to FST items that can be related or formula-type FST items. Shows the individual accounts that make up the FST item.
	Removes the selected dimension.
	Moves the selected dimension up/down.

Drill-through queries

Background

Data originating from external sources can be loaded to CCH Tagetik using ETL (see Extraction, Transformation & Loading). The ETL saves the data in a staging table before importing them into CCH Tagetik. This staging table may contain information on the data at a level of detail that is not present once the data has been loaded to CCH Tagetik.

Function utility

The drill-through query allows you to retrieve the data that makes up an amount that has been imported into CCH Tagetik from the staging table managed via ETL.

Example

An amount of € 978 is present on the Insurance account in CCH Tagetik. This amount has been imported from an ERP system containing the details that make up the amount in question. Setting the drill-through query therefore allows you to retrieve the following data:

Tagetik	ERP System
Insurance = € 978	• Insurance - Auto = 100 • Insurance - Accident = € 159 • Insurance - Fire = € 230 • Insurance - Goods = € 100 • Insurance - Other = € 398 **Total insurance = € 978**

Using the drill-through query

Start using drill-through queries

Excel > CCH® Tagetik > Navigation > select form >

1. Click : the **Drill-through** tab containing a list of drill-through queries defined in the form will open in the left-hand panel.

2. Select the desired cell.
3. Do the following;

If you want to run...	Then...
a specific query	1. From the list, select the query to be run. 2. Click : the system will run the query according to the filters of the selected cell.
all available queries	click : the system will run the query according to the filters of the selected cell.

Enable the automatic running of drill-through queries

Excel > CCH® Tagetik > Navigation > select form >

1. Click : the **Drill-through** tab containing a list of drill-through queries defined in the form will open in the left-hand panel.
2. **Live mode**: the system will automatically run drill-through queries every time a cell is selected.

Pivot tables

Function utility

Pivot tables allow you to simplify and more effectively analyse the data present in a form. The functionality can be used in **Navigation** mode or **Data Entry** mode.

To see a preview of the pivot table in Design mode, click .

Characteristics of the tables

The pivot table's input data are data from the dimensions present on the row, column and tab filters. The table is subdivided into the following areas:

- rows
- columns
- data (table content)
- dimensions not used on the rows and columns.

The dimensions for which a single filter is present are excluded from the table.

The pivot table is positioned in a new work sheet, showing the **List of fields** to allow you to further rearrange the data, adapting it to your own needs using the **Pivot table tools** toolbar.

When opened, the table shows the dimensions present on the matrix rows in the rows area and the dimensions present on the matrix columns in the columns area. Any tab filters are aggregated.

The display layout can be modified by dragging the dimensions present on rows, columns, data areas and filter areas. This provides several types of views of the data. For example, rows and columns can be inverted or data can be aggregated differently, but the values of the cells in the data area cannot be edited.

This functionality is available for users who have function 0003715 enabled in their user role.

Navigation from Web

Contents

This section covers the following topics:

Navigating web forms 350
The web forms page (Navigation) 350
Cell properties panel 352

Navigating web forms

User | <application database>

Forms tile > click grouping of forms > (next to form)

Expand nodes (row drill down)

1. Select a row and click , a number of rows equal to the number of the node's children, or of related accounts, will be shown. The selected row shows the subtotal of the details below.
2. If there are several nodes on the selected row, select the header cell of an individual node to run the drill down on that individual node only.
3. To run the drill down on all nodes, select the cell containing the value to be expanded.

***Note**: column drill down is not available for web forms. Drill down is not available in the presence of OR filters.*

The web forms page (Navigation)

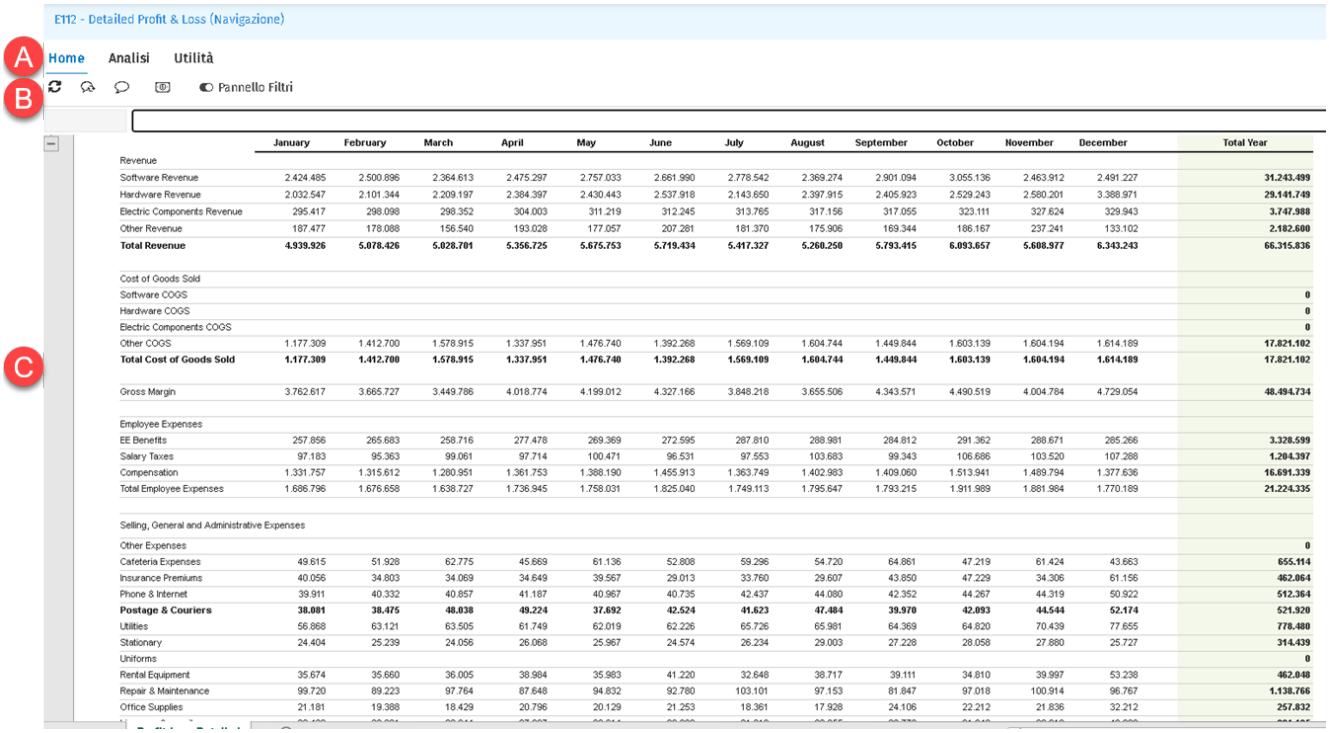
Forms tile > click grouping of forms > (next to form)

Forms tile > Reporting > Forms grouping (next to form)

Introduction

A web form in Navigation mode allows the user to consult the form via the navigation interface described below.

What it looks like



Element	Description
A	Title bar
B	Window management buttons
C	Action buttons
D	Work desktop

Title bar information

- Code and description of the web form
- Web form run mode (Navigation, Data entry)

Page management buttons

Element	Description
	Hides the toolbar.
	Shows the hidden toolbar
	Creates a bookmark for the current view (see "Bookmarks in web forms" on page 12). If the icon is blue, the current form has already been bookmarked.
	Saves the display preferences set in navigation mode (visible columns, column width, minimised areas, etc.). These preferences are automatically applied whenever the form is subsequently opened in navigation mode.
	Closes the form.

Action buttons

Button	Description
	Restore original. Runs the web form again. *Use when one or more filters have been changed.*
.XLS	Allows you to export the web form in Excel format. It is not possible to navigate within the exported file.
Filter Pane	Shows the filters in place on the form under the toolbar. The value of those filters can be edited and the form can be run again via the **Reload** button.
Show rows and columns	Shows/hides the rows/columns on which the **Hide** property was activated when the form was designed.
	Opens the **Cell properties** panel to the right (see "Cell properties panel" below).
	Refreshes the form and updates comments on cells.

Cell properties panel

Forms tile > click grouping of forms > (next to form) > select cell

Application database > Processes tile > select process > select step tab > select entity > (Workflow) right click task form > Run on web > select cell >

Application database > Processes tile > select process > select Detail summary tab > select entity > Actions column > form > Select the form > select cell >

Purpose of the panel

This panel shows all the available information on the selected cell of a web form and its content.

***IMPORTANT**: if the user selects a different cell while viewing the panel, the system automatically updates the panel with the information on the selected cell.*

Panel sections

Element	Description
Cell filters	Shows details of the filters applied to the cell.
Cell information	Shows additional information on the cell.

Cell information

Element	Description
Cell	Cell position in Excel
Workbook type	Form type (output/data entry)
Template	Name of the template containing the cell
Cell type	Name of the matrix containing the cell, matrix row and column number
Editability	Cell editability status. If the status is *non-editable*, it provides information on the reason.
Formula	(Only if the cell contains a formula) it shows the formula syntax.
Dividing factor	Weight applied to the account related to the cell (days, months). Only if applied.
Sign change	Cell data sign status (sign change active/inactive).
Styles	Style applied to the cell

CCH® Tagetik Intelligent Analytics with Microsoft Excel 365 & Ask AI

Introduction

CCH® Tagetik Intelligent Analytics allows analysis and reporting on data CCH® Tagetik working directly in Excel, with an intuitive user experience.

CCH® Tagetik Intelligent Analytics is available for both Excel desktop app and Excel for web.

The user can exploit the full functionality and calculation potential of Microsoft Excel; CCH® Tagetik Intelligent Analytics allows users to work directly on Excel files, or to enter data into Excel:

- by connecting directly to CCH Tagetik data
- from other external sources

***Note**: The file can be saved locally on the user's machine, or on shared folders. By saving the file to SharePoint or OneDrive, you can take advantage of all the native Office 365 features, such as automatic versioning, commenting, and real-time collaboration—allowing multiple users to view, comment on, and edit the same Excel sheet simultaneously.*

***IMPORTANT**: Some features in Excel for web may differ from desktop Excel, or may not be present at all. For differences in the native functionality of Microsoft Excel, please refer to the official Microsoft documentation.*

***IMPORTANT**: to display the data of the CCH Tagetik environments you want to connect to, you need to have an Intelligent Analytics license for each environment*

Prerequisites and Deployment

For information on the prerequisites and deployment of CCH® Tagetik Intelligent Analytics with Microsoft Excel 365, please refer to the technical documentation available on the ***CCH® Tagetik Community***, see CCH® Tagetik Community

Export

The Export feature creates a copy of the Excel workbook, showing only the values in the cells.

Any formulas are not exported: only the calculated results will be visible in the generated document.

Export starts after clicking the Export button.

Once the new file has been created, the download starts automatically.

Connections to CCH® Tagetik

Introduction

The Connection Management panel allows you to specify which CCH® Tagetik environments to connect to in order to retrieve data for analysis.

***Note**: It is possible to connect to several CCH® Tagetik environments at the same time.*

***IMPORTANT:** a configuration file in .json format, specific to each environment, is required to define the connection. If not available, it can be requested from Tagetik via the support portal.*

Each CCH® Tagetik environment can have multiple applications. In the connection panel you also specify, for each environment, the applications you want to connect to.

Applications specified in an Excel sheet are saved with the Excel file, while environments, specified via the .JSON files mentioned above, are saved in the user's Excel configuration.

> ***Note**: If you work in Excel desktop and Excel web, or if you use Excel from another PC, you must specify the environments again.*

Working with Connections to CCH® Tagetik

How to create a Connection to a CCH® Tagetik Environment

Excel > CCH® Tagetik Analytics > Connections

This procedure allows you to set up a new connection to a CCH® Tagetik environment.

> ***Note**: the configuration will still be valid for subsequent Excel desktop or web sessions on the same PC.*

1. In the add-in ribbon, click on the **Connections** button to open the Connections management side panel.
2. Click the **New Connection** button to open the .json file with the configuration for the CCH® Tagetik environment you want to connect to.
3. Click **Add** to create the new configuration.

***IMPORTANT:** the .json file, if not available, can be requested from Tagetik via the support portal.*

Configuring an Application

Excel > CCH® Tagetik Analytics > Connections

This procedure allows a new CCH® Tagetik Application to be set up using an existing Environment to which the user has authenticated.

> ***Note**: the configuration of an application is saved on the current Excel file.*

1. In the add-in ribbon, click on the **Connections** button to open the Connections management side panel.
2. If you are not already authenticated, **LOG IN** first to the Environment where the Application you want to connect to is located. If the environment is not present, it must be configured (see How to create a Connection to a CCH® Tagetik Environment)
3. Click **Add Application** and select the Environment and Application Database to be used.
4. In the **Name** field, specify a mnemonic name for the application, which will be used in the definition of CCH® Tagetik Functions and CCH® Tagetik Pivots.
5. Click **Add** to create the new configuration.

***IMPORTANT:** At least one application must be defined in an Excel sheet in order to use the add-in.*

Working with CCH® Tagetik functions

The add-in provides simplified **wizards** in the side panel for the definition of **TGK.DISPLAY** and **TGK.VAL functions**, which, by means of graphical tools, allow you to create **formulas** to return a value or dimensional elements, respectively.

Create a formula to return a value

Excel > CCH Local > Formula

1. Select an empty cell on the Excel sheet, click on **Formula** in the Excel toolbar and choose the **TGK.VAL** option. The corresponding wizard will be shown in the side panel.

2. Select the desired application from the list of applications. Only those for which a login has been made will be selectable.
3. Select the **Reporting View**
4. Select the **Measure**
5. **Optional:** if necessary, **dimensional filters** can be added to the formula. Click **Add filter** and select the desired dimension from the list or a reference to a cell containing a filter previously defined with TGK.FILTER, TGK.OR or TGK.DISPLAY. A new section for the selected dimension will be added to the panel, allowing filter criteria to be specified.
6. Click **CONFIRM** to generate the formula in the selected cell or **CANCEL** to cancel the operation.

Create a formula to return dimensional elements

Excel > CCH® Tagetik Analytics > Formula

1. Select an empty cell on the Excel sheet, click on **Formula** in the Excel toolbar and choose the **TGK.DISPLAY** option. The corresponding wizard will be shown in the side panel.
2. Select the desired application from the list of applications. Only those for which a login has been made will be selectable.
3. Select the dimension.
4. Select dimensional elements or a cell containing a filter previously defined with TGK.FILTER, TGK.OR or TGK.DISPLAY.
5. **Optional:** if necessary, enter a text to be displayed in the cell containing the formula.
6. Click **UPDATE** to generate the formula in the selected cell or **EXPAND BY ROWS**, **EXPAND BY COLUMNS** to generate multiple formulas across rows or columns—one for each selected item. Click **CANCEL** to cancel operations.

Edit a formula

Excel > CCH® Tagetik Analytics > Formula

1. Select on the Excel sheet a cell containing the TGK.DISPLAY or TGK.VAL **formula** to be modified
2. Click the **formula** button to access the panel for making changes

CCH® Tagetik Functions

Introduction

CCH® Tagetik Intelligent Analytics for Microsoft Excel provides a set of **functions** that can be used in Excel formulas to retrieve data and element lists from CCH® Tagetik.

These functions allow the user to build Excel reports, retrieving data from CCH Tagetik, with a completely free and intuitive approach, without the need for any additional training.

Excel **formulas** can be written to combine several functions and reference the values of other cells, even from other Excel sheets, to build reports and simulations from whatever source the data came from.

CCH® Tagetik Intelligent Analytics functions can be accessed via the Excel formula bar or through its side panelCCH® Tagetik.

In the add-in CCH® Tagetik Intelligent Analytics, the following functions are available for Microsoft Excel:

- TGK.FILTER
- TGK.OR
- TGK.DISPLAY
- TGK.VAL

TGK.FILTER function

The **TGK.FILTER** function allows you to select lowest level elements or nodes from the CCH Tagetik data model and use them in other formulas.

Parameters

- **Dimension_code**: string specifying the Tagetik dimension code.
- **Hierarchy_code**: (optional) string specifying the code of a dimension hierarchy. If omitted, the following parameters indicate the codes of lowest level elements; otherwise they indicate the codes of node elements from the specified hierarchy.
- **Selected_item1...n**: list of strings representing the code of a lowest level element or node. The list must consist of at least one element.

Returned value

A value representing a filter consisting of only the specified lowest level elements or nodes. The filter is represented by an Excel Data Type (official Excel documentation: https://support.microsoft.com/en-us/office/excel-data-types-stocks-and-geography-61a33056-9935-484f-8ac8-f1a89e210877) It can be used as a filter parameter in the functions TGK.VAL, TGK.DISPLAY or TGK.OR.

Examples

```
TGK.FILTER("DIM",, "01", "02")
```

This example returns the lowest level elements '01' and '02' of the analytical dimension with code 'DIM'.

```
 TGK.FILTER("DIM", "H1", "01", "02")
```

This example returns the nodes '01' and '02' of the 'H1' hierarchy of the analytical dimension with code 'DIM'.

TGK.OR function

The **TGK.OR** function allows you to combine multiple filters within the same analytical dimension using the logical OR operator.

Parameters

filter1...n: list of filters to be combined with the OR operator. Two or more filters can be combined.

Returned value

A value representing the combination of the specified filters. The filter is represented by an Excel Data Type (official Excel documentation: https://support.microsoft.com/en-us/office/excel-data-types-stocks-and-geography-61a33056-9935-484f-8ac8-f1a89e210877). It can be used as a filter parameter in the TGK.VAL or TGK.DISPLAY functions.

Example

```
TGK.OR(TGK.FILTER("DIM",, "01"), TGK.FILTER("DIM", "H1", "02"))
```

This example combines in OR the lowest level element filter '01' and the node filter '02' of the hierarchy 'H1' both of the analytical dimension with code 'DIM'.

TGK.DISPLAY function

The **TGK.DISPLAY** function allows you to retrieve the element descriptions defined via the TGK.FILTER and TGK.OR formulas

Parameters

- **application_name**: string specifying the application name.
- **filter**: a filter obtained using the TGK.FILTER or TGK.OR function.
- **text**: (optional) string specifying the text used to overwrite the original description.

Returned value

A value containing the elements selected through the applied filter, enriched with descriptions. The filter is represented by an Excel Data Type (link to official Excal documentation: https://support.microsoft.com/en-us/office/excel-data-types-stocks-and-geography-61a33056-9935-484f-8ac8-f1a89e210877). It can be used as a filter parameter in the TGK.VAL or TGK.OR functions.

Example

```
TGK.DISPLAY("App", TGK.FILTER("DIM",, "01", "02"))
```

This example shows the descriptions of elements '01' and '02' of the analytical dimension with code 'DIM'.

TGK.VAL function

The **TGK.VAL** function allows you to retrieve values from the specified filters.

Parameters

- **application_name**: string specifying the application name.
- **reporting_view**: string specifying the reporting view.
- **measure**: string specifying the measure to be aggregated.
- **filter1...n**: (optional) filters to be combined with the AND operator.

Returned value

The numerical value of the filtered data intersection for the specified measure. The function returns an empty cell if the filter does not match any data.

Examples

```
TGK.VAL("App", "ReportingView", "Measure", TGK.FILTER("DIM",, "01", "02"))
```

This example retrieves the aggregate value from the reporting view ‘ReportingView’ for elements ‘01’ and ‘02’ of the analytical dimension coded ‘DIM’, related to the measure ‘Measure’.

```
TGK.VAL("App", "ReportingView", "Measure", TGK.OR(TGK.FILTER("DIM1",, "01", "02"), TGK.FILTER("DIM1", "H1", "03")), TGK.FILTER("DIM2", "H2", "04"))
```

This example retrieves the aggregate value from the reporting view ‘ReportingView’ for the measure 'Measure' resulting from the application of the following filters based on the logic below:

```
TGK.FILTER("DIM1",, "01", "02") OR TGK.FILTER("DIM1", "H1", "03") ] AND TGK.FILTER("DIM2", "H2", "04")
```

Pivot CCH® Tagetik

Introduction

CCH® Tagetik Pivot **tables** are dynamic tables created using CCH® Tagetik Intelligent Analytics.

This type of table allows you to easily analyse multidimensional data from a **Reporting View**, by organising them in rows and columns, similar to Excel Pivots.

Unlike standard Excel Pivots, CCH® TagetikTagetik Pivots are **integrated** with the company's data model and support advanced functionalities. Specifically, through the Pivot tool, data can be extracted from the **Intelligent Analytics** platform by specifying the measures to be analysed, aggregating them by dimension and displaying them with the required layout.

CCH® Tagetik Pivots consist of a dedicated **Excel formula (TGK.PIVOT)** that uses the **SPILL** mechanism to automatically return the calculation results to the worksheet and supports the main Excel operations:

- **copy and paste** pivot, which creates a new independent pivot;
- **cut & paste** or **drag & drop**, which allow the pivot to be moved while retaining its metadata;
- **duplication of a sheet** within the workbook.

In all these cases, the pivots continue to function correctly once repositioned or duplicated.

Pivot Wizard

The Pivot wizard is the main **interface** for creating and editing CCH® Tagetik Pivot tables and consists of three main sections:

- *Source*: for selecting the data source.
- *Filters*: for defining dimensional filters to be applied globally.
- *Values, Rows & Columns*:
  - *Values*: for selecting the measures to be analyzed.
  - *Selected Values*: List of Selected Values.
  - *Dimensions*: list of available dimensions with their hierarchies and levels.
  - *Rows / Columns*: Containers for arranging hierarchy levels in rows or columns.

Filters definition

It is possible to configure, for each Pivot, one or more dimensional filters that will be applied during data extraction.

The filter can be defined for each dimension by selecting individual elements directly from the panel displaying the entire hierarchical structure of the dimension. Alternatively, the filter can be set by referencing a cell, by selecting a cell that contains an *Excel Data Type*.

Property

It is possible to display and edit the properties of **hierarchy levels** and **measures,** placed in **rows/columns** and in **Selected Values** respectively.

Change sign

This property is **only available for measures in Selected Values** and allows you to define how the sign of the displayed values is to be handled.

Available options

- **None**
  The value is displayed without any sign change.
- **Invert**
  The sign of the value is reversed.
- **Based on account**
  The sign of the value is based on the account nature.

Default value

- For **financial reporting views**: **Based on account**
- For **analytical reporting views**: **None**

Display options

This property defines how elements are displayed in the report.

Available options

- **Code**
- **Description**
- **Code and Description**

Default value:

- **Description**

Order

The **Sorting** property allows you to define the elements sorting criteria; it is only available for elements in the **Columns** and **Rows** containers

Sorting by

- **Code**
  The elements are sorted in alphanumerical order.
- **Amounts**
  Elements are sorted based on the sum of the associated amounts.

Sort type

- **Ascending**
- **Descending**

***NOTE**: The available properties and their default values may vary depending on the type of reporting view and the selected item.*

Management and display of hierarchy levels

When a level of the dimensional hierarchy is placed in *Rows* or *Columns* in the Pivot, the display in the Excel table is based on a multilevel expansion logic, similar but not identical to that of standard Excel Pivots.

- Each selected level of the hierarchy is represented as a separate header level in the table.
- Layers are nested in the same order as they were added to the container (the first element corresponds to the outermost layer).
- The resulting table shows the Cartesian product of the contributing elements for each selected level, generating an expanded tree structure.

Measures management and display

Pivots support the selection of one or more measures from the *Reporting View* and the behaviour of the table varies according to the number of measures selected.

When only one measure is selected:

- If no dimensions are configured in a row or column, the description will be given in the row or column header where there are dimensions;
- If dimensions have been configured in both rows and columns, the description of the measure will appear in the first cell in the top left-hand corner.

When more than one measure is selected:

- A *placeholder* element called *[Selected Values]* is inserted;
- This element may be placed in rows or columns;
- Each selected measure will be represented as a separate header in the table.

Automatic formatting

Two pivot formatting modes are available:

From the Settings menu, the chosen styles are applied to all pivots in the workbook, the first time the sheet or individual pivot is updated.

From the pivot wizard, in the Layout section, by enabling Customize Style.

In this case, styles are applied upon refresh only to the current pivot, overriding those defined in the global settings

From the Settings menu, it is also possible to remove the formatting chosen for all pivots from the wizard using the reset button.

Only the custom styles defined in Excel Cell Styles can be used and assigned to the following pivot areas:

- Column header titles
- Column headers
- Row header titles
- Row headers

- Values
- Default, which includes all areas not covered by a specific style, including the corner area between titles and row and column headers

Error and conflict management

When creating or updating a Pivot, the system checks for cells whose contents could be overwritten by the Pivot. If the area in which the Pivot is generated **contains non-empty cells**, the formula cannot expand and the **#SPILL!** error is displayed in the top left cell of the pivot area.

***IMPORTANT**:In the event of an error, a warning is shown allowing you to either continue (where possible) or cancel.*
In particular, it is not possible to continue in the event of a collision with other Pivots in Excel or Tagetik; whereas, in the case of a collision with other types of content, it is up to the user to choose whether to continue or not.

Working on CCH® Tagetik Pivot

Create a Pivot

This is the procedure to create a new CCH® Tagetik Pivot in the Excel sheet.

Prerequisites

- Logging in to a Tagetik application.
- Have selected an empty Excel cell from which to create the Pivot.

Procedure

Excel > CCH Local > 

Pivot

1. In the Ribbon of the add-in, click on the **Pivot** button. The Pivot side panel will open automatically.
2. In the Source section, select the desired **application** and **reporting view**.
3. **Optional**: if required, dimensional filters can be added to the pivot. Click on the **Add filter** button and select the desired dimension. A new section for the selected dimension will be added to the panel, allowing filter criteria to be specified.
4. Select one or more measures from the **Values** panel by clicking or dragging directly onto the **Selected Values** panel. Added measures can be ordered or deleted via context menu or drag and drop.When multiple measures are selected, a dummy element - Selected values - is added to the Rows and can later be repositioned on the Columns.
5. **Optional**: If required, one or more **hierarchy levels** can be selected from the dimensions panel by clicking or dragging in the Rows or Columns sections according to the desired layout. Selected hierarchy levels and the dummy element Selected Values can be removed or moved between Rows and Columns.

Display or edit a Pivot

This procedure allows you to view or change the Pivot settings.

Prerequisites

- The Pivot must have been added to the Excel sheet;
- The selected cell must belong to the area of the Pivot that is to be edited.

Procedure

Excel > CCH Local >  *Pivot*

1. In the Ribbon of the add-in, click on the **Pivot** button. The Pivot side panel will open automatically.
2. All Pivot parameters are loaded based on the previously saved selections.

Delete a Pivot

This procedure describes how to delete a Tagetik Pivot Table from the Excel sheet.

Prerequisites

- The Pivot must have been added to the Excel sheet;
- The selected cell must belong to the area of the Pivot that is to be deleted.

Procedure

Excel > CCH Local >  *Pivot*

1. In the Ribbon of the add-in, click on the Pivot button; The Pivot side panel will open automatically;
2. Click on the **delete** button in the top right-hand corner; the Pivot will be removed from the sheet, and the cell area will be cleared.

Copy a Pivot from Dashboard

This procedure allows a Tagetik Pivot to be created from a chart in the Intelligent Analytics dashboard.

Prerequisites

- You must have a chart already configured on the Intelligent Analytics dashboard page;
- You must be logged in to the Tagetik application within the Excel add-in;
- You must have selected an empty cell from which you wish to create the Pivot.

Procedure

Excel > CCH Local >  *Pivot*

1. Open the Intelligent Analytics Dashboard containing the chart;
2. Click on the menu of the chart to be copied and select Copy Dashboard;
3. In the Ribbon of the add-in, click on the Pivot button; The Pivot side panel will open automatically;
4. Click on the chart import button in the top right-hand corner; the new Tagetik Pivot Table will be created in the sheet.

Create a new Pivot via Ask AI

This procedure allows you to create a new Tagetik Pivot in the Excel sheet from a conversation with the AskAI chatbot.

Prerequisites

- Log in to a Tagetik application;
- Select an empty cell from which you wish to create the Pivot.

Procedure

Excel > CCH Local > 

Pivot

1. Open the **Pivot wizard**: in the Ribbon of the add-in, click on the **AskAI** button. The Pivot side panel will open automatically.
2. Select the application from which you want to query AskAI.
3. Submit the request to AskAI in natural language in the relevant section, and send the message. If the message is processed successfully, the new Tagetik Pivot Table will created in the sheet, and the Pivot View and Edit panel will open at the same time.

> ***IMPORTANT**: if Ask AI can't process the request, if there are ambiguities in identifying the data, or if technical errors occur, a message will appear in the chat from the bot containing a description of the error.*

Distribution Tools

The Tagetik platform offers advanced features for report automation and distribution, with tools designed to optimize the generation and sharing of content.

Report list

The Report List allows you to aggregate and distribute forms in Excel format, containing financial and non-financial data extracted from the application database. These forms can be exported, printed, or sent via email, even to users who do not have the Tagetik Excel add-in. The Report List also supports the inclusion of ad hoc forms, blank sheets, and external Excel files, ensuring flexibility in the composition and dissemination of content (vedi Report list).

Report bursting

La funzionalità di Report Bursting consente la produzione massiva di report, anche attraverso schedulazione, a partire da documenti Collaborative Office o Form. È possibile definire la struttura del report, le logiche di popolamento con dati specifici per ciascun destinatario, e le modalità di generazione (see Report Bursting)

This tool enables the generation and delivery of customized reports to specific user groups.

The Report Bursting functionality allows for mass production of reports, including scheduling options, based on Collaborative Office documents or Forms. You can define the report structure, the logic for populating it with data specific to each recipient, and the generation methods (see Report Bursting)

Report list

Utility of the function

A Report list allows you to create a set of forms in Excel format that can be exported, printed and sent by email. These forms contain relevant financial data extracted from the CCH Tagetik application database and can be shared within the company with users who do not have the Excel add-in installed. A Report list can also be inserted into ad hoc forms, empty sheets and Excel sheets not generated by CCH Tagetik.

Types of report list

There are two types of Report list:

- Ad hoc: allows you to run a set of forms quickly. It is created on-the-fly and cannot be saved for future use.
- Shared: once set, the settings are saved in a special form that allows you and other users to call up and rerun the same Report list subsequently. The Report list is run simultaneous to the form containing it. Some of Report list run parameters can be set to run when the form containing the Report list is run. This makes the Report list configurable. Forms containing shared Report lists may have a single template and do not support tab filters.

Creating a Report list

Report lists are created via the **Action List Wizard** window. This window can be opened from the **CCH® Tagetik** toolbar or the **Design** toolbar. It has the following distinctive features:

- Ad hoc and shared Report lists can be created from the **CCH® Tagetik** toolbar.
- From the **Design** toolbar, a shared report list can be created and saved on the open form or, alternatively, the settings of a Report list that is already present can be changed. In fact, there can be just one Report list in a form.

***Note**: a Report list's Office colour palate must be consistent with the one used in the forms.*

When the Report list is being created, it is also possible to customise the initially assigned default settings for each individual form.

Publication and saving

The system asks you to set the publication method and, if the **Disk or portal publication** method is chosen, to specify the type of file to be generated (Excel or PDF). If the user decides to generate separate files, by default the system will name the file with the code of the form being exported, i.e. **${Report.code}**. If there is just one file, by default the system will name the file with the Report list code, i.e. **${ReportList.code}**.

By default, the system saves the various files in the Tagetik subfolder in the user's Documents folder, with the syntax **${getEnv(<USERPROFILE>)}\Documents\Tagetik\My Report List\${report.path}**. The Tagetik folder is organised into subfolders corresponding to the form grouping structure.

The following functions are available to allow the publication path to be set:

- **getEnv(<NOME_VARIABILE_AMBIENTE>)** which returns the value of a client environment variable.

  *For example, **${getEnv("USERPROFILE")}\Documents** returns the current user's Documents folder.*

- **report.path** which returns the path of the report within the hierarchy. The nodes are indicated by the code and separated by '\'.

The system creates a single file or one file for each of the Report list's forms, depending on the settings in use, and saves it in the specified folder. Those files are visible to everyone.

If the user has the rights to load documents to SharePoint, the generated files can be published in shared folders in a SharePoint portal using the syntax **\\server\share**.

Role settings

Functionalities related to Report lists are only available to users who, in the role with which they are accessing the application database, have the following functions enabled:

- **0002761 - Reporting .NET - Report List - Report List**. If enabled in read/write mode, this allows the user to insert/edit/delete Report lists. If enabled in read only mode, this allows the user to access the wizard in view only mode on existing Report lists.
- **0002762 - Reporting .NET - Report List - Output to Printer**. This function allows you to print the output of a Report list.
- **0002763 - Reporting .NET - Report List - Output to mail**. This function allows you to send the output of a Report list via email.
- **0002764 - Reporting .NET - Report List - Output to system file**. This function allows you to save the output of a Report list on the disk.
- **0002765 - Reporting .NET - Report List - Store Report list in Reports**. This function allows you to save a Report list on a report to be used later.
- **0003692 - Client. NET - Design - Report list - Password protection**. This function allows you to set a security password.
- **0004158 - Reporting - Generate a Form from the Form Navigation tree and save on a default End Point** This function allows you to add the Report List tile to the home page.

Report Lists from the web

A dedicated tile for managing report lists can be added to the CCH Tagetik web interface (User Home page). Excel and PDF files required by report lists created by the Excel add-in can also be generated from this page. Depending on how the Report List has been configured from the Excel add-in, the system will generate a single Excel file or a single PDF with as many sheets as there are forms included in the Report List, or the same number of individual Excel or PDF files.

Managing Report lists from Excel

Create a Report list

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

1. Select the desired options.

   ***Note**: the Print and Disk or portal publication options are alternatives.*

2. If **Print** has been selected, set the printer to be used. If **Disk or portal publication** has been selected, set the publication to be generated, and the file name and path.
3. If useful, set the recipients of the emails (see "Indicate the email recipients." below).
4. Click **Next**.
5. Select the desired form or node of forms and click .

   ***Note**: each form can be inserted into the Report list several times.*

6. Click **Next**.
7. Defining and managing parameters for selected forms.
8. To override tab filters, select **Override tab filters**, click , override the filters as desired and click **OK**: **Ignore** will appear next to the overridden parameters.
9. Click **Next**.
10. To override the individual form's default settings, select the desired form and the desired settings, and customise them.
11. Click **Next**.
12. If **Allow to specify output sheets sorting** was selected in the previous window, a window will open to specify how the output sheets will be sorted. Select the desired sheet and drag it to the desired position, or use the buttons.
13. Click **Next**.
14. Set the Report list's run and save options and click **End**.
15. If **Schedule the run** was selected in the previous window, the **Schedule Wizard** window will open. Set the parameters as desired and click **Next**.
16. If the window for setting the parameters defined as **Run**, set them and click **Next**.

If the Report list scheduling is not successful, to define an Excel-type TGKService, see Pagina TGK Service.

Indicate the email recipients.

In **Send mail** > **Recipients**, choose one of the following options:

- Type the desired email addresses, separated by commas.
- Type the desired user's username from the relevant CCH Tagetik list.
- By selecting and the desired user, and clicking . If the user is not on the list, click and type the desired email address.
- To indicate a reference user, type **<name of reference user>**. The email will be sent to all users linked to the reference user.

  ***IMPORTANT**: the characters < and > are part of the syntax for inserting the name of the reference user.*

Changing the Report list's security password

In Navigation mode, click **Change Report List Password** on the multifunction bar.

Run a Report list

1. In Navigation mode, open the desired form containing the Report list.
2. If requested, enter the value of the necessary parameters: the system will run the Report list.
3. To update the entire Report list, click . To update specific parts of the Report list, click and select them.

Once the Report list has been run, the system prints the forms according to the print areas defined within the various forms.

Manage the index

Create an index

Excel > CCH® Tagetik > Design

1. Open the desired form.
2. In the multifunction bar, click . Otherwise, in the **Open items** tab, click the desired form > : the system will create an index of the existing sheets.

Edit an index

1. To open the **Index Wizard**, choose one of the following methods:
   - With the index open, double click one of its cells.
   - In Design mode with the index open, click in the multifunction bar.
   - In the **Open items** tab, double click below the item containing the index.
2. To enter cell fields, click **Report** or **Sheets** as desired, select the desired predefined functions from the list and click .
3. Click on the desired button.

Report List Wizard window

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Purpose of the Wizard

This Wizard takes the user through a guided procedure to create and set up a Report list. It is made up of six steps:

Phase	Description
1	Allows you to set the publication method.
2	Allows you to define the forms/Excel sheets to be included in the Report list.
3	Allows you to define and manage the run parameters for the selected forms. Parameters shared by several forms are only required once. System parameters or ad hoc parameters can be added.
4	Allows you to override, for individual forms, certain settings assigned as defaults in the previous phases. Specifically, this allows you to override the print or save settings, parameters, templates and output sheets if you do not want to publish an entire form. Allow to specify output sheets sorting.
5	Only present if **Allow to specify output sheets sorting** was selected in the previous phase. Allows you to define the order of the sheets generated by the Report list.
6	Allows you to set the Report list run and save options.

Report List Wizard window: first tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Fields

Element	Description
Print	Allows you to select the printer on which to print the forms. • **<Default Printer>**: the device's default printer • **<Current PDF Printer>**: any PDF printer in use Once the Report list has been run, the system prints the forms according to the print areas defined within the various forms. *Configure the print areas of each form before creating the Report list.*
Disk or portal publication	Allows you to specify the type, format and name of the file to be generated and where it will be saved.
Send mail	Allows you to send the reportlist.zip file containing a file for every form in the Report list to a list of recipients. This is only possible after the fields related to the email have been set in the Repository configuration (see Pagina impostazioni Email). A reference user from among the Tagetik users of the environment in which you are working can also be inserted as the recipient. This is identified by the user name between angle brackets, e.g. <ferrari>. In this case, the email is automatically sent to all users linked to it (see Users) The user must specify the name of the files to be attached to the email. To send several files, the default option **${Report.code}.xlsx** must not be changed. In this way, the system gives each file a name equal to the form code to be sent

Report List Wizard window: second tab

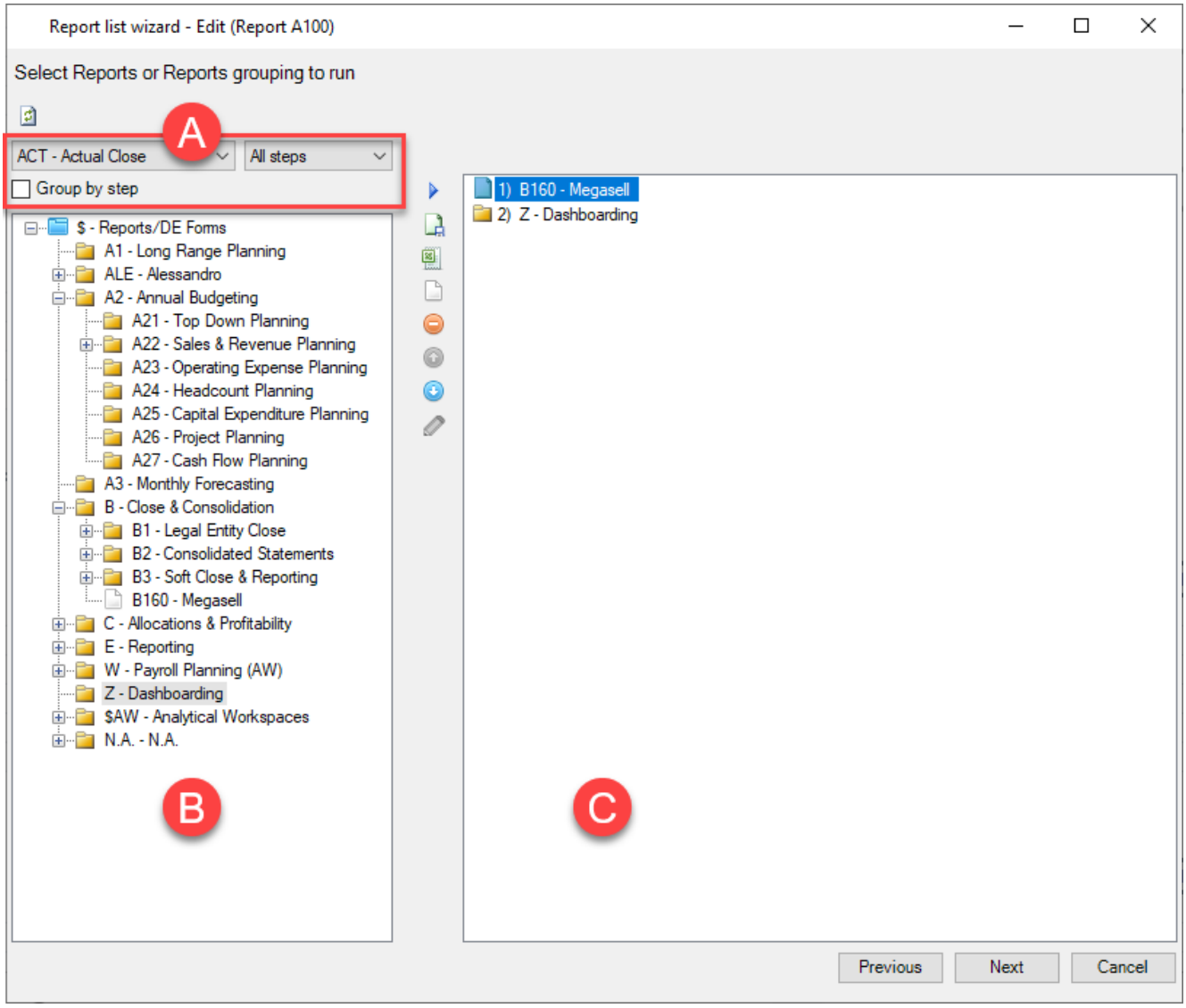
Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Purpose of the second step

This step allows you to define the forms/ad hoc forms/Excel sheets to be excluded from the Report list.

What it looks like



Element	Description
A	Form search filters by process and step to which they belong. **Group by step**: groups forms by step if the process includes several steps.
B	List of grouped forms based on the form grouping nodes
C	Forms in the Report list

Buttons

Element	Description
	Refreshes the list of forms.
	Adds the form selected in the list to the Report list.
	Adds an ad hoc form to the Report list.
	Adds an Excel file to the Report list.

Element	Description
	Adds an empty sheet to the Report list.
	Removes the selected element from the Report list.
	Moves the element selected in the Report list up or down the list.
	Renames the element selected in the Report list.

Report List Wizard window: third tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Purpose of the tab

This tab is dedicated to choosing parameters. If the Report list contains a single form, then the required parameters are those defined in the form. If the Report list includes several forms, then the parameters that are shared by several forms are only required once.

Parameter options

One of the following input modes must be set for every parameter.

Option	Description
Run	Allows you to insert the parameter value directly on running the Report list. If at least one parameter is set to "Run", it is not possible to treat the Report list as "ad hoc", but it must be saved within a form.
Select	Allows you to select the parameter value from the list immediately in the definition step. Therefore, this value will not be required when run.

Buttons

Element	Description
+ System parameter	Allows you to add system parameters.
+ adHoc parameter	Allows you to add ad hoc parameters.
	Only with **Override tab filters** selected. Opens a window to override tab filters. See "Override tab filters window" on the next page.

Fields

Element	Description
Override tab filters	Allows you to override tab filters present in the forms in the Report list.

Override tab filters window

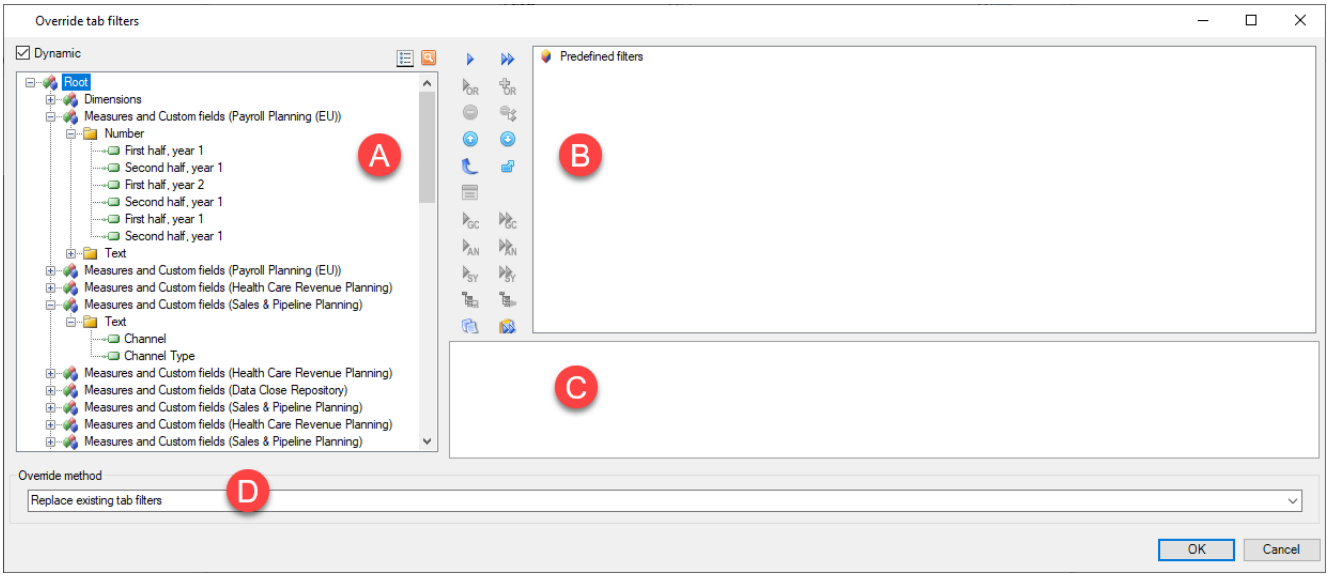
Excel > CCH® Tagetik > Report list > (third tab of the Wizard) select Override tab filters > ...

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list > (third tab of the Wizard) select Override tab filters > ...

Purpose of window

This window allows you to override tab filters. On closing the window, the system assigns **Ignore** as the definition method for the parameters shown in the run parameters configuration window. In this way, the user does not need to specify the value of the parameter, as it is overridden.

What it looks like



Element	Description
A	List of overridable tab filters (static, dynamic, parametric lists)
B	List of chosen tab filters to override
C	Details of the selected tab filter
D	Selected override method

Override method

Option	Description
Applies override only if there are no pre-existing tab filters.	The system adds the overridden tab filters only if it does not identify pre-existing tab filters.
Nest under existing tab filters	The system adds the overridden tab filters to any existing filters. If the overridden filter concerns a dimension already present in the existing filters, the system does not consider the overridden filter and returns a non-blocking warning message when running/refreshing the Report List.
Replace existing tab filters	The system inserts the tab filters, replacing the pre-existing ones.

Buttons in window

Element	Description
	Changes the display mode.
	Allows you to search for the desired tab filter.
	Adds one or several elements or nodes selected in the dimensions tree under the selected node in the filters tree.
	Only with filters already defined, adds one or several elements or nodes selected in the dimensions tree to all filters already defined. ***Note**: if there are no pre-defined filters in the column, it functions like the  button.*
	Adds one or several filters selected in the dimensions tree to a selected filter, creating a OR type logical relationship. OR type logical relationships can only be created with elements or nodes belonging to the same dimension as the filters already defined. The added filter is labelled with the concatenation of the individual element codes separated by ",". The value displayed is equal to the sum of the elements entered.
	Adds one or several elements selected in the dimensions tree belonging to the same filter dimension to a node of filters linked by an OR logical relationship. The added filter is labelled with the concatenation of the individual element codes separated by ",". The value displayed is equal to the sum of the elements entered.
	Only with filters already defined. Deletes the selected element or node.
	Only with filters already defined. Deletes the entire level of the selected filter.
	Only with filters already defined. Moves the selected element up one position.
	Only with filters already defined. Moves the selected element down one position.
	Only with filters at several levels already defined in the row/column columns. Moves the selected level up one level.
	Only with filters already defined. Replaces en bloc the selected elements of a filter with elements selected in the dimensions tree. Useful for reusing a filter by replacing its elements, but not its properties.
	Only with filters already defined in the rows/columns with an OR logical relationship. Deletes one or several elements from the filter containing the OR logical relationship.
	(Only for control groups) Adds the control group selected in the dimensions tree to the filter. See I gruppi di controllo: concetti di base.
	(Only for control groups) Adds the control group selected in the dimensions tree to all filters already defined. See I gruppi di controllo: concetti di base.
	(Only for FST items) Analytically adds, under the selected filter, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	(Only for FST items) Analytically adds, under all filters, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.
	(Only for FST items) Synthetically enters the FST under the currently selected filter. See "Financial statement templates" on page 83.
	(Only for FST items) Synthetically adds, under all filters, the FST selected in the dimensions tree. See "Financial statement templates" on page 83.

Report List Wizard window: fourth tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Fields

Element	Description
Customize print settings	Allows you to define a different printer for the individual form.
Customize parameters	Allows you to customise the parameters defined in the third step for the individual form.
Customize templates and output sheets	Allows you to indicate templates and output sheets to be exported for the individual form. If not specified, the system exports all templates and output sheets in the Report list by default. **Override sheet name**: allows you to rename the selected template/output sheet, to which a progressive number is assigned by default. This may be inserted in any of the managed table languages. In the Report list run step, the system uses the name of the table language of the user that performed the run.
Allow to specify output sheets sorting	Allows you to define an arbitrary order for all sheets generated by the Report list, regardless of their order in the source forms or Report list. This is not possible in the following cases: • if the selected publication method is **Print** • if publication/send email takes place on different files • there is an override on the file name set on the element, template or tab filters. • if there is an override on the email recipients set on element, template or tab filters.

Report List Wizard window: fifth tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Purpose of the tab

This tab shows a list of the sheets of all the forms present in the Report list. It is only visible if the **Allow to specify output sheets sorting** option has been selected on the fourth tab.

Buttons

Element	Description
	Only with filters already defined. Moves the selected element up one position.
	Only with filters already defined. Moves the selected element down one position.
	Opens a window to change the order of the selected sheets. *Select the sheets of which you want to change the order and then click* .

Report List Wizard window: sixth tab

Excel > CCH® Tagetik > Report list

Excel > CCH® Tagetik > Design > (Left-hand panel) select form > > Report list

Fields

The available fields vary depending on whether the Report list configuration window has been opened from **CCH® Tagetik**'s multifunction bar or the left-hand **Design** panel.

Element	Description
Run	Allows you to set the Report list run settings. See "Run" below.
Saving	Allows you to set the Report list saving methods. See "Saving" on the next page.
Scheduling	Allows you to schedule the running of saved Report lists or Report lists that need to be saved on a shared form. See "Scheduling" on the next page.
Protect generated files	Only if the file format is not .PDF. Allows you to set a security password for Report lists generated in **Disk or portal publication** or **Send mail** modes. See "Protect generated files" on the next page.

Run

Element	Description
Include in the Report list	Allows you to indicate what to include in the Report List: • **All** • **Only Templates that are defined**: the template of a form if at least one cell of a matrix is defined. • **Only Templates with not null matrix cells**: the template of a form if at least one cell of a matrix is defined with a value other than 0. • **Only Sheets that are defined**: the sheet of a form if at least one cell of a matrix is defined. • **Only Sheets with not null matrix cells**: the sheet of a form if at least one cell of a matrix is defined with a value other than 0.
Always consider templates without any matrix as valued	Disabled with **All** selected. Also include templates without matrices in the Report list. Such templates could contain headers, queries or other information that would otherwise be discarded.
Open the generated files at the end	Disabled if .PDF format has been chosen. If the Report list has **Single file** as its publication file type, the system will open all the generated files when the run finishes.
Always show the data processing logs when finished	After the Report list has been run, the log window will appear even if no errors have been detected.
Create the sheets index	If the Report list has **Single file** as its publication file type and is saved in a container form that does not have an index, the system will automatically create an index of the forms included in the Report list. It is created after the Report list has been defined.
Run now	The Report list is run once its definition is complete. If at least one parameter is set to "Run", it is not possible to treat the Report list as "ad hoc", but it must be saved in a form.
Generate non-refreshable Excel files disconnected from Tagetik	The Report list is saved as a standard Excel document. Consequently, it is not linked to the data in the Tagetik database and cannot be updated. This option is selected and cannot be changed if .PDF file format has been chosen.
Use form code for sheet names	The Report list uses a code generated on the basis of the code of the form for the name of the sheets that it contains.

Saving

Element	Description
Do not save	This prevents the Report list from being saved and called up in the future.
Save as a new shared form:	Saves the Report list in a form of the application database and allows other users to call it up. The code and description to be assigned to the new form must be specified. **Save and close the new form**: when the Report list has been defined, the system does not run the new form and, by default, saves the Report list under the node of the hierarchy of N.A. reports.
Save as a new ad hoc form	Saves the Report list locally in a form on the computer, preventing it from being called up by other users. The form appears after the Report list has been defined.
Save to the current form	Saves the Report list in the current form.

Scheduling

Element	Description
Schedule the run	Enables scheduling for the Report list and, after **End** has been selected, opens a window to set the relative parameters. See "Schedule Wizard window" below.

Protect generated files

Element	Description
Password	Allows you to enter the password to protect the generated files. The syntax of the cell fields can be used, such as the scenario code chosen in the Report list parameters, for example.
Confirm password	Allows you to enter the **Password** to confirm it.

Schedule Wizard window

Excel > CCH® Tagetik > Report list > define Report list via Wizard

Excel > CCH® Tagetik > Design > (left-hand panel) select form > > Report list > define Report list via Wizard

Purpose of window

This window allows you to set the Report list scheduling parameters. Scheduling is only possible for Report lists that have already been saved or those which need to be saved on a shared form.

Report list scheduling can only be enabled if the **Schedule the run** option has been selected in the sixth tab. The window opens when the Report list definition sequence has ended.

Fields

Element	Description
Description	Description associated with the data processing to be scheduled. Makes it easier to identify data processing from the list of Scheduled items.
Recipient at start	List of users to be notified of the start of data processing. A user can be specified using their email address or user code.
Email note at start	Note to be added to the standard wording of the notification email
Recipient on success	List of users to be notified of the outcome of data processing completed successfully. A user can be specified using their email address or user code.
Email note in case of success	Note to be added to the standard wording of the notification email when the data processing is successfully completed

Element	Description
Recipients on failure	List of users to be notified of the outcome of data processing not completed successfully and not to be rerun. A user can be specified using their email address or user code.
Email note in case of failure	Note to be added to the standard wording of the notification email when the data processing fails and is not to be rerun.
Validity start date	Date from which the scheduled data processing is run. **Define Start date validity**: allows you to set the start date.
Minutes	Minute(s) in which to run the data processing. • **Every minute**: runs the data processing every minute. Equivalent to using ***** in the Crontab syntax. • **[*minute*]**: from 00 to 59 minutes
Hours	Hour(s) in which to run the data processing. • **Every hour**: run the data processing every hour. Equivalent to using ***** in the Crontab syntax. • **[*hour*]**: from 0 to 23 hours.
Days of the month	Day(s) of the month in which to run the data processing. • **Every day of the month**: runs the data processing every day. Equivalent to using ***** in the Crontab syntax. • **[*day of the month*]**: from 1 to 31 days.
Month	Month(s) in which to run the data processing. • **Every month**: runs the data processing every month. Equivalent to using ***** in the Crontab syntax. • **[*month*]**: from 1 to 12 months
Days of the week	Day(s) of the week in which to run the data processing. • **Every day of the week**: runs the data processing every day. Equivalent to using ***** in the Crontab syntax. • **[*day of the week*]**: days from Monday to Sunday
Limit number of repetitions	Limits the number of times the system must run the data processing. If the limit is not specified, the data processing is repeated ad infinitum according to what is specified in the Crontab syntax. **Number of repetitions**: number of time the system must run the data processing starting from the specified date/time
Maximum wait for booking	Allows you to set the wait time for booked data processing that cannot be run immediately due to a system overload or other data processing being run. The system assigns status **B** (Booked) to the booked data processing and automatically reattempts to run the data process every minute until the system accepts it or the maximum wait time set by the user has lapsed. **Maximum wait for the run (min)**: maximum wait time for running the data processing of the prebooked Report list.
Rerun of the data processing in the case of an exception	Allows you to configure a new run of the scheduled data processing that was not completed successfully. This is particularly useful when the failure is due to a temporary problem. • **Maximum number of iterations**: the maximum number of times that the data processing must be rerun in case of failure. • **Minutes to wait before the next attempt**: the number of minutes of wait time before the scheduled date processing is rerun.
Allowed servers	List of the servers on which the scheduling can be run (**All** or specific servers).

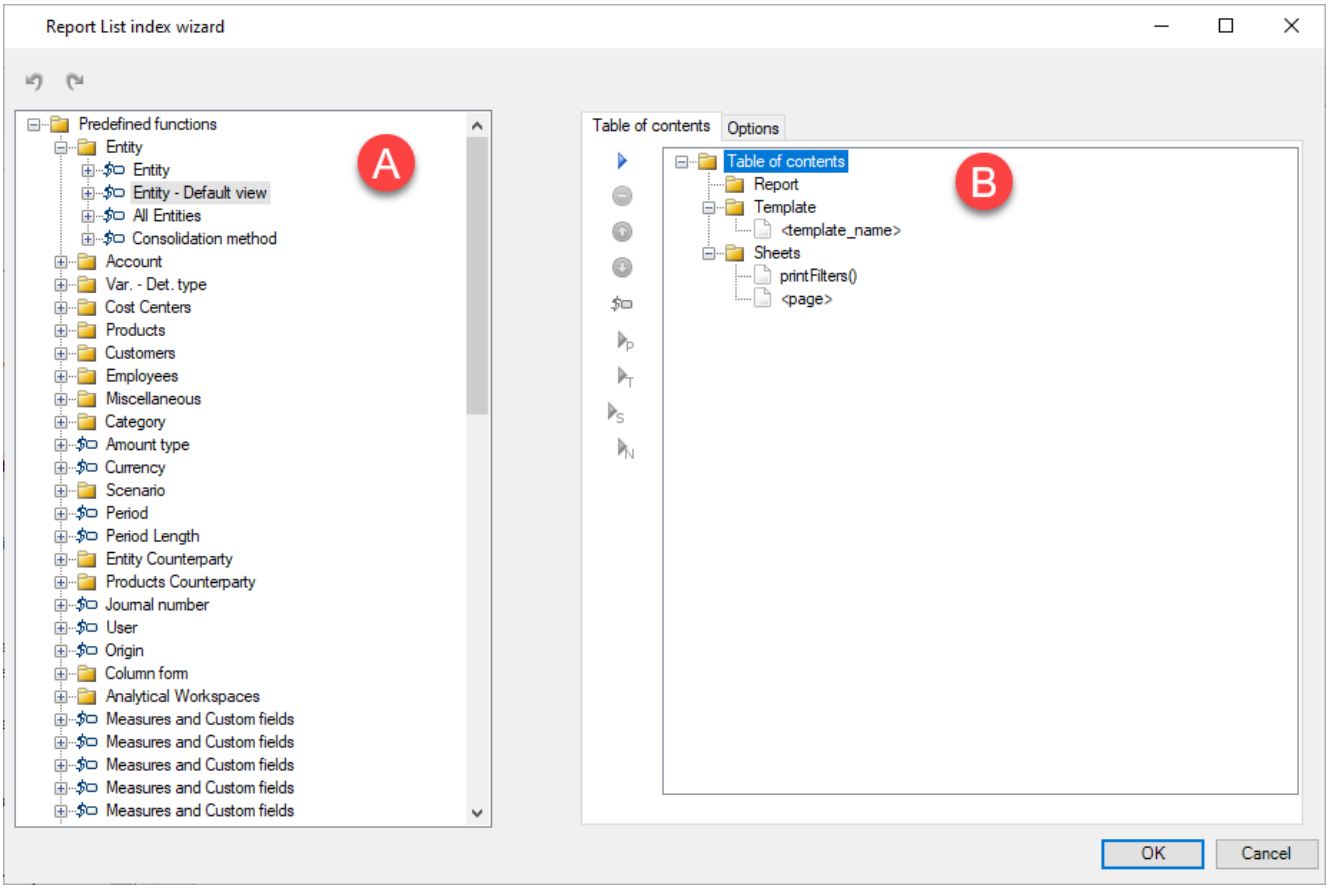
Report List Index Wizard window

Excel > CCH® Tagetik > Design > (Left-hand panel) click form > > Index

Purpose of window

This window allows you to create and then customise the structure and layout of the index of sheets contained in a form with several templates or a Report list that publishes the results in a single file.

What it looks like



Element	Description
A	List of predefined functions
B	Configuration tab

Table of Contents tab

In this tab, you can customise the structure of the table of contents, which always comprises three levels.

Element	Description
Report	• If the table of contents is a Report list, it shows the code and the description of the forms that comprise the Report list. • If the table of contents is a form comprising several templates, it does not show anything.
Template	Shows the name of the template.
Sheets	Shows the following: • the content of the **print Filters** cell fields • the report's tab filters • **<page>** the page number Hyperlinks to access the report's various sheets are created at this level.

Table of Contents buttons

Element	Description
	Adds the report selected in the list.
	Removes the selected element.

Element	Description
	Moves the selected element up or down.
	Adds an advanced cell field. See "Configuring cell fields" on page 42.
	Adds the specific cell field for the **Page number** tables of contents.
	Adds the specific cell field for the **Template name** indices.
	Adds the specific cell field for the **Sheet name** tables of contents.
	Adds the specific cell field for the **Page name** indices. Shows the actual name and any overridden name or the number of the sheet in the Report list.

Options tab

This tab allows you to customise the layout of the table of contents.

Element	Description
Show hyperlink in sheets	When the cursor is positioned over the table of contents page number, it shows a tooltip with the name of the sheet to which the hyperlink is linked.
Insert separator dots before the last column	Inserts dots to separate the content of the last column of the table of contents from that of previous columns.
Autofit columns	Automatically adapt the width of columns.
Format automatically	Applies the appropriate style sheet to the table of contents. See "Tagetik styles " on page 230. If a style sheet with the **TGK_TOC_PAGE_COLUMN, TGK_TOC_REPORT_ROW** or **TGK_TOC_TEMPLATE_ROW, TGK_TOC_SHEET_ROW** styles is associated with the report on which the table of contents is defined, the system will apply the defined style to the table of contents.
Also show the template name	Keeps the **Template** level in the table of contents. If deselected, this level will be removed.
In the case of the report list with sheets that are sorted manually, in the index:	Only with **Allow to specify output sheets sorting** selected in the Report list definition step. Allows you to specify the impacts of the sheet sorting override on the table of contents. • **Keep the sheets sorting and duplicate the rows related to reports/templates**: keeps the progressive order of the table of contents' page numbers and rows. The order corresponds to that which the user gives each sheet. If the table of contents allows you to move from one report/template to another, any report/template headers will be replicated. • **Keep the order of reports/templates and show unordered sheets**: the table of contents will show the **Report**, **Template** and **Sheets** levels, if active. The order of the reports is subject to the override set during the definition of the Report list. The order of templates and sheets is that of the respective reports to which they belong and could result in unsorted page numbers in the table of contents.
Rows	Allows you to configure the method by which the rows below the table of contents must be generated. • **Shift cells down**: insert the new cells, moving the existing cells down. • **Entire row**: inserts a new entire row into the Excel sheet. • **Don't expand**: prevents new rows from being inserted.

Managing Report Lists from the Web

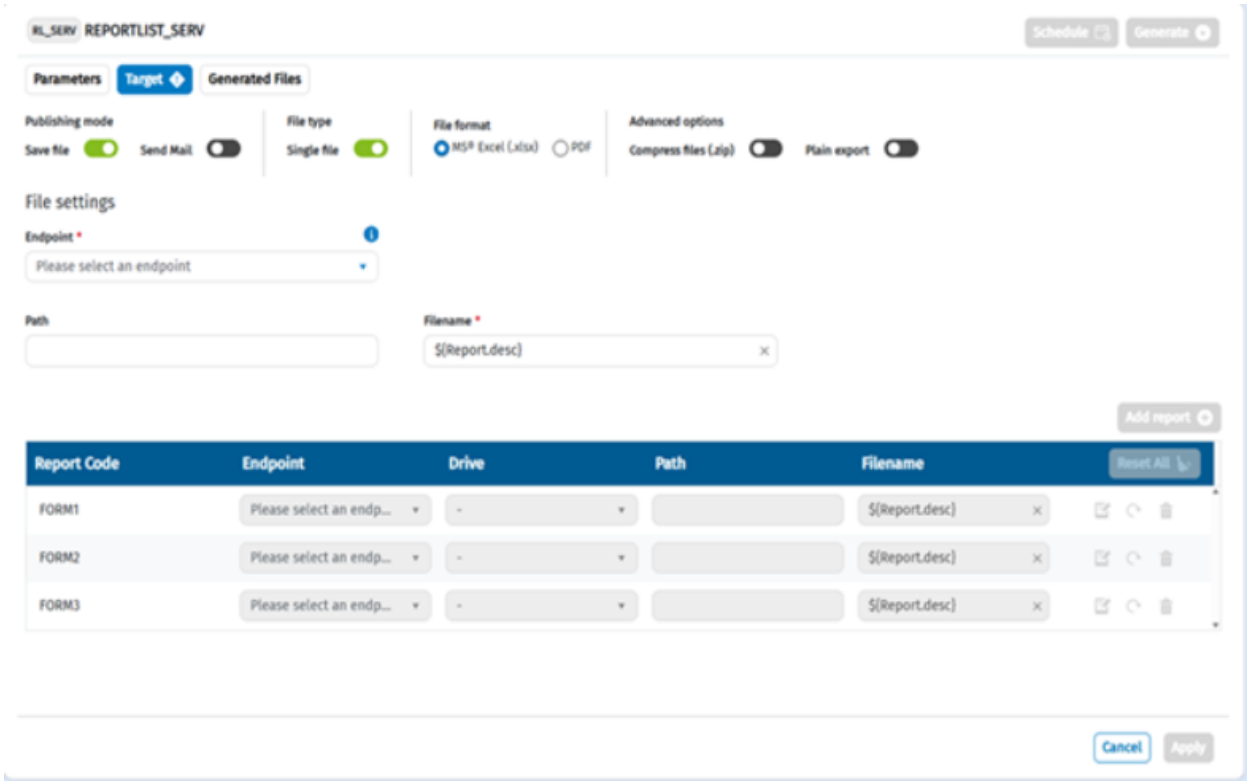
Configuration sets

Run Report List from the Web

Tile Report List

Navigation > Reports & Analysis > Report List

1. Select the desired Report List.
2. A new window is displayed with 3 tabs: 'Parameters', 'Target', 'Generated files':



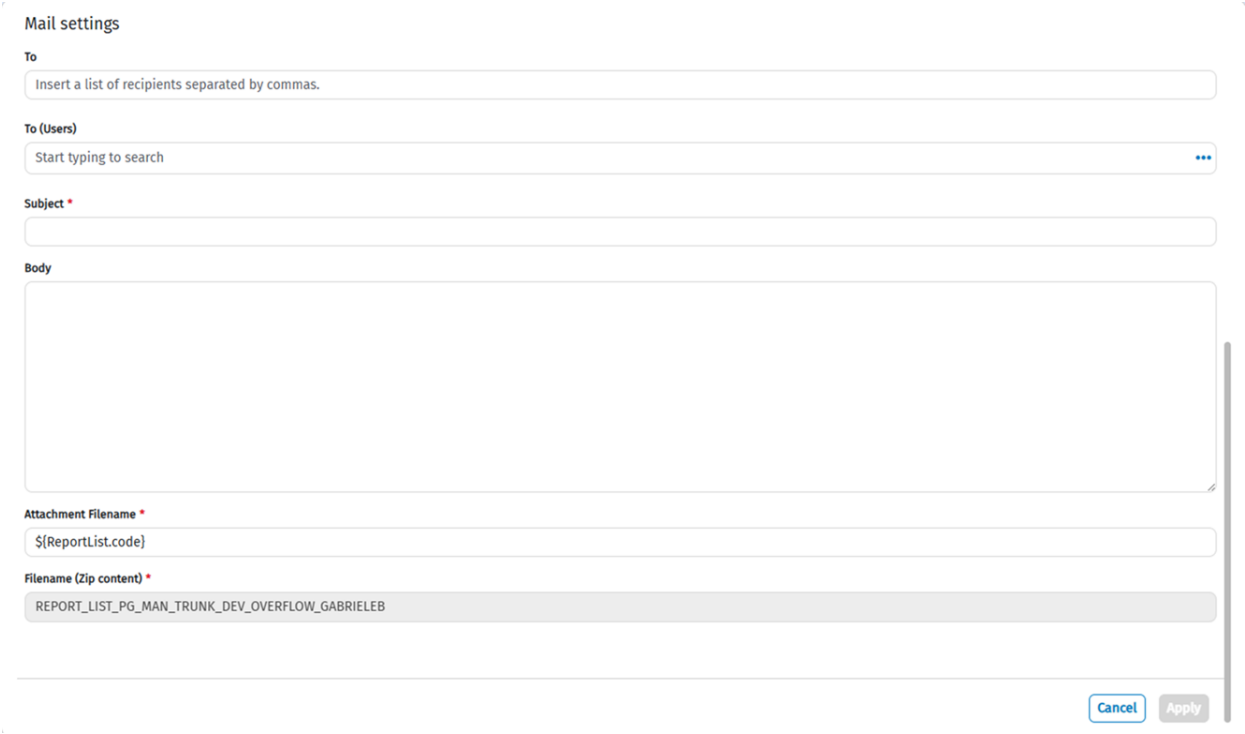
3. | If... | Then... |
|---|---|
| The Report List requires parameters | the 'Parameters' section will show the selectable ones, then click on ... next to each dimension to select the desired items or nodes. |
| the Report List does not require parameters | the 'Parameters' section will remain empty. |

4. In the 'Target' section, the user must enter all the necessary information to generate the output file, such as:
   - Publication mode (**Save file** - option selected by default - or Send mail)
   - File type (**Single file** - selected by default)
   - File format (**MS Excel .xlsx** - selected by default - or **PDF**)
   - Advanced options (**Compress file** - zip - and **Plain export**)
   - Endpoint
   - Path
   - File name
5. In the 'File Settings' section, the user must enter the desired Endpoint, Path and File Name (set by default to ${Report.desc}).

Admin will thus create Endpoints in the Repository on which the report will be saved:

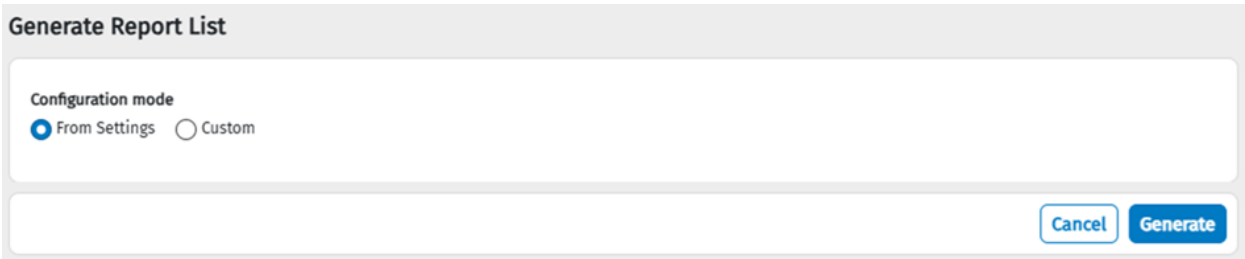
- Automatic creation of sub-folders in the endpoint when saving the RL
- Handling of the automatic creation of a subdirectory in the endpoint when saving the RL via a dynamic script
- Server-side support for generating a PDF or Excel
- Generating files without .ZIP
- Files generated at the end of LAUNCH_REPORT_LIST data processing

When the user activates the 'Send mail' option, the e-mail settings are displayed below the report list.

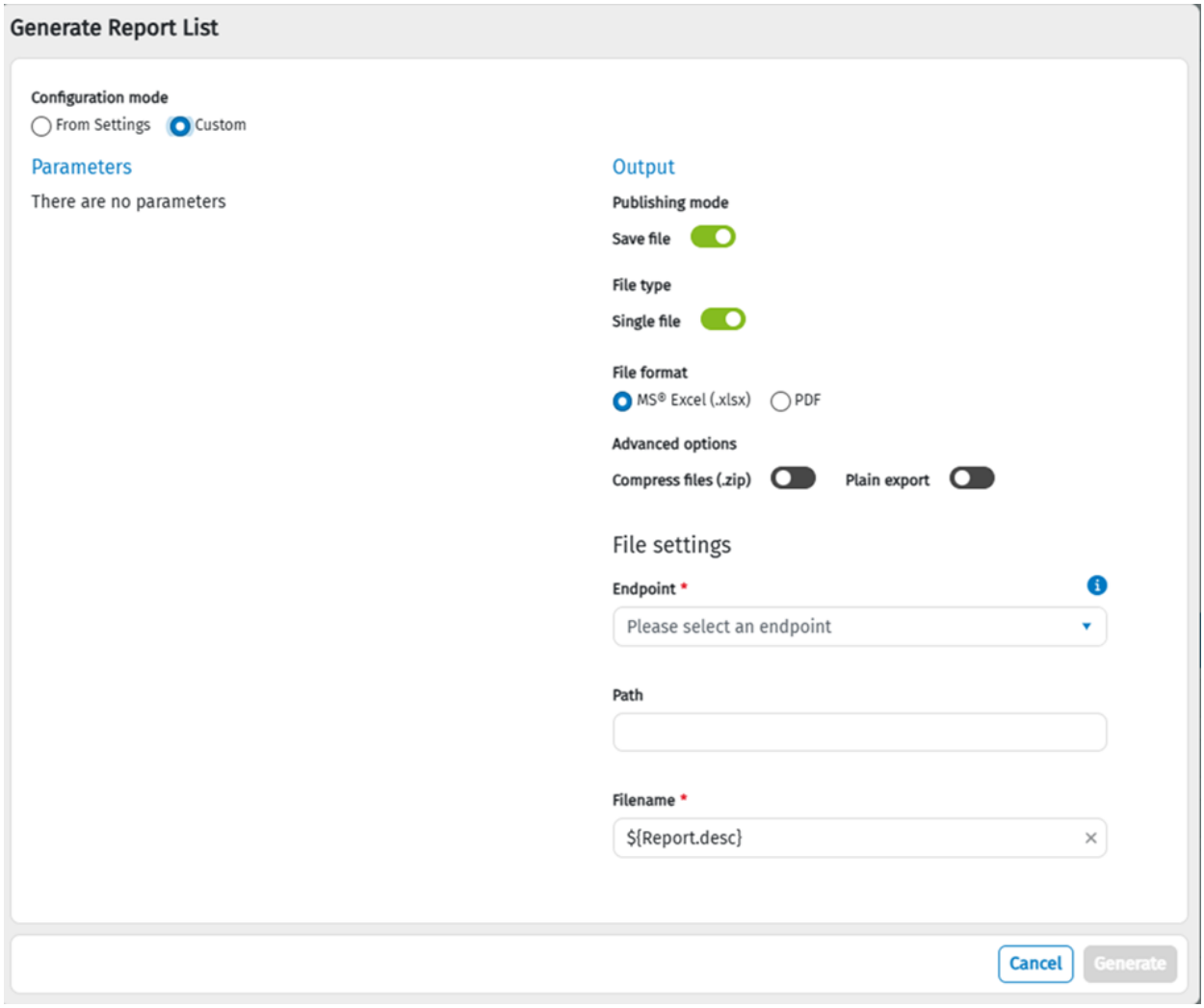


6. Clicking **Generate** will start the data processing that will produce the file for the selected Report List and the user will see a notification of data processing in progress.

   The files can be generated using the saved settings:



Alternatively, ad-hoc settings, valid for the specific execution only, may be used:



7. data processing:

If...	Then...
the user is still on the Report List page	A 'Results' pop-up window will appear containing two tabs: 'Exported files management' and 'Audit'. The 'Audit' tab will show the audit details of the data processing just performed.
the user is no longer on the Report List page	a notification of completed data processing will be displayed in the bottom right-hand corner, and via the **Show Result** button, the 'Results' window can be displayed.

***IMPORTANT**: If a Report List is generated several times with the same code and by the same user, it will be overwritten by the most recent file.*

Server-side support for generating a PDF or Excel

The configuration to generate only PDF files or only Excel files is also supported for the new Report List based on the server engine, instead of always generating both file types.

Types of parameters and functionalities/services supported on the Web

The types of parameters supported when launching a Report List from the Web are: standard dimensional parameters, AIH parameters, NON-dimensional parameters (Text, Flag, Date, Dictionary) and Multi-hierarchy parameters.

The features/services supported on the Web instead are:

- The mandatory flag

- Restrictions related to user rights
- Multi-selection within the parameter tree
- Quick search, on the dimension tree of the selected parameter, only active in the window containing the parameters in use of the Report List
- Lowest level elements enabling flag
- Nodes enabling flag
- Default Value, explicitly selected from the parameter definition, the default value defined via advanced dynamic script is not supported.
- Default Value, explicitly selected from the parameter definition, the default value defined via advanced dynamic script is not supported.
- Advanced selection (insert lowest elements, insert children elements)
- Advanced selection script
- Dependent parameters, where the dependency may be implicit (between two parameters belonging to the same dimension) or explicit (where the dependency is defined within the script)

Automatic creation of sub-folders in the endpoint when saving RL

The user is able to automatically generate sub-folders from a selected Endpoint, based on the definition of the Report List file name. Each section of the filename delimited by '\' will be a subfolder of the selected endpoint, the last section being the file name.

Managing the automatic creation of a subdirectory in the endpoint when saving the RL using a dynamic script

When launching a Report List from the web, the user can define a dynamic file name and path using the 'Cell field' language. In this way, the system generates a folder for each element divided by '\' recognising the script in the file name and translating it according to the Report configuration, within the Report List.

Generate files without .ZIP

When the user generates or schedules, the server side of the Report List will have the option of compressing the generated files. By default generated files are not compressed. This option can be accessed by a flag in the generation and scheduling window.

Tab for Managing Files Generated on Personal Report Type Endpoints

The Generated Files section, which has been renamed Personal Reports, is supported. This section, accessible from the Report List tile on the User Home Page, allows the user—after selecting a Report List—to view, download, or delete the latest file(s) generated for that Report List through the endpoint marked as Personal Report, by the currently logged-in user.
To activate this section, it is first necessary to create an Endpoint, and save it in the Settings of the application database, in the section *Form> Report List > Endpoint for Personal Reports*, and then generate the selected Report List using this Endpoint, which will be the first in the list.
The file name will follow a standard non-modifiable logic: ReportList_database ID_user code_ ReportListcode if it is a single file or ReportList_database ID_user code_ReportListcode_Form_code in the case of separate files.

Files generated at the end of LAUNCH_REPORT_LIST data processing

If you choose to generate Excel type files then the files produced by the data processing will be browsable, unless you have chosen the option "Generate non-refreshable Excel files disconnected from Tagetik" in report list design mode, in which case the files will be simple excel files.

Opening a file generated by the Report List data processing that does not have the option "Generate non-refreshable Excel files disconnected from Tagetik" will prompt the user to log in to the Excel add-in CCH Tagetik, if the user is not logged in. If the report list has the 'Single File' option, then the tools enabled in navigation mode will be 'Cell Properties' and 'Drill Down'.

***IMPORTANT**: For the navigation tools to work properly, the use of the prefix 'TMP_' in 'Name Management' in reports contained in a Report List is not recommended.*

Override options and override run Parameters

We introduce advanced options, applicable from the Report List settings in the Excel add-in, which allow you to override the run parameters for each of the reports used in the Report List. This functionality allows the user to change one or more of the run parameters of a report compared to the parameter settings of the Report List. It also allows you to run, server-side, a Report List that includes the same report several times, but with different run parameters set.

Filename and Endpoint Override

This function makes it possible to manage the replacement of one or more file and/or path names in the new Report List, even within several Reports contained in the Report List:

- The replacement options are already available in **Excel** and only apply to Report Lists generated with the Excel engine.
- The replacement options have been extended to the Tagetik **web** application, in this case the Report Lists are generated with the Tagetik server engine, which unlike Excel involves the use of Endpoints and does not support GetEnv() functions and saving files to local folders.

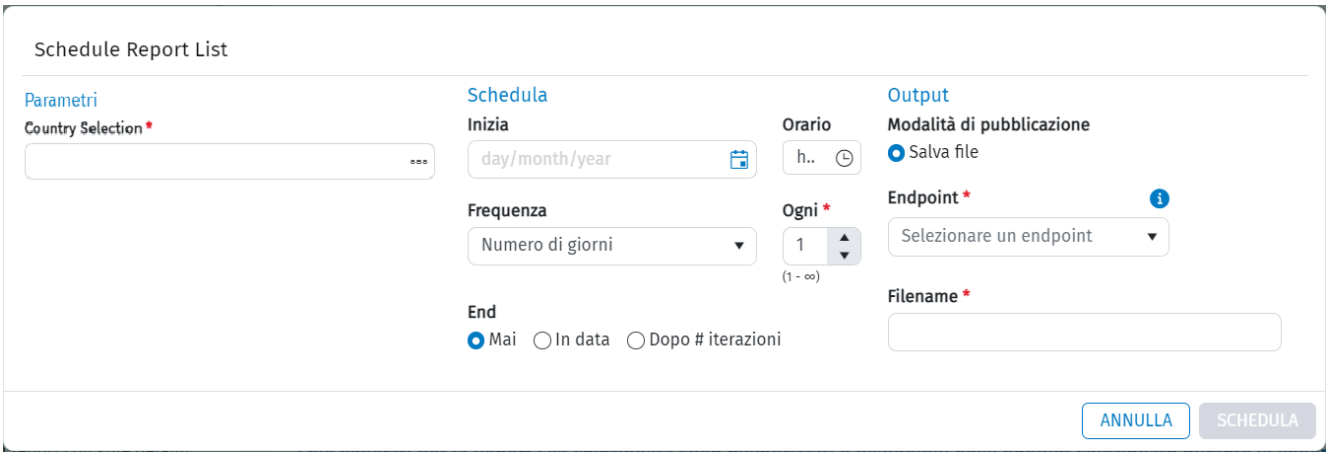
When the user selects a specific report included in the Report List and enables file name and/or target replacement:

If...	...then
A Report List generates a **single file**	reports that have replaced one or more save settings (Endpoint, File Name and Path) will be separated as single files and generated separately.
A Report List generates several **separate files**	the replacement affects only the target or the name of the overwritten files.

Scheduling a Report List from the Web

Tile Report List

Navigation > Reports & Analysis > Report List



1. Select the desired Report List, then click on **Schedule**.
2. A new window consisting of three sections will pop up: in addition to 'Parameters' and 'Output', similar to those in the generation window, there will also be a 'Schedule' section where the user can set the settings required for scheduling. The 'Start' field indicates the date on which the scheduling is to start. The 'Time' field indicates at what time each report list is to be launched. The 'Frequency' field indicates how often the scheduling is to be repeated:
    - Number of days -> The field 'Every' appears, in which the user indicates how often the scheduling is to be run

- Days of the week -> A list of days of the week appears below where the user can select on which days the scheduling is to be run
- Day of the month -> An 'On day' field will appear in which the user must indicate the day of the month on which to run the scheduling
- Day of the year -> Two fields will appear in which the user must enter the month and day respectively on which to run the scheduling

The 'End' field allows the user to indicate when the scheduling should end:

- Never -> Scheduling will be repeated without an expiry date
- On Date -> Scheduling will end on the day indicated below
- After # iterations -> Scheduling will end after # times

3. When clicking on **Schedule** (only active if all the previously described sections have been filled in correctly) the generation of the selected Report List will be scheduled, based on the parameters entered during the scheduling phase.

   ***Note**: Mandatory fields will be marked with a red asterisk and the 'Schedule' button will remain disabled until the user has filled them all in.*

4. It is possible to **set up** a notification e-mail for each possible execution outcome directly from the scheduler.

   ***Note**: Currently, in the event of total or partial data processing success, the content and recipients of the e-mail are the same, but the subject line differs. It will be 'Data processing completed successfully' if the operation was successful, or 'Data processing completed with warnings' if there were warnings.*

Index explanation

***Note**: The following applies to both scheduling and generation.*

If the **Generate Index** option was selected in Excel during the creation of a single-file report list, the ZIP created following generation will contain an Excel file incorporating an Index as the first sheet marked with a table to navigate to all the forms in the report list.
Each table element on the **Sheet Description** section is in fact a hyperlink which, once clicked, navigates to the selected form.
Similarly, the ZIP file will also contain a Pdf file that also contains a table of contents on the first page with hyperlinks on the **Sheet Description** section, but in addition will contain the new item **Page Number** indicating the page on which the form in question is hyperlinked, respectively.

Report List Index on web

In the new Report List, the **Table of** Contents, which was automatically generated by adding the index to the list, is not supported. The index is still available, but in a different format and without **table of contents**.
The format is tabular and includes a first column for the **Form Code**, a second for the **Form Description**, a third for the **Sheet Description**, and a fourth column (for PDF files only) for the **Page Number.**

***Note**: The **Sheet Description** and **Page Number** contain a hyperlink to the corresponding page.*

Features not supported

When adding a form node to the Report List and not individual forms one by one, it will not be possible to override the filename and file path on this same node

Manage notifications when one or more reports fail in the Report List

When one or more reports, contained in a Report List, fail during data processing, a red light will light up in the end-of-processing notification, highlighting the fact that data processing has ended with errors. Further details on the error are then provided by the Audit.

Support of Explicit Restrictions on the New Report List

Explicit restrictions on the web, defined manually in the parameter definition performed on the Excel add-in, are supported.

Report List Page (web)

User | <application database>

Tile Report List

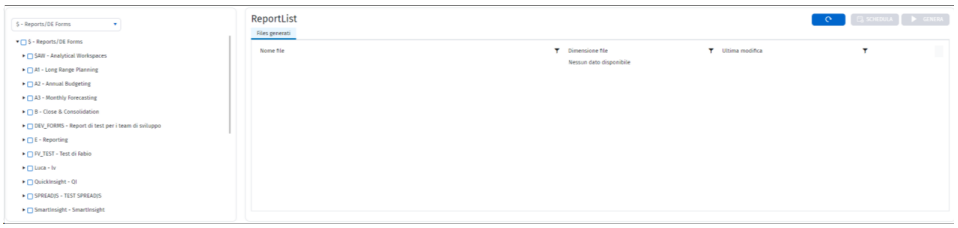
Navigation > Reports & Analysis > Report List

Purpose of the page

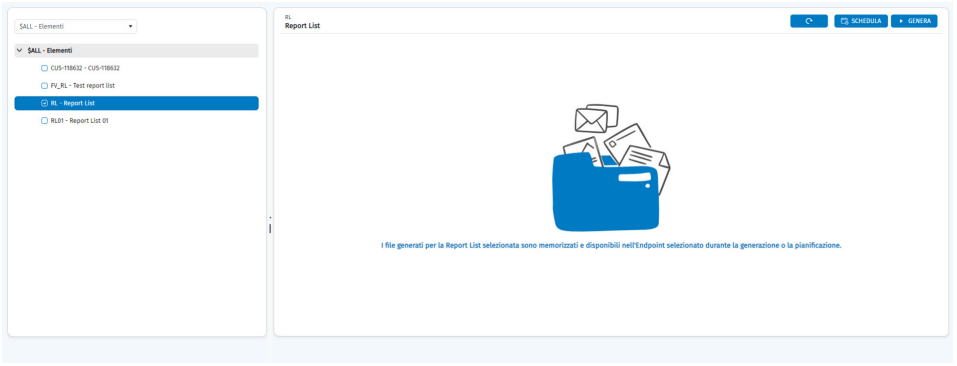
This page displays a list of Report Lists previously prepared in Excel and allows you to start the processes that generate the required files.

What it looks like

Before selecting a Report List from the list on the left, the page looks like this:



After selecting a Report List, the page looks like this:



Buttons on the page

Element	Description
	Updates the list of Report Lists shown on the page.
Schedule	Allows you to schedule the generation of the selected Report List
Generate	Starts the data processing that generates the required files from the selected Report List.

Parallelization of Report list launch

The system offers the user the possibility of choosing how to parallelize the launch of a form. Parallelization on a single node is always guaranteed. However, a specific server configuration is required to parallelize on multiple nodes. A Kafka service is available in your cluster and the variables are set: JAVA_OPTS="$JAVA_ OPTS -Dit.grupposervizi.easy.kafka.bootstrapserver='broker:29092'" JAVA_OPTS="$JAVA_OPTS -Dit.grupposervizi.easy.reportlist.distributed.enabled=true"

1. "Single-node parallelization" on the facing page
2. "Multi-node parallelization" on the facing page.

Single-node parallelization

Singl-node parallelization allows up to a maximum of 5 forms to be run in parallel for each report list. Therefore, if the user launches 2 report lists, then the server will launch 10 forms in parallel.

Multi-node parallelization

Parallelization on multiple nodes is a feature that allows you to scale the launch of a report list on multiple nodes within the same cluster. This is made possible by the Kafka service which connects the nodes via a message queue. As stated previously, to enable the functionality it is necessary to have a compatible cluster and set the following properties:

1. **JAVA_OPTS="$JAVA_OPTS -Dit.grupposervizi.easy.kafka.bootstrapserver='broker:29092'"**: Deve contenere l'indirizzo del server kafka
2. **JAVA_OPTS="$JAVA_OPTS -Dit.grupposervizi.easy.reportlist.distributed.enabled=true"**: Deve essere settato a true altrimenti la scalabilità sarà single node

Note: Both in the multi-node and single-node case, each application server can launch a maximum of 5 forms for each report list and 2 report lists simultaneously.

Report Bursting Overview

The Report Bursting functionality allows for the massive creation of CCH® Tagetik reports, both interactively (from user interface) and by scheduled data processing.

The reports are created from a **Collaborative Office** document or from a **Form**.

The Report bursting function allows defining:

- the structure of the report to generate
- the logics the report will be filled with, with specific data for each recipient
- how the report should be generated (in what format, if at a specific date or according to a cyclic scheduling, etc.)

For example, if you want to generate a report of sales for the managers of various regions or a P&L account for all subsidiaries or for all cost centres of a big organization, you just have to define the report's structure once and then use the "Report Bursting" function to generate the reports for all recipients automatically, providing only the relevant information

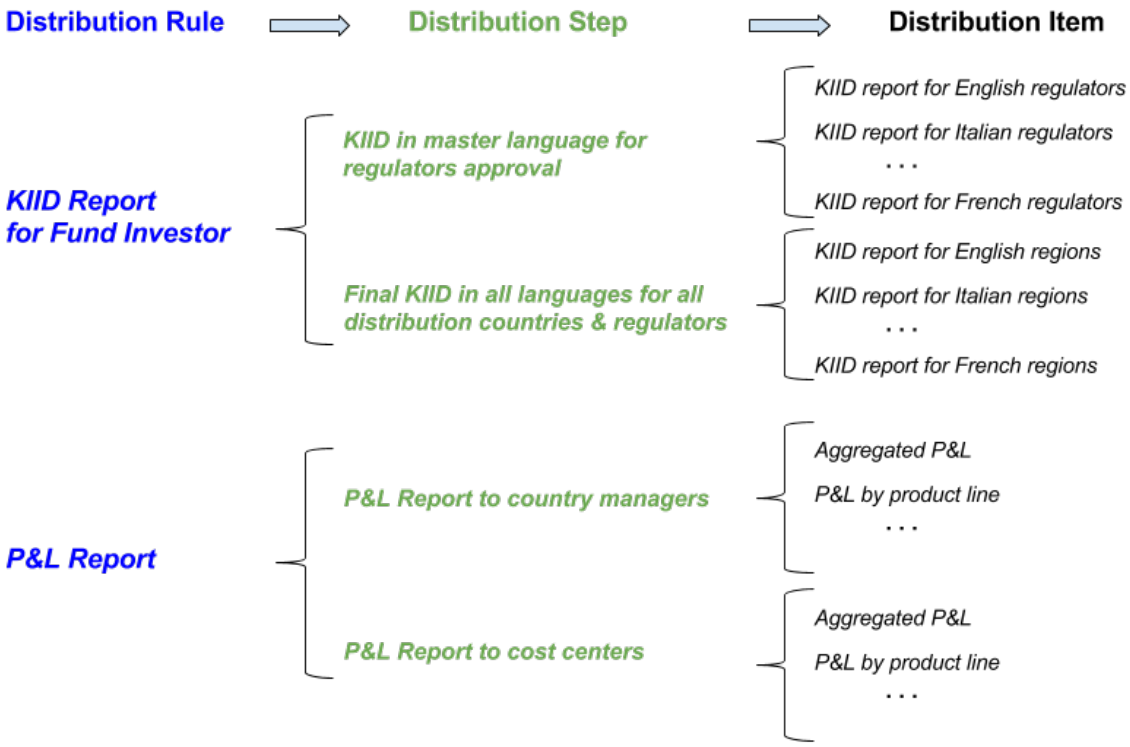
In order to define which reports have to be generated and the logics with which they should be filled and saved, it is necessary to create a **Distribution Rule** with the specific data for each recipient. It is possible to define several distribution rules for various reporting needs or for various recipients types. For example, we could have a distribution rule for the generation of the KIID report for fund investors, another rule for the P&L Account report, for subsidiary and for cost centre.

A distribution rule consists of several **Distribution steps** that allow distinguishing different reports generation and distribution rules.

With a single data processing, it is possible to generate all reports defined for a distribution step but it is not possible to generate all steps of a distribution rule.

The **Distribution items** are defined in a distribution Step and represent the highest level of detail of the Report Bursting setting. A Distribution item defines:

- the structure of the form or of the Collaborative Office document to use to produce the report
- the filters to apply to data in order to generate the final report. Those filters can be dynamic, meaning that it is possible to specify a CCH® Tagetik hierarchy's node in the distribution Item and then generate a different report for each lowest level element associated with that node or to the descending nodes
- the report's format
- any validations
- the name of the file that will contain the final report. Also in this case, it is possible to specify the name with a dynamic syntax that then will result in a different name for each generated report



Configuration

In order to define which reports have to be generated and the logics with which they should be filled and saved, it is necessary to create:

- a **Distribution Rule**. The distribution rule represents the process that is to be implemented and is the first step in the distribution process.
- one or more **Distribution steps.** Each step represent a phase of the distribution process and allows dividing the rule, for example, according to the frequency of the generation and distribution of the reports. For each distribution step, it is possible to define a distribution list (list of items)
- one or more **Distribution lists.** The distribution list consists of one or more items in which the user can specify the report to generate/ distribute, the set of filters to apply to the report and the recipients.

Distribution rule

The distribution rules table can be accessed from the user home page by clicking on the "Report bursting" tile.

Each rule is identified by a **Code** and a **Description** and by:

Property	Description
Endpoint	External source in which the files to distribute are saved. You can choose one of the external sources configured in the Repository of CCH® Tagetik

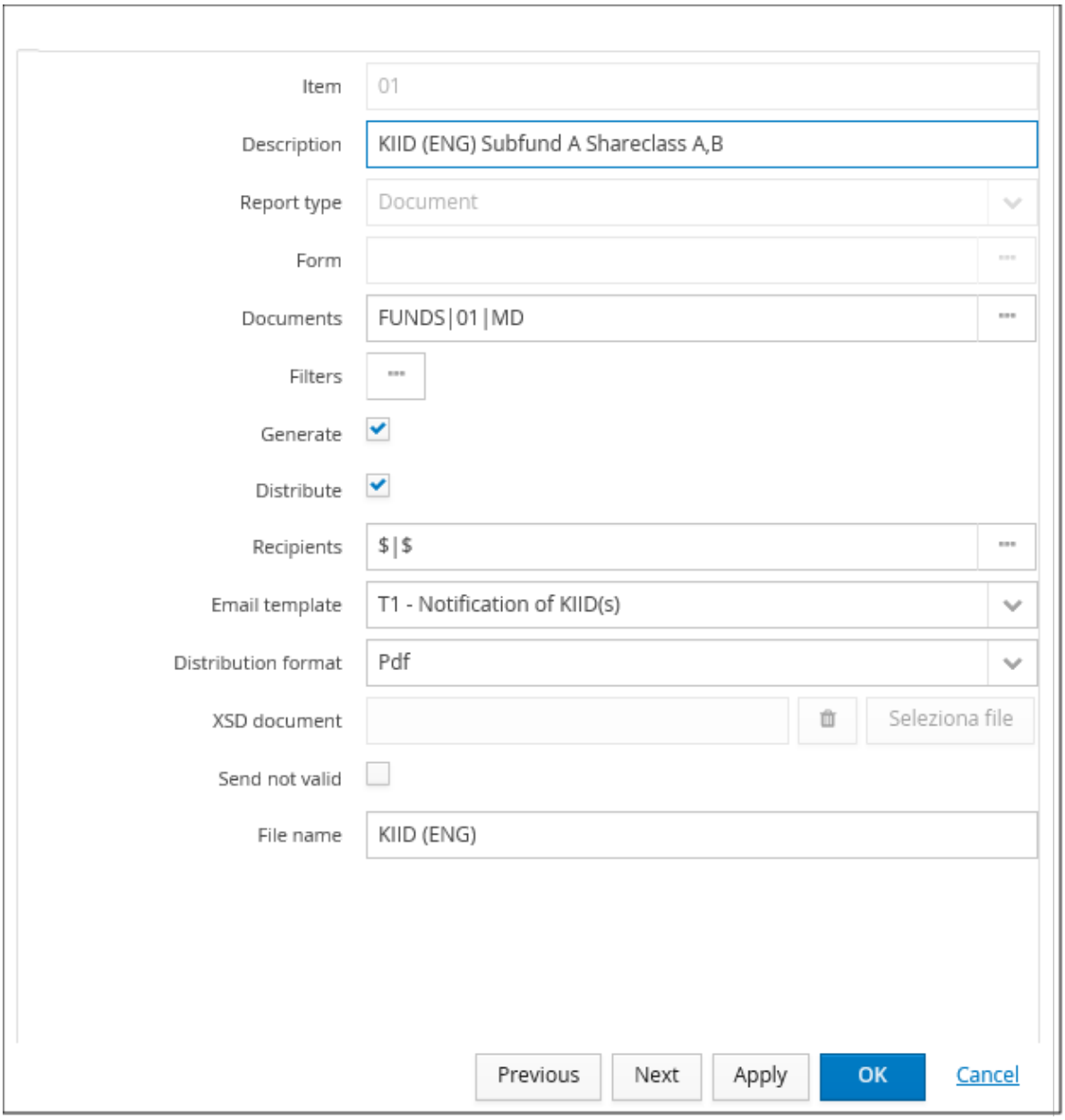
Property	Description
Directory	Path to the folder shared with the external source. For example, in order to save the files distributed for that rule into a folder named TAGETIK "Quarterly Reports" , the description of the directory should be TAGETIK/Quarterly Reports/, with a slash "/" that separates the names of the various folders. If no directory is specified, the system will save the distributed files into the root.  The path can be defined also parametrically by suing a specific syntax in order to report some specific information about the generated report and / or the value of the filters configured when defining the item. For further details on the syntax, see paragraph "Parametric elements" on page 395.  During the definition phase, the system does not check the validity of the characters inserted into the "Directory". If the Directory contains one of the characters listed below, then the system will replace such characters with the Underscore (_) symbol during the definition and/or distribution phase. ◦ Asterisk (*). ◦ Vertical bar (|). Backslash (\). ◦ Colon (:). ◦ Quote ("). ◦ Less than (<). ◦ Greater than (>). ◦ Question mark (?). ◦ Space. This character is replaced only during the distribution phase.

Distribution step

The distribution step table can be accessed from the *Distribution step* menu that appears after clicking on the *User actions* ⋮ menu in the distribution rules definition window. Each step is identified by a **Code** and a **Description**.

Distribution list

The distribution items table can be accessed from the *Distribution list* menu accessible after clicking on the *User actions* ⋮ menu in the distribution steps definition window.



Each distribution item is identified by a **Code** and a **Description** and by :

Property	Description
Report Type	Type of report , possible values: ◦ *Form (.xlsx)* ◦ *Document (.docx)* After saving the item, this field is no longer modifiable.
Form	Form ID code. This option is enabled only if *Report type* is *Form (.xlsx)*. Some form features may not be supported. For further details, refer to the section "Unsupported features".

Property	Description
Documents	ID string containing the code of the document type, the code of the document and that of the document activity selected in the accessible documents tree. This option is enabled only if the *Report type* is *Document (.docx)*
Filters	Allows setting specific filters. Upon generation, the reports will inherit such filters that could be different from those selected when defining the document or the form. so that the report inherits them upon generation, such filters If the Report type is: • *Document*, it is possible to select all necessary filters in terms of *Scenario, Period, Entity, Category, Custom dim1, ...,5 and Currency*. For each dimension, except the Currency, it is possible to select a single element or a node. It is not mandatory to define them and, if a user decides not to select any value, the reports will be generated using the filters set in the *Document properties*. • *Form*, it is possible to select the value to be assigned to the dimensions inserted as parameters in the filters definition. In the presence of nodes, it is possible to deploy the node, generating as many reports as there are elements belonging to the node, by enabling the "**All lowest elements**" option.  The "All lowest elements" option cannot be enabled more than once in case of Forms and the multiple activation may result in the JVM's resources being overloaded, causing performance degradation until the critical state is reached.  Report Bursting does not support forms with more than one parameter on the same dimensional hierarchy or group. For dimensions for which the **Ask at launch** option has been enabled, the system will prompt the user through a window to select the value to be assigned to the parameter when the item or step containing it is executed.
Generate	Specifies if the report has to be generated
Distribute	indicates whether the report is to be distributed. Furthermore, by activating the option **Delete distributed files**, the files will not be saved in the distribution archive but will be deleted after sending.  In the event of errors during distribution, correctly generated but not sent files will be available in the generated files window for later distribution.
Recipients	Shows the list of the email's recipients separated by comma. To insert a recipient, you can either: • enter an email address (name@email.com), or • select the user or a group of users among those available in the list

Property	Description
Distribution format	Format of the reports to be distributed. Possible values : • *Pdf (default value)* • *Original*. the report format is *.xlsx* for "Form" report type and of *.docx* type for "Document"report type • *Txt*. Valid only for "Form" Report type. • *XML*. Valid only for "Form" Report type.  The *Txt* format exports only the first sheet of the Form. The columns are separated by tab characters, the cell formats are kept for the numbers and the dates, and also any hidden rows or columns are exported. The *XML* format works in a similar way but the first sheet of the generated Form must contain a text corresponding to a valid XML. The generated XML can be validated by a XSD file. If the validation has negative result, it is possible to choose not to distribute the report.
XSD document	Name of the necessary XSD file for the validation of the XML file. In order to upload the XSD document, click on the *Select file* button and navigate the file system to select the file. After uploading the file, it is possible to remove it by using the **Delete** button. This option is enabled only if the **Distribution format** is *XML*.
Send not valid	The validation process does not affect the generation of the document. Therefore, a correctly generated document might not be a valid XML document. By default, it is not possible to distribute an invalid document. By enabling this option, it is possible to override the distribution of the document. This option is enabled only if the **Distribution format** is *XML*.

Property	Description
File name	Name of the generated files (it must be a value comprised between 1 and 255). The inserted value can be: ◦ *constant* (File's name), or ◦ *calculated* automatically by the system in order to show in the file's name some specific information about the report and / or the configured filters. For this purpose, the user must use a specific syntax for each information he wants to display. For further details about the syntax of the parametric elements that can be included in the file name, please see paragraph **Parametric elements syntax**  When defining the file name, the system does not check: • **the number of inserted characters:** If this value exceeds 255 characters, during the distribution phase different reports may have the same name and, therefore, could be overwritten • **the validity on the inserted characters:** If the file name contains one of the characters listed below, then the system will replace each character with the Underscore symbol (_) during the generation and/ or distribution phase. • ◦ Asterisk (*). ◦ Vertical bar (|). Backslash (\). Forward slash (/) ◦ Colon (:). ◦ Quote ("). ◦ Less than (<). ◦ Greater than (>). ◦ Question mark (?). ◦ Space. This character is replaced only during the distribution phase.

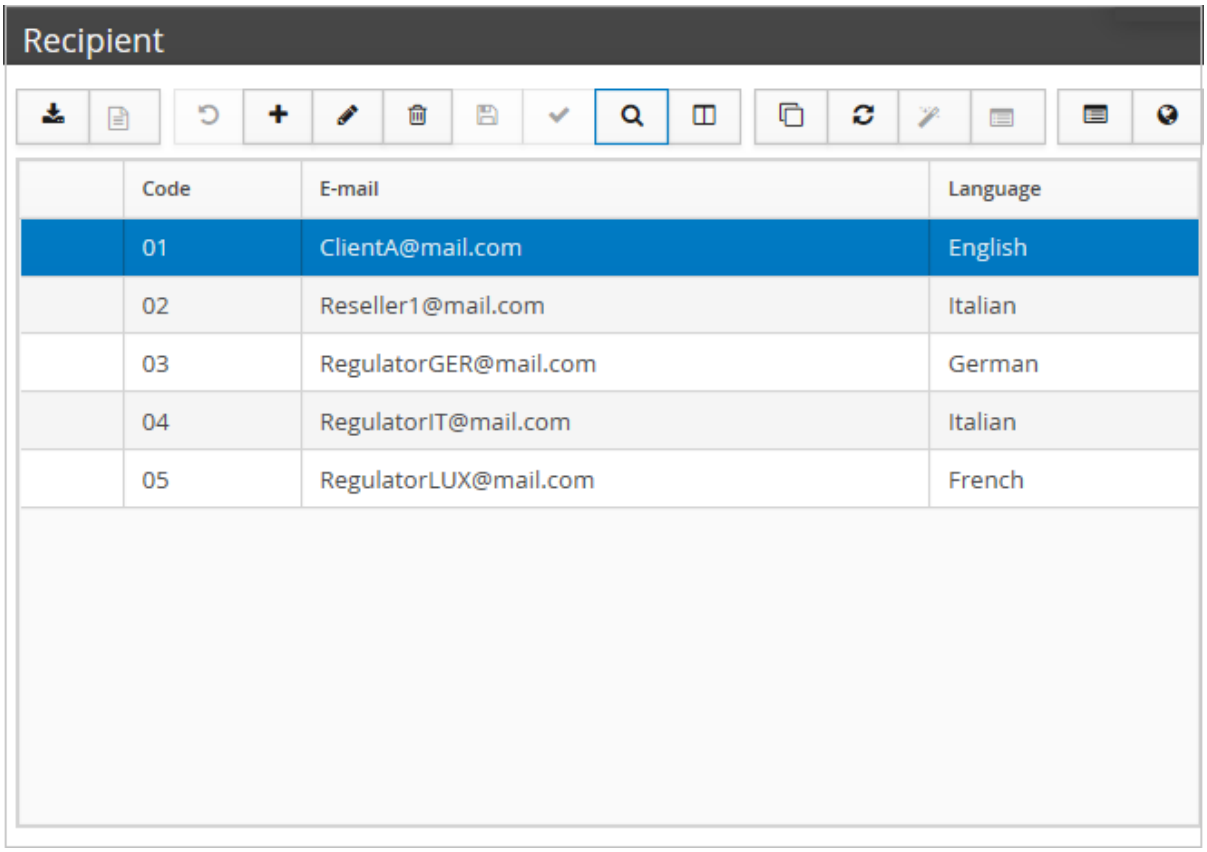
Definition

After creating the report, it is possible to distribute, either by step or by item, the generated files by an automatic email. For this purpose, it is necessary to define:

- the email recipients
- a possible group of recipients, in order to organise the defined recipients
- the body and subject of the email (Email template) the recipients will receive upon completion of the distribution data processing

Recipients

The definition of the recipients can be accessed from the *Recipient* menu available after clicking on the "Report bursting" button on the user Homepage.



By default the window displays the list of recipients sorted by code and no other types of sorting can be used. The toolbar in the top of the window allows running the functions listed below:

Button	Description
	Export Excel Exports data from the grid to an Excel file
	Add This button opens a window containing two tabs: • **Attributes** in which it is possible to add the following details for the recipient: ◦ a *Code* ◦ a *Description*, ◦ an *Email* address, and ◦ the *Language* of the email body and subject • **Grouping** in which it is possible to relate the recipient to a node of recipients groupings defined in the window
	Edit Sets the window in "edit" mode and allows the user to modify the attributes of an existing recipient
	Delete Deletes the selected recipient When a recipient is deleted, its email and code will be replaced with "------" in the Audit tables (online and offline)

Button	Description
	**Search** Opens a filter window for defining which data have to be displayed  The columns you can filter are: *Code* and *Language*
	**Select columns** The button opens a window containing all the fields that can be displayed as columns. The fields with a tick on the left are displayed in the window, while those without the tick are hidden. Click on the left hand corner of the field to hide it or display it.
	**Duplicate** Duplicates the selected record

Recipients group

CCH® Tagetik allows you to define an aggregation structure of "grouping" type in order to sort the recipients according to your needs. The system provides a predefined aggregation structure with code $. Then the user can customize the grouping structure according to his needs and specify the nodes to which each new recipient must be linked. The grouping's nodes are used to simplify:

- the management of the distribution of the reports
- the selection of a group of recipients

The definition of the grouping's nodes can be accessed from the *Recipients group* menu that becomes visible after clicking on the "Report bursting" tile on the home page.

Email template

The definition of the email template can be accessed from the *email template* menu available after clicking on the "Report bursting" button on the Homepage.

Each template is identified by a **Code**, a **Subject** and an **Email body**.

In the email's subject and body it is possible to insert some parametric parts of text in order to report the values of the filters of the item that generated the Email or insert, as in the case of the email's body, a table summarizing the content of the attached files. For further details about the syntax of the parametric elements that can be inserted in the email's subject and body, please see "Parametric elements" on the facing page

 The body of the email can be formatted according to the Html specs (in this case (in this case, it is possible to include only some Html tags in the email's body) or unencrypted. The accepted Html tags are the following: <b>, <i>, <u>,
, <ol>, <ul> and <li> (it is important to use always an opening tag <> and a closing tag </>).

 If not allowed tags are used, when sending the email, the system will delete such tags and keep the text between them.

Multilanguage management

When a template is used for several recipients having different languages, it is possible to send an email with subject and body translated in the specific language to each recipient. In order to define more languages in the same template, just select the template in the email template definition window and then access the **User Actions > Multilanguage Management**menu. After clicking on the menu, the system opens

the inestra "Multilanguage management" window in which it is necessary to define one template (*Object* and *Body*) for each language of the various recipients that will receive the template in header by email.

Parametric elements

The system allows you to use parametric text sections, in order to display the values of the filters configured when defining the item, and some specific pieces of information of the generated reports. In order to display such information, the user must use a specific syntax.

The following table shows the accepted parametric elements and their syntax:

Parameter	Syntax	Replaced value
Entity	${Entity}	Code of the deployed entity filter . For hierarchy node: hierarchy code - node code **EXAMPLE** For a deployed Entity node (e.g. 010, 020, 030) by writing ${Entity}, the system will translate this instruction into '010', '020', '030'
Entity description	${Entity.desc}	Description of the deployed entity node. For a hierarchy node: description of the hierarchy node
Entity attributes	${Entity.attrib1} .. ${Entity.attrib5}	Value of attribute 1,..,5 of the deployed entity filter. For a hierarchy node, the resulting value is null
Category	${Category}	Code of the deployed category filter. For a grouping node: grouping code - node code
Category description	${Category.desc}	Description of the deployed category filter. For a hierarchy node: description of a hierarchy node
Custom dimensions	${Dest1} ... ${Dest5}	Code of the custom dimension 1,..,5 deployed filter. For a hierarchy node: hierarchy code - node code
Custom dimensions description	${Dest1.desc} ... ${Dest5.desc}	Description of the custom dimension 1,..,5 deployed filter For a hierarchy node: hierarchy node description
Custom dimension attributes	${Dest1.attrib1} ... ${Dest5.attrib5}	Value of attribute 1,..,5 of the custom dimension 1,..,5 deployed filter For a hierarchy node the resulting value is null
Currency	${Currency}	Code of the Currency filter

Parameter	Syntax	Replaced value
Currency description	${Currency.desc}	Description of the currency filter
Scenario Year	${Year}	Year of the Scenario filter
Scenario Code	${Scenario}	Code of the Scenario filter
Period Code	${Period}	Code of the Period filter
Period end date	${Period.YYYY} ${Period.YY} ${Period.MM} ${Period.dd}	Period end date. The resulting value depends on the syntax used; more in detail: • ${Period.YYYY} the resulting value is yyyy (four digits year) • ${Period.YY} the resulting value is (two digits year) • ${Period.MM} the resulting value is MM (two digits month) • ${Period.dd} the resulting value is dd (two digits day)
Attributes of generated Filesi		
Attribute	**Syntax**	**Replaced value**
Language	${Language}	Language of the selected report
Date	${Date} ${Date.YYYY} ${Date.YY} ${Date.MM} ${Date.DD} ${Date.hh} ${Date.mm}	File generation and / or distribution date. The resulting value depends on the syntax used; more in detail: • ${Date} the resulting value is YYYY_MM_hh_mm (YYYYyear, MM month, hh hours, mm minutes) • ${Date.YYYY} the resulting value is YYYY (four digits year) • ${Date.YY} the resulting value is YY (two digits year) • ${Date.MM} the resulting value is MM (two digits month) • ${Date.DD} the resulting value is DD (two digits day) • ${Date.hh} the resulting value is hh (two digits hour) • ${Date.mm} the resulting value is mm (two digits minutes) Only in "File name", if the syntax used is ${Date}, the resulting value will be: DD_MM (DD day, MM month)
File name	${Filename}	Name chosen for the generated files The file name must contain less than 255 characters
Recipients	${Recipient}	Code of the email recipient

<table>
<tr><th>Parameter</th><th>Syntax</th><th>Replaced value</th></tr>
<tr><td>Table</td><td>{Table(${},..,${})}</td><td>Within the body of the email, it is possible to insert a table whose rows and columns depend on the parameters "${}" between brackets. It is possible to insert all parameters described above but the attributes and their descriptions.

EXAMPLE
For a deployed entity filter (e.g. 010, 020, 030) and for a Custom dimension 2 node (e.g. 01,01) with its description, the table will have three columns(one for Entity, one for Custom dimension 2 and one for its description) as shown in the image below:

<table>
<tr><th>Entity</th><th>CustDim</th><th>CustDim 2. desc</th></tr>
<tr><td>010</td><td>01,01</td><td>Account - Default view</td></tr>
<tr><td>020</td><td>01,01</td><td>Account - Default view</td></tr>
<tr><td>030</td><td>01,01</td><td>Account - Default view</td></tr>
</table>
</td></tr>
</table>

Where to insert the parametric elements

The parametric elements described above can be inserted in: Email body and subject, Table, File name and Directory. Exceptions to this are the following parameters:

- **Year** of the scenario; it cannot be used in "Tabla" and "Directory"
- **Recipient** of the email; it cannot be used in "File name"

and some attributes of the generated file. For the file's attributes, see the following table:

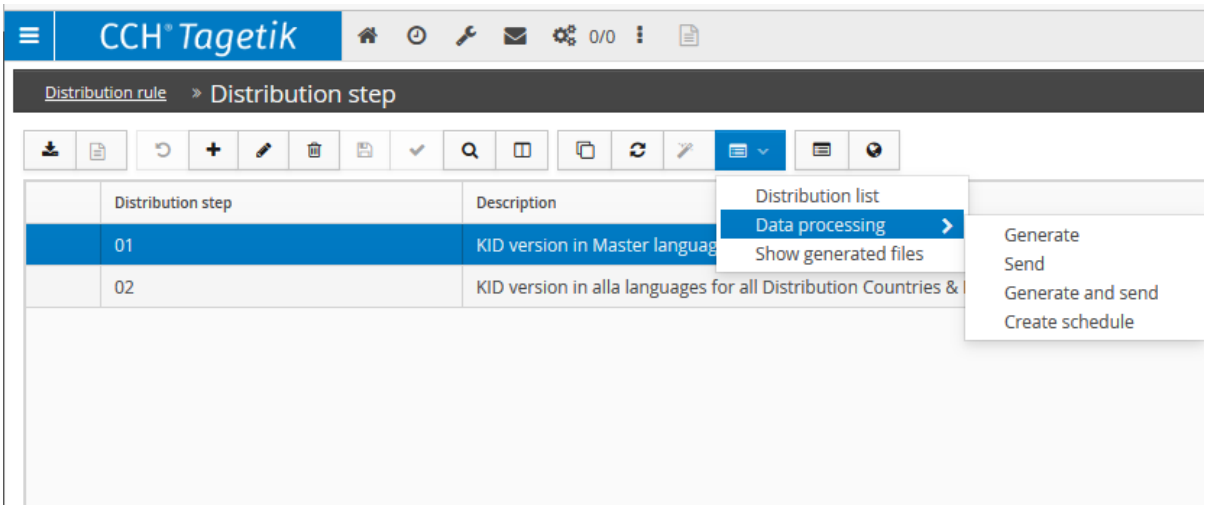
File attributes	Email object	Email body	Table	File name	Directory
Language	✓	✓	✗	✓	✗
Date	✓	✓	✗	✓	✓
File name	✗	✗	✓	✗	✓
Recipients	✗	✗	✗	✗	✓

Run

Three types of data processing can be run, by step or by item, once the configuration of the "Distribution process" is completed:

- **"Generate" on the facing page**This data processing generates massive amounts of reports from a Collaborative Office Document or a Web Form previously defined. The massive generation of reports from a standard template, allows for a dynamic creation of many reports. For example, it is possible to generate a report by Entity without the need to prepare an "ad hoc" copy per Entity and then modify its filters and update its content. In this example, the "Generate" data processing runs the templates on the basis of a predefined list of entities, configuring every time the filters, updating data and saving the reports. This reduces the time of the necessary activities for the generation and maintenance of reports
- **"Send" on the facing page** This data processing sends, by email or FTP, the generated files
- **Generate and send.** This data processing generates and sends in a single run the two above

The data processes can be scheduled. For further details on the scheduling, see paragraph "Create schedule" on page 401



Below, we describe only the Generate and Send data processes by item since they are equal to the data processes by step.

Generate

In order to generate the reports related to an item, it is necessary to:

- access the distribution list window
- select the item for which the reports must be generated
- click on the **Generate** button accessible from *User actions* > *Data processing* > *Generate*

The system will open the *Generate* window in which it is necessary to:

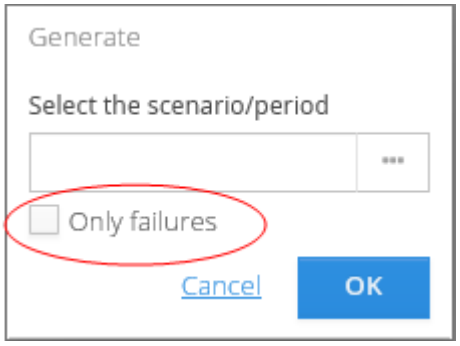
- select the values for the dimensions requested at runtime for which the reports will be generated
- click **OK** to run the data processing

After clicking on **OK**, the system runs the Generate data processing that:

- creates a new report as a copy of the template selected when defining the item
- calculates the filters for the report update
- updates the document with the calculated filters
- generates one report for each deployed filter of the distribution list
- validates the XML document according to the XSD document (only if the distribution format is XML and at least one XSD document is linked to the distribution Item)
- saves the generated report displayed in the window

Once the generation of the reports is completed, an error may have occurred while running the data processing and, consequently, some reports may have not been generated. In order to generate again ONLY the reports failed in the first execution, it is possible to enable the **Generate only failures** option in the Run window

In case of Web Form, if filters are applied, the form does not generate any sheet; no error is generated and the report is not saved. This is highlighted in the audit.



Send

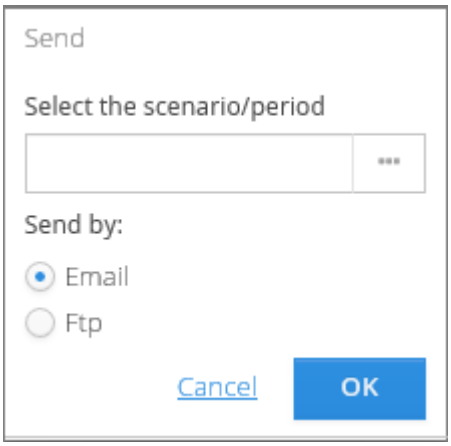
In order to send the reports for an item, it is necessary to:

- access the **Distribution list** window and:
- select the item for which the generated files shall be sent, and
- click on the **Send** from *User Actions* > *Data processing* > *Send*. The system displays the *Send* window in which you have to select:

- the values for the dimensions requested at runtime for which you want to send the reports
- chose the file sending mode:
  - *Email*, to send the zip files generated via email. Before sending the files via email, define the recipients, the body and subject of the email.

    In the configuration of the Repository database (Mail section - Maximum size of attachments in MB from "Report Bursting module" option), it is possible to limit the maximum size (in MB) of the zip file generated by the system (0 means unlimited). If such size is exceeded, the system compresses the original files in several archives (in zip format), so that they can be sent separately without exceeding the set limits.

  - *Ftp*, to save the generated files into an external source (the files are saved individually and not in zip format). Before sending the file via Ftp, define the external source in which to save the files when creating the rule
- click **OK** to run the data processing



After clicking on **OK**, the system runs the send data processing that:

- if the selected send mode is **Email**:
  - calculates the list of email recipients
  - according to the recipients' language, calculates the templates (email's subject and body) to use in the emails
  - calculates the parameters to fill the templates, and
  - sends an email with a zip file attached containing the files generated for each set of recipients/template language/parameters
- if the selected send mode is **Ftp**, it saves the generated files into the specified external source

Attention:

- the sender of all emails is the executor of the distribution data processing. The email of the sender is determined by the system according to the values of the E-mail information defined in the users table of the Repository. If this information has not been defined, then the system does not send any email and inserts a detail row with the message *"The email address of the sender wasn't defined"*
- the distribution of an invalid XML document is allowed only if, when defining the distribution list, you have chosen to override its distribution by enabling the *Send not valid* option
- with equal files, each new distribution data processing on FTP overwrites the files saved into the external source

Create schedule

Tagetik includes a scheduling mechanism which allows planning the execution of any data processing that must be run according to a predefined frequency. Thereby:

- the user won't have to run any time-consuming data processing that will be carried out, for example, in a period of lower activity
- frequent data processes will be run on schedule

The generate / send report data processing can be scheduled by accessing the distribution steps definition window through the *User actions* > *Schedule*path

After selecting the desired values for the dimensions requested at runtime and choosing the data processing type to schedule, the system inserts the scheduled data processing in the list of Scheduled items (accessible from the administrator user web interface by following the path *Repository* > *Scheduling* > *Scheduling* > *Scheduled items*). In order to set all the scheduling parameters, it is necessary to access the "Actions /Edit" window (for further details, see **PDF Scheduling**).

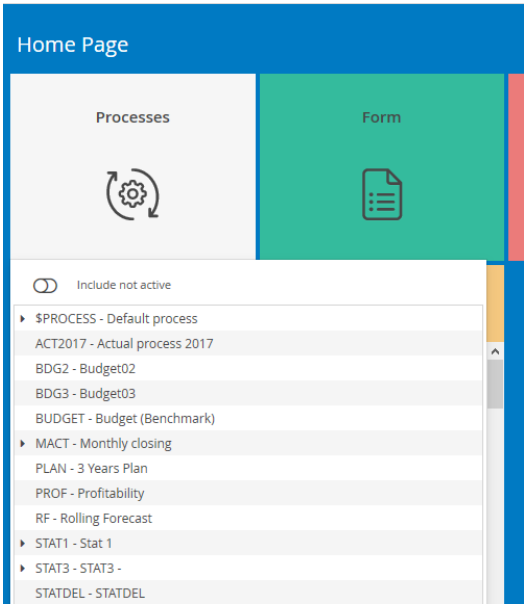
The data processing is labelled by default as "Invalid". In order to make it valid, it is necessary to access the "Scheduled items" window and then enable the "Valid" option; only after this modification, the scheduler will be able to take charge of the data processing execution.

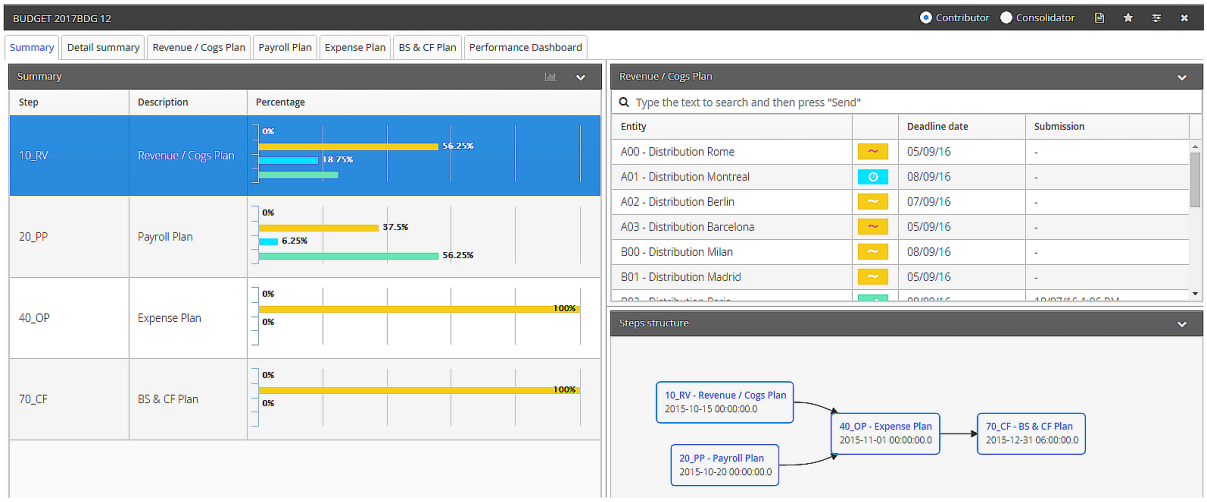
Run report from process cockpit

Like any other process of Tagetik, the distribution process can be structured in multiple steps which allow the user to:

- run all the activities under his responsibility
- monitor the progress status of the activities of the entities assigned to him
- generate / distribute all documents for each entity

In order to generate / distribute a document from the process cockpit, first the user must access the end-user web interface, click on the **Processes**activity button on the home page and then select the desired process by double clicking on it. After selecting the process, the system displays the process cockpit (for further details on the cockpit, read the manual "Process cockpit").





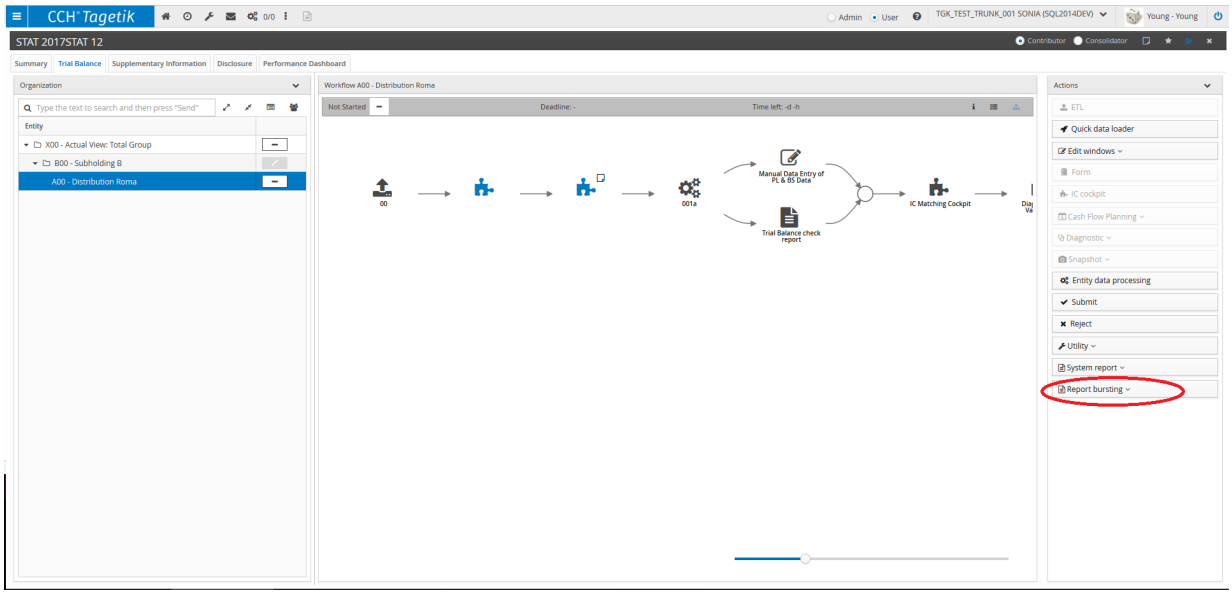
The process cockpit is divided into several tabs. The "Overview" tab shows the summary information about the status of the data collection / submission and a "tab for each step". A user that accesses the process cockpit can see ONLY the set of entities (entities and nodes) for which he/she is authorized according to the user rights defined by the administrator. This means that the cockpit has a different structure according to the user who runs it.

- The type and the number of entities (entities and nodes) displayed inside the cockpit depend on the user rights and on the process settings
- the type and the number of activities / data processes available depend on the process rules settings

Once the process cockpit has been displayed, in order to generate a report it is necessary to:

- select a tab of the process step in order to open the related area
- select the desired entity (hierarchy node or single lowest level element) among those available in the *Organization* area
- click an option among **Generate**/**Send**/ **Generate and Send** available from the *Report bursting* drop down menu 

The system will open a window in which the user can select the step or list of items for which the selected data processing shall be run. It is possible to select only steps/items belonging to the same rule. If the process is not single submission, then the reports will be filtered for the scenario/period selected when opening the cockpit.



At this point the system opens a wizard in which as a first step the system asks the user to select for which step or list of items belonging to the same rule to execute the selected data processing.

In the next step, the user must select the values with which to filter the dimensions required at launch when defining the element or, if no definition is required at launch, execute the data processing directly, specifying whether there is need to generate only files that failed in a previous run.

If the process is not a single submission and the dimensions of Scenario, Period and Entity are required at launch, the system displays the default values for the dimension filters, those selected in the navigation of the cockpit and the user can then decide to modify them and launch with new values. In cases where the default values are not part of the restrictions defined for the parameter, then no report is generated with these filters applied and this case is highlighted in the Audit.

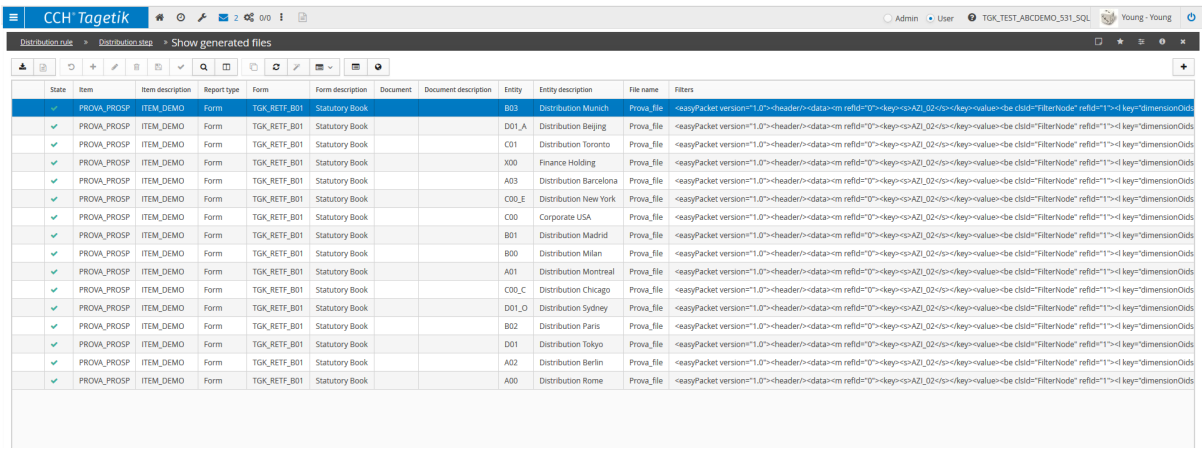
Reports check

After running the generation data processing for rule, step or item, it is possible to:

- display the generated files (this display can be accessed by Rule, Step and Item)
- check the result of the data processing. In fact, each operation is tracked, therefore it is possible to reconstruct the result of the data processing by scenario / period, as well as what has been done, when and by whom. This allows you to:
  - analyse any warning messages / errors
  - analyse the modifications made and the makers of such operations
  - make a summary of what happened during a given period of time

Show generated files

The *Show generated files* window can be accessed from the *User actions > Show generated files* menu. This menu is available the Rule, Step and Item definition windows.



The information available on the window varies according to the access mode.

For example, if it is displayed by rule *(Distribution rule > User actions > Show generated files)* then, in addition to the above mentioned information, the system will display the **code** and **description** of the generated steps and items.

In detail, such information includes

Information	Description
Status	Displays through a traffic light the status of the data processing: ◦ *Green*, Data processing completed without errors ◦ *Yellow*, Data processing successfully completed but with Warning messages, therefore some checks may have not been passed
Validation date	Date in which the validation of the XML file was run
Validation state	Result of the validation check on the XML file. Possible statuses: ◦ *Valid.* Document successfully validated ◦ *Warnings are present.* Document validated but one or more warning have been reported ◦ *Invalid.* The document did not pass the validation
Validation message	Message containing the result of the validation check in case the document did not pass the validation or one or more warnings have been reported
Code and **Description** Distribution step	Code and description of the distribution step
Item **Item description**	Code and description of the item
Report type	Type of generated report. Possible values: ◦ *Form* ◦ *Document*
Form **Form description**	ID code and description of the form
Document **Document description**	ID code and description of the Collaborative Office document(document type code|document code | document activity code
Scenario	Code of the generation scenario
Period	Code of the generation period

Information	Description
Entity **Entity description**	Code and description of the entity used as a filter in the generated file
File name	Name of the generated files
Filters	*Entity, Category, Custom dimensions and Currencyused as filters in the generated file*
Creation date	Date in which the file was generated
Sent e-mail	Specifies that the created file has been sent
Inconsistent filter	Specifies that the filters of the created file differ from the filters of the generating item
Warning	Detail of the warning message

Whenever an item or a list of items is regenerated and the Send option is enabled for the regenerated item, the system:

- stores offline the files present in the archive of generated files whose date is prior to or equal to the date of the last generation
- removes the offline files from the generated files storage
- saves the regenerated files

Running the regeneration multiple times may lead to having many old files. To reduce the usage of the disc and simplify and speed up the files management, it is possible to delete the operations in the generated files storage by using the **Delete** button on the button bar of the storage window.


The deletion of the files from the storage causes the deletion of the log of old generated files whose date is prior to or equal to the date of the last generation.

Actions

Show files storage

The storage of the old generated files is accessible from the *Show files storage* window *(Show generated files > User actions > Show files storage)*. This window displays, sorted from the most recent to the oldest regeneration date, the list of old regenerated files according to the selected file.

Download

The generated files can be downloaded by the **Download** option in the *User actions* menu accessible from the *Show generated files*window. After selecting the files to download and then clicking on download, it is necessary to select one of the following distribution formats:

- *Original (default value).* The system downloads the file in ".xlsx" format if *Report type* is *Form* or in ".docx" format if *Report type* is *Collaborative Office Document*
- *Pdf*
- *Txt.* Valid only if *Report type* is *Form*

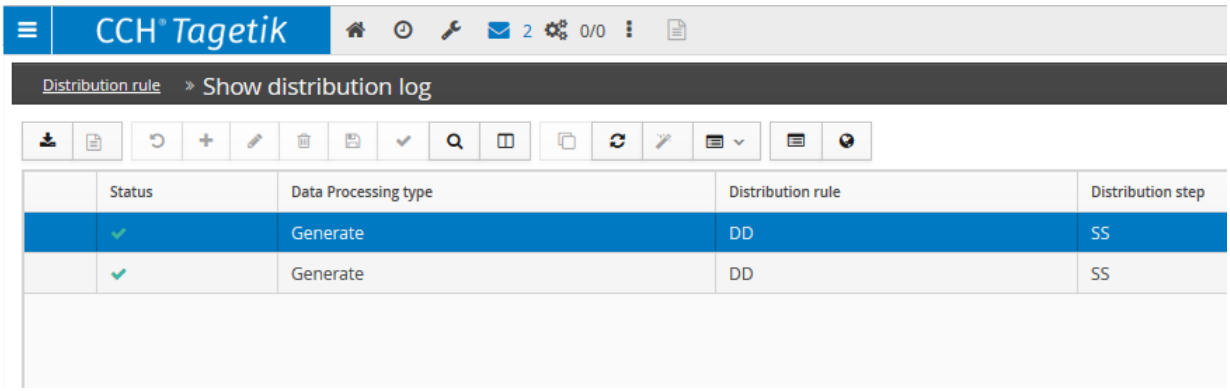
After selecting the format, by clicking the **Download** button, it can be saved locally. The system displays a dialog window in which you can choose whether to open it immediately (*Open*) or save it into the Downloads folder (*Show downloads*).


The file's name is specified in the **File name** field and is defined when creating the item. In case of multiselection, the system downloads one single file in .zip format named "GeneratedReport_scenario code_period code_YYmmDDhhMM.zip". This file must be extracted by using Winzip or a similar tool. Once the extraction is completed, the folder set for the copy will contain one file for each report, each one identified by *File name*.

The Txt format exports only the first sheet of the Form. The columns are separated by tab character, the cell formats are kept for numbers and dates and any hidden rows or columns are exported as well.

Show log

The check of the distribution data processing can be run by distribution rule by accessing the *User actions (* *) > Show distribution log* menu.



For each step of the selected rule, the window displays the following information:

- **State**. Displays through a traffic light the status of the data processing:
  - *Green.* Data processing completed without errors
  - *Yellow.* Generation successfully completed but with warning messages, therefore some checks may have not been passed
  - *Red.* Error occurred during the data processing

  In case of warning/error, by accessing the **Detail** window (*User actions > Detail*) it is possible to display for each warning/error some information about the data processing run and the warning/error **Message**.
- **Data processing type.** Data processing type run (*Generate*, *Send: Email* and *Send: Ftp*)
- **Distribution rule.** ID code of the rule
- **Step.** ID code of the step
- **Item.** ID code of the item
- **Scenario.** Code of the scenario defined when generating the report
- **Period.** Code of the period defined when generating the report
- **Data Processing date.** Data processing run date

In the *Show distribution log* for *Generate* data processing, the system inserts a row for each item run (in case of run by step, one for each item of the step). On the other hand, in case of *Send* data processing, the system inserts only one row per execution and the **Item** information will be empty in case of run by Step.

Features not supported in Web Form 2.0 compared to Excel add-ins

Spreading

Spreading does not support advanced syntax (e.g. "+10%").

Matrix Layout options

The column autofit function is supported 'by value' but not 'by template'. The 'from template' option is ignored in web forms, the 'from value' option is always used.

Drill Down

In Drill Down, only row/column drill down is supported.

The Cell Drill down is partly covered by the ad-hoc analysis.

Slicer

Slicers have not been implemented in Web Forms 2.0.

It is recommended to use the filter panel of web forms to change parameter values at runtime, and the Cell Field objects to display parameter values within the form.

Autofit rows

The row autofit function is supported for LBL forms. However, it is not supported in multidimensional forms.

Autofit Column

Autofit rules for columns 'from template' ('From Value' is supported)

Validation checks

Watch Expressions in matrices (both outside and inside the matrix) are supported but are implemented differently than in Excel add-ins.

The *when saving* and *when closed* flags do not work in Web Forms 2.0 as expressions are continuously evaluated and messages about violated expressions are displayed in a box in the top right-hand corner. Breach of the blocking expressions, shown in red in the box, prevents data from being saved.

Tagetik Hyperlink

Tagetik Hyperlinks are supported.

The reference to the Excel cell must only refer to cells in the current sheet.

When opening the form, transferring parameters via Excel cell is not supported.

Features not supported in Web Forms

Below is a list of functionalities that are not supported by Web Forms.

This is not to be intended as n exhaustive list. in general, Web forms are based on a completely different technology, architecture and codebase than Excel add-ins. For this reason, **it is not guaranteed that forms executed as Web forms will support the same features and behave the same way as they do when executed using the Excel add-in.**

- Macros or VBA objects.
- Work offline.
- Report list.
- Reporting Cell-based (TGKSend, TGKRet, ecc.).
- Hidden comments.
- Hidden sheets.
- Save / Load formulas.
- Show formulas used.
- Reference to Spreadsheet Cell in Query Form (e.g. using the syntax ${A1} to refer to the spreadsheet cell A1 in the selection).

Excel formulas

List of Excel Formulas supported in Web forms 2.0:
http://help.grapecity.com/spread/SpreadJSWeb/webframe.html#FormulaFunctions.html

Cell Formats and Locale

The decimal and thousand separators for numeric formats depend on the nationality set in the Tagetik user configuration.

The date/time formats depend on the country set in the Tagetik user configuration. It cannot be guaranteed that the data and number format **will always be the same as that applied by MS Excel**. Here are some examples of such differences:

- For Japan, the 'short date' format, unlike MS Excel, always uses two characters for the month (e.g. Excel '1976/8/30', but Web form 2.0 '1976/08/30');
- For the U.K. and Italy, the format '30/8/76 16:28', unlike MS Excel, always uses two numbers for the year (e.g. Excel '30/8/1976 16:28', but Web form 2.0 '30/8/76 16:28');
- For Poland, the 'long date' format may use alternative names for months and days of the week, different from those used by MS Excel;
- For Turkey, the thousand separator is a space instead of a dot;
- The 'short date' and 'long date' formats with '*' are not dependent on the user's locale.
- Formats with a fixed locale set from Excel's Design area (such as 14-Mar-18) are adapted to the user's locale, except when the fixed locale is American (EN_US), Chinese (ZH_CN), or Japanese (JA_JP)

Validation Query

If elements generated by the query that can be selected in cells include numbers only (e.g. "01", "02", etc.), the cell formatting type must be "General" or "Text", otherwise, when the value is selected, the system could remove the zero and treat the selected value as a number. In contrast, the Excel client always treats the selected value as text. .

Excel table defined in the Query result area

If an Excel table is defined within the query result area, the Excel table will not be present in web forms.

Single Template and Multi-Template

The use of 'named ranges' in formulas or charts is not recommended when using sheet filters, as they may not be correctly evaluated in spreadsheets following the first one in each template (e.g. formulas with names generated by Tagetik using custom queries).

Charts may lose some of their formatting and colour palette (if the standard palette is not in use) in the sheets following the first one.

Managing Dictionary Items Outside the Matrix

Dictionary items have not been implemented, outside the Matrix, in web forms 2.0

Edit monthly ICs in Web Forms

The modification of monthly ICs for AIH matrices was not implemented in Web Forms 2.0.

Multidimensional forms - saving

Saving data in multidimensional forms when non-incremental saving is active (all dimensions are at the highest level of detail)

Download PDF

Web form can only be downloaded as .xlsx file; downloading PDFs has not been implemented in Web Forms 2.0.

Matrix Cell Properties for overridden contents

The Cell Properties functionality is not supported if the cell content is overridden.

Intelligent Analytics

Introduction

Intelligent Analytics(AI) allows users to use Excel and web dashboards to easily navigate CCH® Tagetik data

***IMPORTANT**: AI is available on the current and previous two versions of CCH Tagetik. On earlier versions, backward compatibility is not guaranteed.*

Numerical data is stored in AI structures in double precision floating point format.

Functional Requirements

To access data via Intelligent Analytics, you must create a **Reporting View** from the Dataset containing the data you wish to view (see Reporting View Analitical Information Hub) or from the Financial Workspace (see Financial Reporting View).

***IMPORTANT**: AIH Reporting Views can only be created on Analytical Workspaces that have been updated according to the Version Update Window of the partitions of an Analytical Workspace*

To access dimensional hierarchies via Intelligent Analytics, the option **convert to flat structure** must be activated on the desired hierarchies, see Add new hierarchy window.

Restrictions

Reporting views cannot be created on AIH Datasets that:

- contain at least one dimension field with Personal data protection option
- contain at least one field relating to the dimension Azienda_CTP and/or DEST2_CTP
- contain only text and/or date fields

***Note**: text and date fields within datasets that also contain other field types will not be part of the Reporting View.*

The Reporting View Registry

Introduction

Reporting View Registry (RVR) allows you to create, update and delete financial reporting views.

You can create a Reporting View from the Financial Workspace.

***IMPORTANT**: see Reporting View for information on managing analytical reporting views*

Reporting View Registry page

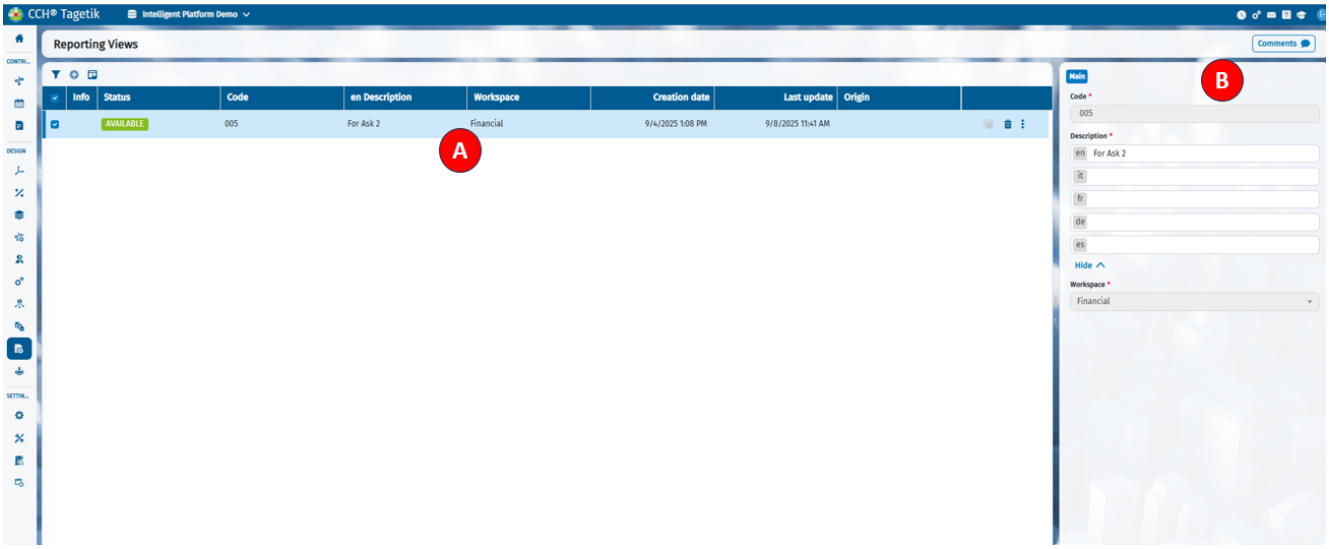
User with Reporting View management rights

CCH® Tagetik > Reporting > Reporting views

Purpose of the page

This page shows the data of existing reporting views and allows you to manage update operations.

What it looks like



Element	Description
A	List of existing reporting views
B	Main section with reporting view fields

Buttons on the page

Element	Description
	This filters the data displayed on the page.
	Adds a row in order to insert a new element. Disabled in this version as there cannot be more than one financial reporting view
	Opens the pop-up to manage the columns displayed on the page
	Selects a reporting view and opens the main section
	Deletes the reporting view
	Refresh: updates the data of a reporting view **Retry**: restarts the last failed operation for a reporting view **Reset**: restores a reporting view in case of error

Page fields

Please find below the most important fields.

Field	Description
Main Section	
Code	Unique reporting view code
Description	Description of the reporting view
Workspaces	Reporting view workspace
Reporting Views Section	
Status	It can take the following values: • **online** • **creating** • **deleting** • **updating** • **offline** • **timeout** • **error**

Managing Reporting Views

CCH® Tagetik > Reporting > Reporting views

Editing a reporting view

This procedure allows you to edit a reporting view.

1. Select the reporting view check box  : the Main section pops up
2. on the right of the screen Edit the desired fields

Note: the edit operation is only available when a reporting view is in **ONLINE** status.

Deleting a Reporting View

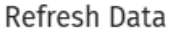
This procedure allows you to delete a reporting view.

1. Click  : the confirmation pop-up opens
2. Click  to confirm the operation.

   ***Note**: The delete operation is available when the reporting view is in ONLINE or OFFLINE status*

Updating the data of a reporting view

This procedure allows you to update the data of a reporting view.

1. Click
2. Select 

   ***Note**: The update operation is only available when a reporting view is in ONLINE status*

Rerunning a reporting view in case of errors

This procedure allows you to rerun the last operation performed on a reporting view, if an error or timeout has occurred.

1. Click
2. Select 

   ***Note**: The Retry operation is available when the reporting view is in ERROR or TIMEOUT status*

Restoring a reporting view in case of errors

This procedure allows you to restore a reporting view if an error occurs during the update. This procedure is useful for errors that cannot be fixed by retry.

1. Click on
2. Select **Reset Data**

   ***Note**: The operation is available when the reporting view is in ERROR status and a restore is expected.*

Dashboard: introduction

Define

A **dashboard** is a workspace where **visuals** can be added to make data analysis easier.
Within the workbench section, a new dashboard can be created, see Working on a Dashboard

Homepage dashboard

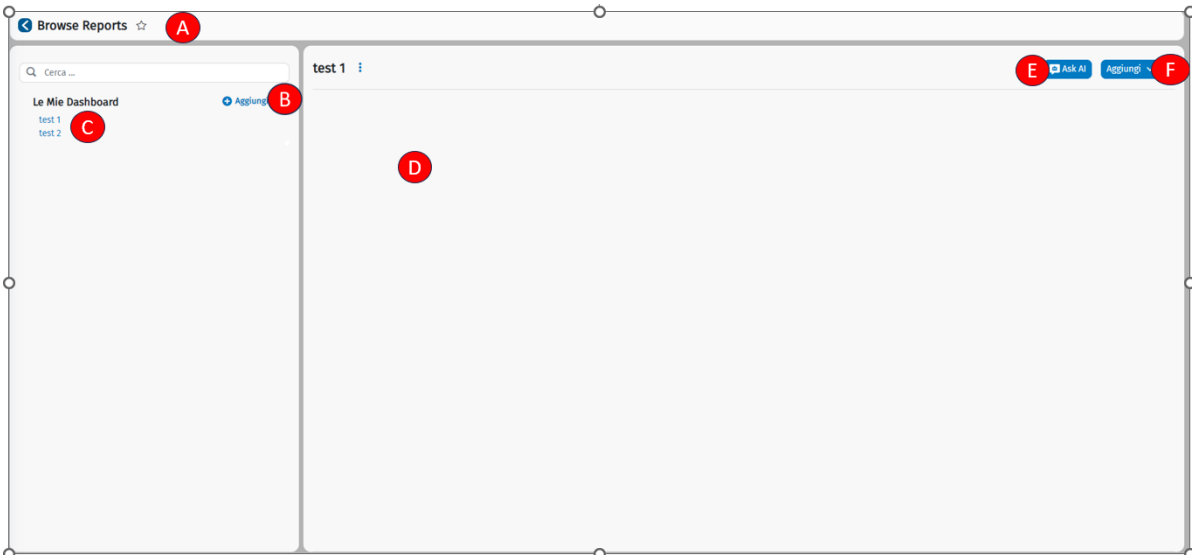
User User with display rights on workspace, Reporting view, Dimensions

CCH® Tagetik > Reports & Analysis > Dashboard Browse All > Add New > New Dashboard

Purpose of the page

A dashboard can be managed on this page.

What it looks like



Element	Description
A	Title of the page "Browse Reports"
B	'Add new' button
C	List of dashboards created
D	Section of the dashboard where all charts are visible
E	ASK AI button
F	Add button

Dashboard tab

Field	Description
Name	Textbox containing the name of the dashboard to be created

Element	Description
A	Dashboard
B	Dashboard name
C	Cancel button
D	Save button

Working on a dashboard

Create a dashboard

CCH® Tagetik > Reports & Analysis > Browse All > Add Add New

CCH® Tagetik > Reports & Analysis > (My Dashboards section) > Create Dashboard

1. Enter the **name of** the dashboard in the Name * en field of the dashboard creation window
2. Click Save

   Note: the same operation can be performed by following the path CCH® Tagetik > Reports & Analysis > Dashboard Browse All
3. Clicking Browse All takes you to the workbench/dashboard section where you can see the list of all dashboards and the button

Edit a dashboard

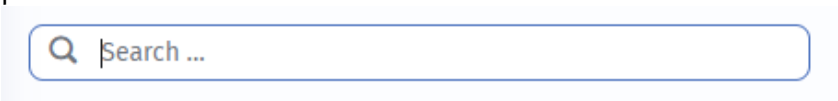
CCH® Tagetik > Reports & Analysis > Browse all dashboards > Select a dashboard > ⋮ > Edit

1. Edit the name of the dashboard in the field Name * en
2. Click Save

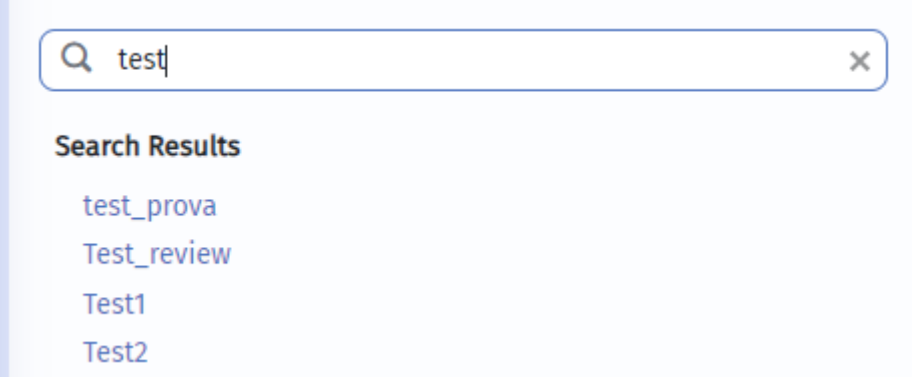
Search for a dashboard

CCH® Tagetik > Reports & Analysis > Browse all dashboards > Search a dashboard

1. place the mouse in the **search** field of the dashboard in the left section of the workbench



2. type the **name** to be searched for to display the search result

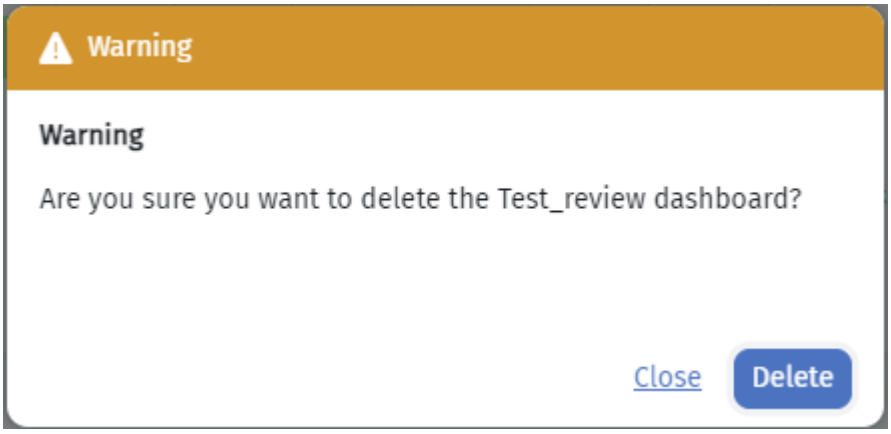


Delete a Dashboard

CCH® Tagetik > Reports & Analysis > Dashboard Browse All > select Dashboard > ⋮ > Delete dashboard > pop-up confirm delete

1. Select a dashboard from the list
2. click ⋮

3. Click **Delete**
4. In the pop-up window, confirm the deletion by clicking **Delete**



Visual: introduction

Define

The visual is a graphic object that allows data to be displayed by retrieving them from the reporting view. For each dashboard in each user's workspace, it is possible to create, edit and display visuals.

***IMPORTANT**: Each visual must be set up by choosing configuration parameters from existing reporting views. see Setting up a visual*

Home page Visual

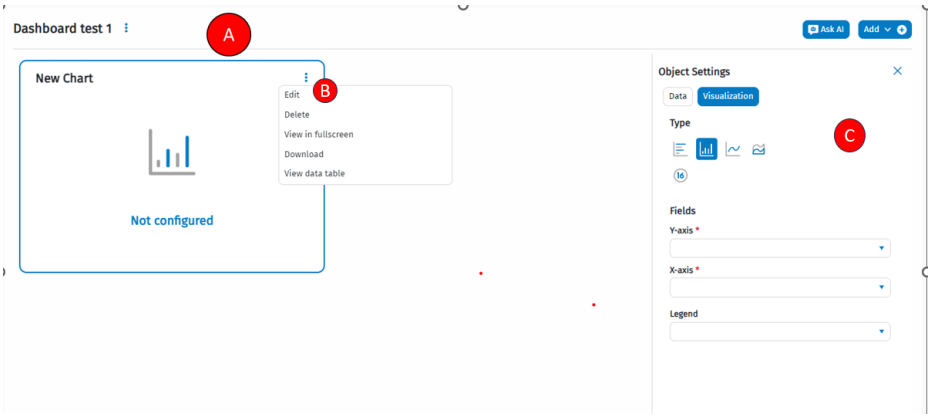
User User with display rights on workspace, Reporting view, Dimensions

CCH® Tagetik > Reports & Analysis > Dashboard Browse All > select Dashboard >

Purpose of the page

On this page you can manage the visual menu and set the Object Settings

What it looks like



Element	Description
To	Dashboard section
B	Visual action menu
C	Object setting

Actions on this page

Actions can be reached via the icon
For further details on the procedures, see Working on visuals

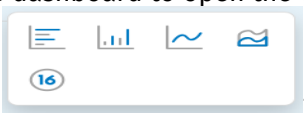
Element	Description
Edit	allows editing the visual
Delete	allows deleting the visual
View in fullscreen	allows displaying the visual in full screen
Download	allows downloading the visual in png, jpeg, pdf, svg, csv, xls format **IMPORTANT: only available for visuals of Bar/Area/Line /Column type already correctly configured.**
Display Data table	allows displaying the Data table of a chart

Working on visuals

Create a visual

CCH® Tagetik > Report & Analysis > Browse all > Select a dashboard > Add > visual button

1. Click on **Add** button 

on the dashboard to open the **visuals list** pop-up page
2. Click on the **icon of the desired visual**  to open the Object settings window
3. Configure Object Settings Tab Data, see Object Settings Tab Data
4. Configure Object Settings Visualization Tab , see Object Settings Tab Visualization

Moving a visual

CCH® Tagetik > Report & Analysis > Browse all > Select dashboard > select visual

1. Click on the visual
2. *drag and drop* it to the desired position

   ***Note**: Each visual created will be 4x4 in size and will occupy the first available space of that size*

Resize a visual

CCH Tagetik > Report &Analysis > Browse all > Select a dashboard > Select a visual

1. Select a **visual**
2. Click on corners or sides of the visual to enlarge or reduce the size

***IMPORTANT**: the visual size adapts to the zoom of the web page*

Edit a visual

CCH Tagetik > Report &Analysis > Browse all > Select a dashboard > Select visual > ⋮ > Edit

1. Click **Edit**
2. Add link **Object Setting Data Tab** and add link **Object Setting Visualization Tab**

Delete a visual

CCH Tagetik > Report &Analysis > Browse all > Select a dashboard > Select visual > ⋮ > Delete

1. Click **Delete** to open the visual deletion window
2. Click again on **Delete** to confirm the deletion
3. Click **Close** to cancel

View a chart in full screen

CCH Tagetik > Report &Analysis > Browse all > Select a dashboard > Select visual > ⋮ > View in full screen

1. Click **View in Full Screen**
2. Click **Esc** to exit full screen mode

***IMPORTANT**: only available for already configured visuals of Bar/Column/Area/Line type*

Download a visual

CCH Tagetik > Report &Analysis > Browse all > Select a dashboard > Select visual > ⋮ > Download

1. Choose the file format
2. Click on the format type

***IMPORTANT**: only available in **png, jpeg, pdf, svg, csv, xls formats**.*

***IMPORTANT**: only available for already configured visuals of Bar/Column/Area/Line type*

Displaying the Data Table of a Chart

CCH Tagetik > Report &Analysis > Browse All > Select Dashboard > Select Visual > ⋮ > View Data Table

1. Click on **view data table**
2. Click on **Hide Data Table** to hide the data table and display the chart

***IMPORTANT**: only available for already configured visuals of Bar/Column/Area/Line type*

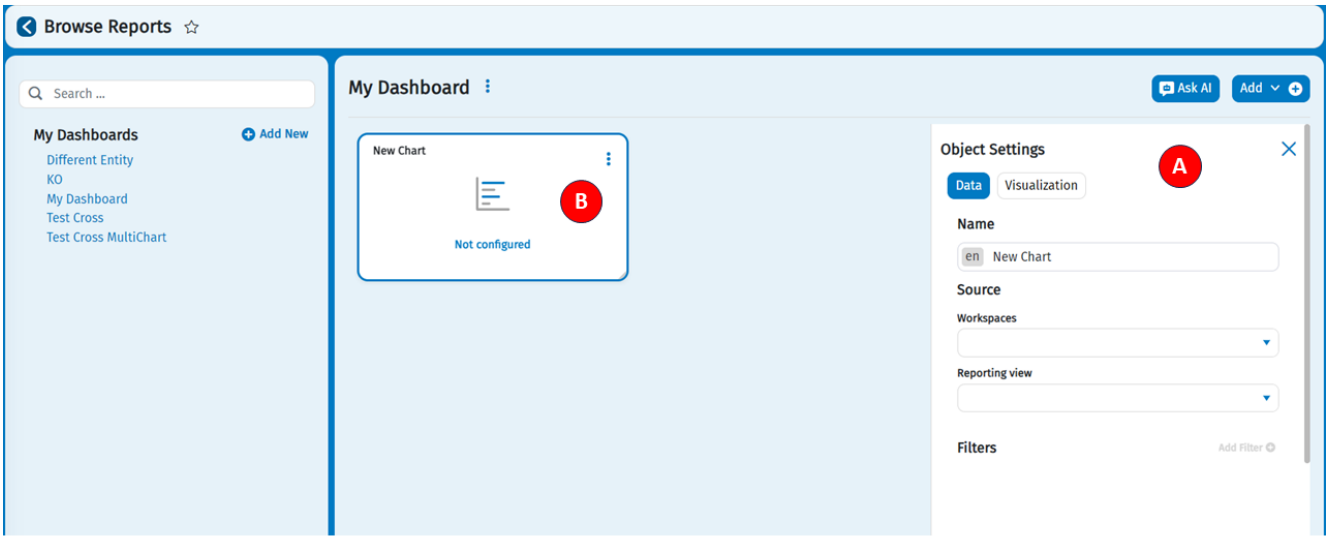
Object Setting: Data Tab

CCH® Tagetik > Report & Analysis > Select a dashboard >  *> visual > Data Tab*

Purpose of window

The Object Setting section allows you to configure the chosen visual. In the **Data** tab, you can enter the **workspace**, the **reporting view** and any **filters.**

What it looks like



Element	Description
A	Visual configuration tab
B	Card visual

Fields in window

Please find below the most important fields.

Element	Description
First name	Includes the title of the visual
Workspaces	Workspace selection menu
Reporting view	Reporting view selection menu
Filters: 	Filter section open button

Configure Object Settings: Data Tab

Configure Data Tab Object Settings

CCH® Tagetik > Report & Analysis > Browse all > Select a dashboard > Add > visual button

1. Enter name of **visual**
2. Expand **workspace** list if desired
3. Expand **Reporting View** list and choose one
4. Add any filters, see procedures on visual filters

Working on visual filters

Adding a filter on the dimension

CCH® Tagetik > Reports & Analysis > Dashboard Browse All > select Dashboard or Add New > Add > Chart Type > Add Application Database Filter

1. click Add Filter
2. select the dimension from the pop-up window that opens

Select filter nodes and/or lowest elements

CCH® Tagetik > Reports & Analysis > Dashboard Browse All> select Dashboard or Add New > Add > Chart type > Add Application Database filter > ⋮

1. Click ⋮
2. select **hierarchy**
3. Select **nodes and/or lowest level elements**

Note: Selections within the same filter are combined, while selections across different dimensions intersect

Filters Tree lookup for nodes and lowest elements selection

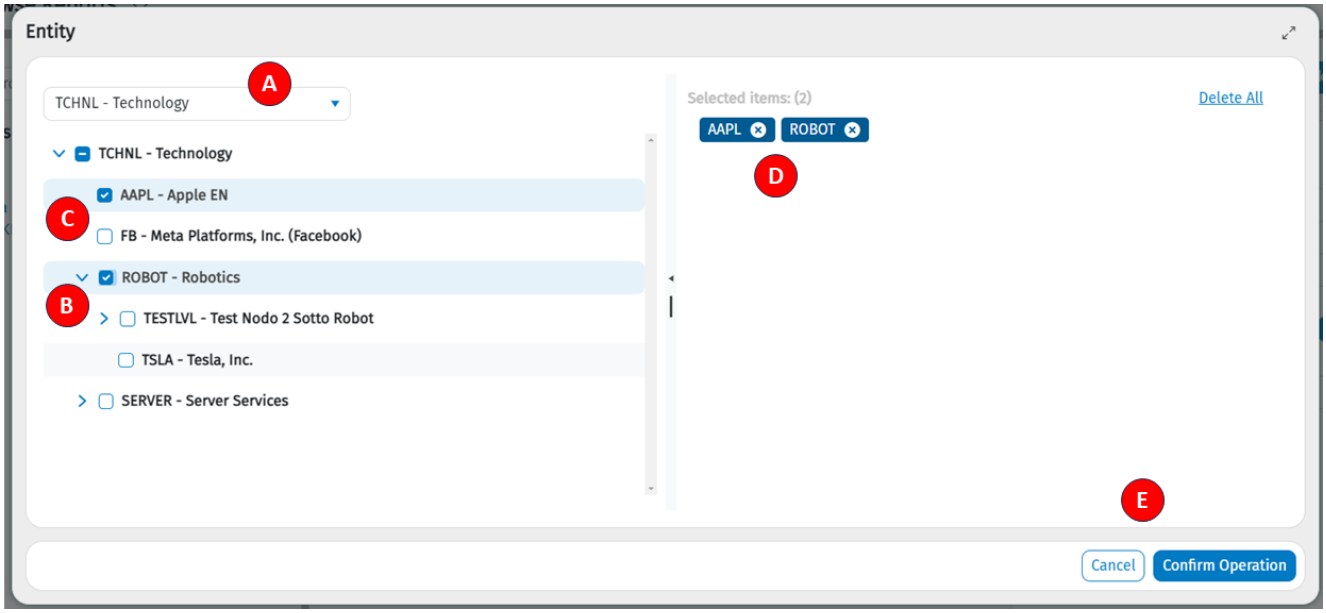
User

CCH® Tagetik > Report & Analysis > Dashboard Browse All > Select a dashboard or Add new dashboard > Add button on the dashboard > Chart type button > Add filter > Three dot button lookup

Purpose of the page

The filters tree look up allows you to select the nodes and lowest elements of a filter hierarchy.

What it looks like



Element	Description
A	Selected Hierarchy
B	Hierarchy node
C	Hierarchy lowest level element
D	Nodes and lowest elements already selected for the filter
E	Confirmation or deletion buttons

Buttons and/or links on the page

Element	Description
FB - Meta	Lowest elements and Nodes selection Check-box
AAPL	Button to delete previously entered filters
Confirm Operation	Operation confirmation button
Cancel	Operation cancellation button
Delete All	Removes all added filters

Object Settings: tab Visualization bar chart

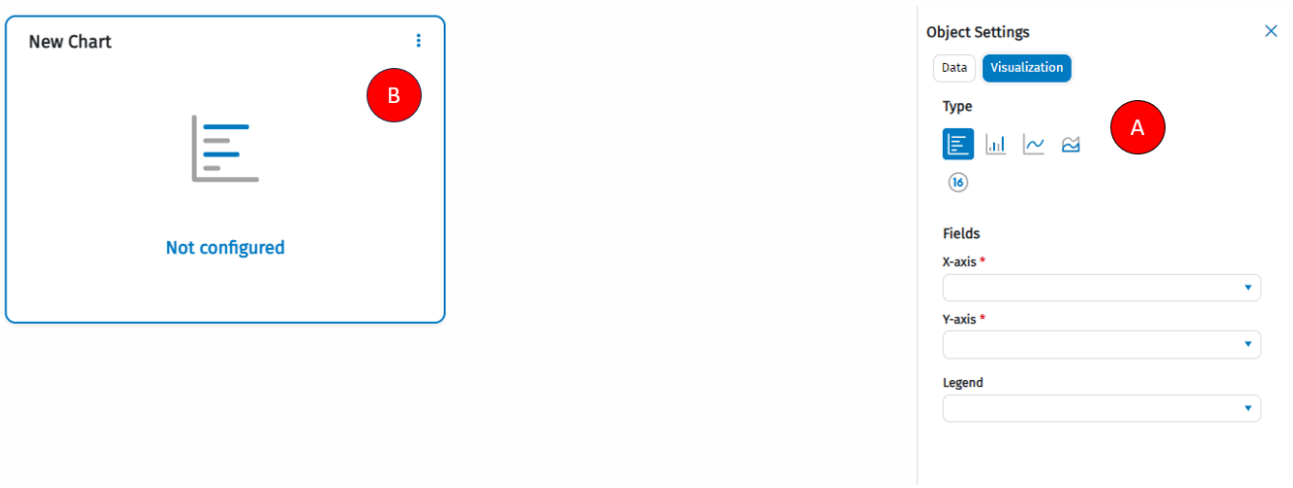
User with display rights on workspace, Reporting view, Dimensions

CCH® Tagetik > Report & Analysis > Select a dashboard > Add button on the dashboard > visual button >

Purpose of window

The Object settings section allows you to configure the chosen visual. In the bar chart Visualization bar, you can select the X-axis, the Y-axis and the legend.

What it looks like



Element	Description
A	Visual configuration tab
B	Visual card

Fields in window

Field	Description
X-axis *  X-axis is mandatory	Bar Chart Object Measure
Y-axis *  Y-axis is mandatory	Bar Chart Object Dimension
Legend	Dimension to be inserted into the legend

Configure Visualization Tab Object Settings

Configure Visualization Tab Object Settings: Bar Graph

CCH® Tagetik > Report & Analysis > Browse all > Select a dashboard > Add > visual button > Visualization Tab

1. Select the **measure** from the **x-axis** field list
2. Select the **dimension** from the **y-axis** field list

***IMPORTANT**: The **x-axis** and **y-axis** fields are mandatory: if they are not selected, the system displays a warning message and a red asterisk.*

***Note**: the legend field is optional: you can select the dimension to be entered from the list in the menu.*

Configure Visualization Tab object settings: columns/rows/area chart

CCH® Tagetik > Report & Analysis > Browse all > Select a dashboard > Add > visual button > Visualization Tab

1. Select the **measure** from the **y-axis** field list
2. Select the **dimension** from the **x-axis** field list

***IMPORTANT**: The **x-axis** and **y-axis** fields are mandatory: if they are not selected, the system displays a warning message and a red asterisk.*

***Note**: the legend field is optional: you can select the dimension to be entered from the list in the menu.*

Configure Visualization Tab Object Settings: KPIs

CCH® Tagetik > Report & Analysis > Browse all > Select a dashboard > Add > visual button > Visualization Tab > KPIs

1. Select the **measure** from the list
2. Insert **symbol** field if desired
3. Insert **Decimals** field if desired
4. Insert **Scale Factor** field if desired
5. Insert **Scale Notation** field if desired

Window Object Settings column / line / area charts

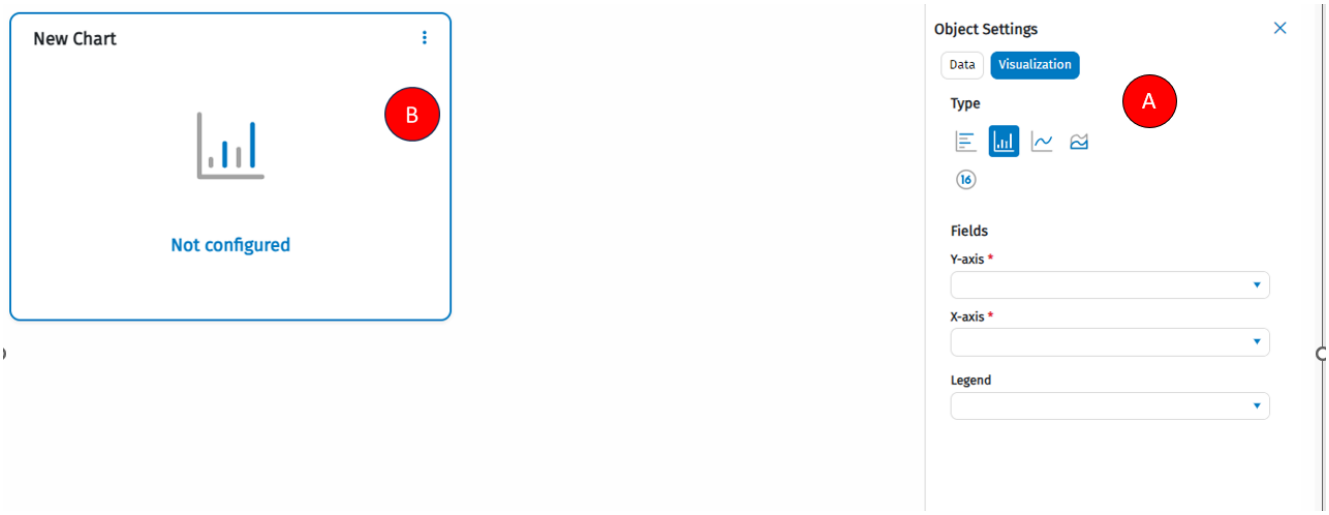
User with display rights on workspace, Reporting view, Dimensions

CCH® Tagetik > Report & Analysis > Select a dashboard > Add button on the dashboard > visual button

Purpose of window

The Object settings section allows you to configure the chosen visual. In the Visualization tab of the chart, the Y-axis, X-axis and legend are chosen.

What it looks like



Element	Description
A	Visual configuration tab
B	Visual card

Fields in the page/section

Field	Description
Y-axis * Y-axis is mandatory	Bar Chart Object Measure
X-axis * X-axis is mandatory	Bar Chart Object Dimension
Legend	Dimension to be inserted into the legend

Setting the Display Tab Object: KPI

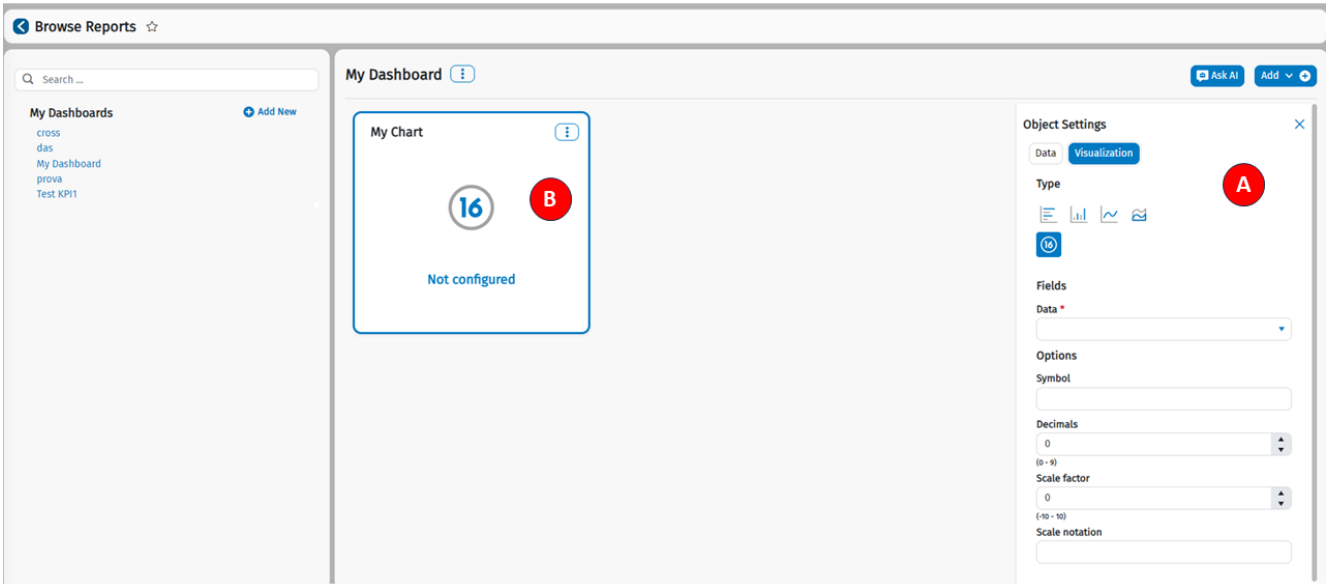
User with display rights on workspace, Reporting view, Dimensions

CCH® Tagetik > Report & Analysis > Select a dashboard > Add *> visual > Display tab*

Purpose of window

The Object Setting section allows you to configure the chosen visual. In the KPI Display tab, you can select the measure and, optionally, the data fields, **symbol, number of decimals, scale factor, and scale notation**.

What it looks like



Element	Description
A	Visual configuration tab
B	Visual card

Fields in section

Field	Description
Data	KPI Measurement
Symbol	Possible symbol to precede the measure in the visual
Decimal	Number of decimal digits to be displayed in the visual. Only whole numbers between 0 and 9 inclusive are allowed.
Scale factor	Scale factor for measure display. Only whole numbers from -10 to 10 are allowed. The value from the measure in the visual is shown by dividing by 10^n where n is the value of the scale factor
Scale notation	Possible text to be placed after the measure in the visual. Up to 5 characters are allowed.

Measure Properties Page

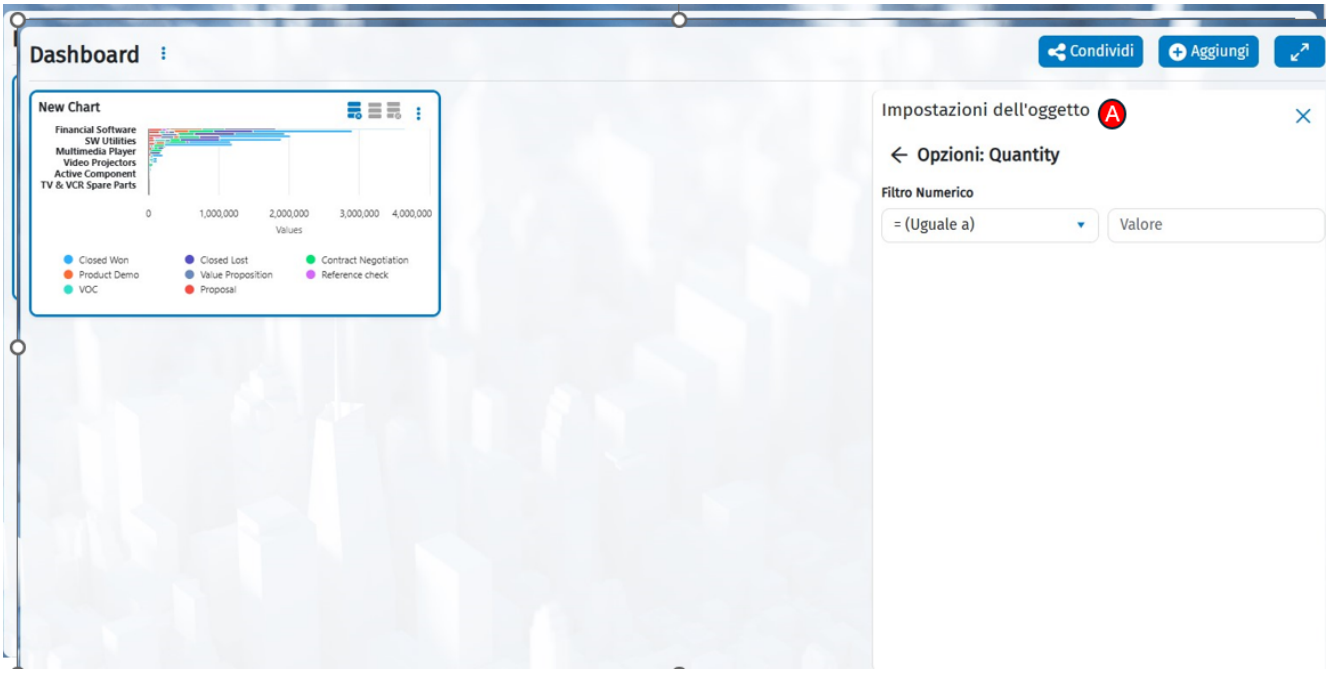
User with display rights on Workspace, Reporting view, Dimensions

CCH® Tagetik > Report & Analysis > Select a dashboard > Add button on the dashboard > select a chart > click on tab Data > click on a wheel

Purpose of the page

On this page, you can select the 'measure' filter

What it looks like



Element	Description
A	Measure Setting Configuration Panel

Buttons on the page

Element	Description
	This filters the data displayed on the page.
	Adds a row in order to insert a new element.

Page fields

Please find below the most important fields.

Field	Description
	Sets the filter type
	Sets the numeric value

Working on measure properties

Setting the Measure Filter

CCH® Tagetik > Report & Analysis > Select dashboard >  *> select a chart*

1. To set a measure filter, click on : the Object Settings panel pops up
2. Select a Filter type from the Measure Filter field
3. Set one or more values

***Note**: you can choose between the following operators (=, <>, >=, >, <=, <, not in, in)*

***Note**: the filter set will be displayed below the measure in the Visualisation tab*

Dimension Properties Page

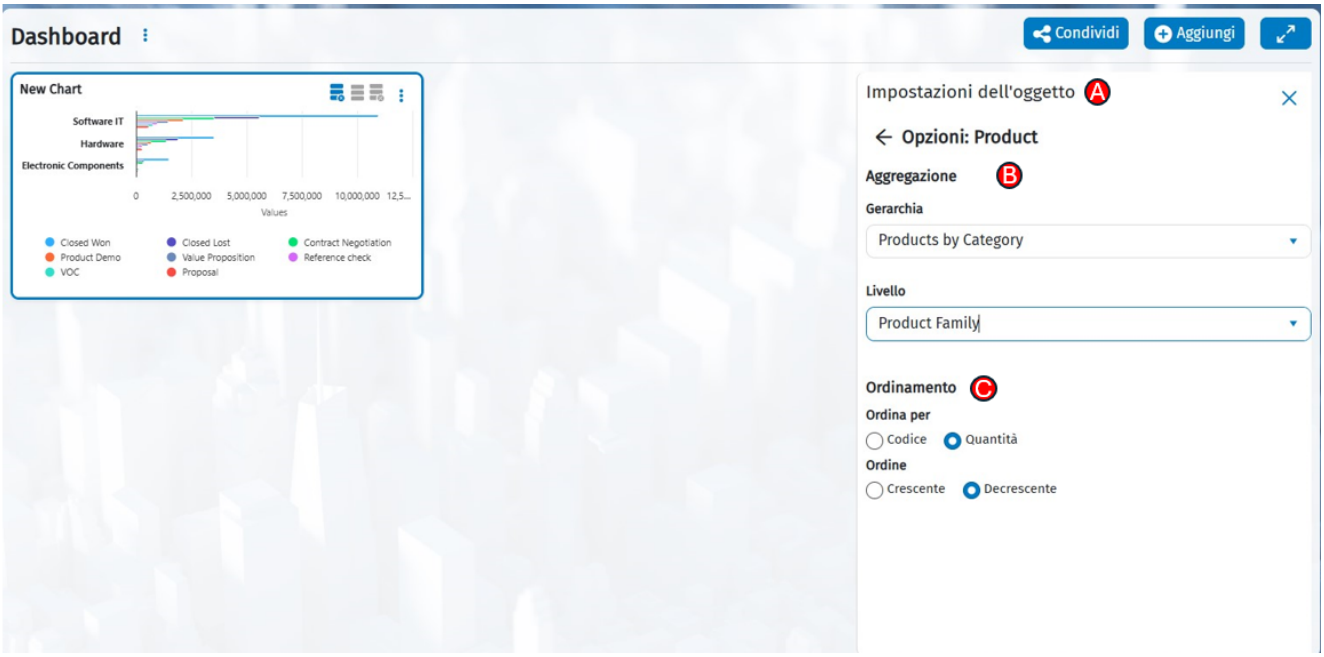
User with display rights on Workspace, Reporting view, Dimensions

CCH® Tagetik > Report & Analysis > Select a dashboard > Add dashboard button > select a chart > Visualisation tab > wheel

Purpose of the page

This page allows you to configure dimension properties such as hierarchy, sorting.

What it looks like



Element	Description
To	Dimension settings configuration page
B	Hierarchy Configuration Section
C	Sorting Configuration Section

Page fields

Please find below the most important fields.

Field	Description
Hierarchy	Hierarchy selected on dimension
Level	Hierarchy Level selected on dimension
Sort by	Sort selection by Code or Value
Order	Ascending and Descending Sort Selection

Cross Filtering

Cross Filtering allows you to select a node/lowest element on a chart and apply the filter to other visuals in the same dashboard for a quick access to the information

It is possible to interact with a graph as follows:

- Selecting one or more elements of the legend
- Selecting one or more bars/columns (bar chart and column chart)
- Selecting one or more lines/areas (Area Chart, Line Chart)
- Select one or more Chart Labels

Interaction with legend elements

The selected legend element(s) on a chart are applied as a temporary filter on all other visuals on the same dashboard along with already active filters.

Interaction with bars/columns

The bars and columns selected on a chart are applied as temporary filters on all other visuals on the same dashboard along with other temporary filters.

Interaction with Areas/Lines

The lines and areas selected on a chart will be applied as a temporary filter on all visuals on the same dashboard along with the temporary filters already applied.

Interaction with chart labels

The label selections on a chart will be applied as a temporary filter on all visuals on the same dashboard along with the filters already applied.

Example 1

Examples	*Description*
Dimensional element	*Legend element Line*
	Area
	Columns/Bars (if no legend)
	Chart Label
Element consisting of a pair	*Columns/Bars (in presence of legend)*

***IMPORTANT**: Selected elements are **highlighted and the ot**hers are greyed out.*

***IMPORTANT**: You can select one or more elements by holding down the **CTRL key and des**elect them by clicking on a different element or on the elements themselves.*

***IMPORTANT**: The system shows an **icon** on the visuals on which cross-filtering is applied. Visuals that do not have the dimension within the reporting view are not affected by cross filtering.*

Drill Down

Introduction

The Drill down is a procedure for displaying information at a greater level of detail by deploying the children of the selected hierarchy node.

When configuring the visual, it is therefore necessary to specify a hierarchy for the dimension(s) to drill down.

This interaction has a temporary effect, whereby reloading the page returns to the initial state on all visuals

Drill Down on a Single Element

To drill down, double-click on a hierarchy node within a visual, such as the element description on the axis or legend, a bar, a line or an area, depending on the type of visual selected.

If both a dimension and a hierarchy are set on the axis and the legend, drilling down on the element representing the dimension pair (e.g. a column or a bar) will deploy the nodes of both dimensions.

For each visual, you can return to a previous state by navigating the entries that are added to the top left of the visual after each drill-down operation. Use the Home button to return to the initial state.

Working on cross-Filtering procedure

CCH Tagetik > Report &Analysis > Dashboard Browse all> select dashboard or add new dashboard> select chart

Interacting with dimensional element or matching a dimensional pair

This procedure is useful for interacting with **dimensional elements** or **elements that correspond to a pair of elements** and applying the temporary filter to all **visuals** on the dashboard.

1. Click on a **dimensional element** or an element that corresponds to a dimensional pair to apply the filter.

   ***Note**: the selected element is highlighted and all others are deselected (grey)*

2. Click on the same or a different element to **deselect** an already selected element or dimension pair.

***Note**: an icon will appear on all visuals with the temporary filter applied*

***Note**: by Default all elements will be highlighted, no filter will be applied to the chart*

Interacting with multiple elements that correspond to a pair of elements or dimensional elements simultaneously

This procedure is useful for interacting with one or more **dimensional elements** or **elements that correspond to a pair of elements** and applying the temporary filter on all **visuals** on the dashboard.

1. Click on **elements** or **elements that correspond to a pair of elements**
2. Click on a new element by pressing the **CTRL** key to **simultaneously** select one or more dimensional elements or dimensional pairs

   ***Note**: the selected elements are highlighted and all others are deselected (grey)*

1. Click on the selected item or on a deselected item **without pressing CTRL** to **deselect** an item.

***Note**: an icon will appear on all visuals with the temporary filter applied*

***Note**: by default all elements or pairs of elements will be highlighted, no filter will be applied to the chart*

Ask AI for Reporting

Ask AI, by means of specific user requests, allows users to display reporting data via visuals from input data contained in existing Reporting View, see Reporting View

Ask AI returns a visual type

From a user request made using Ask AI chat, the most plausible type of visuals can be obtained based on the question asked.

The type of visual that is identified and displayed is one of the visuals supported by the application.

***IMPORTANT**: Ask AI chat is only available for Reporting if function **0004257 - ASK AI into intelligent analytics dashboard** is enabled (only for system roles **$ALL**, **$AUDIT** and **$USER**).*

***IMPORTANT**: If the function is disabled, the Ask AI button for opening the chat is not visible (see Working with Ask AI Chat)*

When the user's request, entered either manually or by voice command, is sent, a new panel pops up showing the identified and configured visual type. See Making a textual request in Ask AI chat and Making a request in Ask AI chat with voice command.

Identified visual types

The visual is configured and displayed, in different types, only if the measure (Data field) requested, or one measure and at least one dimension (X-Axis or Y-Axis field) belonging to an existing Reporting View are identified. The possible cases are shown in the table below:

If...	...then
only one measure, or a measure and filters are identified	the Visual identified and displayed will be of the **KPI** type
a measure and at least one dimension, or the measure, filters and at least one dimension are identified	the Visual identified and shown will be one of the other types of **visuals supported** by the application

Fields in the 'Object Settings' panel after the Ask AI answer

Once the user's question, entered either manually or by voice command, has been submitted, and an existing Reporting View is identified containing elements (measure, dimensions and dimension values) required in the question asked, Ask AI returns a valid answer, the content of which is displayed through the fields of the **Object Setting** window

If...	...then
a Reporting View is identified and contains at least one required measure	the Analytical Workspace and Reporting View fields are configured in the **Object Settings** window in the **Data tab** and in the **Visualization** tab the X or Y-Axis field is filled with the measure description
a Reporting View is identified and contains at least one required measure used to fill the X-Axis or Y-Axis field	in the **Object Settings** window in the **Visualization** tab the identified visual type is selected
a Reporting View is identified containing at least one required dimension, which is a flat Dimension	the X or Y-Axis field is filled in the **Object Settings** window in the **Visualization** tab with the description of the identified Dimension
a Reporting View is identified containing at least one required dimension, which is a Hierarchy Dimension	the X or Y-Axis field is filled in the **Object Settings** window in the **Visualization** tab with the description of the identified Dimension

If...	...then
A Reporting View is identified containing at least one measure and contains a Hierarchy Dimension as well as its required level in the question The levels of the hierarchy required in the question asked and identified refer to the level descriptions in the window, see Defining levels of a hierarchy	the X- or Y-Axis field is filled in the **Object Settings** window in the **Visualization** tab with the description of the identified Dimension, and the hierarchy description and the required hierarchy level are entered in Options
A Reporting View is identified containing at least one dimension as well as a second required Dimension	the Legend field is configured with it in the **Object Settings** window in the **Visualization** tab with the same rules as in the previous cases for the first Dimension in the Reporting View depending on whether it is a flat or hierarchy Dimension
A Reporting View is identified containing at least one measure but no required Dimension	in the **Object Settings** window in the **Visualization** tab neither the X or Y Axis field nor the Legend field are filled with the description of the Dimension
A Reporting View is identified containing at least one measure and one or more Dimensions with one or more required values	the Filters field in the **Object Settings** window on the **Data** tab is configured with the identified Dimension(s) and its (their) values, see Setting Filters

Fields in the 'Object Settings' panel for Financial Reporting View or containing the 'Account' dimension after the Ask AI response

When a user query requesting a value for the 'Account' dimension is submitted, and an existing Reporting View is identified that contains elements such as the Amount measure, Entity currency amount, Consolidation currency amount, dimensions including the Account dimension, and the dimension values requested in the query, Ask AI returns a valid response, which is displayed through the fields in the **Object Settings** window

If...	...then
a Reporting View is identified with a measure that is a value of the **Account** dimension	In the **Object Settings** window, under the **Data tab**, the Workspace and Reporting View fields are configured, and the measure that represents a value of the **Account** dimension is set as a filter in the Filters field for the Account dimension. In the **Display** tab, the X- or Y-Axis field is populated with the description of the Amount measure, Entity currency amount, or Consolidation currency amount (depending on what is requested in the query)
a Reporting View is identified with a measure that is a value of the **Account** dimension andcontains a required dimension	in the **Object Settings** window under the **Data tab,** the Workspace and Reporting View fields are configured, and the measure that represents a value of the **Account** dimension is set as a filter in the Filter field for the Account dimension .In the **Visualization** tab, the X- or Y-Axis field for the measure is populated with the description of the Amount measure, Entity currency amount, or Consolidation currency amount (depending on what is requested in the query). The X- or Y-Axis field for the dimension is populated with the description of the identified dimension, and the Legend field is populated with the **Account** dimension
a Reporting View is identified with a measure that represents a value of the Account dimension and contains a second required Dimension	in the **Object Settings** window under the **Data tab**, the Workspace and Reporting View fields are configured, and the measure that represents a value of the Account dimension is set as a filter in the Filter field for the Account dimension . In the Visualization tab, the X- or Y-Axis field for the measure is populated with the description of the Amount measure, Entity currency amount, or Consolidation currency amount (depending on what is requested in the query). The X- or Y-Axis field for the dimension is populated with the description of the identified dimension, and the Legend field is populated with the second requested dimension

IMPORTANT: If the identified Reporting View contains dimensions (other than Account) that are hierarchical and/or include their required levels, they are handled and configured in the same way as in the previous case (see Fields in the panel: 'Object Settings' after Ask AI's response)

IMPORTANT: If the identified Reporting View contains one or more Dimensions with one or more requested values, they are handled and configured in the same way as in the previous case (see **Fields in the panel: "Object Settings" after the Ask AI answer)**

User choice through multiple RV options

When the user's query, entered manually or via voice command, is submitted and multiple existing Reporting Views are identified that contain the elements (measure, measure as an account value, dimensions, and dimension values) requested in the query, Ask AI interacts with the user through the chat by displaying several selectable options or the possibility to rephrase the request:

- Each option is displayed with a Title containing a code 'ReportingViewCode' and a description 'ReportingView description'
- Up to three selectable options are shown, although more Reporting Views are identified
- Each option can be selected, allowing the user to choose which Reporting View's graph to display
- When one of the available options, each associated with an identified Reporting View, is selected by the user by clicking on it, the corresponding graph is displayed and the 'Object Settings' panel opens and is configured as described in the previous sections (see links to sections: 'Fields in the panel: "Object Settings" after Ask AI's response' and 'Fields in the panel "Object Settings" for financial Reporting Views or those containing the Account dimension after Ask AI's response')
- A further option 'Try asking again' is available, allowing the user to rephrase the request
- When the user selects the 'Try asking again' option by clicking on it, the **message** prompting the user to enter a request and indicating the **language** in which a request can be recorded via voice command is shown again, and the text box is enabled to allow the user to type a new request.

Chart title

After the user enters and submits a question containing one or more elements between measures, dimensions and filters, when AskAI returns a valid answer, the title of the chart shown is constructed based on the identified elements

If in ASK AI's reply AI is identified...	...then the title of the chart will be
at least one measure or one measure and one or more filters	**Measure**
at least one measure and also one Dimension and one or more filters	**Measure for Dimension1**
at least one dimension or measure, a hierarchy and one or more filters	**Measure for *Dimension1 Hierarchy***
at least one measure and one level of a hierarchy Dimension or one measure, one level of a hierarchy and one or more filters	**Measure for *hierarchy level***
at least one measure and two Dimensions or one measure, two Dimensions and one or more filters	**Measure for Dimension1 and Dimension2**

***IMPORTANT**: the same applies to the hierarchies and levels of Dimension2 as for Dimension1.*

In the **Object Settings window**, the system displays the title of the chart in the **Name** section:

- In the first field, relating to the first configuration language set Configuration languages, the title is shown in the corresponding language for both the measure and the dimension(s)
- In the next three fields, if any, relating to the other configuration languages set, the chart title is shown for both the measure and the dimension(s) in the corresponding language only if the respective descriptions in that language have been set. If the description of the measure and/or dimension(s) in the configuration language(s) following the first one have not been set, they are shown in the chart title in the first configuration language
- From the fifth field onwards, as for any Reporting languages set, the title of the graph is shown:
  - in the corresponding language for the measure, only if the respective description in that language has been set. If the measure description in one of the reporting languages is not set, the measure is displayed in the first configuration language
  - in the first configuration language for the dimensions.

Error Management in Ask AI for Reporting

When using Ask AI for Reporting, there may be cases where no visuals can be configured and displayed.

Error types and associated messages

In such cases, the system displays error messages to describe the detected issue and help the user identify it.

Condition	Error message displayed
The reporting project for Ask AI has not yet been populated with data from CCH Tagetik	"It is not yet possible to ask questions to ASK AI, please try again later. If the problem persists contact support."
no measure is identified in the question asked...	"From your request it was not possible to tell what measure you are referring to"
There is no reporting view containing the measure identified and found in the reporting project for Ask AI...	From your request, I understood you want the measure "measure_name". I can't find a reporting view with the measure "measure_name" or there is no data.
The AskAI chat session expires after a period of inactivity and the page must be reloaded...	"Your session has expired due to inactivity. Please reload the page."

SmartInsight

Introduction

SmartInsight is an ad hoc reporting tool that allows information from the CCH Tagetik database to be extracted via a browser. SmartInsight complements the structured reporting of CCH Tagetik and allows reports to be created quickly and intuitively in an interface with Excel-like spreadsheets.

The SmartInsight sheet

Smartinsight looks like an Excel worksheet, with the following specificities:

- It can contain several matrices. A *matrix* is a table of rows and columns of values extracted from the application database of CCH Tagetik (see "The structure of the matrix" on page 440). Each matrix can be updated individually.
- It can read data from different workspace datasets.
- It can be imported/exported in CSV and XLSX format.
- It can be saved and shared with other users.

Activation and roles

Only users with specific roles can access the SmartInsight module.

- *Power user* or *Contributor*: users with function 0004078 enabled (see Roles) .
- *Admin*: users with function 0004160 enabled. The administrator can manage the Smartinsight sheets of other power users (e.g. no longer active or that have changed roles). In particular, the administrators can delete them or assign them to themselves for editing.

Features not supported

Below are some features not supported by this tool:

- It is not possible to realise multi-template forms in the same SmartInsight sheet.
- It is not possible to perform cancel/repeat actions on the matrix with the key combinations CTRL+Z and CTRL+Y.
- Advanced scripting is not supported; it is not possible, for example, set a parameter based on other parameters using a script.
- Cut/copy/paste actions can be performed in two ways:
  - with the key combinations CTRL+X / CTRL+C / CTRL+V
  - via the pop-up menu that appears when right-clicking on the matrix.

  It is not possible to combine the two modes.
- The update function has the following limitations:
  - Does not update the formulas in the SmartInsight sheet unless they are enabled in the matrix by the user
  - Does not propagate styles

Currency conversion

SmartInsight allows currency conversion on the multidimensional matrix either by entering the currency from the filter panel or from the matrix (see"Currency conversion" on page 458). Moreover, unlike forms, with SmartInsight it is possible to enter the currency dimension in a matrix on the rows or columns, even though this same dimension is already present in the filter panel, provided the currency conversion is defined.

Activities log

Saving, deleting, sharing, renaming, opening a SmartInsight or performing a refresh on a matrix are saved in the Application Audit.

SmartInsight page

User | <application database>

SmartInsight tile

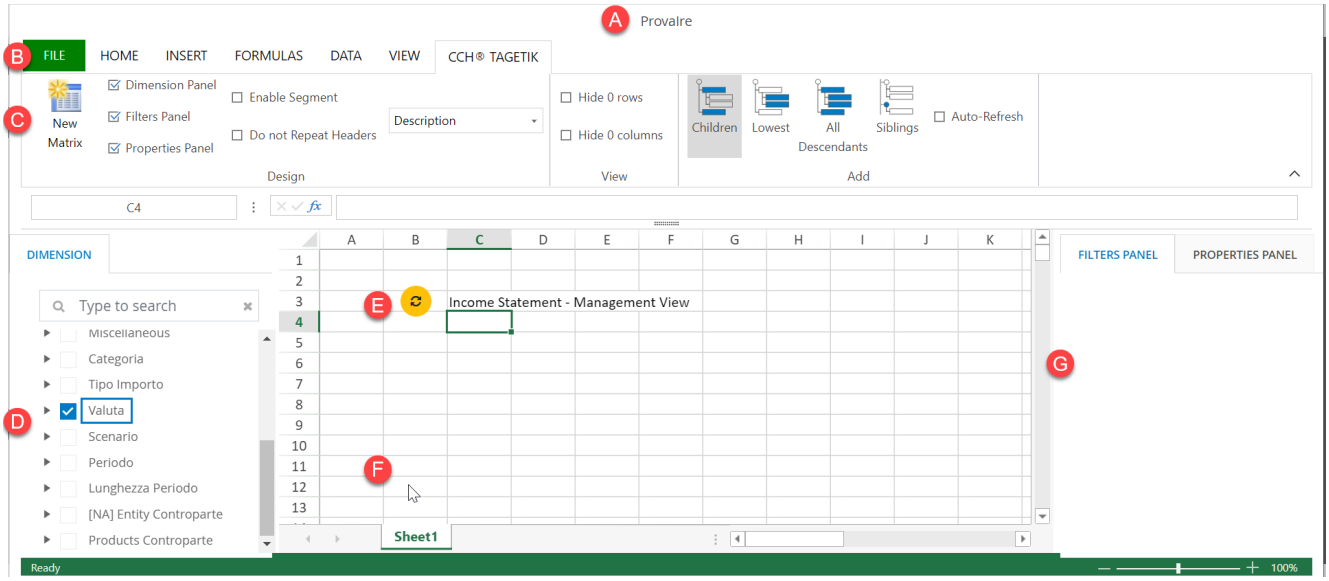
Navbar > Reports & Analysis > SmartInsight

Navigation Panel > Reports & Analysis > SmartInsight

Purpose of the page

This page allows you to extract and display data from the CCH Tagetik application database.

What it looks like



Element	Description
To	SmartInsight sheet name and, for shared sheets, the name of the user owning the sheet
B	Menu bar. In addition to the standard Excel menus, it also opens the toolbar of CCH Tagetik.
C	Tool bar
D	Dimensions panel
E	Matrix update button, one for each matrix. It also indicates its status: • : modified matrix. View detailed data. • : unmodified matrix. ***Note**: this button does not update the applied styles or edits to formulas.*
F	Spreadsheet. It can contain several matrices.
G	Configuration Tab: • **Filter panel**: shows the filters applied to the matrix by dimension/hierarchy. • **Properties panel**: shows information about the selected cell in the matrix and any conversion information.

Buttons in the toolbar CCH Tagetik

Element	Description
Design	
New Matrix	Creates a matrix at the selected cell in the work area. When enabling the matrix, you can select either the financial workspace or a dataset, virtiìual dataset or a AIH datsource of your choice.
Dimension panel	Shows/ hides the dimensions panel.
Filters panel	Shows/Hides the filter panel where the filters applied to the selected matrix are listed by dimension/hierarchy.
Properties panel	Shows/hides the property panel of the selected cell.
Enable segment	Shows/hides intercompany data related to Entities within a chosen Entity node. Intercompany data are displayed in the filter panel and in rows/columns containing a hierarchy: • entity • custom 2 • counterparty entity • custom dimension 2 ***IMPORTANT**: Be sure to update the matrix to display current data.* ***Note**: function not available if there are multiple matrices or for matrices drawn on the Analytical Workspace.*
Do not repeat headers	If enabled, it shows the headings of rows/columns of nested dimensions only when one of the dimensions changes and not on all rows/columns. This improves the readability of data.
Display mode	Selects which dimension information (code, description, code and description) to display for all matrices in the following places: • in the dimensions panel • in the filter panels • in the row/column headers By default, an empty SmartInsight sheet displays the information defined in the form display settings (see Finestra Impostazioni).
View	
Hide 0 rows	If enabled, hides rows with a value of zero in all cells. If the SmartInsight sheet is the result of a web form 2.0 conversion, the form setting is inherited. ***Note**: not available if there are several matrices.* ***IMPORTANT**: Be sure to update the matrix to display current data.* ***IMPORTANT**: The function also applies to changing or inconsistent matrices. Then all rows and columns could be hidden and only the refresh button would remain visible.*
Hide 0 columns	If enabled, hides columns with value zero in all cells. If the SmartInsight sheet is the result of a web form 2.0 conversion, the form setting is inherited. ***Note**: not available if there are several matrices.* ***IMPORTANT**: Be sure to update the matrix to display current data.* ***IMPORTANT**: The function also applies to changing or inconsistent matrices. Then all rows and columns could be hidden and only the refresh button would remain visible.*

Element	Description
Remove 0 values	If enabled (button with dark grey background), deletes all rows and columns with value zero in the data update operation on a matrix. ***Note**: the operation only affects the matrix being updated* ***IMPORTANT**: Deletion of rows and columns is irreversible.* ***IMPORTANT**: If the matrix has no data, only the refresh button is shown on the screen without any header.*
Add	
Children/Lowest/All descendants/ Siblings	Set the drill-down. If enabled, when a row/column header is double-clicked, it adds as many rows/columns to the matrix as the following elements: • the children of the aggregate node in the row/column header **(Children)**. • the lowest level elements of the aggregate node in the row/column header **(Lowest)**. • the leaves of the aggregate node in the row/column header **(All descendants)**. • the siblings of the aggregate node in the row/column header **(Siblings)**. ***Note**: The **Siblings** option does not apply for dimension root nodes.*
Auto-refresh	Enables automatic matrix update after a drill-down.

Manage SmartInsight sheets

Create a SmartInsight sheet

SmartInsight tile > File > New

Navbar > Reports & Analysis > SmartInsight > File > New

***IMPORTANT**: when a new sheet is created, unsaved changes in the current sheet are lost.*

Save a SmartInsight sheet

SmartInsight tile > File

- To save the current SmartInsight sheet, click **Save**.
- To save a copy of the current SmartInsight sheet, click **Save as**, enter the desired name and click **Save**.

The settings of the **do not repeat headers**, **enable segment**, **display mode**, **hide 0 rows**, **hide 0 columns** and **auto-refresh** options are saved together with the contents of the sheet.

It is also possible to save a SmartInsight sheet that does not contain matrices.

Export a SmartInsight sheet

SmartInsight tile > File > Export

1. To export in .xlsx format, click **Excel File**, to export in .csv format, click **CSV File**.
2. Set the export parameters and click **Export Excel File** or **Export CSV File**.

***IMPORTANT**: even if the exported file is subsequently reimported into SmartInsight, it will not retain the information on the matrices present.*

Open a SmartInsight sheet

SmartInsight tile > File > Insights

1. Open a SmartInsight sheet as indicated below:

To open a SmartInsight sheet...	Then click...
of which you are the proprietary user	**My Insights**
shared with the user	**Shared Insights**
of any other user ***Note**: only possible for the Admin. user*	**All Insights**
previously added to the list of **Pinned Insights**	**Pinned Insights**

2. Select the desired SmartInsight sheet, using filters if necessary.

***IMPORTANT**: when a new sheet is created, unsaved changes in the current sheet are lost.*

Import a SmartInsight sheet

SmartInsight tile > File > Import

1. To export a file in .xlsx format into a SmartInsight sheet, click **Excel File**, to export a file in .csv format, click **CSV File**.
2. Set the import parameters and click **Import Excel File** or **Import CSV File**: the contents of the current SmartInsight sheet will be replaced with the imported file.

Manage the sharing of a SmartInsight sheet

SmartInsight tile > File > Insights > My Insights

- To share a SmartInsight sheet, click the relevant button or click and then **Share**: the SmartInsight sheet will be shared with all users.
- To cancel the sharing of a SmartInsight sheet, click the relevant button or click and then **Share**: the sharing of the SmartInsight sheet will be cancelled.

***Note**: each user can only share SmartInsight sheets of which they are the proprietary user.*

Add/remove a SmartInsight sheet from Pinned Insights

SmartInsight tile > File > Insights > My Insights/Shared Insights

- To add a SmartInsight sheet to the list of **Pinned Insights** to be opened easily, click the relevant button or click and then **Pin**: the SmartInsight sheet will be displayed in the **Pinned Insights** tab.
- To remove a SmartInsight sheet from the list of **Pinned Insights** to be opened easily, click the relevant button or click and then **Pin**: the SmartInsight sheet will be removed from the list displayed in **Pinned Insights** tab.

***Note**: both SmartInsight sheets of which you are the proprietary user and SmartInsight sheets shared by other users can be added.*

Rename a shared SmartInsight sheet

SmartInsight tile > File > Insights > My Insights/Shared Insights

1. To rename a SmartInsight sheet, click the relevant button then click **Rename**.
2. Enter the desired name and click **OK**.

Delete a shared SmartInsight sheet

SmartInsight tile > File > Insights > My Insights/Shared Insights

To delete a SmartInsight sheet, click the relevant button then click **Delete**.

IMPORTANT: *deleted SmartInsight sheets cannot be recovered.*

Managing the SmartInsight module

User | <application database>

Activate SmartInsight

1. Click at the bottom-right of the page.
2. Select **SmartInsight**.
3. Click .

The SmartInsight tile appears on the page.

Remove a user from a SmartInsight project

1. Click the **SmartInsight** tile.
2. In the **All Insight** list, click next to SmartInsight.
3. Click **Assign to me** to remove the proprietary user and assign the project to yourself.

The current user is the new proprietary user of SmartInsight.

Design

This section provides comprehensive guidelines for working with matrices and their components. It includes detailed instructions on how to manage the matrix structure, work with rows and columns, and use the filter panel to optimize data visualization. Additionally, it explores the functionality and use of the dimensions panel, offering tips on how to make the most of its features. The section also covers the use of formulas and how they work, as well as explaining how to use the properties panel.

Contents

This section covers the following topics:

The structure of the matrix 440
How to work with matrices 441
Working with rows and columns in a matrix 442
How to work with the filters panel 444
Dimension Panel 445
How the dimension panel works 445
Working with the dimension panel 449
Working with formulas 450
Functioning of formulas 451
Properties panel 452

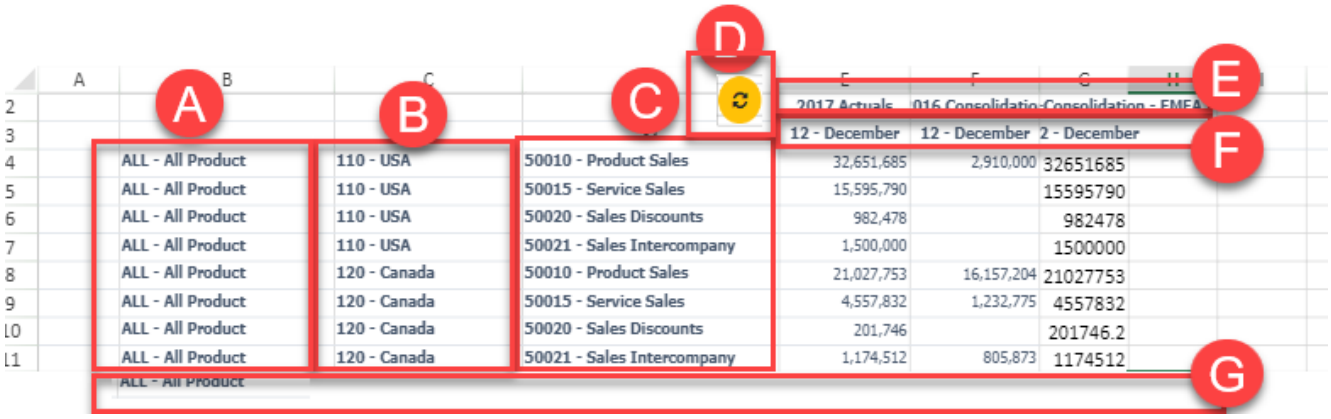
The structure of the matrix

Introduction

Below are the main features of the matrices created and used by SmartInsight. For further details on the matrix concept, see "Form matrices" on page 100.

Structure of a matrix

The matrix described below represents a typical matrix.



Element	Description
A	Third level of row header
B	Second level of row header
C	First level of row header
D	Refresh button
E	Second level of column header
F	First level of column header
G	Row placeholder

Rows and columns management

The matrix data are defined via **the dimension tree**. The rows and columns of the matrix can be populated with dimensional elements in three main ways:

- dragging one or more elements from the dimensions panel to the spreadsheet
- with rows/columns that have already been created, using drill-down
- with rows/columns that have already been created, by typing the dimension code or description

Matrix formulas

SmartInsight supports the handling of formulas in a matrix, enabling the user to write formulas instead of cells of values in a matrix. The user also has the option of extending a formula over an entire row or column.

For further information, please refer to the relevant page.

Multimatrix Smartinsight

In the SmartInsight, several matrices can be created and linked to different workspaces.

In the presence of several matrices, the flags *Segment*, *Hide 0 rows* and *Hide 0 Columns* cannot be used. Whereas, the *Display mode* and *Do not repeat headers* flags are applied to all existing matrices.

Matrix filters can be managed through the dedicated panel, even in the case of SmartInsight containing several matrices. When you select a matrix cell, you will be able to display the general filters associated with it in the dedicated panel.
The user can add or remove filters at the matrix level, for each available dimension. Moreover, filters can be moved at row or column level and vice versa.

How to work with matrices

SmartInsight tile

Navbar > Reports & Analysis > SmartInsight

Create a matrix

1. In the spreadsheet, select an empty cell not adjacent to an existing matrix and then click **New Matrix**

   ***Note**: The selected cell is occupied by the matrix update button. The matrix starts in the cell at the bottom right of the selected cell.*

2. In the **New Matrix** window, select or search for the desired dataset and click **OK**. The matrix development cell (containing the update button) and the dimensions panel appear on the spreadsheet.

   ***Note**: Virtual Datasets and Datasources are now also supported to increase performance and be more flexible. Now it is also possible to search for a specific dataset/workspace by simply typing its name after clicking on the drop-down selector. The search will start after typing at least two characters.*

3. Click on the matrix development cell and place the cursor in the bottom right-hand corner. The cursor changes to +.
4. Click and, while holding down, select the desired column area and repeat the action for the row area.
5. Drag the dimensions to the desired row/column headers.

For instructions on how to work on rows, see "Working with rows and columns in a matrix" on the next page.

Moving a matrix

1. In the spreadsheet, select the entire matrix area.
2. Drag the matrix to the desired position.

Cut, copy and paste a matrix

1. In the spreadsheet, select the entire area of a matrix.
2. To cut, press CTRL+X.
3. To copy, press CTRL+C.
4. Select the cell in which to paste the matrix and type CTRL+V.

> ***IMPORTANT**: if the original matrix is copied and pasted, the original will become a static copy. The first copy and paste generates an updatable matrix. Subsequently, the pasted matrices are only static copies of the matrix. Therefore they will not have an update button.*

Cutting and pasting is also possible with the matrix context menu options or the **Home> Cliboard** buttons on the toolbar. However, pairs of commands from the same group must always be used, e.g. always from the context menu, or always by key combination.

Deleting the contents of a matrix

It is possible to delete the contents of a matrix in three different ways:

- Select all row and column headings and press the DELETE key.
- Select all row and column headings, right-click, then click on **Clear Contents**.
- Select all columns and rows containing column and row headers, right-click, then click **Delete**.

Working with rows and columns in a matrix

SmartInsight tile

Navbar > Reports & Analysis > SmartInsight

Add rows/columns from the dimension panel

1. From the dimension panel, select one or more elements belonging to the same hierarchy and drag them into the spreadsheet: the valid cells for releasing the dragged elements are highlighted in the spreadsheet.
2. Drop the elements in the desired position chosen from among those available.
3. To update the matrix, click .

Add rows/columns by entering a code or description

1. In an empty header cell in the matrix, enter the code or description of a dimension element: after the third character is entered, the list of elements corresponding to the entered characters will appear.

   ***Note**: it is not possible to select elements of dimensions other than the one set in the header.*

2. Select the desired element: if the matrix has several levels of headers, the other headers are completed automatically by copying the trees of the levels below and the headers of the levels above from the row above/column to the left.

   ***Note**: if the cells of all row/column header levels are defined manually, only the row/column with the inserted elements will be added.*

   ***Note**: if the selected element corresponds to the element of the cell above/to the left, it is considered to be a placeholder.*

3. To update the matrix, click .

Add rows/columns using drill-down

1. In the **CCH Tagetik** tab, select the desired option in the **Add** section.
2. Double click the header on which to run the drill-down: the rows/columns with the elements deriving from the chosen option will be filled automatically.

   Note: *drill-down operations can still be performed on notes positioned below placeholders. The rows/columns generated by the drill-down inherit the placeholder from the source rows/columns and therefore they are not populated with values corresponding to the placeholder*

   Note: *the style of the source cells is not inherited by the rows/columns generated by the drill-down.*

3. If the **Auto-Refresh** option is disabled, the user must click to update the matrix.

   Note: *the parent element is not recalculated as the sum of the child elements but remains unchanged.*

Delete a row/column

1. Select a row/column header, press the DEL key or select a row/column, right click and select **Clear contents**.
2. To update the matrix, click .

Add new header levels

The only way to add a new header level to a SmartInsight matrix is by dragging a dimension from the dimension panel as described below.

1. To create a new row/column, select the row below or the column to the right, right click and select **Insert**.
2. From the dimension panel, select an unused dimension, drag it to the spreadsheet and drop it on the empty row/column.
3. To update the matrix, click .

Add a row or column to the matrix

Select the entire row below/entire column to the right of where the new row/column is to be added, right click and select **Insert**.

Change a row into a column or vice versa

1. Select the row/column to be changed, right click and select **Move to first empty column/row**: the row/column will be moved to the first empty column/row.
2. To update the matrix, click .

Change a row/column into a matrix filter

1. Select the row/column to be changed, right click and select **Move to filters panel**: the row/column will be moved from the matrix to the filters panel.
2. To update the matrix, click .

Transform a matrix filter into a row/column

SmartInsight tile > CCH Tagetik > Filters Panel

Navbar > Reports & Analysis > SmartInsight >CCH Tagetik > Filters Panel

1. Click the button for the filter to be transformed and select **Move to first empty row/column**: the filter will be moved to the first empty row/column.
2. To update the matrix, click .

Move a row up or down or a column right or left

1. Select the header of the row/column to be moved, right click and select **Move up/down** or **Move to left/Move to right**.
2. To update the matrix, click .

Working on row/column headers

1. Select one or several row/column header cells.

   ***IMPORTANT**: the cells must be adjacent and only header cells without numerical values.*

2. Proceed as follows:

To...	Then...
copy	CTRL+C or right click +**Copy**.
cut	CTRL+X or right click +**Copy**.
drag	position the cursor on the edge of the cells and click with the  cursor.

3. Select the first empty row header below/empty column to the right, or the headers to be overwritten. Proceed as follows:

To...	Then...
paste copied/cut headers	CTRL+V.
drop dragged headers	release the mouse button.

Particular cases

- The paste/drop operation is also applied to hidden cells.
- The insert copied/cut cells function is not enabled.
- The headers are multiplied and show the sub-trees only when the copied cells are pasted/dropped in empty cells at the end of the matrix.

Select cells

To select...	Then select...
the first populated cell in the desired row/column	an empty cell from the row/column and type CTRL+ARROW KEY (in the desired direction)
the last populated cell in the desired row/column	a populated cell from the row/column and type CTRL+ARROW KEY (in the desired direction)
all empty cells up to the first populated cell (included in the selection)	an empty cell from the row/column and type CTRL+SHIFT+ARROW KEY (in the desired direction)
all populated cells up to the last populated cell (included in the selection)	a populated cell from the row/column and type CTRL+SHIFT+ARROW KEY (in the desired direction)

How to work with the filters panel

How to display the filters panel

SmartInsight tile > CCH Tagetik

Navbar > Reports & Analysis > SmartInsight >CCH Tagetik

To hide/display the dimension panel, click on **Filters** in the **Panels** section.
To display filters for a specific matrix, an individual matrix cell must be selected.

How to add filters to the matrix

SmartInsight tile > CCH Tagetik > Filters Panel

Navbar > Reports & Analysis > SmartInsight >CCH Tagetik > Filters Panel

1. Select a single cell of the matrix to which you want to add filters.

   ***Note**: If you select more than one cell or a cell outside the matrix, the filters will not be shown.*

2. In the dimensions panel, select the elements to be added to the matrix filters and drag them into the filter panel.

   ***Note**: only one filter can be entered for the currency dimension*

3. To apply filters to the matrix, click on .

Ho to remove filters from the matrix

SmartInsight tile > CCH Tagetik > Filters Panel

Navbar > Reports & Analysis > SmartInsight >CCH Tagetik > Filters Panel

To remove an individual filter, click on the **X** next to the element.

To remove an entire section of filters, click on .

Transform a matrix filter into a row/column

SmartInsight tile > CCH Tagetik > Filters Panel

Navbar > Reports & Analysis > SmartInsight >CCH Tagetik > Filters Panel

1. Click the button for the filter to be transformed and select **Move to first empty row/column**: the filter will be moved to the first empty row/column.

2. To update the matrix, click .

Dimension Panel

Introduction

The dimensions shown in the dimension panel depend on the dataset selected when the matrix was created and on the user's permissions. From this panel, it is possible to search for and drag individual elements or nodes of a hierarchy onto the matrix to add row/column headers.

The dimensions tree is very similar to the one used in the Matrix Wizard ("Dimensions tree in the Matrix Wizard" on page 110), although it shows fewer dimensions. The Account dimension supports the sign class, so the elements with inverted sign are shown in orange (see "Sign class" on page 102).

Available dimensions

The following dimensions are displayed on the dimension panel:

- CCH Tagetik's system dimensions
- the custom dimensions (e.g. products, cost centres)
- certain specific dimensions for reporting such as period length (see Periods: basic concepts).

How the dimension panel works

Dragging elements into target cells

When one or several elements are dragged from the dimension panel, SmartInsight highlights the *valid* target cells, namely those in which elements can be dropped.

The main behaviours are detailed below:

When...	Then...
the matrix is empty, or has no rows/columns defined	• the cell below the update button will be the first row header • the cell to the right of the update button will be the first column header.
a dimension already in use is dragged	• the cells occupied by other elements of the same hierarchy will be valid (the cell into which it is dropped will be overwritten). • the first available empty cell at the headers level of the dimension belonging to the same hierarchy will be valid.
an unused dimension is dragged	• the cells occupied by other dimensions will be valid, for row headers and column headers (the cell into which it is dropped will be overwritten). • empty cells of the first free level of headers will be valid.
several dimensions are dragged simultaneously with auto-complete	the cells will filled in on the basis of the information provided with the drag operation. The system will duplicate and relate the dragged elements to those present in the matrix.

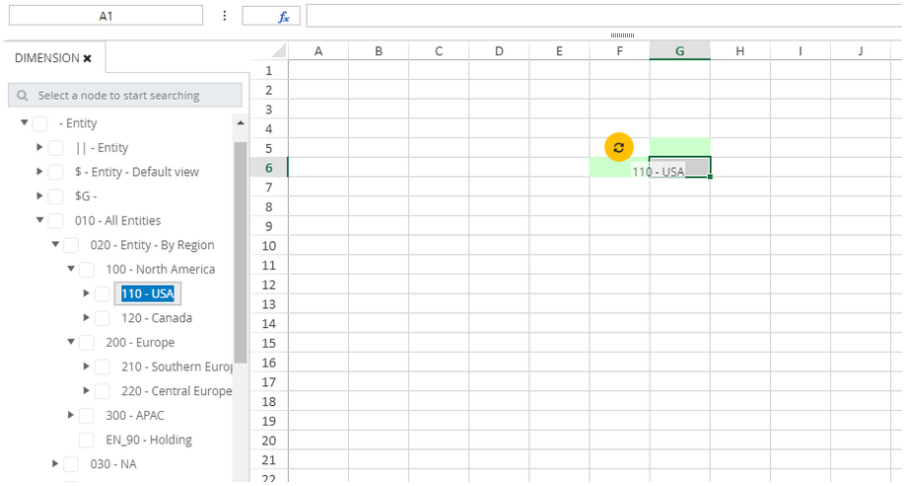
***IMPORTANT**: it is not possible to drag a dimension already present on the filters panel.*

***IMPORTANT**: it is not possible to drag the dimension's elements node, if hierarchies rather than groups are present on the dimension. It is never possible to drag "period", "period length", "amount type" or "currency" dimension nodes to the matrix.*

Example of header setting in empty matrix

In this example, the position of element "110 - USA" is defined as a header (Entity dimension, hierarchy 010).

By dragging this element into cell F6, this cell will be set as the first-level of row headers. By dragging it to cell G5, this cell will be set as the first level of column headers.



Example of valid cells with populated matrix

In this example, the row headers have two levels:

- Level -1: Scenario hierarchy
- Level -2: Entity hierarchy

The column only has one level:

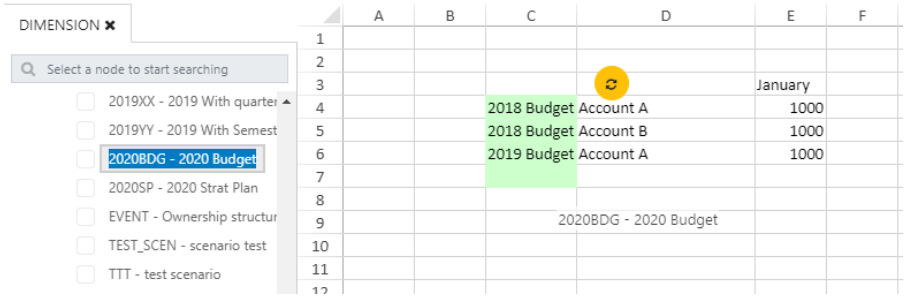
- Level -1: Period hierarchy

	A	B	C	D	E
1					
2				January	
3		2017 Budget	Account A	1000	
4		2017 Budget	Account B	1000	
5		2018 Budget	Account A	1000	
6					

Example of valid cells for a dimension already in use

In this example, by dragging 2019BDG, which belongs to the same hierarchy as 2018BDG and 2017BDG, only the following cells will be valid:

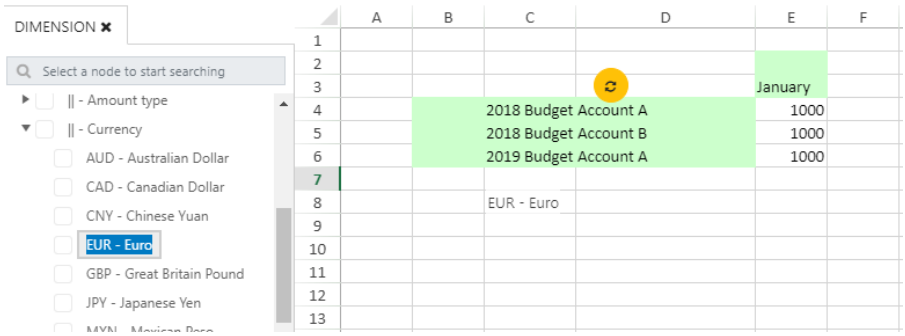
- The cells occupied by other elements of the same hierarchy, i.e. B3:B5.
- The first available empty cell at the headers level of hierarchy 2019BDG, i.e. B6.



In these cases, the cell into which 2019BDG is dragged will be overwritten.

Example of valid cells for an unused dimension

In this example a dimension that is not being used is dragged.



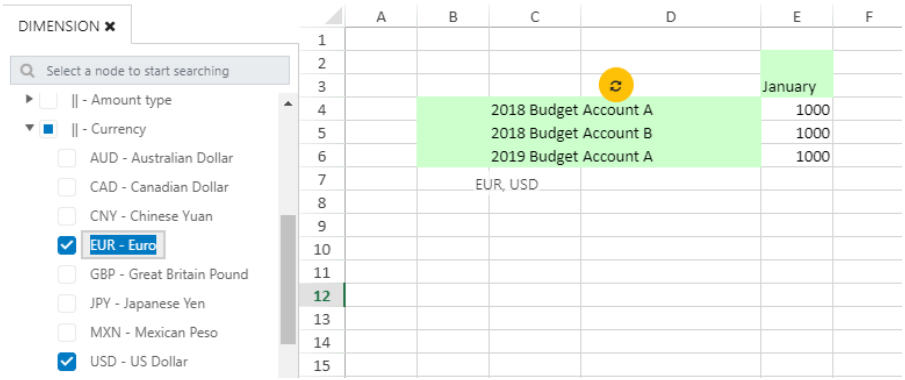
By dragging EUR, which belongs to the Currency hierarchy, the following cells are highlighted:

- The cells occupied by other dimensions, i.e. C4:D6 for the row headers and E3 for the column headers.
- The empty cells at the first free level of headers, i.e. B4:B6 for row headers and E2 for column headers.

For example, by dragging EUR to C5 (occupied by 2018 Budget), the entire entity hierarchy column will be replaced with EUR.

Example of valid cells for several dimensions with auto-complete

In this example, the EUR and USD dimensions (Currency hierarchy) are dragged to row headers level -3. SmartInsight recopies the highlighted headers to add EUR and USD to the left:



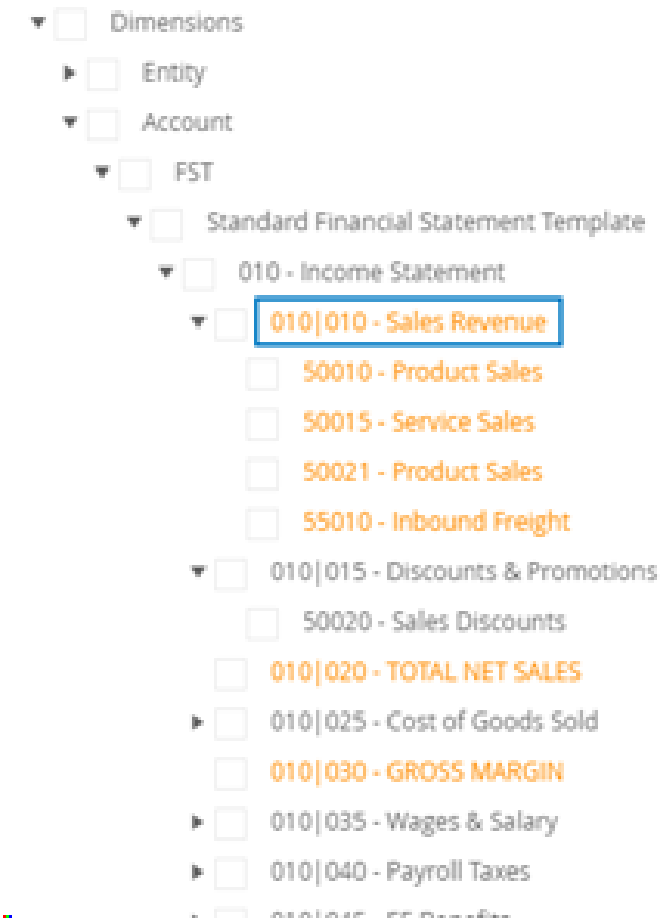
In this example, however, EUR and USD have been added to row header level -1 after a new column has been added to the spreadsheet. SmartInsight recopies the rows one at a time to add EUR and USD to the right:

	A	B	C	D	E	F
1						
2						
3					January	
4		Euro	2018 Budget	Account A	1000	
5		Euro	2018 Budget	Account B	1000	
6		Euro	2019 Budget	Account A	1000	
7		US Dollar	2018 Budget	Account A	1000	
8		US Dollar	2018 Budget	Account B	1000	
9		US Dollar	2019 Budget	Account A	1000	
10						

Restriction for accounts

The first account dragged from the dimensions tree and dropped on an empty row (or column) is considered the *reference element* for that row (or column). Then, by manually entering or editing the code of an account in a cell below (or to the right, for filters on column headers), that code will be searched for in the subtree of the reference account.

Example reference account



In this example, the following behaviours are obtained:

Reference account	Behaviour
010|015	By entering code 50020, the descendant element of node 010 will be entered.
010|015	By entering code 30010, which is not present in node 010, a placeholder not present in the subtree will be entered.
50010 from node 010|010	By 50021, the descendant element of node 010 will be entered.

Particular cases

Some particular cases are detailed below:

- Reference elements from the FST (financial statement template) and Control groups can also be dragged as reference elements. The system searches for the elements included in the hierarchy to which the reference element belongs.
- By entering, in the matrix, a formula element contained in the tree of the FSTs or control groups in a cell from the first row/column, the subsequent elements will be interpreted as placeholders.
- If the manual entry generates multiple results, the first element to be found will be entered.

Working with the dimension panel

View the dimension panel

SmartInsight tile > CCH Tagetik

Navbar > Reports & Analysis > SmartInsight >CCH Tagetik

The dimension panel opens automatically when a matrix is created.

To hide/show the dimension panel, click **Dimension Panel**.

Search for an element in the dimensions tree

SmartInsight tile > CCH Tagetik > Dimension

Navbar > Reports & Analysis > SmartInsight >CCH Tagetik> Dimension

1. Select a hierarchy node: the search box above will be activated for the selected hierarchy.
2. Type the text to search and then press Enter: the dimensions panel will show the hierarchy nodes and sheets that contain the searched text in their code or description.
3. To end the search, click **X** in the search window.

Select one or several elements in the dimensions tree

SmartInsight tile > CCH Tagetik > Dimension

Navbar > Reports & Analysis > SmartInsight >CCH Tagetik> Dimension

1. Click the control box or on the name of the desired element.
2. To select other elements of the same hierarchy, click the control box of the individual elements or select them by pressing CTRL or SHIFT

   ***Note**: elements of different hierarchies cannot be selected using CTRL or SHIFT. If an element has already been selected, it will be automatically deselected when an element of another hierarchy is selected with CTRL or SHIFT.*

Working with formulas

Enabling the use of formulas on a matrix

To enable the use of formulas on a row/column of a matrix:

1. Select a single matrix header cell and open the pop-up menu.

   ***Note**: It is also possible to select a cell at the end of the matrix.*

2. From the pop-up menu, select **Enable formulas on row** if the cell is a row header, or **Enable formulas on column** in the case of a column.

Disabling the use of formulas on a matrix

To disable the use of formulas on a row/column of a matrix:

1. Select a single matrix header cell in which formulas are enabled.
2. From the pop-up menu, select **Disable formulas on row** if the cell is a row header, or **Disable formulas on column** in the case of a column.

Extending a formula within a row/column

To extend a formula over a row/column of a matrix:

1. Select a single value cell of the matrix in which a formula is present.
2. From the pop-up menu, select **Fill row with formula** if the cell is on a formula-enabled row, or **Fill column with formula** in the case of a column.

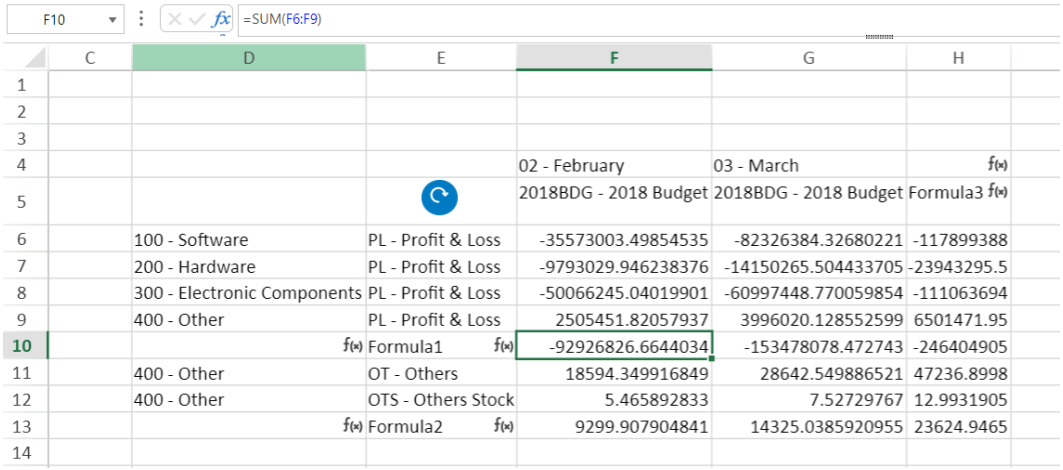
   ***Note**: If the cell is part of both a row and a column that are both enabled for formulas, then both commands will be displayed from the pop-up menu.*

   ***Note**: the formula is not really extended over the entire row/column, but is extended to the right in the case of a row and downwards in the case of a column*

Functioning of formulas

Introduction

Below are the main features for using formulas within SmartInsight. An example of a matrix with enabled formulas is shown below.



	C	D	E	F	G	H
1						
2						
3						
4				02 - February	03 - March	f(x)
5				2018BDG - 2018 Budget	2018BDG - 2018 Budget	Formula3 f(x)
6		100 - Software	PL - Profit & Loss	-35573003.49854535	-82326384.32680221	-117899388
7		200 - Hardware	PL - Profit & Loss	-9793029.946238376	-14150265.504433705	-23943295.5
8		300 - Electronic Components	PL - Profit & Loss	-50066245.04019901	-60997448.770059854	-111063694
9		400 - Other	PL - Profit & Loss	2505451.82057937	3996020.128552599	6501471.95
10		f(x)	Formula1 f(x)	-92926826.6644034	-153478078.472743	-246404905
11		400 - Other	OT - Others	18594.349916849	28642.549886521	47236.8998
12		400 - Other	OTS - Others Stock	5.465892833	7.52729767	12.9931905
13		f(x)	Formula2 f(x)	9299.907904841	14325.0385920955	23624.9465
14						

Matrix formulas

SmartInsight supports the handling of formulas in matrix, row or column, even after refresh.

In order to use formulas in a matrix, the rows or columns in which the formulas are to be entered must be specially enabled. Otherwise, the formulas and text entered in the matrix cells would be cleared at each matrix refresh.
When enabling or disabling formulas, all cells related to the specific row/column are cleared.

Enabling formulas

In order to use formulas in a matrix, you must first select one and only one row or column header cell, and through the pop-up menu, using the **Enable formulas** command, the action will be applied to the entire related row or column.
It will also be possible to insert formulas on the first empty row or column immediately following the last matrix header; it will still be possible to add new dimensional elements in the rows or columns following these formulas.

Formula-enabled rows and columns are distinguishable by the symbol f(x) in the relevant header cells. Moreover, it is not possible to **drag&drop** dimensional elements onto a formula header cell.

When enabling a formula in a matrix, whether in a row or column, the system will clear the descriptions of the relevant headers and values in the matrix cells, allowing the user to manually enter free text in the header and formulas in the value cells.

The user can complete the formula (to the right if set on rows or downwards if set on columns) by using the **Fill row with formula/Fillcolumn with formula** command in the pop-up menu, to be activated on a cell that already contains a formula.

Once formulas have been enabled on a row or column, you can:

- write formulas in the value cells within the enabled row/column;
- extend a formula on the enabled row/column, to the right for rows and downwards for columns;
- write free text in the header cells within the enabled row/column;

- keep formulas and text intact after matrix refresh;
- keep the formulas and text entered when saving the SI.

Disabling Formulas

The user may decide to remove a formula in several ways:

1. Deletion of an entire row or column of the sheet.
2. Removal of the formula from the pop-up menu (**Disable formulas** option) to be used only on the header cells of a formula.

The latter option clears the values contained in relative cells at both the value and header level. The user can then use the related row or column as a placeholder to add new dimensional elements or turn it into a formula with the command in the pop-up menu **(Enable formulas**).

By manually deleting the contents of one or more cells, the row or column will remain enabled when writing formulas.

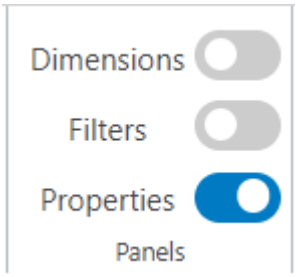
Disclaimer

The perfect functioning of the formulas is not guaranteed when the following features co-exist:

- **Remove 0 values**
- **Do not repeat headers**
- **Copy/Paste** and **Cut/Paste** from cells containing a formula to cells without a formula and vice versa
- **Drill up**
- **Hide 0 rows** and **Hide 0 columns**
- In the case of **Move to left/Move to right** within rows or **Move up/Move down** within columns, the contents of formula headings are not moved.

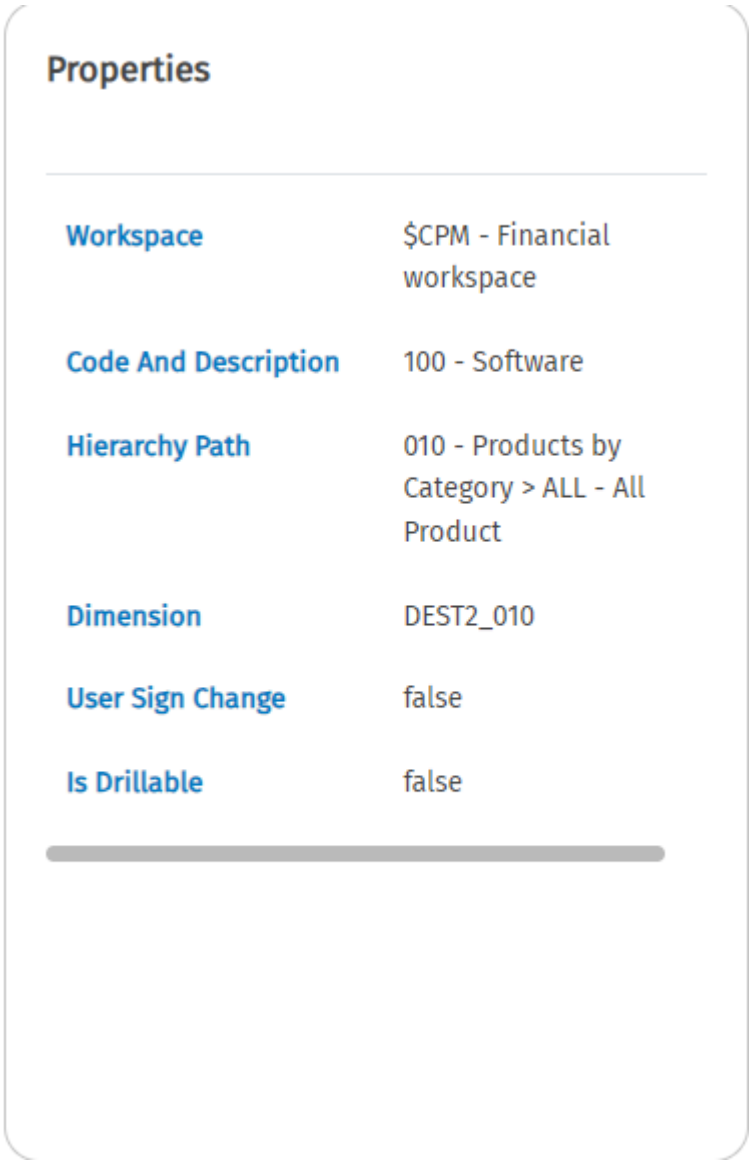
Properties panel

In the Panels section of the CCH Tagetik ribbon there is a **Properties Panel** button to display the list of information of a selected cell from a SmartInsight matrix.



It is possible to display the information of the selected dimension:

- Workspace: indicates the workspace the matrix belongs to
- Code and Description: code and description of the dimension
- Hierarchy Path: this field contains Code and Description of the hierarchy path to the parent node of the selected element. If the selected element is a root node, this field will be empty.
- Dimension: name of the dimension
- User Sign Change: specifies whether or not sign change is possible
- Is Drillable: specifies whether the dimension is drillable or not
- Conversion: if present as an entry, specifies what type of conversion was performed on a selected currency dimension.


Properties
Workspace
$CPM - Financial workspace
Code And Description
100 - Software
Hierarchy Path
010 - Products by Category > ALL - All Product
Dimension
DEST2_010
User Sign Change
false
Is Drillable
false

Navigation

Contents

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How to work with limits 456
Currency conversion 458
Currency conversion procedure 458

The drill-down function

Function utility

The drill-down function allows the user to view the components that make up an aggregated figure.

This functionality is available on rows/columns containing nodes of aggregated data and is useful for speeding up data analysis.

Drill-down example

			2018BDG - 2018 Budget 01 - January	2018BDG - 2018 Budget 12 - December
012 - PL Account Accounts	100 - North America	100 - Software	-1830679.683771229	-8682858.190982908
012 - PL Account Accounts	100 - North America	200 - Hardware	-2377053.473932437	-27157772.419237938
014 - Balance Sheet Accounts	100 - North America	100 - Software	1147927.23	21198825.02
014 - Balance Sheet Accounts	100 - North America	200 - Hardware	302503.430689655	3535209.43



			2018BDG - 2018 Budget 01 - January	2018BDG - 2018 Budget 12 - December
012 - PL Account Accounts	100 - North America	100 - Software	-1830679.683771229	-8682858.190982908
012 - PL Account Accounts	100 - North America	200 - Hardware	-2377053.473932437	-27157772.419237938
014 - Balance Sheet Accounts	100 - North America	100 - Software	1147927.23	21198825.02
014 - Balance Sheet Accounts	100 - North America	200 - Hardware	302503.430689655	3535209.43
014 - Balance Sheet Accounts	110 - USA	100 - Software		
014 - Balance Sheet Accounts	110 - USA	200 - Hardware	-81.379310345	
014 - Balance Sheet Accounts	120 - Canada	100 - Software	1147927.23	21198825.02
014 - Balance Sheet Accounts	120 - Canada	200 - Hardware	302584.81	3535209.43

In this example, the “100 - North America” node can have the following behaviours:

- **Add children**: the system adds as many rows as there are children, multiplied (by Cartesian product) by the elements to the right of the node on which the operation was requested.

- **Add all descendants**: the system adds as many rows as there are descendants,multiplied (by Cartesian product) by the elements to the right of the node on which the operation was requested.
- **Add all lowest level elements**: the system adds as many rows as there are lowest level elements, multiplied (by Cartesian product) by the elements to the right of the node on which the operation was requested.
- **Add siblings**: the system adds as many rows as there are siblings (elements of the same level), multiplied (by Cartesian product) by the elements to the right of the node on which the operation was requested.

In all cases, the added nodes inherit the root of the original node.

Working with drill-down

SmartInsight tile > CCH Tagetik

Navbar > Reports & Analysis > SmartInsight > CCH Tagetik

Activate drill-down

Select the desired type of drill-down in the **CCH Tagetik** tab, select the desired option and click the **Add** section. In the work desktop, double click the header of the row/column containing a node, then click to show the aggregated data.

The value of the parent node remains unchanged, but the system displays the details of the child values in rows below the parent node cell.

Activate automatic updating for drill-down matrix

To enable the automatic updating of the matrix when running drill-down, select **Autorefresh.**

Setting Limits

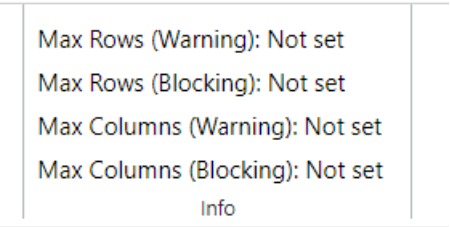
From SmartInsight > File > Settings it is possible to reach the section dedicated to displaying or setting limits for rows and columns of smartinsights matrices. The 'Settings' section is in fact available to all types of users, but only editable by administrators. These limits apply to all SmartiInsight users and to all SmartiInsights in that application database.



They are divided into warning limits and blocking limits:

- When a matrix reaches a warning limit, the user is informed with a message but can continue working on the matrix or update it.
- When a matrix reaches a blocking limit, the users are informed with a message and cannot continue to perform certain operations on the matrix or update it unless they first reduce the number of elements in a row or column or both until the limits are no longer exceeded.

The user can also view information about the set limits from the Ribbon Bar in the 'Info' section.



If one of the fields remains empty, that limit will be ignored.

It is not mandatory to set all 4 fields.

How to set limits

Once in the Settings section (see User manual page Settings section), it will be possible for Admin users to set limits for matrix rows/columns for all smartinsights in an application database.

There will be the 4 fields concerning block and warning limits for rows and columns.

If at least one of the fields is changed, the 'Save' button is enabled. Bear in mind that the warning limits cannot exceed the relevant blocking limits,

- It is not possible to set a limit less than 1 in any of the fields. Should the user try to enter a value less than this, the value 1 will still be displayed.
- It is not possible to set a limit greater than 1000000 for rows. Should the user try to enter a value greater than this, the value 1000000 will still be displayed and the user will receive the following warning: 'Attention, the maximum number of rows is 1,000,000'.
- It is not possible to set a limit greater than 10000 for the columns. Should the user try to enter a value greater than this, the value 10000 will still be displayed and the user will receive the following warning: 'Attention, the maximum number of columns is 10,000'.

This information is visible under each field in the Settings section.

If a warning limit is entered that exceeds the relevant blocking limit, the user will be warned by the following message "Warning, column warning limit exceeds column blocking limit" with regard to columns, and by the following message "Warning, row warning limit exceeds row blocking limit" with regard to rows, and will not be able to select the "Save" button.

When the user selects the 'Save' button, the following confirmation message is displayed: 'Changes saved successfully'.

The limit change operation is recorded in the Audit table with the description 'SmartInsight limits settings change'.

How to work with limits

LAUNCH

When opening a SmartInsight, if it contains a single matrix that exceeds one of the blocking limits in a row or column, only the headers will be displayed but not the matrix values, and the user will display the following warning: 'Cannot retrieve values of a matrix violating blocking limits'.

When opening a SmartInsight, if it contains a single matrix that exceeds one of the warning limits in a row or column, but none of the blocking limits, a window pop up showing this message: "This matrix violates one or more limits, are you sure you want to refresh it?". The user can choose to

- continue and then update the matrix by selecting the 'Continue' button
- cancel and display only the matrix headers and not the values, by selecting the 'Cancel' button

UPDATE

When the user tries to update a matrix that exceeds one of the blocking limits in a row or column, the operation is blocked and the following warning will pop up: 'This matrix cannot be refreshed because of a limit violation'.

When the user tries to update a matrix that exceeds one of the warning limits in a row or column, but not the blocking limits, a window pops up showing the following message: 'This matrix violates one or more limits, are you sure you want to refresh it?'. The user can choose to

- continue and then update the matrix by selecting the 'Continue' button
- cancel the operation and return to the previous situation by selecting the 'Cancel' button

MATRIX OPERATIONS

The proper handling, in coexistence with the limits, of the following operations is guaranteed:

- copy and paste single dimension at end of matrix
- drill down
- Drill Down To Children
- Drill Down To Lowest
- Drill Down To Descendants
- Drill Down To Siblings
- auto-complete at end of matrix
- addition, from the dimension panel, of new elements of an already existing dimension in the matrix
- addition, by dimension panel, of elements of a dimension not existing in the matrix
- "Move to first empty row / column" within the matrix
- "Move to first empty row/column" from filter panel
- addition of empty rows/columns

When the user performs one of these operations, if warning limits are exceeded, the warning 'This matrix violates one or more warning limits' will be displayed, but the operation will still be performed.

When the user performs one of these operations, the warning 'The action cannot be performed due to a limit violation' will be displayed and the operation will be blocked.

The correct handling, in coexistence with the limits, of the following operations is not guaranteed:

- copy and paste several elements of the same dimension simultaneously inside and outside the matrix
- cut and paste several elements of the same dimension simultaneously inside and outside the matrix
- drag and drop multiple elements of the same dimension simultaneously inside and outside the matrix
- cut and paste single dimension at end of matrix
- single dimension drag and drop at end of matrix

- copy and paste an entire set of dimensions at the end of the matrix
- cut and paste an entire set of dimensions at the end of the matrix
- drag and drop an entire set of dimensions at the end of the matrix
- replacement of a dimension with more elements than previously in the matrix

Currency conversion

Introduction

Currency conversion on the multidimensional matrix can be performed in SmartInsight in two ways: from the **Filters panel** or from the matrix itself (see "The structure of the matrix" on page 440). Some particular cases which may arise depending on the type of dataset used in the matrix are outlined below.

Conversion on financial matrix

For multidimensional matrices defined on the financial dataset, the following elements can also be customised: FX rate type, Reference Scenario and Reference Period.

The following FX rate types are supported:

- final FX rate
- average FX rate
- period average FX rate
- account-based FX rate

Conversion on analytical matrix

For multidimensional matrices defined on the analytic dataset, it is only possible to apply the conversion on the fly if the currency dimension is present in the dataset.

The following FX rate types are supported:

- measure-based FX rate
- account-based FX rate, only if the reference analytic dataset contains the account dimension.

Currency conversion procedure

SmartInsight tile

Navbar > Reports & Analysis > SmartInsight

Convert currency from filters panel

1. From the dimension panel, drag a currency onto the filers panel.
2. Select **For conversion**, then choose the type of conversion to perform. The filters active as conversion are identified with ⇄.
3. Click the ⋮ button for the currency filter and select **On-the-fly conversion**: the data of the entire matrix will be converted into the currency entered as the filter and follow the algorithms used by the reporting engine.

   ***Note**: the currency dimension can be entered in the matrix as a row/column header even if it is also present in the filters panel, but only if the relative currency conversion has been defined.*

Convert the currency from the matrix

1. In the matrix, right click on the cell containing the currency to be converted, then select **On-the-fly conversion**.
2. Select **For conversion**, then choose the type of conversion to be performed: the cells in which a conversion has been performed are identified with ⇄.
3. To remove the conversion, right click on the cell, select **On-the-fly conversion**, then **As filter**.
4. Click to view the updated data.

***Note**: it is possible to perform conversions based on other scenarios/periods apart from those in the matrix.*

Other Reports

The Tagetik platform offers a variety of flexible tools to support different needs related to the creation, analysis, and distribution of both operational and strategic reports.

Ad-hoc Analysis

It allows users to independently extract any numerical data available in the financial model of a CCH Tagetik application and analyze it using a pivot table—either directly from the web interface or by downloading it as a local Excel file (see Ad-hoc Analysis).

Ad-hoc Report

An Ad-hoc Report is an output form saved locally on the user's computer. As the name suggests, it is created to meet the specific needs of an individual user and is not intended for centralized or shared use (see Report ad hoc).

System Reports

System Reports display the same information entered through data entry pages, but in read-only mode. These reports do not allow data editing and can also display data from locked scenarios. Their purpose is to show only the data from active scenarios and for the dimensions the user has access to (e.g., scenarios, companies, categories, custom dimensions, accounts). As a result, they typically do not display historical data by default, although such data remains accessible (see System Reports).

XBRL

XBRL (eXtensible Business Reporting Language) is an XML-based language that represents an international standard for the representation and transmission of accounting and financial information (see XBRL).

Magazine

Magazine is a CCH Tagetik reporting solution which allows users to create periodic publications on specific topics (e.g. KPI Analysis, revenues analysis), using forms (see Magazine).

Ad-hoc analysis

Ad-hoc analysis is a functionality that allows the user to independently extract any numeric data present in the financial model of a CCH Tagetik application, and to analyze it via pivot table, directly from the web interface, or to download it as a local MS Excel file.

The Ad-hoc analysis functionality can be run:

- from the *System report* tile in the User home page;
- from the *Process report* menu in the **Actions** panel of the **Detail summary** tab of a process cockpit;
- from a web form, by using the **Analysis** menu

In the first two cases, it is possible to define a bookmark to the Ad-hoc Analysis window, which can be used to create a Bookmark type tile on the user's Home page. By accessing from the created tile, the user can run an ad-hoc analysis with a series of pre-defined filters.

NOTE In the third case, if you define a bookmark when you are on an Ad-hoc Analysis window run from a web form, the bookmark will open the source web form and not the Ad-hoc analysis.

Filters window

When opening the Ad-hoc analysis (except when it is run from the ***Analysis > By cell*** or ***Analysis > By cube*** menu of the web form), the user must specify the set of data for the purposes of the analysis.

The Ad-hoc Analysis filter window includes a series of rows, one for each Hierarchy or Grouping to be displayed or filtered.

In the case of the Account dimension, in addition to hierarchies, you can use the Control Group, the Financial statement templates or the Forms to specify which accounts you want to display or filter.

In each row, the user must define the following fields:

- **Dimensions** in this field you select the Dimension and the Hierarchy (Control Group, Financial statement template or Form) which is of interest for the analysis;
- **Filters** in this field you can select the elements you want to include in the analysis;
- **Detail** if this option is enabled, all the selected elements, limited to those for which there is data in the system, will be reported as an analysis dimension, up to the last level of detail. If the option is disabled, data will still be filtered on the selected elements only, and will be added up without having the split for each element of lowest level;
- **Display mode** allows you to indicate whether the elements, if the **Detail** option is enabled, must be represented with the code only, the description only, or with code and description.

When the user opens the Ad-hoc analysis window from a process cockpit, the filters for the Scenario, Period and Entities dimensions are predefined with those selected in the cockpit.

The **Next** button closes the filters window and runs the ad-hoc analysis, by displaying on the pivot table the selected data, based on the set up dimensions and options.

The **Download** button does not close the filters window and extracts an .xlsx file that is downloaded locally with the selected data, based on the set up dimensions and options.

The extracted .xlsx file contains two sheets: The **data** sheet, in which all the records representing the selected data are shown in tabular form. The **pivot tabl**e sheet, in which the data sheet's **data** is represented by a pivot table which is pre-set by the system.

In the data sheet, in addition to the column with the amount, there are one or two columns for each level of each dimension present in the filters window having the **Detail** option enabled.

For each level there will be a column for the code or for the description of the elements, or two columns, one for the code and one for the description, depending on how the **Display Mode** field is set up. The name of the column is given by the name of the level - if set up in the data model - namely by a technical name provided by the system. If there are two columns, one for the code and one for the description, then the two columns will show "(Code)" or "(Description)" at the end of the name.

The order of the columns is set up automatically by the system and the user cannot modify it without first opening the file in Excel.

NOTE if the user runs the Ad-hoc analysis utility from the ***Analysis > By cell*** or ***Analysis > By cube*** menu of a web form, the initial window for the selection of the dimensions and of the filters for the data to analyze is not available, and the system opens directly the pivot table on the web interface; this is pre-filtered with the filters that operate respectively on the cell selected in the initial web form, or on the matrix of the selected cell.

System reports

Contents

This section covers the following topics:

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System reports

Introduction

System reports present, in a single view, the same information that is inserted from data entry pages. You cannot edit data in these reports; however, you can view the data in locked scenarios.

The purpose of these reports is to show only data of active scenarios and for the dimensions which the user can view (scenarios, entities, categories, custom dimensions, accounts). Consequently, by default they do not typically show data stored offline, which are accessible in any case.

To navigate the data easily, set the desired dimensions as the user's Point of view. See User settings in the General Introduction section. You can also save the set filters, to consult them later.

Types of report on gross and IC amounts

The system reports that can be generated for gross and IC amounts are listed below.

Report	Description	Path
Original gross amounts	Shows the original gross amounts. With the same dimensions, and for data entered via a data entry form, there can be several records because the system can be set to generate a row for every change made to the data.	*System reports tile > Gross and IC amounts > Original gross amounts* *Navigation panel > Reports & Analysis > System Reports > Original gross amounts*
Original IC amounts	Shows the original intercompany amounts. With the same dimensions, and for data entered via a data entry form, there can be several records because the system can be set to generate a row for every change made to the data.	*System reports tile > Gross and IC amounts > Original IC amounts* *Navigation panel > Reports & Analysis > System Reports > Original IC amounts*

Report	Description	Path
IC reconciliation accounts	Shows IC reconciliation accounts.	*System Reports tab > Gross and IC amounts > IC reconciliation accounts* *Navigation panel > Reports & Analysis > System Reports > IC reconciliation accounts*
Intercompany reports	Collects data on the intercompany entries (IC amounts and entity journals) that exist between entities belonging to the same group.	*System reports tile > Gross and IC amounts > Intercompany report* *Processes tile > select process > Consolidator > Overview tab or Step tab > (Actions) Process reports > Intercompany report* *Processes tile > select process > Contributor > Detail summary tab > (Actions) Process reports > Intercompany report* *Navigation panel > Processes > select process > Consolidator > Overview tab or Step tab > (Actions) Process reports > Intercompany report* *Navigation panel > Processes > select process > Contributor > Detail summary tab > (Actions) Process reports > Intercompany report*
Consolidated amounts	Shows the amounts present on the consolidation scenarios. You can view proportional, converted and consolidable data. The consolidable data represent the entity's contribution to the group consolidated financial statements, for the amount categories.	*System reports tile > Gross and IC amounts > Consolidated amounts* *Navigation panel > Reports & Analysis > System Reports > Consolidated amounts*

Types of reports on journals

The system reports that can be generated on journals are listed below.

Report	Description	Path
Original journal rows	Shows all the rows of the journals on the original scenario. You can view both the rows of entity journals and the rows of consolidation journals.	*System reports tile > Journals > Original journal rows* *Navigation panel > Reports & Analysis > System Reports > Original journal rows*
Consolidated journal rows	Shows all the rows of the journals on the consolidation scenarios. You can view both the rows of entity journals and the rows of consolidation journals.	*System reports tile > Journals > Consolidated journal rows* *Navigation panel > Reports & Analysis > System Reports > Consolidated journal rows*
Entity journals - Headers	View all entity journal headers The following information can be viewed from the individual header: • the original and consolidated rows, with the relative balancing • any documents attached to the journal ***Note**: you can download the documents attached to the journals locally.*	*System reports tile > Journals > Entity journals - Headers* *Navigation panel > Reports & Analysis > System Reports > Entity journals - Headers*
Consolidation journals - Headers	View all consolidation journal headers The following information can be viewed from the individual header: • the original and consolidated rows, with the relative balancing • any documents attached to the journal ***Note**: you can download the documents attached to the journals locally.*	*System reports tile > Journals > Consolidation journals - Headers* *Navigation panel > Reports & Analysis > System Reports > Consolidation journals - Headers*

Other types of reports

The other system reports that be generated are listed below.

Report	Description	Path
FX rates	Shows data on FX rates.	*System Reports tile > FX Rates > FX Rates* *Navigation panel > Reports & Analysis > System Reports > FX Rates > FX Rates*
Overridden FX rates	Shows data on overridden FX rates. Only available under the following conditions: • if the in memory consolidation engine is being used • if the consolidation process is activated and enabled	*System Reports tile > FX Rates > Overridden FX rates* *Navigation panel > Reports & Analysis > System Reports > FX Rates > Overridden FX rates*

Report	Description	Path
Notes	Shows the explanatory notes on the values entered during data entry activities. If the **Save data in form incrementally** option is enabled in the process's gross and IC amount rules, the history of the operations carried out on every note will be available. See Amount & Intercompany rules page in the Processes section.	*System Reports tile > Other > Notes* *Navigation panel > Reports & Analysis > System Reports > Other > Notes*
Comments	Shows a list of comments on the data entered during data entry.	*System Reports tile > Other > Comments* *Navigation panel > Reports & Analysis > System Reports > Other > Comments*
Ad-hoc analysis	Allows you to extract any numerical data present in the financial model of a CCH Tagetik application and analyse it using a pivot table, directly from the web interface, or by downloading it as a local MS Excel file.	*System Reports tile > Gross and IC amounts > Ad-hoc analysis > Ad-hoc analysis* *Processes tile > select process > Contributor > Detail summary tab or Step tab > (Actions) Process reports > Ad-hoc analysis* *Processes tile > select process > select scenario and period (if the process is not a single submission process) > select step tab > select entity > (Workflow) if present, click task form > Open Form in Web > Analysis tab* *Forms tile > select form > ▶ > insert parameters > Analysis tab* *Navigation panel > Reports & Analysis > Reporting tools > Ad-hoc analysis* *Navigation panel > Processes > select process > Contributor > Detail summary tab or Step tab > (Actions) Process reports > Ad-hoc analysis* *Navigation panel > Processes > select process > select scenario and period (if the process is not a single submission process) > select step tab > select entity > (Workflow) if present, click task form > Open Form in Web > Analysis tab* *Navigation panel > Reports & Analysis > Browse all > select report > Analysis tab*

System reports - Original gross amounts page

User

System reports tile > Gross and IC amounts > Original gross amounts

Navigation panel > Reports & Analysis > System Reports > Original gross amounts

Report contents

Shows the original gross amounts.

With the same dimensions, and for data entered via a data entry form, there can be several records because the system can be set to generate a row for every change made to the data.

Filters and fields on the page

For a description of all the available filters and the fields of this report, see the page dedicated to gross amounts (Gross Amounts page).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

System reports - Original IC amounts page

User

System reports tile > Gross and IC amounts > Original IC amounts

Navigation panel > Reports & Analysis > System Reports > Original IC amounts

Report contents

Shows the original intercompany amounts.

With the same dimensions, and for data entered via a data entry form, there can be several records because the system can be set to generate a row for every change made to the data.

For example, if the value 100 is initially present in a cell and the user overwrites that value with 110 first, and then with 150, the system inserts the initial amount record of 100, an amount record of 10 and another amount record of 40.

***Note**: the way in which the amount inserted from a form is updated is defined in the process's gross and IC amount rules, see Amount & Intercompany rules page in the Processes section.*

Filters and fields on the page

For a description of all the available filters and the fields of this report, see the page dedicated to intercompany gross amounts (IC Amounts page).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

System reports - Intercompany reconciliation accounts page

User

System Reports tab > Gross and IC amounts > IC reconciliation accounts

Navigation panel > Reports & Analysis > System Reports > IC reconciliation accounts

Report contents

Shows IC reconciliation accounts.

Filters on the page

For a description of all the available filters, see the description on the fields on the insert IC reconciliation accounts page (Reconciliation accounts page).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

System Reports - Consolidated amounts page

User

System reports tile > Gross and IC amounts > Consolidated amounts

Navigation panel > Reports & Analysis > System Reports > Consolidated amounts

Report contents

Shows the amounts present on the consolidation scenarios.

You can view proportional, converted and consolidable data. The consolidable data represent the entity's contribution to the group consolidated financial statements, for the amount categories.

Filters and fields on the page

Fields	Description
Consolidation scenario	Consolidation scenario for which the data are reported
Period	Period of the indicated scenario
Entity	Entity of the consolidation scenario for which the data are reported
Account	Account on which the data are present
Custom dimension	Custom dimension on which the data are managed
Category	Category of the reported data
CTP entity	Counterparty entity from the two entities involved in the journal.

Fields	Description
Amount Type	Type of data to show: • **Original** • **Consolidable** • **Converted** • **Proportional**
Consolidation currency	Currency in which the consolidation was carried out
Consolidation currency amount	Amount expressed in the consolidation currency
Transaction currency	Currency in which the transaction was carried out
Transaction currency amount	Amount expressed in the transaction currency
Entity currency	Currency of the entity in the process
Entity currency amount	Amount expressed in the entity currency
Notes	Any explanatory notes
Origin	Origin of the reported amount
Conversion	FX rate applied to change from the proportional amount to the converted and then consolidated amount
CTP Custom dimension 2	Counterparty custom dimension 2, if the data type is Intercompany.

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

System reports - Original journal rows page

User

System reports tile > Journals > Original journal rows

Navigation panel > Reports & Analysis > System Reports > Original journal rows

Report contents

Shows all the rows of the journals on the original scenario. You can view both the rows of entity journals and the rows of consolidation journals.

Filters and fields on the page

For a description of all the available filters and the fields of this report, see the page dedicated to the rows of original journals (Original journal rows page).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

System reports - Consolidated journal rows page

User

System reports tile > Journals > Consolidated journal rows

Navigation panel > Reports & Analysis > System Reports > Consolidated journal rows

Report contents

Shows all the rows of the journals on the consolidation scenarios. You can view both the rows of entity journals and the rows of consolidation journals.

You can view proportional, converted and consolidable data. The consolidable data represent the entity's contribution to the group consolidated financial statements, for the entity and consolidation journal categories.

Filters and fields on the page

For a description of all the available filters and the fields of this report, see the page dedicated to the rows of consolidated journals (Consolidated journal rows page).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

System reports - Entity Journals - Headers page

User

System reports tile > Journals > Entity journals - Headers

Navigation panel > Reports & Analysis > System Reports > Entity journals - Headers

Report contents

View all entity journal headers

The following information can be viewed from the individual header:

- the original and consolidated rows, with the relative balancing
- any documents attached to the journal

***Note**: you can download the documents attached to the journals locally.*

Filters and fields on the page

For a description of all the available filters and the fields of this report, see the page dedicated to the headers of entity journals (Entity journal headers page).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

System reports - Consolidation journals - Headers page

User

System reports tile > Journals > Consolidation journals - Headers

Navigation panel > Reports & Analysis > System Reports > Consolidation journals - Headers

Report contents

View all consolidation journal headers

The following information can be viewed from the individual header:

- the original and consolidated rows, with the relative balancing
- any documents attached to the journal

***Note**: you can download the documents attached to the journals locally.*

Filters and fields on the page

For a description of all the available filters and the fields of this report, see the page dedicated to the headers of consolidation journals (Consolidation journal headers page).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

System reports - FX Rates page

User

System Reports tile > FX Rates > FX Rates

Navigation panel > Reports & Analysis > System Reports > FX Rates > FX Rates

Report contents

Shows data on FX rates.

Filters on the page

For a description of all the available filters, see the description of the fields in the define FX rates page (FX Rates page in the Consolidation section).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

Page fields

If the **Show reciprocal FX rates** option is active (see the details on general configuration in the Settings window in the General Introduction page), the page will also show the following fields:

- Reciprocal Average FX rate
- Reciprocal Final FX rate
- Reciprocal Period Average FX rate

The values will be the reciprocal amounts of the values inserted in the respective FX rates.

System reports - Overridden FX rates page

User

System Reports tile > FX Rates > Overridden FX rates

Navigation panel > Reports & Analysis > System Reports > FX Rates > Overridden FX rates

Report contents

Shows data on overridden FX rates.

Only available under the following conditions:

- if the in memory consolidation engine is being used
- if the consolidation process is activated and enabled

Filters on the page

For a description of all the available filters, see the description of the fields in the define overridden FX rates page (Overridden FX rates in-memory page in the Consolidation section).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*

Page fields

If the **Show reciprocal FX rates** option is active (see the details on general configuration in the Settings window in the General Introduction page), the page will also show the **Reciprocal FX rate** field, whose value will be the reciprocal amount of the value inserted in the FX rate.

System reports - Notes page

User

System Reports tile > Other > Notes

Navigation panel > Reports & Analysis > System Reports > Other > Notes

Report contents

Shows the explanatory notes on the values entered during data entry activities.

Filters on the page

For a description of all the available filters, see the description of the fields on the insert notes page (Edit Notes page).

Actions on this page

Element	Description
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*
Detail	Only available if the **Save data in form incrementally** option is enabled (see Amount & Intercompany rules page). Shows the history of operations run on the note.

System reports - Comments page

User

System Reports tile > Other > Comments

Navigation panel > Reports & Analysis > System Reports > Other > Comments

Report contents

Shows a list of comments on the data entered during data entry.

If the **Form** field contains a code, this indicates the form from which the comment has been inserted.

You can also insert a comment from the IC cockpit, using the Edit IC function. (IC entry management page in the consolidation section).

Page fields

Element	Description
Comment	Comment inserted during data entry
Scenario	Scenario to which the comment relates
Period	Period to which the comment relates
Entity	Entity to which the comment relates
Account	Account to which the comment relates
Financial Statement Template	FST containing the item to which the comment relates
Description of the document	Description of any documents attached to the comment
Document file name	Name of any documents attached to the comment
Form	If the comment was inserted from a form, this indicates the code of that form.
Custom dimensions	Custom dimensions to which the comment relates
CTP entity	Any counterparty entity to which the comment relates
CTP Custom dimension 2	Any counterparty custom dimension 2 to which the comment relates
Data type	The type of data on which the comment was inserted. This can be one of the following: • Original • Original amount in other currency • Proportional • Converted • Consolidable • Description
Active	Indicates whether the comment is still visible from the form or it has been deleted.
Navigation	Indicates whether the comment is also visible in navigation mode.
Currency	Currency to which the comment relates
Period length	Length of the period to which the comment relates
Origin	Where the comment was inserted from

Actions on this page

Element	Description
> **Download**	Allows you to download the attachments to several selected comments (with multiple edits) in a single .zip file.
Also show archived data	Allows you to also download data stored offline. ***Note**: this operation may take several minutes.*
	Allows you to download any attachments associated with individual comments.

System reports - Intercompany Report window

User

System reports tile > Gross and IC amounts > Intercompany report

Processes tile > select process > Consolidator > Overview tab or Step tab > (Actions) Process reports > Intercompany report

Processes tile > select process > Contributor > Detail summary tab > (Actions) Process reports > Intercompany report

Navigation panel > Processes > select process > Consolidator > Overview tab or Step tab > (Actions) Process reports > Intercompany report

Navigation panel > Processes > select process > Contributor > Detail summary tab > (Actions) Process reports > Intercompany report

Purpose of window

Allows you to set filters to export the desired data related to intercompany entries in a single Excel file.

Report contents

Collects data on the intercompany entries (IC amounts and entity journals) that exist between entities belonging to the same group.

Reports on consolidation scenarios show the proportional amount type data. They show the IC entries that flow into the consolidation area, before the consolidation process. These are therefore still in the entity/transaction currency rather than the consolidation currency. This also allows you to view the IC entries of the entities consolidated at equity which, as they are not consolidated, would not be shown on the consolidated amount type.

Filters in the window

Element	Description
Original scenarios	Allows you to select only scenarios/periods related to original scenarios.
Consolidation scenarios	Allows you to select only scenarios/periods related to consolidation scenarios.
Scenario /Period	Allows you to select an original or consolidation scenario that is active and visible to the user. Contributor users can only select original scenarios/periods.
Report type	• **Synthetic**: the data are grouped by category, declaring entity, custom dimension 2, account and, if sorting by Category/Declaring entity/CTP Entity/Account is selected, also by counterparty entity. A row with the subtotals is inserted for every grouping. • **Analytic**: all the rows of data are reported, without any grouping.

Element	Description
Account nature	Allows you to include in the report only intercompany entries on accounts of the indicated nature: • Balance Sheet • Profit & Loss • Memorandum • Other Variations • Other Stocks
Account type	Allows you to include in the report only intercompany entries on accounts of the indicated type: • Normal • Variation • Detail
Account	Allows you to include in the report only intercompany entries of the selected accounts: ***Note**: only accounts that are consistent with the filters set in the **Account nature** and **Account type** fields are proposed.*
Transaction currency	Allows you to include in the report only intercompany entries in the indicated transaction currency.
Declaring entity	Allows you to include in the report only intercompany entries for the indicated declaring entity or entities. ***Note**: if a consolidation scenario/period is selected, the entities must belong to the consolidation area of the selected scenario.*
CTP entity	Allows you to include in the report only intercompany entries for the indicated counterparty entity or entities. ***Note**: you can also select an entity that is not visible to the user.* ***Note**: if a consolidation scenario/period is selected, the entities must belong to the consolidation area of the selected scenario.*
Also show inverted relationships	If a declaring entity has been selected, the report will include both the intercompany entries in which this entity is the declaring entity and those in which it is the counterparty. If a counterparty entity has been selected, the report will include both the intercompany entries in which this entity is the declaring entity and those in which it is the counterparty.
Category	Allows you to include in the report only intercompany entries on the indicated category. ***Note**: You can select a amount or entity journal category that is visible to the user.*
Elimination/Matching Rules	Allows you to include in the report only intercompany entries on accounts belonging to the indicated matching logic. ***Note**: only one IC matching logic can be selected.*
Sorting	Allows you to set the sorting criteria for the data shown in the report: • Category/Declaring entity/Account/CTP Entity • Category/Declaring entity/CTP Entity/Account

System reports - Ad Hoc Analysis page

User

System Reports tile > Gross and IC amounts > Ad-hoc analysis > Ad-hoc analysis

Processes tile > select process > Contributor > Detail summary tab or Step tab > (Actions) Process reports > Ad-hoc analysis

Processes tile > select process > select scenario and period (if the process is not a single submission process) > select step tab > select entity > (Workflow) if present, click task form > Open Form in Web > Analysis tab

Forms tile > select form > ▶ > insert parameters > Analysis tab

Navigation panel > Reports & Analysis > Reporting tools > Ad-hoc analysis

Navigation panel > Processes > select process > Contributor > Detail summary tab or Step tab > (Actions) Process reports > Ad-hoc analysis

Navigation panel > Processes > select process > select scenario and period (if the process is not a single submission process) > select step tab > select entity > (Workflow) if present, click task form > Open Form in Web > Analysis tab

Navigation panel > Reports & Analysis > Browse all > select report > Analysis tab

Purpose of the page

Allows you to extract any numerical data present in the financial model of a CCH Tagetik application and analyse it using a pivot table, directly from the web interface, or by downloading it as a local MS Excel file.

It may be useful to bookmark this page to be able to run this report with the pre-set filters (see Prepare the work environment in the General Introduction section).

***Note**: if created from a web form, the bookmark opens the source form, not the Ad-hoc analysis form.*

Page content

The page shows several rows, one for each Hierarchy or Grouping to be viewed or filtered.

For the Account dimension, in addition to hierarchies, you can use Control groups, Reclassification FSTs or Forms to specify the accounts to be shown or filtered.

If opened from a process cockpit, the fields for the **Scenario**, **Period** and **Entity** dimensions will be pre-set using those selected in the cockpit.

***IMPORTANT**: this page will not be shown if the report is run from a web form. In this case, the report is generated directly with the filters set in the selected cell or the relative matrix.*

Report contents

The report is generated in Excel format and contains two spreadsheets: data and pivot table.

Sheet	Description
Data	Table of records extracted for the report. In addition to the column containing the amount, one or two columns are presented for every dimension that has been set to generate the report, for which the **Detail** option is active. The number of columns shown depends on how the **Display mode** field is set. The column name is the name of the level, if set in the data form, or a name defined by CCH Tagetik. If the display mode is set to columns, the columns will report the word “(Code”) or “(Description)”. ***Note**: the order of the columns can only be changed in Excel, i.e. only once the report has been generated.*
Pivot table	Pre-set pivot table which displays the extracted data.

Page fields

If not run from a web form, the following fields must be set in order to generate the report:

Element	Description
Dimensions	Allows you to select the dimension and hierarchy (Control Group, Reclassification FST or Form) of the data to include in the report.
Filters	Allows you to select the other elements to include in the report.
Detail	If selected, all the selected elements for which data are registered will be reported to the last level of detail. Otherwise, the data are shown as totals, without being detailed for each individual element.
Display mode	If the **Detail** option is selected, you can specify whether the elements must be shown with the code only, with the description only, or with the code and the description.

Buttons on the page

Button	Description
Next	Closes the window and runs the report generation according to the set filters.
Download	Generates and locally downloads an Excel file with the data extracted based on the set filters.

eXtensible Business Reporting Language

XBRL Reporting

The XBRL format (eXtended Business Reporting Language) is an XML-based language that represents an international standard for the representation and transmission of accounting and financial information.

An XBRL instance is a file in XBRL format. To create an XBRL file, each piece of business data must be linked with the 'XBRL concept' that describes that data. For this purpose, each type of XBRL reporting is accompanied by a taxonomy: the taxonomy is a sort of 'vocabulary' containing all metadata that can make up the economic-financial document.

An XBRL file consists mainly of facts and contexts.

Facts can be divided into two categories:

- *items*: are facts containing a single value
- *tuples*: are facts containing several values. In fact, a tuple consists of several nested items, each containing its own value.

In an XBRL file, each item must be assigned to a context.

The context defines the entity to which the fact applies (e.g. Entoty or group of Entities), the period for which the fact is relevant and an optional scenario. The scenario provides additional information about the event such as the geographical area to which it relates.

 Hereafter, for the sake of simplicity, the facts may be referred to generically as *Item*.

In order to be able to generate an XBRL instance, Tagetik allows the basic information of the taxonomy to be entered into the system and business data to be associated with the XBRL concepts that represent them.

Tagetik allows you to manage the entire logical flow of data production and the final reclassification in XBRL format of the information.

In particular, we can divide the production of the XBRL file into three macro-activities:

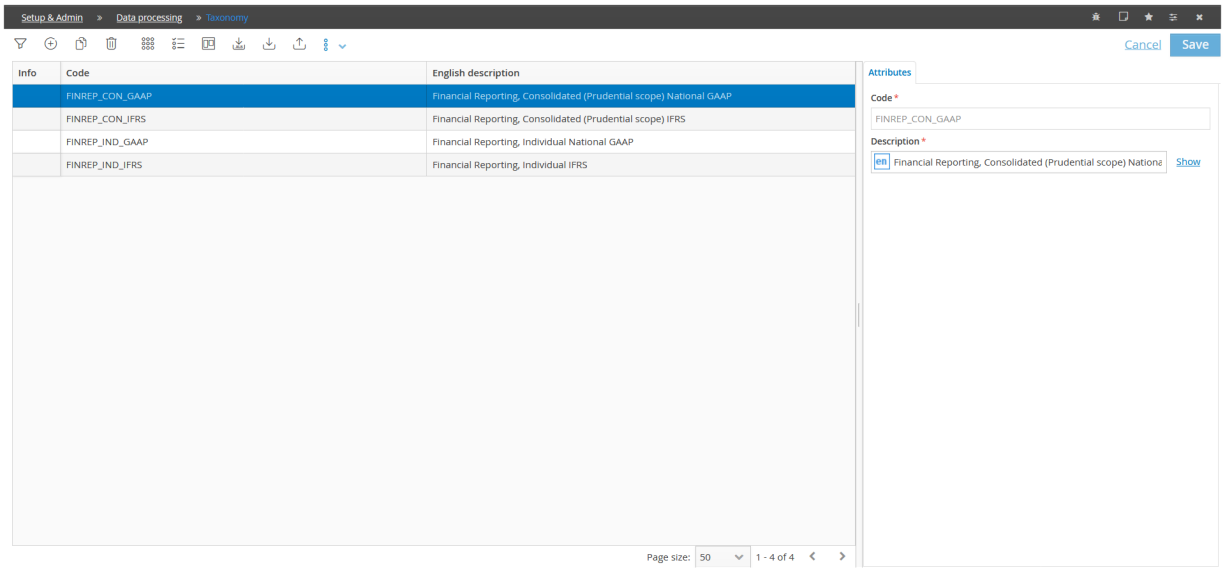
1. the parameterisation of the taxonomy (through the use of the import utility)
2. the preparation of the data to be reported: for example, the balance sheet data in Tagetik can be reclassified in Item/Context pairs, aggregating them in an appropriate manner. There is no built-in process within Tagetik to do this: it depends on the complexity of the specific taxonomy to be reported and the organisation of dimensions within the Tagetik domain. Tagetik provides some pre-package solutions, e.g. for the EBA and EIOPA taxonomies: the former uses the Tagetik APM engine to prepare the data to be reported, while the latter makes use of a stored procedure and ETL.
3. the generation of the XBRL file from the table (XBRL_DATA) containing the data from point 1.

Taxonomy

In Tagetik, the taxonomy represents the set of items, tuples and MD scenarios (multidimensional, also referred to as sub-contexts). These scenarios are used when creating contexts by combining them with the period and entity information.

 In the case of complex taxonomies, such as EBA and EIOPA taxonomies, that include the management of several entrypoints, it is necessary to define a taxonomy code for each entrypoint to be

reported. For example: the taxonomy EBA 2.4.1.1 includes the modules FINREP_IND_GAAP, FINREP_IND_IFRS, FINREP_CON_GAAP, FINREP_CON_IFRS, ecc. For each module it is necessary to define a different taxonomy code.

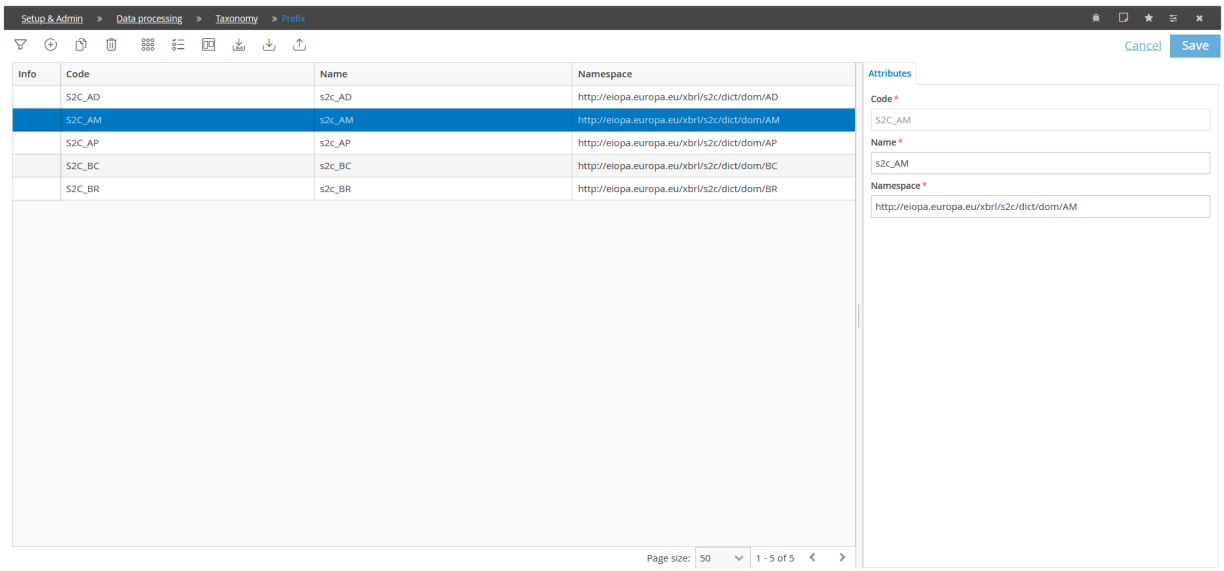


Tagetik can automatically import from specific files any information related to items and MD contexts necessary for the production of XBRL files. Please see import taxonomy and contexts.

The only manual parametrization related to the taxonomy consists in the insertion of prefixes.

Each element of the taxonomy has an "Universal Resource Identifier" (URI) that identifies the organization that maintains the definition of each element: this URI is named *namaspace*. For instance, in the taxonomy EIOPA, the namespaces starts by *http://eiopa.europa.eu/*: in particular, an element of the AM domain has the namespace *http://eiopa.europa.eu/xbrl/s2c/dict/dom/AM*.

The *prefix* (or namespace prefix) is a short sequence that indicates a namespace. In the example shown above, the prefix for the namespace is *S2C_AM*.



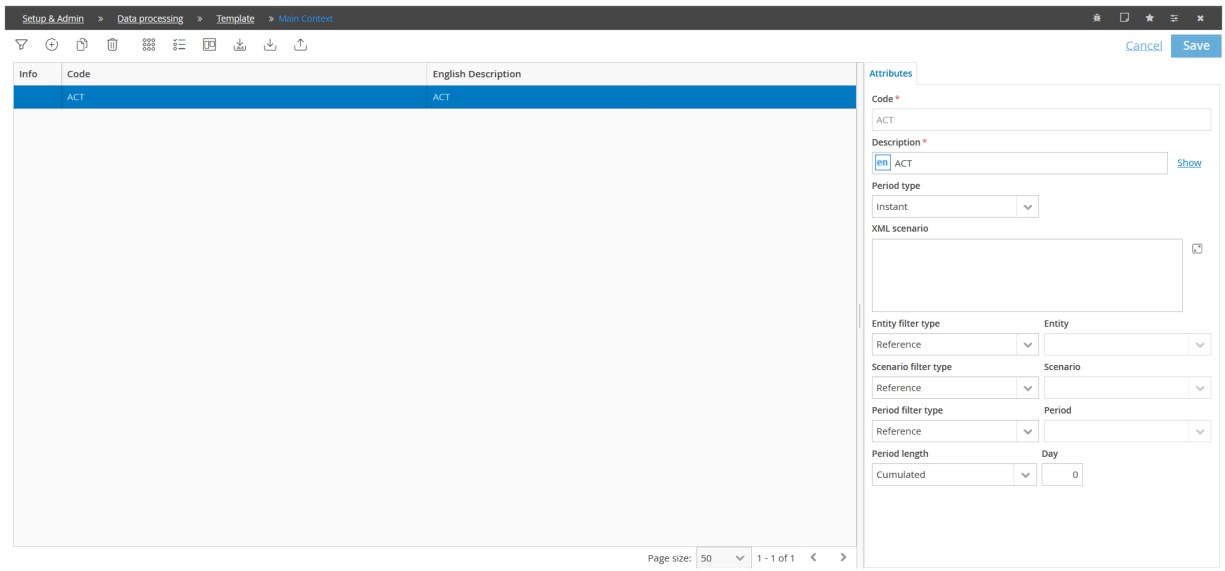
Instance template

The instance templates specify which XBRL contexts can be used and link the Item / context pairs with the metadata of Tagetik.

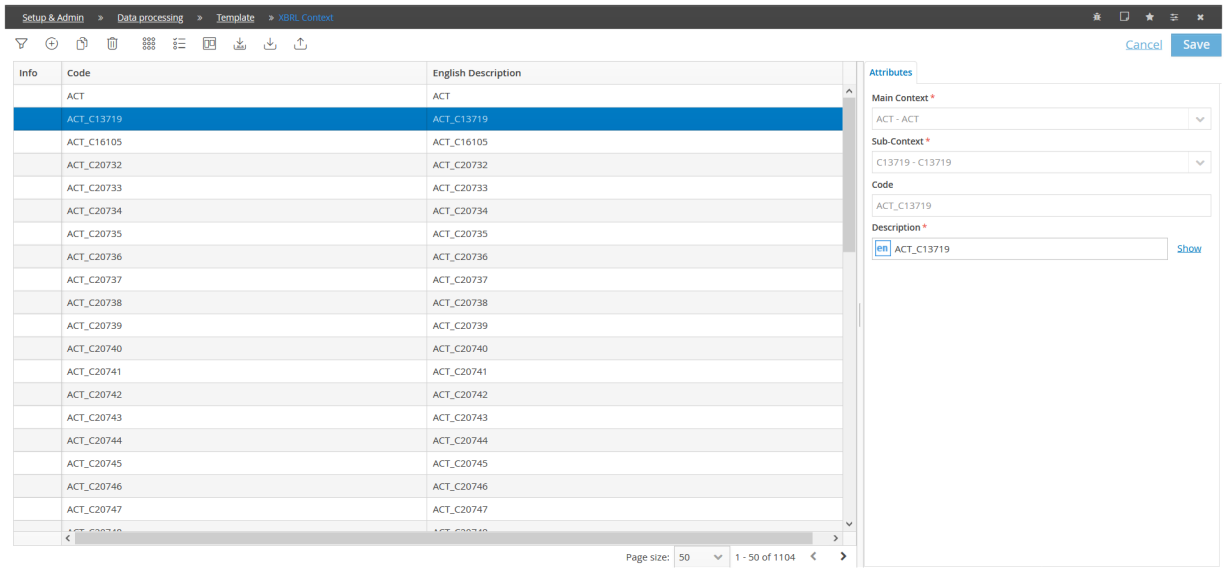
A taxonomy must be related to each instance template.

After creating the instance template, it is possible to import the XBRL contexts by using the import utility. Please see import taxonomy and contexts.

A taxonomy can manage one or more *MAIN*contexts: these contexts are characterized only by the period (that can be *instant* or *duration*) and entity information. A main context doesn't contain any other dimension.



The combination of MD contexts (or sub-contexts) with the main contexts allows for the creation of all possible contexts that can be used for the taxonomy module.



In the MAPPING area, it is possible to associate the item / context pairs, that must be reported when creating the file, with the Tagetik dimension.

Definition of an XBRL instance

In Tagetik, the XBRL file is generated by instance.

The instance, defined in the XBRL Instance table, represents the relationship between the items and the contexts of a given taxonomy and the Tagetik dimensions.

In particular, the instance links the template code (and the related taxonomy) to the scenario, period and entity information of Tagetik.

When creating the XBRL file, the system selects data linked to the scenario/period/entity/template specified on the instance and reclassified in the appropriate pairs item/context (available in the table XBRL_DATA).

 Some important information necessary for the filling of the XBRL file must be specified on the instance, such as:

- CIK: represents the Entity's ID code (for the European taxonomies, it represents the LEI code). This code indicates the reporting Entity to the authority that receives the file.
- CIK Namespace: indicates the URI that identifies the type of code used. For example, in the case of the EIOPA reporting, the CIK Naspace is: *http://standard.iso.org/iso/17442*.
- XBRL taxonomy file name: it must indicate the entrypoint to be reported (file .xsd, for example *http://www.eba.europa.eu/eu/fr/xbrl/crr/fws/corep/its-2015-04/2016-01-31/mod/corep_lcr_da_con.xsd*). When generating the XBRL file, the system inserts this information in header as *SchemaRef*.

If the taxonomy includes the insertion of long parts of text into the file(for example, some sections of the Notes), it is possible to specify on the instance the Collaborative Office document from which this information can be retrieved.

In this case, it is necessary to define on the Collaborative Office document free text objects, where to insert the text that must be reported on the XBRL file. These objects must be tagged with the item and context information.

So, when generating the XBRL file, the system can retrieve the facts to be reported from Tagetik's data (XBRL_DATA) and from the Collaborative Office document.

Generate Xbrl instance

Once data that need to be reported (in the XBRL_DATA table) have been arranged and reclassified in the appropriate item/context, it is possible to generate the XBRL file.

This can be done from the Data Entry & Reporting - XBRL - Instance menu: just select the instance you want to run and then the "Generate XBRL Instance" window menu.

Once the data processing is completed, it is possible to download the xbrl file from the File Management tab.

 In the event that the number of rows to report is very high (approximately over one million), the system may need too much memory space in order to run the file, more than is available in the application. In this case, the generation of the file will fail.

In order to avoid this problem, it is possible let the system generate the xbrl file by applying the "split" procedure: the system splits the archive's data into "separated" blocks and generates one file for each block obtained. Once the procedure is completed, the system gathers automatically the files obtained in one final instance.

In order to run the split procedure, it is possible to set the maximum value of rows in which data must be arranged. This option can be set up in the configuration, in the "Advanced": "Limit for XBRL split" tab. The default value is 1 million of rows.

Import Items and Contexts

This utility enables the import into Tagetik of the list of Items and Contexts of a given Taxonomy.

Import taxonomy

The import of a taxonomy allows for the import of the list of items. In the case of complex taxonomies, such as EBA and Eiopa, the import must be performed by entrypoint (or module, foe example, by EBA, FINREP_IND_GAAP, FINREP_IND_IFRS, FINREP_CON_GAAP, FINREP_CON_IFRS, etc.).

For the purposes of import , it is necessary to provide the system with:

- the cache file, namely the zip file containing all the folders that compose the taxonomy. Typically, this file can be downloaded from the website of the competent authority (for example, the website of EBA or EIOPA or of the Chamber of commerce for the Italian taxonomy).
- the taxonomy file, namely the xsd file of the taxonomy (or of the module), conveniently modified. This file is provided by Tagetik.

Import contexts

The import of contexts allows for the import of the list of sub-contexts (or MD contexts), Main contexts and XBRL contexts related to the taxonomy (or module of the taxonomy).

Moreover, it is necessary to provide the system with an appropriate xml file, which is supplied by Tagetik.

Tuple management

Some taxonomies include the report of tuples.

A tuple is a fact containing several values. Specifically, a tuple is made up of several nested items, each one containing its own value.

Two types of typle are available:

- the same tuple appears in more occurrences, with a dimension which serves as discriminating factor. In **Example 1**, the tuple 1 has three occurrences in which the discriminating factor is the geographic area, represented by the item 0:

```
<itcc-ci:tuple1>
<itcc-ci:item0 contextRef="Context-3" id="Item-390">Italy</itcc-ci:item0>
<itcc-ci:item1 contextRef="Context-3" decimals="0" id="Item-391" unitRef="U001">103</itcc-ci:item1>
<itcc-ci:item2 contextRef="Context-3" decimals="0" id="Item-392" unitRef="U001">303636</itcc-ci:item2>
<itcc-ci:item3 contextRef="Context-3" decimals="0" id="Item-393" unitRef="U001">3306285</itcc-ci:item3>
</itcc-ci:tuple1>
<itcc-ci:tuple1>
<itcc-ci:item0 contextRef="Context-3" id="Item-398">Japan</itcc-ci:item0>
<itcc-ci:item1 contextRef="Context-3" decimals="0" id="Item-399" unitRef="U001">100</itcc-ci:item1>
<itcc-ci:item2 contextRef="Context-3" decimals="0" id="Item-400" unitRef="U001">200</itcc-ci:item2>
<itcc-ci:item3 contextRef="Context-3" decimals="0" id="Item-401" unitRef="U001">0</itcc-ci:item3>
</itcc-ci:tuple1>
<itcc-ci:tuple1>
<itcc-ci:item0 contextRef="Context-3" id="Item-398">USA</itcc-ci:item0>
<itcc-ci:item1 contextRef="Context-3" decimals="0" id="Item-399" unitRef="U001">300</itcc-ci:item1>
<itcc-ci:item2 contextRef="Context-3" decimals="0" id="Item-400" unitRef="U001">200</itcc-ci:item2>
<itcc-ci:item3 contextRef="Context-3" decimals="0" id="Item-401" unitRef="U001">10</itcc-ci:item3>
</itcc-ci:tuple1>
```

Example 1

- if the tuple needs one occurrence only: for example, it contains information such as the address of the reporting company. In this case, the file will contain one single tuple, like in **Example 2**:

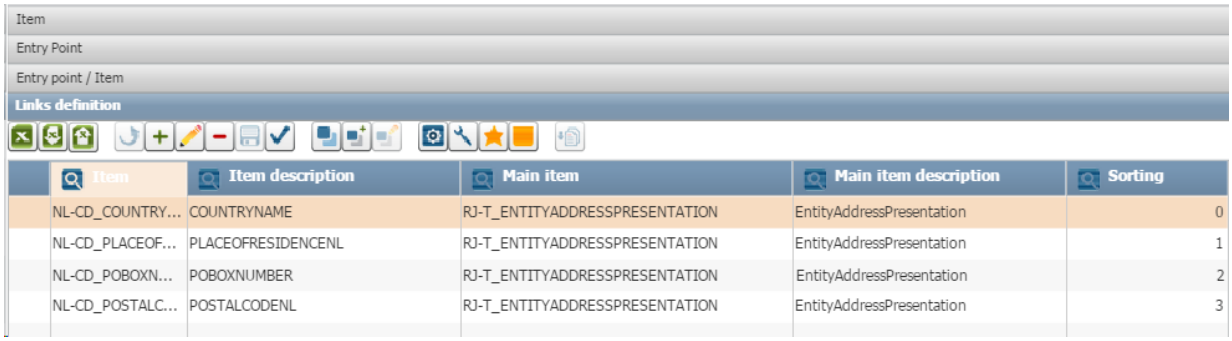
```
<rj-t:EntityAddressPresentation>
    <nl-cd:POBoxNumber contextRef="FY16d">2312</nl-cd:POBoxNumber>
    <nl-cd:PostalCodeNL contextRef="FY16d">2501CD</nl-cd:PostalCodeNL>
    <nl-cd:PlaceOfResidenceNL contextRef="FY16d">Den Haag</nl-cd:PlaceOfResidenceNL>
    <nl-cd:CountryName contextRef="FY16d">Nederland</nl-cd:CountryName>
</rj-t:EntityAddressPresentation>
```

Example 2

The system allows the management of both cases. Below we describe how to manage them.

The tuples are defined manually within the system. After importing the items that compose the taxonomy, the list of tuples can be found by setting the filter "substitution group" to "tuple" in the "Data Entry & Reporting - XBRL - Taxonomy- Item" window menu.

From the usual menu, the "Link definition" tab allows you to define the items that compose each tuple and their sorting (sorting according to which the system will insert the items into the tuple when generating the file).



Item	Item description	Main item	Main item description	Sorting
NL-CD_COUNTRY...	COUNTRYNAME	RJ-T_ENTITYADDRESSPRESENTATION	EntityAddressPresentation	0
NL-CD_PLACEOF...	PLACEOFRESIDENCENL	RJ-T_ENTITYADDRESSPRESENTATION	EntityAddressPresentation	1
NL-CD_POBOXN...	POBOXNUMBER	RJ-T_ENTITYADDRESSPRESENTATION	EntityAddressPresentation	2
NL-CD_POSTALC...	POSTALCODENL	RJ-T_ENTITYADDRESSPRESENTATION	EntityAddressPresentation	3

After defining the items that compose the tuple, it is necessary to prepare data that will fill the XBRL_DATA table.

This table includes the PARAM_TUPLA field.

If the tuple has only one occurrence (without discriminating factor), this field won't have to be defined. Just insert the value of all items that compose the tuple.

On the contrary, if the tuple has more than one occurrence, then it will be necessary to define the PARAM_TUPLA field with the discriminating value of each occurrence, for each item of the tuple, except the item with sorting 0 (this item represents the information that distinguishes the occurrence of the tuple). The system will find the item with sorting 0 of the tuple and will assign it the value shown in the field PARAM_TUPLA, for each occurrence. So, in the case of Example 1, it is necessary to insert into the XBRL_DATA table the Item1, item2 and item3 with their values for the Italy occurrence and in the PARAM_TUPLA field the "Italy" value for each one.

For the second and third occurrence, it is necessary to proceed in the same way, by setting to PARAM_TUPLA, the values "Japan" and "USA".

Analytics

The Analytics allow you to create Microsoft SQL Server Analysis Services (SSAS) cubes configured and generated via CCH Tagetik.

The Microsoft SQL Server Analysis Services (SSAS) cubes provide On-Line AnalyticalProcessing and data mining functions allowing for a simple and quick analysis of huge amount of data.

Such a simple and quick analysis of huge amount of data is made possible using a dedicated OLAP database in which data contained in the system at a specific time are duplicated and transformed into pre-aggregated multidimensional structures called cubes.

Thus a cube is a set of data organised in a specific structure defined by a set of measures and dimensions.

The analytics can be created through the definition of "Tagetik Models" or the launch of "XMLA Commands" that can be managed by accessing the "Analytics" tile in the **Home page - Administrator**.

Tagetik Models

A Tagetik Model corresponds to a Microsoft SQL Server Analysis Services (SSAS) Model which is created and managed through the admin interface of CCH Tagetik.

The Tagetik models configuration page can be accessed via the "Analytics" tile in the **Home page - Administrator**Homepage.

A model is actually created as a SSAS model as soon as the first deploy is run.

When a deployed model is removed from CCH Tagetik, it will be removed from the SSAS database as well.

Each model is identified by a Code and a Description and is characterized by a set of attributes defining how the SSAS model has to be generated.

Attributes

The Attributes panel contains the generic parameters that define the model:

- Code. ID code of the model.
- Description. Description of the model.
- Name of the SSAS Database that corresponds to this model
- Cube. Name of the cube for this SSAS model
- Service Analysis Services. This is the TGK Service resource to be used for deploying the model. The TGK Service is set up in the Repository.

The following details have to be specified in the "Advanced settings" area of the panel:

- **Language**. Language of the description of the dimensions' elements.
- **Template**. Specifies if the model must be based on the CCH Tagetik standard template or on a custom template (from Analysis Services): in this case, it is necessary to specify the Template's name, namely the name of the SSAS database for the creation of the model.
- **Users definition**. When the standard template is used (see above), it is possible to see a list of the domain users who can access the cube. The user must be shown as domain@user; to allow all users to navigate the cube, it is possible to add “everyone”, namely the keyword of the active directory meaning "all users". This option is not available when a template from Analysis Services is in use: in this case, the user access can be set up directly inside the template.
- "**Disable Optimize sum formulas in Financial Statement Templates**".

This option indicates the calculation method of the formula FST items whose formula is a sum of elements of account type.

If this option is disabled, such items are calculated as aggregations of elements of account type (typically, this calculation has a better performance).

If this option is enabled, such items are calculated via MDX script of SSAS cube.

The management of the time dimension in the SSAS cube can be set up in the "Cube type" area.

- **Time – Version**. With the Time – Version cube, the management of the time dimension is based on a calendar dimension based on the "End date" defined inside the periods of CCH Tagetik and on the "Version" defined on the Scenario of CCH Tagetik.
- **Scenario - Period**. The time dimension is managed through the Scenario and Period dimensions set up in CCH Tagetik. The cube contains also the "Reference Scenario" and "Period length" dimensions.
- **Aggregation over time** for Stocks and BS accounts. This is enabled only if the selected cube is Time – Version type and determines the aggregation type on the Stocks and BS accounts. This can get the following values:
  - First Last period. The values taken into account are those of the first and of the last selected periods (even though they are not filled in).
  - Defined First Last period – Null values filled in with 0 on the periods opened in Tagetik. The values taken into account are those of the fist and of the last periods that are not empty (by filling the null values with a zero for all the periods opened in Tagetik)
- **Partitions advanced management**. Selecting it enables the advanced management of the partitions. The user can define the partitions by setting the filters on the scenario and the data types (original, consolidated and offline) that will populate the cube. When this is selected, it is no longer possible to insert the filter on the Scenarios grouping as they will be specified in each partition.
- **Scenarios Grouping**. This is enabled only if the Partitions Advanced Management is not enabled (see above). It determines which scenarios grouping contained in the Tagetik list will populate the time dimensions and the subset of data. Only grouping's nodes that do not contain multiple relationships can be chosen.
- **Category grouping**. Categories grouping contained in the Tagetik's List that will populate the categories dimensions and the sub-set of data.

The Attributes panel shows some additional information on the model:

- **Last change**. Date of the last modification to the model.
- **User**. User who made the last modifications to the model.
- **Last Deploy**. Date of the last deploy of the model.

Partitions advanced management panel

This panel is available only if the "Partition Advanced management" option has been enabled in the "Attributes" panel when saving the model.

You can add further partitions according to the version of SSAS in use: with the Standard Edition you can create up to 3 partitions, whereas with the Enterprise edition you can create as many partitions as you like. For each partition, it is necessary to specify a code, a description, a filter on the scenarios groupings and the type of data to be inserted, by selecting the following flags:

- Original Open. Original open data will be inserted in the partition.
- Original Closed. Original closed data will be inserted in the partition.
- Original Archive. Original offline data will be inserted in the partition.
- Consolidated Open and Closed. Consolidated, open and closed data, both amounts and adjustments, will be inserted in the partition.
- Consolidated Archive. Consolidated offline data, both amounts and adjustments, will be inserted in the partition.
- Open Original data adjustments. Adjustments' open original data will be inserted in the partition.

- Closed Original data adjustments. Adjustments' closed original data will be inserted in the partition.
- Archive Original data adjustments. Adjustments' offline original data will be inserted in the partition.

A specific check will prevent data from being duplicated inside the cube, therefore it is not possible to insert the same type of data for the same scenario in different partitions.

Calculations and Measures panel

In this panel you can set up the dimensions for the calculations (e.g. for the On-the-fly Conversion) and the measures you want to be included in the model, as well as their name.

It is also possible to insert a MDX script that will be added to the MDX script of the SSAS cube: the script must be written with the syntax requested by SSAS, by following the names given to the various dimensions and measures of the model.

Conversion Panel

In the Conversion Panel you can set up some dimensions for the currency conversion of data.

The settings contained in the panel vary depending on the "Cube type" displayed in the "Attributes" panel.

A "Time Version" model contains the following dimensions:

- The "Conversion" dimension that specifies the currency to which the amounts have to be converted
- The dimensions for the "Conversion at specific FX rate", namely "Reference currency", "Reference time" and "Reference version".

Only the "Conversion" dimension is used in the cube to convert the analysed data to another currency.

The "Reference Currency", "Reference Time" and "Reference Version" dimensions are used in the cube to convert the analysed data to another currency but at a FX rate that refers to a specific moment in time.

A "Scenario Period" Database contains the following dimensions:

- The "Conversion" dimension that specifies the currency to which the amounts have to be converted
- The dimensions for the "Conversion at specific FX rate", namely "Reference Scenario" and "Reference Period"

Only the "Conversion" dimension has to be used in the cube to convert the analysed data to another currency.

The "Conversion", "Reference Scenario" and "Reference period" dimensions are used in the cube to convert the analysed data to another currency but at a FX rate that refers to a specific moment in time.

To convert the analysed data to the data's currency at a FX rate that refers to a specific moment in time , it is necessary to use only the "Reference Scenario" and "Reference Period" dimensions and not the "Conversion" dimension.

Dimensions panel

In the remaining panels you can set up the dimensions you want to be included in the model, as well as the hierarchies, their names and other specific attributes for each dimension.

Deploy Action

The deploy data processing creates the model on the SSAS server when it is run for the first time for a given Tagetik model. It updates the SSAS model when it is launched following the modification of the Tagetik model.

Any modification made in the Tagetik models administration page won't affect the SSAS model until the execution of the deploy data processing.

Process Action

The process data processing of a model updates the SSAS cube of that specific model.

It can be run only after at least one execution of the deploy data processing.

If the "Only partitions" option is selected, only the values in the cube are updated but not the dimensions. This reduces the process time but it is achievable only if no changes have occurred in the dimensions contained in the cube since the last time the process data processing has been run.

If the model has the "Partitions Advanced Management" option enabled in the "Attributes" panel, the process data processing can be run also for just one or a specific set of partitions.

Download MDXScript Action

The model process data processing, besides updating the cube's data, creates the corresponding MDX script. Through the "Download MDXScript" action it is possible to download to a text file the MDX script of the cube generated by the process data processing, for example for test purposes or for the development of custom MDX calculations to be added to the cube through the "Customized Script MDX" field available in the "Calculations and Measures" panel.

Magazine

Contents

This section covers the following topics:

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Magazine

Introduction

Magazine is a CCH Tagetik reporting solution which allows users to create periodic publications on specific topics (e.g. KPI Analysis, revenues analysis), using forms. It works according to a subscription model, in which publications are automatically sent to interested users.

Publications

The forms are *published* by manually or automatically running the **Magazine_issue_creator** data processing.

In particular, this data processing makes it possible to perform the following activities:

- periodically check the magazines to be published
- run the forms on which the magazines are based and make them available, for each subscribed user

The publication start and end dates and specific issue parameters can be set in the data processing settings.

User roles

The user roles associated with the Magazine solution are outlined below.

User	Description
Standard user	Users authorised for the *MAGAZINE_USER* role can access the **Magazine Library** page, where users can view the catalogue of available Magazines and subscribe to them. Based on their user rights, users will only see publications for which they are authorised.
Magazine Administrator	User authorised for the *MAGAZINE_ADMIN* role. This user can create and edit magazines and delete issues of them. The Administrator user can edit the structure of the magazine forms, but only until publication.

For more details on the functions of user roles, see I ruoli.

Working with magazines

Add the Magazine Library tile

User | <application database>

Click > > **Magazine Library**: the tile will be added to the homepage.

View the list of magazines

User | <application database>

Magazine Library tile

Subscribe to a magazine

Magazine Library tile

1. Click the **Catalogue** tab: the list of available magazines will appear.
2. Click the magazine to subscribe to: the **SUBSCRIBED** label will appear and a new tile with the title and cover of the magazine to which the user has subscribed will appear on the homepage.

Consult magazine publications

User | <application database>

Do the following:

If you want to consult...	Then...
the most recent publication	on the homepage, click the desired magazine's tile: the most recent publication will open.
a previous publication	1. On the homepage, click the lower part of the desired magazine's tile: the list of available publications will appear. 2. Click the desired publication.

Cancel a subscription

User | <application database>

1. Click . The tile management buttons will appear on the tiles.
2. Click ✖ on the specific magazine's tile.

Manage existing magazines

User | <application database>

Magazine Library tile

1. Click the **Administration** tab: the list of magazines will appear.
2. To manage an individual magazine, use the buttons in the corresponding box.

Create a magazine from the interface

User | <application database>

1. Click the **Administration** tab: the list of magazines will appear.
2. Click the box to create a magazine; the **New Magazine** window will appear.
3. Complete the fields and click **OK** to save.

Create a magazine in web forms via the Contributor Cockpit

User | <application database>

Processes tile > select process > Contributor

Navigation panel > Contribute > Processes > select process > Contributor

A single magazine can be created from a form opened in browse mode. For forms opened in data entry mode, the create button is disabled.

1. Select the tab for the desired step.
2. In **Organization**, select the desired entity.
3. To open the form:

If the desired form...	Then...
is associated with a task	click the task containing the form, then right click and select **Open Form in Web**.
is not associated with a task	In **Actions**, click Form and, next to the desired form, click to open it in the web interface.

4. Click the **Utility** tab and then click : the window to set the magazine's properties will open.

Create a magazine in web forms via the Forms tile

User | <application database>

Forms tile > click grouping of forms > (next to form)

Forms tile > Reporting > Forms grouping (next to form)

Click the **Utility** tab and then click : the window to set the magazine's properties will open.

Magazine Library page

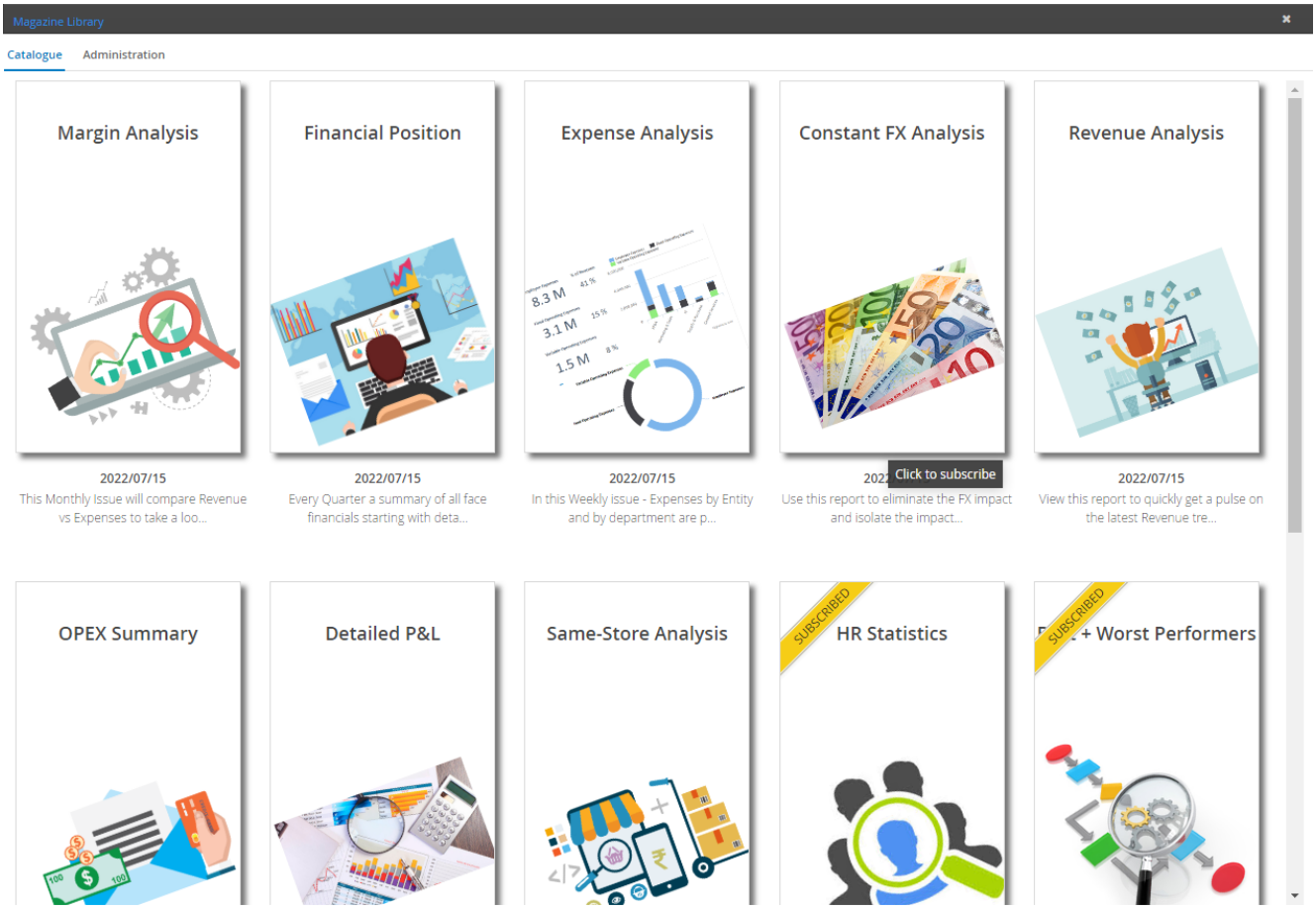
User | <application database>

Magazine Library tile

Purpose of the page

This page allows the user to access the catalogue of magazines and also allows **Magazine Administrator** users to manage magazines.

What it looks like



Element	Description
Catalogue	• This shows the available magazines and those to which the user has already subscribed. • It allows the user to subscribe to other magazines and cancel current subscriptions. See "Catalogue tab in Magazine Library" below.
Administration	(Only for **Magazine Administrator** users) It allows the user to manage magazines. See "Administration tab in Magazine Library" on the next page.

Catalogue tab in Magazine Library

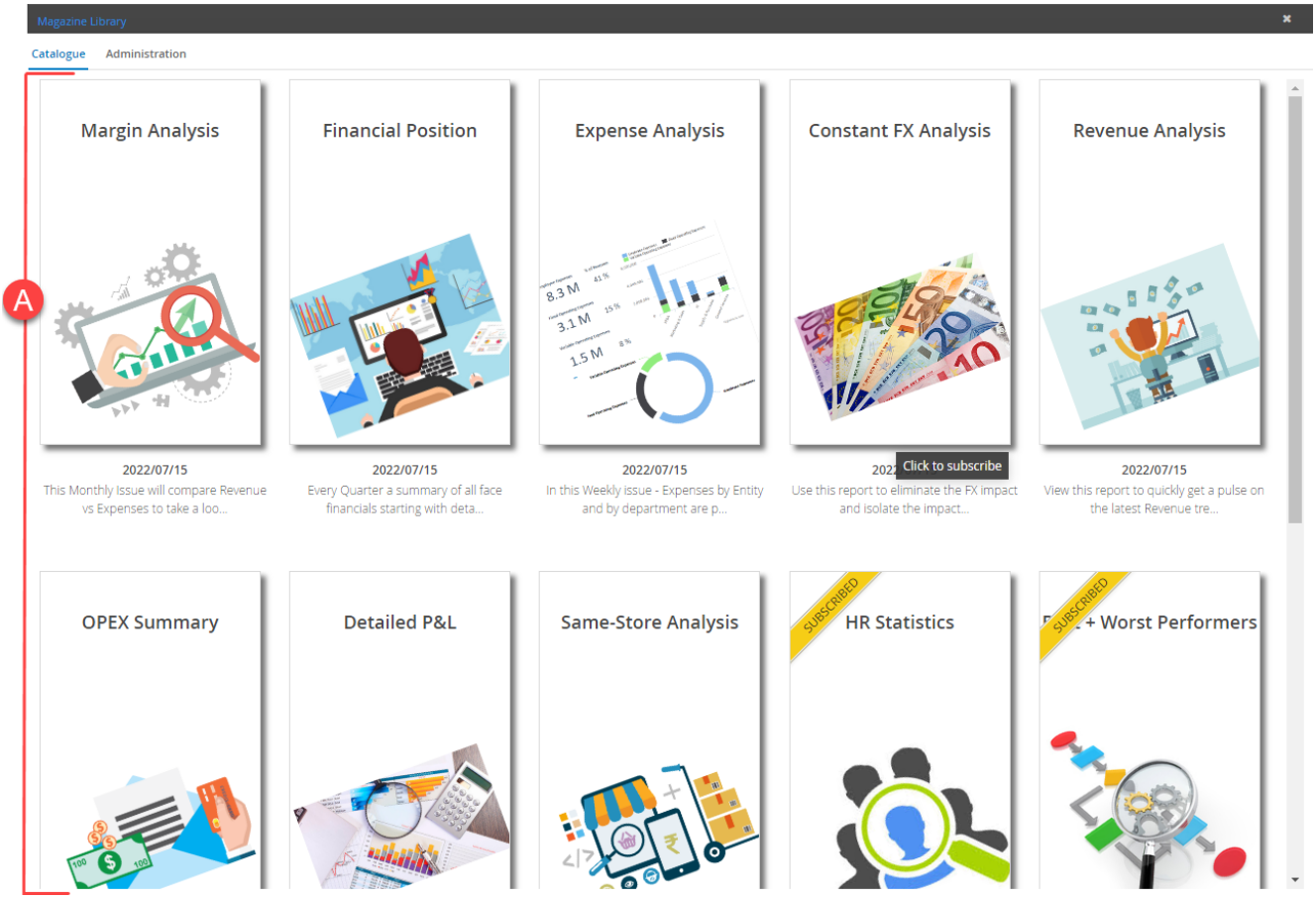
User | <application database>

Magazine Library tile

Tab purpose

This tab displays the available magazines and allows the user to create or cancel subscriptions.

What it looks like



Element	Description
A	Available magazines
SUBSCRIBED	Label to identify the magazine to which the user has subscribed

Administration tab in Magazine Library

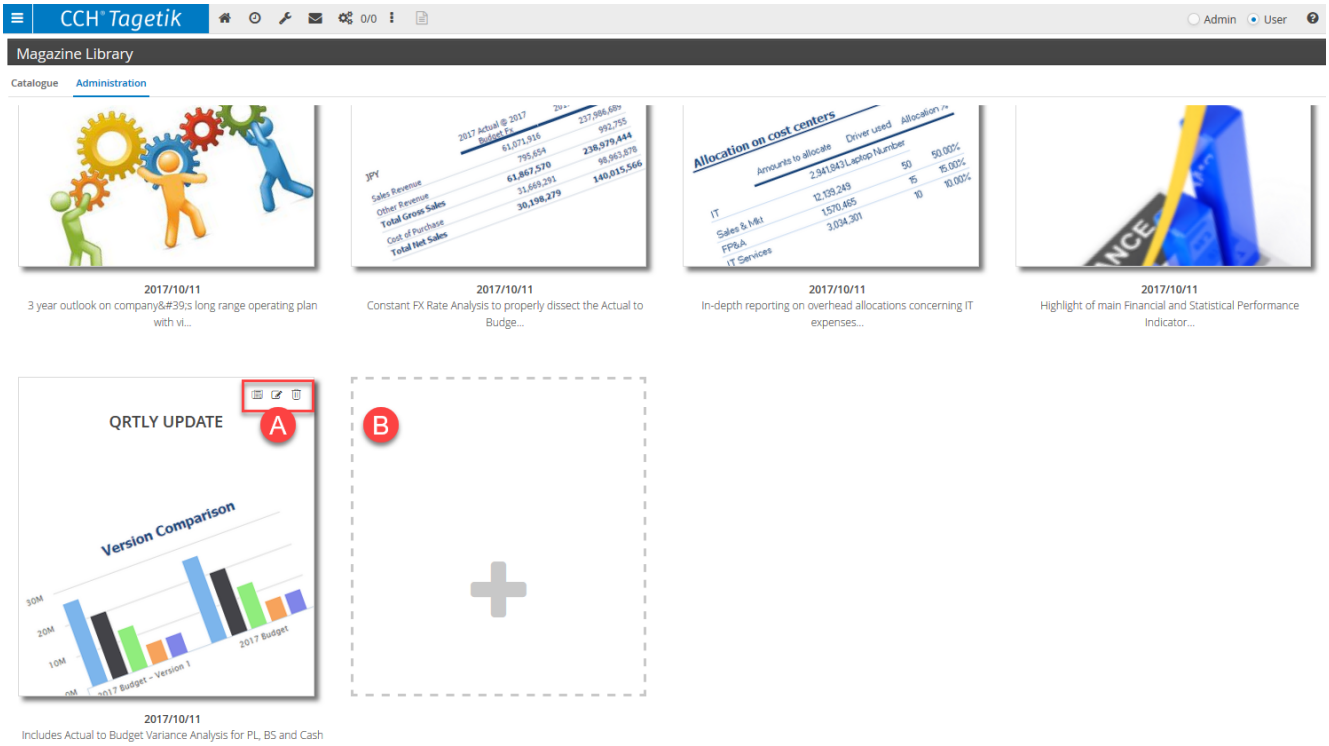
User | <application database>

Magazine Library tile > Administration tab

Tab purpose

This tab allows administrator users to manage magazines.

What it looks like



Element	Description
A	Buttons to manage the magazine
B	Box to create a magazine

Buttons

Element	Description
	**Manage issues**. Allows the user to manage magazine publications. See "Manage issues page" below.
	**Edit magazine**. Allows the user to edit the properties of the magazine. The title, description and cover image can only be edited if no subscribed users are active. The associated report can only be edited if there are no publications in the release phase.
	**Delete magazine**. Allows the user to delete a magazine, including the publications associated with it an any scheduled processing still pending completion. Users subscribed to the magazine can no longer consult any publication.

Manage issues page

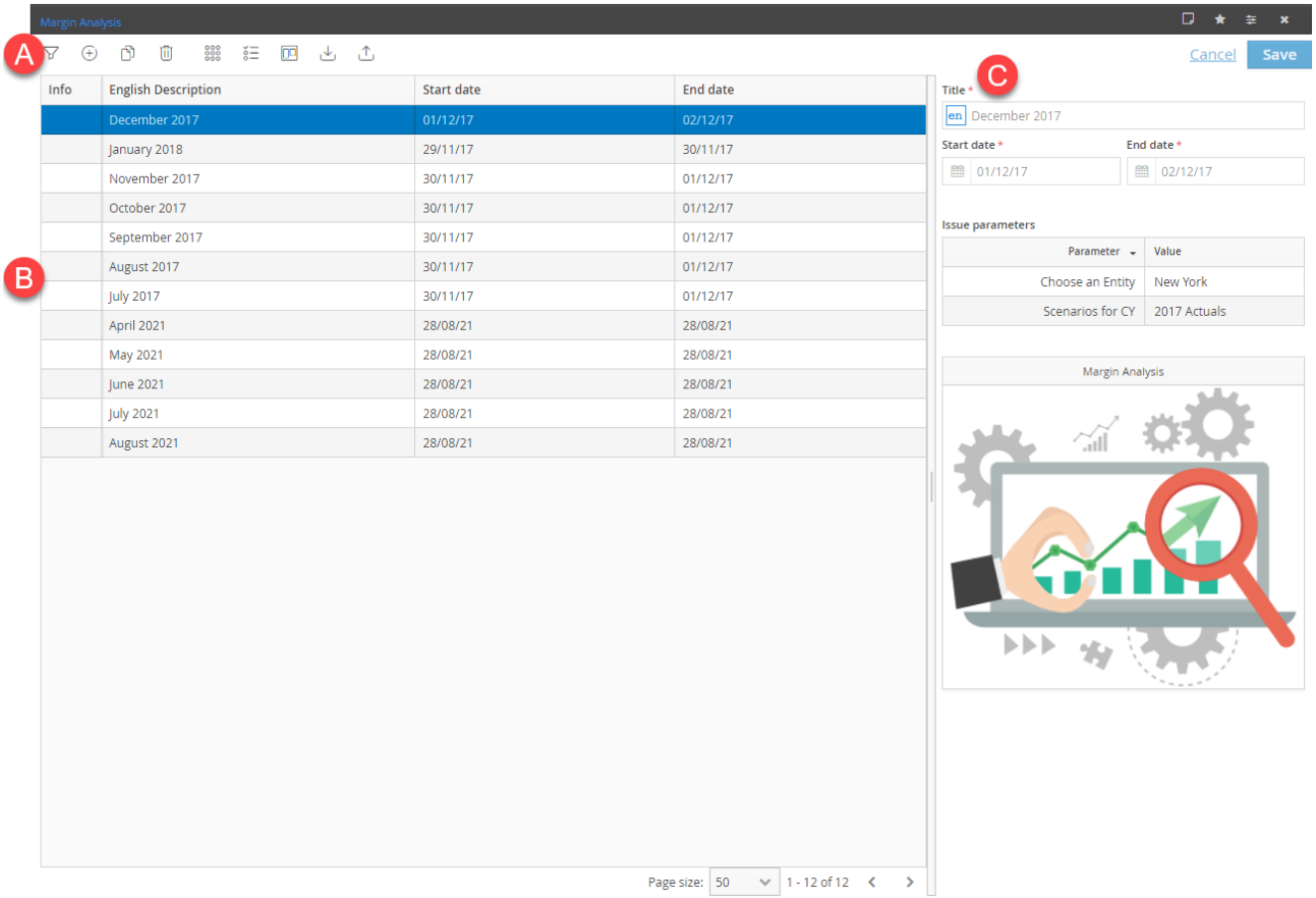
User | <application database>

Magazine Library tile > Administration tab > on the desired magazine

Purpose of the page

This page allows the user to view, create, edit and delete magazine publications.

What it looks like



Element	Description
A	Tool bar
B	List of available publications
C	Details of selected publication

Specific actions on the page

Element	Description
	Delete the selected publication and previous issues. Users subscribed to the magazine can no longer consult the publication, even if it has already been published.

Page fields

***Note**: every time the data processing is run in the period from the start date to the end date, the system overwrites the result of the previous data processing.*

***IMPORTANT**: when the publication end date is reached, a new issue will need to be created to edit a magazine.*

Field	Description
Title	Publication title
Start date	Date on which the publication creation data processing starts to run the magazine form for each subscribed user.
End date	Issue date of the definitive version of the publication for all subscribed users.
Choose parameters	Allows the user to set any form parameters. Whether parameters are obligatory or not will depend on whether they are set as obligatory in the form associated with the magazine. The button is no longer available when the parameters have been set.
Issue parameters	Lists the chosen parameters. They are not editable, but it is possible to create a new issue with different parameters.